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Draft Preparatory study on iron and steel – ecodesign measures under the ESPR

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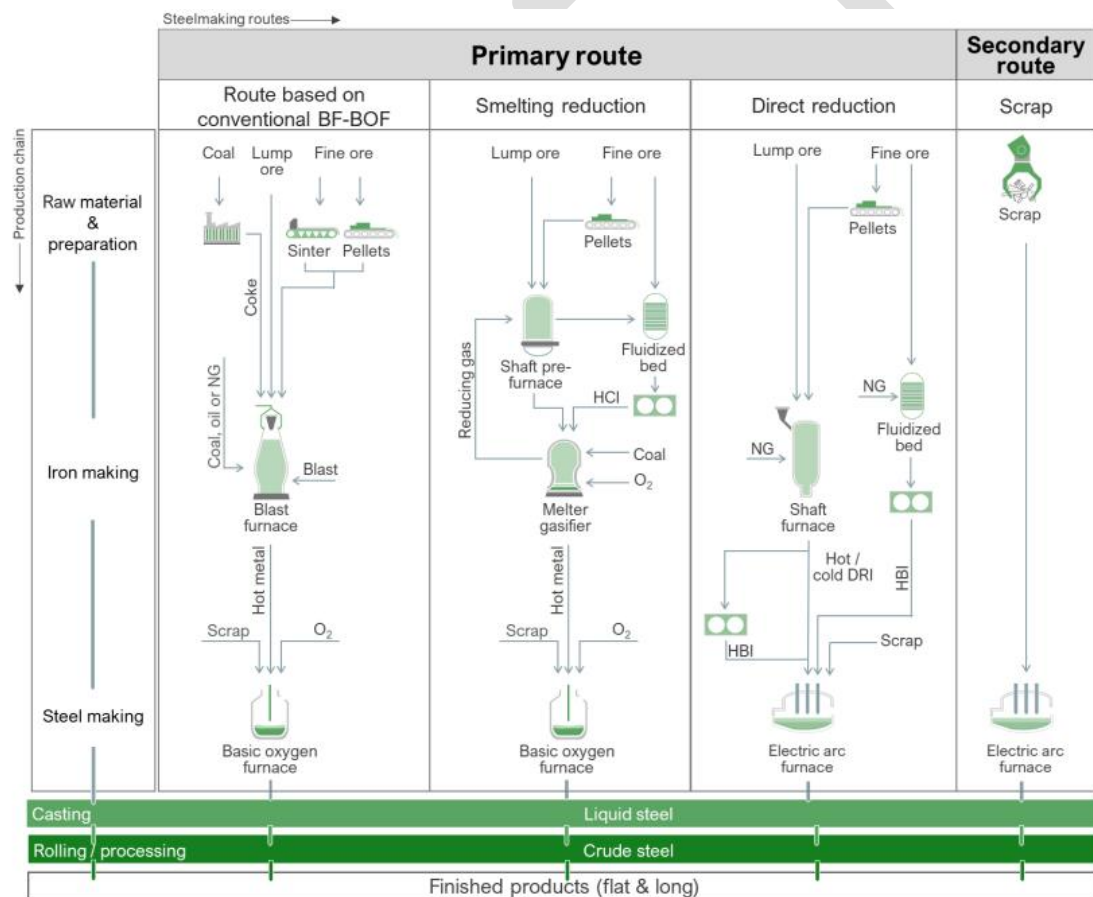
TASK 1: SCOPE, LEGISLATION AND STANDARDISATION

1.1 Product scope and definition

Steel is an alloy of iron and carbon, it contains by mass more iron than any other single element, having a carbon content generally less than 2.11% and containing other elements (when the carbon content is over 2.11%, the material is defined as cast iron). A limited number of chromium steels may contain more than 2.11 % of carbon, but 2.11 % is the usual dividing line between steel and cast iron (CEN, 2000) (Wente, Wondris, and Nutting, 2023) (Pero-Sanz Elorz, Quintana Hernández, and Verdeja González, 2017).

Steel is made from iron ore, a mineral of iron, oxygen and other elements. The raw materials used in steelmaking are extracted through mining and subsequently processed to create steel. Traditional steelmaking encompasses two main routes, as illustrated in Figure 1: the **primary route**, where steel is produced from virgin iron ores, and the **secondary route**, where steel is made from scrap melting. The primary route comprises the classic integrated blast furnace and basic oxygen furnace route (BF-BOF), smelting reduction (smelting reduction plant and BOF) and direct reduction (DR), which requires a direct reduction plant and an electric arc furnace (EAF). Conversely, the secondary route produces crude steel by recycling steel scrap in the EAF (Draxler et al., 2021).

Figure 1. Traditional Steelmaking Routes



Source: Green Steel for Europe – Technology Assessment and Roadmapping (Draxler et al., 2021)

The output resulting from the processes described above is **liquid crude steel**, which is the immediate hot molten steel product of primary metallurgy in the steel melting shop process. In a second stage of the steelmaking process, liquid crude steel is adjusted to the customers' requirements by specific secondary metallurgy processes such as e.g. alloying or degassing and subsequently cast either into ingots or by means of continuous casting, obtaining **crude steel**¹, which is defined as steel in its first solid (or usable) form: **ingots**,

¹ Crude steel is the term used to describe the first solid state of steel after it has solidified from its liquid state. For statistical purpose, crude steel covers ingots, continuously cast semi-finished products as cast and liquid steel for production of casting (EUROFER, s.d.).

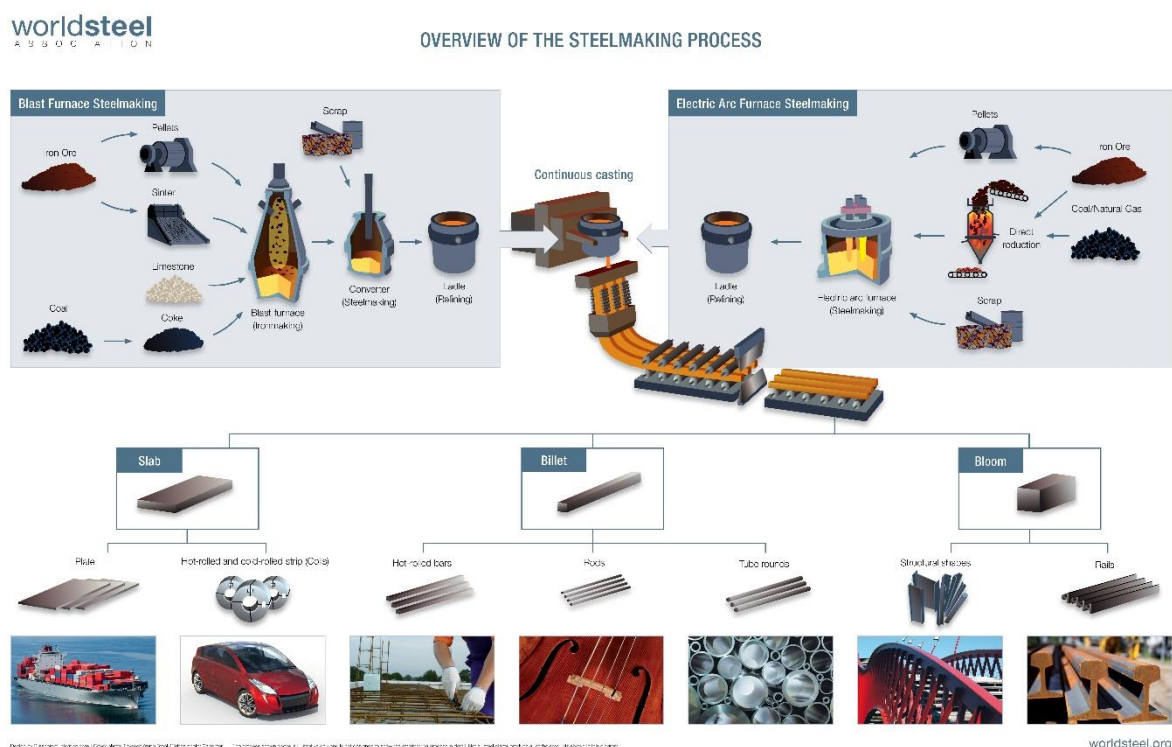
semi-finished products (billets, blooms, slabs), and **liquid steel for castings** (World Steel Association, 2022).

Afterwards, steel is subsequently processed in rolling mills and finishing lines, obtaining **finished products** ready to be sold in the market (Remus et al., 2013). There are distinct finishing processes involved in shaping and treating steel products. These processes encompass, inter alia, hot rolling, cold rolling, wire drawing (wire mill), continuous hot dip coating. Task 3 of the present study is dedicated to an in-depth analysis of the production routes for steel finished products, providing a more comprehensive description of each process within the ferrous metal processing sector.

Finished products are usually classified in two broader categories, **flat and long products**, each characterized by specific dimensions, shapes, and applications. Flat products, including sheets, plates, and coils, have a width equal to or greater than their thickness and are often produced through hot and/or cold rolling processes, resulting in defined dimensions with smooth and consistent surfaces. On the other hand, long products, such as bars, rods, wires, and structural sections like beams and rails, have lengths significantly exceeding their cross-sectional dimensions. They are produced via continuous casting or rolling techniques. Flat products may undergo additional treatments like continuous hot dip coating to enhance their properties or resistance to corrosion. Some additional treatments, e.g. batch galvanising or painting, may be applied to the end-use product that is manufactured from the finished steel product.

Figure 2. presents some representative finished products for the three categories of semi-finished products, including its end-use application sectors, inter alia, ship building, automotive, general engineering, building and construction, railway, and tubes.

Figure 2. Semi-finished and finished steel products



Source: Overview of the steel production process (WorldSteel, s.d.)

In conclusion, the world of steel products is incredibly diverse, encompassing a vast array of shapes, surface treatments, and mechanical or chemical properties. To illustrate this diversity, only considering their chemical properties, there are over 3,500 different grades of steel. Annex 1 will delve deeper into the subject, exploring different classifications and categorisations systems and enabling a more comprehensive understanding of the steel products available within the EU market.

1.1.1. Glossary of terms

Table 1 provides a glossary of steel-related terminology curated to serve as a valuable reference tool within the context of this preparatory study. This compilation of terms spans a wide spectrum of concepts integral to the steel industry, offering clear and concise definitions.

Table 1. Glossary of steel-related terms

Term	Definition
Iron ore	The primary raw material in the manufacture of steel, typically containing iron oxides mixed with other elements.
Pig iron	The product that results from smelting iron ore with a high-carbon fuel such as coke.
Sponge iron	It is the product of reducing iron oxide in the form of iron ore and steel plant wastes into metallic iron, below the melting point of iron and typically in the range of 800–1200 °C. Also known as direct reduced iron (DRI).
Steel	Steel is an alloy of iron and carbon, containing by mass more iron than any other single element, having a carbon content generally less than 2% and containing other elements (when the carbon content is over 2.11%, the material is defined as cast iron). A limited number of chromium steels may contain more than 2.11 % of carbon, but 2.11 % is the usual dividing line between steel and cast iron.
Crude Steel	Crude steel is defined as steel in its first solid (or usable) form: ingots, semi-finished products (billets, blooms, slabs), and liquid steel for castings. This is not to be confused with liquid steel, which is poured steel. For statistical purpose, crude steel covers ingots, continuously cast semi-finished products as cast and liquid steel for production of casting.
Ingot	A form of semi-finished type of metal. Liquid metal is teemed (poured) into moulds, where it slowly solidifies. Once the metal is solid, the mould is stripped, and the ingot is then ready for subsequent rolling or forging.
Semi-finished product	Steel products such as billet, blooms and slabs. These products can be made by direct continuous casting of hot steel or by pouring the liquid steel into ingots, which are then hot rolled into semi-finished products.
Slabs	A semi-finished steel product obtained by rolling ingots on a rolling mill or processed through a continuous caster and cut into various lengths. The slab has a rectangular cross section and is used as a starting material in the production process of flat products, i.e. hot rolled coils, plates, sheets, strips, etc. While the width is at least 2 times of the thickness, slabs of up to 600mm thickness may be continuously cast.
Blooms	Bloom is a semi-finished long product with a square or rectangular cross section usually above 155 mm x 155 mm. This product is either rolled or continuously cast and is then transformed by rolling to obtain finished products like wire rod, merchant bars and other sections.
Billets	Billet is a semi-finished steel product (similar to blooms) with a cross-section typically up to 155mm x 155mm. Billets can be rectangular (including square), circular or polygonal in cross-section, with the width measuring less than twice the thickness. This product is either rolled or continuously cast and is then transformed by rolling to obtain finished products like wire rod, merchant bars, rebar, tube rounds and other sections. The range of semi-finished products above 155 mm x 155 mm are called blooms.
Continuous casting	A process for solidifying steel in the form of a continuous strand rather than individual ingots. Molten steel is poured into open-bottomed, water-cooled moulds. As the molten steel passes through the mould, the outer shell solidifies.
Finished product	It is the term used to describe products obtained upon hot rolling/forging of semi-finished steel products, such as blooms, billets or slabs - colloquially known in the industry as 'semis'. These cover two broad categories of products, namely 'long' products and 'flat' products.
Flat product	A type of finished rolled steel product usually produced from slabs or ingots. These are generally sold either as flat sections, as is the case with 'quarto plate', or coils, either hot or cold, which are rolled lengths coiled up to make transportation easier.
Long product	A type of finished rolled steel product usually derived from billets, blooms or ingots. They are, as the name suggests, often long or narrow in their dimension. They include for instance rail, wire rod and steel bars.

Scrap	Steel scrap is one of the steel industry's most important raw materials. It comes from all steel-containing products that reach the end of their life (post-consumer scrap), from demolished structures to end of life vehicles, packaging, white goods and machinery, and the yield losses in the steelmaking and manufacturing processes (pre-consumer scrap). It can also include iron scrap. All steel can be recycled into new steel. All new steel contains some steel scrap.
Cast steel products	Finished steel goods produced by casting liquid steel (< 2% carbon) into a mould of net- or near-net shape (incl. both non-alloy and alloy steels)
Cast iron products	Finished iron goods produced by casting liquid iron (> 2% carbon) usually directly produced from pig iron into a mould of net- or near-net shape
Galvanized Coil	Hot rolled or cold rolled steel sheet or strip that has received a hot dip coating applied by continuous immersion of steel through a bath containing molten zinc or applied by electrolytic methods. For the purpose of this study, this category includes coatings of other metals (e.g. aluminium, lead, lead-tin, aluminium-silicon, aluminium-zinc)

Source: JRC elaboration based on World Steel Association, EUROFER and American Iron and Steel Institute, Treatise on Process Metallurgy (Barati, 2014) and recommendations gathered through stakeholder consultation.

1.1.2. Existing definitions and categories for steel

Steel can be subject to classification using numerous diverse systems. Annex 1 presents an overview of existing definitions of steel products using the most common categorisation based on physical and chemical properties, products characteristics, and economic statistics:

- Steel grades, designated according to the chemical composition, mechanical and physical properties
- Application and mechanical or physical properties
- Shape and surface conditions
- European Statistics
- Combined Nomenclature
- Other categorisations

1.1.2.1 Steel grades

The steel grade is linked to the chemical composition, the structure and the crystallography of the product. According to the standard EN 10020:2000 "Definition and classification of grades of steel" (CEN, 2000), steel can be classified into:

- Non alloy, stainless steel and other alloy steels by chemical composition
- Main quality classes defined by main property or application characteristics for non-alloy, stainless and other alloy steels.

In addition, a classification is also provided by the European Commission JRC study "Best Available Techniques (BAT). Reference Document for the Ferrous Metals Processing Industry" (Aries et al., 2022). Steel grades are defined in four principal categories:

- Carbon steel non alloy, for which the content of alloy elements is lower than 0.4%, except for silicon (< 0.6%) and manganese (<1.65%);
- Carbon steel low alloy, for which the content of each alloy element is lower than 5%;
- Stainless steel, which is also considered as a high alloy steel;
- High alloy steels, other than stainless steel for which the content of some alloy elements is over 5%.

Electrical steels and high strength steel should also be considered as special steel grades.

Finally, the Regulation (EU) 2019/331 determining rules for harmonised free allocation of emission allowances (European Commission, 2019) defines three product benchmark for steel, **hot metal**, **EAF carbon steel** and **EAF high alloy steel** (see Table 2). The definitions for the EAF categories include reference to the grade of steel and alloy content. These two categories are the same considered by the German Steel Association in the

Low Emissions Steel Standard (LESS) (WV Stahl, 2024), with the caveat that terms used are structural and reinforcing steel for carbon steel and quality steel for high-alloy steel.

Table 2. Categories of steel according to EU-ETS classification

EU – ETS Product Benchmark	Definition
Hot metal	Liquid iron saturated with carbon for further processing, considered as product of blast furnaces, and expressed in tonnes of liquid iron at the exit point of the blast furnace. Similar products such as ferroalloys are not covered by this product benchmark. Residual material and by-products are not to be considered as part of the product. It covers all processes directly or indirectly linked to the process units blast furnace, hot metal treatment units, blast furnace blowers, blast furnace hot stoves, basic oxygen furnace, secondary metallurgy units, vacuum ladles, casting units (including cutting), slag treatment unit, burden preparation, BF gas treatment unit, dedusting units, scrap pre-heating, coal drying for PCI, vessels preheating stands, casting ingots preheating stands, compressed air production, dust treatment unit (briquetting), sludge treatment unit (briquetting), steam injection in BF unit, steam generation plant, converter BOF gas cooling and miscellaneous are included.
EAF carbon steel	Steel containing less than 8 % metallic alloying elements and tramp elements to such levels limiting the use to those applications where no high surface quality and processability is required and if none of the criteria for the content of the metal alloying elements and the steel quality for high alloy steel are met
EAF high alloy steel	Steel containing 8 % or more metallic alloying elements or where high surface quality and processability is required. EAF steel should be regarded as high quality steel if at least one of the following criteria is met: 1. A maximum hydrogen content of 0.0003 per cent 2. A maximum sulphur content of 0.003 per cent 3. A maximum phosphorus content of 0.01 per cent 4. Micro cleanliness: a) K3 (oxide) < 40; K4 < 50 according to DIN 50602 (or any equivalent international standard) b) Sulphide: A/thin 2.0; A/thick 1.5 according to ISO 4967 c) Oxide: B/thin 1.5; B/thick 0.5 according to ISO 4967 d) ASTM E45: Procedure B, C, D maximum 2 e) SEP 1920: ultrasonic testing: core test – KSR maximum 2 mm 5. Macro cleanliness: blue brittleness break test: maximum 2.5 mm/dm² 6. It can also be assumed that QST has a high surface quality and processing capability if one of the following non-destructive technological tests is required for more than 10 per cent of the outgoing product stream: a) Ultrasonic testing according to ASTM E213 or EN 10246-6, 7.14 b) Magnetic particle testing according to ASTM E709 or EN 10246-12 c) Dye penetrant inspection according to ASTM E165 d) Electromagnetic testing i. Eddy current, ASTM E309: ii. Flux leakage, ASTM E570:

Source: Guidance Document n°9 on the harmonised free allocation methodology for the EU-ETS post 2020 (European Commission, 2019)

1.1.2.2 Classification according to application

An alternative method for categorising steel products involves grouping them based on their intended application. One method is to utilise the standard EN 10027-1:2016 – Designation systems for steels – Part 1: Steel names, as detailed in Table 3. Another approach is to classify the products using as categories (application groups) the sub-committees under CEN/TC 459, which are in charge of standardising a wide range of steel products with many different applications (see Table 4).

Table 3. Classes of application and their relevant mechanical properties. Mandatory requirement for nomenclature according to EN 10027-1:2016

Application	Letter	Relevant mechanical property
Structural steel	S	Minimum Yield Strength
Pressure purposes	P	Minimum Yield Strength
Line pipe	L	Minimum Yield Strength
Engineering Steel	E	Minimum Yield Strength
Steel for reinforcing concrete	B	Characteristic yield strength
Steels for prestressing concrete	Y	Nominal tensile strength
Steel for or in the form of rails	R	Specified minimum Brinell Hardness (HBW)
Flat products for cold forming	D	Symbol for Hot/cold rolling
High strength steel flat products for cold forming	H	
Tin mill products (steel products for packaging)	T	Nominal yield strength
Electrical steel	M	Max specified loss

Source: Advancing on comparability aspects for product and organisation environmental footprint (European Commission, Joint Research Centre, et al., 2022)

Table 4. Application groups according to the sub-committees under CEN/TC 459

Application groups
Structural steels other than reinforcements
Concrete reinforcing and prestressing steels
Steels for heat treatment, alloy steels, free-cutting steels and stainless steels
Wire rod and wires
Steels for pressure purposes
Steel sheet and strip for electrical applications
Coated and uncoated flat products to be used for cold forming
Steel tubes, and iron and steel fittings
Steel castings and forgings

Source: CEN

1.1.2.3 Steel products grouped by shape and surface conditions

Regarding steel products grouped by shape and surface conditions, the standard EN 10079:2007 also provides a comprehensive framework for classifying steel products. In addition, categories used by the World Steel Association and EUROFER statistics also respond a shape criteria, although they are also partly based on the intended use. Further explanation can be found in Section 1.3.1 of Annex 1 of this preparatory study.

1.1.2.4 European statistics

The EU's industrial production statistics play a pivotal role in tracking and analysing economic activities within the EU. These statistics are compiled through the PRODCOM² (PRODUCTION COMMUNAUTAIRE) survey and also in the EUROPROD database, which incorporates external trade statistics. The PRODCOM survey captures data on various economic activities, which are then classified according to the Statistical Classification of Economic Activity (NACE).

Within this comprehensive framework, iron and steel products are specifically categorised under subsectors of NACE Sector C24:

- C24.1 Manufacture of basic iron and steel and of ferro-alloy
 - 24.10 Manufacture of basic iron and steel and of ferro-alloy
- C24.2 Manufacture of tubes, pipes, hollow profiles and related fittings, of steel

² <https://ec.europa.eu/eurostat/web/metadata/classifications>

- 24.20 Manufacture of tubes, pipes, hollow profiles and related fittings, of steel
- C24.3 Manufacture of other products of first processing of steel
 - 24.31 Cold drawing of bars
 - 24.32 Cold rolling of narrow strip
 - 24.33 Cold forming or folding
 - 24.34 Cold drawing of wire
- C24.5 Casting of metals
 - 24.51 Casting of iron
 - 24.52 Casting of steel

1.1.2.5 Combined Nomenclature

The **Combined Nomenclature (CN)** is a comprehensive system used for the classification of goods in accordance with both the Common Customs Tariff and the external trade statistics of the European Union. It represents a notable extension of the Harmonised System (HS) nomenclature, tailored to encompass specific EU-related subdivisions, employing eight-digit codes for precise classification.

Within the CN framework, iron and steel products find their classification in two primary chapters: Chapter 72, dedicated to iron and steel, and Chapter 73, focusing on articles made from iron and steel. Section 1.5 of Annex 1 presents the HS headings at four-digit level of the main categories of these chapters.

1.2 Relevant legislation and policy initiatives

The policy background for the steel industry and its associated products is characterised by a multi-tiered regulatory framework that spans across different levels of governance. This section provides an overview of the policy landscape related to steel at the EU (Section 1.2.1), Member States (Section 1.2.2.) and third-country (Section 1.2.3) levels.

1.2.1 EU legislation and policy initiatives

This section delves into the expansive landscape of EU legislation and policy initiatives directly impacting both the EU steelmaking industry and steel products sold in the EU market. These EU policies are considered the most relevant for this preparatory study, and Table 5 presents a summary of the initiatives, categorised within distinct domains. It is worth noting that some legislation is still under discussion, in which case a certain level of uncertainty regarding the final implications must be accepted.

Table 5. Classification of EU policy initiatives

Domain	Policy Initiatives
Circular Economy and Sustainable Product Policy	Circular Economy Action Plan (CEPA) Ecodesign for Sustainable Product Regulation Construction Products Regulation Waste Framework Directive Waste Electrical and Electronic Equipment Directive Waste Shipment Regulation End-of-Life Vehicles Directive Packaging and Packaging Waste Directive Registration, Evaluation, Authorisation, and Restriction of Chemicals (REACH) Classification, Labelling, and Packaging Regulation (CLP)
Sustainable Industry and Investments	European Industrial Strategy Green Deal Industrial Plan European Critical Raw Material Act Net Zero Industry Act European Taxonomy EU Open Strategic Autonomy

GHG Emissions and Pollution Reduction and Energy Transition	European Emissions Trading System Carbon Border Adjustment Mechanism Industrial Emissions Directive EU Hydrogen Strategy
Trade Policy	EU-US Global Agreement on Sustainable Steel and Aluminium Trade Defence Instrument
Consumers Rights	Green Claims

Source: JRC elaboration

1.2.1.1. Circular Economy Action Plan (CEAP)

A new **Circular Economy Action Plan (CEAP)**³ was adopted in March 2020. It consists of a set of initiatives to establish a coherent product policy framework that will help stimulate Europe's transition to a circular economy, decoupling economic growth from resource use. The plan encompasses the entire life cycle of products and key value chains.

Overall, the CEAP encapsulates the European Union's commitment to achieving a regenerative economic model that not only minimises environmental harm but also promotes innovation, job creation, and sustainable growth. Through its multifaceted approach, the CEAP aims to reshape production, consumption, and waste management patterns, paving the way for a more resilient and circular economy that safeguards natural resources while enhancing the well-being of both society and the environment.

The CEAP's strategic framework rests on three fundamental pillars. The first revolves around designing and producing goods that prioritise longevity, repairability, and recyclability. This includes promoting eco-design principles and setting ambitious standards for product durability, ensuring that items can be easily repaired, and maximizing the recyclability of materials. The second pillar emphasizes fostering a strong market for secondary raw materials, incentivising recycling, and enabling remanufacturing and refurbishment. This entails creating a demand-driven market for recycled materials, streamlining waste management practices, and facilitating the return of products to the market after their initial use. The third pillar centres on empowering consumers to make sustainable choices through improved information, fostering awareness, and integrating circularity into public procurement.

1.2.1.2. Ecodesign for Sustainable Product Regulation

On 2022, the European Commission adopted a **Proposal for a Regulation on Ecodesign for Sustainable Products (ESPR)**⁴. This proposal, commonly referred to as the 'ESPR', aligns with the overarching objectives outlined in the **EU Green Deal**⁵ and the **Circular Economy Action Plan (CEAP)**. Its primary purpose is to reshape the regulatory landscape of the EU, ensuring that products placed in the EU internal market become increasingly sustainable. The ESPR entered into force in July 2024.

At its core, the ESPR aims to reduce the environmental burden of products across their entire lifecycle, while simultaneously enhancing the efficiency of the EU's internal market. Building upon the existing successful approach of the **Ecodesign Directive**⁶, which predominantly applies to energy-related products, the ESPR seeks to broaden its scope beyond energy-related products to make it applicable to the wider spectrum of products, including **intermediate** ones. The integration of intermediate products such as steel into the framework of the ESPR represents a pivotal step towards fostering a more resilient and sustainable industrial landscape. This inclusive approach acknowledges the interconnectedness of supply chains and recognises the pivotal role that sustainable design and production of intermediate products play in achieving broader sustainability goals.

The ESPR envisions a roadmap with concrete actions to improve product durability, reusability, upgradability and repairability; address the presence of hazardous chemicals; enhance energy and resource efficiency; increase recycled content, enable remanufacturing and high-quality recycling; reduce carbon and environmental footprints; counter premature obsolescence; incentivise product-as-a-service; and harness the potential of digitalisation.

³ [COM/2020/98 final](#)

⁴ [COM/2022/142 final](#)

⁵ [COM/2019/640 final](#)

⁶ [Directive 2009/125/EC](#)

The Regulation creates a framework to set concrete **information and performance requirements** for the placing on the market of specific products related to their most relevant impacts and sustainability aspects. In terms of information requirements, the initiative also envisages the introduction of a **Digital Product Passport** to provide information about products' environmental sustainability, adding transparency and helping consumers and business to make informed choices, and creates the opportunity to collected information on different steps of the entire value chain.

1.2.1.3. Construction Products Regulation (CPR)

The **Construction Products Regulation**⁷ establishes harmonised rules for the marketing and use of construction products within the EU, including structural and non-structural steel components and products used in construction. Its primary aim is to ensure that construction products placed on the EU market meet essential performance requirements related to safety, health, environmental protection, and other critical aspects are accompanied by a declaration of the performance of the product for which the manufacturer assumes the responsibility.

Construction, buildings and engineering works are a major customer of the steel industry as they use around 50% of all steel consumed in Europe (European Commission, 2021). Steel products placed in the EU market must comply with the CPR provisions. Key implications encompass the obligatory CE marking for steel products, aligned with harmonised EU standards that define performance declaration requirements based on harmonised assessment methods.

In November 2024, the **revised CPR**⁸ entered into force. It aligns closely with the ESPR and develops, in addition to safety and functionality aspects, environmental performance for construction products with a view to provide reliable information to be used in the environmental assessment of construction works. Furthermore, the CPR may also establish rules for the recycled content or reuse of products.

It is worth noting that the CPR primarily regulates final construction products, while intermediate products and energy-related construction products (heaters, boilers, heat pumps, water and space heating appliances, fans, cooling and ventilating systems and photovoltaic products, for example) are mainly regulated under the ESPR.

Under the CPR, steel is considered as either as (final) construction product or as intermediate product (defined as "key part"⁹). All products containing steel which are permanently installed in construction works (buildings and civil engineering works) are in the scope of CPR. Nearly all of these products are already regulated by harmonised standards adopted under the current CPR which however do not contain provision on environmental sustainability yet. However, the scope of the CPR does not include steel products which are used during construction activities but are not permanently installed in the construction works such as scaffolding, falsework (temporary support beams), and other construction tools such as wrenches, hammers, spatulas, trowels, ladders, road plates and temporary trench covers.

According to ESPR, recital 51, the ESPR should, in addressing construction products, set requirements for final products only where the obligations created by the CPR are unlikely to sufficiently achieve the environmental sustainability objectives pursued by ESPR. Additionally, according to recital 49 of ESPR, when prioritising intermediate products, the Commission should also take into account the consequences for final products that are made from such intermediate product.

1.2.1.4. Waste Framework Directive (WFD)

The **Waste Framework Directive (WFD)**¹⁰ is a fundamental piece of legislation that governs the management of waste, including industrial waste generated by sectors such as the steel industry. Its primary objectives are

⁷ [Regulation \(EU\) No 305/2011](#)

⁸ [COM/2022/144 final](#)

⁹ a part which is used as a component or spare part for a product and which has been specified by a harmonised technical specification as essential for the characterisation, safety or performance of a product;

¹⁰ [Directive 2008/98/EC](#)

to promote the efficient use of resources, reduce the environmental impacts of waste generation and management, and ultimately contribute to a circular economy.

The Directive establishes a clear waste hierarchy, emphasising waste prevention, followed by reuse and recycling, with disposal as a last resort. For the steel industry, this means a compelling mandate to minimise waste generation and prioritise recycling and reuse practices during manufacturing processes.

As part of the WFD's implementation, the **SCIP database** was introduced to enhance the transparency of information on hazardous substances in products. Steel producers, along with other manufacturers, are required to report any articles they place on the market containing Substances of Very High Concern (SVHCs) above a 0.1% weight by weight (w/w) concentration to the SCIP database. The submission to the SCIP database ensures that information on the presence of SVHCs is available throughout the product's lifecycle, including the waste stage, thus supporting safe waste management and recycling processes.

Additionally, the WFD requires stringent reporting and documentation, compelling steel manufacturers to maintain records on waste generation, disposal methods, and recycling rates. These stringent reporting obligations ensure compliance with the Directive while underscoring the industry's commitment to environmental stewardship and the principles of the circular economy.

1.2.1.5. Waste from Electrical and Electronic Equipment (WEEE)

The **Waste Electrical and Electronic Equipment (WEEE) Directive**¹¹, which sets minimum requirements with regard to recycling of PV modules, which Member States may adjust when they transpose the Directive into their own legislation, e.g. defining more stringent requirements or target quotas. Regarding recycling of inverters, while the WEEE Directive does not explicitly refer to them, is still understood to fall under its scope.

Many electrical and electronic devices contain steel components, particularly in their casings, frames, and structural elements. These steel components are subject to the recovery and recycling targets set by the directive. Manufacturers and producers of EEE are generally responsible for ensuring that the steel and other materials in their products are managed in an environmentally sustainable way when the products become waste.

The WEEE Directive is currently under evaluation in order to assess whether it is still fit for purpose, explore opportunities to simplify it, and determine whether a review is needed.

1.2.1.6. Waste Shipment Regulation

The **Waste Shipment Regulation**¹² implements the Basel Convention on the control of transboundary movements of hazardous wastes and their disposal, as well as the OECD Decision on the control of transboundary movements of waste destined for recovery operations.

Its primary aim is to ensure that waste is transported and treated in a way that protects the environment and human health, with specific provisions to prevent illegal shipments and to facilitate the tracking and control of waste movements. To this end, it sets out procedures and controls for the transfer of waste between EU member states and also for the import and export of waste between the EU and non-EU countries. The regulation is designed to ensure that waste is managed in an environmentally sound manner throughout the process, from its point of origin to its final disposal or recovery.

The regulation differentiates between various types of waste, setting out distinct procedures and controls for hazardous and non-hazardous waste. Hazardous waste, due to its potential risks, is subject to more stringent controls, including prior written notification and consent before being shipped.

Given that **ferrous metal scrap is the largest category of waste exported outside the EU**, this regulation has a significant impact on the iron and steel industry, particularly concerning the handling and export of waste materials and other by-products of their production processes (e.g. dross, scaling, ash, slag, dust, powder, sludge, etc.). When these waste materials are being exported to third countries, the regulation requires the industry to properly document and track the waste shipment. This includes the completion of waste transfer notes,

¹¹ [Directive 2012/19/EU](#)

¹² [Regulation \(EC\) No 1013/2006](#)

consignment notes, and other paperwork to accompany the movement of waste materials. The regulation also requires companies to classify their waste according to specific codes and descriptions, ensuring that it is correctly labelled and packaged for transportation.

In November 2021, the Commission adopted a **proposal for a new Regulation on waste shipment**¹³. A political agreement between the Parliament and the Council was reached in November 2023. This revision responds to the need of ensuring that the EU takes greater responsibility of its waste and does not export its environmental challenges to third countries and tackle illegal waste shipment.

In addition, it will contribute to unlocking the potential of EU waste market to transition to a circular economy, securing the supply of raw materials. This waste often has a positive economic value, notably as secondary raw materials that can replace and reduce dependence on primary materials, contributing to the open strategic autonomy of the EU.

The proposed regulation will establish a list of non-OECD countries to which exports of non-hazardous waste from the Union for recovery are authorised. In addition, the export of green-listed waste outside the OECD will be subject to the notification procedure. These measures are anticipated to result in a decrease in the volume of exported waste, leading to the retention in the EU of between 20% and 50% of the expected exports of waste to non-OECD countries, equivalent to 0.6 to 2 million of tonnes of ferrous metals. As a result, the use of steel scrap retained within the EU under this measure is projected to contribute to a reduction in CO2 emissions for the steel industry in the EU, estimated to be between 1 and 2.9 million tonnes per year (European Commission, 2021).

1.2.1.7. ELV Directive

The **End-of-Life Vehicles Directive**¹⁴ (Directive 2000/53/EC) lays down measures to prevent waste from vehicles, promoting the recycling, recovery, and proper disposal of vehicles that have reached the end of their life cycle. The ELV Directive establishes stringent recycling and recovery targets. It mandates that no later than 1 January 2015, for all end-of life vehicles, the reuse and recovery shall be increased to a minimum of 95 % by an average weight per vehicle and year. Within the same time limit, the re-use and recycling shall be increased to a minimum of 85 % by an average weight per vehicle and year.

Additionally, the Directive introduces the legal obligation of remove metal components containing copper, aluminium and magnesium if these metals are not segregated in the shredding process (Annex I, point 4 element 2), and hazardous materials.

In 2023, the European Commission proposed a new Regulation concerning end-of-life vehicles (ELVs) as part of its commitment to the European Green Deal and the Circular Economy Action Plan. This **proposed ELV Regulation**¹⁵ builds on and replaces two existing Directives: Directive 2000/53/EC, which addresses end-of-life vehicles, and Directive 2005/64/EC¹⁶, focusing on the type-approval of motor vehicles concerning their reusability, recyclability, and recoverability.

The automotive manufacturing industry is a significant consumer of steel, with Europe's automotive sector alone accounting for a substantial 19% of the EU's steel industry demand (EUROFER, 2024). However, the industry uses a limited amount of recycled steel in new vehicles, primary due to the limited quality of ferrous scrap coming from end-life of vehicles. The quality characteristics of many automotive steel grades shown lower tolerance for tramp elements than long products used e.g. in construction and buildings. Nevertheless, it is worth noting that there are currently numerous initiatives from automotive manufacturers aimed at increasing the recycled content of steel in vehicles.

In order to address these challenges, the proposed ELV regulation foresees the possibility to establish a **minimum share of steel recycled from post-consumer steel waste to be present and incorporated into vehicles**. However, the establishment of a future target will be based on a dedicated feasibility study covering all relevant technical, environmental and economic factors linked to the feasibility of such target. In

¹³ [COM/2021/709 final](#)

¹⁴ [2000/53/EC](#)

¹⁵ [COM/2023/451 final](#)

¹⁶ [Directive 2005/64/EC](#)

addition to this measure, which would be realised through a specific Delegated Act, the proposal includes provisions for a declaration of the recycled content of steel present in vehicles.

1.2.1.8. Packaging and Packaging Waste Directive (PPWD)

The PPWD¹⁷ covers both packaging design and packaging waste management of all types of packaging placed on the EU market (all types of materials and all type of functions: primary, secondary and tertiary packaging). It aims to deal with the increasing impact of packaging and packaging waste on the environment and to remove barriers in the internal market caused by diverging national rules on packaging design.

Among other things, the PPWD requires Member States to take measures aimed at the prevention of packaging waste and to **meet recovery and recycling targets**. It also requires them to set up packaging extended producer responsibility schemes by end of 2024 in line with the minimum requirements of Art. 8a of Directive 2008/98/EC on waste as amended in 2018.

In the context of steel products, the PPWD has implications primarily for packaging made from steel or containing steel components. Packaging materials that include steel, such as steel drums or containers, are subject to the recycling and recovery targets specified by the directive.

1.2.1.9. Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)

Regulation (EC) No 1907/2006 - **Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)**¹⁸ aims to ensure the safe use of chemicals in the EU. More specifically, REACH aims to ensure a high level of protection of human health and the environment, including the promotion of alternative methods for assessment of hazards of substances, as well as the free circulation of substances on the internal market while enhancing competitiveness and innovation. REACH regulates, amongst many other aspects, the restriction of the placing on the market and use of certain substances listed in its Annex XVII, including in some cases, the incorporation of substances into articles. Consequently, restrictions also cover recycled substances and the presence of restricted substances in recovered materials. The authorisation title of REACH applies to the placing on the market and use of substances of very high concern (SVHC), aiming at their progressive substitution by less hazardous substances or technologies and by subjecting their use to specific conditions.

REACH manage the processes of registration, evaluation, authorisation and restriction of chemical substances and mixtures. Any manufacturer or importer of a substance, in quantities exceeding 1 tonne per year, shall submit a registration to the European Chemicals Agency (ECHA), who assess compliance with the regulation.

Authorisation provisions apply to substances of very high concern to assure that risks are properly controlled, and restriction provisions are applicable to certain substances that cause an unacceptable risk to human health or environment (List of substances in Annex XVII of the Regulation).

Steel is an alloy and therefore considered a mixture under REACH, While steel itself is not subject to REACH authorisation, the use of certain chromium compounds listed in Annex XIV to REACH, such as chromium trioxide, in the surface treatment of steel is subject to authorisation under REACH.

It should be noted that according to the definition of “article” in Article 3 (3) of REACH, and as further illustrated for the case of aluminium in the relevant ECHA guidance¹⁹, most steel semi-finished products may qualify as articles under REACH, rather than as mixtures.

1.2.1.10. EU Regulation for the Classification, Labelling and Packaging of substances and mixtures (CLP)

The **CLP Regulation (CE) 1272/2008**²⁰ governs the classification, labelling, and packaging of chemical substances and mixtures. It serves as an integral component of the EU's broader framework for chemical management and safety.

The CLP Regulation aims to ensure a high level of protection of human health and the environment. It achieves this through a harmonised system of classifying chemicals based on their intrinsic properties, such as their

¹⁷ [Directive 94/62/EC](#)

¹⁸ [REGULATION \(EC\) No 1907/2006](#)

¹⁹ Guidance on requirements for substances in articles Version 4.0. ECHA 2017. Annex IV, example 16.

²⁰ [Regulation \(EC\) No 1272/2008](#)

toxicity, flammability, and environmental impact. The regulation mandates the use of standardised labels and pictograms to communicate these hazards effectively, enhancing the safety of workers, consumers, and the environment.

Regarding steel products, while the CLP Regulation primarily targets chemical substances and mixtures, steel manufacturers and downstream users must adhere to CLP requirements when their products contain chemicals that fall under its scope. This regulation is relevant to the manufacturing of steel products to the extent it involves the use of alloying elements, coatings, treatments, or surface finishes which are substances or mixtures subject to CLP.

1.2.1.11. European Industrial Strategy

In 2020, the European Commission presented the **European Industrial Strategy**²¹. This strategy stands as a pivotal framework that underlines the European Union's commitment to fostering a resilient, innovative, and sustainable industrial landscape. Building upon the foundation laid in its initial release, the strategy was updated²² in 2021 to reflect the dynamic shifts in global dynamics and emerging challenges. It emphasises a holistic approach that blends innovation, digital transformation, and the green transition to forge a competitive and inclusive European industry.

Central to the European Industrial Strategy is the vision of enhancing Europe's strategic autonomy in key sectors. By promoting open strategic autonomy, the EU aims to fortify its position in critical value chains, ensuring a robust response to disruptions and fostering a level playing field for its industries. This entails bolstering innovation through research and development, investing in digital capabilities, and nurturing a circular and sustainable economy. The strategy prioritises sectors such as clean energy, digital technologies, and the health sector, envisioning a harmonious synergy between innovation and environmental sustainability.

A cornerstone of the strategy is the ambition to align industrial growth with the overarching climate and sustainability objectives. With a commitment to achieving climate neutrality by 2050, the **European Industrial Strategy accentuates the importance of decarbonisation and circular economy in industrial operations**. It seeks to incentivise sustainable practices and facilitate the transition to green technologies. Furthermore, the strategy encourages collaborative efforts among industries, research institutions, and governments to pave the way for a green, competitive, and secure industrial ecosystem.

The European Industrial Strategy is not only a blueprint for strengthening Europe's economic resilience and competitiveness but also a testament to its commitment to social inclusivity. It underscores the importance of a skilled workforce, innovation clusters, and strategic investments in regions with potential for industrial growth. Through fostering synergies between public and private stakeholders, this strategy aspires to harness the transformative power of industry for the betterment of societies, economies, and the environment, positioning the European Union as a beacon of sustainable, forward-looking industrial progress.

Additionally, the updated European Industrial Strategy is accompanied by the document **"Towards competitive and clean European steel."**²³ This document, tailored to the steel sector, outlines a comprehensive roadmap for fostering competitiveness while advancing sustainability goals. By aligning with the core principles of the European Industrial Strategy, this document emphasises the steel sector's transition to cleaner and more innovative production processes. For this to happen, Europe needs a deeper and more **digital single market, greater resilience and open strategic autonomy** in line with the adopted Commission Trade Policy Review.²⁴

1.2.1.12. Green Deal Industrial Plan

In February 2023, the European Commission presented the Green Deal Industrial Plan for the Net-Zero Age²⁵, a strategic blueprint designed to bolster the competitiveness of EU's net-zero industry and support the fast transition to climate neutrality. Central to this vision is the Commission's commitment to prioritize net-zero technologies within the framework of Ecodesign working plans.

²¹ [COM/2020/102 final](#)

²² [COM/2021/350 final](#)

²³ [SWD/2021/353 final](#)

²⁴ [COM/2021/66 final](#)

²⁵ [COM\(2023\) 62 final](#)

The plan rests upon two pivotal pillars, the **Net Zero Industry Act (NZIA)** and the **Critical Raw Materials Act (CRMA)**. The NZIA focuses on strengthening the European manufacturing capacity of net-zero strategic technologies while overcoming barriers to scaling up the manufacturing capacity in Europe. On the other hand, the CRMA aims to ensure the EU's access to a secure and sustainable supply of critical raw materials.

1.2.1.13. European Critical Raw Materials Act

Critical raw materials (CRMs) are crucial to the twin green and digital transition and other EU priorities, such as resilience and security. According to current projections, global demand for some CRMs, such as rare earths and lithium, will soon outstrip global supply. As a result, other leading economies are strategically stepping up efforts to secure supply, increasing demand for critical raw materials and triggering potential trade restrictions.²⁶ Thus, the global race for supply and recycling of CRMs makes it imperative to build a more resilient supply chain, securing a reliable and unhindered access to CRMs.

In response to these challenges, the EU adopted in May 2024 a **European Critical Raw Materials Act**.²⁷ The Regulation includes a list of critical and strategic raw materials and sets measures throughout the value chain - including extraction, processing and recycling- to ensure that, by 2030, no more than 65% of the Union's annual consumption of each strategic raw material at any relevant stage of processing is from a single third country. Furthermore, the Regulation foresees the possibility to introduce requirements to make information on the environmental footprint of certain CRMs available when placed on the EU single market. While steel itself does not directly fall under the classification of critical or strategic raw materials, stainless steel and other-alloy production relies on certain alloys that hold such a distinction. As an example, at least 11 of the materials on the Critical Raw Materials (CRM) list are key inputs for steel production. Notably, alloys such as silicon, manganese, nickel, and niobium play pivotal roles in this context (European Commission et al., 2023).

1.2.1.14. Net Zero Industry Act

The **Net Zero Industrial Act Regulation (NZIA)**²⁸ entered into force in June 2024. This regulation aims to enable the conditions for setting up net-zero technology manufacturing in Europe, inter alia, including improving investment certainty, lowering administrative burden, facilitating access to markets by means of encouraging public and private demand through supportive schemes, supporting innovation, enhancing skills for quality job creation in net-zero technologies, and coordinating industrial partnership.

The steel sector plays a pivotal role in the manufacturing of net-zero technologies, as it is at the foundation of the value chains of renewables technologies, hydrogen, wind turbine towers, solar panels, electric vehicles, etc. To achieve the ambitious objectives set forth in the **REPowerEU Plan**²⁹, 74 million tons of steel will be required. To illustrate, for every newly deployed megawatt (MW) of solar power, a substantial 35 to 45 tons of steel are indispensable, while each MW of wind power relies on a substantial 120 to 180 tons of steel (EUROFER, 2023). Recognising this importance, the NZIA includes energy-intensive industries like iron and steel production within its scope. This inclusion ensures that steel plants producing components for net-zero technologies and investing in decarbonisation are eligible for the Act's provisions.

1.2.1.15. EU Taxonomy for Sustainable Activities

The **EU Taxonomy** (Regulation (EU) 2020/852 "Taxonomy Regulation" (TR) adopted on 18 June 2020.)³⁰ is a classification system for sustainable economic activities that aims to facilitate sustainable investment. The TR is focused on six environmental objectives: climate change mitigation, climate change adaptation, sustainable use and protection of water and marine resources, transition to a circular economy, pollution prevention and control, and protection and restoration of biodiversity and ecosystems.

The TR establishes a **harmonised criterion**³¹ to determine whether an economic activity qualifies as environmentally sustainable for the purposes of establishing the degree to which an investment is environmentally sustainable.

²⁶ [Call for evidence for an impact assessment - European Critical Raw Materials Act](#)

²⁷ [Regulation \(EU\) 2024/1252](#)

²⁸ [PE/45/2024/REV/1](#)

²⁹ [SWD/2022/230 final](#)

³⁰ [PE/20/2020/INIT](#)

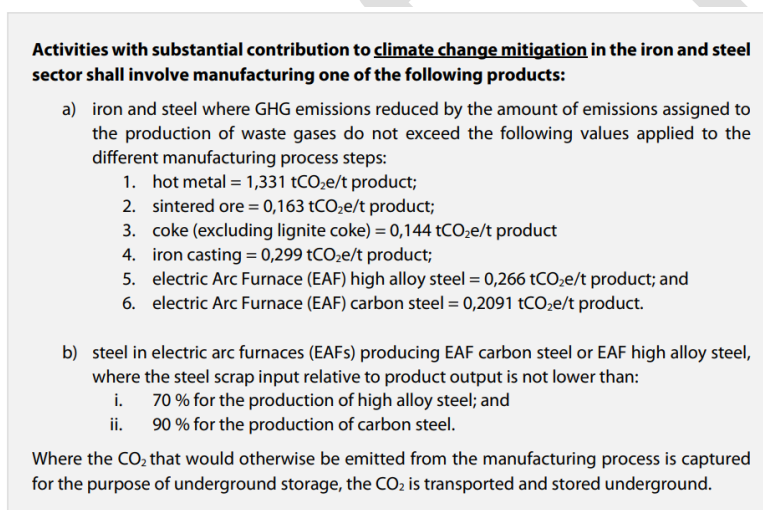
³¹ [COM/2021/2800](#)

An economic activity shall qualify as environmentally sustainable when: a) contributes substantially to one or more of the environmental objectives; b) does not significantly harm any of the environmental objectives; c) respect minimum social safeguards; and d) complies with technical screening criteria that have been established.

In the context of the manufacture of iron and steel, the EU sets out specific criteria that steel production processes must meet to be categorised as environmentally sustainable as a transitional activity, as defined in the article 10.2 of the TR (see Figure 3.). These criteria encompass various aspects, such as greenhouse gas emissions, resource efficiency, and compliance with best available techniques (BAT). Since it is defined as a transitional activity, the criteria has to be reviewed every 3 years as defined in the Article 19 of the TR. Therefore, the thresholds and criteria are subject to revision as technologies and best practices evolve. The European Commission, with input from the Platform on Sustainable Finance and other stakeholders, periodically reviews and updates the technical screening criteria to reflect advancements in science, technology, and policy.

The EU Taxonomy serves as a guide for investors, businesses, and policymakers in identifying sustainable economic activities and directing investments towards them. It reinforces the EU's ambitions for a more sustainable and circular economy by encouraging steel manufacturers to adopt cleaner and more efficient production processes. It is important to highlight that criteria are subject to revision and updating.

Figure 3. Technical Screening Criteria for Manufacture of Iron and Steel



Note, Thresholds reflects the average value of the 10% most efficient installations in 2016 and 2017 (t CO₂ equivalents/t) as set out in the Annex to the Implementing Regulation (EU) 2021/447

Source: COM/ 2021/ 2800 (European Commission, 2021)

1.2.1.16. EU Open Strategic Autonomy

As defined in the study on new product priorities for the ESPR (Faraca et al., 2024), the EU Open Strategic Autonomy refers to the ability to act autonomously (i.e., without being dependent on other countries) in important policy areas, reflecting its strategic interests and values. Strengthening the resilience and sustainability of the EU economy while de-risking its supply chains is a pillar of the EU's drive towards Open Strategic Autonomy.

The concept of Open Strategic Autonomy was initially introduced in the Commission Communication outlining a major recovery plan post-COVID-19 crisis – **Europe's moment: Repair and prepare for Next Generation**³². This policy document emphasises that it is imperative to enhance resilience and prevent, protect and withstand future shocks, advocating to reduce dependency and strengthen security of raw material supply. Recognising that global trade and its integrated value chains are pivotal for Europe's recovery and growth, the strategy envisions an Open Strategic Autonomy model. This involves shaping a new global economic governance system, fostering mutually beneficial bilateral relations, and safeguarding against unfair practices.

³² COM/2020/456 final

Building on this foundation, the Commission reiterated this idea in its new trade strategy for the coming years – **An Open, Sustainable and Assertive Trade Policy**.³³ This strategy underscores the EU's commitment to open and fair trade, emphasizing well-functioning, diversified, and sustainable global value chains.

Furthering this vision, the strategy **Resilient EU2030**,³⁴ introduced by the Spanish Presidency in the European Council, outlines a roadmap to strengthen the EU's resilience and competitiveness in the coming decade. It emphasizes the need to adopt new production methods in the industry to reduce CO₂ emissions and highlights the role of the steel industry, underscoring the importance of electric arc furnace technology and the utilisation of scrap-based steel, particularly in the construction sector.

1.2.1.17. Industrial Emissions Directive (IED)

Iron and steel production brings a number of environmental impacts including air emissions (such as CO, SO_x, NO_x and dust), emissions to water, hazardous, solid or liquid waste. In 2017, 222 iron and steel installations were operating under the **Industrial Emission Directive (IED)**.³⁵ The Commission regulates emissions falling under the IED mainly through Commission Implementing Decisions establishing Best Available Techniques (BAT) conclusion for Iron and Steel Production and Ferrous Metal Processing (European Commission, 2021).

Although regulating GHG emissions is not a primary objective of the Industrial Emissions Directive (IED), standards set by the Directive impose emissions prevention or reduction techniques to steel producers, which can also impact CO₂ levels. Recent evaluation concluded that the IED effectively mitigate pollution from industrial installations, especially to air, but demonstrated less efficacy in areas related to decarbonisation, efficient use of energy, raw materials or waste management (European Commission, 2021).

In line with the **Green Deal**³⁶ and the **Zero Pollution Action Plan**³⁷, the Commission introduced a proposal in 2022 to **revise the IED**³⁸, aiming to enhance the Directive's effectiveness by emphasizing energy, water, and material efficiency, alongside advocating for safer chemical usage within industrial processes. The agreement was formally adopted by Parliament in March 2024 and by the Council in April 2024. The revised IED EU 2024/1785 was finally published on the 15 July 2024³⁹. The Commission proposal expands the scope of the IED **with the inclusion of large-scale battery production and mining** (extraction of industrial minerals and metals). In addition, the revision seeks to accelerate the uptake of innovative techniques for decarbonisation, depollution, resource efficiency (energy and water consumption) and circular economy with the setting-up of the new EU Innovation Centre for Industrial Transformation and Emissions (INCITE), based in the Joint Research Centre in Seville, who will play a key role in this endeavour, overseeing technological innovations and accelerating the identification of novel techniques that may be taken up in the future Best Available Techniques (BAT) conclusions.

1.2.1.18. European Emissions Trading System (EU ETS)

The European Trade System (EU ETS)⁴⁰ is the world's first major carbon market and plays a pivotal role in reducing GHG emissions from power and manufacturing industries, including the iron and steel sector. It operates on the '**cap and trade**' principle, imposing a ceiling on total GHG emissions that can be emitted by installations covered under the EU ETS system, where installations can buy or receive emissions allowances that can be traded. Installations must surrender enough allowances that cover their emissions. Conversely, if an installation reduces its emissions, surplus allowances can be utilised for future emissions or exchanged with installations running short on allowances.

As part of the EU's increased emissions reduction ambition (Fit-for-55 package⁴¹), in 2021, The European Commission released a legislative proposal to review the ETS directive⁴². The reform, adopted in May 2023, has increased the emissions reduction target for ETS sectors, including iron and steel, by 61% by 2030. This implies

³³ [COM/2021/66 final](#)

³⁴ <https://futuros.gob.es/sites/default/files/2023-09/RESILIENTEU2030.pdf>

³⁵ [Directive 2024/1785/EU](#)

³⁶ [COM/2019/640 final](#)

³⁷ [COM/2021/400 final](#)

³⁸ [COM/2022/156 final/3](#)

³⁹ [PE/87/2023/REV/1](#)

⁴⁰ [Directive 2003/87/EC](#)

⁴¹ [COM/2021/550 final](#)

⁴² [COM/2021/551 final](#)

a one-off reduction of the overall emissions cap by 117 million allowances (90 million in 2024 and 27 million in 2026), accompanied with a stronger linear reduction factor of 4.2% per year instead of the 2.2% under the current system. Free allowances to industries will be phased out from 2026 and totally disappear by 2034.

Furthermore, the allocation of free allowances under the EU ETS will have to reflect technological progress in innovative low-carbon technologies. Additionally, strengthening the Market Stability Reserve (MSR) is linked to the review of the EU ETS. Increasing EU ETS ambitions will require MSR rules to remain fit to their purpose to tackle structural imbalances in the allowances.

1.2.1.19. Carbon Border Adjustment Mechanism (CBAM)

The **Carbon Border Adjustment Mechanism (CBAM)**⁴³, introduced in the Fit-for-55 package, is a new policy instrument to address the risk of carbon leakage and ensure that the price of imports of products under CBAM scope reflects their carbon intensity. It prevents the EU's emissions reduction efforts from being offset by increasing emissions outside its borders through relocation of production to non-EU countries (with lower climate policy standards than those of the EU) or increased imports of carbon-intensive products. The CBAM will start to be implemented in 2026, following a 2023-2025 transitional period which imposes reporting obligations for EU importers.

This mechanism complements the EU Emissions Trading System (ETS). Thereby, imported products would be subject to a regulatory system that applies carbon costs equivalent to those that would have been borne under the EU ETS. Prior to CBAM, carbon leakage risks have been managed by allocating tradeable allowances under the EU ETS to plants that produce those goods at no cost. The CBAM phase-in plan gradually cease the free allocation of EU ETS allowances for CBAM-covered sectors, spanning nine years from 2026 to 2034.

The EU steel industry stands out as one of the sectors facing one of the greatest risk of carbon leakage, facing the challenge of potentially relocating production to countries with less stringent climate policies. The CBAM aims to ensure that carbon costs are harmonized for steel products entering the EU market, irrespective of their origin, creating a level playing field and minimizing the potential for competitive imbalances.

1.2.1.20. EU Hydrogen Strategy

The **EU Hydrogen Strategy**⁴⁴, introduced in 2020, emerges as a transformative roadmap for the European Union's energy landscape. With its sights set on a sustainable and resilient future, this strategy positions hydrogen at the forefront of the clean energy transition. By harnessing the potential of both green and low-carbon hydrogen, the strategy addresses energy security and environmental imperatives, aligning seamlessly with the EU's commitment to achieving climate neutrality by 2050.

Acknowledged as one of the high-emission industry, the steel sector is poised to derive substantial benefits from hydrogen integration. The strategy recognises hydrogen's potential as a key player in decarbonising sectors like steelmaking, acting both as a clean energy carrier and feedstock. Given the steel sector's substantial carbon footprint, innovative approaches are imperative to align with the EU's climate objectives.

1.2.1.21. EU-US global arrangement on sustainable steel and aluminium

In late 2021, the EU and the US committed to start negotiations on a sectoral agreement, the **Global Arrangement on Sustainable Steel and Aluminium (GSA)**⁴⁵, to address the issues of carbon intensity and global overcapacity in these two sectors. The initial deadline was October 2023. After several rounds of negotiation, the Joint statement on US-EU Summit in Washington DC on 20th October 2023, the parties committed to progress on the topic in the coming months, but no new information has been released in this respect so far.

⁴³ <https://data.consilium.europa.eu/doc/document/ST-16060-2022-INIT/en/pdf>

⁴⁴ [COM/2020/301 final](https://ec.europa.eu/commission/presscorner/detail/fr/ip_21_5724)

⁴⁵ https://ec.europa.eu/commission/presscorner/detail/fr/ip_21_5724

1.2.1.22. Trade Defence Instruments

Trade Defence Instruments (TDIs) are strategic measures used by the EU to protect its domestic industries from unfair trade practices by third countries. These measures comply with World Trade Organisation (WTO) rules and are designed to ensure a level playing field for EU producers. The main TDIs used by the EU include anti-dumping duties, countervailing duties (anti-subsidy), and safeguards.

Specifically for steel products, the EU has consistently employed TDIs in response to practices such as dumping or subsidies by third countries, or in reaction to disruptions caused by sudden surges in imports. The global steel market is highly competitive and has been a focal point for trade disputes and protective measures.

The EU monitors steel imports and may initiate investigations to determine whether dumping or subsidised imports are taking place. Should such an investigation conclude that these practices are detrimental to the EU steel industry, the EU is poised to enact trade defence measures targeting particular steel products from specific countries. For instance, in 2023, the European Commission renewed the anti-dumping measures on imports of stainless steel tube and pipe butt-welding fittings from China and Taiwan for an additional five years and expanded the measures to include Malaysia (European Commission, 2023).

Unlike subsidies and dumping, safeguards are not taken to address unfair practices. Instead, they are temporary measures applied when a surge in imports of a particular product threatens to cause serious injury to the EU industry. In 2019, the Commission imposed **definitive safeguard measures against imports of certain steel products**.⁴⁶ The measures concern 26 steel product categories and are composed of tariff-rate quotas; imports exceeding these quotas are subject to a duty of 25%. The tariff rate quotas are designed to preserve the historical import levels into the EU and will be increased progressively. These safeguards are scheduled to remain effective until their expiration in June 2026. Moreover, the European Regulation on Common rules for imports⁴⁷ provides for safeguard measures to protect the European market from excessive imports of low-cost steel that could damage EU producers. The Conflict Minerals Regulation⁴⁸ can also be considered within the trade policy domain, since it requires companies to ensure that the raw materials used, such as minerals and metals, do not come from conflict zones or do not fuel human right violations. The steel industry is directly affected as it must ensure the traceability and compliance of the raw materials used in steel application.

TDIs undergo regular reviews and may be subject to modifications, extensions, or terminations, contingent upon market conditions and their continuing impact on the EU industry.

1.2.1.23. Legislative Proposal on the Substantiation and Communication of Explicit Green Claims

The **Green Claims Initiative**⁴⁹ was announced by the Circular Economy Action Plan and the New Consumer Agenda⁵⁰. It aims to ensure that environmental claims are substantiated based on reliable, comparable and verifiable information across the EU, allowing market actors to make more sustainable decisions and increase consumer reliance in green labels and information.

The proliferation of misleading environmental claims is significant. Currently, there are 456 voluntary environmental labels worldwide⁵¹ that are often poorly defined and rely on non-comparable methods to assess environmental impact. To address this issue, the **Legislative Proposal on the Substantiation and Communication of Explicit Green Claims**⁵², presented in March 2023, requires environmental claims to be substantiated by using harmonised methods, in particular the EU Product and Organisation Environmental Footprint (PEF and PEO) methods, developed by the Commission's Joint Research Centre.⁵³

Furthermore, the proposal would also include provisions on environmental labels and labelling schemes. In particular, it would introduce a validation procedure for new schemes established by private operators (that should be assessed by national authorities). These schemes would undergo a thorough assessment by national authorities. To ensure the credibility of labelling schemes, the proposal mandates an ex-ante verification

⁴⁶ [Commission Implementing Regulation \(EU\) 2019/159](#)

⁴⁷ [Regulation\(EU\) 2015/478](#)

⁴⁸ [Regulation\(EU\) 2017/821](#)

⁴⁹ [Initiative on substantiating green claims](#)

⁵⁰ [COM/2020/696 final](#)

⁵¹ [Ecolabel Index](#)

⁵² [COM/2023/166 final](#)

⁵³ [Inception impact assessment on a legislative proposal on substantiating green claims](#)

process. This verification is entrusted to an officially accredited independent body, adding an extra layer of reliability to environmental declarations and labelling initiatives.

1.2.2 Member States legislation and policy initiatives

Notably, it is worthy to mention Green Public Procurement (GPP) schemes. These programs are particularly directed at sourcing low-carbon construction materials, including steel and concrete. Leading the charge in the adoption of such initiatives are countries like France, Sweden, the Netherlands and Spain, which employ a variety of instruments such as requiring embodied carbon disclosure, setting carbon limits, and providing economic incentives. These voluntary schemes, often applied at the project level, have been documented in the literature (Mission Possible Partnership, 2022), (Jernkontoret, 2023), (Kadefors A. et al, 2019). Some examples of Member States' initiatives are presented in

Table 6.

Table 6. Exemplary initiatives across EU Member States that directly or indirectly target steel-products

Country	Description
France	In the construction sector, France has set targets for the embedded carbon footprint of the whole construction (so-called "RE 2020" ⁵⁴). The regulation sets maximum emission thresholds, reduced over time and expressed in kg of CO ₂ per square metre per year, over a 50-year building lifetime. By targeting the total embedded emissions in construction materials over the entire life cycle of the building, RE2020 is expected to trigger a progressive transformation towards the production and use of low-carbon construction materials.
Sweden	The Swedish Transport Administration (Trafikverket) in its infrastructure projects applies procurement requirements related to mitigation of embodied carbon. The system applies GWP limits on a project level, and sets the criteria on maximum carbon emission levels for reinforcement steel (for the projects of a certain budget).
Netherlands	In the Netherlands an environmental scoring system is used in the procurement of the construction projects. The project-level Environmental Performance of Buildings (known as MPG in Dutch) and Economic Cost Indicator (ECI in English or MKI in Dutch) systems use shadow costs for materials to calculate a single score for environmental impacts at a project and product level respectively.
Spain	The draft Industry Law incorporates provisions for tenders under the General State Administration. Notably, it mandates "an increasing percentage of the materials and basic industrial products involved shall be required to be low carbon. The technical specifications of the tender shall determine which products and materials are affected by the provisions of the preceding paragraph, [...]". ⁵⁵ The draft is intended to be approved in the third quarter of 2024.
Germany	In the Klimaschutzprogramm 2030 ⁵⁶ , the role of public procurement is identified as a tool to promote low-emission products, including steel materials.
Italy	Some examples of policy initiatives have been identified in Italy: -Decreto Legge 162/2019, on urgent provisions on the extension of legislative deadlines, the organization of public administrations, as well as technological innovation ⁵⁷ -Decreto Legge 50/2016 on Public Procurement Code -The Minimum Environmental Criteria (CAM) ⁵⁸ including the Decree of the Ministry of the Environment of 11 October 2017 which encourage steel companies to supply products compliant with environmental requirements by promoting the use of sustainable steel in

⁵⁴ <https://www.ecologie.gouv.fr/reglementation-environnementale-re2020>

⁵⁵ https://industria.gob.es/es-es/participacion_publica/paginas/detalleparticipacionpublica.aspx?k=562

⁵⁶ <https://www.bmel.de/SharedDocs/Downloads/DE/Landwirtschaft/Klimaschutz/Klimaschutzprogramm2030.html>

⁵⁷ <https://www.normattiva.it/uri-res/N2Ls?urn:nir:stato:decreto.legge:2019-12-30:162>

⁵⁸ <https://www.ecosurvey.it/en/2018/05/minimum-environmental-criteria/>

	public works and subsequent amendments. It requires a minimum of up to 75% recovered content for each product placed in a public building or infrastructural project (ECOS, 2024) -Decreto Legge 77/2021 ⁵⁹ , which encourages the adoption of sustainable technologies in the steel sector with direct funding for technological innovation and emission reduction
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Source: Stakeholder's questionnaire

1.2.3 Third-country legislation and policy initiatives

The following section presents non-exhaustive list of the third country legislation and policy initiatives that directly or indirectly address iron and steel products.

1.2.3.1. California Buy Clean Program

The **Buy Clean California Act (BCCA)** of 2017 specifically targets carbon emissions associated with construction materials such as structural steel, carbon steel rebar, flat glass and wood insulation. When used in public works projects, these eligible materials must have a GWP that does not exceed the limit set. Following this, multiple state governments, including Washington, Colorado, Oregon, Minnesota, New York and New Jersey, have also enacted Buy Clean or related policies for construction materials (Gangotra et al., 2023) (California Government, n.d.).

Under the BCCA, all construction products used in state-funded building projects must provide an environmental product declaration (EPD). This requirement extends to the structural steel industry, including hot-rolled sections, hollow structural sections, plates, and concrete reinforcing steel. To qualify for use in these projects, these materials must meet GWP thresholds and hold independently certified EPDs, following ISO 14025 standards. Emissions included within the boundary are those associated with the production of the steel at a given facility and do not include the upstream emissions or the downstream emissions such as fabrication.

These thresholds (Table 7) were calculated as an average value collected over two years plus a tolerance to account for uncertainty set a 35%, 20%, and 15% for hot rolled sections and plates, hollow structural sections, and concrete reinforcing steel respectively (Mantle Developments, 2021). While these thresholds are expected to become more stringent over time, there are no explicit provisions addressing the use of scrap in the steel production process (Hasanbeigi and Sibal, 2023).

Table 7. GWP thresholds for steel products under the Clean California Act

Eligible Material	Description of eligible materials	Maximum Acceptable GWP Limit ^[1] For Unfabricated Product (Cradle-To-Gate) ^[2]	Maximum Acceptable GWP Limit ^[1] For Fabricated Product (A1 Module Only) ^[3]
Hot-rolled structural steel sections	Hot-rolled sections consisting of wide flange beams (W-shape and HP-shape used in structural applications), standard beams (S-shape), misc. beams (M-shape), channels, angles, and tees.	1,010 kg CO ₂ eq. ^[4] or 1.01E+03 kg CO ₂ eq. for one metric ton of structural steel.	1,080 kg CO ₂ eq. or 1.08E+03 kg CO ₂ eq. for one metric ton of structural steel.
Hollow structural sections	Hollow structural sections with round, square, or rectangular cross-section.	1,710 kg CO ₂ eq. or 1.71E+03 kg CO ₂ eq. for one metric ton of structural steel.	1,830 kg CO ₂ eq or 1.83E+03 kg CO ₂ eq for one metric ton of structural steel.
Steel plate	Plate materials	1,490 kg CO ₂ eq. or 1.49E+03 kg CO ₂ eq. for one metric ton of structural steel.	1,590 kg CO ₂ eq. or 1.59E+03 kg CO ₂ eq. for one metric ton of structural steel.
Concrete reinforcing steel	ASTM A615/A615M ASTM A706/A706M ASTM A767/A767M ASTM A775/A775M	890 kg CO ₂ eq. or 8.90E+02 kg CO ₂ eq. for one metric ton of bar.	920 kg CO ₂ eq. or 9.20E+02 kg CO ₂ eq. for one metric ton of bar.

⁵⁹ <https://www.fao.org/faolex/results/details/es/c/LEX-FAOC214946/>

Note: An unfabricated product is a basic construction material produced by a manufacturer and needs additional processing by a fabricator before use in a construction project. A fabricated product is a basic construction material that was processed by fabricators before use in a construction project. The current limits exclude biogenic carbon which is derived from organic matter.

[1] GWP limit is based on a 100-year lifetime impact.

[2] Use this column to determine compliance when an EPD declares unfabricated product GWP. Compare manufacturer cradle-to-gate GWP (i.e., the sum of information modules A1-A3) to the limit. For a manufacturer's unfabricated product EPD: A1 represents GWP due to raw material supply; A2 represents GWP due to transport; and A3 represents GWP due to manufacturing

[3] Use this column to determine compliance when an EPD declares fabricated product GWP (compare GWP from information module A1 to the limit). These limits are derived from the unfabricated product GWP and account for the waste in the fabrication process. For a manufacturer's fabricated product EPD: A1 represents GWP due to raw material supply (adjusted for waste during the fabrication process); A2 represents GWP due to transport (typically represents impacts of transport to fabrication); and A3 represents GWP due to manufacturing (typically represents impacts due to fabrication).

[4] Kilogram carbon dioxide equivalent.

Source: Buy Clean California Act Environmental Product Declaration (EPD) Compliance Guide (Department of General Services of California, 2023)

1.2.3.2. Border Carbon Adjustments for Canada

The Government of Canada announced in the 2021 its plans to develop a Border Carbon Adjustment (BCA) as an element of Canada's Climate Plan (Government of Canada, 2021). This proposed BCA, just like the EU CBAM, serves to equate the costs of imports with those of domestically produced goods by imposing carbon-related charges, thus mitigating the risk of carbon leakage (Government of Canada, 2021) (Cosbey, Bernstein, and Stiebert, 2021).

The government initiated public consultation exploring the use of the BCAs for a variety of fossil fuel and emissions-intensive trade-exposed (EITE) sectors, including steel, which account for more than 70% of Canada's exports (Chen et al., 2023).

The role of a BCA on the Canadian economy is indeed heavily dependent on whether BCAs are applied in the US, and the degree with which the US pursues climate action (about 56% of Canada's imports in EITE sectors in 2020 come from the US). In addition, concerns exist regarding the compliance of BCAs with the existing free trade agreement between US, Mexico and Canada. Although the Biden administration has expressed interest in an instrument like a BCA to complement its climate ambition, to date, there have been no concrete indication that it is planning to impose any kind of carbon price (Chen et al., 2023).

The consultation process concluded in mid-2023. Feedback gathered through it will play a pivotal role in guiding the government's subsequent actions in the context of implementing a BCA.

1.2.3.3. Canada Standard on Embodied Carbon in Construction

The objective of this standard⁶⁰ is to establish requirements to disclose and reduce the embodied carbon footprint of construction projects in accordance with the commitments in the *Greening Government Strategy*. Procurements commenced after 31st December 2022 are required to apply to this standard. It applies to organizations described in section 3 of the Policy on Green Procurement and to any new construction or renovation of real property in Canada at or above the thresholds listed in Appendix A at the time of solicitation of design services.

Among other requirements, organizations involved in the procurement of construction services must disclose the embodied carbon footprint of structural materials measured in global warming potential (GWP) with Environmental Product Declarations (EPDs). Where EPDs are not readily available, they must provide a life cycle assessment report that complies with international standards and that is verified by an LCA reviewer as such a report is considered an acceptable equivalent form of disclosure.

1.2.3.4. India

In 2019, the Government of India introduced the **Steel Scrap Recycling Policy**, a strategic framework aimed to develop a globally competitive industry reducing its reliance on imported scrap to become self-sufficient by 2030. This policy aligns with the objectives outlined in the **National Steel Policy**, which aim to produce 300 million tonnes per annum (MnTPA) of steel by 2030 with a contribution of 35-40% from Electric Arc Furnace/Induction Furnace (EAF/IF) route.

⁶⁰ <https://www.tbs-sct.canada.ca/pol/doc-eng.aspx?id=32742>

At its core, the Steel Scrap Recycling Policy is a framework to create an appropriate ecosystem to increase collection, processing and recycling of steel scrap. To this end, the initiative envisages the creation of metal scrapping centres across India. Concurrently, it addresses the need of alleviate urban congestion resulting from ELVs while creating an effective mechanism for the management of waste streams and residues generated during the dismantling and shredding processes, ensuring full compliance with the Hazardous & Other Wastes Rules (Government of India, 2019).

1.2.3.5. United Kingdom CBAM

Between March and June 2023, the UK Government launched a consultation on “addressing carbon leakage risk to support decarbonisation” covering among others, the iron and steel sector. This process aimed to gather insights across stakeholders on the best possible mix of policies to effectively tackle carbon leakage. Among the diverse range of potential policies under consideration, the options include a **carbon border adjustment mechanism (CBAM)**, **mandatory product standards (MPS)**, and **demand side policies such as product labelling and GPP initiatives** (UK Government, 2023).

As a result of this consultation, the UK government has decided to establish a UK CBAM by 2027. This mechanism, similar to the EU CBAM, will impose a carbon price on the most emissions intensive industrial goods imported to the UK from the aluminium, cement, ceramics, fertiliser, glass, hydrogen, iron and steel sectors. The UK CBAM will target Scope 1, Scope 2 and select precursor product emissions embodied in imported products, ensuring equivalent coverage with the UK Emission Trading Scheme (ETS) (UK Department for Energy Security & Net Zero, 2023).

1.2.3.6. United States Federal Buy Clean Initiative

The US, being the largest domestic buyer of construction materials, has set Green Public Procurement (GPP) criteria, including steel and other materials such as concrete. To align with its ambitious goal of achieving net-zero emissions procurement by 2050, it introduced the **Federal Sustainability Plan**⁶¹, which is realised through the **Federal Buy Clean Initiative**⁶².

Under this initiative, the General Services Administration (GSA), an agency that manages federal buildings and construction projects, has established standards for procuring low-carbon construction materials to be used on federal construction projects and that qualify for funding under the **Inflation Reduction Act (IRA)**⁶³ (Gangotra et al., 2023).

The IRA will be the instrument supporting this green shift. It provides the GSA with USD 2.15 billion for acquisition and installation of construction materials and products with substantial lower levels of embodied GHG emissions. In addition, an allocation of USD 250 million is dedicated to grants and technical assistance for Environmental Product Declarations. (U.S. General Services Administration, 2023).

For a construction material to be eligible for IRA funding, it must be accompanied by an EPD verified by a third external party. The EPD's reported Global Warming Potential (GWP) should fall below the prescribed GSA IRA limits, which are detailed in Table 8 (GSA, 2023). These limits are categorized in four columns: the second column defines maximum embodied carbon limits for the top-performing 20% of products within the same category, taking into consideration shared functional requirements like strength. The third column presents limits for the next 40%, which can be employed when the top 20% products are not accessible at the project site. The fourth column shows industry average or better limits, which may only be applied where top 20% and 40% are not available (GSA, 2023).

As in previous schemes analysed, the GSA recognises the necessity to differentiate GWP limits for integrated mills and EAF steelmaking technology routes. This distinction is grounded in the belief that it will drive innovation across various manufacturing processes and consequently lead to reduced emissions across all technology routes.

Table 8. GSA IRA Limits for Low Embodied Carbon Material Requirements – May 16, 2023

⁶¹ <https://www.sustainability.gov/federalsustainabilityplan/>

⁶² <https://www.sustainability.gov/buyclean/>

⁶³ <https://www.whitehouse.gov/cleanenergy/inflation-reduction-act-guidebook/#~:text=The%20Inflation%20Reduction%20Act%20specifies,mode%20or%20condition%2C%20low%20or>

Steel Product Category	Top 20% Limit	Top 40% Limit	Better Than Average Limit
Rebar (fabricated)	728	794	850
Rebar (unfabricated)	611	716	760
Hollow Structural Sections (fabricated)	1,778	1,854	1,898
Hollow Structural Sections from EAF (unfabricated)	1,580	1,620	1,652
Hollow Structural Sections from Integrated Mills (unfabricated)	TBD	TBD	TBD
Hot-Rolled Sections (fabricated)	1,022	1,128	1,163
Hot-Rolled Sections (unfabricated)	686	713	869
Cold-Formed and Galvanized (stud, track, framing, etc.)	2,228	2,324	2,408
Structural Steel Plate from EAF (unfabricated)	987	1,152	1,190
Structural Steel Plate from integrated mills (unfabricated)	TBD	TBD	TBD

Note. EPD-Reported GWPs, in kilograms of carbon dioxide equivalent per metric ton - kgCO₂e/ t). Requirements are provisional. GSA is proceeding with an IRA Pilot Program for a limited number of construction projects. GSA intends to update its interim requirements using lessons learned from the Pilot Program. Construction product assemblies (such as fabricated steel products including handrails, conduit, ductwork, pipes, metal raceways, etc.) qualify for the IRA funding if at least 80% of the assembly's total cost or total weight comprises materials that meet the requirements set in this table.

Source: Interim IRA Low Embodied Carbon Material Requirements (U.S. General Services Administration, 2023)

These limits have been developed base on the **UL's PCR Guidance for Building-Related Products and Services, Part B (8/2020, version 2.0)** (UL, 2020) and the **SCS Global Services' PCR for Designated Steel Construction Products (5/2015, version 1.0)** (SCS Global Services, 2015). GSA'S GWP limits are built on industry-average EPDs and actual products' publicly-available EPD data.

The American Iron and Steel Institute (AISI) has raised significant concerns about the methodology and underlying implications of the proposed standards, strongly advocating for a re-evaluation of the current approach. The institute expressed reservations about the consistency of methodology and data sources, the categorisation of steel products, the exclusion of relevant data, and the influence of transportation impacts. Furthermore, the AISI is apprehensive about the potential implications of the proposed standards, particularly regarding their impact on the sustainability goals of the steel industry (American Iron and Steel Institute, 2023).

In August 2024, the Environmental Protection Agency (EPA) announced its plan for implementing a **new label programme**⁶⁴ based on the Federal Buy Clean Initiative. This program will define what constitutes "clean" construction materials.

1.2.3.7. United States CBAM

During the 117th Congress, various legislative initiatives were introduced to use trade policy to address carbon emissions. In July 2021, Senator Chris Coons and Representative Scott Peters introduced companion bills in the Senate and the House of Representatives. These bills proposed a border carbon adjustment (BCA) fee on imports of carbon-intensive goods into the US, including steel, aluminium, cement, and fossil fuels. (Smith, 2023) (Cohen, 2023).

Finally, the last Congress did not approve any legislation on this specific issue, but some proposals have been reintroduced in the 118th Congress. This is the case for the PROVE IT Act⁶⁵, which requires the Department of Energy to study and determine the emissions intensity of nearly two dozen products (including steel) made in the US and by G-7 countries, free-trade agreement partners, foreign countries of concern and countries that hold a substantial global market share for a covered product. This is considered as an intermediate step to establish a future BCA, since it lays the ground for an informed discussion around potential emissions-based BAC tax (Dumain, 2023) (Latham & Watkins, 2023).

Whitehouse is another sponsor of CBAM legislation, proposing bills in previous Congress sessions that would have imposed a domestic price on carbon — something that might be necessary for ensuring a BCA's

⁶⁴ https://www.epa.gov/system/files/documents/2024-08/lpa_final_8-6-24.pdf

⁶⁵ https://www.coons.senate.gov/imo/media/doc/prove_it_act_118th_congress_bill_text2.pdf

enforceability by the World Trade Organization (WTO) but has long been politically divisive and not currently a part of the conversation in Congress (Dumain, 2023).

Countries developing their own CBAM

Some countries reacted to the European CBAM with the announcement of plans to introduce their own CBAMs⁶⁶. This is the case of Australia, where the government initiated feasibility assessment of introducing a local CBAM in the summer of 2023. The second round of consultations is currently (December 2024) underway, and a final decision has not yet been made. Taiwan plans to develop its own version of CBAM by mid-2025. South Korea began consultations on CBAM introduction in August 2024.

⁶⁶ <https://gmk.center/en/infographic/how-different-countries-around-the-world-reacted-to-cbam-introduction-in-the-eu/>

1.3 Standards

The following sections aim to provide an overview regarding the most relevant existing standards which are applied to steel. The chapter addresses terminology standards (Section 1.3.1) established by different Standards Development Organisations (SDOs) and the European Standards (Section 1.3.2) on steel considering both the steel application (Section 1.3.3) and the mechanical testing (Section 1.3.4).

The list of Steel Terminology Standards is reported in Table 9.

Table 9. List of Steel Terminology Standards

Code	Title
EN 10020:2000	Definition and Classification of Grades of Steel
EN 10027-1:2005	Designation systems for steels—Part 1: Steel names
EN 10052:1993	Vocabulary of Heat Treatment Terms for Ferrous Products
EN 10079:2007	Definition of Steel Products
EN 10169-1:2003	Continuously Organic Coated (Coil Coated) Steel Flat Products—Part 1: General Information (Definitions, Materials, Tolerances, Test Methods)
EN 10266:2003	Steel tubes, fittings and structural hollow sections—Symbols and definitions of terms for use in product standards
ISO 6929:1987	Steel products—Definitions and classification
ISO 2532:1974	Steel wire ropes—Vocabulary
ISO 3252:2000	Powder metallurgy—Vocabulary
ISO 4885:1996	Ferrous products—heat treatments—Vocabulary
ISO 8954-1:1990	Ferroalloys—Vocabulary—Part 1: Materials
ISO 8954-2:1990	Ferroalloys—Vocabulary—Part 2: Sampling and sample preparation
ISO 8954-3:1990	Ferroalloys—Vocabulary—Part 3 Sieve analysis
ISO 17893:2004	Steel wire ropes—Vocabulary, designation, and classification
GB/T 13304-1991	Steels—Classification
GB/T 15574-1995	Steel products classification and definitions
GB/T 15575-1995	Steel products standard designation
GB/T 341-1989	Steel wire—Classification and vocabulary
JIS G 0201:2000	Glossary of terms used in iron and steel (Heat treatment)
JIS G 0202:2013	Glossary of terms used in iron and steel (Testing)
JIS G 0203:2009	Glossary of terms used in iron and steel (Products and quality)
ASTM A941-13b	Standard Terminology Relating to Steel, Stainless Steel, Related Alloys, and Ferroalloys
ASTM A644-14	Standard Terminology Relating to Iron Castings
ASTM A751-14a	Standard Test Methods, Practices, and Terminology for Chemical Analysis of Steel Products
ASTM A902-15	Standard Terminology Relating to Metallic Coated Steel Products

Source: Handbook of Comparative World Steel Standards. 5th Edition (Bringas, 2016)

1.3.1 Standards by steel application

List of standards from EN 10027-1:2016

For the different steel application, defined in Section 1.1.1.2, the list of related EN standards is reported in Table 10.

Table 10. Steel application categories and related EN standards

Categories	Standard
Structural steel	EN 10025-2:2004 Hot rolled products of structural steels - Part 2: Technical delivery conditions for non-alloy structural steels
	EN 10025-3:2004 Hot rolled products of structural steels - Part 3: Technical delivery conditions for normalized/normalized rolled weldable fine grain structural steels

	EN 10025-4:2004 Hot rolled products of structural steels - Part 4: Technical delivery conditions for thermomechanical rolled weldable fine grain structural steels
	EN 10025-5:2004 Hot rolled products of structural steels - Part 5: Technical delivery conditions for structural steels with improved atmospheric corrosion resistance
	EN 10025-6:2004 Hot rolled products of structural steels - Part 6: Technical delivery conditions for flat products of high yield strength structural steels in the quenched and tempered condition
	EN 10149-2:2013 Hot rolled flat products made of high yield strength steels for cold forming - Part 2: Technical delivery conditions for thermomechanically rolled steels
	EN 10149-3:2013 Hot rolled flat products made of high yield strength steels for cold forming - Part 3: Technical delivery conditions for normalized or normalized rolled steels
	EN 10210-1:2006 Hot finished structural hollow sections of non-alloy and fine grain steels - Part 1: Technical delivery conditions
	EN 10248-1:1995 Hot rolled sheet piling of non-alloy steels - Part 1: Technical delivery conditions
	EN 10346:2015 Continuously hot-dip coated steel flat products for cold forming - Technical delivery conditions
Pressure purpose	EN 10028-2:2009 Flat products made of steels for pressure purposes - Part 2: Non-alloy and alloy steels with specified elevated temperature properties
	EN 10028-3:2009 Flat products made of steels for pressure purposes - Part 3: Weldable fine grain steels, normalized
	EN 10028-4:2009 Flat products made of steels for pressure purposes - Part 4: Nickel alloy steels with specified low temperature properties
	EN 10028-5:2009 Flat products made of steels for pressure purposes - Part 5: Weldable fine grain steels, thermomechanically rolled
	EN 10028-6:2009 Flat products made of steels for pressure purposes - Part 6: Weldable fine grain steels, quenched and tempered
	EN 10028-7:2009 Flat products made of steels for pressure purposes - Part 7: Stainless steels
	EN 10120:1996 Steel sheet and strip for welded gas cylinders
	EN 10207:2005 Steels for simple pressure vessels - Technical delivery requirements for plates, strips and bars
	EN 10213:2007 Steel castings for pressure purposes
	EN 10216-1:2013 Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
	EN 10216-2:2013 Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 2: Non-alloy and alloy steel tubes with specified elevated temperature properties
	EN 10216-3:2013 Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 3: Alloy fine grain steel tubes
	EN 10216-4:2013 Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 4: Non-alloy and alloy steel tubes with specified low temperature properties
	EN 10216-5:2013 Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 5: Stainless steel tubes
Steel for line pipe	EN ISO 3183:2012 Petroleum and natural gas industries - Steel pipe for pipeline transportation systems
	EN 1123-1:1999 Pipes and fittings of longitudinally welded hot-dip galvanized steel pipes with spigot and socket for waste water systems - Part 1: Requirements, testing, quality control
	EN 1123-2:2006 Pipes and fittings of longitudinally welded hot-dip galvanized steel tube with spigot and socket for waste water systems - Part 2: Dimensions
	EN 1123-3:2004 Pipes and fittings of longitudinally welded hot-dip galvanized steel pipes with spigot and socket for waste water systems - Part 3: Dimensions and special requirements for vacuum drainage systems and for drainage systems in ship-building
	EN 1124-1:1999 Pipes and fittings of longitudinally welded stainless steel pipes with spigot and socket for waste water systems - Part 1: Requirements, testing, quality control
	EN 1124-2:2014 Pipes and fittings of longitudinally welded stainless steel pipes with spigot and socket for waste water systems - Part 2: System S, forms and dimensions
	EN 1124-3:2008 Pipes and fittings of longitudinally welded stainless steel pipes with spigot and socket for waste water systems - Part 3: System X - Dimensions
	EN 1124-4:2008 Pipes and fittings of longitudinally welded stainless steel pipes with spigot and socket for wastewater systems - Part 4: Components for vacuum drainage systems and for drainage systems on ships
	EN 1759-1:2004 Flanges and their joints - Circular flanges for pipes, valves, fittings and accessories, Class designated - Part 1: Steel flanges, NPS 1/2 to 24
	EN 10241:2000 Steel threaded pipe fittings
	EN 10253-1:1999 Butt-welding pipe fittings - Part 1: Wrought carbon steel for general use and without specific inspection requirements

	EN 10253-2:2007 Butt-welding pipe fittings - Part 2: Non-alloy and ferritic alloy steels with specific inspection requirements
	EN 10253-3:2008 Butt-welding pipe fittings - Part 3: Wrought austenitic and austenitic-ferritic (duplex) stainless steels without specific inspection requirements
	EN 10253-4:2008 Butt-welding pipe fittings - Part 4: Wrought austenitic and austenitic-ferritic (duplex) stainless steels with specific inspection requirements
	EN 12007-1:2012 Gas infrastructure - Pipelines for maximum operating pressure up to and including 16 bar - Part 1: General functional requirements
	EN 12007-2:2012 Gas infrastructure - Pipelines for maximum operating pressure up to and including 16 bar - Part 2: Specific functional requirements for polyethylene (MOP up to and including 10 bar)
	EN 12007-3:2015 Gas infrastructure - Pipelines for maximum operating pressure up to and including 16 bar - Part 3: Specific functional requirements for steel
	EN 12007-4:2012 Gas infrastructure - Pipelines for maximum operating pressure up to and including 16 bar - Part 4: Specific functional requirements for renovation
	EN 12007-5:2014 Gas infrastructure - Pipelines for maximum operating pressure up to and including 16 bar - Part 5: Service lines - Specific functional requirements
	EN ISO 11960:2014 Petroleum and natural gas industries - Steel pipes for use as casing or tubing for wells
Engineering Steel	EN 10025-2:2004 Hot rolled products of structural steels - Part 2: Technical delivery conditions for non-alloy structural steels
	EN 10293:2015 Steel castings - Steel castings for general engineering uses
	EN 10296-1:2003 Welded circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions - Part 1: Non-alloy and alloy steel tubes
	EN 10296-2:2005 Welded circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions - Part 2: Stainless steel
	EN 10297-1:2003 Seamless circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions - Part 1: Non-alloy and alloy steel tubes
	EN 10297-2:2005 Seamless circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions - Part 2: Stainless steel
	EN 10223-3:2013 Steel wire and wire products for fencing and netting - Part 3: Hexagonal steel wire mesh products for civil engineering purposes
	EN 10250-1:1999 Open die steel forgings for general engineering purposes - Part 1: General requirements
	EN 10250-2:1999 Open die steel forgings for general engineering purposes - Part 2: Non-alloy quality and special steels
	EN 10250-3:1999 Open die steel forgings for general engineering purposes - Part 3: Alloy special steels
	EN 10250-4:1999 Open die steel forgings for general engineering purposes - Part 4: Stainless steels
	EN 10277-2:2008 Bright steel products - Technical delivery conditions - Part 2: Steels for general engineering purposes
Steels for reinforcing concrete	EN 10080:2005 Steel for the reinforcement of concrete - Weldable reinforcing steel - General
	ASTM A615/A615M Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
	BS 4449:2005+A3:2016 Steel for the reinforcement of concrete. Weldable reinforcing steel. Bar, coil and decoiled product. Specification
	DIN 488-1 Reinforcing steels - Part 1: Grades, properties, marking
	ISO 6935-1:2007 Steel for the reinforcement of concrete
	EN 1992-1-1 Eurocode 2: Design of concrete structures - Part 1-1: General rules and rules for buildings
Steels for prestressing concrete	EN 10138-2:2019 Prestressing steels - Part 2: Wire.
	EN 10138-3:2006 Prestressing steels - Part 3: Strands
	EN 10138-4:2019 Prestressing steels - Part 4: Bars
	EN 523:2003 Steel strip sheaths for prestressing tendons - Terminology, requirements, quality control
	EN 524-1:1997 Steel strip sheaths for prestressing tendons - Test methods - Part 1: Determination of shape and dimensions
	EN 524-2:1997 Steel strip sheaths for prestressing tendons - Test methods - Part 2: Determination of flexural behaviour
	EN 524-3:1997 Steel strip sheaths for prestressing tendons - Methods of test - Part 3: To-and-fro bending test
	EN 524-4:1997 Steel strip sheaths for prestressing tendons - Test methods - Part 4: Determination of lateral load resistance
	EN 524-5:1997 Steel strip sheaths for prestressing tendons - Test methods - Part 5: Determination of tensile load resistance

	EN 524-6:1997 Steel strip sheaths for prestressing tendons - Test methods - Part 6: Determination of leaktightness (determination of water loss)
Rails	EN 13674-1:2011 Railway applications - Track - Rail - Part 1: Vignole railway rails 46 kg/m and above
	EN 13674-2:2006 Railway applications - Track - Rail - Part 2: Switch and crossing rails used in conjunction with Vignole railway rails 46 kg/m and above
	EN 13674-3:2006 Railway applications - Track - Rail - Part 3: Check rails
	EN 13674-4:2006 Railway applications - Track - Rail - Part 4: Vignole railway rails from 27 kg/m to, but excluding 46 kg/m
	EN 13262:2004 Railway applications - Wheelsets and bogies - Wheels - Product requirement
Flat products for cold forming	EN 10111:2008 Continuously hot rolled low carbon steel sheet and strip for cold forming - Technical delivery conditions
	EN 10130:2006 Cold rolled low carbon steel flat products for cold forming - Technical delivery conditions
	EN 10152: 2009 Electrolytically zinc coated cold rolled steel flat products for cold forming - Technical delivery conditions
	EN 10209:2013 Cold rolled low carbon steel flat products for vitreous enamelling - Technical delivery conditions
	EN 10346:2015 Continuously hot-dip coated steel flat products for cold forming - Technical delivery conditions
High strength steel flat products for cold forming	EN 10268:2006 Cold rolled steel flat products with high yield strength for cold forming - Technical delivery conditions
	EN 10346:2015 Continuously hot-dip coated steel flat products for cold forming - Technical delivery conditions
	EN 10131:2006 Cold rolled uncoated and zinc or zinc-nickel electrolytically coated low carbon and high yield strength steel flat products for cold forming - Tolerances on dimensions and shape
	EN 10149-2:2013 Hot rolled flat products made of high yield strength steels for cold forming - Part 2: Technical delivery conditions for thermomechanically rolled steels
	EN 10149-3:2013 Hot rolled flat products made of high yield strength steels for cold forming - Part 3: Technical delivery conditions for normalized or normalized rolled steels
Tin mill products (packaging)	EN 10202:20022 Cold-reduced tinplate and electrolytic chromium/chromium oxide coated steel
	EN 10244-4:2001 Steel wire and wire products - Non-ferrous metallic coatings on steel wire - Part 4: Tin coatings
Electrical Steel	EN 10106:2007 Cold rolled non-oriented electrical steel sheet and strip delivered in the fully processed state
	EN 10107:2014 Grain-oriented electrical steel strip and sheet delivered in the fully processed state
	EN 10341:2006 Cold rolled electrical non-alloy and alloy steel sheet and strip delivered in the semi-processed state
	EN 10251:1997 Magnetic materials - Methods of determination of the geometrical characteristics of electrical steel sheet and strip

Source: own elaboration based on EN 10027-1:2016 and Handbook of Comparative World Steel Standards. 5th Edition (Bringas, 2016)

1.3.2 Mechanical Properties Testing Standards

Table 11 compiles the Relevant International Standards related to the different mechanical tests for steel

Table 11. Mechanical tests

Mechanical Test	Description	Relevant International Standards
Tensile Test	It measures the material's response to axial stretching forces to determine properties like ultimate tensile strength, yield strength, elongation, and reduction in area.	ASTM E8/E8M, ISO 6892-1, EN 10002-1
Compression Test	It subjects the material to axial compressive forces, assessing properties such as compressive strength and deformation under compression.	ASTM E9, ISO 6892-1, EN 10002-1
Bend Test	It evaluates ductility and resistance to cracking under bending forces, often used for welds.	ASTM E290, ISO 7438, EN ISO 5173

Impact Test	It measures a material's ability to absorb energy under dynamic loading; Charpy and Izod tests assess toughness and notch sensitivity.	Charpy Impact Test: ASTM E23, ISO 148-1, EN 10045-1 Izod Impact Test: ASTM D256, ISO 180, EN ISO 180
Hardness Test	It determines resistance to deformation, indentation, or scratching; Brinell, Rockwell, and Vickers test are common.	Brinell: ASTM E10, ISO 6506-1, EN ISO 6506-1 Rockwell: ASTM E18, ISO 6508-1, EN ISO 6508-1 Vickers: ASTM E92, ISO 6507-1, EN ISO 6507-1
Fatigue Test	Simulates cyclic loading to assess fatigue behavior, including endurance limit and fatigue life.	ASTM E466, ISO 12106, EN ISO 12106
Creep Test	Evaluates deformation under sustained loads and elevated temperatures.	ASTM E139, ISO 204, EN 10302
Torsion Test	Applies twisting forces to determine shear strength and torsional behavior.	ASTM E143, ISO 16811, EN ISO 16811
Fracture	Measures resistance to crack propagation and brittle fracture.	ASTM E399, ISO 12135, EN ISO 12135
Toughness Test	Evaluates a material's resistance to fracture under impact and static loads.	ASTM E23, ISO 148, EN 10045-1
Shear Test	Measures shear strength and behavior under shear forces.	ASTM E384, ISO 178, EN ISO 178
Creep Rupture	Evaluates time-dependent deformation and failure under sustained loads at elevated temperatures.	ASTM E139, ISO 204, EN 10302

Source: own elaboration based on "Handbook of Comparative World Steel Standards. 5th Edition" (Bringas, 2016) and "A study of the mechanical properties of steel/aluminium/GRP laminates" (Khalili, Mittal, and Kalibar, 2005).

1.3.3 Relevant standards for the decarbonisation of the steel products

This section presents an overview of relevant standards that establish unified methods for calculation and assessment of environmental performance of steel products. Table 12 presents standards for product life cycle assessment, environmental product declarations and product category rules, and measuring and reporting GHG emissions from the steelmaking process.

Table 12. Relevant standards for the decarbonisation of the steel products

Standard	Description
Product LCA/EPD/PCR Standards	
ISO 14040:2006	Environmental management — Life cycle assessment — Principles and framework
ISO 14044:2006	General LCA
ISO 14021:2017	Labels Type II
ISO 14025:2006	Labels Type III
ISO 14026:2017	Principles, requirements and guidelines for communication of footprint information
ISO 14067:2018	Series of standards that specifies the requirements for quantifying the carbon footprint of products; the standard can be used in multiple sectors and is not sector specific
ISO 21930:2017	Series of standards that establishes the principles, specifications and requirements to develop an Environmental Product Declaration (EPD) for construction products and series. EPD are widely used in the construction sector, although at an early stage of development and use in the other steel end use sectors.
EN 15804:2012+A2:2019	Sustainability of construction works — Environmental Product Declarations — Core rules for the product category of construction products
prEN 17662	LCA for steel and aluminium c-PCR under EN15804
ISO 20915:2018	Series of standards that specifies the life cycle (LC) inventory calculation methodology for steel products
ISO 14067:2018	Carbon footprint of products
EN 45557:2020	Recycled content in energy related products
Organisation or site GHG emissions quantification and reporting	
ISO 14064-1:2018	Quantification and reporting org
ISO/TR 14069:2013	Guidance to ISO 14064-1
ISO 14064-2:2019	Quantification and reporting project
ISO 14064-3:2019	Verification and validating statements
ISO 14694-1:2021	Stationary emissions GHG in energy intensive industries.
ISO/AWI TS 19694-2	Stationary source emissions. GHG emissions in energy-intensive industries. Part 2: Iron and Steel Industry (under development)

SS-EN 19694-1:2016	GHG Energy intensive industries general
SS-EN 19694-2:2016	GHG Energy intensive industries, iron and steel industry
ISO 14404-1:2013	CO2 Calculation method Steel complimentary calculation rules BF-BOF
ISO 14404-2:2013	CO2 Calculation method Steel complimentary calculation rules EAF
ISO 14404-3:2017	CO2 Calculation method Steel complimentary calculation rules DR-EAF
ISO 14404-4:2020	CO2 Calculation method Guidance on 14404
ISO 14065:2020	Principles and Requirements for validating and verifying environmental information

Source: Initiatives and standards for the transition to near zero steel production (Wei, Gyllenram, and Östman, 2023) and Emissions Measurement and Data Collection for a Net Zero Steel Industry (IEA, 2023)

1.4 Initiatives towards decarbonisation of steel industry

This section provides a concise overview of various standards and protocols, as well as industry or internationally driven initiatives for decarbonisation of steel industry. Table 13 offers a summary of exemplary initiatives described in this chapter, drawing from existing literature (Hasanbeigi and Sibal, 2023) (Jernkontoret, 2023; IEA, 2023).

Table 13. Overview of exemplary initiatives relevant to steel industry

Type	Exemplary initiatives
Emissions Measurement Methodologies (Section 1.4.1)	• ISO/EN standards – see Section 1.3.3 (C, S, P) • World Steel Association's CO2 Data Collection User Guide (v. 11) (C) • World Steel Life Cycle Inventory (LCI) methodology (P) • WRI's GHG Protocol for Steel (C, P) • American Iron and Steel Institute Steel Production Greenhouse Gas Emissions Calculation Methodology Guidelines (C, P) • Horizon Zero Initiative (P) • EU PEF methodology (P) • EN 15804:2019 • NetZero Steel Pathway Methodology Project (C) • Responsible Steel Standards & Certification (S) • Science Based Target Initiative (SBTi) (C, P) • The Steel Climate Standard - Global Steel Climate Council (GSCC) (P)
Demand Side Initiatives (Section 1.4.2)	• Industrial Deep Decarbonization Initiative (IDDI) (C) • SteelZero Initiative (C) • First Movers Coalition (public-private partnership) (C)
Clean finance initiatives and venture capital funding (Section 1.4.3)	• Climate Bonds Initiative's Criteria for the Steel Industry (C, P) • Climate Action 100+steel strategy (C) • Sustainable Steel Principles (C) • Glasgow Financial Alliance for NetZero (C) • Equator principles (C) • UN Convened Net Zero Asset Owner Alliance (C) • Centre for Climate Aligned Finance (C)
Initiatives and standards that set thresholds to define "green steel" (Section 1.4.4)	• IEA's definition of Near-Zero Emission Steel and Low Emission Steel (S) • Responsible Steel Performance Levels (S) • Low Emission Steel Standard (LESS) - WV Stahl (German Steel Association) (P) • Climate Bonds Initiative's Criteria for the Steel Industry (C, P) • Global Steel Climate Council (GSCC) (P) • EUROFER Stainless Steelmakers Proposal (P)

Note. C = Company level, S = Site level, P = Product level

Source: JRC elaboration

1.4.1 Emissions Measurement Methodologies

This section presents an outline of measurement methodologies for accounting greenhouse gas (GHG) emissions embedded in steel products or in the steel industry. Alongside the ISO and EN standards discussed in Section 1.3.3, numerous organisations have developed their own approaches.

1.4.1.1 World Steel Association's CO₂ Data Collection User Guide (v.11)

The World Steel Association (worldsteel) set up a methodology for collecting and reporting on the CO₂ emissions intensity of steel production among its members. The CO₂ emissions data are collected on voluntary basis and involve currently approximately 200 sites each year⁶⁷. The data collection and verification methodology is described in the **CO₂ Emissions Data Collection User Guide** (World Steel Association, 2023), which is in line with an international standard ISO 14404:2013 – calculation method of carbon dioxide emission intensity from iron and steel production. The data is confidential and only reported to members that contribute to the benchmarking system. The only industry-wide global averages for the entire industry or processes are made publicly available.

1.4.1.2 World Steel Life Cycle Inventory (LCI) methodology

The Worldsteel LCI methodology (World Steel Association, 2017) is used to measure the environmental impact of 17 finished steel products. This type of impact assessment adheres to ISO 14040:2006 and ISO 14044:2006, which establishes comprehensive requirements, guidelines and principles for generic LCA. Notably, the Worldsteel LCI methodology is the basis for ISO 20915, which is a steel-specific LCA measurement methodology (IEA, 2023)

1.4.1.3 WRI's GHG Protocol for Steel

The World Resources Institute's (WRI) GHG Protocol is a framework⁶⁸ for the companies and organisation to measure and report their GHG emissions. Among the tools from the Protocol Initiative's website (www.ghgprotocol.org), those relevant for steel industry are: GHG Protocol for Corporate accounting and Reporting Standard (WRI, 2001 - revised in 2004) and the guidance on GHG emissions calculations from iron and steel production (WRI, 2008). The GHG Protocol is widely used by the private sector, with 9 in 10 Fortune 500 companies⁶⁹ using their methodology; the Corporate Standards is currently under review (IEA, 2023).

1.4.1.4 American Iron and Steel Institute (AISI) Steel Production Greenhouse Gas Emissions Calculation Methodology Guidelines

American Iron and Steel Institute proposes its members own calculation approach for GHG emissions from steel production; it focuses on product-level disclosures and corporate-level reporting and is aligned with ISO standards. AISI guidance is designed to be a living document to gather best practices for GHG emission calculations. It is meant to be used by AISI and its members when engaging in initiatives and policy-related discussions pertaining to the development of carbon steel and stainless steel production GHG emissions calculations. (American Iron and Steel Institute, 2022).

1.4.1.5 Horizon Zero Initiative

A private sector initiative that focuses on product-level GHG accounting and a technical architecture to digitally track these emissions. The initiative aims to provide a harmonized GHG accounting framework — one that can be compared across supply chains and products and to promote the exchange of product-level GHG information. Horizon Zero's Steel Accounting Guidance (Wright et al., 2023) details the emissions calculation and reporting that can be used for the comparison of products across the supply chain.

⁶⁷ As reported on the initiative website in October 2023.

⁶⁸ The WRI is not a standards body and does not verify or certify measurements based on the GHG Protocol methodologies.

⁶⁹ The Fortune 500 refers a list of the largest companies (by revenue) in the United States compiled by Fortune magazine.

1.4.1.6 EU PEF methodology

The European Commission proposed the Product Environmental Footprint (PEF) method as a standardised approach to assess environmental performance⁷⁰. The PEF is the EU recommended Life Cycle Assessment (LCA) based methods to quantify the environmental impacts of products.

The rules provided in the PEF method are designed to conduct PEF studies that are more reproducible, comparable and verifiable, than those produced by alternative existing approaches. However, comparability is only possible if the results are based on the same Product Environmental Footprint Category Rules (PEFCR). The primary objective of a PEFCR is to establish a uniform and precise set of rules to calculate the relevant environmental information of products within a defined category, thereby enabling their comparison (L and R, 2019).

As of now, a specific PEFCR (Eurometaux, 2019) has been developed for steel metal sheets, which is applicable to the construction and appliances sectors. This PEFCR details the standards for evaluating the environmental impact of steel sheets used within these industries, ensuring that assessments across different producers and products are consistent and based on a common framework.

1.4.1.7 EN 15804:2019

The EN 15804:2019 is a European standard that applies to all construction products. This standard provides a method to assess the impacts of products and services in order to obtain a Type III environmental declaration of Construction products (EPD), providing quantified environmental information on a harmonized and scientific basis. The purpose of an EPD in the construction sector is to provide the basis for assessing buildings and other construction works, and identifying those which cause less stress to the environment.

EN 15804+A2 is used for the environmental assessment of steel in the context of the Construction Products Regulation (Regulation (EU) 3110/2024) as part of the mandatory declaration of performance and conformity.

1.4.1.8 The Net-Zero Steel Pathway Methodology Project

The Net Zero Steel Pathway Methodology Project⁷¹ proposes a framework to establish consistency by outlining key principles to underpin steel sector guidance for net zero targets. These principles are designed to promote consistency among companies in measuring and setting GHG reduction targets, in alignment with the sector's required contribution (Bonaplata et al., 2021).

1.4.1.9 Responsible Steel

Responsible Steel is a voluntary certification initiative for steel producers that is based on the International Standard V2.0 (ResponsibleSteel, 2022). The standard covers 13 principles of the steel production, including environmental, social and governance issues; the latest edition of the standard, from May 2024, incorporates additional requirements on GHG emissions and the sourcing of input materials. For the approval of the certificate, a producer must achieve a minimum threshold level of performance for the intensity of GHG emissions for the production of **crude steel**. The scope of the GHG emission calculations includes direct (Scope 1), indirect (Scope 2), and upstream indirect (Scope 3) emissions associated with the extraction, processing and transportation of input materials. The results define and distinguish 4 levels of performance, where level 1 is a reference for the responsible steel threshold, and level 4 set the ambitious of “nearly zero GHG emission” steel production (Hasanbeigi and Sibal, 2023).

1.4.1.10 Science Based Target Initiative for Steel

The Science Based Targets initiative (SBTi) is a joint initiative established by the UN Global Compact, WWF, the World Resources Institute (WRI) and Carbon Disclosure Action (CDA). It works with companies to first benchmark emissions, before setting emission reduction targets for different levels of ambition. There is a specific **Steel Science-Based Target- Setting Guidance** (SBTi, 2023) that specifies how much and how quickly a company would need to reduce its GHG emissions to align with the goals of Paris Agreement.

⁷⁰ [EU Commission Recommendation 2021/2279](#)

⁷¹ <https://www.netzerosteelpathwayproject.com/>

1.4.1.11 The Steel Climate Standard - Global Steel Climate Council (GSCC)

The Global Steel Climate Council (GSCC), an international coalition of steel producers, developed the Steel Climate Standard for measuring and reporting steel carbon emissions. The scheme requires independent third-party verification of emissions calculations and reduction targets (GSCC, 2023).

This methodology has developed a decarbonisation glidpath aligned with the IEA's Net Zero Emissions by 2050 Scenario. To attain certification, each company is required to calculate their average steelmaking emissions intensities and establish a company-specific trajectory for GHG reduction targets in line with this glidpath (GSCC, 2023).

1.4.2 Demand side initiatives

Industrial Deep Decarbonisation Initiative (IDDI)

The Industrial Deep Decarbonisation Initiative (IDDI)⁷² is the initiative coordinated by United Nations Industrial Development Organisation (UNIDO) and focused on **green public procurement**, standards and data, with an initial focus on the steel and cement sectors. The goal is to standardise carbon assessments, help setting public and private sector procurement targets, incentivise investment into low-carbon emissions product development and design industry guidelines. The initiative is based on the IEA's definition of Low-Carbon Steel (see Section 1.4.1).

SteelZero initiative

SteelZero⁷³ is a bottom up initiative aiming to speed up the transition to a net zero steel industry. Organisations that join SteelZero make a public commitment to buy and use 50% low emission steel by 2030, setting a clear pathway to using 100% net zero steel by 2050. The idea is to send a strong demand signal (for a low carbon steel) to shift global markets and policies towards responsible sourcing of steel.

First Movers Coalition (public-private partnership)

The First Movers Coalition⁷⁴ set commitments for purchasers of steel; its members commit to purchasing volumes of near-zero emission steel by 2030. Steel purchasers set a target that at least 10% of their annual steel procurement volumes by 2030 meet or exceed the First Movers Coalition definition for near-zero emissions (see also Section: 1.4.1)

1.4.3 Clean finance initiatives and venture capital funding

Climate Bonds Initiative's Criteria for the Steel Industry

The Climate Bonds Standard and Certification Scheme is a labelling scheme for bonds and loans. The standard set a sector-specific eligibility Criteria and benchmarks, aiming to ensure a transition to net-zero grown (with respect to the 1.5 °C warming limit in the Paris Agreement). A dedicated criteria is available for the steel industry (Climate Bonds, 2023) and includes eligibility criteria for credible investments and opportunities for policymakers to facilitate the rapid transition of the industry to meet global climate targets (Climate Bonds, 2022).

To assess GHG emissions from projects, the Steel Criteria uses ISO 14404 EN 19694 and the GHG Protocol. The standard uses the same system boundary as the Sustainable STEEL Principles (IEA, 2023).

Sustainable Steel Principles

The Sustainable STEEL Principles (SSP)⁷⁵ is a disclosure framework for the financial sector that enables banks to assess the alignment of their steel lending portfolios with the goal of net zero emissions by 2050, in line with the requirements of the UN-convened Net-Zero Banking Alliance⁷⁶. The initiative aims for gathering

⁷² <https://www.unido.org/IDDI>

⁷³ <https://www.theclimategroup.org/steelzero>

⁷⁴ https://www3.weforum.org/docs/WEF_FMC_Sector_One_pagers_2023.pdf

⁷⁵ https://climatealignment.org/wp-content/uploads/2022/06/sustainable_steel_principles_framework.pdf

⁷⁶ <https://www.unepfi.org/net-zero-banking/>

information on the emissions intensity and scrap charge in the steel sector' clients. The banks use SSP to gather and report on emissions that cover all scope 1-3 emissions (not including indirect emissions from fossil fuel transport and supply). The Sustainable STEEL Principles association signed a Memorandum of Understanding with ResponsibleSteel to support further harmonisation between the two organisations' standards (IEA, 2023).

Climate Action 100+ steel strategy

Climate Action 100+⁷⁷ is an investor-led initiative that makes their signatories to commit to provide enhanced corporate disclosure in line with the final recommendations of the Task Force on Climate-related Financial Disclosures (TCFD) and sector-specific Global Investor Coalition on Climate Change (GIC) Investor Expectations on Climate Change guidelines (when applicable), to enable investors to assess the robustness of companies' business plans against a range of climate scenarios.

Glasgow Financial Alliance for NetZero

The Glasgow Financial Alliance for Net Zero (GFANZ)⁷⁸ is a global coalition of financial institutions that are committed to aligning their lending and investment portfolios with net-zero emissions by 2050. It aims to mobilize the financial industry to support the transition to a low-carbon economy and address climate change. GFANZ focuses on accelerating the decarbonisation of the financial sector and promoting sustainable, climate-friendly investments.

Equator principles

The Equator Principles (EP)⁷⁹ is a risk management framework for financial institutions to determine, assess, and manage environmental and social risk in project financing. It is primarily applied to large-scale infrastructure and industrial projects. The EP provide a set of guidelines based on international standards for responsible banking, including the International Finance Corporation (IFC) Performance Standards and the World Bank Group Environmental, Health, and Safety Guidelines. The principles aim to ensure that projects funded by the financial sector are developed in a manner that is socially responsible and environmentally sustainable, contributing to the promotion of sustainable development.

UN-convened Net-Zero Asset Owner Alliance

The UN-convened Net-Zero Asset Owner Alliance⁸⁰ is a global coalition of institutional investors committed to transitioning their investment portfolios to net-zero greenhouse gas emissions by 2050. The Alliance members are the finance industry's first to set intermediate targets, which include CO2 reduction ranges for 2025 (22 – 32%) and for 2030 (40% – 60%). This coalition, comprising leading asset owners from around the world, aims to align their investments with the goals of the Paris Agreement and support the transition to a low-carbon economy. By joining the alliance, institutional investors demonstrate their dedication to addressing climate change and contributing to sustainable, long-term investment practices.

Centre for Climate Aligned Finance

The Centre for Climate Aligned Finance⁸¹, created by the Rocky Mountain Institute (RMI), is dedicated to aligning financial strategies with climate goals. This initiative serves as a collaborative platform for financial institutions, policymakers, and climate experts to drive capital towards sustainable, climate-friendly investments and integrate climate considerations into financial decision-making. The centre's mission is to accelerate the transition to a low-carbon, climate-resilient economy.

⁷⁷ <https://www.climateaction100.org/approach/global-sector-strategies/steel/#>

⁷⁸ <https://www.gfanzero.com/>

⁷⁹ <https://equator-principles.com/>

⁸⁰ <https://www.unepfi.org/net-zero-alliance/about/>

⁸¹ <https://climatealignment.org/>

1.4.4 Initiatives and standards that set performance thresholds to define “green steel”

Several initiatives have been launched to establish performance thresholds that serve as a **benchmark for the definition of green steel**. A consistent focus among these initiatives is the assessment of the carbon intensity of steel products, thereby defining what constitutes a low-embedded emissions steel product, often referred to as “green steel,” “low-carbon steel,” or “near zero steel”, among others.

Despite the efforts of numerous stakeholders to propose definitions, a consensus has yet to be reached. A recent study by Zaccaro (2024), provides an exemplary overview of the various definitions, which are summarized in **alongside additional** definitions for related terms such as low-carbon steel, climate-neutral steel, clean steel, and others.

Table 14 alongside additional definitions for related terms such as low-carbon steel, climate-neutral steel, clean steel, and others.

Table 14. Definitions of green steel in the literature

Source	Name	Definition
Opportunities and challenges for decarbonizing steel production by creating markets for ‘green steel’ products (Muslemeni et al., 2021)	Green Steel	‘green steel’ will hereinafter be used to refer to steel products manufactured using less GHG-intensive production processes, and not to refer to products with lesser amounts of physical carbon content.”
Advances in Green Steel Making Technology -A Review (Singh and Rout, 2018)	Green Steel	“green steel is a new steelmaking process lowers greenhouse gas emission, cuts costs and improves the quality of steel.”
The prospects for ‘green steel’ making in a net-zero economy: A UK perspective (Griffin and Hammond, 2021)	Green Steel	‘green steel’ often refers to steelmaking that results in lower air pollutant emissions (e.g., CO, SO _x , NO _x , or PM _{2.5}), wastewater contaminants, hazardous wastes (including arsenic, lead, and zinc), and other solid wastes.”
Green steel: Synergies between the Australian iron ore industry and the production of green hydrogen (Wang et al., 2023)	Green Steel	“green steel (i.e., steel produced using hydrogen from renewable sources as the reducing agent).”
Kloekner Metals (Kloekner Metals, s.d.)	Green Steel	Right now, there is no universal definition of green steel. It is often used as an umbrella term to describe any steel with reduced carbon emissions during production. We think only steel with the lowest possible carbon emissions should be called “green,” meaning steel produced using direct reduced iron + green hydrogen or an electric arc furnace + 100% scrap and renewable electricity to eliminate emissions as far as possible. Other steel with lower emissions is “carbon reduced.”
Can the cars we buy drive green steel production? (CEPS, 2023)	Green Steel	[...] “green steel will simply mean ‘steel produced in a way that is compatible with climate neutrality in the long run, i.e. close to zero GHG emissions’. Likewise, decarbonised steel means close to zero GHG emissions embedded in the steel, not the carbon grade related to steel quality characteristics.”
SSAB (SSAB, s.d.)	Green steel	Green steel refers to the strategy aimed at making the steelmaking process greener and more sustainable.
SSAB (SSAB, s.d.)	SSAB fossil-free	SSAB Fossil-free steel is based on iron-ore produced without fossil fuels, using HYBRIT® technology, which uses hydrogen instead of coal in the ore reduction process, emitting water instead of CO ₂ .

SSAB (SSAB, s.d.)	SSAB zero	SSAB Zero™ is made of recycled steel and produced with fossil-free electricity and biogas – resulting in steels with virtually no fossil carbon emissions.
Swiss Steel Group (Swiss Steel Group, n.d.)	Green Steel	Green Steel can be manufactured through the use of optimized raw materials, implementation of energy efficiency practices, and by employing low-CO2 energy sources in the electric steel production method.
IEA (IEA, 2022)	Near-zero emissions steel production	100% iron: 400 kg CO2eq. per tonne of crude steel 100% scrap: 50 kg CO2eq. per tonne of crude steel Including a progressive threshold according to the scrap share
WV Stalh (WV Stahl, 2023)	Green or climate-neutral Steel	Green or climate-neutral steel is described within the context of the international climate policy agenda as "near zero emission" (net zero) because the current state of technology results in (low) residual emissions and (significant) upstream emissions being unavoidable throughout the supply chain
H2 Green Steel (H2 Green Steel, s.d.)	Green Steel	[...] "actual CO2 reduction and improving circularity are both key. Therefore, green steel must be produced from a combination of a significant amount of green virgin iron and scrap in a production process that uses electricity from renewable energy sources."

Source: From niche to mainstream: Shaping demand for Green Steel (Zaccaro, 2024); Green Steel (Kloekner Metals, s.d.); SSAB (SSAB, s.d.); What we understand by Green Steel (Swiss Steel Group, n.d.); and WV Stalh (WVStahl, 2023).

The extensive proliferation and fragmentation of measurement standards and methodologies for establishing thresholds for green steel pose a significant challenge in the realms of climate, trade, and ensuring fair competition. With over 150 sustainability standards and initiatives in place across the entire steel value chain (Zaccaro, 2024), the lack of coherence among these initiatives hinders consumers' ability to discern what constitutes a steel product with lower carbon-embedded emissions. Therefore, promoting alignment is crucial to send an appropriate signal into the market, decrease transition cost, lower verification costs, and foster investment in decarbonisation technologies (ResponsibleSteel, 2022; IEA, 2022; WTO, 2022). However, achieving this alignment may be challenging, as the purpose of these initiatives and standards is not always the same, and each scheme responds to the motivations of its proponents, tailoring its approach to address their specific needs.

Despite the challenges mentioned, there is a growing interest in establishing a commonly accepted standard and methodology at the international level to create consistent benchmark for setting emissions thresholds. To this end, various initiatives have emerged to promote interoperability and comparability. For instance, many organisations endorsed during the COP28 the **Steel Standards Principles**⁸² on common emissions measurement methodologies to accelerate the transition to near zero. Also, since 2023, the **Climate Club** members⁸³, supported by the Interim Secretariat (OCDE & IEA), have been working towards an international common understanding on comparable and interoperable standards for defining low and near-zero emissions steel. Additionally, the World Trade Organisation (WTO), through the WTO's Agreement on Technical Barriers to Trade, promotes the alignment and coherence of decarbonisation standards for the iron and steel sector (WTO, 2022).

Table 15 presents some of the initiatives that propose emissions thresholds for defining "green steel". These thresholds are typically set in line with climate science and a modelling target or decarbonisation trajectory to achieve Paris Agreement commitments by 2050. A critical factor in any definition is **how ferrous scrap is considered**, given that this contributes to a significantly lower carbon footprint with the use of Electrical Arc Furnace (EAF) steel production compared to iron ore-based steel production in a Blast Oxygen Furnace (BOF) (WTO, 2022; IEA, 2022). Additionally, a key differentiating aspect among these thresholds is their temporal

⁸² https://www.wto.org/english/tratop_e/envir_e/steel_standards_principles_e.pdf

⁸³ A high-level forum for cooperation on accelerating climate action and increasing ambition, particularly in the field of industry decarbonisation. <https://climate-club.org/>

dimension. Specifically, thresholds can be either **fixed**, remaining unchanged over time and serving as a stable reference point, or **progressive**, incorporating time-variable targets that evolve over time.

One of the most popular approaches is the “sliding scale” method, which GHG emissions as a function of the proportion of scrap used. The IEA, the First Mover Coalition (FMC), ResponsibleSteel, the WV Stahl – German Steel Federation (Low Emission Steel Standard – LESS), and the Low Carbon Emission Steel Evaluation Method Standard developed by the China Iron and Steel Association (CISA) follow this approach. These initiatives promote the inclusion of scrap-content as a key parameter in the equation on the grounds that it focuses on comparability in the efforts made by steelmakers for decarbonisation, regardless of the manufacturing route used. These organisations argue that there is limited room for an increase of ferrous scrap availability, at least in the short term. However, it is important to note that at the EU level, ferrous scrap collection reaches nearly 100 Mt, of which approximately 18 Mt are exported to third countries (EUROFER, 2023), against a total crude steel production of 136 Mt. Some forecasts estimate that the amount of scrap available in Europe will increase, enabling it to cover 80% of the EU’s steel demand by mid-century (Material Economics, 2018).

In contrast, at the international level, the situation is likely to differ. According to the IEA's Sustainable Development Scenario, which projects a total steel demand of around 2 billion tonnes annually by 2050, there is currently insufficient ferrous scrap to meet demand, and this shortage is expected to persist in 2050 (IEA, 2020). Therefore, ore-based steelmaking will continue to play an important role to meet the current and future steel demand.

Table 15. Main initiatives and standards that propose emission thresholds for the definition of green steel

Initiative	Methodology Approach	Thresholds	Level	System Boundaries
IEA's definition of Near-Zero Steel and Low-Emission Steel	Sliding scale	Fixed thresholds	S	Cradle-to-gate (Scope 1 + Scope 2 + Scope 3.1 partially)
Responsible Steel (Site-level Certification)	Sliding scale	Fixed thresholds	S, P	Cradle-to-gate (Scope 1 + Scope 2 + Scope 3.1 + Scope 3.3 + Scope 3.4 + Scope 3.5)
LESS - WV Stahl (German Steel Association)	Sliding scale	Fixed thresholds	P	Cradle-to-gate (Scope 1 + Scope 2 + Scope 3.1 + Scope 3.3 + Scope 3.4)
Climate Bonds Initiative	Pathway by technology route	Progressive thresholds	C	Scope 1 + Scope 2 + Scope 3.3
Steel Climate Standard - Global Steel Climate Council (GSCC)	Company-specific trajectory based on decarbonisation pathway for flat and long product	Progressive thresholds	P, S	Cradle-to-gate (Scope 1 + Scope 2 + Scope 3.1 + Scope 3.3 + Scope 3.4)
Low CO2 Stainless Steel - EUROFER Stainless Steelmakers	Grade Specific	Fixed thresholds	P	Cradle-to-gate (Scope 1 + Scope 2 + Scope 3.1)
Chinese Low Carbon Emission Steel Evaluation Method Standard	Sliding scale	Fixed thresholds	S, L, P	Cradle-to-gate (Scope 1 + Scope 2 + Scope 3.1 partially + Scope 3.3.)

Note: Level: C- company; S- Site; L- Production line; P: Product

Source: JRC elaboration

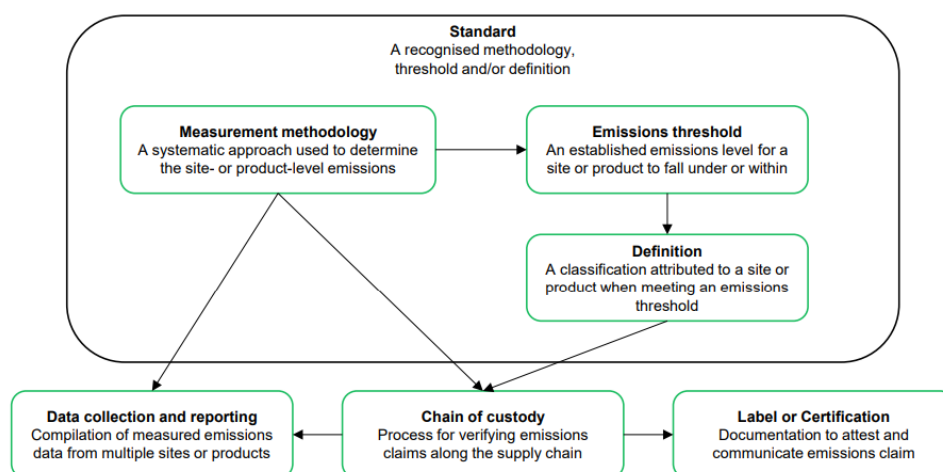
It is important to highlight that this approach is not supported by the entire steel industry. The **Global Steel Climate Council (GSCC)**, for instance, advocates for a different methodology, where performance classes are indexed based on product carbon footprint, not discriminating the labelling process with regards to the scrap content. The GSCC argues that using the sliding scale approach to classify a product may result in categorising two steel products as equally 'green' despite one generating significantly more carbon emissions than the other. For that reason, they advocate for a uniform global approach that applies to all steelmaking routes and covering emission's scope 1, 2 and 3 (GSCC, 2023). It is worth mentioning that the standard focuses on carbon intensities of semi-finished products so that a distinction between flat and long products is proposed. Although the current standard only certifies carbon steel, this approach could be extended to cover speciality steels or stainless steels.

The GSCC has a dynamic approach, setting a glide path with interim targets over time until 2050 as tied to a steel producer's overall corporate-wide emissions reduction targets. This approach differs significantly from other dynamic approaches, such as the one developed by the Science Based Targets Initiative (SBTi) and the Climate Bond Initiatives, which set differentiated thresholds for the iron ore-based and scrap-based technology routes. Additionally, the Low CO2 Stainless Steel, an alternative approach developed by the stainless

steelmakers who are members of EUROFER, establishes grade-specific dynamic threshold. This approach is designed to address the specificities of stainless steel, which encompasses a large variety of grades, leading to a wider carbon footprint range based on the alloy content.

To establish a coherent system that ensures credibility and comparability of environmental performance of iron and steel products in the EU market several elements are essential. Figure 4. illustrates the key building blocks proposed by the IEA to develop a standard that enables comparability. These include a well-defined measurement methodology with clear scope and system boundaries, established emission thresholds, defined classes of environmental performance, a comprehensive data collection and reporting scheme, and a transparent chain of custody to verify emissions claims along the supply chain. Additionally, a label or certification is necessary to attest to the emissions claim.

Figure 4. Key building blocks for defining “low carbon” and “near zero” steel



IEA. CC BY 4.0.

Note: Arrows indicate flow of information between the different processes.

Note. This definition of green steel is based only on the carbon intensity of products and disregard other kind of environmental impact or attributes.

Source: IEA 2024; Standards for a net zero iron and steel sector in India, <https://www.iea.org/reports/standards-for-a-net-zero-iron-and-steel-sector-in-india>, Licence: CC BY 4.0, License: CC BY 4.0

Despite the large number of initiatives and standards that set emission targets, potentially serving as thresholds for the definition of “green steel”⁸⁴, this section describes in detail those that comply with the following criteria:

- Well-documented methodology that is accessible
- Establishes a measuring methodology for the calculation of carbon footprint that produces coherent results
- Allows comparability of environmental performance at corporate or product level
- Provides a comprehensive and reliable methodology for setting thresholds, such as using science-based decarbonisation pathways to define emissions intensity targets/thresholds.

Therefore, the analysis includes the **seven initiatives and standards** presented in Table 15. In addition to the above-mentioned methodologies, other relevant approaches that do not meet the specific criteria have been included in Annex 5 for completeness. These include the Science Based Target Initiative (SBTi), proposals from Federacciai and UNESID, among others. Furthermore, initiatives led by private companies that offer “green steel” products as part of their portfolio have also been considered.

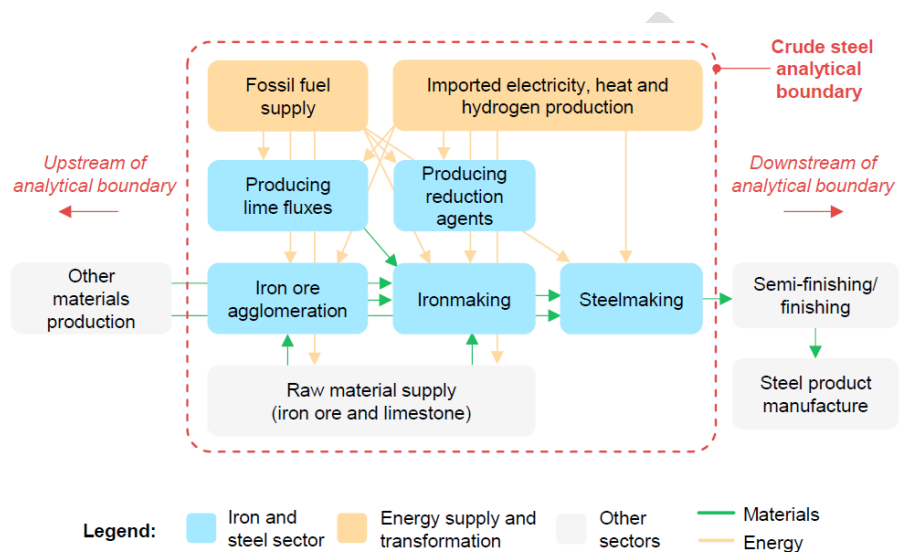
⁸⁴ This definition draws inspiration from the IEA’s definition (IEA, 2022). “Green or climate-neutral steel is described within the context of the international climate policy agenda as “near zero emission” (net zero) because the current state of technology results in (low) residual emissions and (significant) upstream emissions being unavoidable throughout the supply chain” (WV Stahl, 2023). The proposed definition aligns with German Steel Association (WV Stahl)’s definition of green steel as outlined in their classification system for green steel proposal (WV Stahl, 2023).

1.4.5.1 IEA's definition of Low-Emissions Steel

The International Energy Agency (IEA) proposes a common definition for **“near zero emissions production”** for the consideration of the G7 countries and also for **“low-emissions production”**, which recognised interim measures that substantially lower emissions intensity of steel. These measures may include natural gas-based direct reduced iron-electricity arc furnace (N-DRI-EAF), electric kiln or kiln with calcination of CCUS, among others.

Due to the heterogeneity of steel products, the IEA's has only developed its proposal at the crude steel production stage. It includes scope 1, scope 2 and partial upstream scope 3 carbon dioxide emissions (CO₂), as well as methane (CH₄) emissions from fossil fuel supply (see Figure 5.). The sorting and transportation of steel scrap and production of refractory linings for furnaces and ferroalloys are excluded.

Figure 5. System boundaries of IEA's definition of near zero emission steel production



IEA. All rights reserved.

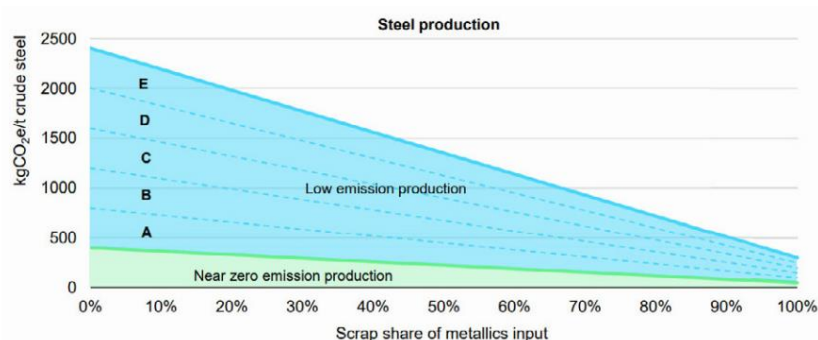
Notes: "Other materials production" refers to the production of material inputs to the iron and steel sector besides iron ore and limestone, including electrodes, alloying elements and refractory linings.

Source: Achieving Net Zero Heavy Industry Sectors in G7 Members (IEA, 2022)

The proposed threshold for “near zero emissions production” definition is established on a **sliding scale** between zero and 100% scrap use, with the quantitative thresholds based on the IEA's scenario analyses of potential pathways to achieve these goals, particularly the IEA's Net Zero by 2050 Roadmap (IEA, 2021). Under this framework, the proposed threshold for crude steel production with zero scrap use is set at 400 kg of CO₂ equivalent per tonne (kgCO₂eq/t), while for crude steel production with 100% scrap inputs, the proposed threshold value is 50 kgCO₂eq/t of crude steel.

In addition, the IEA also has established **various performance levels** that recognised the importance the interim measures that can substantially reduce emissions on the way to achieve near zero production. To this purpose, the intensity range between low emission production and near zero emission production is divided into five increments, or “bands,” labelled from Class A to E, as illustrated in Figure 6. The maximum emissions intensity allowed to qualify as low emission production is set at six times the near zero emission threshold. Setting the threshold at six times the near zero emission level places it approximately 10–20% lower than the emissions intensity of dominant conventional process routes (IEA, 2022).

Figure 6. IEA's near zero emission crude steel production threshold and proposed classification system



Source: Achieving Net Zero Heavy Industry Sectors in G7 Members (IEA, 2022)

1.4.5.2 Responsible Steel performance levels

The Responsible Steel initiative is an international, non-profit multi-stakeholder membership organisation which aims to enhance the responsible sourcing, production and recycling of steel. Its certification is only granted to those sites that are committed to comply with a wide range of requirements under thirteen principles (see Figure 7.) , including GHG emissions, responsible sourcing, human rights, and labour standards compliance (ResponsibleSteel, 2022).

Figure 7. Principles of the ResponsibleSteel Standard Version 2.0



Source: ResponsibleSteel International Standard Version 2.0 (ResponsibleSteel, 2022)

As explained in the Figure 8., under the ResponsibleSteel standard, certification can be obtained for both sites and products. The site-level certification proposes thresholds to assess the decarbonisation progress of steelmakers from a cradle-to-crude steel perspective. On the other hand, product-level certification enables the tracking of the total embodied carbon associated with a specific steel product from cradle-to-gate.

Figure 8. Product-level carbon footprint and site-visit level decarbonisation progress levels

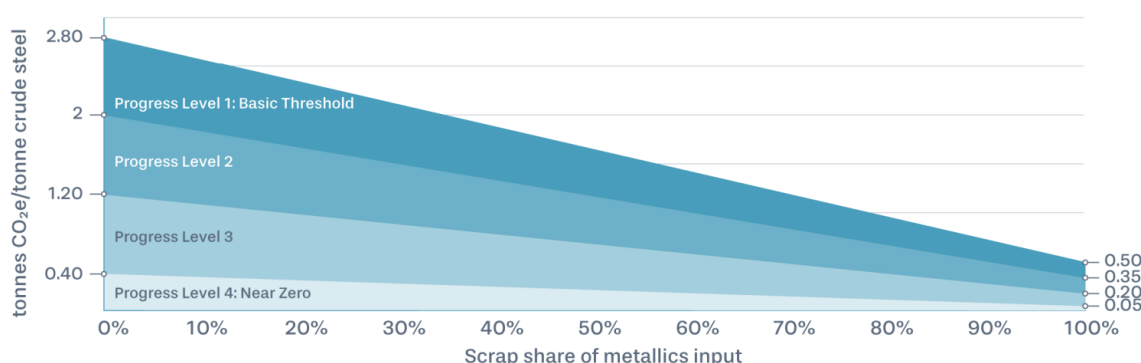
	Decarbonisation Progress Levels	Product Carbon Footprint/EPD
GHG measurement	From cradle-to-crude steel, at site level	Lifecycle emissions, at product level
Comparability	✓ Consistent and comparable unit of measurement across entire global industry	✗ Not comparable across the industry due to different products and accounting methodologies
Emissions inclusivity	✗ Stops at crude steel (to ensure comparability across all steel products)	✓ Cradle-to-gate production and end-of-life emissions considered
Accounts for scrap constraints	✓ Scrap-variable scale prevents carbon leakage	✗ No mandatory scrap declaration
Support for push to near zero	✓ Can drive global, sector-wide progress to near zero, ensuring equity and impact	✗ Useful to the customer to help reduce scope 3 emissions, but on its own will not deliver climate outcomes we all seek
Measurement rules	ResponsibleSteel V2 Criterion 10.4, which builds on the worldsteel CO2 methodology and ISO 14404 series	Internationally/regionally-recognised standards, e.g. PAS 2050, EN 15804, ISO 14067/14025/20915
Required for ResponsibleSteel Progress Level Certification	✓	

Source: ResponsibleSteel Progress Level Certification Requirements for Decarbonisation. [Unpublished]. (ResponsibleSteel, 2024)

In accordance with the standard, it is necessary to account for all relevant greenhouse gas emissions, including carbon dioxide (CO₂), nitrous oxide (N₂O), methane (CH₄), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF₆), and nitrogen trifluoride (NF₃). This encompasses Scope 1 (direct emissions), Scope 2 (indirect emissions), and Scope 3 upstream emissions, which include material extraction, material preparation and processing, waste generated in operations, and transport-related emissions.

ResponsibleSteel specifies four performance levels for both **GHG emissions performance**, and for progress on the **responsible sourcing of input materials**. For GHG emissions performance, as illustrated in Figure 9., the Level 1 Basic Threshold starts at 2.8 tons of CO₂e per ton of crude steel if 0% scrap is utilised and slides to 0.5 tons of CO₂e per ton of crude steel if 100% scrap is utilised. Performance Level 2, Performance Level 3, and Performance Level 4, the hardest to obtain, starts at 2000, 1200, and 400 Kg of CO₂e per ton of crude steel for 0% scrap utilization and slide to 250, 150, and 50 Kg of CO₂e per ton of crude steel for 100% scrap utilisation respectively (Hasanbeigi and Sibal, 2023).

Figure 9. Environmental Performance Levels defined by Responsible Steel



Source: ResponsibleSteel International Production Standard. Version 2.1 (ResponsibleSteel, 2024)

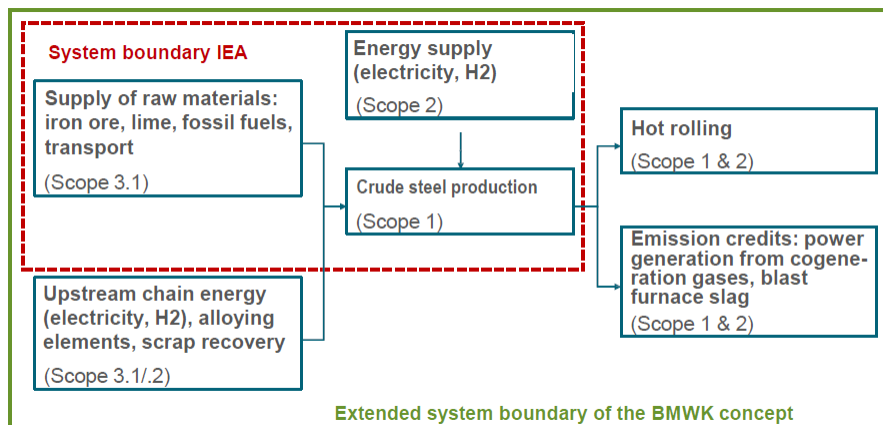
1.4.5.3 Low Emission Steel Standard (LESS) - WV Stahl (German Steel Association)

The WV Stahl (German Steel Association) has developed the **Low Emission Steel Standard (LESS)** based on the definition for climate friendly materials developed by a broad stakeholder process led by the German Federal Ministry for Economic Affairs and Climate Action (BMWK). This definition classifies emissions based on the scrap used in steel production, in line with the basic principles of the IEA definition (WV Stahl, 2024).

However, the LESS standard has a broader scope compared to the IEA, with notable differences in its approach (see Figure 10.). For instance, it expands the scope of the balance sheet to include steel refining, including casting and hot rolling as the first further processing stage⁸⁵. Additionally, the LESS standard takes into account additional emission sources in Scope 3, particularly those related to the use of alloying agents. As a result, this methodology applies partial surcharges compared to the IEA approach. Moreover, the extended scope up to hot-rolled steel, requires further partial surcharges, considering the additional production processes.

⁸⁵ The system boundaries proposed by BMWK are much broader than the IEA one. The use of alloying elements, the preparation of scrap materials and the production of graphite electrodes used are not taken into account by the IEA. The hot rolling process is also not included. Furthermore, the approach of the IEA draws narrower boundaries when it comes to scope 3 emissions.

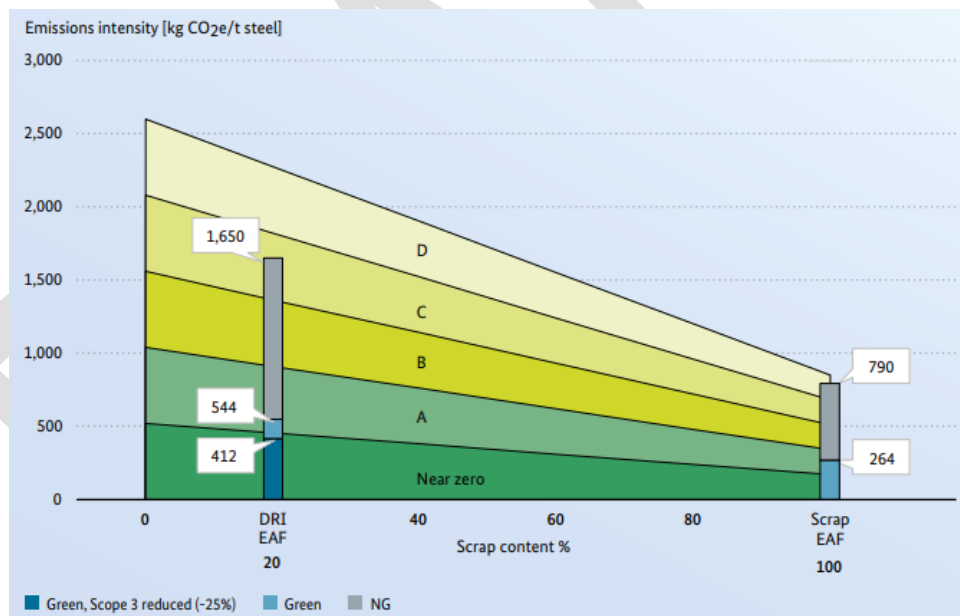
Figure 10. Comparison of LESS and IEA system boundaries



Source: Stakeholder process for "Lead markets for climate-friendly basic materials" (BMWK, 2024)

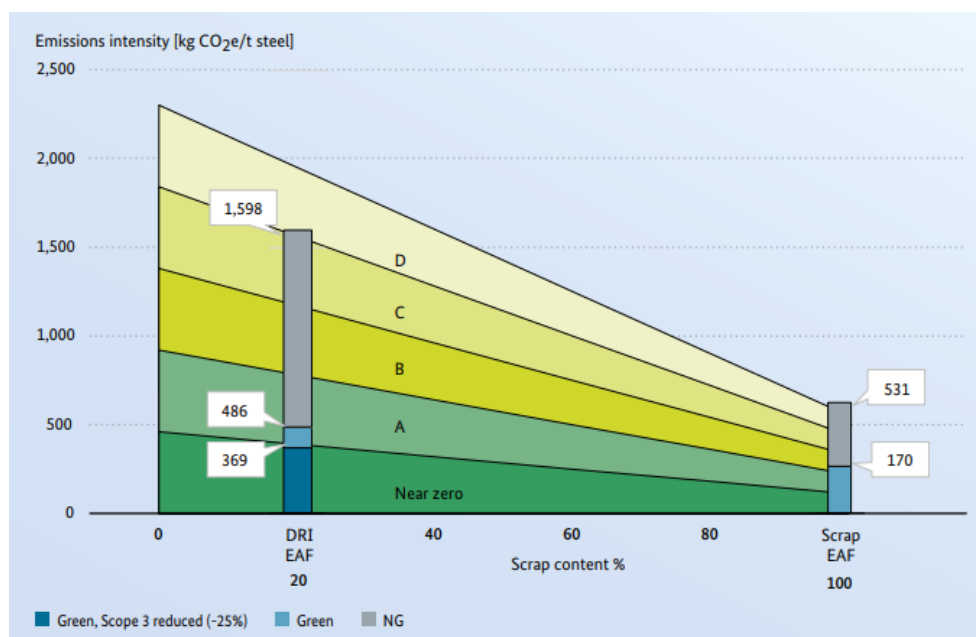
This surcharge was determined during the stakeholder dialog for green lead markets led by the German Federal Ministry for Economic Affairs and Climate Action (BMWK). The technical committee of the BMWK has calculated the surcharge according to the mentioned method. As shown in Figure 11. and Figure 12., the total impact calculated for quality steel (QST) is estimated at 120 kg CO₂e/t of rolled steel and 70 kg CO₂e/t of rolled steel for structural and reinforcing steel (BST). Based on this target value, multiplications are made by a factor for each of the levels A - D; all values above this are qualified as "E" (WV Stahl, 2024).

Figure 11. Definition for Near Zero and Low emissions quality steel



Source: Lead markets for climate-friendly basic materials (Guidehouse GmbH, Fraunhofer ISI, and Wuppertal Institute, 2023) on behalf of the German Federal Ministry for Economic Affairs and Climate Action (BMWK)

Figure 12. Definition for Near Zero and Low emissions structural steel



Source: Lead markets for climate-friendly basic materials (Guidehouse GmbH, Fraunhofer ISI, and Wuppertal Institute, 2023) on behalf of the German Federal Ministry for Economic Affairs and Climate Action (BMWK)

In terms of emissions under scope and system boundaries, the classification system adopts a **cradle-to-gate** approach, excluding downstream emissions. It encompasses Scope 1 and Scope 2 carbon dioxide emissions (CO₂) in line with the EU ETS. For Scope 3, it covers only the upstream energy chain, as well as essential raw materials (scrap, ore, alloy materials, slag formers, refractories, technical gases, and other consumables) and transportation of these materials to the production site. Additionally, Scope 3 emissions include all GHG emissions: CO₂, CH₄, N₂O, HFCs (partial), HFCs, SF₆. In addition, the methodology allows to apply carbon credits for substances and energies that are dispensed beyond the accounting limits, such as slag, use of domed gases for generation of electricity or heat and energy supplied to third parties.

There are two noteworthy aspects of this methodology. Firstly, it considers two distinct categories of steel: reinforcing and structural steel, and quality steel, aligning with the definitions provided by the EU Emissions Trading System (EU ETS)⁸⁶. Moreover, the methodology also considers **different steel grades by implementing alloying adjustment rules**, allowing for the establishment of distinct limit values for all existing steels. In the future, the rulebook will be extended to include additional manufacturing processes (e.g. ingot casting and forged products) and other categories of products as stainless steel.

Building upon this comprehensive methodology, WV Stahl has developed a LESS label⁸⁷ (see Figure 13. LESS Label) to provide information on the class of performance of the steel product, which reflects the gradual progress towards climate neutrality, the percentage of scrap share, and either the product carbon footprint (PCF), or the global warming potential (GWP) in accordance with the environmental product declaration (EPD) of the delivered finished steel product. As no uniform standard for calculating the product carbon footprint currently exists, the PCF is verified to the standard required by the customer (for example: Greenhouse Gas Protocol (GHG), ISO 14064/14067, PAS 2050 or EPD according to ISO 14025 and EN 15804). In addition to the PCF value, the standard (e.g. GHG protocol) according to which the PCF or GWP was certified, and the certificate number are indicated (WV Stahl, 2024). This labelling system aims to provide stakeholders with clear and transparent information, enabling them to make informed decisions.

⁸⁶ Guidance Document n°9 on the harmonised free allocation methodology for the EU-ETS post 2020 (European Commission, 2019)

⁸⁷ The LESS was officially launched in April 2024 and further development is necessary to establish the system owner and the certification system, which is expected by the end of the third trimester of 2024

Figure 13. LESS Label



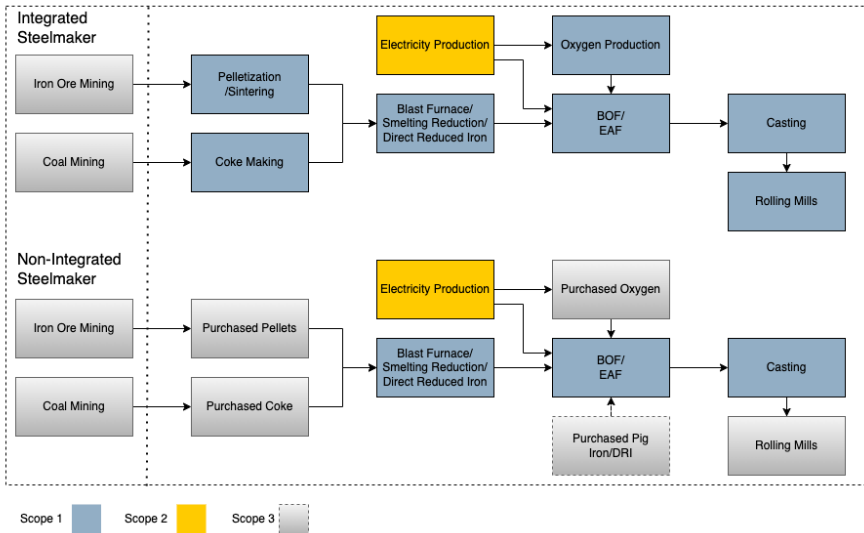
Source: Introduction of a Low Emission Steel Standard (LESS) to support the transformation of the steel industry (WV Stahl, 2024)

1.4.5.4 Climate Bonds Initiative

The Climate Bonds initiative certifies climate bonds, enabling companies to screen assets and capital projects in terms of their climate integrity (i.e. their contribution to climate mitigation and/or to adaptation and resilience). The initiative has developed a specific Steel Criteria (Climate Bonds, 2023) that sets a climate change benchmark for decarbonisation measures and steel production facilities..

The Steel Criteria aligns with the **Sustainable STEEL Principles**⁸⁸ (RMI, 2022) scope and boundaries emission approach. According to these principles, only CO2 emissions are to be accounted for within the fixed system boundaries shown in the Figure 14., regardless of process ownership.

Figure 14. System Boundary of GHG emissions in Climate Bonds Initiative

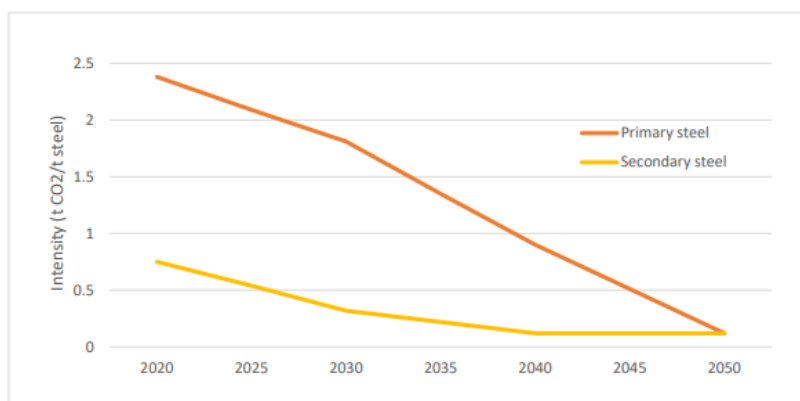


Source: Steel Criteria. The Steel Eligibility Criteria of the Climate Bonds Standard & Certification Scheme (Climate Bonds, 2023)

⁸⁸ The Sustainable STEEL Principles provides a methodology for banks to measure and report the climate alignment of steel lending portfolios compared to net-zero emissions pathways.

In the same way, the Climate Bonds Initiative utilise the split trajectory approach (primary and secondary pathways) of the Sustainable Steel Principles (RMI, 2022). The emissions intensity threshold values are set in alignment with the **IEA Net Zero by 2050 Scenario**. As shown in Figure 15., for primary steel, the intensity threshold in 2020 was 2.38 t CO₂/t steel, after which the threshold value gradually goes down every five years to 0.12 by 2050. For scrap-based steel products, the threshold in 2020 was 0.75 t CO₂/t steel, and then it gradually decreases to 0.12 by 2050.

Figure 15. The emissions pathway for all steel production companies (scope 1 & 2 emissions combined)



Source: Steel Criteria. The Steel Eligibility Criteria of the Climate Bonds Standard & Certification Scheme (Climate Bonds, 2023)

1.4.5.5 Global Steel Climate Council (GSCC)

The **Steel Climate Standard**, developed by the Global Steel Climate Council (GSCC), is an ambitious framework aiming to guide the steel industry towards a more sustainable and low-carbon future. The standard has three core objectives: to establish a technology-agnostic framework for steel product certification and corporate science-based emissions targets, to ensure transparency for customers on the carbon emissions associated with their steel purchases, and to align the industry with the emissions reduction goals of the Paris Climate Agreement by 2050.

In contrast with other methodologies presented before, which establish absolute thresholds, this methodology has established a **decarbonisation glide path aligned with the IEA's Net Zero Emissions (NZE) by 2050 Scenario** (IEA, 2021) to achieve a 1.5°. As the NZE scenario excludes indirect emissions from electricity, the GSCC incorporates the adjustment applied by the Science-Based Target Initiative (SBTi) (SBTi, 2022) and Rocky Mountain Institute (RMI). Therefore, the carbon intensity of steel products in 2050 is targeted to be 0.12 tCO₂e/tonne of hot rolled steel.

Figure 16. The Steel Climate Standard Sector Decarbonisation Glide path



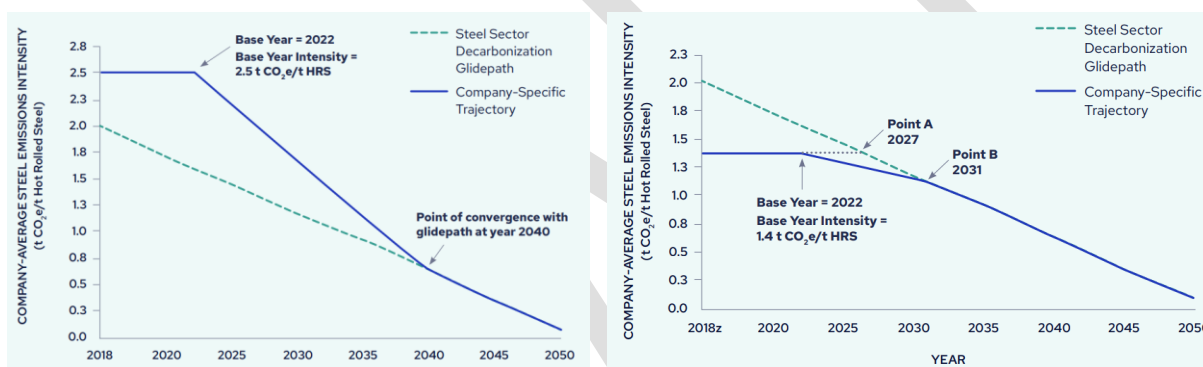
Source: The Steel Climate Standard (GSCC, 2023)

As shown in Figure 16, the Steel Climate Standard establishes steel product-based emissions intensity pathway that distinguish between long products and flat products. This distinction acknowledges that, currently, it is technically unfeasible to manufacture flat and long products with equivalent carbon intensities due to differences in their chemical composition. Similarly, high alloy specialty steel and stainless steel are expected to follow a different glide path to capture their key differences from other product types, however they are not covered by these standards currently (GSCC, 2023).

Companies seeking certification must develop a customized trajectory based on their starting point (base year) and the Steel Climate Standard Decarbonisation Glidpath. This trajectory must adhere to a minimum linear GHG intensity reduction progress, as illustrated in Figure 17, is required to certify flat and long steel products as **lower-carbon emissions steel**.

To obtain the certification, companies are required to undergo third-party verification of their baseline and set science-based emission target (SBET) within 2 years of joining the GSCC and establish interim targets to be achieved within 5-10 years from the base year. Once the GSCC has determined that a product meets the standard, it issues a certification which is valid for 3 years, although the carbon intensity has to be calculated annually and reported to the GSCC.

Figure 17. Company-specific trajectory examples



Source: The Steel Climate Standard (GSCC, 2023)

1.4.5.6 EUROFER Stainless Steelmakers' Proposal

EUROFER's stainless steelmakers members have developed a specific methodology to define "**Low CO₂ Stainless Steel**". The fact that stainless steel has their own specificities (large variety of grades) and depending on alloy content and recycled content the footprint of stainless steel products produced in Europe could range between 1 and 5 ton of CO₂/ ton of stainless steel.

Notably, in Europe, all stainless steel is primarily produced through EAF, resulting in high average scrap content of over 80% (stainless steel + carbon steel). However, the actual input of stainless scrap varies due to limited availability of scrap with the correct alloy content, averaging around 60% (EUROFER, 2023).

Based on the above, it could be thought that European stainless steel products have a low carbon footprint. However, it is noteworthy that the production of ferroalloys (upstream Scope 3) still significantly contributes more than 70% of the Product Carbon Footprint (PCF). EUROFER Stainless maintains that these particular challenges make it impractical to employ a sliding scale approach, thus advocating for a more suitable **grade-specific dynamic threshold**.

The methodology incorporates an alloy content dependent equation known as the calculated Model Footprint (MF_n), enabling the calculation of the MF_n for each grade (n) of a hot rolled product⁸⁹. To establish the threshold, the MF_n is multiplied by a reduction factor (100% - R). This reduction factor is open for debate but for example

⁸⁹ The calculation is based on the average grid mix (376 kg CO₂ / MWh) and a fixed scrap input (70% stainless scrap) in Europe. The equation is derived by linear regression (Kobolde ©) from theoretically calculated footprints of 20 different grades with fluctuating carbon footprint. It contains variables for the different alloying constituents (C, Mn, Cr, Ni and Mo), correction-factors for the melt-shop/casting yield (CY), emissions of the auxiliary processes (EAux) and emissions of Hot Rolling (EHR) and Hot Rolling yield (HRY). In order to report the GHG emission of the products, the standard 'ISO 14067:2018' should be used as a minimum requirement (EUROFER, 2023).

$$Tn = (100\% - R) \text{ MFn}$$

In addition to the threshold for defining low CO2 stainless steel, EUROFER Stainless has proposed a system for categorisation of performance based on the following distribution:

Label A: $MF_n < T_n$

Label C: $2 \cdot T_n < MF_n < 3 \cdot T_n$

Label E: $4 \cdot T_n < MF_n$ or unknown

The China Iron and Steel Association (CISA) has established the Low Carbon Emission Steel Evaluation Method (CISA, 2024), a standardized framework for assessing the carbon emissions of steel products. Developed in collaboration with Baowu Steel Group, 44 industry entities, and over 100 experts, this standard provides a classification system to evaluate the environmental profile of steel products. **The standard aligns with the sliding scale approach** proposed by the International Energy Agency (IEA), Responsible Steel, BMWK, and WV Stahl in their Low Carbon Emission Steel (LESS) standard.

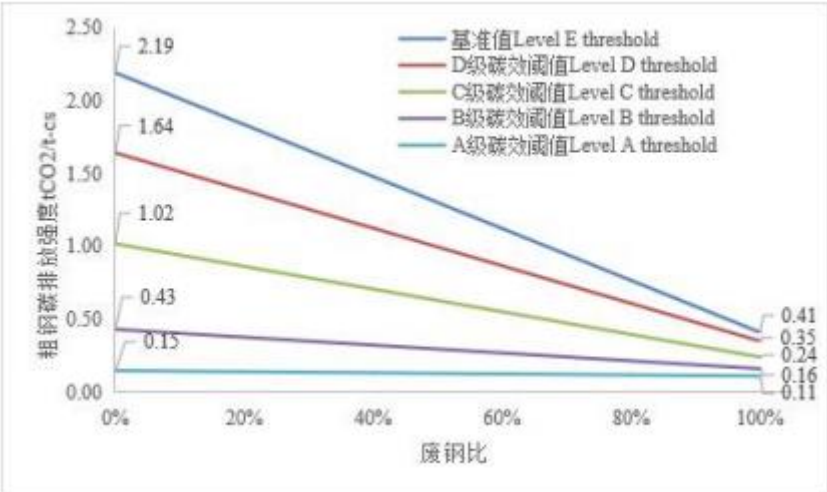
Figure 18. System boundaries of the Low Carbon Emission Steel Evaluation Method



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Building on these GHG accounting rules, the classification system establishes a sliding scale with 5 classes of performance to assess environmental performance of steel, see Figure 19. The assessment can be conducted at two levels: crude steel and hot-rolled product. At the crude steel level, the carbon efficiency thresholds outlined in Table 16 serve as the benchmark, whereas at the hot-rolled product level, additional thresholds presented in Table 17 are applied in conjunction with the crude steel thresholds. Notably, the standard acknowledges the diversity of hot-rolled products and provides specific carbon efficiency thresholds for various product categories, including hot-rolled coils, thick plate products, wire rod, and seamless pipes.

Figure 19. Crude steel boundary carbon efficiency threshold division



Source: Evaluation methodology of low carbon-embodied emission steel (CISA, 2024)

Table 16. Crude steel boundary carbon efficiency threshold curve

Class	a: 0% scrap steel to crude steel carbon performance threshold (tCO ₂ /t crude steel)	b :gradient	100% scrap steel to crude steel carbon performance threshold (tCO ₂ /t crude steel)
E	2.19	1.78	0.41
D	1.64	1.29	0.35
C	1.02	0.78	0.24
B	0.43	0.27	0.16
A	0.15	0.04	0.11

Source: Evaluation methodology of low carbon-embodied emission steel (CISA, 2024)

Table 17. Carbon efficiency threshold of a hot rolled products (tCO₂/t hot-rolled steel)

		All iron ore smelting	All scrap steel smelting
E Carbon Efficiency	Hot rolled coil	0.040	0.110
	Thick plate products	0.065	0.140
	Wire rod	0.045	0.110
	Seamless pipe	0.160	0.340
D Carbon Efficiency	Hot rolled coil	0.030	0.080
	Thick plate products	0.050	0.105
	Wire rod	0.035	0.085
	Seamless pipe	0.120	0.250
	Hot rolled coil	0.020	0.050

C Carbon Efficiency	Thick plate products	0.030	0.070
	Wire rod	0.020	0.055
	Seamless pipe	0.075	0.170
B Carbon Efficiency	Hot rolled coil	0.010	0.027
	Thick plate products	0.015	0.035
	Wire rod	0.010	0.030
	Seamless pipe	0.040	0.085
A Carbon Efficiency	All Products	0	0

Source: Evaluation methodology of low carbon-embodied emission steel (CISA, 2024)

To ensure compliance with the standard's requirements, steelmakers are expected to calculate the carbon emissions of their steel products and determine the corresponding carbon efficiency level in accordance with the GHG accounting requirements. If the steel products meet the specified thresholds presented in Table 16 and Table 17, they may be certified as low-carbon emissions steel, **subject to third-party evaluation and certification**. As part of this process, steelmakers must prepare an evaluation report, which provides a verifiable account of their carbon emissions and reduction strategies.

Finally, the standard introduces a "carbon reduction technology label" to recognise and incentivize the adoption of clean energy and low-carbon technologies in steelmaking. This label serves as a distinction for steelmakers that have successfully integrated these technologies into their production processes.

1.5 Survey on scope

Between 28th September and 22nd October 2023, the JRC conducted a stakeholders' survey. The objective was to collect early feedback on the "Preparatory study on iron and steel – ecodesign measures under the ESPR". The survey focused on five key areas:

- Definition and classification of steel products, which seeks feedback on definitions used and appropriateness of classification
- Scope, which seeks feedback on a preliminary version of the scope.
- Market analysis, which seeks for sources of market-related data.
- Ecodesign aspects, which collects opinions on the most effective and applicable potential ecodesign actions.
- EU GPP, which collects opinions on the current experience of users of voluntary EU GPP criteria.

The feedback has been incorporated into Task 1, resulting in appropriate amendments and updates.

1.6 Recommendations and conclusions on scope

Based on the analysis conducted to understand the different categorisations systems available for steel products, there are a few high level **findings for the considerations, in particular for a definition of scope**:

1. The product group of iron and steel presents a **high degree of heterogeneity**, making it challenging to establish uniform ecodesign requirements. Steel, being a notably complex product, is categorised under many different classification systems, which are based on diverse steel grades, applications, shapes, surface conditions, combined nomenclature, etc. To illustrate the breadth of the product range within scope, it is noteworthy that, according to the World Steel Association, there are over 3,500 grades of steel.

Moreover, the grade of steel is intricately linked to the chemical composition, structure and crystallography of each specific product. These factors are pivotal (but not the only influencing parameter) in determining key functional characteristics, inter alia, strength, elongation, weldability, and hardness. Ultimately, it is these performance characteristics that define the application of the products. Specific treatment e.g. during production processes may provide e.g. higher tensile strength than conventionally achieved with provided grade classification.

2. Following the stakeholders' feedback and review of the existing initiatives, it is apparent that the most appropriate way to cluster products seeking to establish ecodesign measures for steel products should be connected with the product **application/functionality**. This has been confirmed through several means, as explained below, which point out that clustering steel products based on their application/functionality is critical to ensure that product within a category are comparable among them.

- According to the Methodology for the Ecodesign of Energy-using Products (**MEErP**) (European Commission, 2011):

Product grouping has played a very important role. The overriding principle for life cycle assessment - and thereby the clustering products - should be a quantifiable "functional unit". Ideally this means that the function of e.g. a refrigerator is the preservation of perishable food stuffs, a lamp should give light and a dishwasher should clean dishes. This should be the yardstick for clustering products in one preparatory study and apply specific Ecodesign measures that are technology-neutral. Therefore, when clustering products, the functionality of the products is the main criteria to be considered.

In case of multiple functionalities, the Ecodesign measures should address and set minimum requirements for each functional parameter individually.

- According to the **Product Environmental Footprint** (European Commission, 2021):

In Section A.3.2 of the PEF method, it is specified that **comparisons between products may only be made if they are fulfilling the same main function**, expressed through the functional unit (FU). Thus, the FU describes qualitatively and quantitatively the function(s) of the product(s) in scope.

For intermediate products, the FU is more difficult to define because they can often fulfil multiple functions and the whole life cycle of the product is not known (use and end of life phase). Therefore, a material-based approach (or declared unit) may be chosen (mass (kilogram) or volume (cubic meter)).

In addition, as stated in Section A.5.1 of the PEF method, intermediate products are excluded from the benchmarks results, due to the lack of a function-based approach and an incomplete life cycle of the product.

— According to the Product Environmental Footprint Category Rules (**PEFCR**):

In case of intermediate products, the PEFCR is considered as a “module” to be used when developing PEFCRs for products further down the same supply chain. The development of “modules” allows for a higher level of consistency among different supply chains that are using the same modules as part of their LCA. Currently, the PEFCRs available for intermediate products are:

- Metal sheets
- Intermediate product paper
- Animal feed
- Leather

In each PEFCR one or more Representative Products are defined, which, represent the average product on the European market. For the PEFCR on metal sheet (European Commission, Joint Research Centre, et al., 2022), the following list of representative intermediate products has been selected:

- For the subcategory construction: • Copper sheet • Lead sheet • Aluminium sheet • Steel sheet
- For the subcategory appliances: • Aluminium sheet • Steel sheet

The grade of metal, thickness of the sheet and the surface finish depend on the specific end use. Examples are given in the PEFCR, considering two potential end applications (building and appliances) but it is stressed that these examples do not specify exact technical parameters, and should not be used as criteria for benchmarking. It is also underlined that each representative product has its own environmental footprint at the intermediate level and cannot be compared to any of the other representative products at this level. Only when a function is well-defined, and the life cycle assessment is performed on the entire life cycle of this final function (including the use phase and the end of life stage), benchmarking and comparison analysis could be considered. As matter of fact, this PEFCR is applicable to metal sheets as intermediate and therefore all information shall be transferred to the user in the compilation of the PEF profile of the final product produced from the intermediate metal sheet.

— According to **standards**:

ISO 14044, ISO 14025 and EN 15804 which states that the comparison between products shall be done on the basis of the same function (European Commission, Joint Research Centre, et al., 2022).

— According to **stakeholders**:

According to the views expressed in the survey, application and functionality of products are key when clustering products to establish ecodesign measures. Several stakeholders expressed an opinion stating that: *“Customs classification, shape and appearance of products may not always be a sufficient way to categorise products for environmental performance. Whilst in several categories are used in a specific application or have specific functionality, e.g. rebar, others may be too broad e.g. cold rolled flat. Therefore, a combination of categorisation approaches maybe necessary to incorporate the functionality. For this purpose the categorisation in EN 10027 maybe relevant to incorporate, since the end application and functionality are considered in preference to shape. This has relevance for instance in alloyed steel that gain their properties from specific alloy additions that also dominate the environmental impact. Therefore, it would not make sense to compare those products with steel products of different composition, functionality and performance”*

3. While coherence with the CBAM regulation is desirable, it is not clear that the categories used by the Combined Nomenclature (CN) are appropriate for clustering of products for the purpose of this preparatory study. Product quality, use of alloy elements and energy requirements during different treatment process in steelmaking are not totally captured by the CN categories, but influence on the specific emissions associated to the products.
4. The availability of reliable and raw data, with sufficient granularity and temporal and spatial coverage, as well as accessibility, are essential considerations when selecting product categories for the preparatory analysis.

For this reason, the elaboration of the reference list of the product categories, including correspondence between existing classification systems was addressed as a part of Task 2 (see Section 2.1.2).

DRAFT

TASK 2: MARKET ANALYSIS

The objective of the market data analysis presented in this chapter is to provide a comprehensive overview of the value chain for iron and steel products sold within the EU market. The analysis encompasses the following key aspects:

- **Demand Analysis:** Evaluation of Europe's demand for steel-containing products, with consideration of their specific applications.
- **Supply Chain Overview:** Examination of the supply chain for steel and steel products within the EU Single Market, including local production, imports, exports, and trends in secondary flows (e.g., scrap).
- **Market Information:** Insights into the steel market in Europe, including manufacturing costs and employment trends in the sector.

The data collection and analysis were conducted in collaboration with the Fraunhofer Institute for Systems and Innovation Research ISI, engaged as an external contractor to the Joint Research Centre (JRC). This work includes a historical analysis of data spanning at least 10 years (2012–2022), with projections of market trends detailed in **Annex XXX**, which outlines baseline scenarios. Data for the analysis were provided by the European Steel Association (EUROFER).

The base year for the analysis is 2022, reflecting the most recent data available during the development of Task 2 in Q1 2024. The report also incorporates an evaluation of the latest trends prior to publication and includes insights gathered from the first stakeholders' meeting held in June 2024.

2.1 Ex-ante data collection

2.1.1 Criteria for the ex-ante data availability analysis

The starting point for market analysis was an ex-ante data availability check aiming to determine the key sources of market data on iron and steel products for the EU27 market, including the following:

- **PUBLIC:** For the public sector, these include international institutions such as Eurostat or the UN as well as national institutions such as geological surveys or ministries from the European Union Member States and beyond.
- **PRIVATE:** Industry associations representing different parts of the steel industry such as steel industry, stainless steel industry or foundry industry were checked regarding their activities in preparing and publishing statistics.
- **SCIENTIFIC:** Scientific publications such as Scopus and Google Scholar, databases for scientific publications, were used with appropriate search strings, in particular, to find material flow analysis (MFA) studies covering steel flows in Europe.

Those sources were assessed in terms of content (type of data collected), quality, completeness, accessibility, and others. The process was mainly based on desk literature research and included the list of sources identified by stakeholders in the initial questionnaire.

The ex-ante data analysis aimed to address the following questions:

- What are the key sources of (market) data on steel products?
- What type of information is being collected, and what is the granularity level?
- What methods are used for data collection and analysis, what is the frequency of data collected, and the last year available?
- What is the quality, availability, and accessibility of product data?
- What restrictions apply to the publication of data?

As an outcome, a comprehensive list of sources was gathered, allowing for identification of the potential data gaps in the early stage of the market analysis.

2.1.2 Overview of findings

This section provides a detailed overview of the available data sources, organized by the life cycle stages of the steel industry. These stages include mining, iron and ferroalloy production, crude steel production, finished steel products manufacturing, scrap generation and recycling, as well as trade activities for each stage. It includes information on data providers, type of data collected, spatial and temporal coverage, methodological approach (including classification method), frequency of updates and accessibility.

Table 18 presents an extract from the data availability results, highlighting the key data sources for each life cycle stage. Detailed descriptions of these sources are provided in Annex 3. The table includes specific references that were available at the time of the research, conducted in Q1 2024, of which the majority is being updated in the regular basis.

Table 18. Overview of available data sources and their coverage of life cycle stages

		Iron ore	Iron	Ferroalloys	Crude steel	Finished steel products	Scrap and Recycling	Trade
<u>World Steel Association (2023a)</u>	r	Total	Granular		Granular	Granular		Yes
<u>BGS 2024</u>	r	Total	Partial	Granular	Total			Yes
<u>USGS 2021</u>	r	Total	Granular	Granular	Total			
<u>Minerals4EU (2021)</u>	r	Total	Partial		Total			Yes
<u>eurostat (2024b)</u>	r	Granular	Granular	Granular		Granular	Partial	Yes
<u>EUROFER (2023)</u>	r				Granular	Granular	Partial	Yes
<u>WV Stahl (2023)</u>	r				Granular			Yes
<u>Dworak et al. (2021)</u>	p				Granular	Granular	Granular	Yes
<u>CAEF (2021)</u>	r					Partial		
<u>WFO (2021)</u>	r					Partial		
<u>Rostek et al. (2022)</u>	p						Granular	
<u>UN Comtrade (2024)</u>	r							Yes
<u>ISSB (2024)</u>	r							Yes
<u>worldstainless (2023a)</u>	r	Unknown	Unknown	Unknown	Unknown	Partial	Unknown	Unknown
<u>CRU (2024)</u>	r	Yes	Yes	Unknown	Yes	Yes	Unknown	Unknown
<u>eurostat (2024a)</u>	r							Yes

Source: own elaborations

Notes:

"r" - Data updated on regular basis, minimum once a year

"p" - Outcomes of the research project (one time data publication)

"Yes" - Total number for the life cycle stage is available.

"Granular" - Granular data covering the total life cycle stage is available.

"Partial" - Granular data is available only some life cycle stages (but now all)

The ex-ante analysis of data availability revealed that no single data provider covers all life cycle stages comprehensively. As a result, the market data analysis must draw on multiple data sources, applying necessary harmonization efforts to ensure consistency and reliability of the analysis.

The best data availability has been identified for foreign trade. This might be due to the presence of dedicated databases for trade data such as UN Comtrade or Eurostat Comext. However, the relatively good data availability has to be taken with caution; trade statistics tend to contain some inconsistencies in reporting, i.e. outliers, missing values, and bilateral inconsistencies (Chen et al., 2022). Reporting of trade values can be expressed both in quantity (weight of product or a specific contained material) or monetary value. In some cases, the data preparation and cleaning includes recalculation between data provided in volumes and values, which further increases the uncertainty.

A second point of consideration is the availability and granularity of data for the life cycle stage of finished steel products. Due to the high complexity of the categorisation for iron and steel products, data collection for finished steel products is more difficult and effort-intensive, as it requires merging several sources. In this case,

the correspondence between product categories needs to be established. Further descriptions of the process of categorisation of finished products can be found below in Section 2.1.2.2.

The fewest accessible data sources have been identified for scrap generation and recycling, which is the most difficult to survey. The European Steel Association (EUROFER) is among a few organisations that report on scrap recycling (EUROFER, 2023). Some scientific literature that quantifies scrap availability (Dworak et al. 2021; Dworak et al. 2022; Rostek et al. 2022; BIR, 2022) has also been identified; since this quantifies a theoretical potential, the numbers are entirely based on the modelling results.

2.1.2.1. Key data sources for the analysis

Table 19 summarises data sources, including their strengths and weaknesses, used in the market data analysis for the preparatory study. The analysis of the production and trade of iron ores and concentrates, iron, and ferroalloys is based on PRODCOM data (Eurostat, 2024). While the World Steel Association (excluding iron) and EUROFER do not report these figures, they are fully available within PRODCOM.

Data on crude steel and finished products' production and trade are based on EUROFER (2023), which provides appropriate granularity and good transparency regarding the data collection process. Although this data is potentially available within the PRODCOM database, issues such as outliers and missing entries may partially occur. Moreover, the PRODCOM database provides a significantly higher level of granularity, which is not required for the categories of the consolidated list of finished steel products selected for the market analysis. This introduces complexity and a higher risk of error when aligning product categories.

Nevertheless, when compared with data from the World Steel Association, the figures from EUROFER are congruent and show a similar granularity. However, the open accessibility of EUROFER data facilitates its use in publicly available reports, such as the one presented, thus making it the preferred source for this report.

Finally, while data on scrap is rarely available, the EUROFER statistics on the consumption and trade of steel scrap are applied. Monetary values are only available from the PRODCOM database, as the remaining sources focus solely on volumes.

Table 19. Key sources of information on the iron and steel market in the EU27

	Strengths	Weaknesses
EUROFER <u>EUROFER</u> <u>(2023)</u>	• The data collection addresses the specific needs of the EU market. • It provides data on important stages of the steel value chain, such as crude steel, finished steel products, and recycling, including detailed trade data. • It is regularly and reliably published based on a direct industry survey. • It uses a high level of product disaggregation (different but similar to worldsteel). • The type of product categorization is well-known in the industry and therefore well-suited for the process of setting up base cases in cooperation with stakeholders during the preparatory study. • The product categorization has very high compatibility with available LCA studies. • Information on the distribution of finished steel products to end-use sectors is available.	• Not all information collected by EUROFER is made publicly available (e.g., more detailed product level). • EUROFER gathers information through two separate questionnaires, one for production and another for deliveries. The nomenclature of finished steel product categories in the questionnaires is distinct, making the analysis challenging. • There is potential bias in presenting data in the interest of the EU steel industry.
worldsteel <u>World Steel</u> <u>Association</u> <u>(2023a)</u>	• Access to a large number of data providers (~85% of global production), enhancing data credibility. • Provides data on all life cycle stages except for ferroalloys and scrap. • Offers global trend analysis and projections.	• Paywall and restrictions on publishing the latest data. • Potential bias in presenting data in the interest of the global steel industry

	• Regularly and reliably published based on a direct industry survey. • Uses a high level of product disaggregation. • The type of product categorization is well-known in the industry and therefore well-suited for setting up base cases in cooperation with stakeholders during the preparatory study. • The product categorization has very high compatibility with available LCA studies. • Data from before 2018 is freely available.	
PRODCOM <u>eurostat (2024b)</u> Comex <u>eurostat (2024a)</u>	• Developed and operated by a European institution, with open access to all. • Uses an individual product classification system (PRODCOM codes) with a very high level of disaggregation. • Based on a harmonized methodology that increases accuracy and enables comparability between different countries and regions. • Provides the timeliest publication of data among all sources. • Disaggregation of data by country is available. • The only source that provides consistent tonnages and values.	• Data gaps, outliers, and bilateral asymmetries may be present in the dataset since no consistency checks are done. • Some codes are not reported for certain years, countries, or at all. • Delays in data publication.
<u>Dworak et al. (2021)</u>	• A one-time scientific publication based on worldsteel data. • Provides numbers on scrap availability and future scenarios, which can complement other main data sources. • Fits well with worldsteel data, as the production model is based on it.	• As an MFA study, it does not include primary data for production stages, referring instead to worldsteel data. • Provides modeled numbers for stages without primary data, mainly recycling. • The last data year used is 2017; projections are based on market analysis until 2017.
<u>CAEF (2021)</u>	• The most comprehensive Europe-wide survey within the casting industry. • Differentiation of data by types of castings and potentially country-level resolution.	• Covers only castings.
<u>BIR (2022)</u>	• Provides comprehensive and country-specific statistics on steel scrap use in steelmaking, including trade volumes for imports and exports.	• The data format is limited, as the report is only available in PDF format.

2.1.2.2. Overview of the steel product categories considered

Given the wide range of classification systems used across different data sources⁹⁰, setting a reference steel product categorisation for market analysis became essential. A unified categorisation and naming of those categories allows precision and transparency in describing the iron and steel market, facilitating informed discussions within the context of the ESPR.

To achieve this, a thorough analysis of existing classification systems has been conducted, along with the establishment of correspondence between them. Additionally, inaccuracies and uncertainties associated with the matching process have been assessed to ensure a rigorous and reliable approach.

⁹⁰ Most of the data sources rely on their own classification scheme; in some cases, it is only the difference in taxonomy, but often the definition behind the categories is different. See a reference to classification systems used in Annex, part A

Table 20 presents the consolidated list of different iron and steel product categories covering the value chain from mining to finished steel products, including the agreed definitions^{91,92}. It was based primarily on the classification schemes used in the data sources listed in Nevertheless, when compared with data from the World Steel Association, the figures from EUROFER are congruent and show a similar granularity. However, the open accessibility of EUROFER data facilitates its use in publicly available reports, such as the one presented, thus making it the preferred source for this report.

Finally, while data on scrap is rarely available, the EUROFER statistics on the consumption and trade of steel scrap are applied. Monetary values are only available from the PRODCOM database, as the remaining sources focus solely on volumes.

Table 19.

The proposed list of categories was reviewed for consistency and relevance with metallurgists and steel industry experts. In addition, it was cross-checked and adjusted as necessary against the taxonomy commonly used by the European steel industry. The list of the reference products, as presented in **Table 20**, has been analysed in terms of correspondence with EN 10027 standard and CN codes. The results are presented in Annex parts B and C, respectively.

It is important to note the potential risk of double counting when working with different classifications of selected product categories. This is particularly relevant for **tubular products**⁹³ which are often, but not always, counted as end-use products of steel. For instance, **welded tubes** are manufactured from finished steel products, making it reasonable to consider them end-use products. However, they are a step further down the value chain from finished steel products. **Seamless tubes**, on the other hand, are made directly from crude steel intermediates and can be both a finished and/or end-use product. For the purpose of this study, seamless tubes are categorised as finished steel products.

A similar case is **forged products**, which are often excluded from common statistics on finished iron and steel products. This is because a significant share of forged products is not produced by the iron and steel sector, but instead by the forging industry, who sells the work pieces to end-use manufacturers such as the automotive industry. In the EU, the total production of forged products is relatively small compared to the size of the iron and steel market (5.3 Mt in 2022 for all metals (EUROFORGE, 2023)). While tubular products were included in the data analysis, forged products were excluded for this reason.

Table 20. List of consolidated product categories for the market analysis.

Product category	Reference definition used	Correspondence to data source
Ores and Concentrates		
Agglomerated	Agglomerated iron ores and concentrates (excluding roasted iron pyrites)	7101020*
Non-agglomerated	Non-agglomerated iron ores and concentrates (excluding roasted iron pyrites)	7101010*
Iron		
Pig iron	Intermediate iron product gained from smelting of iron ore in a blast furnace (incl. spiegeleisen) containing > 2% iron and < 30% manganese, < 10% chromium, < 8% silicon, < 3% phosphorus, < 10% other elements	24101100*, 24101410*, 2410T110*, EUROFER, EN 10001
Direct reduced iron	Ferrous intermediate products obtained by direct reduction of iron ore by a reducing gas in solid state and hot briquetted iron (HBI or other sponge iron products)	24101300*
Ferroalloys		

⁹¹ There is no dataset in the literature that provides numbers or definitions for exactly the categories of the consolidated list since they were constructed for this study. They can be backed with numbers by merging different sources and matching categories.

⁹² The EUROFER definitions of product categories, used in the data collection, served as the foundation. These definitions were subsequently enriched with additional information, e.g. the full names of the matching PRODCOM codes and the standard EN 10020 supported by a comparison with the descriptions of production processes by AIST (Association for Iron and Steel Technology, 2023).

⁹³ Tubular products are the umbrella term for all kinds of tubes and pipes. They can be differentiated according to how they are produced into welded tubes and seamless tubes.

Ferro-manganese		Alloy of iron and manganese with usually > 2% carbon and > 65% manganese by weight	24101210*, 24101215*, 24101220*, 24101225*	
Ferro-silico-manganese		Alloy of iron, silicon and manganese	24101245*	
Ferro-silico-magnesium		Alloy of iron, silicon and magnesium with > 2% magnesium	24101285*	
Ferro-silicon		Alloy of iron and silicon with usually 15-90% of silicon (incl. up to 10% magnesium)	24101230*, 24101235*, 24101236*	
Ferro-tungsten and ferro-silico-tungsten		Alloy of iron and tungsten with usually 70-80% tungsten (incl. ferrosilicon containing tungsten)	24101250*	
Ferro-titanium and ferro-silico-titanium		Alloy of iron and titanium with usually 30-75% titanium (incl. ferrosilicon containing titanium)	24101255*	
Ferro-nickel		Alloy of iron and nickel with usually > 30% of nickel	24101240*	
Ferro-chromium		Alloy of iron and chromium with usually > 45% chromium	24101260*, 24101265*	
Ferro-niobium		Alloy of iron and niobium with usually 60-70% of niobium	24101270*	
Ferro-molybdenum		Alloy of iron and molybdenum with usually 65-70% molybdenum	24101275*	
Ferro-phosphorus		Alloy of iron and phosphorus with usually 18-25% phosphorus	24101280*	
Other ferroalloys		Alloys exceeding the limits of pig iron and not otherwise covered in the ferroalloy categories (e.g. ferro-vanadium, ferro-boron, ferro-aluminium)	24101290*	
Crude steel total				
by alloy composition	Non-alloy crude steel	Steel grades with < 1.65% manganese, < 0.60% silicon, < 0.40% copper or lead, < 0.10% bismuth, selenium, tellurium, vanadium, < 0.08% molybdenum, < 0.06% niobium, < 0.0008% boron, < 0.10 of any other element except carbon, nitrogen, phosphorus or sulphur	EUROFER, EN 10020	
	Stainless crude steel	Steel grades with >= 10.50% chromium and <= 1.20% carbon with or without other elements	EUROFER, EN 10020	
	Other alloy crude steel	Steel grades not covered by the categories non-alloy steel and stainless steel	EUROFER, EN 10020	
by process	BOF	Crude steel produced in a basic oxygen furnace or via other oxygen based steelmaking routes		
	EAF	Crude steel produced in an electric arc furnace		
by semi-finished products	Ingots	Crude steel product obtained by casting liquid steel into a mould	EUROFER	
	Continuously cast products	Intermediate steel products made by solidifying molten steel in the shape of billets, blooms or slabs from a continuous casting process	EUROFER, AIST	
Finished iron and steel products				
Non-alloy steel and other alloys	Flat products	Hot rolled plate (e.g. Quarto plate)	Hot rolled plates and sheet not coiled but supplied in flat square and rectangular shapes.	EUROFER, AIST
		Electrical sheet	Cold rolled strip with specific watts loss and peak magnetic flux density; grain oriented and non-oriented.	EUROFER, AIST
		Hot rolled coil	Hot rolled flat products of different width that are immediately after the final rolling pass wound into coils.	EUROFER, AIST
		Cold rolled coil	Cold rolled sheets and strip not otherwise covered	EUROFER, AIST
		Tinplate and tinned products	Cold rolled non-alloy low carbon strip and sheet coated with tin. Incl. ECCS: strip coated with duplex film of metallic chromium (on the steel) and chromium oxide or hydroxide (outer layer) by a cathodic process.	EUROFER, AIST
		Galvanized coil	Sheets hot dipped or electrolytically coated with zinc. Incl. sheets coated with other metals (e.g. aluminium, lead, lead-tin, aluminium-silicon, aluminium-zinc)	EUROFER, AIST
		Organically coated and	Sheets, whether or not already metal coated, subsequently coated with an organic material or a mixture of metal powder	EUROFER, AIST

		other coated products	and organic material, by a continuous process. Incl. sheets coated with inorganic material, e.g. vitreous enamel, in a continuous process	
	Long products	Railway track material and other long products	Heavy and light railway and tramway rails or other railway system products. Incl. long products not otherwise covered.	EUROFER, AIST
		Heavy sections and mining frames	Hot rolled parallel flanged heavy sections having a cross section shape resembling the letter H; U sections; special heavy sections with shapes similar to I, H or U and particular features (e.g. unequal or unsymmetric flanges); mining frames resembling the letters H, I or Omega.	EUROFER, AIST
		Other hot rolled bars (e.g. merchant bars)	Bars of solid cross sections with rectangular, circular or other polygons shapes; bars of special shapes; hollow mining drill bars, L and T sections; bulb flats	EUROFER, AIST
		Light sections	Light U, I and H sections of height up to 80 mm, L and T sections; special sections.	EUROFER, AIST
		Bars for reinforcement of concrete	Deformed long products with a cross section that is round or almost round, hot rolled in lengths to produce a crenelated or ribbed surface, and destined for reinforcement of concrete. Also incl. bars and mesh for reinforcement of concrete	EUROFER, AIST
		Rod and wire rod	Hot rolled long products of nominal transverse dimensions generally 5 mm or over wound into coils with round, oval, square, rectangular, hexagonal, octagonal, half round, crenelated or ribbed round or similar cross section	EUROFER, AIST
Stainless steel		Flat products	Hot rolled strip mill products	Stainless flat products produced in a strip mill
	Hot rolled plate (e.g. Quarto plate)		Hot rolled plates and sheet not coiled but supplied in flat square and rectangular shapes.	EUROFER, AIST
	Cold rolled flat products		All flat stainless steel products that are cold rolled.	EUROFER, AIST
	Other flat products		Flat products that are not covered by the other categories.	EUROFER, AIST
	Long products	Wire rod	Hot rolled long products of nominal transverse dimensions generally 5 mm or over wound into coils with round, oval, square, rectangular, hexagonal, octagonal, half round, crenelated or ribbed round or similar cross section	EUROFER, AIST
		Merchant bars	Bars of solid cross sections with rectangular, circular or other polygons shapes; bars of special shapes; hollow mining drill bars, L and T sections; bulb flats; (incl. hot and cold rolled bars)	EUROFER, AIST
		Bright bars	Hot rolled bars that are subjected to a drawing process at room temperature; a die reduces and shapes the cross section and produces a bright, polished finish	EUROFER, AIST
		Drawn wires	Drawn wire is produced by pulling wire rod through one or several dies that reduce the cross-section	EUROFER, AIST
		Other long products	Long products that are not covered by the other categories.	EUROFER, AIST
Foundry products	Cast steel products		Finished steel products produced by casting liquid steel (< 2% carbon) into a mould of net- or near-net shape (incl. both non-alloy and alloy steels)	CAST
	Cast iron products		Finished iron products produced by casting liquid iron (> 2% carbon) usually directly produced from pig iron into a mould of net- or near-net shape	CAST
Forged products			Forgings are steel blanks shaped by a hydraulic press from a solid cast block (ingots, billets or bars)	AIST
Tubular Products	Welded tube		Welded Pipe is one of the end products that can be made from Plates.	AIST

	Seamless tube	Seamless Pipe is produced without a seam or weld in the pipe's circumference by a variety of methods. The most common method is by piercing a solid round steel blank (tube round), which has been prepared and heated to the proper temperature.	AIST
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Note: (*) Asterisk indicate PRODCOM code

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2.1.2.3. Assumptions for the data analysis

This subsection outlines the key assumptions underlying the data analysis, which were necessary to address the limitations and uncertainties inherent in the available data. Following a thorough examination of data availability and the establishment of a consolidated product classification, several assumptions were made to ensure a consistent and reliable approach to data analysis. These assumptions, summarised in Table 21, provide a foundation for the interpretation and discussion of results presented in subsequent sections.

Table 21. Key assumptions for data analysis

Assumption	Description
The average composition of product within category	A major uncertainty arises as there is little information about the exact composition of the material (ores, concentrates, and steel in general) reported within databases. In the consolidated list of products, a high variety of material compositions is considered within the individual product categories. As thousands of specific steel alloys exist, this complexity cannot be represented in production and trade databases nor would integrate this complexity into stock models be practicable or helpful. Consequently, when evaluating the elemental content of flows or stocks, it is necessary to rely on average compositions as a simplifying assumption.
No monetary values for crude steel production	EUROFER data was used as the main source of data on crude steel production volumes, as information in the public sources (i.e. PRODCOM) lacked comprehensive data for many product categories within the scope. However, EUROFER data does not cover monetary values for production, which are available in PRODCOM for all other product categories. As a result, monetary values for crude steel production and trade cannot be reported.
Steel demand per end-use sector	Information on the steel-consuming sectors, where steel is ultimately applied, is limited. This is due to the vast array of final products that use steel, making data collection challenging. While production quantities for finished steel products can be gathered through questionnaires sent to steelmaking companies, this approach is not feasible for end-use products, as numerous companies use finished steel products and the production chain is often complex and multi-step. However, literature provides some insights into the end-use sectors, although these estimates are based on assumptions and should be viewed with caution. EUROFER (2023) reports aggregated numbers for steel usage in eight sectors (Construction, Automotive, Mechanical engineering, Metalware, Tubes, Domestic appliances, Other transportation and Miscellaneous). More detailed information on the distribution of finished steel products to specific sectors can be found in studies by Dworak and Fellner (2021) and Harvey (2022). These assumption-based distributions provide a useful starting point for analysing sector demand for steel, despite their limitations.
Market supply vs. European deliveries	EUROFER reports steel flows in two parallel systems: one comprising production and trade, and the other focusing on domestic supply. The first system includes total production, imports and exports. The other system reports domestic deliveries, imports from third countries and the supply, which is the sum of the two. Since deliveries exclude the domestic production that is exported and lack information on exports, this analysis relies on production, import and export data. However, both reporting systems use slightly different product classifications with a partially higher granularity in the deliveries data, which is sometimes used for disaggregation purposes. Notably, using production data introduces a small degree of double counting, as import numbers may contain receipts, which are products further processed and therefore reported again by European steelmakers. Experts from EUROFER have validated the assumptions made at this stage. For stainless steel, EUROFER reports the total stainless steel crude production but for finished stainless steel product categories EUROFER only reports figures for European deliveries. Imports and exports are available separately. While for most products production data is the basis for market analysis, in the case of stainless steel European deliveries had to be used.
Trade of scrap (monetary values)	At the steel scrap stage, data is available on consumption and trade quantities. However, information on steel scrap generation is based on modelling and therefore subject to high levels of uncertainty. Additionally, there is little information on the quality, origin and composition of scrap flows. Dworak and Fellner (2021) have made a pioneering effort to model steel scrap qualities. While this study adds value to a better understanding of steel scrap characteristics, a regular quantification, e.g. via questionnaire, is missing. While such a comprehensive data collection is missing, the approach by Dworak and Fellner (2021) can serve as a useful approximation of steel scrap qualities.

Tubular products	Tubular products are defined as steel-consuming goods by EUROFER, so these are excluded from finished steel products. Since tubular products are counted here as finished steel products, it has to be taken into account that welded tubes are made of flat steel. Experts estimate that 20% of the welded tubes are made of hot rolled plates and 80% of hot rolled coils. To accurately combine the data, the welded tube quantities reported by the World Steel Association must be deducted from the corresponding EUROFER data categories. In contrast, seamless tubes, which are directly produced from semi-finished products, must be added to the finished steel products reported by EUROFER, as they are not accounted for in their finished steel product data.
Castings based on multiple sources	For castings, a combination of data sources is used. The production is based on data by CAEF (2021), as this is directly derived from industry queries. As they do not report trade, the imports and exports as reported by Eurostat (2024) are used. However, the trade of castings plays only a minor role.
Analysis for the finished steel products required matching data from different sources	For the market analysis of finished steel products, PRODCOM codes were matched with relevant product categorisation to CN codes as well as concordance tables (Magerman, 2022), as detailed in Annex 4, part D. However, for products that could not be allocated through this procedure, manual assignment to the appropriate consolidated product category was necessary. Due to the complexity and partial manual nature of this multi-step process, errors in assignment cannot be ruled out. Consequently, misallocated monetary values may occur, leading to uncertainty in the distribution of total monetary values across specific finished steel product categories.
Data gaps may cause underestimates of values	Not all relevant product codes representing finished steel products are reported and published within the PRODCOM database. For this reason, data gaps may lead to underestimation of the values of finished steel products production and trade.

Also, the processes shown in the Sankey diagrams (Figure 26. and Figure 27.) are not fully balanced but show the flows as quantified by the data sources used in the analysis⁹⁴. This discrepancy arises for three main reasons:

1. Flows shown in the Sankey diagram correspond to the products as defined within this preparatory study. Through the processes, conversion of the material takes place, so the composition of inflows and outflows are different. This is especially visible in iron-making. While iron concentrates contain 50-70% iron (IEA, 2020), the iron-making processes produce metallic iron in the form of pig iron or DRI with significantly higher iron concentrations. In steelmaking, the carbon concentration between inflows and outflows of the quantified flows varies. Furthermore, alloying elements are partially added, especially in the case of stainless steel.
2. Within the production processes, material losses occur due for example to incomplete metallisation of iron concentrates, generation of slags, dusts and dross in metallurgical processes as well as potential cutting losses. These process material efficiencies are outside the scope of the presented studies. Therefore, these material loss flows, which are rather small compared to the total production, are not shown in the Sankey diagram.
3. Stocking and destocking of material may occur at any stage of the life cycle. As no data is available on the stocks, this effect is not quantified within this study.
4. Part of the outflows may be for non-steel-related uses, such as the use of iron ore for the cement or chemical industry.

⁹⁴ The production and trade of iron ores is based on data by eurostat (2024) . The same data source is used for the quantification of the iron flows (both pig iron and DRI). For steel and iron castings, production data by CAEF (2021) and trade data by (eurostat, 2024) are applied. The remaining flows, namely the production and trade of steel and finished steel products (both carbon and stainless steel) as well as the consumption and trade of steel scrap are based on (EUROFER, 2023).

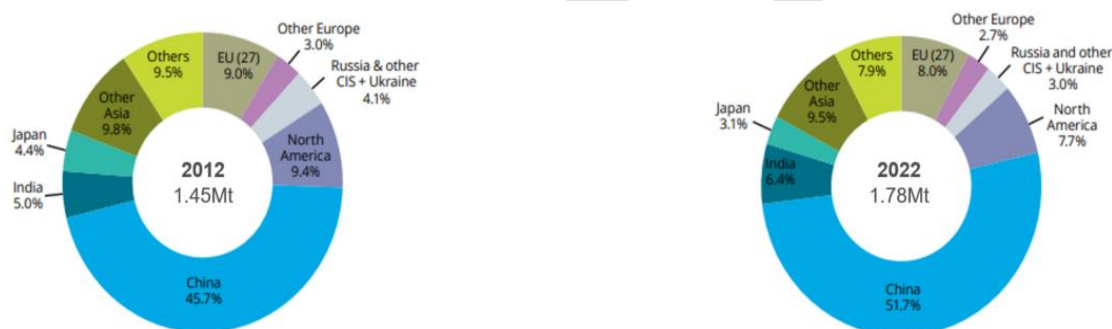
2.2 Demand for steel products worldwide and in the EU27

2.2.1 Demand for steel products

2.2.1.1 Demand by region

In 2022, the reference year for the market data analysis, the global steel demand reached 1.8 billion tonnes; this includes carbon steel, cast products, and stainless steel products. As shown in Figure 20., the EU accounts for approximately 8% of this global demand (apparent consumption⁹⁵), while Asia dominates with two thirds of the total, largely driven by China. North America ranks third, with about 7.7% of global demand (World Steel Association, 2023; worldstainless, 2023). Following two years of decline and post-pandemic market disruptions, experts predict a 1.2% increase in global demand by 2025, reaching 1.815 billion tonnes, with India expected to be the primary driver of growth as Chinese demand continues to decline (Reuters, 2024).

Figure 20. Apparent consumption of finished steel goods worldwide in 2012 and 2022



Source: 2023 World Steel in Figures (World Steel Association, 2023)

2.2.1.2 Demand in the EU

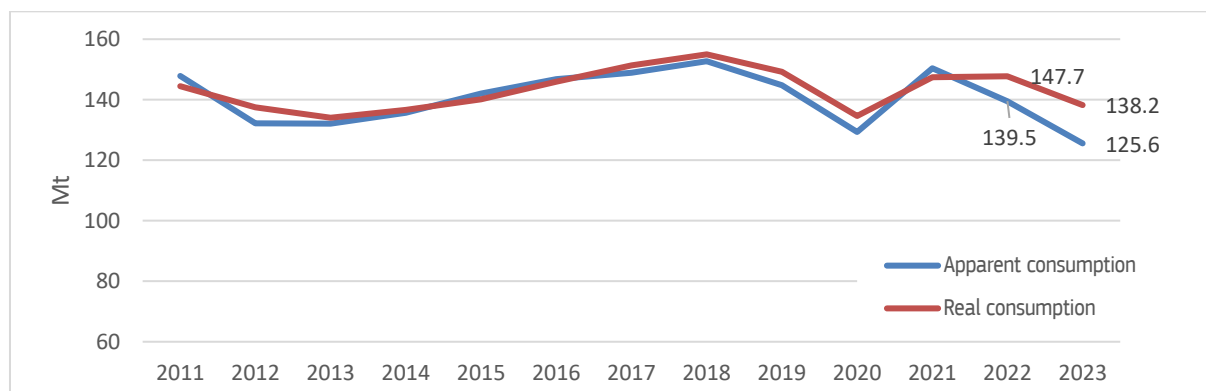
Europe's demand for steel has remained relatively stable over the long term, with a slight increase in consumption from 2012 to 2018, reflecting positive economic growth during that period. However, apparent steel consumption declined in 2020 due to the COVID-19 pandemic, which disrupted value chains, introduced economic uncertainty, and altered consumer behaviour. Although apparent consumption recovered in 2021, it decreased again in 2022 by 7.2%, reaching 140 Mt of steel (EUROFER, 2023). In 2023, apparent consumption recorded its lowest value ever at 126 Mt, representing another significant drop of 9% compared to the previous year (EUROFER, 2024).

The ongoing downturn in apparent EU steel consumption, which began in the second quarter of 2022, is attributed to war-related disruptions, unprecedented energy prices resulting from sanctions on Russia, and rising production costs linked to the broader macroeconomic situation, including persistent high inflation (EUROFER, 2024). This negative trend is expected to continue until at least 2025 due to rising global economic uncertainty, high interest rates, and overall weakness in manufacturing (GMK, 2024).

As shown in Figure 21., which illustrates the evolution of apparent and real consumption between 2011 and 2023, real consumption generally exhibits less fluctuation than apparent consumption. This suggests that the stocking of finished steel goods helps smooth out consumption patterns. Consequently, production and trade are more strongly influenced by the global developments described than the demand for steel by consumer sectors. The decrease in production volumes is partially compensated by destocking steel products.

⁹⁵ Apparent consumption, also referred to as "steel demand," represents the total deliveries of all steel products and qualities by EU producers, plus imports, minus receipts (imports by EU producers for further processing) and exports to third countries. In other words, it is the sum of deliveries by EU producers and imports, less receipts and exports. EUROFER's definition of apparent consumption includes all steel qualities, including stainless steel, as well as finished and semi-finished products. When apparent consumption exceeds real steel consumption, the surplus is stocked in the distribution chain. Conversely, if apparent consumption falls below real steel consumption, inventories are being drawn down. (EUROFER, s.d.)

Figure 21. Apparent and real consumption of steel in the EU27

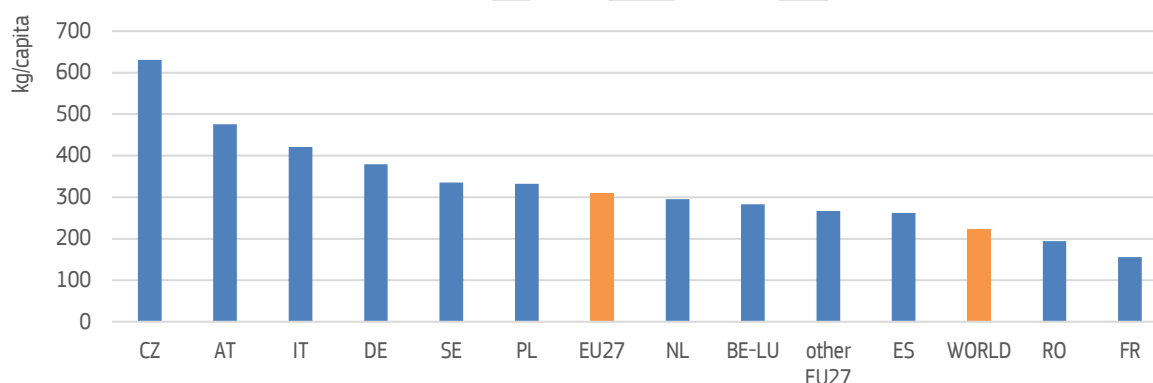


Source: European Steel in Figures 2023 (EUROFER, 2024)

When it comes to apparent steel consumption per capita⁹⁶, Czechia is the largest consumer in the EU, with 631 kg/capita in 2022, compared to the EU27 average of 310 kg/capita (Figure 22.). This is comparable to steel use in other developed economies, such as the United States (280 kg/capita), Canada (350 kg/capita), and Russia (290 kg/capita). However, several Asian countries have significantly higher per capita consumption, with China at 650 kg/capita and South Korea at 990 kg/capita.

Globally, the World Steel Association reports that apparent steel use has increased by 3% over the past five years and by 14% over the past decade (World Steel Association, 2023; World Steel Association, 2020).

Figure 22. Per capita consumption of finished steel products in the EU27 in 2022



Source: Steel Statistical Yearbook 2023 (World Steel Association, 2023)

2.2.1.3 Demand by end-use sector

As illustrated in Figure 23., the demand for steel is driven by four steel-using sectors⁹⁷.

The **construction sector**, the largest consumer of iron and steel products, accounts for more than **50%** of the total demand, with **infrastructure** and **buildings** making up 24% and 28% of this share, respectively. The majority of steel demand in the construction sector is driven by **long products**, such as rebar, wire rod, merchant bars, and sections. In contrast, flat products use is limited to building applications (e.g., roofing sheets, cladding panels, door frames) (Zaccaro, 2024).

The **transportation sector**, which accounts for **11%** of EU steel demand, is the largest consumer of flat products in the EU. This sector can be disaggregated into three sub-categories: **light vehicles** (LDV, LCV), which accounts for 8% of demand; **heavy vehicles and buses** (HCV), which accounts for 1%; and **other transportation**,

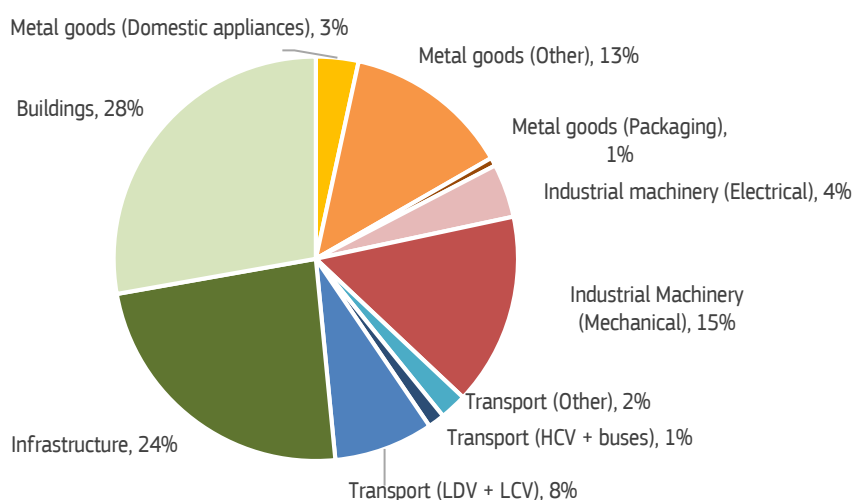
⁹⁶ WorldSteel provides apparent consumption per capita figures for carbon steel. They do not include stainless steel consumption. However, stainless steel is only about 3% of global steel consumption so the per capita numbers can still provide valuable insight.

⁹⁷ The demand by end-use sector information presented uses the sector classification used by Dworak and Fellner, 2021; EUROFER, World Steel Association, 2023, which allows us for further disaggregation of the data into demand for specific finished iron and steel products, as illustrated in the Sankey chart in Figure 24.. The sectors classification used in other sources, such as EUROFER is slightly different, although the correspondence between classifications can be established.

which accounts from the remaining 2%. The automotive industry utilises various types of flat steel for different vehicle components, including the body, drive train, suspension, and numerous smaller parts. The car body, which comprises around 40% of the total steel used in vehicle manufacturing, often employs **Advanced High Strength Steel (AHSS)** and **galvanized steel** (CEPS, 2023). In addition, long products such as bars and rods are used in various applications, including powertrain components (engine, gearbox, transmission), chassis parts (steering, suspensions), and safety and comfort features (tyres, seats), among others. Approximately, long products account for 12% of car's weight (ArcelorMittal, s.d.).

Industrial machinery accounts for **20%** of overall demand, with mechanical machinery making up 15% and electrical machinery accounting for 4%. Finally, **metal goods** sector represents **17%** of total demand, of which 3% are domestic appliances and 1% is packaging; the remaining 13% is not further specified metal goods, such as racks, hangers, carts, and similar products.

Figure 23. Estimated demand for finished iron and steel products by end-use sector in 2020 and 2021.



Note: Due to the high amount of data merging and restructuring the information on end use demand should be read as estimates not exact numbers. The classification of sectors slightly different categorisation compared to EUROFER statistics.

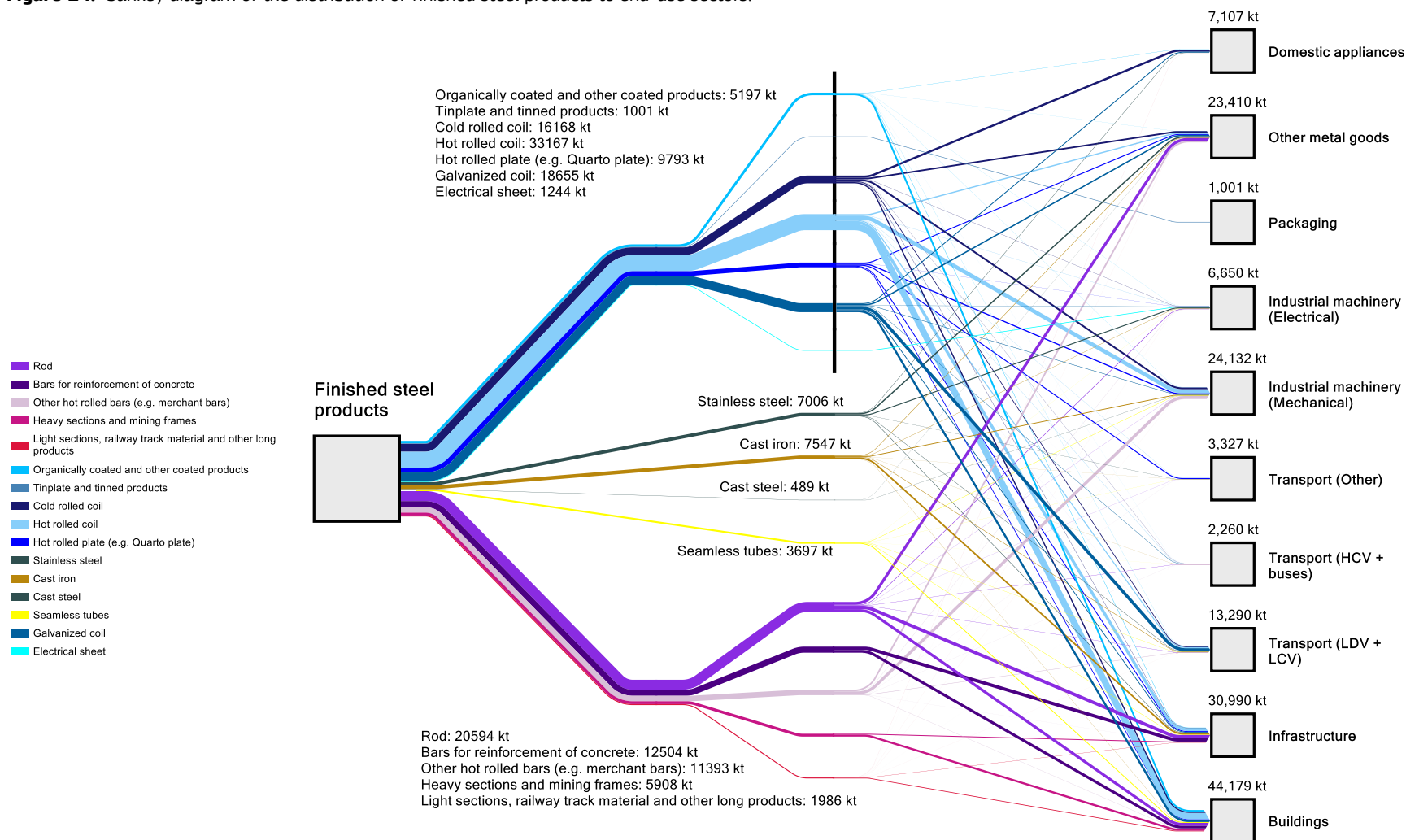
Source: own estimates based on (Dworak and Fellner, 2021; EUROFER, 2023; eurostat, 2024; Harvey, 2022; WFO, 2021; World Steel Association, 2023; worldstainless, 2023)

These numbers can be further disaggregated into demand for specific finished iron and steel products, as illustrated in the Sankey chart in Figure 24.. The detailed distribution of finished iron and steel products across end-use sectors is complex, with many finished steel products being used in multiple sectors. However, some products have more specialised applications. For instance, heavy sections and mining frames, light sections, and bars for reinforcement of concrete are only used in the construction sector in infrastructure and buildings; tinplate and tinned products only go into packaging. Electrical sheets consumption, on the other hand, is driven by demand from electrical industrial machinery and the automotive industry, in particular for electric motors in hybrid and electric cars. Finally, other finished iron and steel products have diverse applications across multiple sectors.

Despite the complexity, some studies have analysed the historical trends in demand for specific iron and steel products by end-use sectors. For instance, Dworak and Fellner (2021) and Rostek et al. (2022) examined the stock dynamics of steel products, providing valuable insight. Both studies cover carbon steel and cast iron and steel, which accounts for approximately 95% of the EU market for steel products, but not stainless steel products.

Notably, the overall steel demand has remained stable over the last two decades, with similar trends observed across end-use sectors. It is worth highlighting that Europe is shown as a major market where demand is largely driven by the need to replace existing products rather than purchase new ones.

Figure 24. Sankey diagram of the distribution of finished steel products to end-use sectors.



Note. Rod refers to the category rod and wire rod, as per Table 20.

Source: own estimates based on merging of several data sources (CAEF, 2021; Dworak and Fellner, 2021; EUROFER, 2023; Harvey, 2022; WFO, 2021; World Steel Association, 2023; worldstainless, 2023), see also Annex , Part B. Due to the significant data merging and restructuring, the information on end-use demand should be read as estimates and not exact numbers. The classification of sectors slightly different categorisation compared to EUROFER statistics.

2.3 Steel supply for the EU27

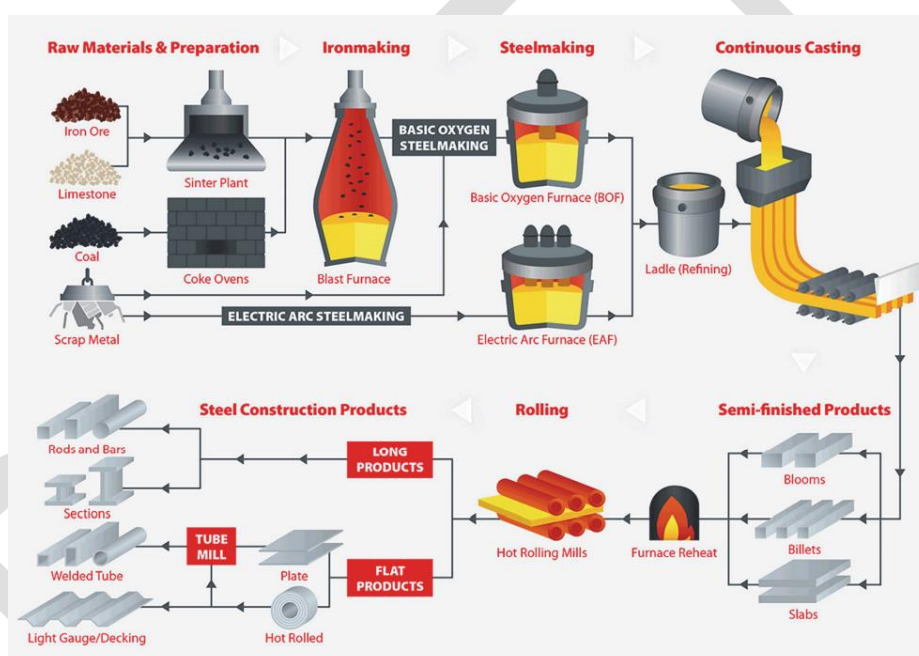
This section provides an in-depth analysis of steel supply for the EU27 market, focusing on domestic production (Section 2.3.1) and trade directions (Section 2.3.2). The Sankey diagrams shown in Figure 26. and Figure 27. illustrate the flows across all the life cycle stages of semi-finished and finished steel products, from mining to finished steel products (see Figure 25.).

The steel production process begins with **iron ores and concentrates**, which are used to produce pig iron via the blast furnace (BF) and direct reduction of iron ore (DRI) via direct reduction process. While a small fraction of pig iron is used for iron casting, the majority is used in **steelmaking processes**. Additionally, steel scrap is used in steelmaking, both in the basic oxygen furnace (BOF) and the electric arc furnace (EAF), where secondary refining can take place to produce the desired steel alloy.

Ferroalloys, used as a source of alloying elements in steelmaking, are excluded from the Sankey diagram, as only a portion of their consumption is dedicated to steel production.

Once the steel is **cast**, it can be further processed to achieve the desired shape, resulting in semi-finished products. After passing through the rolling mill, these semi-finished products are transformed into finished steel products.

Figure 25. Steelmaking process

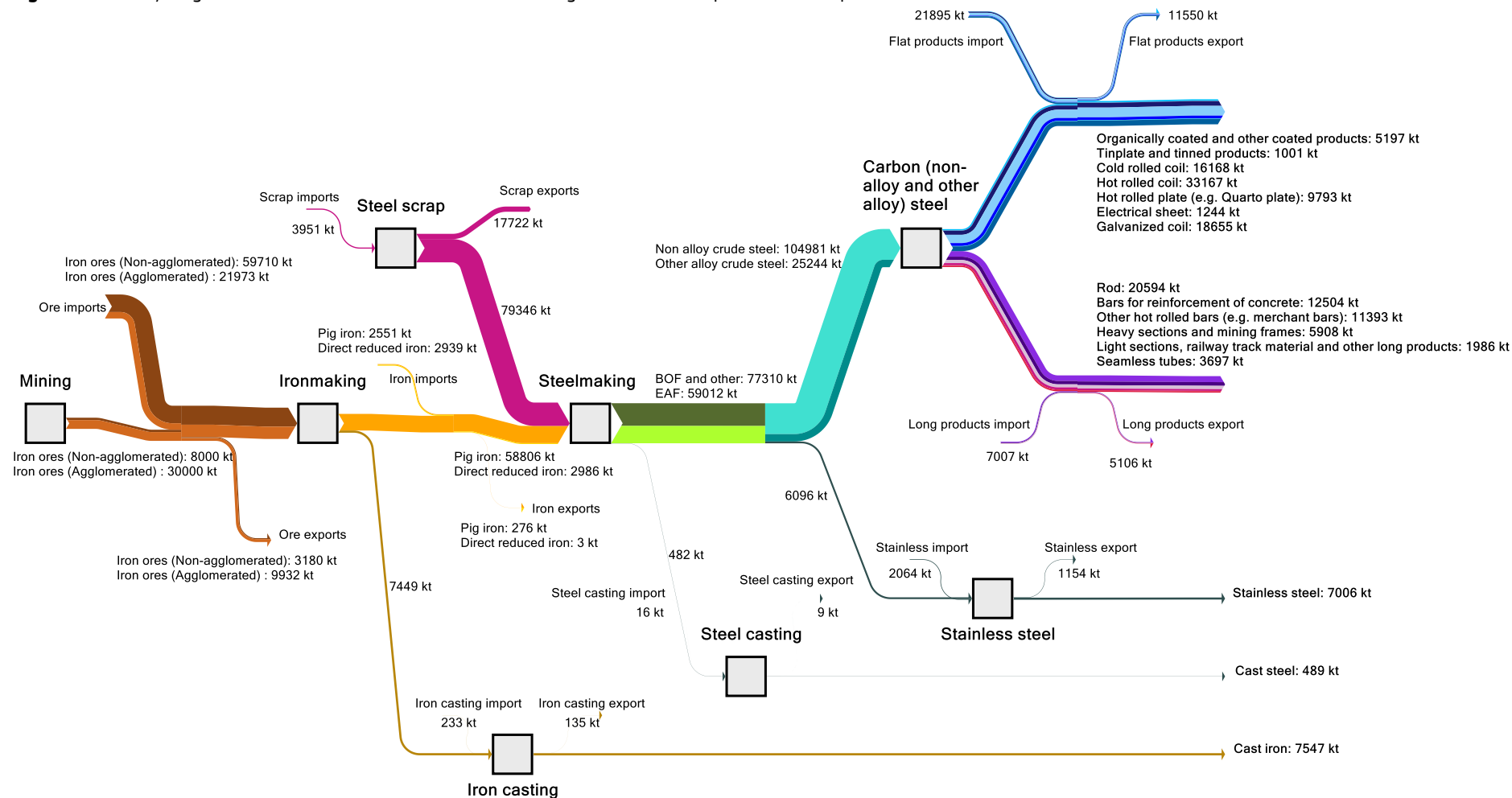


Source: An introduction to steelmaking (NSC, 2017)

The subsequent sections provide detailed production, import, and export data for between 2013 and 2022 for each of the intermediate stages of the steelmaking process, including mining, iron-making, steelmaking, casting, and finished steel production. The inflows consist of domestic production and imports, while outflows comprise apparent consumption (remaining in the EU) and exports. The diagrams also show the disaggregation of material flows based on the consolidated product classification in **Table 20**.

While Figure 26. presents carbon non-alloy and alloy steel qualities, Figure 27. focuses exclusively on stainless steel. Stainless steel production is significantly lower than carbon steel production and is often considered separately due to its distinct characteristics and production process.

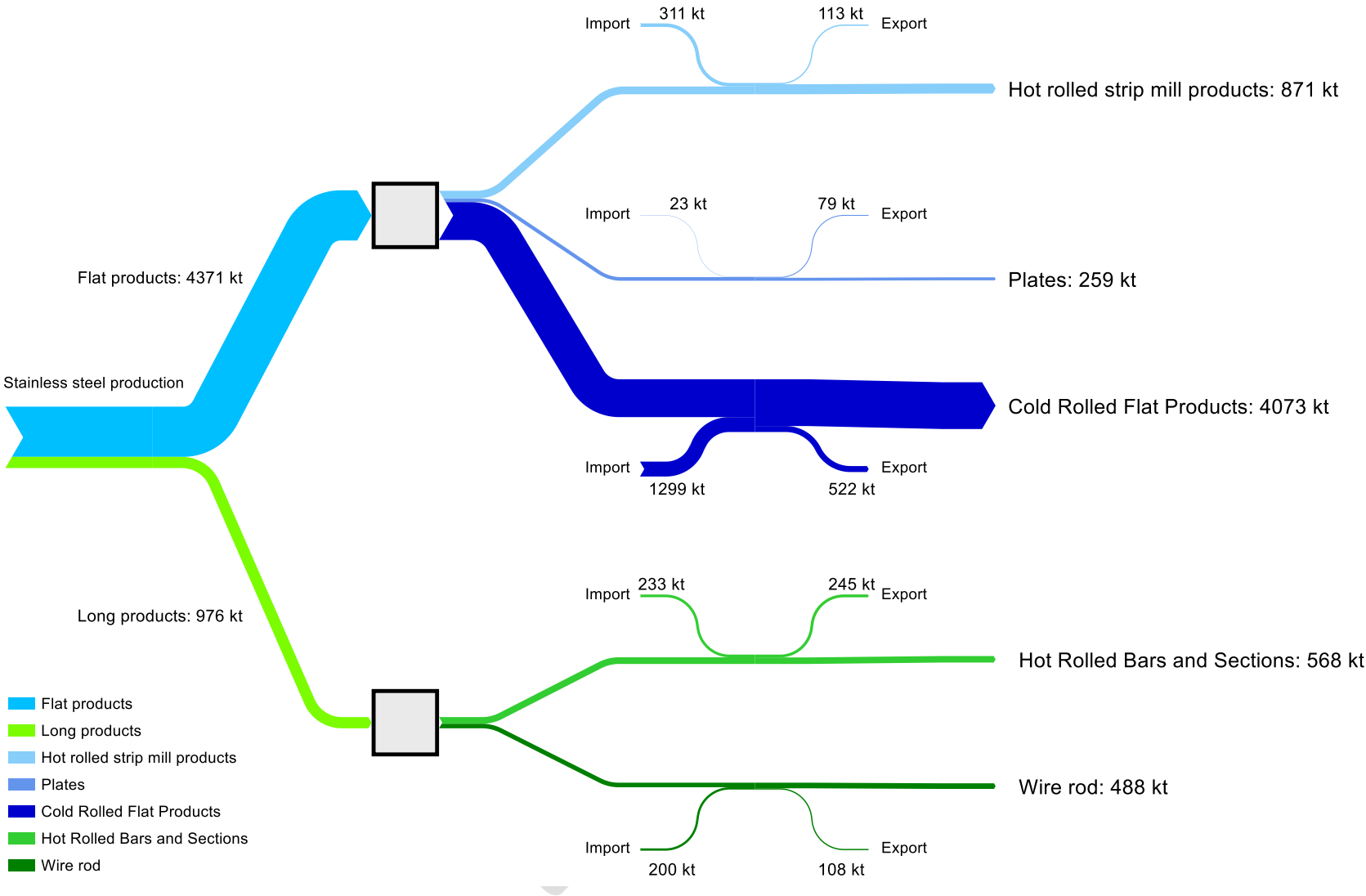
Figure 26. Sankey diagram of the EU27 steel value chain from mining to finished steel product consumption



Note: Rod refers to the category rod and wire rod, as per Table 17

Source: own calculations based on (eurostat, 2024; EUROFER, 2023; CAEF, 2021)

Figure 27. Sankey diagram of stainless steel finished products.



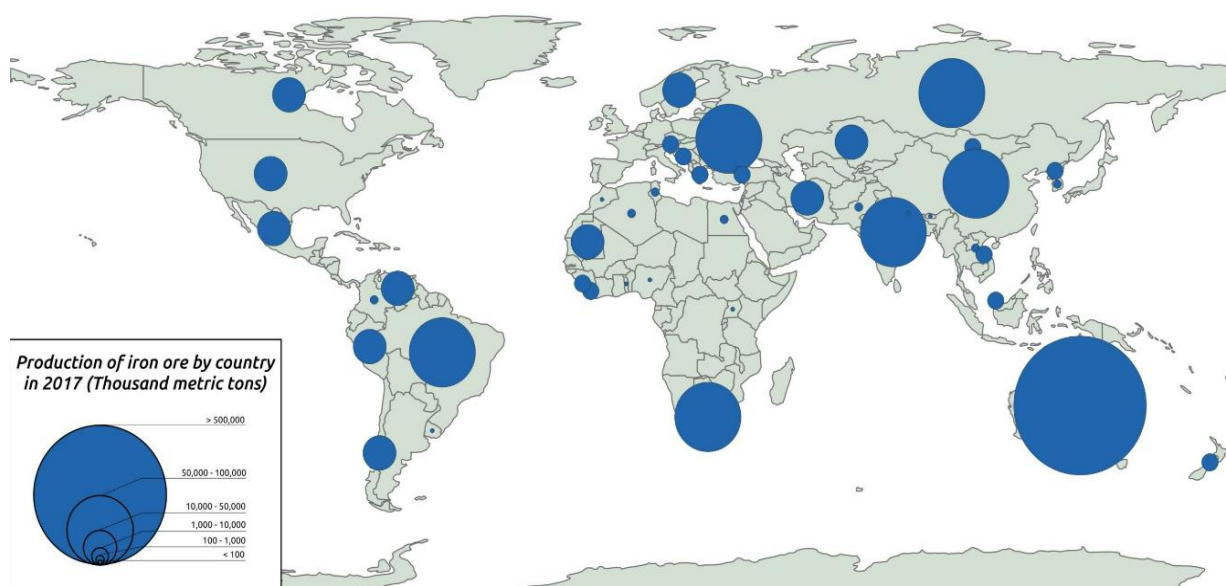
Source: European Steel in Figures 2023 (EUROFER, 2023)

2.3.1 Domestic production by life cycle stage

Iron ore

In 2022, global production of iron ore was estimated to be over 2500 million tonnes (Mt)⁹⁸, a 4.2% increase from 2020 (USGS 2024a; Küblböck K., 2022). As the primary consumer of iron ore, steelmaking accounts for 95% of its supply. Over the past 20 years, iron ore extraction has nearly tripled, with the top global producers (including Australia with 37% of global production, Brazil, China, India and Russia) dominating the market, collectively accounting for 75% of global production, see also Figure 28.. Iron ore production worldwide is recovering after a 3% decline in 2020, due to the COVID-19 crisis. According to Global Data's estimates, production is expected to steadily increase, reaching 2700 Mt by 2025 (Global Data, 2023).

Figure 28. Production of iron ore by country



Source: USGS (USGS, 2024)

In the same year (2022), the EU's sold production⁹⁹ of iron ore amounted to 38 Mt, although total production may be higher due to internal company use (Eurostat, 2024). Only a few Member States in the EU27 produce iron ore, including Sweden, Austria and Germany, with Sweden being the largest producer (Table 22).

Table 22. Production of iron ore in EU per country

Country	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Austria	2 320	2 400	2 775	2 773	2 975	2 800	3 300	3 043	3 311	3 311
Germany	413	456	468	514	447	448	534	529	529	754
Sweden	27 285	28 144	24 600	26 900	27 200	27 530	29 200	29 180	28 687	28 100
EU (27)	30018	31000	27843	30187	30622	30778	33034	32753	32708	32166

Source: worldsteel

Figure 29. shows the development of the EU's sold production of iron ore in both volume and value, based on Eurostat data¹⁰⁰. Between 2013 and 2016, a significant increase in sold production of iron ores and concentrates can be observed, reaching 40 Mt per year. Since then, sold production has fluctuated around 35 Mt, with a

⁹⁸ Estimated global resources of crude ore exceed 800 billion tonnes, containing over 230 billion tonnes of iron. Global reserves are estimated at 170 billion tonnes, containing 81 billion tonnes of iron. (USGS, 2020)

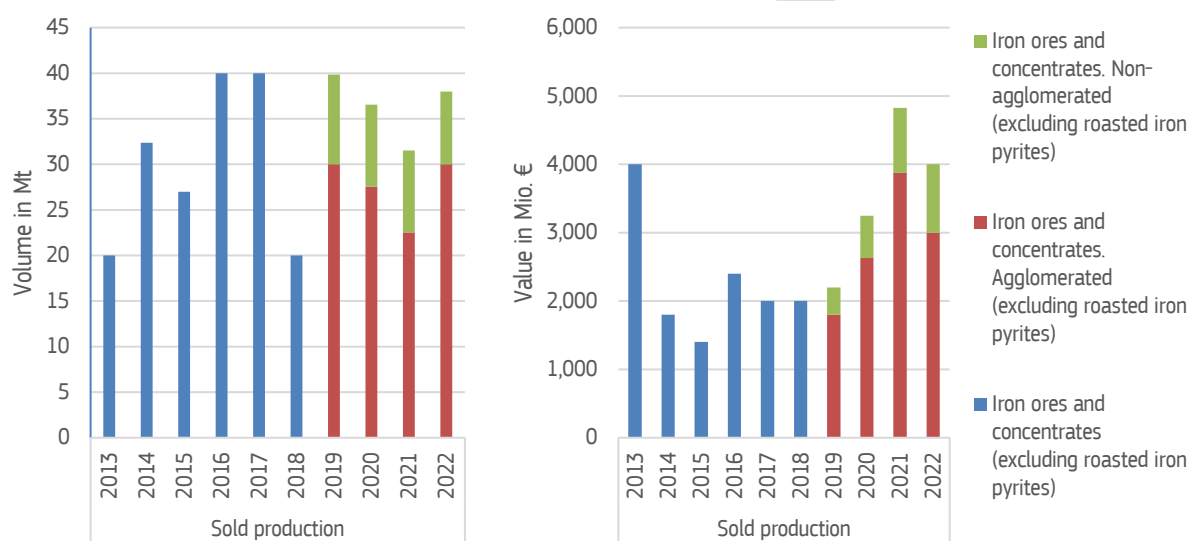
⁹⁹ While statistics for production give the entire amount produced in a certain year for a specific product, the sold production puts a number on the amount of a product produced and sold to a different company. The difference between the two gives an idea that the product is usually further processed by the same company or is sold on the market for the next step of processing handled by different actors.

¹⁰⁰ Table 17 provides Eurostat codes for the iron ore mining in Europe.

notable 50% decline in 2018. The exact reason for this drop is unknown, as the market situation was reportedly stable or even improving, and overall European production did not experience significant changes (Küblböck K., 2022; Löf, Löf, and Ericsson, 2019). The monetary value of sold production did not display a corresponding decline in 2018, suggesting a possible reporting error.

Meanwhile, mine production has remained steady in recent years. Agglomerated ores and concentrates account for approximately 75% of the EU's sold production. The value of the sold production increased over the observed period, reaching EUR 4 billion in 2022. The increase in value with simultaneous stagnation of volumes correlates with the price development of iron ores and concentrates within the observed period. Prices were very high in 2013 (and even higher in preceding years) but decreased and stabilised between 2014 and 2018. Since 2018, prices have become volatile again and have increased significantly, which is reflected in the values of sold production (Hall, 2020).

Figure 29. Iron ore mining in EU27



Note:

- Before 2019, one code was reported representing all types of iron ores and concentrates except roasted iron pyrites, which are usually not used for ironmaking. From 2019 on, more granular data is available, as two codes for iron ore and concentrates are reported, including the information on whether the material is agglomerated or not.
- Unlike the sold production, the total production of iron ore is not reported by PRODCOM.

Source: PRODCOM (eurostat, 2024)

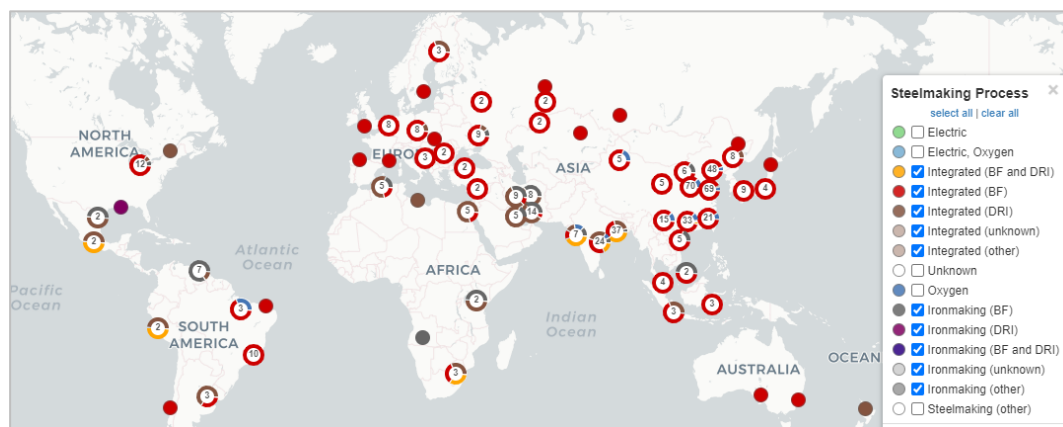
Iron-making

Around 98% of global mined iron ore is used to make pig iron, which is an input material for steelmaking, while the remaining 2% is employed in various applications such as cement production, cosmetics, pigments, and others.

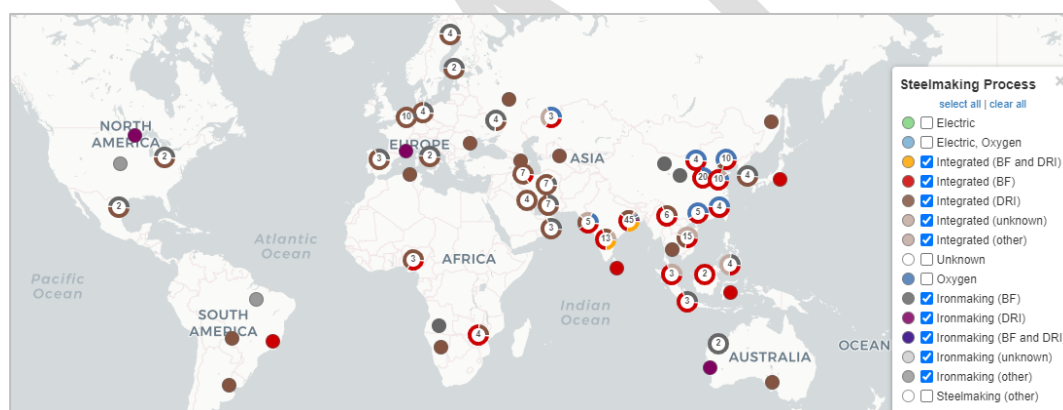
In 2022, global pig iron production reached 1300 Mt, with the EU accounting for 5.8% of this total (USGS, 2024). The global production of direct reduced iron (DRI) reached 126 Mt in 2021, with Europe holding a mere 0.5% share (World Steel Association, 2022). There is only one DRI plant in the EU, owned by ArcelorMittal and located in Hamburg, producing 0.6 Mt of steel per year (Bellona, 2021). However, looking into investment plans globally, and in particular in Europe, the significance of DRI is expected to increase.

Figure 30. presents existing and planned (under construction and planned) iron-making facilities around the globe, with consideration of the technology route, based on Steel Plant Tracker¹⁰¹. The largest pig-iron-producing countries are China, India and Japan.

Figure 30. Iron-making facilities around the world by technology



A) Operating facilities



B) under construction and announced

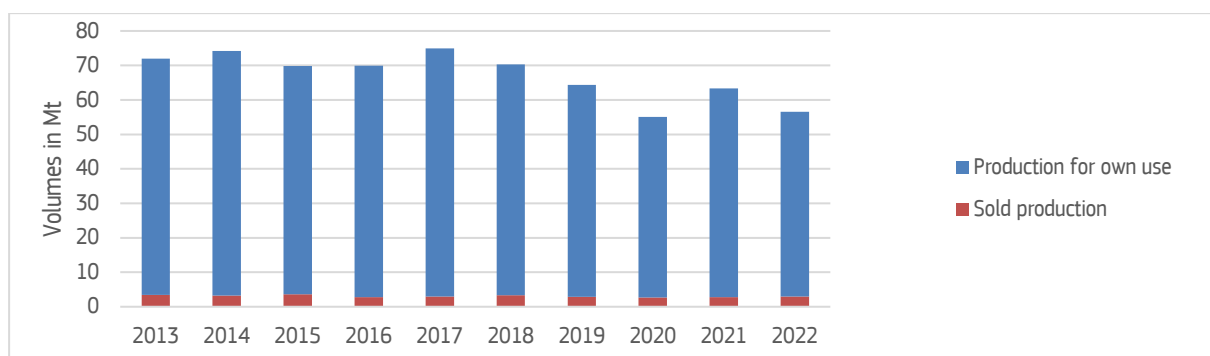
Source: Steel Plant Tracker (Global Energy Monitor, s.d.) Accessed in September 2024

Figure 31. shows the total production and sold production of iron in volumes within the EU, based on Eurostat data. The sold production of iron is significantly lower than the overall production, accounting for only around 5% of the total. The difference between total production and sold production represents the volumes that are further processed internally by companies. In integrated steel mills, pig iron is immediately utilised for steelmaking, which means that most of produced pig iron is not sold to third companies.

The total iron production in the EU decreased from 72 Mt/y in 2013 to nearly 57 Mt/y in 2022. While production volumes remained stable until 2019, significant declines occurred in 2020 and 2022. These can be attributed to the economic challenges posed by the global pandemic in 2020, which led to a decrease in demand for steel products.

Figure 31. Production (sold production +production of own use) of iron in the EU27

¹⁰¹ Global Steel Tracker: <https://globalenergymonitor.org/projects/global-steel-plant-tracker/tracker-map/>; The Tracker covers 92% of global operating crude steel making capacity. Additionally, it covers all crude iron and steel plants under development as of March 1, 2023.



Source: PRODCOM (eurostat, 2024)

Table 23 and Table 24 present trends in pig iron and direct reduced iron production in Europe, per member states. The pig iron production presents declining trends, while DRI remains stable.

Table 23. Production of pig iron in the EU per country between 2013 and 2022

	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Austria	6.2	6.0	5.8	5.6	6.3	5.3	5.7	5.3	6.1	5.8
Belgium-Luxembourg	4.3	4.4	4.2	4.9	4.8	4.8	4.8	3.6	4.2	4.4
Czechia	4.0	4.2	4.0	4.2	3.7	4.0	3.6	3.5	3.9	3.4
Finland	2.1	2.2	2.3	2.3	2.2	2.4	2.3	1.9	2.5	1.9
France	10.3	10.9	10.1	9.7	10.7	10.5	9.9	7.7	9.5	8.2
Germany	26.7	27.3	27.8	27.2	27.8	27.3	25.5	22.5	25.7	23.7
Hungary	0.6	0.8	1.2	0.9	1.3	1.4	1.2	1.0	0.6	0.5
Italy	6.9	6.4	5.1	6.0	5.1	4.8	4.6	3.4	3.9	3.5
Netherlands	5.7	5.9	6.1	6.1	6.1	6.2	5.9	5.4	5.9	5.5
Poland	4.0	4.6	4.8	4.7	5.2	4.9	4.4	3.5	3.6	3.1
Romania	1.6	1.6	2.0	2.0	1.9	2.0	2.1	1.8	2.1	1.5
Spain	3.9	4.0	4.5	4.1	4.5	4.5	3.9	2.9	4.0	3.4
Sweden	2.9	3.1	2.9	3.1	3.1	2.9	3.2	2.9	3.0	2.8
EU (27)	82.9	85.2	84.5	84.8	86.8	84.8	79.8	68.2	79.0	70.9
World										1 301.3

Source: WorldSteel

Table 24. Production of direct reduced iron between 2017 and 2022

	2017	2018	2019	2020	2021	2022
Germany	0.5	0.5	0.5	0.5	0.5	0.5
Sweden	0.1	0.1	0.1	0.1	0.1	0.1
EU (27)	0.6	0.6	0.6	0.6	0.6	0.6
World						125.1

Source: worldsteel

Ferroalloys

While the largest application of ferroalloys is in steelmaking (85-90%), significant amounts are used for non-steel-related applications (Fichte, 2003). Ferroalloys are also employed in the production of super-alloys, non-ferrous alloys, pure metals, and chemicals (Fichte, 2003).

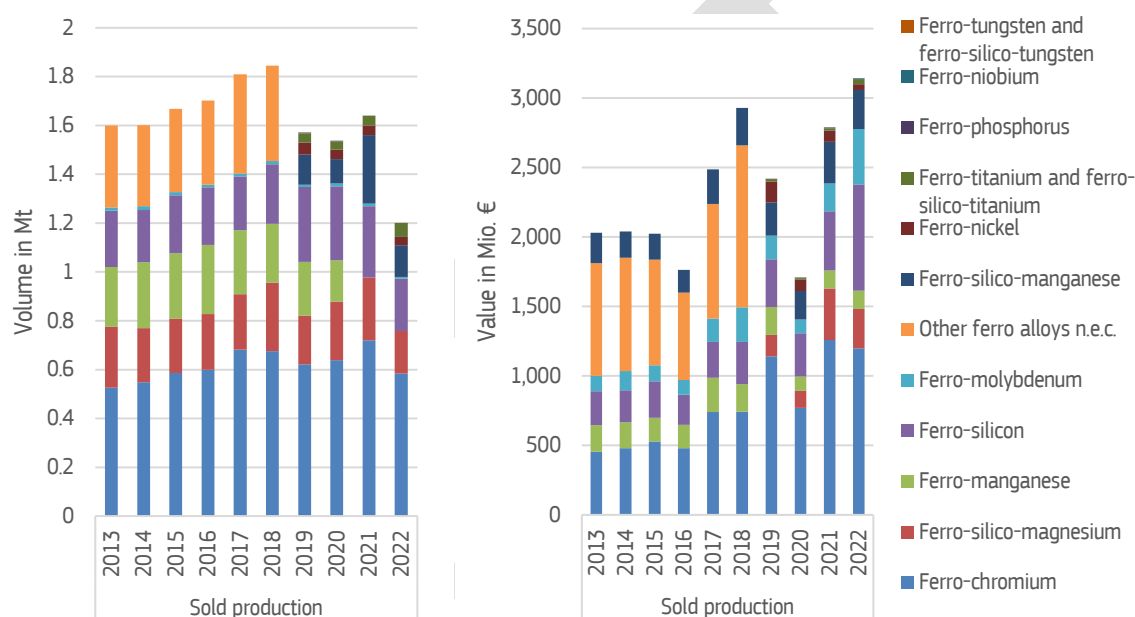
Consequently, from a material flow perspective, it is essential to subtract the non-steel-related fractions of ferroalloy demand in Europe when analysing steel flows and stocks (they have been subtracted from the Sankey chart presented in Figure 26.). Since yearly data for non-steel applications was unavailable, total numbers are presented. Globally, a total of 60.1 Mt of ferroalloys were produced in 2021 (USGS, 2024), while the EU's sold production of ferroalloys reached 1.2 Mt in 2022 (eurostat, 2024). The ferroalloy production leaders are typically located in countries with abundant mineral resources, power availability, and highly developed industries. The biggest producers are China, South Africa, Ukraine, Russia, Kazakhstan, Brazil and India, which together account for 84% of global production.

The EU's ferroalloy sold production comprises a variety of ferroalloys, as illustrated in Figure 32.. In terms of volume, the sold production reached a peak of 1.84 Mt in 2018, followed by a decline to 1.2 Mt/y in 2022. In 2021 and 2022, no sold production of ferro-manganese was reported in the PRODCOM database. [Argus \(2021\)](#) states declines in imports and production; however, the overall drop seems to be a reporting issue by PRODCOM, which may have occurred due to low production and trade numbers.

The most important ferroalloy type by volume for European sold production is **ferro-chromium**, followed by **ferro-silicon**, **ferro-silico-manganese** and **ferro-manganese**. Since 2019, the reported PRODCOM classification has provided more granular information, allowing for a more detailed breakdown of the "other ferroalloy" category into specific types, including ferro-nickel, ferro-phosphorus, ferro-niobium, ferro-silico-magnesium, ferro-titanium and ferro-silico-titanium as well as ferro-tungsten and ferro-silico-tungsten.

The total value of sold ferroalloy production in the EU shows an increasing trend. Interestingly, the lesser-produced alloy types tend to exhibit proportionally higher values compared to more widely used ferroalloys like ferro-chromium.

Figure 32. Sold production of ferroalloys in the EU27



Source: PRODCOM (eurostat, 2024)

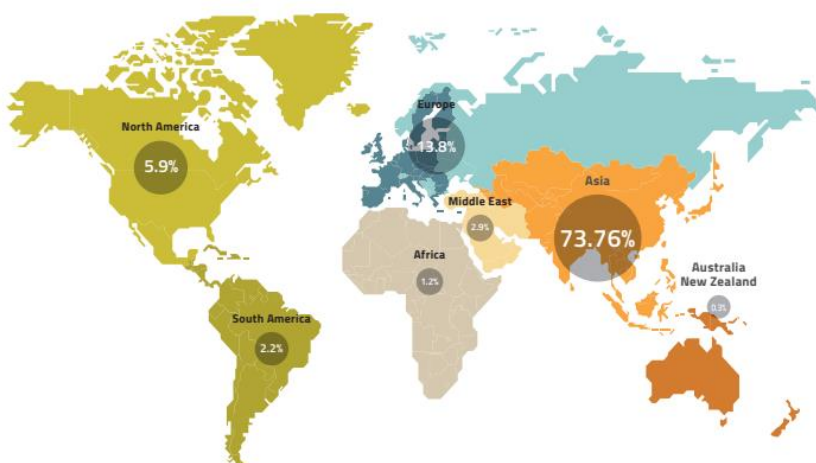
Crude steel

By geographical location

Global crude steel production reached 1850 in 2023 and 1885 Mt in 2022, nearly 5% decrease compared to 2021. China remains the world's leading producer, accounting for 1019 Mt, followed by India (141 Mt), Japan (87 Mt), the United States (81 Mt). Europe (EU27) with 126 Mt of crude steel in 2023 contributed to 6.8% of the global production, see Figure 33. (EUROFER, 2024).

Figure 33. Crude steel production worldwide (in Mt)

		% shares
	2023	2023
Asia and Oceania	1,361	73.6%
of which China	1019	55.1%
of which India	141	7.6%
of which Japan	87	4.7%
Europe	256	13.8%
of which European Union 27	126	6.8%
of which Ukraine and CIS	88	4.8%
North America	110	5.9%
of which United States	81	4.4%
South America	42	2.2%
Middle East	53	2.9%
Africa	22	1.2%
Australia/New Zealand	6	0.3%
WORLD	1.850	100%



Source: European Steel in Figures 2023 (EUROFER, 2024)

Also, the [Energy and Industry Geography Lab](#) (EIGL)¹⁰² and [Global Steel Plant Tracker](#) (GSPT) provide a visual representation of the existing and planned iron and steel plant capacities worldwide, broken down by technology route.

In 2022, global steelmaking capacity increased by 32.1 Mt, reaching a record high of 2,459 Mt, the highest global capacity figure in history (OECD, 2023)¹⁰³. While China and India account for roughly half of the world's steelmaking capacity, the fastest growth is currently observed in the Middle East (e.g., Iran and Iraq), ASEAN countries (e.g., Vietnam, Indonesia), the so-called "other Europe," and more recently in Africa (e.g., Algeria, Zimbabwe)¹⁰⁴. In contrast, the EU is the only region globally to experience a decline in steel production capacity in recent years, with a significant loss of 2.6 million tonnes of steel production capacity, see Table 25 (OECD, 2024)

Table 25. Trends in steelmaking capacities by region (in Mt)

	2019	2020	2021	2022	2023	2019-2023
World (total)	2407	2418.6	2425.2	2441.5	2498.6	3.80%
Non-OECD/EU economías	1757.3	1765.5	1777.3	1787.4	1836.5	4.50%
OECD/EU economies	649.7	653.1	647.8	654.1	662.1	1.90%
Asia	1616.5	1622.5	1622.6	1626.1	1656.6	2.50%
China	1148.3	1147.9	1146.5	1149.9	1173.3	2.20%
India	128.7	128.7	133.9	133.9	138.4	7.50%
ASEAN	74.6	78.7	80.4	80.4	82.9	11.10%
Europe	287.4	287.5	288.1	289.3	294.3	2.40%
EU27	216	213.4	213.4	213.4	213.6	-1.10%
Other Europe	71.4	74.1	74.7	75.9	80.7	13.00%
North America	154.2	157.5	157.7	162.8	165.8	7.50%
CIS	143.4	142.6	143.9	145	145.1	1.20%
Middle East	80.7	84.1	89	92.3	103.3	28.00%
Latin America	73.9	73.4	73.9	73.9	75.9	2.70%
Africa	44.6	44.7	43.5	45.8	51.4	15.10%
Oceania	6.4	6.4	6.4	6.4	6.4	0.00%

Note: green – countries/regions with the fastest grow in steelmaking capacities. Red- countries/regions with negative growth

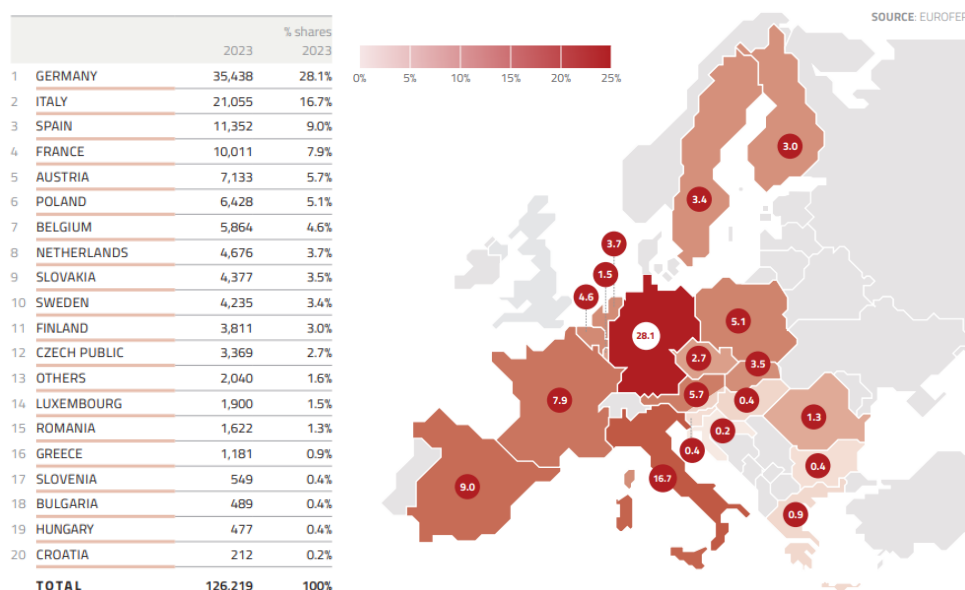
¹⁰² Energy and Industry Geography Lab (v.1.6) <https://energy-industry-geolab.jrc.ec.europa.eu/> has been developed by the European Commission's Joint Research Centre. Among others, it includes the Industrial infrastructure mapping, including the iron and steel asking facilities.

¹⁰³ The ratio between the global crude steel capacities to crude steel is estimated for around 70-80%. Although both have increased, production growth has slowed relative to capacity. As a result, the gap between global steelmaking capacity and crude steel production could fall by 1.7% Mt in 2023 (compared to 2022) and reach 75.6% (OECD, 2024)

¹⁰⁴ While steelmaking capacity in China has remained relatively stable in the last few years, Chinese steel companies are investing heavily in capacity projects overseas. Following OECD data, Chinese companies are involved in 13 cross-border investments and participate in nine joint venture investments abroad. ASEAN, which is the top region among cross-border investments, is expected to increase capacity to levels that far exceed the region's steel demand.

Source: Latest developments in steelmaking capacity 2024 (OECD, 2024)

Figure 34. Crude steel production in Europe



Source: European Steel in Figures 2023 (EUROFER, 2024)

Across the EU, there are over 500 steel production sites spread across 20 Member States (see Figure 34.). Germany is the largest steel-producing country in the EU, accounting for 28.1% of total crude steel production (35.4 Mt in 2023). Italy, Spain and France follow, contributing 16.7%, 9%, and 7.9% of the EU's crude steel production, respectively. The remaining production is distributed across 16 other countries, while seven Member States lack steel production capacity altogether. Notably, the shares of crude steel production by country have remained relatively stable over the past decade, with only minor fluctuations observed.

By technology route (primary vs secondary)

As previously explained, there are two main technologies for crude steel production, blast furnace and basic oxygen furnace (BF-BOF) and electric arc furnace (EAF). Globally, the majority of crude steel capacity (built before 2021), approximately 67%, relies on the BF-BOF route, while 32% of capacity is from EAFs. The less than 1% remaining represents other technologies, such as open hearth furnace (OHF), see also Table 26.

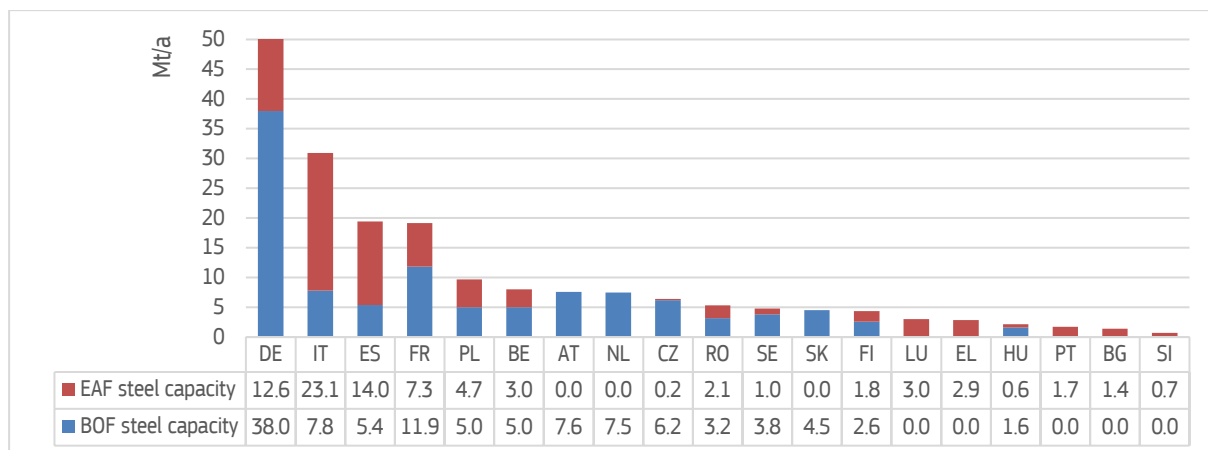
Table 26. Nominal steel production capacities per region and technology

Region	Total crude steel capacity (Mt/a)	Share of BOF capacities	Share of EAF capacities
Asia Pacific	1508.5	78%	20%
EU27	190.4	58%	42%
North America	146.7	30%	70%
Eurasia	97.2	62%	38%
Other Europe	93.7	39%	61%
Middle East	75.6	8%	90%
Central & South America	56.6	67%	33%
Africa	38.3	18%	81%
Grand Total	2206.9	67%	32%

Source: Global Steel Plants Tracker, accessed in 05/2024

In the EU, the ratio between BOF and EAF follows a similar trend, with BF-BOF accounting for 58% and EAF 42%. On country to country level the ratio differs significantly, as shown in Figure 35.; for example for Germany the BOF-EAF ratio is 75-25, while for Italy (second largest steel producer in Europe) it is 25-75.

Figure 35. Iron- and steel-making capacities in Europe by technology route.



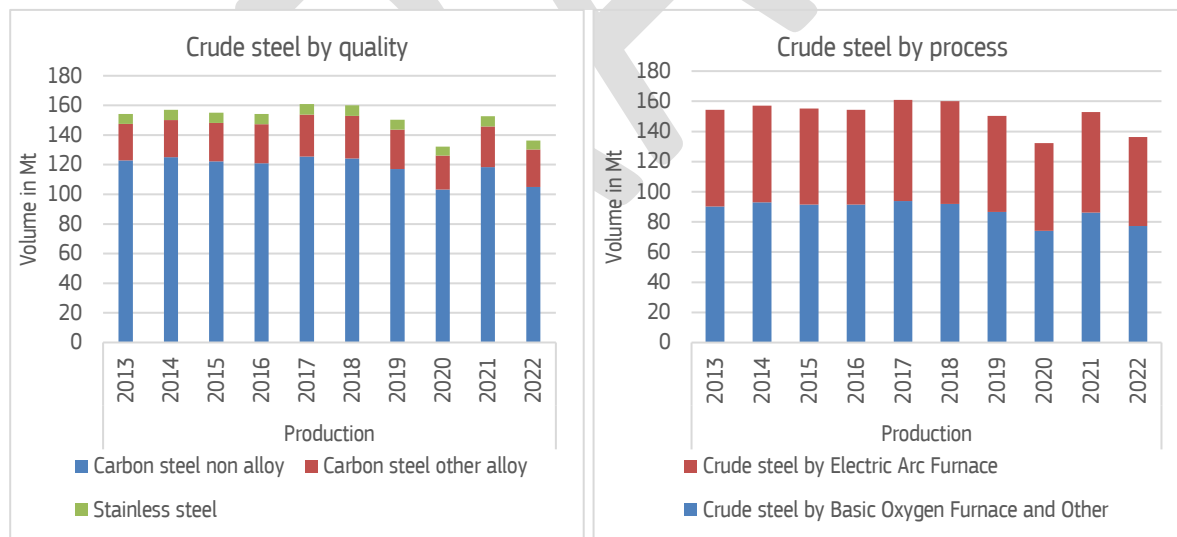
Source: Global Steel Plants Tracker, accessed in 05/2024

Figure 36. presents the trends in crude steel production in Europe, broken down by steel quality and production technology route. Between 2013 and 2018, crude steel production remained relatively stable, but subsequently declined slightly, particularly in 2020 and 2022, reaching 136 Mt/y in 2022. This decline can be attributed to the global COVID-19 pandemic, the subsequent challenging economic situation, and significant supply chain disruptions. Additionally, the start of the war in Ukraine in 2022 and the resulting changes in the energy market also contributed to this decline.

Looking into technology of steel production there is a gradual shift towards scrap-based production, driven by the increasing adoption of EAF technology, which has led to a nearly 2% increase in crude steel production over the past decade.

In terms of steel quality, non-alloyed carbon steel dominates, accounting for 77% of overall steel production. Alloyed steel other than stainless steel makes up 18.5% of crude steel production in the EU27, while stainless steel volumes represent 4.5% of total steel production.

Figure 36. Crude steel production in the EU27

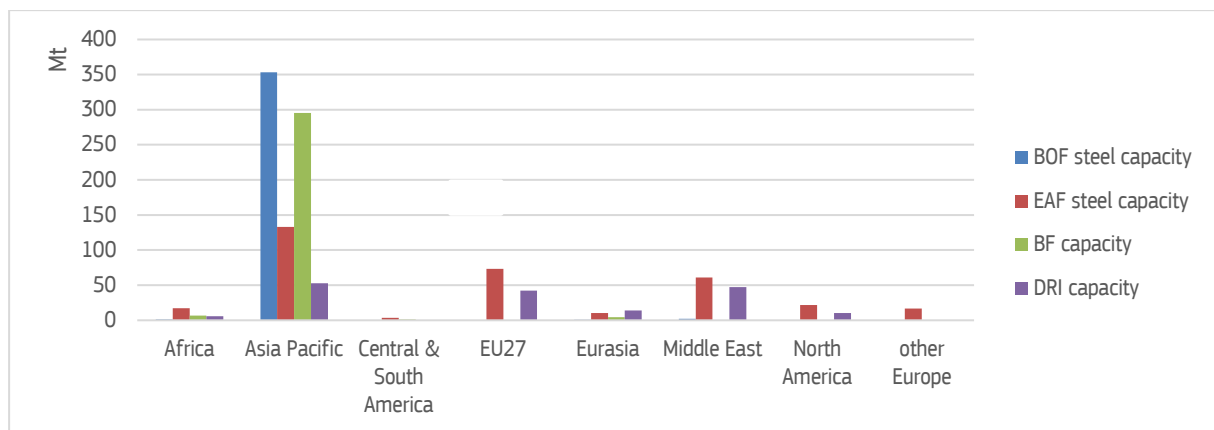


Source: European Steel in Figures 2023 (EUROFER, 2023)

Figure 37. summarises new nominal capacities per type of equipment in the region across the world. The information is based on Global Steel Plant Tracker (version 1.0 from April 2024) and the capacities that are already under construction and announced to start before 2026.

The largest investments are planned in Asia Pacific region, and include over 350 Mt in BOF technology and over 130 Mt in EAF. In the region also 52Mt capacities in DRI are considered In Europe (EU27) the announcements include 73Mt in EAF and 42 Mt in DRI technology.

Figure 37. New (announced and under construction) capacity additions by region by 2026.



Source: Global Steel Plants Tracker, accessed in 03/2024

Finished steel products

The market data analysis presented in this section of the report follows the product classification outlined in Table 20, in Section 2.1.2. Due to limitations in the reported data, certain products are aggregated into the following categories: hot rolled coil and electrical sheet; iron and steel castings (up to 2017); galvanized coil, tinplate, and tinned products; and other hot rolled bars (including merchant bars) and light sections.

Table 27 shows the production volumes of finished steel products in the EU27 based on EUROFER and PRODCOM data. The total production of finished steel products peaked in 2018 at 162.5 Mt/y, before decreasing to 137.7 Mt in 2022 due to the global pandemic and the war in Ukraine.

Table 27. Production volumes [in Mt] of finished alloyed and non-alloyed steel products in the EU27

[volumes in Mt]		2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	Rank
Stainless steel long products		0.6	0.6	0.6	0.6	0.6	0.7	0.7	0.6	0.7	0.6	13
Stainless steel flat products		3.4	3.5	3.6	3.9	3.8	3.8	3.6	3.4	3.9	3.6	11
Long	Total: long	54.4	54.7	55.1	55.1	56.0	57.0	54.1	48.9	56.2	50.5	n/a
	Rod and wire rod	19.1	19.2	19.8	19.5	20.5	21.1	20.2	18.4	21.7	18.9	2
	Railway track material and other long products	3.0	2.9	3.0	2.6	2.6	2.7	2.7	2.5	2.7	2.5	12
	Other hot rolled bars and light sections	11.4	11.9	11.7	11.5	12.1	12.3	10.8	9.5	11.5	10.4	6
	Heavy sections and mining frames	8.1	8.0	8.2	8.6	8.6	8.6	8.0	7.2	8.1	7.2	9
	Bars for reinforcement of concrete	12.9	12.7	12.5	12.9	12.2	12.3	12.5	11.3	12.2	11.5	5
Flat	Total: flat	87.8	89.3	87.7	88.5	91.7	91.5	85.1	74.7	84.5	74.9	n/a
	Tinplate and tinned products and galvanized coil	19.8	21.4	22.0	22.1	22.7	22.2	21.0	17.7	19.3	17.2	3
	Organically coated and other coated sheets	4.2	4.3	4.3	4.5	4.6	4.6	4.8	4.7	5.3	4.4	10
	Hot rolled plate	10.2	10.9	10.4	10.4	11.0	11.0	10.1	9.1	10.1	10.0	7
	Hot rolled coil and electrical sheet	36.4	36.1	34.8	34.6	36.1	36.8	33.7	29.5	33.1	29.2	1
	Cold rolled coil	17.3	16.7	16.4	17.0	17.3	16.8	15.5	13.6	16.7	14.0	4
Castings		0.0	0.0	0.0	0.0	0.0	10.0	9.3	7.1	8.0	8.1	8
Cast steel		0.3	0.3	0.3	0.3	0.0	0.0	0.0	0.0	0.0	0.0	15
Cast iron		9.4	9.3	9.2	9.2	10.3	0.0	0.0	0.0	0.0	0.0	
Total		156.0	157.7	156.6	157.6	162.5	162.9	152.7	134.7	153.3	137.7	

Source: European Steel in Figures 2023 (EUROFER, 2023) and PRODCOM (eurostat, 2024)

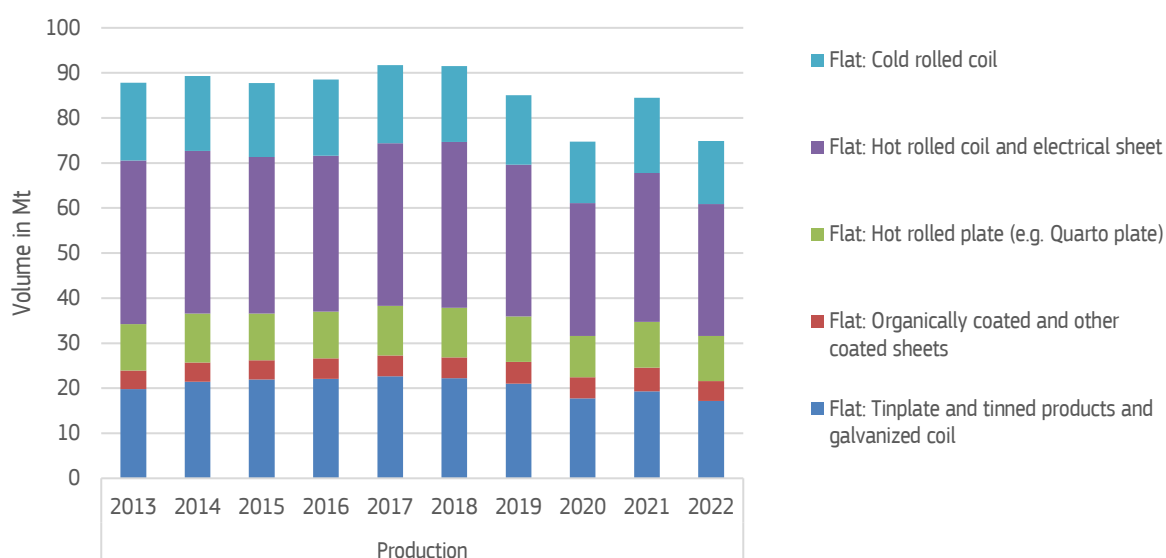
The majority of finished carbon steel products produced within the EU27 are **flat products** (made of non-alloy and other alloy steel), which accounted for 74.9 Mt (54.4% of total finished steel product production) in 2022. Within this group, hot rolled coil and electrical sheet are produced in the highest quantities (29.2 Mt, equivalent to 21.2% of total finished steel products). The second most produced finished steel product is galvanized steel, tinplate, and tinned products, cumulatively accounting for 17.2 Mt (12.5% of total production). Cold rolled coil, hot rolled plate (e.g. Quarto plate), and organically and other coated sheets account for 10.2%, 7.3%, and 3.2% of total production respectively – see also Figure 38.

Non-alloy and other alloy **long products**, with a total production of 50.5 Mt, contributed to 36.7% of the total finished carbon steel product production. The most important subcategories are rod and wire rod (18.9 Mt, 13.7%), bars for reinforcing concrete (11.5 Mt, 8.4%), and other hot rolled bars and light sections (together 10.4 Mt, 7.6%). The remaining long products are heavy sections and mining frames (7.2 Mt, 5.2%) and railway track material and other long products (2.5 Mt, 1.8%) – see also Figure 39.

In addition to carbon steel long and flat products, there are **castings** and **stainless steel**. Castings account for 6% of total production. The majority of these castings are iron castings, while steel castings are significantly less relevant, with no reported production after 2016. Stainless steel products account for 3% of total finished steel product production. The majority (85%) of the stainless steel products produced in the EU27 are flat products.

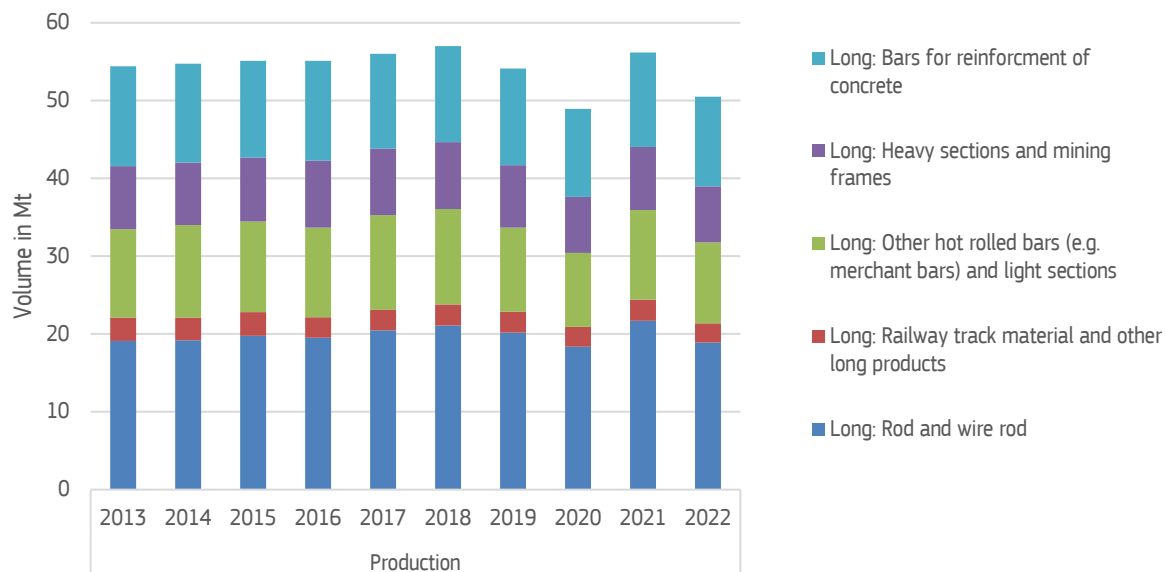
From 2013 to 2022, there were only slight changes in the shares of finished steel products. Long carbon steel products and stainless steel finished products slightly increased in market share, while flat carbon steel products and cast steel decreased.

Figure 38. Production volumes of flat products made from carbon steel in the EU27



Source: European Steel in Figures 2023 (EUROFER, 2023) and PRODCOM (eurostat, 2024)

Figure 39. Production volumes of long products made from carbon steel in the EU27



Source: European Steel in Figures 2023 (EUROFER, 2023) and PRODCOM (eurostat, 2024)

Annex 4 Part A provides additional information on market trends for finished steel products, detailing volumes of domestic production for each product over the last decade.

2.3.2 EU sales and trade by life cycle stage

Building on the analysis of European steel production in the previous section, this chapter incorporates the perspective of trade into the discussion (namely volumes and directions of import and export). Understanding the market size of products sold on the EU market—including both domestically produced goods and imports—is critical for Task 2 of the preparatory study.

A valuable overview of material flows across the life cycle stages is presented in the Sankey diagrams in Figure 26. and Figure 27., offering insights into the broader implications of production and trade that will be described in details in the following section.

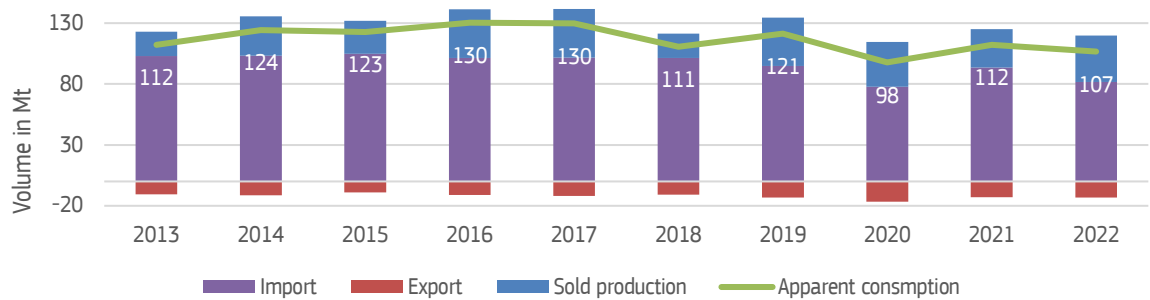
Iron ore

Figure 40. provides a comprehensive overview of the production, trade, and consumption of iron ores and concentrates in the EU27, based on data from Eurostat (eurostat, 2024). The analysis highlights that EU27 imports of iron ores and concentrates significantly exceed domestic production, underscoring the region's reliance on external sources.

Between 2013 and 2022, total imports declined from approximately 103 Mt/y to 82 Mt/y. Conversely, export volumes increased from 10.7 Mt/y to 13.1 Mt/y during the same period. Apparent consumption followed a downward trend after 2017, with a notable decline in 2020 driven by the global pandemic and additional, albeit smaller, impacts from the Russian aggression in Ukraine beginning in 2022.

Although there are no trade restrictions on iron ore imports from Russia, the volume imported from Russia sharply decreased from 10.5 Mt in 2021 to about 2.8 Mt in 2022. This decline reflects a combination of European consumers deliberately reducing reliance on Russian sources and the availability of alternative suppliers (Wood Mackenzie, 2024; Yuriy Grigorenko, 2024). Furthermore, reduced crude steel production via the BF-BOF (Blast Furnace-Basic Oxygen Furnace) route—a primary consumer of iron ores—lowered overall demand for iron ore, facilitating the decrease in Russian imports.

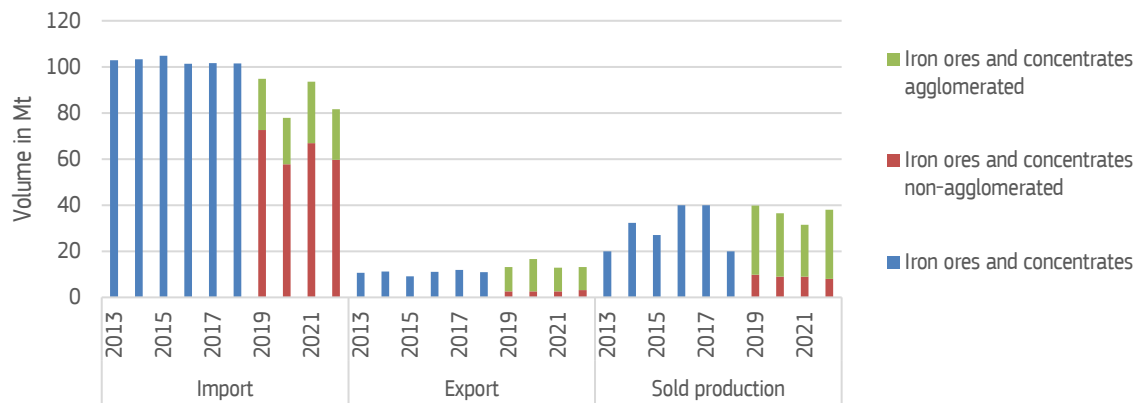
Figure 40. Overview of production, trade and apparent consumption of iron ores and concentrates



Source: PRODCOM (eurostat, 2024)

Figure 41. illustrates the trends in the EU's sold production, import, and export volumes of iron ores and concentrates, disaggregated into agglomerated and non-agglomerated categories from 2019 onwards. This level of detail was unavailable in the PRODCOM database prior to 2019, marking an improvement in data granularity. The data reveals a contrast between production and import patterns. While the majority (74%) of imported iron ores and concentrates are **non-agglomerated**, the composition of sold production differs significantly. Approximately 37% of the sold production is exported, and this exported share predominantly consists of **agglomerated iron ores and concentrates**. This distinction highlights differences in domestic processing capabilities and international market demands for these two product types.

Figure 41. Volumes of iron ore and concentrate imports, exports and sold produced in EU27



Source: PRODCOM (eurostat, 2024)

Figure 42. presents an analysis of the value of iron ores and concentrates, as reported by Eurostat (2024). The data reveals considerable fluctuations, reflecting the dependence of value on raw material prices and the specific characteristics of the ore, such as iron concentration and composition. Notably, agglomerated iron ores and concentrates exhibit a higher value-to-volume ratio compared to non-agglomerated ones. Despite a slight decrease in the volume of imports, the overall value of imports has increased, suggesting a rise in the price of imported ores. In contrast, the value of exports tends to follow the volume trends more closely, indicating that the price escalation for imported ores has been more pronounced than for exported ores. This divergence highlights market dynamics and the premium placed on specific ore qualities, particularly for agglomerated products.

Figure 42. Value to volume ratio (RI) for of iron ore and concentrates



Source: Comext (eurostat, 2024)

Directions of imports/exports to/from the EU27

In 2022, Europe, with 88.6 million tonnes of imported iron ore, was the world's second-largest importer after China, which imported 1086 million tonnes. The main suppliers for Europe are North America (Canada c.a. 30 %; CIS (Ukraine c.a. 18%) and South America (Brazil c.a. 24%). Around 5% of iron ore was imported from Russia; however, following the introduction of trade sanctions against Russia after the invasion of Ukraine, imports from Russia decreased by around 80% in 2023 compared to previous years).

Table 28. Global imports and exports of iron ore in 2022 (in Mt)

Exporting region Destination	European Union (27)	Other Europe	Russia, other CIS and Ukraine	North America	South America	Africa and Middle East	Asia	Oceania	Total imports	Of which extra-regional imports
European Union (27)	31.2	2.6	21.5	22.4	20.5	19.5	1.8	0.3	119.9	88.6
Other Europe	3.5	0.4	1.8	1.6	6.3	2.7	1.2	0.5	17.9	17.6
Russia, other CIS & Ukraine	0	0	0	-	-	0	0	-	0.0	0
North America	1.1	0.1	-	10.0	5.9	0	1.6	0.0	18.7	8.8
South America	0	0	-	0.4	6.9	-	0.1	0.0	7.4	0.5
Africa and Middle East	5.6	0.5	0.4	3.1	28.4	11.2	0.3	-	49.5	38.4
China	1.8	1.0	19.3	17.1	258	56.1	22.2	732	1108	1086
Japan	0.0	-	0.5	7.2	30.2	3.5	0.0	62.9	104	104
Other Asia	1.1	0.1	0.3	6.2	11.6	3.1	45.7	92.2	160	115
Oceania	0	0	-	-	0	1	0	0.0	1.0	1.0
Total exports	44.3	4.6	43.9	67.8	368	97.1	72.9	888	1587	1459
Of which extra-regional exports	13.1	4.3	43.8	57.9	361	85.9	5.1	888	1459	
<i>Net export (export - imp)</i>	<i>-75.5</i>	<i>-13.3</i>	<i>43.8</i>	<i>49.1</i>	<i>360.7</i>	<i>47.6</i>	<i>-1299</i>	<i>887</i>		

Source: World Steel Association (World Steel Association, 2023)

Iron-making

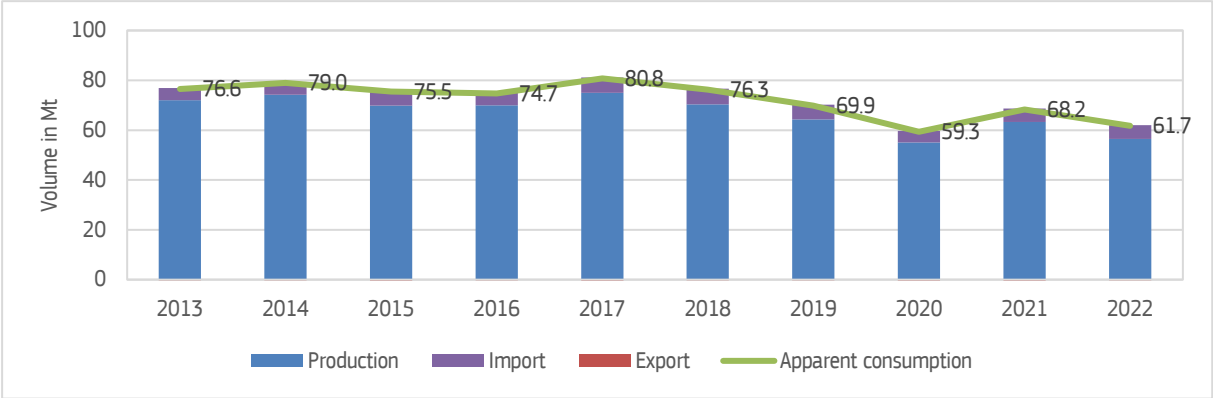
Figure 43. provides an aggregated visualization of the production, trade, and consumption of iron, as reported by Eurostat (2024) highlighting total volumes at this stage of the life cycle. For a more detailed breakdown, including production routes and specific types of iron produced, refer to Figure 44. and Figure 45.

Apparent consumption is calculated as production plus imports minus exports. Since 2017, apparent consumption of iron has generally been on a downward trend, reaching its lowest points in 2020 and 2022. These declines are closely tied to reduced steel production during these periods, which was significantly impacted by the global COVID-19 pandemic and the disruptions caused by the Russian invasion of Ukraine.

Production quantities of iron in the EU significantly surpass imports from third countries, with domestic production meeting 93% of the EU's demand for iron. On average, the EU27 imports approximately 5.5 Mt/y of

iron, a relatively small volume compared to the imports of iron ores and concentrates, which exceed 96 Mt/y annually. Furthermore, less than 1% of domestically produced iron is exported, emphasizing that nearly all iron produced within the EU is retained for further domestic processing. This approach reflects the EU's integrated value chain, wherein raw materials are transformed into higher-value products within its borders, supporting both economic and industrial resilience.

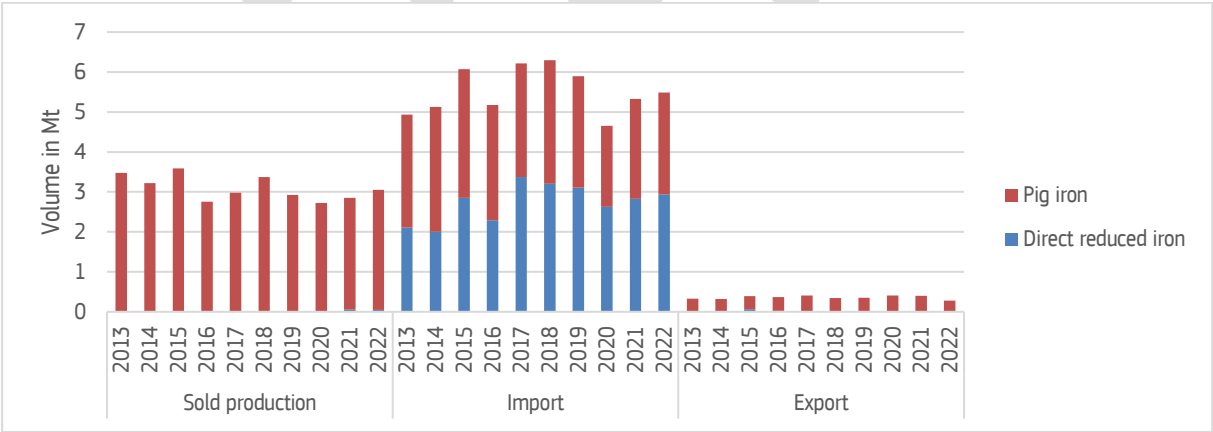
Figure 43. Overview of production, trade and apparent consumption of iron



Source: PRODCOM (eurostat, 2024)

Figure 44. highlights the production and trade flows of iron in the EU27, disaggregated into pig iron and direct reduced iron (DRI). The data underscores the predominance of pig iron in European iron production, with direct reduction playing only a minor role in the current industrial landscape. Nearly all domestically produced iron in the EU27 is pig iron, reflecting the reliance on traditional blast furnace methods. In contrast, imports from third countries show a different composition. Approximately half of the imported iron consists of DRI, indicating that Europe sources a notable portion of its DRI requirements externally. This reliance on imported DRI suggests the potential for further development of direct reduction technologies within the EU, particularly as they align with decarbonisation goals and the transition to greener steel production methods.

Figure 44. Trends in sold production, imports and exports of pig iron and DRI in the EU27



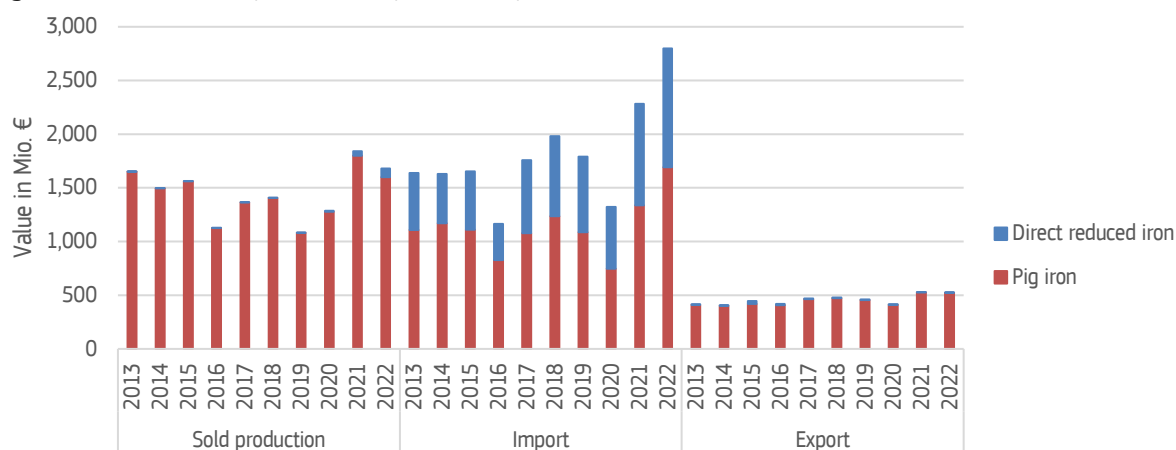
Source: PRODCOM (eurostat, 2024)

Figure 45. presents the values associated with the sold production and trade flows of iron, complementing the volume data discussed earlier. The values of sold production are consistent with the previously analysed production volumes, reflecting the stable role of domestic production in meeting EU demand.

While import quantities have fluctuated around a relatively constant level, the value of these imports has shown a clear upward trend, reaching EUR 2.8 billion in 2022. This increase may be attributed to a shift in the composition of imports, with a higher proportion of DRI—which typically commands higher prices—being imported instead of pig iron.

In contrast, the value of iron exports has remained low and steady over the years. Export quantities are relatively insignificant compared to the total supply and demand, underscoring the EU's focus on retaining domestically produced iron for further processing within the region.

Figure 45. Values of sold production, imports and exports of iron



Source: PRODCOM (eurostat, 2024)

Directions of pig iron and DRI imports/exports to/from the EU27

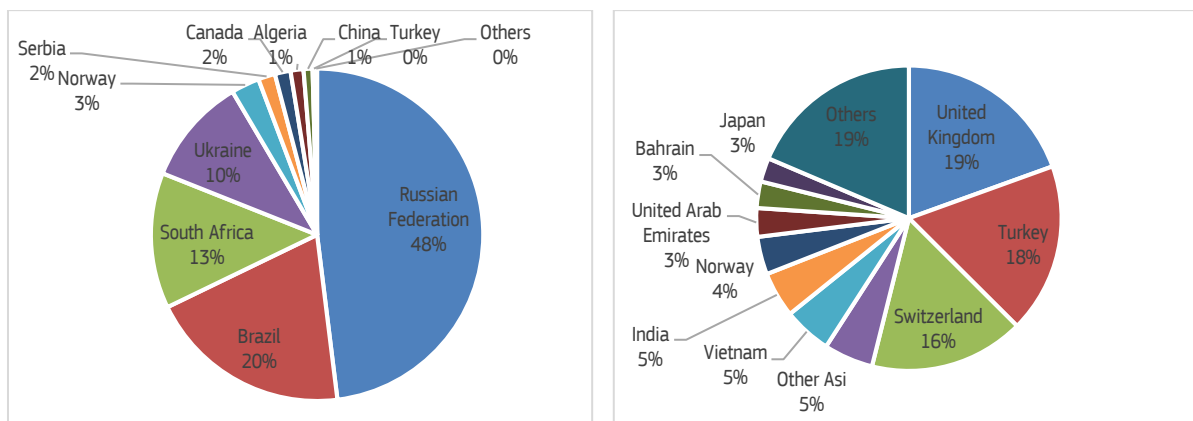
Figure 46. provides an overview of the import and export directions for pig iron, based on data from the UN Comtrade database (code: 7201). In 2022, the EU imported 2.44 Mt of pig iron, primarily from Russia, Brazil, and South Africa, while exporting 58 kt, with the main destinations being the United Kingdom, Turkey, and Switzerland.

The 12th Russia sanctions package introduced by the European Commission in December 2023 will ban the import of Russian-origin pig iron and DRI from 2026¹⁰⁵ (European Commission, 2023), which may impact future supply directions (GMK, 2023). Quotas are in place until 2026 to encourage a stepwise reduction. In the period covered in this report, there are no restrictions in place for this life cycle stage, as evidenced by the import numbers from Russia to the EU. Imports of DRI remained stable (1.6 Mt in 2021 and 1.5 Mt in 2022), but imports of pig iron increased significantly from 0.7 Mt in 2021 to 1.2 Mt in 2022. It is worth mentioning that imports of metallurgical coal (essential as a reduction agent in the blast furnace route) were banned at the beginning of the trade restrictions. Imports decreased by 55% between 2021 and 2022 and stopped entirely by 2023 (Wood Mackenzie, 2024; Yuriy Grigorenko, 2024).

In Ukraine, an estimated 90% of steel production capacity was not operational in 2022. About a third of this capacity is located in the area of Mariupol, which was at the centre of the conflict. Another significant factor was the difficulty in keeping the train transport system running, which is essential for the steel supply chain. As a result of the war, sudden increases in pig iron prices were observed between February and March 2022 (Fastmarkets, 2024; Mainline Metals Inc., 2022).

Figure 46. Directions of pig iron imports (left) and exports (right) to/from the EU27

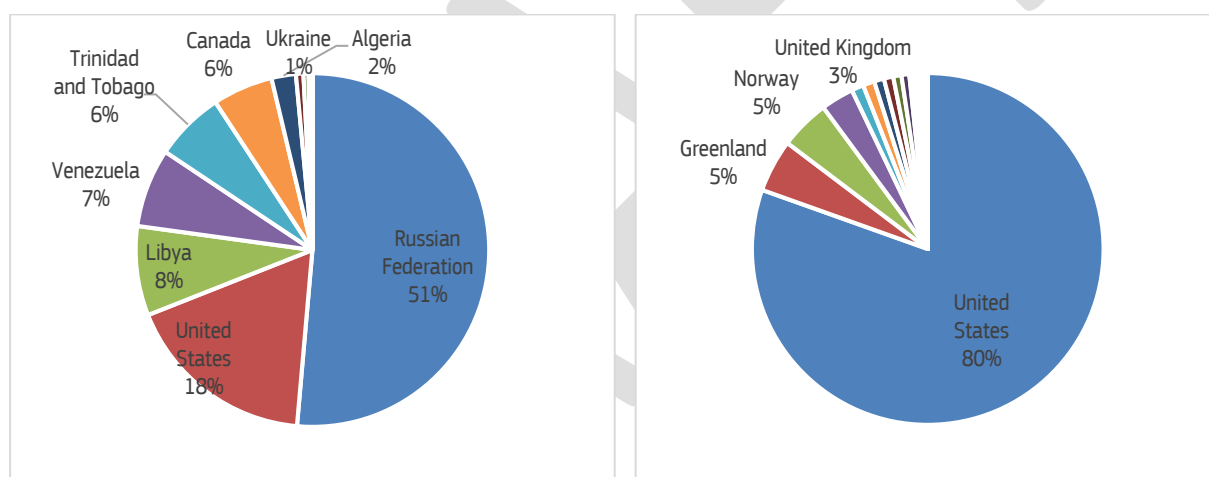
¹⁰⁵ EU restrict the imports of pig iron from Russia by end-2024 will be restricted to a quota of 1.14 million tonnes, and from 1 January 2025-31 December 2025 to a quota of 700,000 million tonnes



Source: UN COMTRADE (UN Comtrade, 2024)

Figure 47. illustrates the import and export directions for direct reduced iron (DRI), based on data from the UN Comtrade database (code: 7203). In 2022, the EU imported 2.98 Mt of DRI, with Russia being the largest supplier, contributing over 51% of the total imports. Other significant suppliers included the United States (18%) and Libya (8%). In contrast, the EU's exports of DRI were minimal, amounting to only 2.2 kt in 2022. The United States was the primary destination, accounting for 80% of these exports. These figures underline the EU's heavy reliance on external sources for DRI while maintaining a negligible role as an exporter in this trade flow. This dependency on Russian DRI imports may face significant challenges in the coming years, as the EU moves towards implementing sanctions on Russian-origin iron products starting in 2026.

Figure 47. Directions of direct reduced iron imports (left) and exports (right) to/from the EU27



Source: UN COMTRADE (UN Comtrade, 2024)

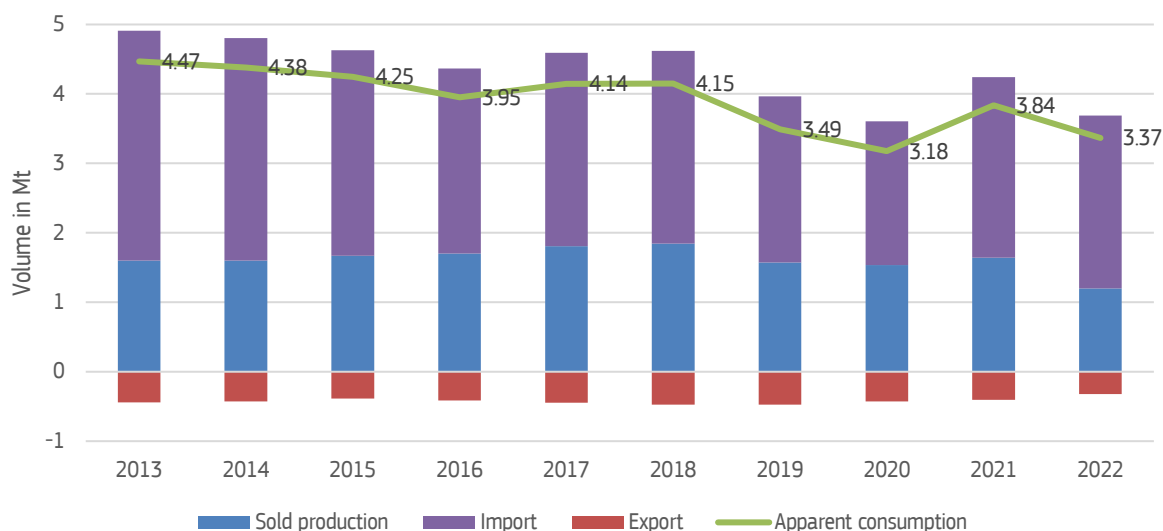
Ferroalloys

Figure 48. resents an overview of the production, trade, and consumption of ferroalloys based on Eurostat (2024). This aggregated visualization provides total volumes at this life cycle stage, while subsequent sections offer more detailed insights into specific types of ferroalloys.

Apparent consumption is calculated as production plus imports minus exports and has exhibited a decreasing trend over the observed period. Imports play a crucial role, meeting 69% of the EU27's demand for ferroalloys alongside domestic production. Despite this, total imports have steadily declined from 3.3 Mt/y in 2013 to 2.5 Mt/y in 2022.

Production of ferroalloys in the EU increased until 2018 but has since shown a declining trend. Exports, though relatively modest at below 0.5 Mt/y, represent about 26% of the EU27's sold production. This export share highlights the EU's limited but significant presence in the global ferroalloy market.

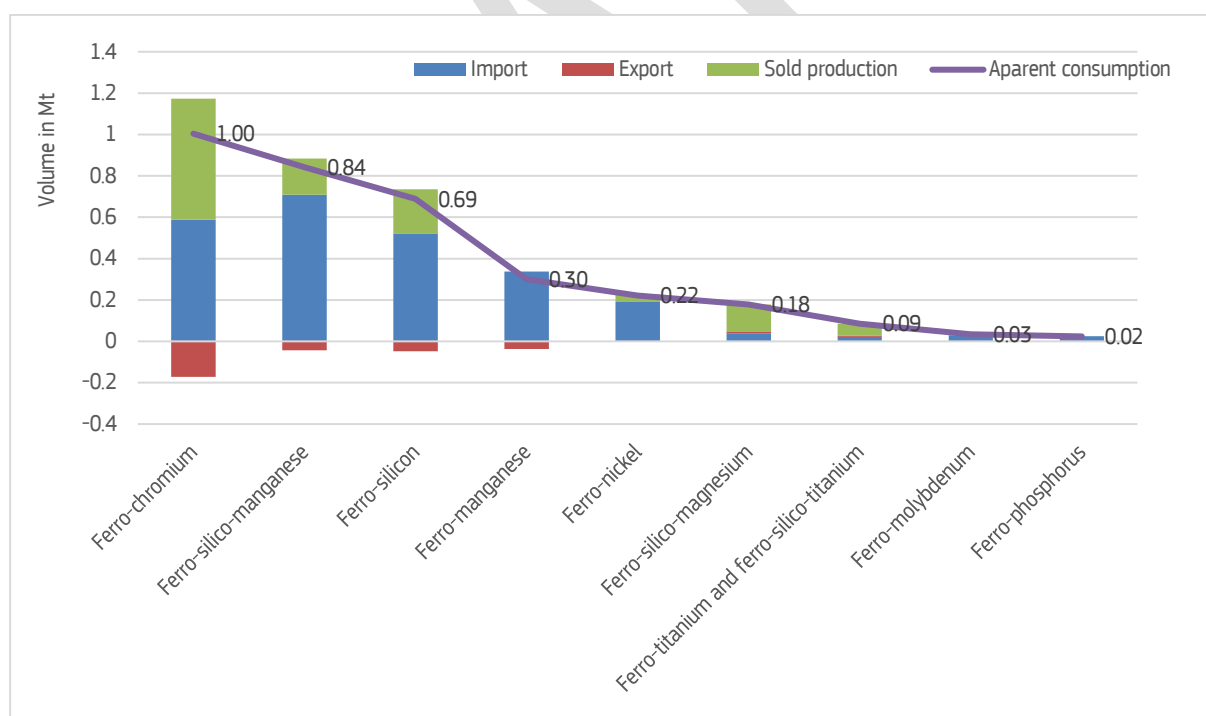
Figure 48. Overview of production, trade and apparent consumption of ferroalloys



Source: PRODCOM (eurostat, 2024)

Figure 49. shows the sold production, imports, and exports for each ferroalloy, based on Eurostat data. The reliance on imports varies for different types of ferroalloys in 2022. While ferro-titanium, ferro-silico-titanium, and ferro-silico-magnesium are predominantly domestically produced, for ferro-chromium—the most important ferroalloy in terms of volume—imports and domestic production are of a similar magnitude. For the remaining ferroalloys, imports are significantly more important than domestic production.

Figure 49. Ferroalloys imports, exports and sold production by ferroalloy in 2022



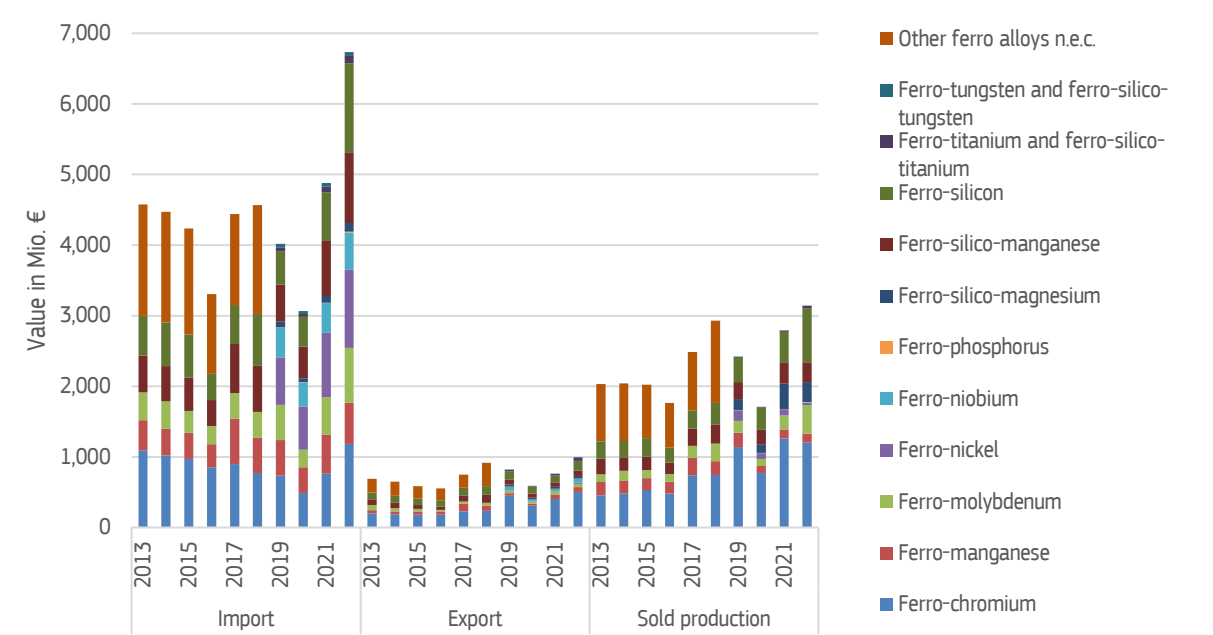
Source: PRODCOM (eurostat, 2024)

The general trends in ferroalloy production and trade volumes align closely with the monetary values of these material flows, as illustrated in Figure 50.. However, a divergent trend has emerged in recent years, particularly where values have risen despite declining volumes. This is largely attributed to rising prices, a trend that became especially pronounced in 2022. During this period, the monetary values of ferroalloy imports, and to a lesser extent domestic production, saw significant increases.

Different types of ferroalloys exhibit varying ratios of volume to value, reflecting their market dynamics and material properties. Ferro-niobium, ferro-molybdenum, and ferro-tungsten stand out for their disproportionately

high values compared to their relatively low production and trade volumes. These high-value ferroalloys are critical for specialized applications, further underscoring their importance in the EU’s material strategy despite their smaller physical quantities.

Figure 50. Ferroalloys imports, exports and sold production of the EU27 in values



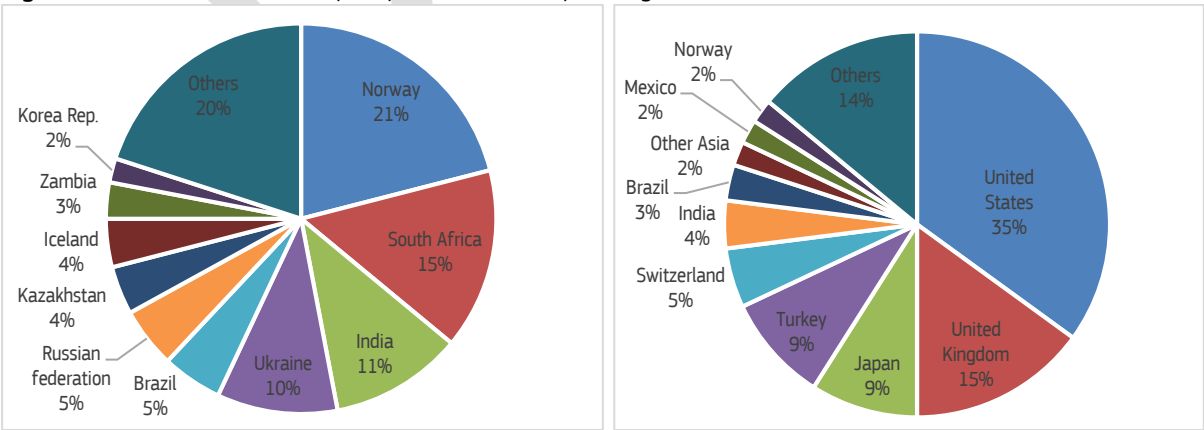
Source: Comext and PRODCOM (eurostat, 2024)

Directions of ferroalloys' imports/exports to/from the EU27

Figure 51 illustrates the import and export directions for ferroalloys, as reported in the UN Comtrade database (code: 7202). In 2022, total imports of ferroalloys into the EU exceeded 2,453 kt, with the primary trade partners being Norway, South Korea, South Africa, India, and Ukraine. During the same period, 322 kt of ferroalloys were exported, with key destinations including the United States, United Kingdom, Japan, and Turkey.

Ferroalloys were included in the EU’s 12th Russia sanctions package, although Russia has historically been a minor supplier of ferroalloys to the EU. Imports from Russia increased from 70 kt in 2021 to almost 130 kt in 2022 before declining again. These imports will now be subject to trade restrictions under the sanctions. Despite the small volumes involved, Ukraine views this shift as an opportunity to reestablish itself as a supplier, provided it can restore its ferroalloy production capacities in the future (Yuriy Grigorenko, 2024). The decrease in consumption can therefore be attributed to a reduction in demand from the steel production sector, similar to what was observed for iron-making.

Figure 51. Directions of ferroalloys imports (left) and exports (right) to/from the EU27



Source: UN COMTRADE (UN Comtrade, 2024)

Finished steel products

Table 29 provides a detailed overview of the European market for finished steel products (excluding stainless steel), distinguishing between flat and long products, over the past decade. The data highlights trends in European deliveries, imports from third countries, and total consumption.

Total consumption of finished steel products peaked at 155.2 Mt in 2018 before declining in subsequent years. The lowest consumption was observed in 2020 (128.5 Mt), reflecting the pandemic's impact, with modest recovery in 2021 and subsequent fluctuations.

European deliveries of flat products remained relatively consistent until 2019, with a decline starting in 2020. Imports of flat products peaked at 20.5 Mt in 2021 but decreased to 17.5 Mt in 2023. Long product deliveries experienced more variability, with a significant peak in 2018 (58.1 Mt), followed by declines in subsequent years. Imports of long products remained lower than flat products, fluctuating around 5–7 Mt annually.

Table 29. Market supply of the flat and long steel finished products [in Mt]

		2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
European deliveries	Flat	66.7	67.5	69.4	70.0	70.4	67.2	59.5	65.8	60.6	59.1
	Long	35.0	38.5	46.3	36.9	58.1	50.9	48.2	39.8	52.7	41.0
	Total	101.7	106	115.7	106.9	128.5	118.1	107.7	105.6	113.3	100.1
Imports third countries	Flat	12.8	16.5	17.8	17.9	19.0	17.1	14.6	20.5	18.6	17.5
	Long	4.5	5.4	6.0	6.0	7.7	6.1	6.2	6.9	5.5	5.3
	Total	17.3	21.9	23.8	23.9	26.7	23.2	20.8	27.4	24.1	22.8
Total consumption		119.0	127.9	139.5	130.8	155.2	141.3	128.5	133.0	137.4	122.8

Source: EU Steel Market Situation and Outlook International Challenges (EUROFER, 2024)

Table 30 provides a comprehensive overview of the market supply for stainless steel finished products, broken down into flat and long products, from 2014 to 2023. The data includes European deliveries, imports from third countries, and total consumption, illustrating trends and shifts in the market.

Total consumption fluctuated between 5.0 Mt and 6.4 Mt, with a peak in 2018 and a low point in 2020 during the COVID-19 pandemic. The decline to 5.0 Mt in 2023 represents the lowest level in the recorded period, suggesting reduced market activity or demand.

The markets for flat and long stainless steel products in the EU exhibit distinct characteristics. Flat products show greater variability, with European deliveries peaking at 3.9 Mt in 2016 and 2021, but declining to 3.4 Mt in 2020 and again in 2023. Imports of flat products are highly dynamic, reaching a peak of 1.6 Mt in 2022 before sharply decreasing to 0.7 Mt in 2023. In contrast, the market for long stainless steel products is notably more stable. European deliveries of long products have consistently ranged between 0.6–0.7 Mt, with only a slight decline to 0.5 Mt in 2023. Imports of long products have also remained steady, typically around 0.3–0.4 Mt throughout the observed period.

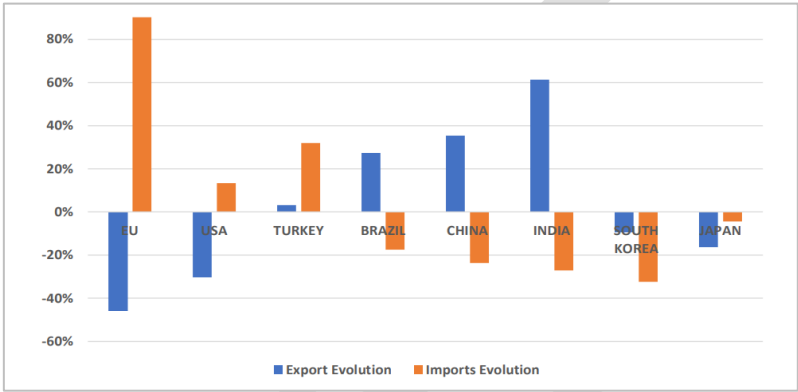
Table 30. Market supply of stainless steel long and flat finished products [in Mt]

		2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
European deliveries	Flat	3.5	3.6	3.9	3.8	3.8	3.6	3.4	3.9	3.6	3.4
	Long	0.6	0.6	0.6	0.6	0.7	0.7	0.6	0.7	0.6	0.5
	Total	4.1	4.2	4.5	4.4	4.5	4.3	4	4.6	4.2	3.9
Imports third countries	Flat	1.3	1.1	1.2	1.4	1.5	1.5	1.0	1.2	1.6	0.7
	Long	0.3	0.3	0.3	0.3	0.4	0.4	0.3	0.4	0.4	0.4
	Total	1.6	1.4	1.5	1.7	1.9	1.9	1.3	1.6	2	1.1
Total consumption		5.7	5.6	6.0	6.1	6.4	6.2	5.3	6.2	6.2	5.0

Figure 52. highlights the percentage change in imports and exports of finished steel products between 2012 and 2022, showcasing a significant transformation in Europe's trade dynamics. Over this period, Europe reduced its exports of finished steel products by over 40% while simultaneously increasing its imports by nearly 90%. This shift is further detailed in Figure 53., which illustrates how the European trade balance has undergone a complete reversal. Figure 54.

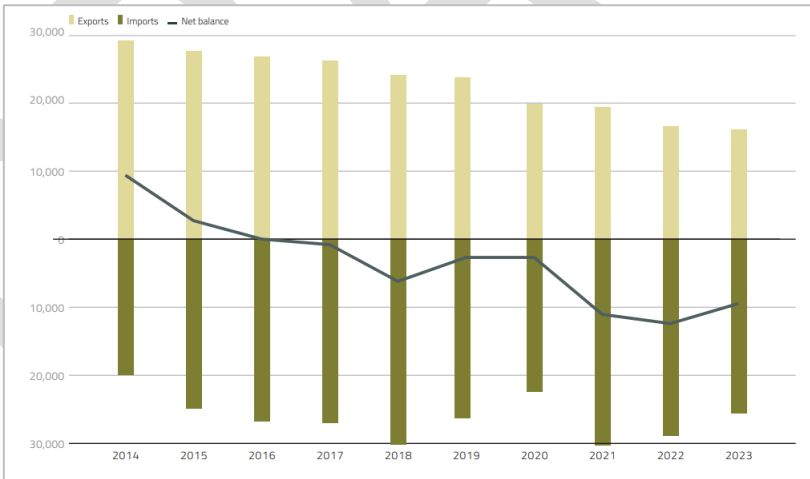
A decade ago, the EU was a net exporter of finished steel products, with annual exports of 30.6 Mt far exceeding imports of 18.9 Mt. By 2022, this relationship had inverted: imports rose to 31.2 Mt/y, while exports fell to 18.0 Mt/y(EUROFER, 2023). This change resulted in the EU losing approximately 30 million tonnes of sales in both domestic and export markets, marking a dramatic shift in its position within the global steel industry. This reversal, coupled with a structural loss in demand and a surge in imports, has significantly impacted the EU's steel sector. Declining capacity utilization has compounded these challenges, creating additional pressure on steel producers and accelerating the industry's decline (McKinsey & Company, 2021).

Figure 52. Percentage change in imports/exports by trade partner (finished steel, %) between 2012 and 2022.



Source: EU Steel Market Situation and Outlook (EUROFER, 2023)

Figure 53. Finished steel products' net trade balance by volume (all qualities in Mt)



Source: European Steel in Figures 2024 (EUROFER, 2024)

Table 31 illustrates the production, import, and export volumes of various types of finished steel products in 2022, highlighting their contributions to the EU’s apparent consumption (AP)¹⁰⁶. The numbers presented in the table results from own calculation that were made in the scope of the preparatory study considering multiple data sources: (Eurostat, 2024; EUROFER, 2023; CAEF, 2021), see also section 2.1.2 to learn more about the assumptions for the analysis.

¹⁰⁶ This measure represents the total steel products available for consumption in the domestic market, and is estimated using the formula: Apparent consumption =Domestic production + Imports – Export. It does not account for changes in inventory levels or unrecorded trade activities but provides a reliable estimate of demand within the region.

Flat steel products (except for stainless steel) account for 56.5% (85.8 Mt) of total consumption, with the most galvanized coil (13.0%, 19.7 Mt), and cold rolled coil (10.7%, 16.2Mt). Long steel carbon products account for 34% (52 Mt) of total consumption, with the most important being rod and wire rod (13.7%, 20.6 Mt).

Table 31. Market size of the finished steel products in 2022

Steel finished product (in kt)	Production	Imports	Exports	Apparent consumption
Flat products	74880	20810	9844	85844
Cold rolled coil	14005	3766	1603	16168
Hot rolled plate (e.g. Quarto plate)	10011	1774	1991	9793
Hot rolled coil and electrical sheet	29249	9008	3226	35030
Organically coated and other coated products	4419	1204	426	5197
Tinplate and tinned products and galvanized coil	17196	5058	2598	19656
Long products	50484	7007	5106	52385
Railway track material and other long products	2459	68	541	1986
Heavy sections and mining frames	7172	295	1559	5908
Other hot rolled bars (e.g. merchant bars) & light sections	10423	1861	891	11393
Bars for reinforcement of concrete	11518	2101	1115	12504
Rod and wire rod	18912	2682	1000	20594
Stainless steel products	4193	2064	1154	5103
Stainless steel long products	623	432	353	702
Stainless steel flat products	3570	1632	801	4401
Castings	8110	249	144	8215

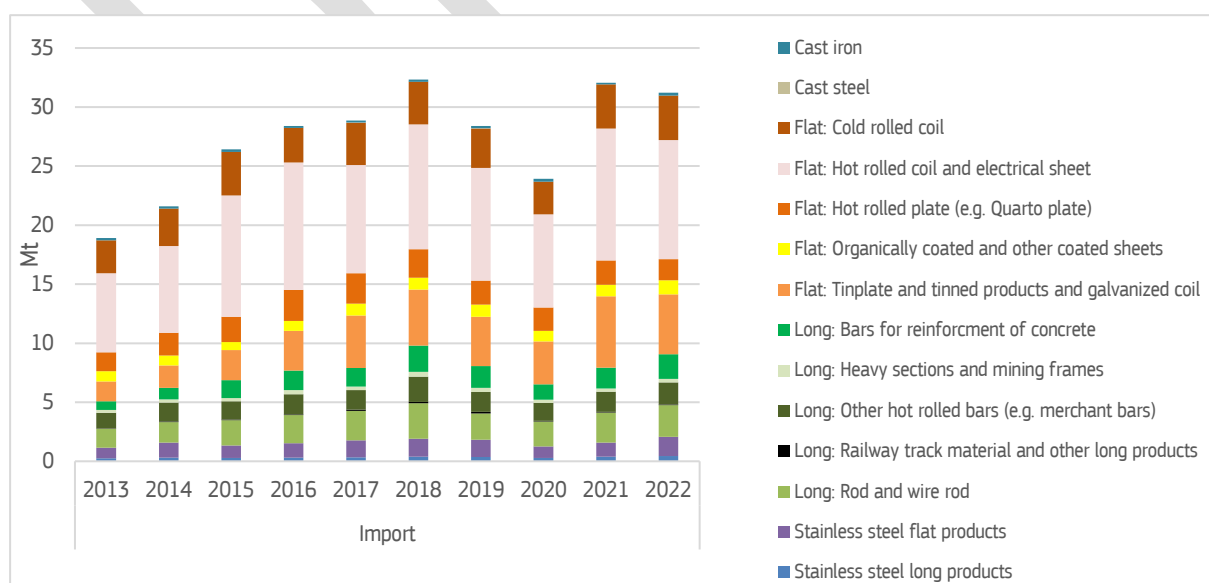
Source: Source: own calculations based on (eurostat, 2024; EUROFER, 2023; CAEF, 2021)

Focusing on the composition of **import flows** (Figure 53., Figure 54. **Figure 54.** Imports of finished steel products to the EU27), flat steel products (excluding stainless steel) emerge as the most traded goods, representing 72% of total imports, equivalent to 11.6 million tonnes (Mt) in 2022. Imports supply approximately 20% of domestic demand across all flat carbon steel categories. Additionally, long products made from non-alloy and alloy steel also contribute significantly to trade, with net imports reaching 7.0 Mt per year in 2022.

The pivotal role of flat steel products is further reflected in **export flows** (Figure 53. and

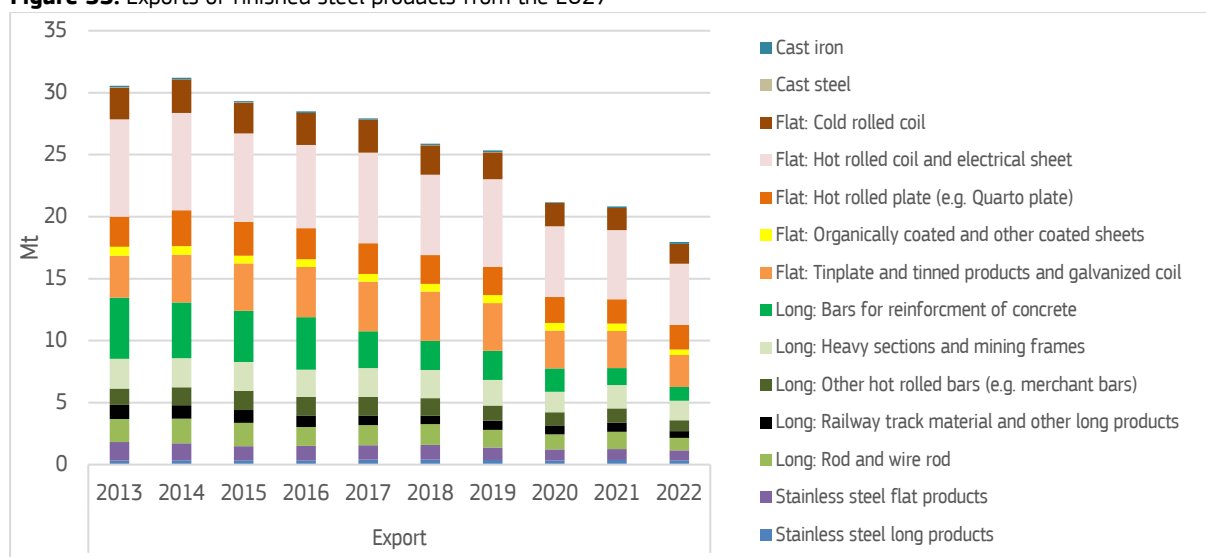
Figure 55.). These products account for around 60% of total steel exports, with hot rolled coils and electrical sheets standing out as the most significant export categories.

Figure 54. Imports of finished steel products to the EU27



Source: Source: own calculations based on (eurostat, 2024; EUROFER, 2023; CAEF, 2021)

Figure 55. Exports of finished steel products from the EU27



Source: Source: own calculations based on (eurostat, 2024; EUROFER, 2023; CAEF, 2021)

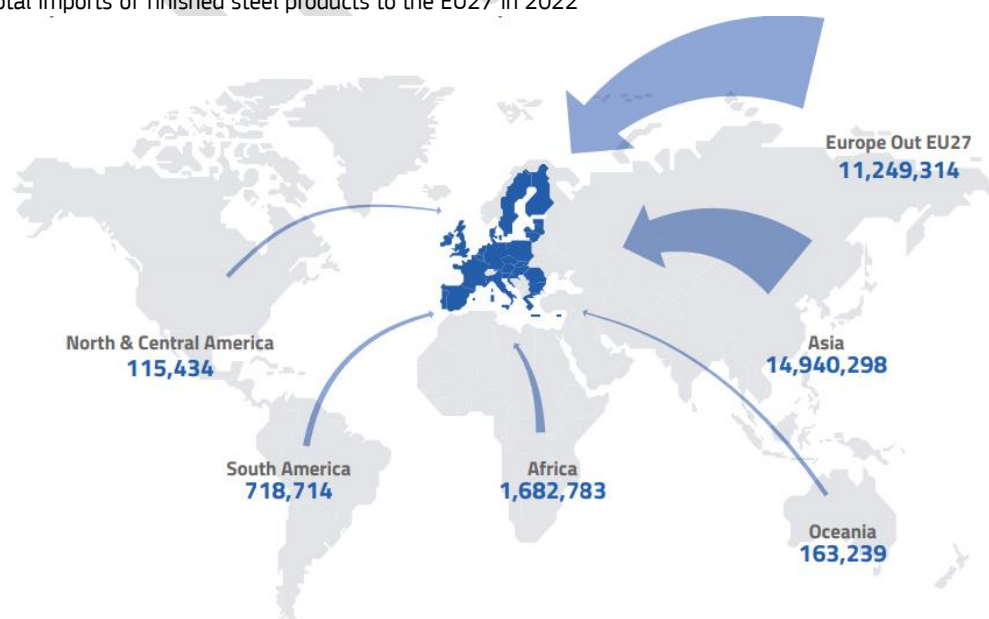
For stainless steel products, trade plays a highly significant role. In 2022, 2.1 million tonnes per year (Mt/y) were imported, while 1.2 Mt/y were exported. These figures are substantial when compared to the total EU27 stainless steel production of only 6.1 Mt/y. Imports meet 33% of the demand for flat stainless steel and 56% of the demand for long stainless steel in the EU27, highlighting a high reliance on imports for these products.

In contrast, the trade of castings is relatively insignificant. Only 4% of cast iron demand is satisfied through imports, and exports remain very low, at just 144 kt in 2022. This indicates that castings demand is predominantly met by domestic production, resulting in minimal import reliance within the EU27.

Directions of imports to the EU27

Figure 56. and Figure 57. illustrate the imports of finished steel products by third countries (trade partners). The EU imported 28.9 million tonnes of finished steel products in 2022, and 26 million tonnes in 2023. 63% of total EU27 imports originate from six major exporting countries. The highest import volumes during the observed decade were from Turkey (13%) and China (12%) of the total imports of finished steel products. Other significant countries of origin include Russia, South Korea, India, and Ukraine.

Figure 56. Total imports of finished steel products to the EU27 in 2022



Source: European Steel in Figures 2023 (EUROFER, 2023)

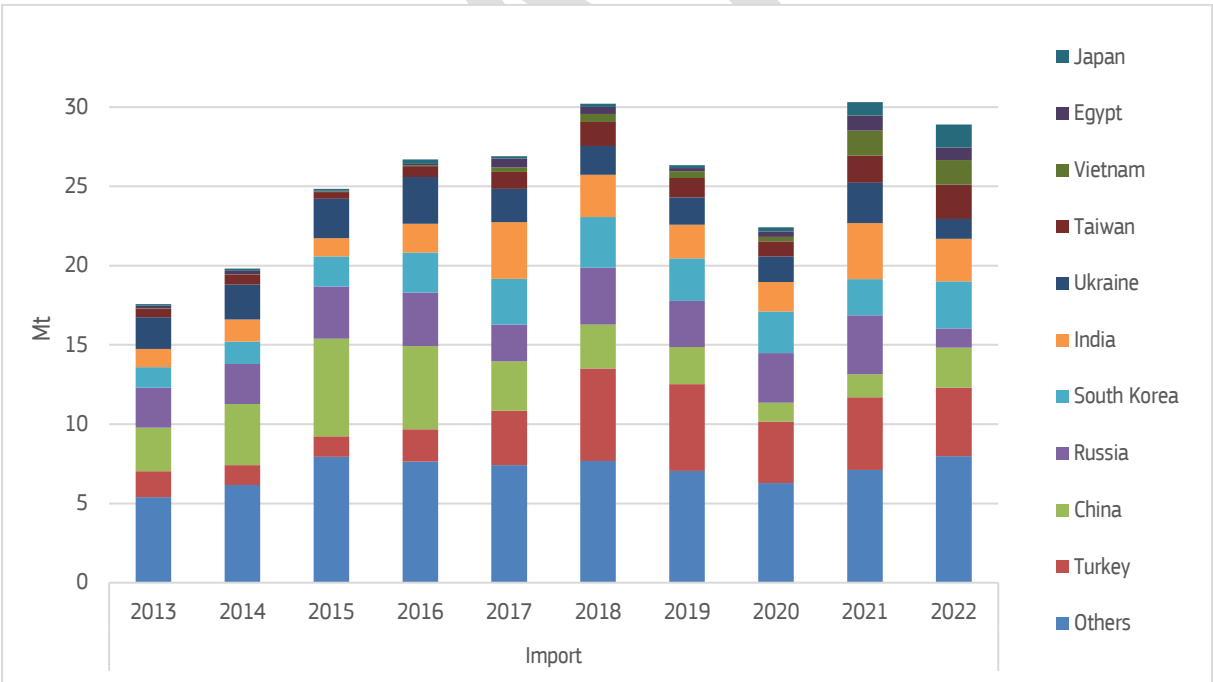
During the period observed, imports from Turkey increased significantly, growing by more than two and a half times. Imports from India and South Korea more than doubled. There was also a notable rise in imports from Vietnam, Taiwan, Japan, and Egypt, which evolved from minor suppliers in 2013 to important trade partners by 2022. In contrast, imports from Russia collapsed after 2021 due to economic sanctions and trade restrictions imposed by the EU27.

Finished and semi-finished products, particularly flat carbon steel, have been identified as important non-food, non-energy imports from Russia (Unguru et al. 2022a). Consequently, the EU's 4th package of restrictive measures, implemented in early April 2022, specifically targeted hot rolled finished steel products and steel pipes. As a result, imports of finished products decreased from 3.7 Mt in 2021 to 1.2 Mt in 2022. This reduction was particularly significant for flat products, which saw a 75% drop in volume, compared to a 50% reduction for long products (Yuriy Grigorenko 2024).

Looking ahead, the 8th package of restrictive measures, implemented at the end of September 2023, enforced a full restriction on trade codes 72 (iron and steel products) and 73 (articles of iron and steel, excluding pig iron, DRI, and ferroalloys). In addition to restricted imports from Russia, significant imports of finished products from Ukraine also declined, particularly flat products, which dropped from 2.5 Mt in 2021 to 1.5 Mt in 2022. This reduction was especially relevant for plates, as Ukraine was a major producer and supplier to the EU. Many of Ukraine's key production facilities were destroyed during the conflict. Russia, previously the second-largest supplier of plates to the EU, also faced restrictions.

While European companies shifted sourcing toward Asia (in 2023 the top trade partner for the finished steel goods was South Korea followed by India and Taiwan, following (EUROFER, 2024), the supply gap could not be filled quickly. The trade of finished steel products with the EU has been highly relevant for Ukraine, with exports to the EU accounting for about one-third to half of Ukraine's total steel exports, depending on the steel grade (Fastmarkets, 2024; Unguru et al., 2022).

Figure 57. Total imports of finished steel products to EU27 between 2013 and 2022.

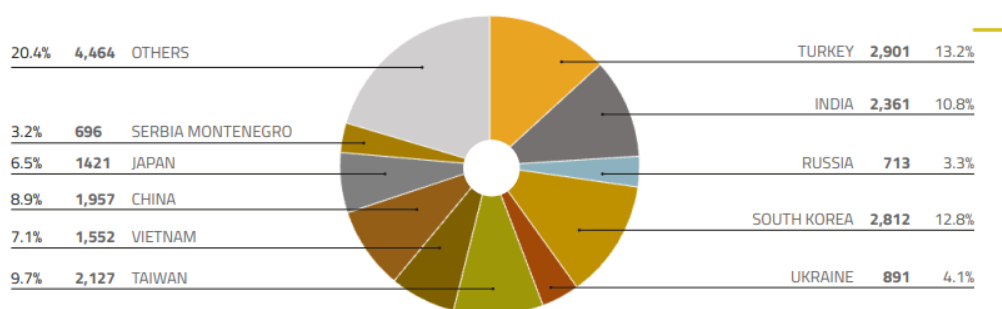


Source: European Steel in Figures 2023 (EUROFER, 2023)

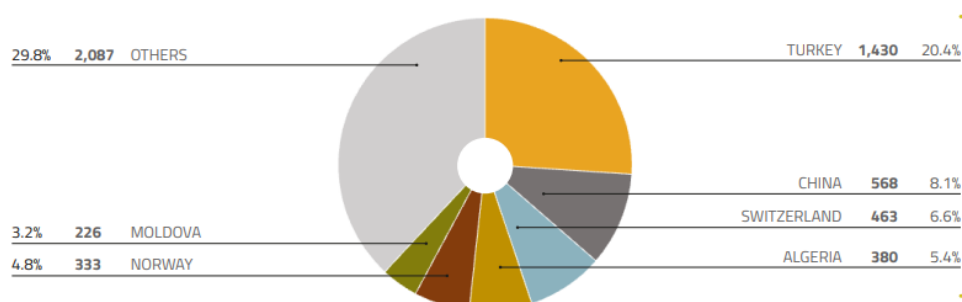
Figure 58 highlights the directions of imports in 2022, distinguishing between flat and long steel products. Turkey emerges as the most significant trade partner for both flat products, accounting for 13.2% of imports, and long products, with a share of 11.6%. However, Switzerland leads as the primary trade partner for long products, representing 20.4% of imports. India also plays a crucial role, holding a 10% share in the import of flat products.

Figure 58. Direction of imports for flat and long steel products in 2022

TOP 10 FLAT PRODUCTS IMPORTS



TOP 10 LONG PRODUCTS IMPORTS



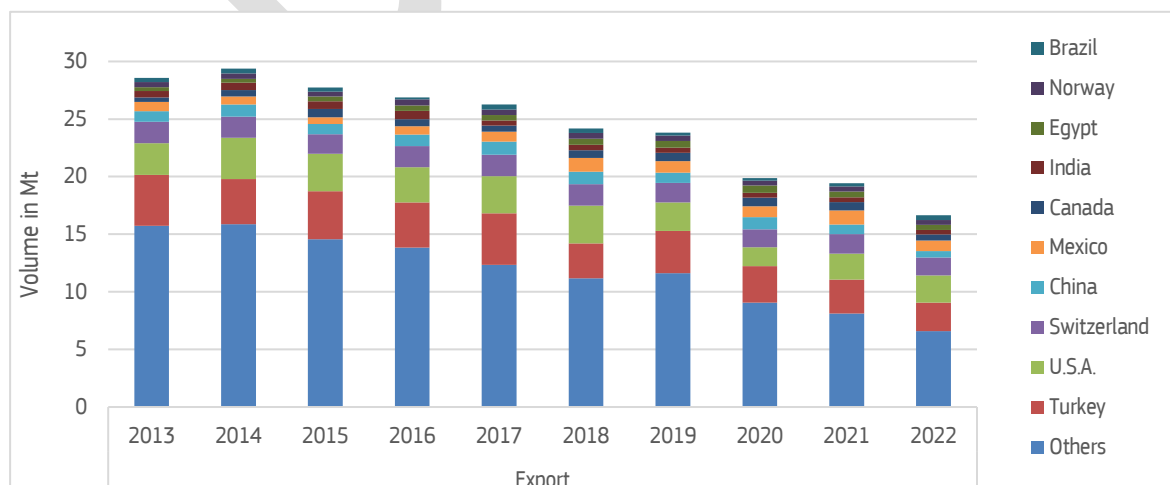
Source: European Steel in Figures 2023 (EUROFER, 2023)

Directions of exports from the EU27

Exports of finished steel products from the EU27 to third countries decreased consistently from 29 Mt/y in 2013 to 17 Mt/y in 2022. The exports of finished steel products from the EU27 are more diverse than the import flows, which are focused on countries with intensive steel industries. In total, 39% of exports go to a diverse group of third countries not specified by EUROFER.

The most important recipient countries are Turkey, the USA, and Switzerland, accounting for 15%, 14%, and 9% of total EU27 exports, respectively. While exports to the USA and Switzerland remained mostly constant, exports to Turkey decreased over the observed period (Figure 59). However, the most significant decrease in exports comes from lower trade flows to minor importing countries that are not further disaggregated.

Figure 59. Total exports of finished steel products from the EU27 between 2013 and 2022

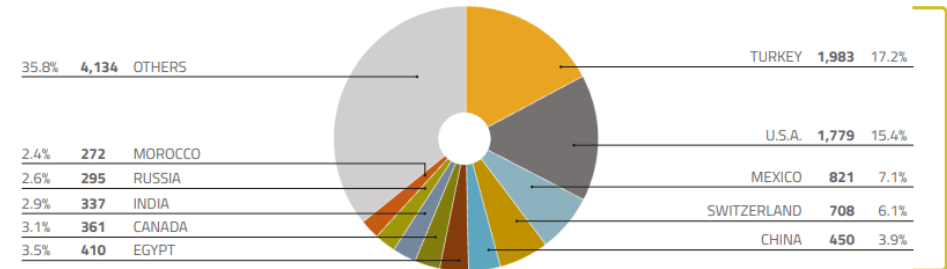


Source: European Steel in Figures 2023 (EUROFER, 2023)

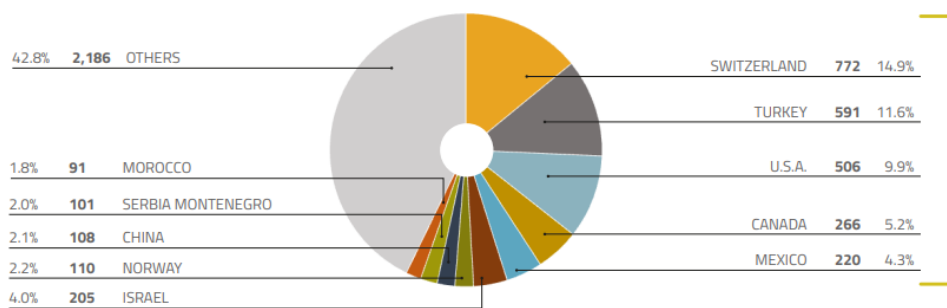
Figure 60. presents the direction of exports by product type, distinguishing between flat and long steel products. Turkey is the most significant trade partner for flat products, accounting for 17.2% of exports, and the second most important for long products, with a share of 11.6%. For long products, Switzerland takes the lead, representing 14.9% of exports. The United States and Mexico also play a key role as important trade partners for the export of both flat and long steel products.

Figure 60. Direction of exports for flat and long steel products in 2022

TOP FLAT PRODUCT EXPORT DESTINATIONS



TOP LONG PRODUCT EXPORT DESTINATIONS



Source: European Steel in Figures 2023 (EUROFER, 2023)

2.4 Market for scrap

2.4.1 Demand for steel scrap

Making steel from scrap rather than using primary raw materials—i.e. iron ore and coking coal—has well-documented economic and environmental benefits. Each recycled tonne of steel scrap saves approximately 1.6 tonnes of CO₂ equivalents (Broadbent, 2016; World Steel Association, 2021).

Steel scrap is utilized in both of the predominant steelmaking routes. In the BF-BOF route, scrap consumption is technically limited to around 25%-30%¹⁰⁷, as it is added to the converter primarily for cooling the melt. Here, scrap acts as a cooling agent, absorbing excess heat generated by the exothermic reactions between oxygen and elements like carbon (C), silicon (Si), manganese (Mn), and phosphorus (P). Scrap also serves as a source of iron units. In the EAF route, steel scrap typically serves as the sole feedstock material¹⁰⁸, allowing for up to 100% scrap utilization in the production of new steel products (Voraberger, 2022).

The use of scrap in steel production varies significantly across countries and depends on various factors, including the quality requirements of the final product, the availability of scrap supplies, and the existing EAF

¹⁰⁷ May vary depending on product and process, e.g. limited due to temperature requirements. Among the facilities that use up to 30% of scrap in the BOF route is the Dutch steel mill Tata Steel.

¹⁰⁸ While in Europe most EAF plants use 100% scrap, the majority of plans can also use DRI or pig iron as feedstock. Especially German and Italian mills experiences using both DRI and pig iron in addition to scrap.

infrastructure. Table 32 highlights the largest consumers of scrap globally in absolute terms, while Figure 61. presents the same data on a per capita basis.

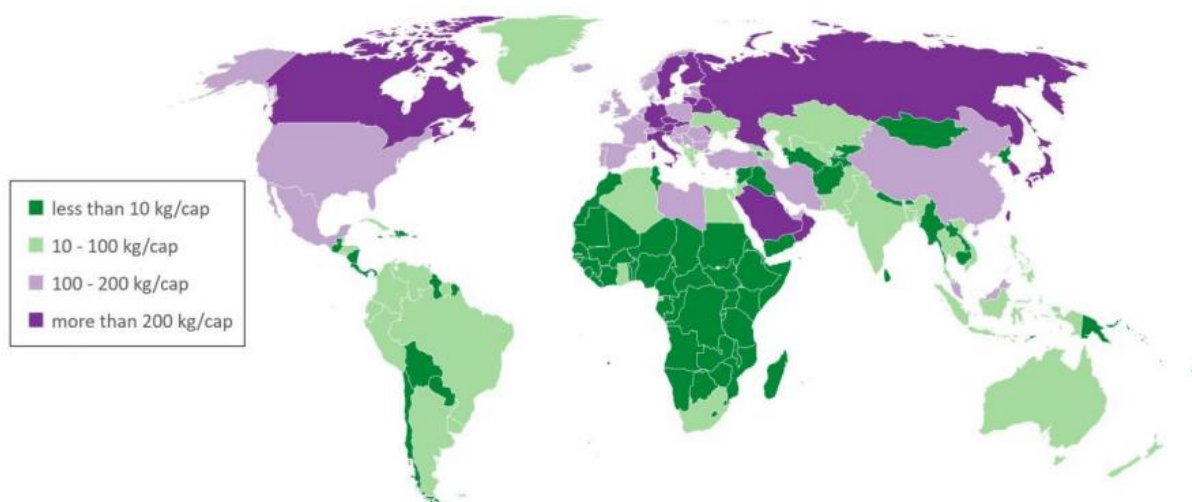
China dominates the global ranking, consuming a remarkable 225 million tonnes (Mt) of scrap in 2022, despite the fact that over 80% of steel production in China is BF-BOF based. Notably, the proportion of scrap used in Chinese steelmaking has increased from 14% in 2010 to 21% in 2022. **The European Union is the second-largest consumer of scrap**, followed by the United States. Turkey, however, leads in terms of scrap utilization rate, with up to 90% of steel production relying on scrap. At the same time, Turkey is the world's largest importer of steel scrap (Pothen and Hundt, 2024).

Table 32. Global steel scrap consumption by country in 2022 (in Mt)

	Scrap consumption (SC) in Mt	Steel production (SP) in Mt	Ratio SC/SP
China	225	1013	22%
European Union	78	137	57%
USA	55	81	68%
Turkey	30	35	86%
Japan	30	89	34%
Russia	28	72	39%
India	26	118	22%
South Korea	25	66	38%
Others	113	261	43%

Source: SteelMint

Figure 61. Estimated apparent domestic scrap supply per capita in 2022



Source: European post-consumer steel scrap in 2050: A review of estimates and modelling assumptions (Pothen and Hundt, 2024)

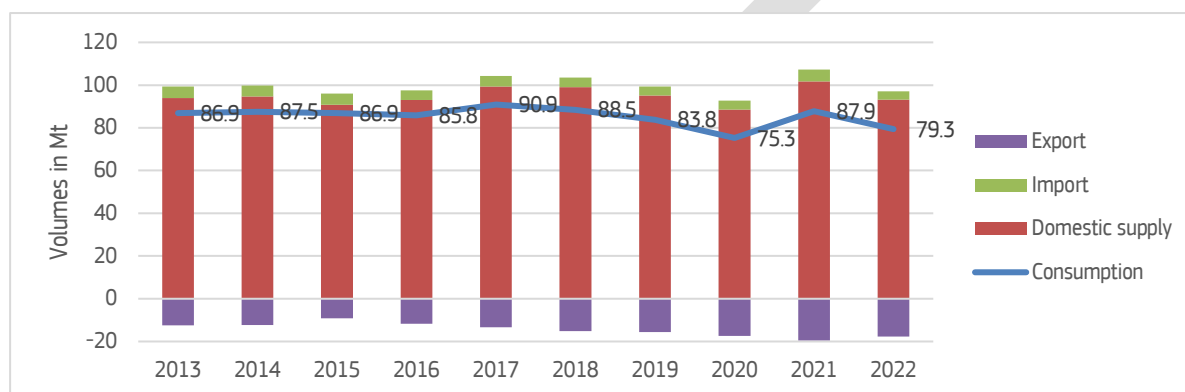
While steel is the world's most circular material, scrap is a finite resource, and future steel production will require both iron ore and scrap-based inputs. The demand for steel scrap is expected to rise significantly as more ambitious climate policies are adopted, a trend supported by studies forecasting increased scrap availability (Pothen and Hundt, 2024). According to the latest projections, global demand for steel scrap is anticipated to reach 778 million tonnes by 2030, a 15.1% increase compared to 2021 levels (GMK, 2023). This growth is primarily driven by expansion plans for new Electric Arc Furnaces (EAFs).

Currently, the global ratio of steelmaking in Basic Oxygen Furnaces (BOFs) to EAFs stands at 70/30, but this is expected to shift to 60/40 by 2030. To meet the additional production capacity enabled by EAF technology, more than 300 million tonnes of scrap per year will be required (GMK, 2023).

2.4.2 Steel scrap supply

Figure 62. shows the scrap supply in the EU, based on data from (EUROFER, 2023), with consideration of domestic supply¹⁰⁹, exports and imports, as well as apparent consumption. In 2022, the consumption of steel scrap in the EU reached 79.3 Mt, with domestic supply of 93 Mt¹¹⁰, 3.9 Mt of imports and 17.6 Mt of steel scrap exports. Today, Europe is a net exporter of scrap, but producers fear a supply gap in the coming years (Varriale, 2023). Opinions on the EU's scrap supply are divided. EURIC, a European association representing the recycling industry, does not foresee any risk of scrap shortages for European steel producers. In contrast, the Association of Rebar Producers and Exporters (IREPAS) estimates that the EU may become a net importer of scrap in less than 5 years due to the transition of European steel producers to EAFs (GMK, 2023).

Figure 62. Steel scrap (domestic) generation, imports and exports in the EU27



Source: European Steel in Figures 2023 (EUROFER, 2023)

Figure 63 illustrate the flow of scrap in the steelmaking process, showcasing the stages when distinct types of scrap are being generated, namely (SBTi, 2023):

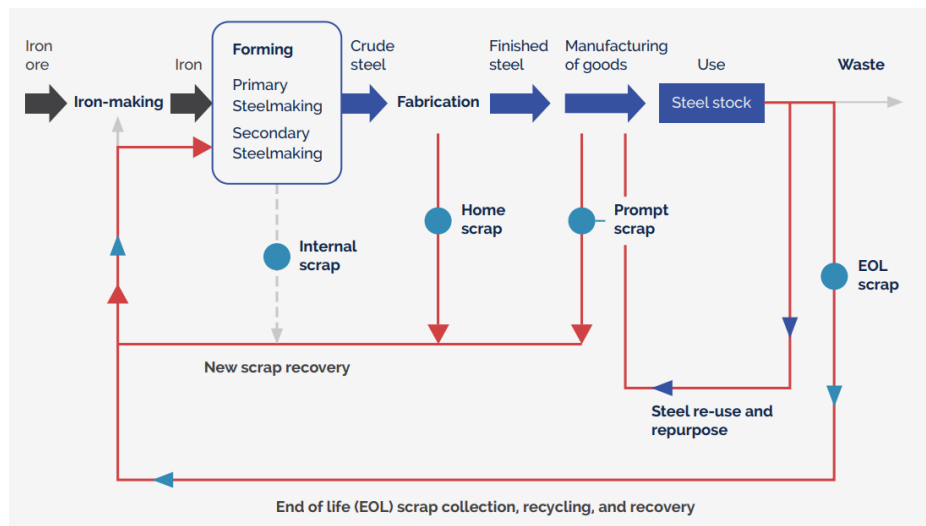
- **Internal scrap**, which is generated during manufacturing of crude steel, up to and including casting. Majority of this scrap is recycled immediately at the same facility it was created.
- **Home scrap**, which is generated during rolling and finishing of steel. This scrap is most often recycled immediately at the same facility it was created.
- **Prompt scrap**, also known as pre-consumer manufacturing scrap, is generated during the manufacturing of steel products by customers.
- **End-of-life scrap**, also known as post-consumer scrap, is generated at the end-of-life of a steel product.

The analysis of the scrap generation per type is critical for understanding the availability of its different qualities. **The composition of new scrap (incl. internal, home and prompt scrap) is well known** and in principle new scrap does not need any pre-treatment process before it is re-melted, although cutting to size might be necessary. Even new scrap with paint or coating (with the exception of cable which does need treatment prior to input into a furnace) does not generally need any waste-related pre-treatment before being sent to the furnaces, since many furnaces can melt such new scrap directly if required (Joint Research Centre, Muchová, and Eder, 2010). **The old scrap (end of life scrap)**, on the other hand it **is often contaminated to a certain degree**, depending highly on its origin and the collection systems used. Since the lifetime of many metal products can be longer than 10 years and sometimes longer than 50 years, for instance products for building and construction, there is an accumulation of metal in use since the beginning of the industry (Joint Research Centre, Muchová, and Eder, 2010, p.)

Figure 63. Flow of scrap in the steel sector

¹⁰⁹ Does not include scrap that has not been collected or separated for recycling, so it does not give any information on the production or generation of steel scrap. Furthermore, the use of domestic scrap neglects the stocking of steel scrap, for which no data is available.

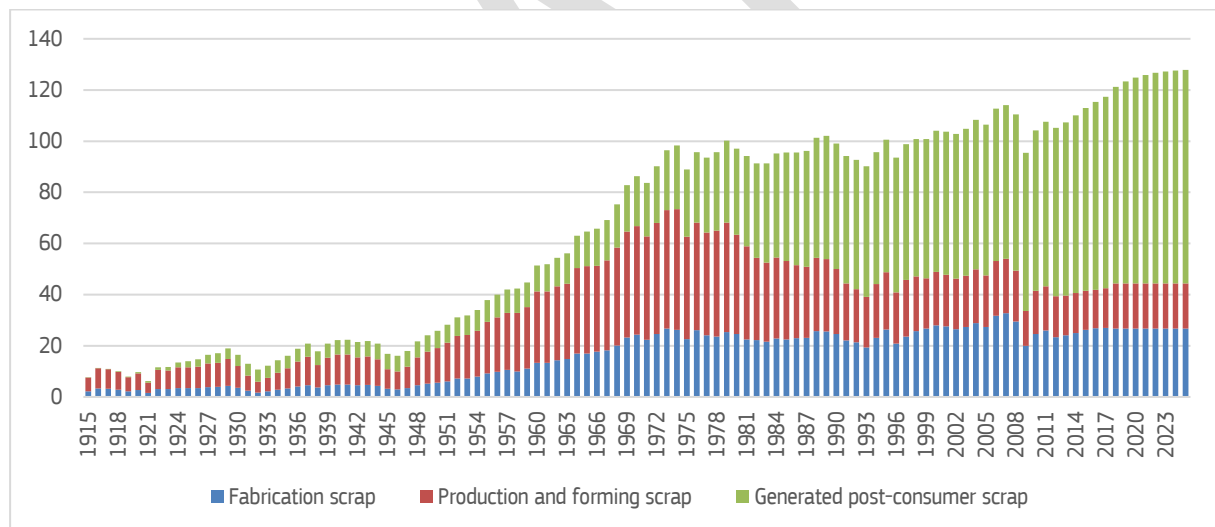
¹¹⁰ Production supply is based on the Eurostat data which is calculated based on the volume of production in EAF and BOF technology route and historical scrap data covering 2006-2015.



Source: ETC and RMI for MPP (SBTi, 2023)

Figure 64. summarises the historical trends of scrap generation, for new and old scrap in Europe (based on : Dworak, Rechberger, and Fellner, 2022). In the past, domestic scrap generation was dominated by new scrap. Today, the ratio of new to old scrap is 50-50 on average and fluctuated depending on economic developments. The study investigates the projection regarding the scrap generation up to 2050, showing continued grow up, in particular for the post-consumer scrap generation. The projection of scrap demand and supply are further elaborated in baseline scenario description in [Annex x](#).

Figure 64. MFA model results for the historic and future generation of steel scrap



Source: How will tramp elements affect future steel recycling in Europe? – A dynamic material flow model for steel in the EU-28 for the period 1910 to 2050 (Dworak, Rechberger, and Fellner, 2022)

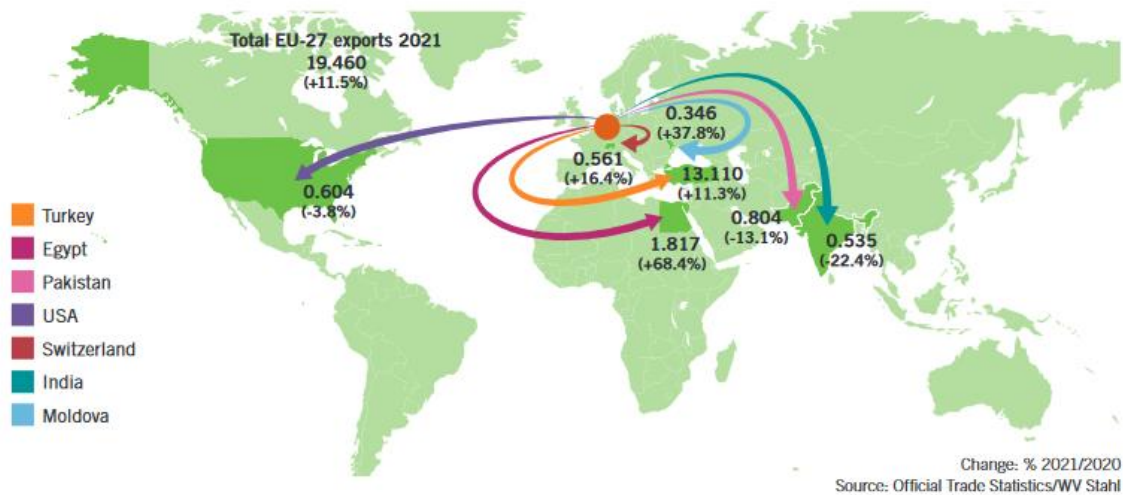
Directions of exports from the EU27

Ferrous scrap is currently the most exported type of waste from the EU, accounting for 59% of all EU waste exports (EUROFER, 2023). In 2021, Europe exported over 19.4 Mt of steel scrap to third countries, an increase of 11% compared to the previous year. According to BRI statistics (Bureau of International Recycling, 2022), the EU countries with the highest exported volumes are the Netherlands (4.6 Mt), Belgium (3.1 Mt), Romania (1.3 Mt), Germany (1.2 Mt), France (1.0 Mt), Sweden (0.8 Mt) and Bulgaria (0.6 Mt). The most rapid increase in scrap exports was noted in Bulgaria, with a 40% increase compared to the previous year.

As shown in Figure 65., over 67% of European scrap was exported to Turkey in 2021 (13.1 Mt), making it by far the most important trade partner for scrap export. Other partners include Egypt (1.8 Mt), Pakistan (0.8 Mt), the USA (0.6 Mt), Switzerland (0.56 Mt), India (0.5 Mt) and Moldova (0.34 Mt).

Scrap is increasingly recognised as a scarce strategic raw material essential for achieving carbon neutrality goals. Some European countries are taking measures to safeguard their domestic markets. For instance, Italy has introduced stricter regulations on scrap exports, declaring it a critical raw material and requiring exporters to submit declarations (GMK, 2023).

Figure 65. Steel scrap exports from the EU27 (in Mt)

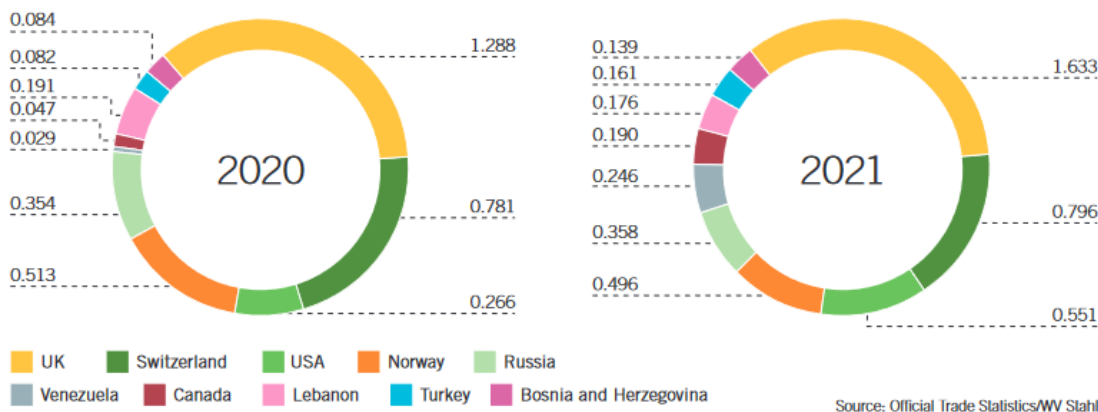


Source: World Steel Recycling in Figures 2017 – 2021 (Bureau of International Recycling, 2022)

Directions of imports to the EU27

The total volume of imports reached 5.3 Mt in 2021 and increased by 31% compared to the previous year. The UK and Switzerland are the main trade partners for the EU27, responsible for over 50% of supply (respectively 1.6 Mt and 0.8 Mt in 2021). The growing importance of scrap supply was noted for Venezuela and Canada, with an increase of 748% and 304% respectively between 2020 and 2021; see also Figure 66. (Bureau of International Recycling, 2022).

Figure 66. Ferrous scrap suppliers of the EU27 [in Mt].



Source: World Steel Recycling in Figures 2017 – 2021 (Bureau of International Recycling, 2022)

2.5 Market for “green” steel

When examining the steel industry, it is important to highlight the latest trends in the so-called “green steel” market, which are discussed in the following chapter. Since there is no harmonized definition of green steel, this introduction has a general character, and considered all existing definition of green steel, as previously described in section 1.4.4 (and in particular Table 14).

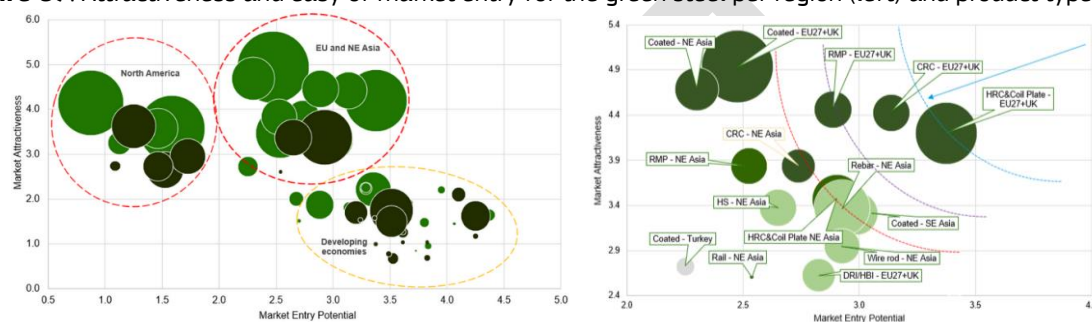
2.5.1 Demand for green steel

Steel accounts for a substantial share of total supply-chain emissions in industries that consume steel, ranging from 32% in the construction of buildings to 50% for domestic white goods (GMK, 2023). Consequently, the procurement of "green" steel presents an opportunity for steel consumer industries, such as automotive, construction and infrastructure sectors, industrial machinery or metal goods, to reduce their Scope 3 emissions.

The global green steel market, valued at \$200 million in 2022, is projected to experience significant growth, with an annual growth rate of 123% by 2030 (Market and Research, 2023). While Europe currently leads in green steel capacity adoption, the Asia Pacific region is expected to exhibit the most rapid growth in consumption until 2030. According to Research and Markets report (2023), Europe's demand for green steel reached 3.68 million tonnes in 2023, with 1.51 million tonnes supplied locally.

An analysis by CRU Consulting (CRU, 2023) assesses the demand for green steel in terms of market attractiveness and ease of market entry. The findings confirm Europe's favourable position in the green steel market today, as well as in the future (see Figure 67).

Figure 67. Attractiveness and easy of market entry for the green steel per region (left) and product type (right)

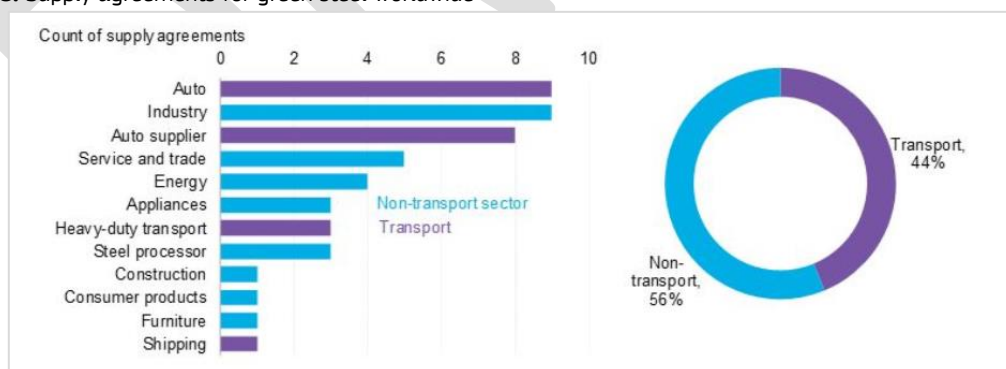


Note: Bubble size are proportional to green steel demand (forecast for the period 2030-2040)

Source: Enabling decisions in the green steel space that will help shape the future of the steel industry (CRU, 2023)

The demand for green products initially comes from sectors where materials represent a relatively small share of the total production cost and can be absorbed or passed on to consumers with little resistance as automotive or household appliance, where the impact would be marginal or negligible (IEA, 2023). According to BloombergNEF (2023), as of May 2023, a total of 48 green steel supply agreements had been announced, with 44% of these agreements signed with the transport sector, as illustrated in **Error! Reference source not found.**

Figure 68. Supply agreements for green steel worldwide



Note: Data as of May 2023

Source: Green steel demand is rising faster than production can ramp up (BloombergNEF, 2023)

The automotive industry stands as a key driver of green steel demand, with numerous car manufacturers pledging to become carbon neutral by 2040 and entering into binding agreements with suppliers to source low-emissions steel. For instance, in 2021, SSAB supplied the first batch of "green steel" produced using 'green' hydrogen to a vehicle manufacturer for truck production. Similarly, Mercedes Benz has signed an agreement with H2 Green Steel for 50,000 tons of steel made with hydrogen (Ricardo, 2024).

BCG (2022) estimates that the demand for green steel could range between 20 to 40 million tons by 2030. Moreover, a recent study elaborated by Ricardo (2024) for Transport & Environment indicates that the projected

demand for green steel from the automotive sector is expected to reach approximately 21 million tons in 2030, accounting for 11% of the total lower-carbon steel production capacity.

According to the BCG (2023), demand could potentially exceed domestic supply by 15 to 20 million tons (Mt) in the next decade. On the other hand, following the CRU analysis (CRU Consulting, 2024), thanks to new investments in the green steel, supply and demand for green steel are projected to reach equilibrium by mid-2030.

2.5.2 Supply for green steel

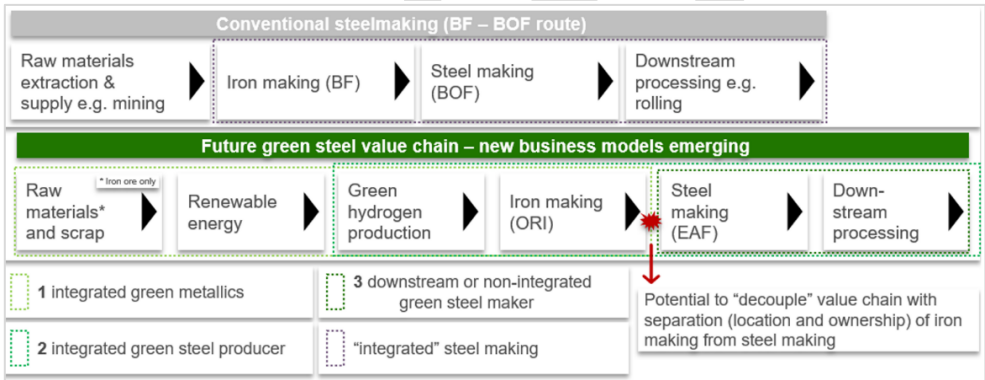
In line with the European transition toward carbon neutrality, there have been several industrial announcements aiming for lower-carbon steel production totalling 107 Mt of production capacity by 2030 across 33 different plants in Europe (Ricardo, 2024).

A significant proportion of future green steel production capacity is expected to rely on the Direct Reduced Iron–Electric Arc Furnace (DRI-EAF) route, which is emerging as the leading decarbonisation technology for the steel industry. This route typically involves a two-phase transition: starting with **natural gas-based NG-DRI-EAF**, which serves as an initial stepping stone, and gradually transitioning to **hydrogen-based H2-DRI-EAF** as green hydrogen becomes more available and cost-competitive.

The first commercial-scale, fully electrolytic hydrogen-based DRI plants are anticipated to become operational in the EU in 2026, marking a significant milestone in the industry’s decarbonisation efforts. Notable examples include the H2 Green Steel project in Sweden, which is expected to produce 2.5 million tonnes (Mt) annually, and the Hybrit project, led by SSAB in Sweden, with a planned capacity of 1.2 Mt annually. (IEA, 2023).

Figure 69 illustrates the range of innovative business models being developed to support the decarbonisation of the steelmaking value chain. These models encompass partnerships across industries, investments in renewable energy integration, and scaling of hydrogen production infrastructure, all of which are essential to achieving cost-effective and sustainable green steel production.

Figure 69 New business models in the industry value chain towards decarbonisation



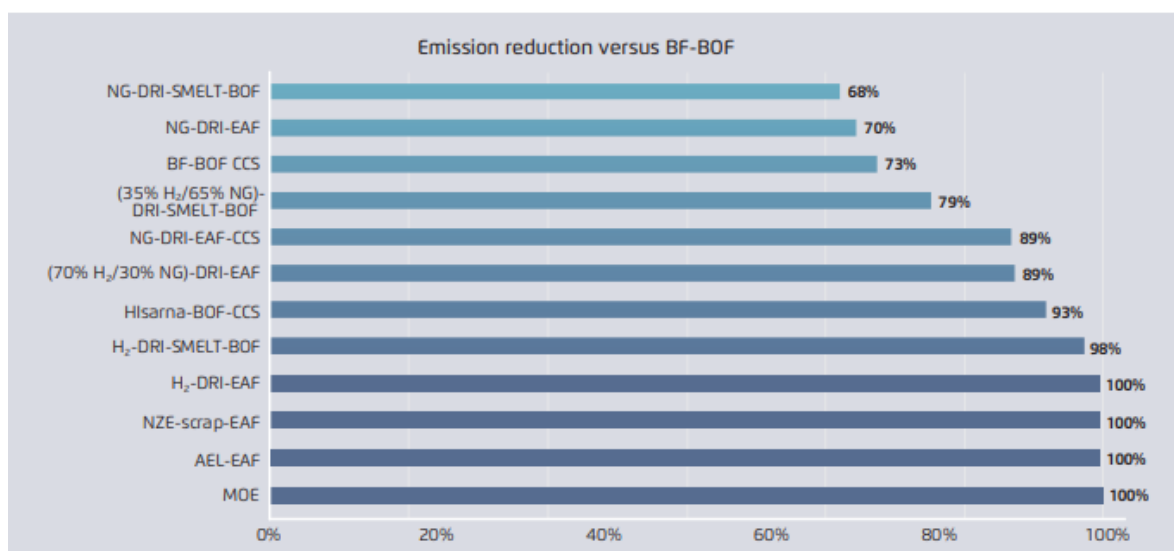
Source: Enabling decisions in the green steel space that will help shape the future of the steel industry (CRU, 2023)

Boston Consulting Group (BCG, 2022) conducted an analysis of the CO₂ emission reduction potential of various steel production technologies, including natural gas-based direct reduced iron with electric arc furnace (NG-DRI-EAF), 100% green hydrogen-based direct reduced iron with electric arc furnace (H₂-DRI-EAF), and carbon capture, utilization, and storage (CCUS). The analysis used the traditional BF–BOF route as the benchmark.

For instance, CCUS technology offers a viable pathway for producing low-carbon steel in existing BF–BOF plants by capturing and storing carbon emissions, achieving up to 73% reduction compared to the integrated route. However, CCUS remains a relatively immature technology, facing significant challenges such as high implementation costs, energy intensity, scalability, and the need for robust infrastructure to transport and store captured carbon.

In contrast, the NG-DRI-EAF route can achieve direct emissions reductions of up to 70%, making it a transitional solution for decarbonising steel production. The 100% green hydrogen-based direct reduction iron (H₂-DRI-EAF) route demonstrates the highest potential, with emissions abatement potential of up to 100%, highlighting its critical role in achieving near-zero carbon steel production. Figure 70 illustrates the emission abatement potential across several break-through technology routes for steelmaking.

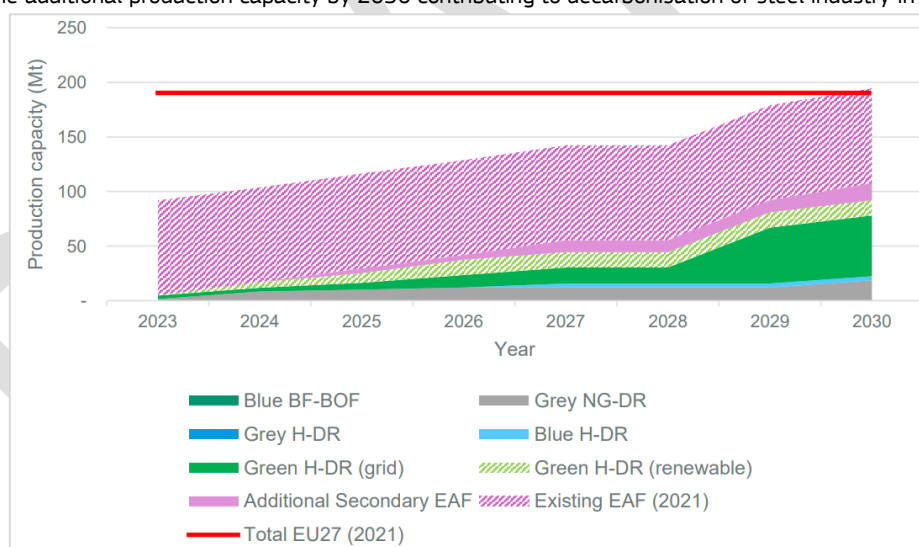
Figure 70. CO₂ emission abatement potential of steelmaking routes compared to BF-BOF



Source: Low-carbon technologies for the global steel transformation. A guide to the most effective ways to cut emissions in steelmaking

When combining the existing and announced DRI and EAF capacities (see also Figure 37), the total “green steel” production capacity in Europe is projected to reach 172 million tons by 2030 (Ricardo, 2024). Taking into account other investments, including natural gas-based DRI or conventional BF-BOF with carbon capture and storage, among others, the total capacity is expected to reach 194 million tons by 2030, as depicted in Figure 71.

Figure 71. The additional production capacity by 2030 contributing to decarbonisation of steel industry in EU27



Note. Blue BF-BOF = conventional BF-BOF + carbon capture and storage (CCS); Grey NG-DR = natural gas-based DRI; Grey H-DR = hydrogen-based DRI, hydrogen produced with gas; Blue H-DR = hydrogen-based DRI, hydrogen produced with gas + CCS; Green H-DR (grid) = green hydrogen-based DRI using electricity from the grid; Green H-DR (renewable) = green hydrogen-based DRI using renewable electricity
Source: The use of Green Steel in the Automotive Industry (Ricardo, 2024)

2.6 Economics and employment in the EU iron and steel industry

2.6.1 Structural changes over the years

The iron and steel sector was one of the main economic drivers when, in 1952, six countries (Belgium, Germany, France, Italy, Luxembourg, and the Netherlands) signed the Treaty of Paris, establishing the European Coal and Steel Community (ECSC). The treaty aimed to promote economic expansion, employment, and improved living standards through the creation of a common market for coal and steel (Treaty of Paris, 1952).

The first global steel crisis struck in the 1970s, following the recession triggered by the global oil shock. This period saw a dramatic collapse in steel demand coinciding with the commissioning of new production capacity. Despite these challenges, EU steel production remained relatively stable between the late 1970s and mid-2000s, with some countries, such as Spain and Italy, experiencing gradual growth. However, the EU steel sector faced two major crises that reshaped the industry.

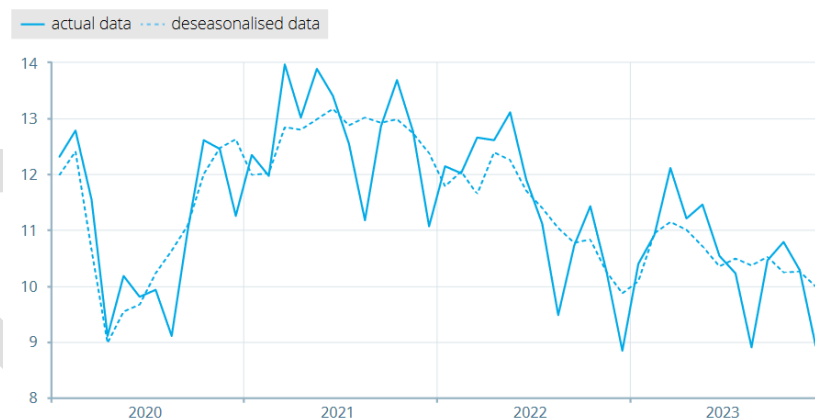
The first was the 2008–2009 financial crisis, which led to a severe drop in steel production (Figure 72). Although production rebounded in the years following the crisis, it entered a long-term decline in the EU. This period marked a turning point for the global steel industry, as it coincided with a persistent issue of global excess production capacity. The second major crisis occurred with the outbreak of the COVID-19 pandemic, which caused steel production to plummet to levels comparable to those seen during the financial crisis (Figure 73) (OECD, 2020).

Figure 72. Evolution of crude steel production 2007-2010 (million tonnes) in the EU27



Source: World Steel in Figures (World Steel Association, 2011)

Figure 73. Evolution of crude steel production 2020-2023 (million tonnes) in the EU27



Source: World Steel in Figures (World Steel Association, 2024)

Structural changes within the EU's iron and steel industry highlight a significant shift from being a robust net exporter to increasingly relying on steel imports (see: Table 31). Several key factors have driven this transition, including:

- **Rising energy costs:** Energy prices in the EU have significantly outpaced those in non-EU markets, creating substantial financial pressure for EU-based steel producers.
- **Market uncertainties:** Price fluctuations and supply shocks for raw materials have exacerbated challenges. Notable price increases occurred during key periods: (i) in 2003–2004 due to insufficient mining capacity, (ii) in 2011 during a commodity boom, and (iii) in 2016 due to market corrections and modest recovery in steel demand (OECD, 2020).
- **Stringent environmental regulations:** The ambitious EU climate agenda and strict environmental policies—often more stringent than those outside the EU—have placed additional burdens on EU steelmakers.

By late 2024, the European steel industry raised concerns about its critical situation (Figure 73), describing it as "the worst crisis since the financial and economic crisis of 2009." Industry stakeholders identified key challenges: global steel overcapacity, unfair trade practices, and high energy prices. These pressures have led to an EU capacity utilization rate of just 60%. The Global Forum on Steel Excess Capacity (GFSEC), an international platform¹¹¹ addressing steel sector challenges, reported that global excess steel capacity reached 551 million tonnes in 2023—four times the EU's annual production (EUROFER, 2024). This figure is projected to increase to 630 million tonnes by 2026 (EUROFER, 2024).

According to GFSEC, non-market policies and practices of certain countries (e.g., additional investment by state-owned companies) are contributing to increasing the global excess capacity issue, distorting markets. In turn, this tilts the playing field towards inefficient producers. Global excess capacity disrupts the global steel market and displaces efficient and more sustainable steel production (OECD, 2024).

In terms of ownership, the European steel industry has undergone significant changes over the past century. According to the OECD, the share of privately owned steelworks in Europe increased from 46% to 92% between 1986 and 1995 (OECD, 2018). By 1996, countries like Italy and Poland, which were largely dominated by state-owned steelworks, had nearly entirely privatised their steel industries. Since the late 1990s, international mergers and takeovers have become widespread in Europe.

Despite a decrease in production, the **European steel industry remains a global leader in manufacturing highly specialised product parts and implementing cutting-edge production technologies** (ESTEP, 2020). European companies seek their strategic competitive advantage through Research & Development and Innovation (R&D&I). The challenges of sustainability and CO₂ reduction present a wide range of opportunities for companies willing to invest in innovation (European Commission, 2021). Additionally, successive waves of consolidation have strengthened the industry's resilience and ensured a competitive environment, benefiting downstream industries reliant on steel.

2.6.2 Employment

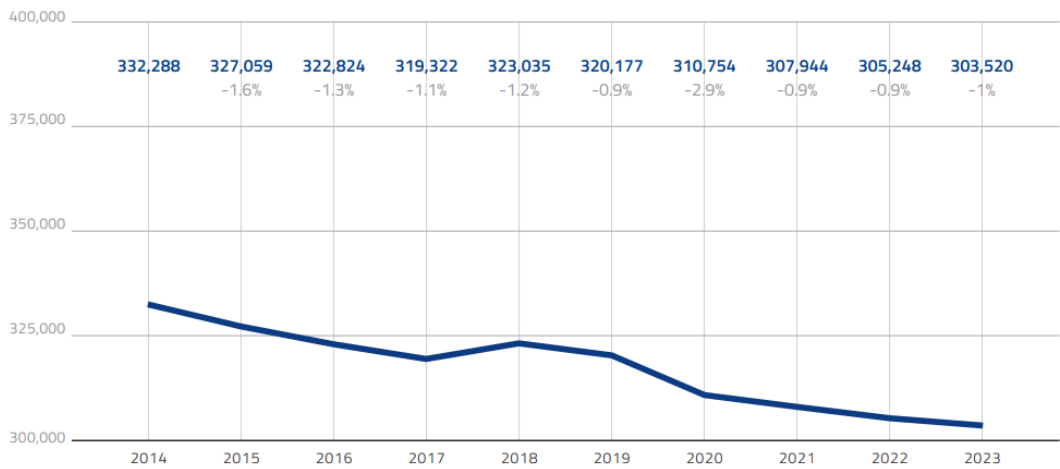
In 2020, the European Commission published a comprehensive blueprint report addressing the employment and competitiveness of the European steel industry. This initiative was part of the broader EU strategy to meet the evolving skill requirements of the steel sector (European Commission, 2020). The report provides a detailed analysis of the European steel industry's historical development and current challenges, focusing on its strategic importance for industrial value chains and employment. It also includes national profiles of the steel industries in seven Member States: Germany, Italy, Spain, France, Poland, the Netherlands, and Finland, which together represent the largest steel-making markets in the EU. This section summarises the trends in employment and workforce development. The analysis is complemented by the most recent data on employment figures in the EU steel industry.

The European steel industry remains a cornerstone of economic progress, industrial growth, and job creation across Europe. However, the industry has undergone significant transformations over the past decades. Beginning in the mid-1980s, restructuring efforts—coupled with increasing levels of overproduction and intensified competition from imports originating in emerging markets—triggered a sharp decline in employment and widespread workforce reductions. Total direct employment in the European steel industry plummeted from 996,000 positions in 1974 to just 278,000 by 2000 (European Commission, 2020).

While the sector experienced a slight recovery in employment around 2010, the overarching trends in European steel-making have been marked by continued declines in both production and workforce numbers (Figure 74.).

¹¹¹ The EU is a member of the GFSEC, including EU Member States (Austria, Belgium, Finland, France, Germany, Greece, Hungary, Italy, Luxembourg, Netherlands, Poland, Slovak Republic, Spain, Sweden, Switzerland) and other countries (Argentina, Australia, Brazil, Canada, Japan, Korea, Mexico, Norway, South Africa, Türkiye, United Kingdom, United States).

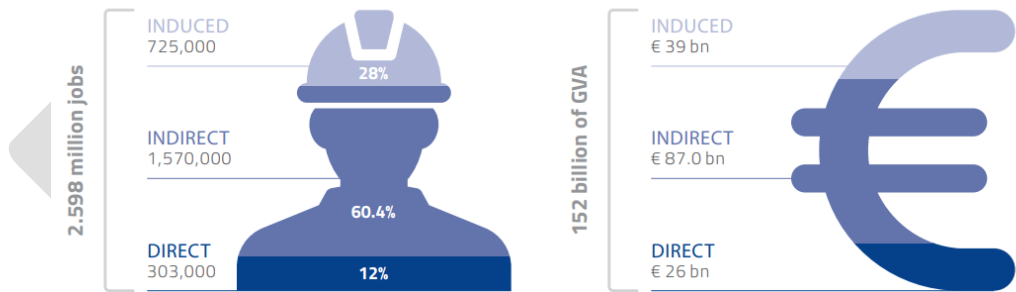
Figure 74. Employment in the European steel industry between 2015 and 2023



Source: European Steel in Figures 2024 (EUROFER, 2024)

In 2023, the EU steel industry directly supported approximately 303,500 jobs, with its total employment impact extending to up to 2.6 million jobs when accounting for indirect and induced employment across related sectors; this demonstrates the significant multiplier effect of the industry, highlighting its critical role in sustaining broader economic activity (see Figure 75.). The gross value added (GVA) of the sector in 2023 is estimated at EUR 152 billion, reflecting an increase from EUR 143 billion in 2022 (EUROFER, 2023).

Figure 75. Employment & GVA in the EU steel industry



Source: European Steel in Figures 2024 (EUROFER, 2024)

Across the EU27, Germany stands out as the largest employer within the European steel industry. In 2022, over 79,500 people were employed in direct jobs within the German steel sector, making it the leading contributor to employment in this industry. Italy ranked second with 30,864 steel workers, followed by France (25,317), Romania (22,121), and Poland (21,500). **Figure 76.** illustrates the distribution of employees across the leading steel companies in the EU27.

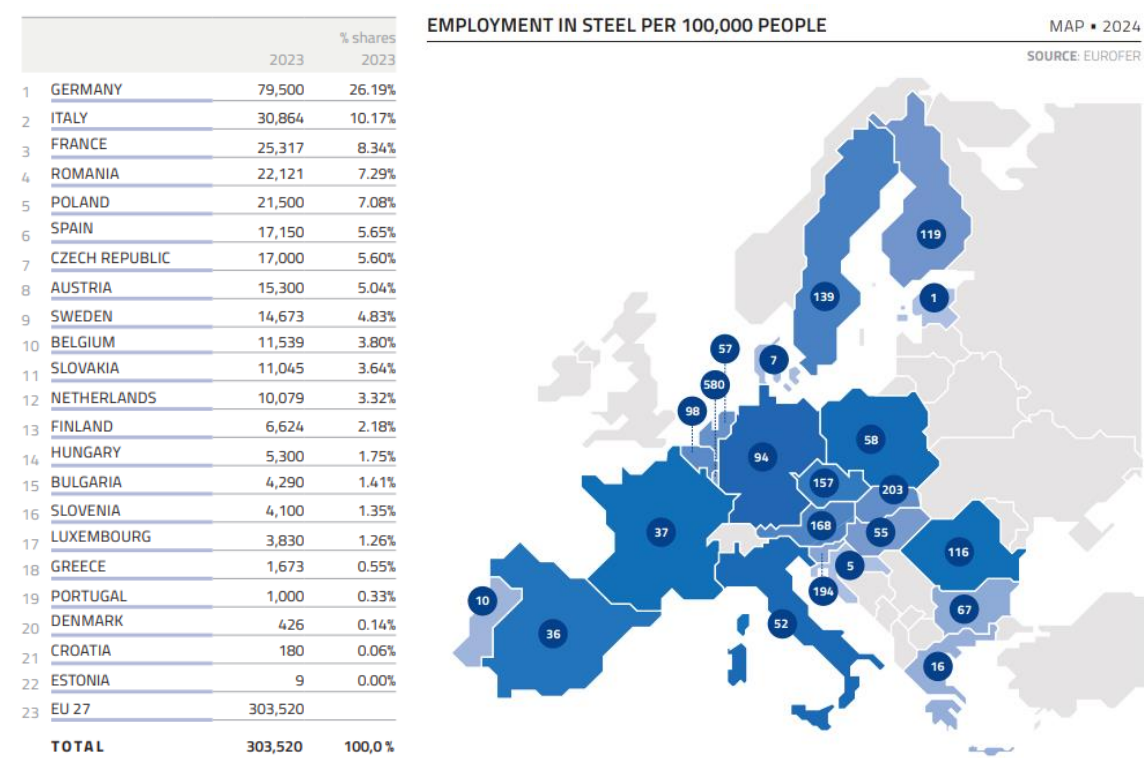
Employment data for the European steel industry is closely monitored by Eurostat, which provides insights into the employment structure within iron and steel manufacturing. According to Eurostat, the total number of employees engaged in basic iron, steel, and ferroalloys production in 2022 was 312,185, with most of these positions concentrated in large enterprises employing 250 or more staff (Table 33).

However, the steel sector also includes a significant number of small and medium-sized enterprises (SMEs)¹¹², which play a vital role in the industry's ecosystem. SMEs, defined according to European standards, make up

¹¹² Small and medium-sized enterprises (SMEs) are defined in the [EU recommendation 2003/361](#). The main factors determining whether an enterprise is an SME are: staff headcount (<250 staff headcount) and either turnover (≤ € 50 m) or balance sheet (≤ € 43 m).

approximately 2,500 enterprises within the European steel sector. Despite their numerical dominance—78% of these companies employ fewer than 10 personnel—SMEs contribute about 6% of all direct jobs in the sector.

Figure 76. Employment in steel per 100,000 people in the European Union in 2023



Source: European Steel in Figures 2024 (EUROFER, 2024)

Table 33. Employment structure in basic iron and steel and ferroalloys production in the EU27

	Number of enterprises	Number of Persons employed	Net turnover in million euro
Total	2,751	312,185	231,297.57
From 0 to 9 persons employed	2,133 (78%)	4,504	774.26
From 10 to 19 persons employed	251	3,399	976.36
From 20 to 49 persons employed	120	3,574	2,078.95
From 50 to 249 persons employed	120	13,304	12,573.06
250 persons employed or more	127	287,402 (92%)	214,894.93 (93%)

Source: EUROSTAT

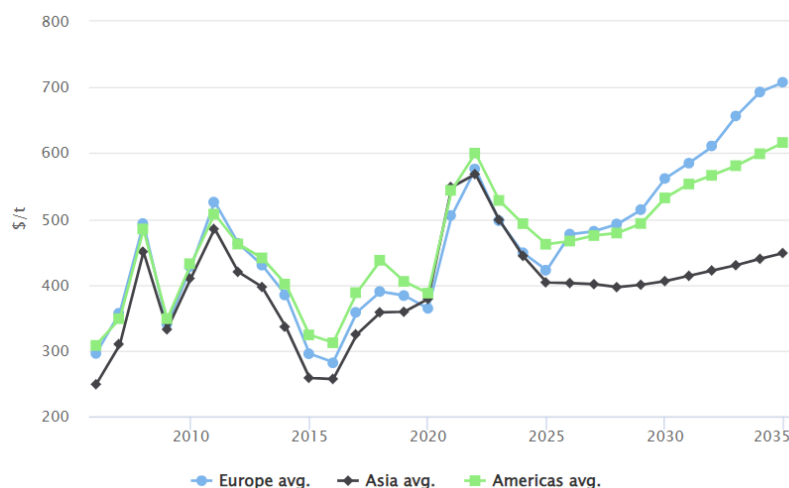
2.6.3 Production costs

Figure 77. illustrates the average site costs of crude steel production across Europe, Asia, and America, presenting historical trends from 2008 and projections extending to 2035 (CRU, 2024). These cost fluctuations are predominantly driven by variations in the price of key inputs, particularly coking coal, which remains a critical raw material in the steel-making process.

The COVID-19 pandemic in 2020 caused severe disruptions to the global steel industry, leading to an estimated 5% decline in global crude steel output (European Commission, 2020). During the early waves of the pandemic, lockdowns led to depleted steel inventories, disrupted supply chains, and idled production capacities. As demand rebounded in the wake of global stimulus packages, supply struggled to keep pace, causing sharp increases in steel prices and exacerbating volatility in raw material costs (McKinsey, 2023). These disruptions highlighted the steel industry’s vulnerability to sudden shocks and underscored the need for more resilient supply chain strategies.

The main reason for a spectacular difference between the costs in Europe and Asia beyond 2025 is related to carbon pricing, and in particular phase out of free allocation under EU-ETS in Europe. The analysis of site costs by different technology routes is addressed in detail later in the text, providing insights into how regional policies and production methods affect competitiveness. Despite regional differences, the cost trends are broadly aligned across these markets.

Figure 77. Crude steel production costs by region



Source: CRU Cost Analysis Tool (CRU, 2024)

Production costs per technology

Figure 78. presents anonymized information on the site cost distribution of crude steel manufacturing facilities in Europe, compared within a global context, based on data from the CRU Cost Analysis Tool¹¹³. The analysis considers two primary technology routes for steel production in 2022: the Blast Furnace-Basic Oxygen Furnace (BF-BOF) route and the Electric Arc Furnace (EAF) route.

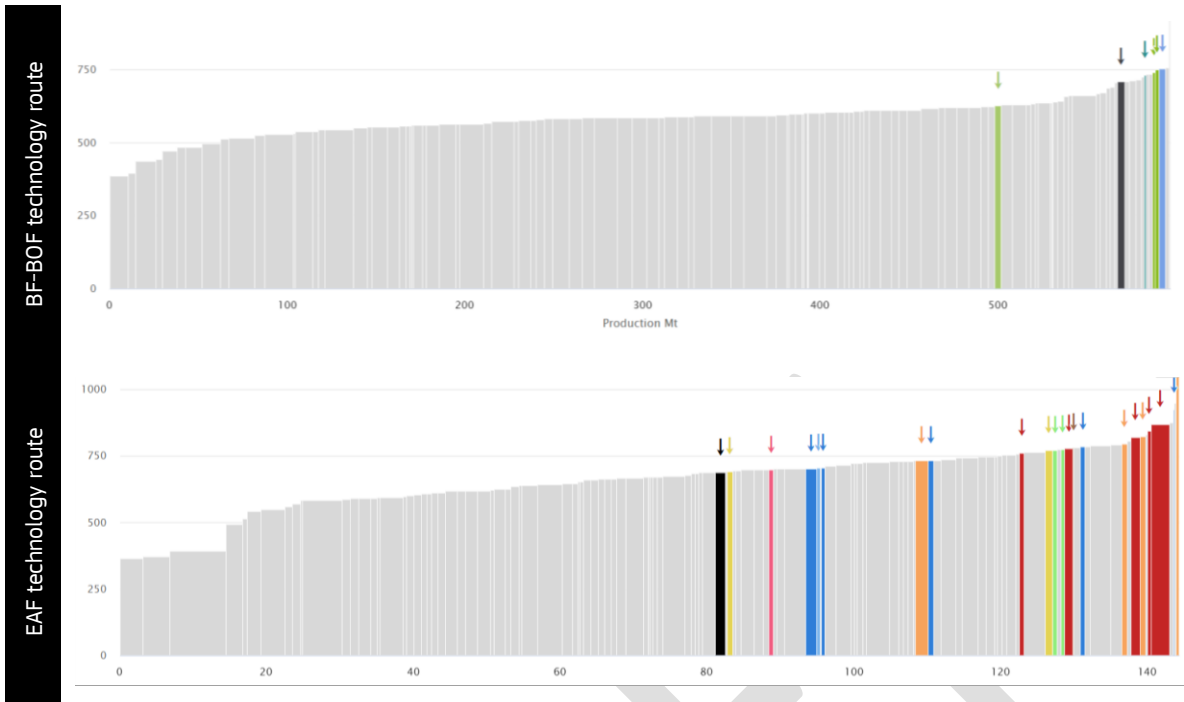
For the BF-BOF technology route, production costs¹¹⁴ ranged from EUR 400 to EUR 750 per tonne of crude steel, depending on the specific facility and its location. Steel production costs using the integrated route in the EU27 are among the highest globally, with the majority of EU27-based facilities positioned on the higher-cost end of the global cost curve.

In contrast, for the EAF technology route, EU27-based production facilities are relatively more competitive. Many EU27-based EAF facilities are positioned closer to the middle of the global cost curve, yet no facility shows the costs that are below the average.

¹¹³ CRU Costs Analysis Data Tool: CRU Group's comprehensive data and analytics platform provides detailed insights into production costs, market dynamics, and competitive benchmarking across global steel and other commodity sectors. The tool enables users to assess site-specific cost drivers, including raw materials, energy, labour, and policy impacts, offering robust support for strategic decision-making and industry analysis. <https://www.crugroup.com/en/data/>

¹¹⁴ The site costs include the following categories of costs: labour costs, raw material costs, (purchased electricity) energy costs, credits (given for by-products that can be used elsewhere in the steelmaking process) and other costs (incl. overheads, sustaining capital, interest on working capital, parts and spares).

Figure 78. Steel manufacturing costs profile across the globe in 2022 (in USD\$ per tonne)

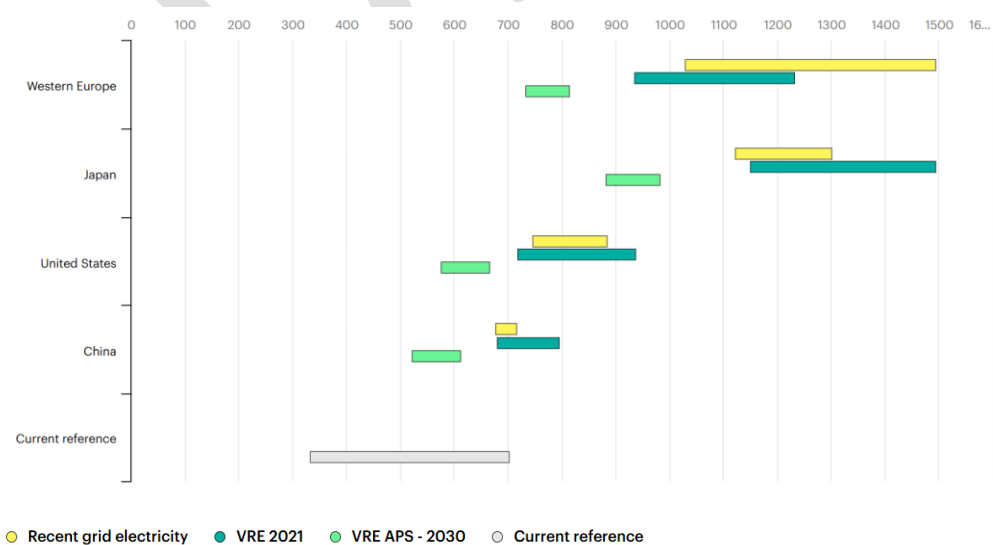


Note: The coloured bars show the entities that are located in the EU27.
Source: CRU Costs Analysis Tool

Numerous studies analyse the costs of decarbonisation technologies, such as those by the (IEA, 2020), (IEA, 2024), (Sandbag, 2024), (Agora Industry, Wuppertal Institute, and Lund University, 2024) and others. While these analyses provide highly valuable insights, they often do not distinguish costs by region of production, which is a key differentiating factor.

The *Energy Technology Perspectives* report (IEA, 2023) addresses this issue by examining crude steel production costs via the H₂-DRI route across regions (see Figure 79). It is important to recognize that these costs can vary significantly depending on the region of production, primarily driven by differing grid electricity prices. For instance, the production cost of steel using low-emission electrolytic hydrogen from grid electricity ranged between 680 and 1,500 USD per tonne in 2021. By 2030, this cost range is expected to narrow to 520–980 USD per tonne. However, regional differences persist, with costs in Japan and Western Europe remaining higher compared to those in the US and China.

Figure 79. Indicative production costs for crude steel via H₂-DRI route per region (2021 vs. 2030) in USD/t.



Note: VRE = variable renewable energy; APS = Announced Pledges Scenario; The VRE cost range represents electrolysis powered by solar PV, offshore wind or onshore wind. An additional hydrogen storage cost to guarantee a minimum load of 80% is considered. 'Current reference' values show production costs using the dominant incumbent means of production today with unabated fossil fuels. Recent prices for natural gas and grid electricity are used to compute the cost ranges in each region where relevant. The cost of capital is assumed at 5%, while the other techno-economic assumptions are sourced from literature.

Source: Energy Technology Perspective 2023 (IEA, 2024)

Production costs per product type

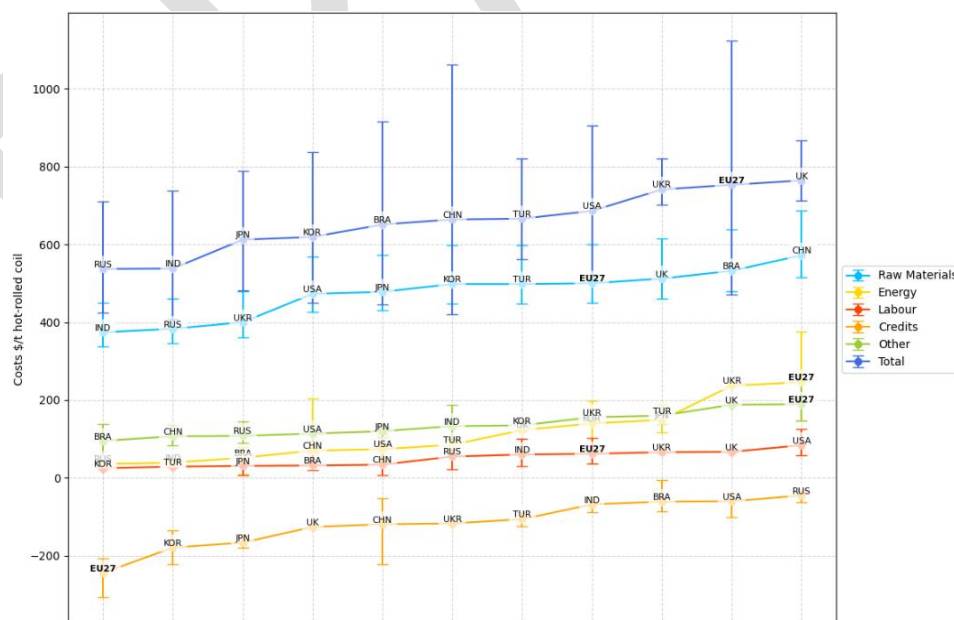
The latest JRC report (Medarac, Moya Rivera, and J, 2020) goes into more details when analysing the production costs of the European iron and steel industry in a global context. It **provides manufacturing costs breakdowns** for two typical products - hot rolled coil (as a proxy for **flat products**) and wire rod (as a proxy for **long products**) in both the BF-BOF and EAF technology routes. The analysis focuses on the site cost structure, particularly the impact of energy prices. The summary of findings, complemented with the latest (2022) data is presented below.

A. Flat steel products

The breakdown of production costs for **hot rolled coil using the BF-BOF technology route** is presented in **Figure 80**. With an average total production cost of USD 750 per tonne of hot rolled coil (EUR 712/t)¹¹⁵, the EU27 ranks second highest after United Kingdom. The main contributors to these costs are raw material costs (66%), energy costs (33%) and 'other costs' (25%). CO₂ costs, included in 'other costs', represent 5% of the total production costs for the EU27. It is worth noting that the EU27 is among the world leaders in creating credits from recycling materials and energy.

Compared to 2020 (Medarac, Moya Rivera, and J, 2020), there has been a 35% increase in production costs in the EU27, rising from EUR 450/t to EUR 712/t in 2022. This trend is observed globally, e.g. in Russia, where costs were the lowest in 2020 and 2022 it rose from EUR 395/t to EUR 511/t, respectively. In Europe, the cost increase is largely driven by a surge in energy prices, which climbed from EUR 76/t in 2020 to EUR 233/t in 2022, and an increase in other costs from EUR 125/t to EUR 179/t during the same period. At the same time, the benefits from credits have significantly grown, rising from EUR 83/t in 2020 to EUR 207/t in 2022.

Figure 80. Hot rolled coil production costs in the BF-BOF integrated technology route



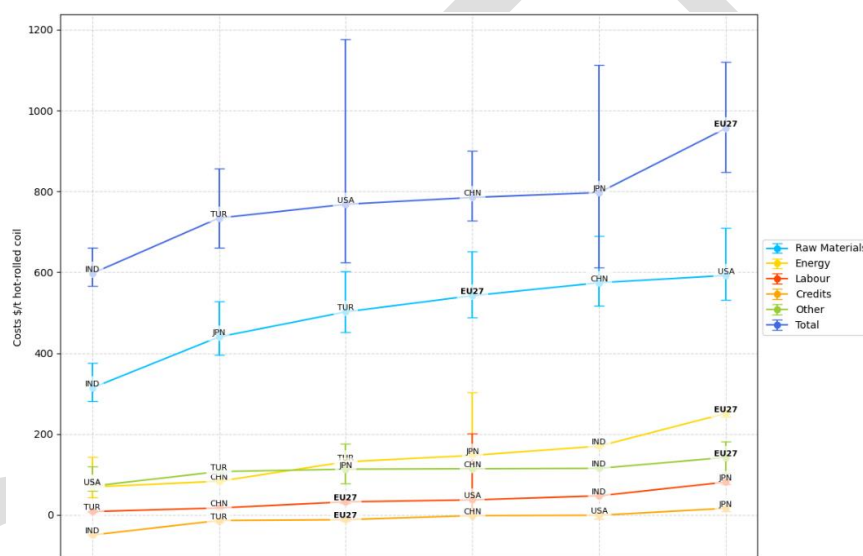
Source: Own estimates based on CRU platform data and JRC report methodology (Medarac et al. 2020)

¹¹⁵ European Central Bank, average EUR/USD conversion in 2020: 1 USD=0.95 EUR

The breakdown of production costs for **hot rolled coil using the EAF technology route** is presented **Figure 81****Error! Reference source not found.** The results indicate that the EU27 has the highest production costs, followed by Japan and China. While raw material costs are relatively consistent across regions, energy costs in the EU27 are significantly higher than in most other areas, except for Ukraine. Additionally, 'other costs' in the EU27 are the highest among all regions.

Compared to 2020 (Medarac, Moya Rivera, and J, 2020), the EU27 has surpassed Japan, which previously had the highest production costs. Over the two-year period, the total production costs in the EU27 increased from EUR 486 per tonne to EUR 956 per tonne. This dramatic rise is primarily attributed to a substantial increase in energy costs, which jumped from EUR 70 per tonne in 2020 to EUR 230 per tonne in 2022. Similar trends have been observed across other regions. For instance, in 2020, Japan, the then-cost leader, had production costs of approximately EUR 550 per tonne, while in 2022, even India experienced costs of around EUR 600 per tonne. This underscores the widespread impact of rising energy prices on global steel production costs.

Figure 81. Hot rolled coil production costs in the EAF process



Source: Own estimates based on CRU platform data and JRC report methodology (Medarac, Moya Rivera, and J, 2020)

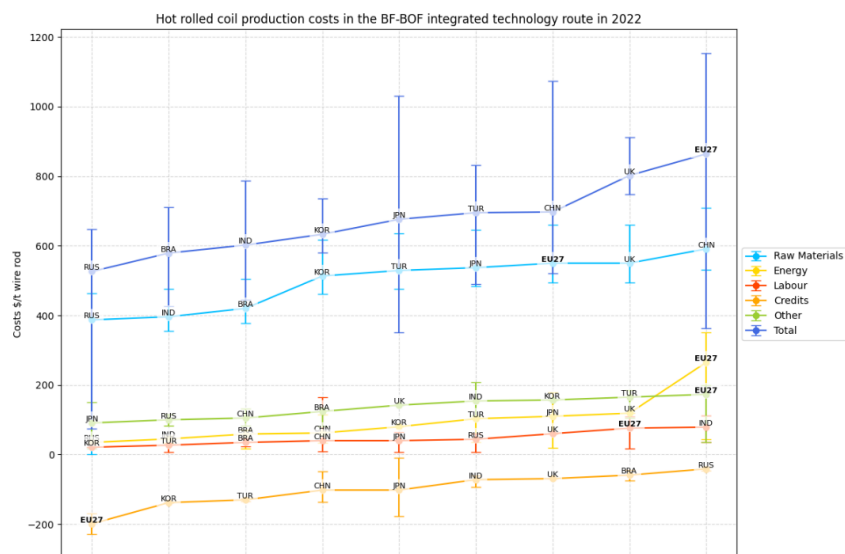
B. Long steel products

The breakdown of production costs for **wire rod** using the BF-BOF technology route is presented in

Figure 82, and for the EAF technology route in Figure 83. In the EU27, the production costs of wire rod are the highest both for the integrated route and the recycling route. Materials costs in the EU27 account for 64% of production costs for the integrated route and 50% for the recycling route. These material cost shares are similar in other countries, in particular in case of integrated route. The share of energy costs in the EU for wire rod production via the integrated route is 31%. In the scrap-based route, energy costs account for 30%, the highest share observed among the countries considered.

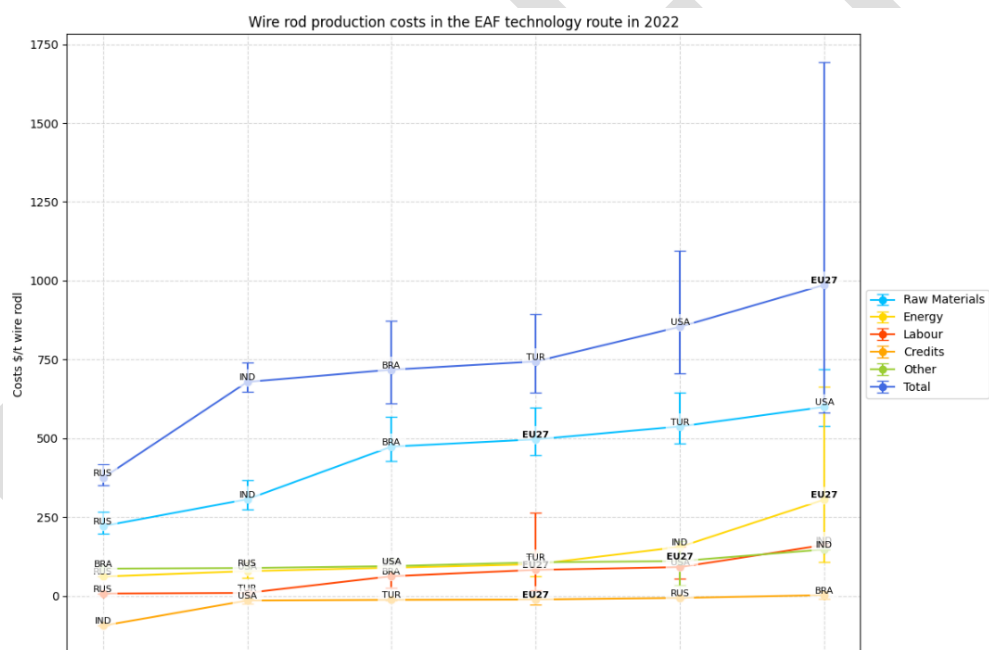
In 2020 (Medarac, Moya Rivera, and J, 2020), the EU27 already ranked first globally in production costs for the BF-BOF route and second, after the USA, for the EAF route. By 2022, production costs had risen sharply, in the EU27 and across regions. For the BF-BOF route, costs increased from EUR 440 per tonne in 2020 to EUR 820 per tonne in 2022. Similarly, for the EAF route, costs grew from EUR 480 per tonne to EUR 930 per tonne over the same period. These increases underscore the impact of rising energy prices and other cost pressures on wire rod production in the EU27.

Figure 82. Wire rod production costs in the BF-BOF process



Source: Own estimates based on CRU platform data and JRC report methodology (Medarac, Moya Rivera, and J, 2020)

Figure 83. Wire rod production costs in the EAF process



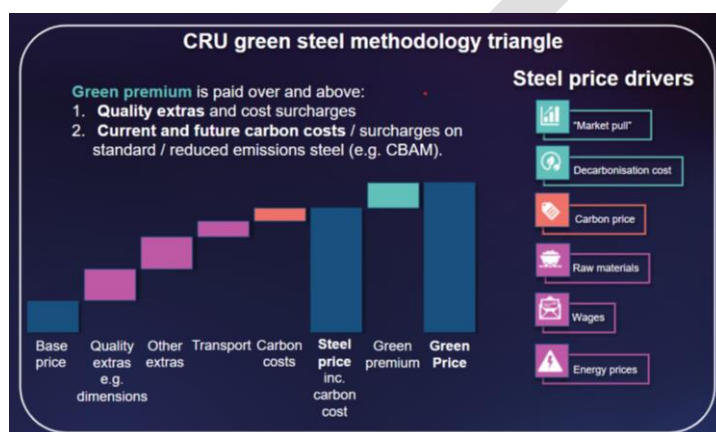
Source: Own estimates based on CRU platform data and JRC report methodology (Medarac, Moya Rivera, and J, 2020)

2.6.4 The green premium

The green premium represents the additional cost paid over and above quality extras and cost surcharges, mainly driven by energy costs, and current and future carbon costs. As shown in Figure 84, transitioning to green steel production incurs higher costs, such as development costs, increased CAPEX to implement new technologies, and higher OPEX due to changes in raw materials and field inputs. Covering these incremental costs through the green premium is essential to incentivize producers to shift their production methods (CRU Consulting, 2024).

Two key drivers will shape the green premium, the **cost of decarbonisation** and the **market pull**. The first sets the lower bound of the range premium, representing the new expenses associated with implementing new technologies for green steel production. On the other hand, the market pull establishes the upper bound of the range premium and is driven by customer demand for green steel, which is influenced by the proportion of cost and carbon footprint that the steel represents in the final product and varies across different end-use sectors.

Figure 84. Green steel price components



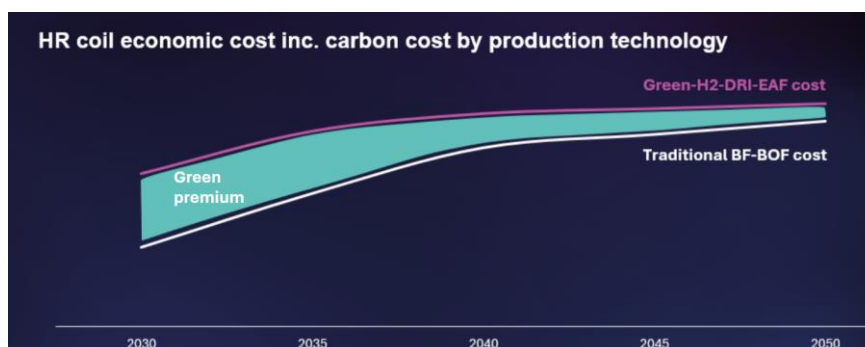
Source: Steel decarbonisation: How will Green Steel be priced? (CRU Consulting, 2024)

CRU analysis study the drivers (Figure 84) and projection of the green premium for the steel production. The conclude that in the next decades, the green premium is expected to undergo changes, and as a consequence of the premium between traditional and green routes for steel making is expected to decrease, as illustrated in **Figure 85** for example of the hot rolled coil produced in the EU27.

As producing steel through the BF-BOF will became increasingly more expensive, in particular in Europe after free allowances in sectors covered by EUETS are dismissed (between 2027 and 2032). The forecast assumes that the carbon price in the EU will rise substantially, reaching 133 EUR per ton in 2030, compared to current carbon price of 90 EUR per ton (GMK, 2023). This increase is anticipated to serve as a strong incentive for the transformation of the steelmaking industry.

On the other hand the cost of steel produced through the H2-DRI-EAF process is expected to gradually increase but stabilize in the long term. This price increase is attributed to the projected trend in hydrogen costs, which are anticipated to rise during the consolidation phase and subsequently decrease, from approximately \$7/kg today to less than \$1/kg by 2050 (BloombergNEF, 2023).

Figure 85. HR coil economic cost including carbon cost by production technology in the EU27

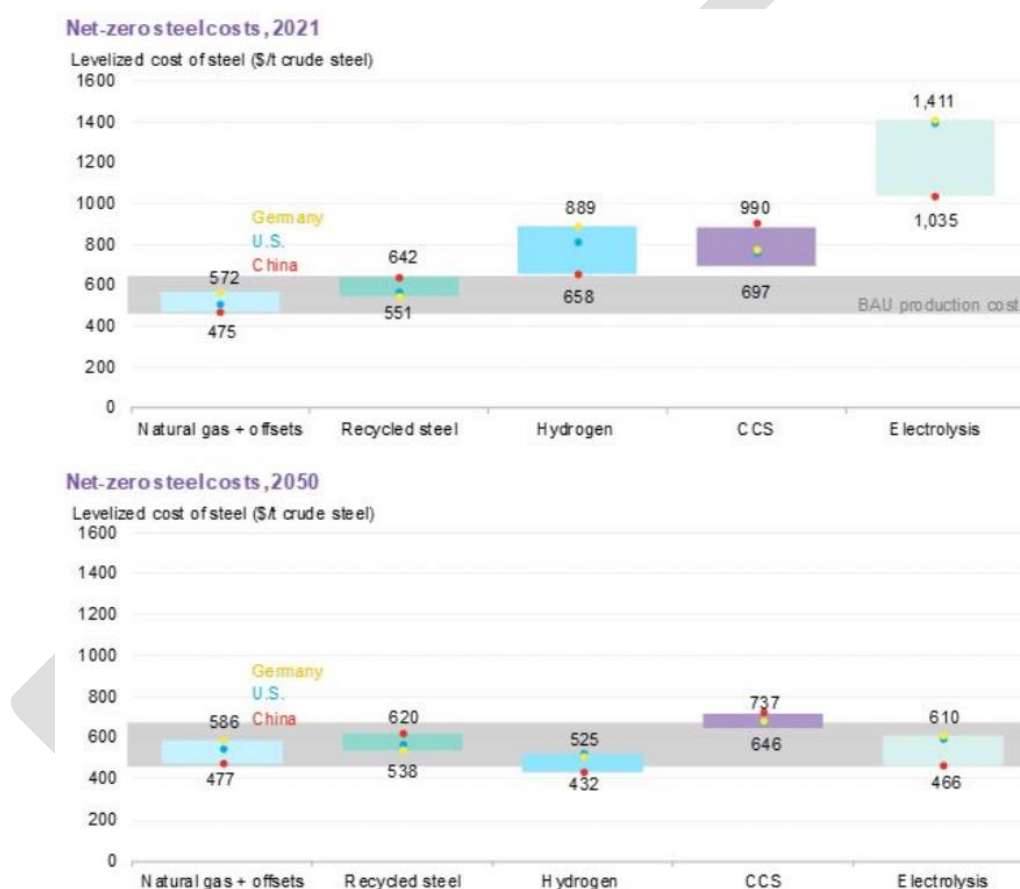


Source: Steel decarbonisation: How will Green Steel be priced? (CRU Consulting, 2024)

Another study by BloombergNEF (2023) analysed the levelised costs of green steel production routes across the globe (see: **Figure 86**). The study found that, on average, green steel currently costs 40% more than standard steel due to the higher expenses associated with low-carbon technologies, renewable energy inputs, and hydrogen-based processes. However, BloombergNEF projects that these costs could significantly decrease by 2050. This decline is attributed to advancements in green steel technologies, scaling of hydrogen production, and the falling costs of renewable energy. By 2050, it is estimated that green steel could be priced 5% lower than fossil-based production routes, effectively eliminating the "green premium."

Despite these optimistic projections, current green steel initiatives still come at a notable premium. For instance, H2 Green Steel has announced that its steel will come at a 25% premium, while SSAB indicates that its fossil-free steel will be priced at 300 EUR per ton higher than conventional steel.

Figure 86. Levelised cost for the major green steel technology routes 2021 and 2050



Source: Green steel demand is rising faster than production can ramp up (BloombergNEF, 2023)

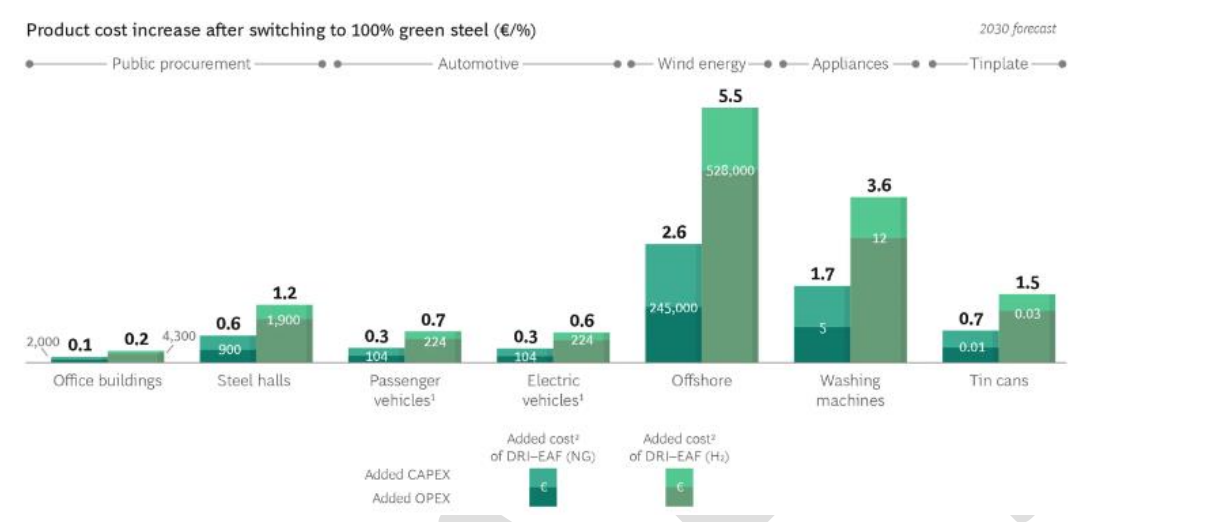
The Boston Consulting Group (BCG, 2022) analysed the cost implications of transitioning to 100% green steel under two technology routes: natural gas-based direct reduced iron with an electric arc furnace (NG-DRI-EAF) and hydrogen-based direct reduced iron with an electric arc furnace (H2-DRI-EAF) (see **Figure 87**). The study provides insights into how the adoption of green steel affects end-product pricing across various industries.

For example, automakers may be more willing to absorb the green premium due to the relatively minor impact on vehicle costs. The projected price increase for a midsize passenger vehicle is estimated to range from 0.3% of the final price for the NG-DRI route to 0.7% for the H2-DRI route. This represents an increase of less than EUR 250 for a vehicle typically valued in the tens of thousands of euros.

For the appliance sector, the cost increase from transitioning to 100% green steel is similarly modest. The price of a standard household appliance is projected to rise by only 2–4%, equating to an additional EUR 12 or less per unit.

In contrast, the impact is more pronounced for offshore wind turbines, where the significant steel requirements make the cost increase more substantial. This additional cost may not be easily or immediately passed on to buyers, potentially creating financial challenges for project developers in the renewable energy sector.

Figure 87. Product cost increase after switching to 100% green steel



Source: Transforming the steel industry may be the ultimate climate change (BCG, 2022)

2.7 Summary and conclusions of the market analysis

A reference finished product categorisation has been established for the purpose of the ESPR preparatory study for steel. The product classification, shown in Table 20 has been created taking into consideration the data available in existing reporting (both public and private) on iron and steel products across the value chain. The reference product classification considers all qualities of steel products (carbon, stainless, and castings) and includes a list of reference definitions. The correspondence between the reference product classification with CN codes and the EN10027 standard has been established in Annex 3, parts B and C.

There is no single data source that covers all the information on the iron and steel supply chain for the EU market. An inventory of the available sources by life cycle stage has been documented in Annex 3, part A, and the most suitable data sources for market data analysis were identified. The primary data sources on finished steel products entering the EU market are data collected by EUROFER from its members, complemented with information from Eurostat/PRODCOM and Comex, UN Comtrade, and others, as shown in Table 19.

The European Union accounts for about 8% of global demand, while the largest consumer is Asia, which accounts for about two thirds of global steel demand. Europe’s demand for steel has been relatively stable in the long term, with a slight increase in consumption from 2012 until 2018, corresponding to positive economic development during that time. Since 2020, steel consumption has been declining in Europe and worldwide, mainly due to the aftermath of the COVID-19 crisis and a prolonged downturn in the construction sector. Despite these trends, the output of steel-using sectors continued to grow, albeit with significant differences across individual European economies and sectors. Additionally, global steel demand is expected to show a mild recovery in 2024 that will continue into 2025.

The construction sector accounts for nearly half of the demand for iron and steel products. However, new markets for steel are being created within the renewable energy sector¹¹⁶. For instance, electrical steel is critical for transformers, wind turbine generators, and other energy technologies. The transition to electric mobility also relies on high-quality steel for car bodies, engines, and charging stations. The food industry is

¹¹⁶ Note construction sector may also include some components used for the renewable installations (such as Reinforced bars and wire rods for concrete reinforcement of the foundations on-shore wind mills and off-shore).

another rising sector where steel is integral to secure food packaging, as it preserves nutritional value well and offers high protection from damage (EUROFER, 2024).

Iron ore mining has almost tripled over the past 20 years, and the iron and steel sector remains highly concentrated among the top global producers (including Australia, Brazil, China, India and Russia), which together account for 75% of global production. In Europe, only a few Member States produce iron ore, with Sweden being a significant producer. To meet demand, Europe imports 75% of its iron ore and concentrates, mainly from Canada, Ukraine and Brazil.

Global iron-making is dominated by pig iron production, totalling 1300 Mt in 2021, compared to direct reduced iron production of only about 126 Mt. Europe supplies 5.8% (and 0.5%, respectively) of the global demand. Looking into investment plans globally, and particularly in Europe, the significance of DRI is expected to increase. The imports (8.9% of total EU demand) are still dominated by imports of direct reduced iron, with the main trade partners being Russia, Brazil and South Africa.

The apparent consumption of ferroalloys in Europe shows a decreasing trend; 69% of the demand for ferroalloys is met by imports, mainly from Norway, South Africa, India and Ukraine. Among the various ferroalloys, ferro-chromium has the highest consumption level, followed by ferro-silico-manganese.

The EU27, with a production of 136.3 Mt of steel per year, accounts for 7.2% of the global output of 1885 Mt. The key global players are China, India, Japan, the USA and Russia. One third of Europe's production takes place in Germany. In Europe, the ratio between BOFs and EAFs is 57.9% to 42.1%, with a slowly increasing shift towards electric arc furnaces (EAFs) (by nearly two percentage points within the last decade). While in Europe, North America and some others invers in EAF and DRI technologies, in Asia the majority of new capacities is expected to be in BF-BOF technology route (Figure 37).

The total production of finished steel products peaked in 2018, reaching 162.5 Mt/y, and then decreased to 137.7 Mt in 2023 (as a direct impact of the global pandemic and the war in Ukraine). In 2022, the imports of steel products met 19.9% of demand. In 2023, although the imports of semi-finished and finished products decreased by 8.5%, their market share in total demand reached 27% (EUROFER, 2024).

Flat steel carbon products account for 56.6% (85.5 Mt) of apparent consumption in 2022, with the most important categories being hot rolled and electric sheet (23.1% or 35 Mt), tinplate, tinned products and galvanized coil (13% or 19.7 Mt) and cold rolled coil (10.7% or 16.2 Mt). Imports account for 25% of the flat product demand, with the key trade partners being Turkey, South Korea, India and Taiwan.

Long steel carbon products account for 34.7% (52.4 Mt) of total consumption, with the most important category being rod and wire rod (13.7% or 20.6 Mt). Imports account for 13.4% of the demand, with the key trade partners being Turkey, China and Switzerland.

The EU, with a steel scrap consumption of 79.3 Mt in 2022, is the second largest scrap consumer worldwide, after China. The production of flat steel products has a dominant role in the generation of new scrap that is used to dominate domestic production. Today, over 60% of demand is met by post-consumer scrap. Europe is a net exporter of scrap, with 17.6 Mt of steel scrap exported, mainly to Turkey (67%).

Table 34. Trends in production and trade by finished steel product category

Finished steel product	Trends in production and trade	Import directions
Rod and wire rod	The production of rod and wire rod has shown slight fluctuations but remains mostly stable. About 13% of the apparent consumption is met by imports, which show a slightly increasing trend. Exports slightly exceeded imports in 2013. Since then, exports have decreased while imports have increased, leading to a net import of rod and wire rods amounting to 2,680 kt/y in 2022.	UK, Malaysia, Turkey, Ukraine
Railway track material and	European production of railway track material and other long products decreased from 2,990 kt/y in 2013 to 2,460 kt/y in 2022. Exports of these products are highly significant, but they have also decreased, from 1,150 kt/y in 2013 to 540 kt/y in	China (>50%), UAE, UK

other long products	2022. The reduced production is attributed to lower demand from outside Europe, while the EU27 apparent consumption has remained constant. Imports of railway track material and other long products are very low compared to production and exports, remaining around 70 kt/y.	
Other hot rolled bars (e.g. merchant bars)	The production and apparent consumption of other hot rolled bars slightly decreased between 2013 and 2022. In 2013, imports and exports were balanced at around 1,320 kt/y. Since then, imports have increased to 1,860 kt/y in 2022, while exports have decreased to 890 kt/y in the same year. Consequently, the reliance on import for other hot rolled bars has increased over the period observed.	China (>50%), UAE, UK
Heavy sections and mining frames	The production of heavy sections and mining frames slightly decreased between 2013 and 2022. Apparent consumption increased until 2018 but returned to the 2013 level by 2022. Exports are significantly higher than imports, although exports continuously decreased from 2,400 kt/y in 2013 to 1,560 kt/y in 2022. Imports remained low, reaching 295 kt/y in 2022, and showed a development aligned with the apparent consumption.	Turkey (c.a.50%), UK, Switzerland
Bars for reinforcement of concrete	The production of bars for reinforcement of concrete slightly decreased from 2013 to 2022. In 2013, exports were high at 4,920 kt/y, while imports were significantly lower at 740 kt/y. Since then, the trade situation has changed substantially. Exports dropped to 1,115 kt/y, while imports increased, leading to a net import situation from 2021. In 2022, imports accounted for 2,100 kt/y. Consequently, the share of domestic supply in the apparent consumption has decreased over time.	Turkey, Algeria, Russia
Hot rolled plate (e.g. Quarto plate)	Compared to other finished steel goods, the situation for hot rolled plate (e.g. Quarto plate) has remained relatively constant. The ratios of production, imports and exports have mostly stayed the same over the observed period, with imports and exports being largely balanced. In 2023 the production was 10,011 kt/y, while import and export were 1,774 kt/y and 1,991kt/y, respectively.	India, Indonesia, South Korea
Hot rolled coil and electrical sheet	The production of hot rolled coil and electrical sheet decreased from 36,360 kt/y in 2013 to 29,250 kt/y in 2022. Apparent consumption increased until 2018 and returned to the 2013 level in 2022. Imports significantly increased from 6,690 kt/y in 2013 to 9,008 kt/y in 2022. During the same period, exports declined from 7,850 kt/y to 3,226 kt/y.	Turkey, Japan, Taiwan
Galvanized coil and tinplate and tinned products	Both production and apparent consumption of galvanized coil, tinplate, and tinned products increased from 2013 until 2017, then decreased. Significant changes in trade occurred during this period. Imports surged from 1,690 kt/y in 2013 to 5,060 kt/y in 2022, while exports decreased from 3,390 kt/y to 2,600 kt/y. The share of apparent consumption met by imports rose from 9% in 2013 to 26% in 2022, highlighting the growing relevance of imports for these products.	China, Turkey, Vietnam, South Korea
Cold rolled coil	The production of cold rolled coil decreased from 17,260 kt/y in 2013 to 14,010 kt/y in 2022. During the same period, apparent consumption within the EU remained mostly constant. The reduction in domestic production was offset by higher imports, which rose from 2,780 kt/y in 2013 to 3,766 kt/y in 2022. In contrast, exports decreased from 2,570 kt/y to 1,603 kt/y.	Turkey, India, South Korea, Taiwan
Organically coated and other coated products	The production of organically coated and other coated products increased from 4,160 kt/y in 2013 to 4,420 kt/y in 2022. Apparent consumption grew even more, driven by higher imports and lower exports, indicating a rising demand for these products. Imports increased from 840 kt/y in 2013 to 1,200 kt/y in 2022, while exports decreased from 720 kt/y to 430 kt/y.	India (>30%), South Korea, Vietnam
Long stainless steel products	The production of long stainless steel products slightly increased from 580 kt/y in 2013 to 620 kt/y in 2022. Trade in both directions is highly relevant and mostly balanced. There is a slight trend towards increasing imports, while exports remain mostly constant at around 360 kt/y. Due to the high volume of imports and exports, more than half of the apparent consumption is met by imports, which increased to 62% in 2022.	n/a

Flat stainless steel products	For flat stainless steel products, trade is more significant than for carbon steel products, though less so compared to long stainless steel products. The production of flat stainless steel products slightly increased from 3,390 kt/y in 2013 to 3,570 kt/y in 2022. Significant shifts towards higher imports and lower exports were observed. Imports rose from 890 kt/y in 2013 to 1,630 kt/y in 2022, while exports decreased from 1,520 kt/y to 800 kt/y. Consequently, apparent consumption increased significantly from 2,780 kt/y in 2013 to 4,400 kt/y in 2022.	n/a
Castings	The production of castings decreased from 9,740 kt/y in 2013 to 8,110 kt/y in 2022. Trade in castings is minimal, indicating that demand is largely met domestically. Only 2% of the apparent consumption is supplied by imports. Both imports and exports remain low and show no clear trend, maintaining levels of around 200 kt/y and 130 kt/y, respectively.	n/a

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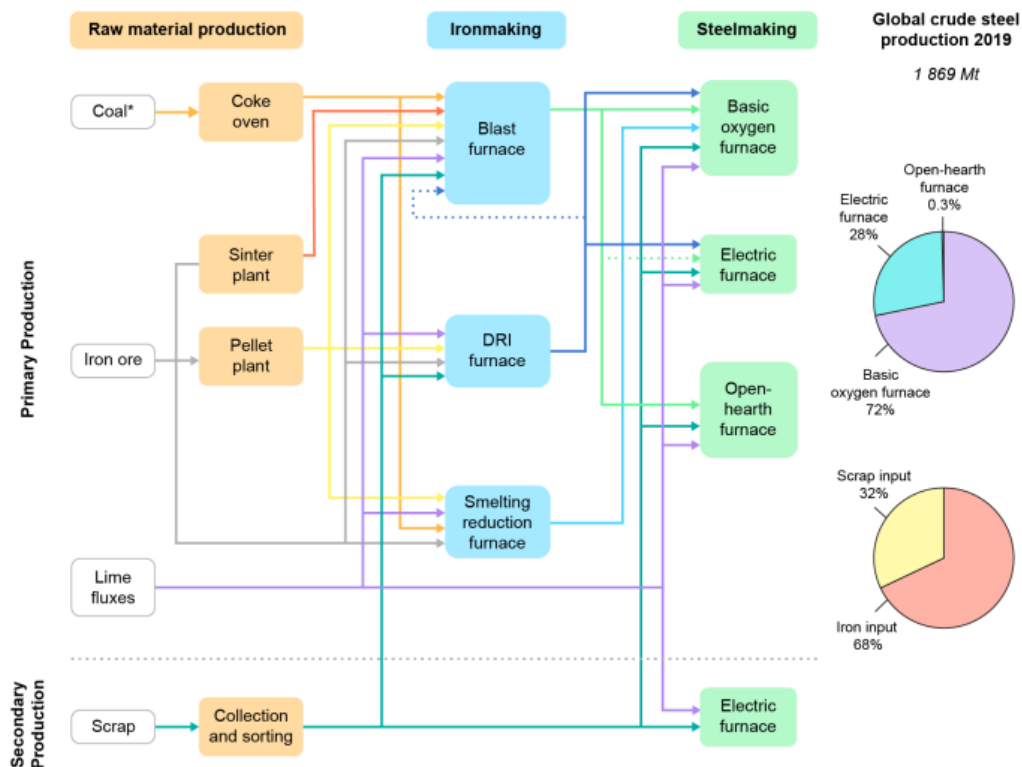
TASK 3: Technical Analysis

Task 3 aims to conduct a technical analysis of the current iron and steel products available on the EU market and provide general insights for defining the Base Cases, which represents the average product sold on the EU market in terms of resource efficiency, emissions and functional performance. A comprehensive understanding of the current and potential future steelmaking technology routes used in steel making in the EU and within the EU imports is crucial for making recommendations on new targets, policies and regulations (see section 3.1 and 3.3). Additionally, since the scope of products under iron and steel as intermediates is broad, a selection of the most relevant “representative products” is carried out (See section **Error! Reference source not found.**). Task 3 involves the identification of BAT (Best Available Technology) and BNAT (Best Not yet Available Technology) in section 3.4, which will be the basis for modelling in Task 6. The environmental and lifecycle cost analyses conducted throughout the study will largely rely on the base cases and technology analysis carried out in this task (see section 3.3), providing a point of reference for Tasks 4, 5, and 6 of the Ecodesign Preparatory Study.

3.1 Overview of steelmaking production routes in the EU

There are two main routes for traditional steelmaking: the primary route, which produces steel from iron ore, and the secondary route, where steel is made from scrap melting. As illustrated in **Figure 88**, the primary route encompasses the integrated blast furnace and basic oxygen furnace processes (BF-BOF). Conversely, the secondary route involves producing crude steel by remelting steel scrap in the electric arc furnace (EAF). Crude steel is then further processed as semi-finished products (slabs, blooms, or billets), while additional thermal, mechanical and/or chemical treatment can be applied depending on the requested properties of the final steel products ready to be sold in the market.

Figure 88. Main steel production pathways and material flows (Scope: Global)



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Source: Iron and Steel Technology Roadmap (IEA, 2020)

Apart from the conventional production routes, current developments propose different technology routes to decarbonize steelmaking and consequently improve the environmental performance of iron and steel products.

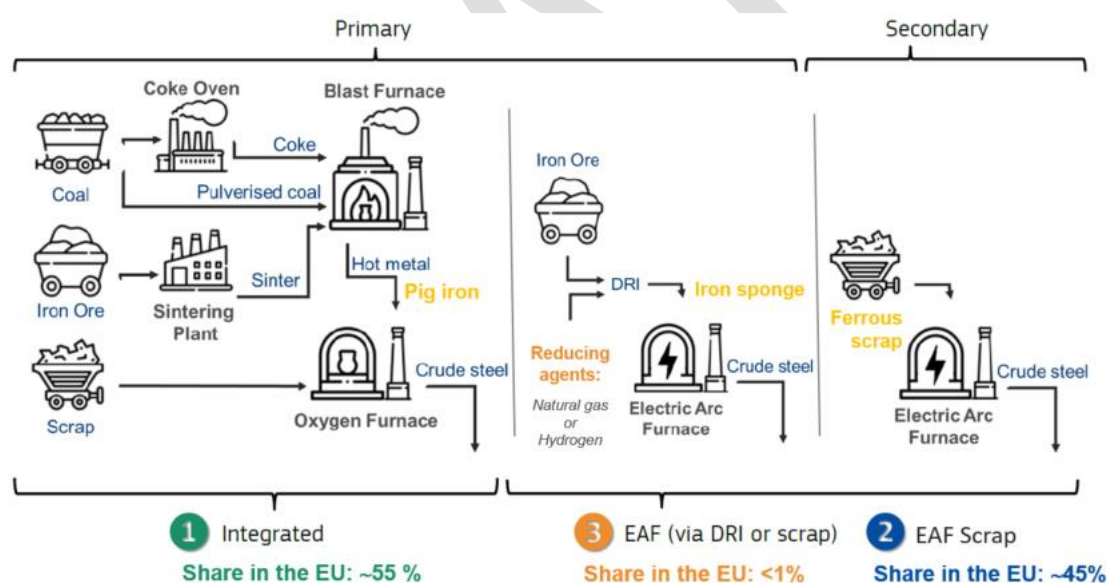
One of the most promising technologies is the direct reduction route. This process produces iron sponge¹¹⁷ thanks to a direct reduction of high-Fe content (>65%) iron ores by reducing agents such as natural gas or hydrogen. The output is then sent to an EAF to produce crude steel. Already significant investments and several projects take place in the EU to promote the hydrogen-based DRI-EAF, some examples include initiatives such as Hybrit, Gravithy and Stegra (previously H₂ Green Steel)¹¹⁸. Going further, new non-conventional technology routes for steelmaking are explored in the section 3.4 on BAT/BNAT.

With the aim of analysing the current state of steelmaking production routes in the EU and explore potential future developments, the analysis will focus on four specific technology routes to produce steel (cf. **Error! Reference source not found.**):

- Integrated route with blast furnace and basic oxygen furnace (BF-BOF)
- Ferrous scrap in the electric arc furnace (EAF),
- Direct reduced iron made from natural gas as the input of the EAF (NG-DRI-EAF),
- Direct reduced iron made from green hydrogen as the input of the EAF (H₂-DRI-EAF).

In addition to having different processes and reaction conditions, these routes also have different material inputs and energy demands. The objective of this section is to perform an evaluation of the current state of play of steel making production routes in the EU. The four routes are illustrated in **Figure 89** and then described in more details.

Figure 89. Main production routes for steelmaking in the EU



Source: adapted from Greenhouse gas intensities of the EU steel industry and its trading partners (Joint Research Centre, Koolen, and Vidovic, 2022)

3.1.1 BF-BOF route

Iron ore is the input for steel making process through the primary route. In Europe, the production of primary steel is integrated, as it includes the following steps: coke production; ironmaking, steelmaking, casting and rolling. This route starts with pre-processing: iron ore reacts with limestone in the sinter plant or the pelletising plant, to produce virgin iron in the form of sinter or pellets. In parallel, coke is made from coal in a coking plant at 1000°C (high temperatures in an oxygen-free atmosphere in order to evaporate coal volatile elements).

¹¹⁷ Sponge iron: It is the product of reducing iron oxide in the form of iron ore and steel plant wastes into metallic iron, below the melting point of iron and typically in the range of 800–1200 °C. Also known as direct reduced iron (DRI).

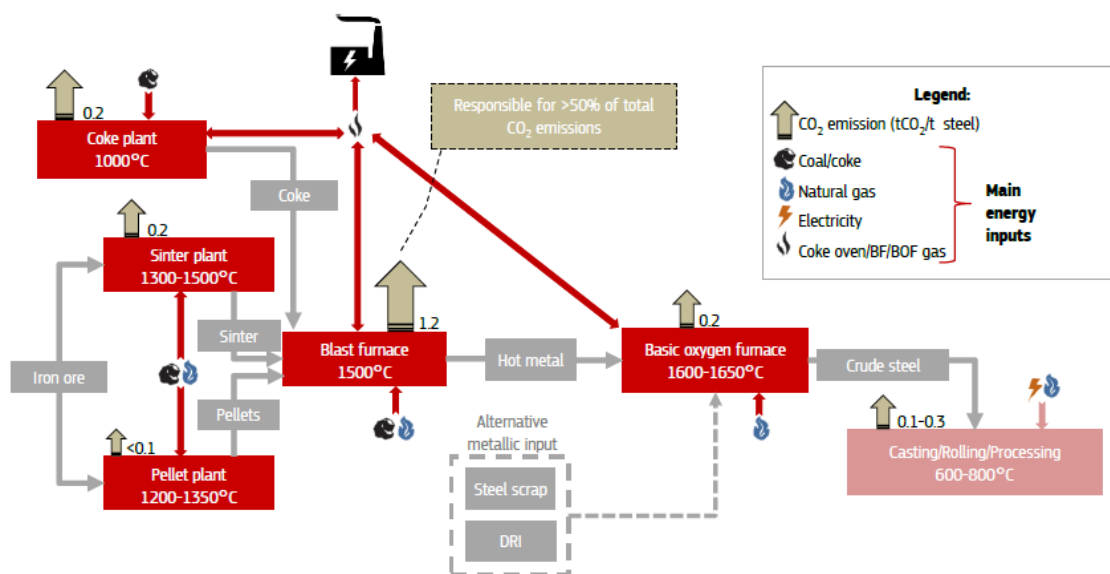
¹¹⁸ MacKinsey & Company (2022): Closing Europe's green-metallurgy gap. <https://www.mckinsey.com/industries/metals-and-mining/our-insights/closing-europes-green-metallurgy-gap>

The iron pellets or sinter are then mixed with the coke in a blast furnace operating at 1500°C, as illustrated in Figure 90. At this step, a certain amount of ferrous scrap can be added as a cooling agent. The objective of this step is to reduce the iron oxides with the carbon in the coke, to form metallic iron, called pig iron. The pig iron coming out of the blast furnace then enters the basic oxygen furnace where the carbon content of the pig iron is reduced thanks oxygen blowing and liquid steel is produced, which will later solidify to become crude steel.

The integrated route is highly energy intensive and emits significant CO₂ amount. The energy demand for this process is typically around 22 GJ/ts (ton of steel) (Strezov, Evans, and Evans, 2013). The blast furnace step is the most energy demanding step of steelmaking, considering it operates at 1500°C to melt the sinter and pellets.

The advantage of this route is its maturity, making it a well-known process easy to operate. It produces steel of great quality, considering iron ore is the main input. Thus, most flat products are made through this route, since they need to comply with more stringent requirement than long products. However, it has multiple drawbacks, such as its environmental impact: this route has significant CO₂ emissions mainly due to the quantity of coal is burnt in the BF. Additionally, blast furnaces are designed for continuous operation and require stable, high production levels to remain efficient. Frequent shutdowns or adjustments to production are costly and technically challenging. Lastly, the infrastructure for BF-BOF steelmaking is expensive to build and maintain, making it difficult to scale production up or down based on demand without incurring significant costs.

Figure 90. Simplified flow diagram and CO₂ emissions of the BF-BOF route



Source: JRC report - Technologies to decarbonise the EU steel industry (Somers, 2022)

3.1.2 EAF route

The second most common steelmaking route is through EAF (also called secondary route), most particularly with steel scrap as the main input of the process. Indeed, reusing scrap enhances the circularity of steel in the industry. Both pre-consumer¹¹⁹ and post-consumer¹²⁰ scrap can be used as inputs in the EAF. The reaction also requires lime, coal and oxygen to operate. The EAF uses graphite electrodes in order to provide electricity for the reaction, as the main source of energy. These electrodes are consumed throughout the reaction and need to be replaced over time. Typically, the consumed graphite is about 3 to 3.5 kg per tonne of liquid steel produced.

This route's energy demand accounts for 2.5 to 3 GJ/tcs (ton of crude steel) (Somers, 2022). EAF route doesn't require the blast furnace step, which represents a big part of the energy consumption of the BF-BOF process.

The EAF route has been developed after the BF-BOF route but still is considered a common process to produce steel. Some countries in the world such as Turkey or the United States have a larger share of steel produced

¹¹⁹ Pre-consumer scrap is scrap generated during the manufacturing of steel, before its end-use.

¹²⁰ Post-consumer scrap is scrap generated during the end-use of the steel product, by the consumer.

from EAF rather than from BF-BOF, which is not the case in Europe with a ratio of 58/42% (2022 data). Again, Germany is the European country with the most plants producing steel through EAF, although the EAF plants are well distributed over multiple countries in the EU. The EAF plants are more numerous than the BF-BOF ones, but with almost systematically lower capacities for each EAF plant.

The use of electricity to heat the iron (instead of fossil fuels to melt the iron in the BF) makes the EAF route much better in terms of environmental impacts – even though the impact then depends highly on the electricity mix of the country of production. Additionally, this process uses steel scrap as an input, requiring less or no iron ore, and thus less raw material extraction. This point also lowers the environmental impacts of this process in comparison to the BF-BOF route.

However, the use of scrap in the process has consequences on the quality of the output products. Indeed, during the end-of-life of the steel and the recycling of scrap, the latter is often mixed with tramp elements such as copper, that are not further removed from scrap sent to the secondary market of steel. These tramp elements then end up in the liquid steel produced through EAF and can have consequences on the mechanical properties of steel. Some industries are demanding specific qualities of steel (e.g. for the production of flat products) and these expectations are not always met with steel from the secondary route, depending on the input scrap quality. A consequence of this is that most of the intermediate products produced from the EAF route in the EU are long products and only very few flat products. Overall, there is not sufficient good quality scrap to project a steel production coming exclusively from this route with 100% scrap input and still producing qualitative steel (Materials Processing Institute, 2021). A more detailed discussion on these aspects is given in section 3.4.1.

3.1.3 NG-DRI-EAF route

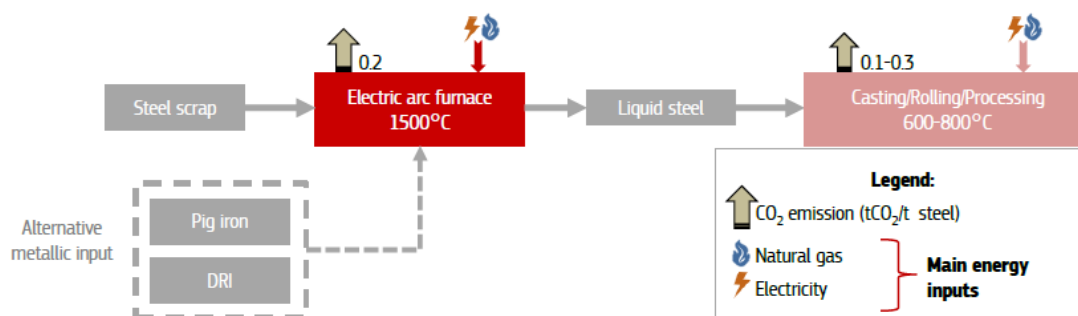
This route is similar, process-wise, to the previous EAF route described. However, the inputs are slightly different: instead of having mostly steel scrap entering the reactor, DRI made from the reaction of iron oxides with natural gas is the main input. The iron oxides enter the process as either iron pellets (cf. description of pelletising in the BF-BOF section) or lump ore. This reaction of reduction produces metallic iron, and the natural gas, in this case, serves as reducing agent. Once reduced, this product made of approximately 97% iron called sponge iron, enters the EAF (Materials Processing Institute, 2021) to go through the same process as explained in the classic EAF section. The energy demand of this route is 10 GJ/ts (Strezov, Evans, and Evans, 2013), which is more than the classic EAF with mainly scrap as input.

This route has been producing steel for the European market since the 70s, but at a very small scale, with its first plant set up in Germany (Lüngen and Schmöle, 2022). Having DRI as an input is relatively recent in comparison to both routes studied previously, and not as common, since the adaptation of old steelmaking plants is not easy; it would require specific new reactors to be set up DRI capacity.

On one hand, an advantage of the DRI-making process is that it does not require the iron to be melted in order to reduce the iron (Spreitzer and Schenk, 2019). This means that the temperature of the reaction tank does not need to be as high as the blast furnace, since it stays below the melting point of iron. The reduction plant can operate under 1000°C. On the other hand, one of the drawbacks is that the DRI process requires high grade iron ore as an input (Materials Processing Institute, 2021), and cannot operate with a very high percentage of scrap entering the reactor.

Figure 91 illustrates the EAF process considering both the steel scrap input and the DRI input. Pig iron input is also represented since it can additionally be added in various quantities to the process, especially to steel scrap, to increase the quality of the output liquid steel.

Figure 91. Simplified flow diagram and CO₂ emissions of the EAF route, considering various inputs

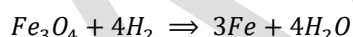


Source: JRC

Source: JRC report - Technologies to decarbonise the EU steel industry (Somers, 2022)

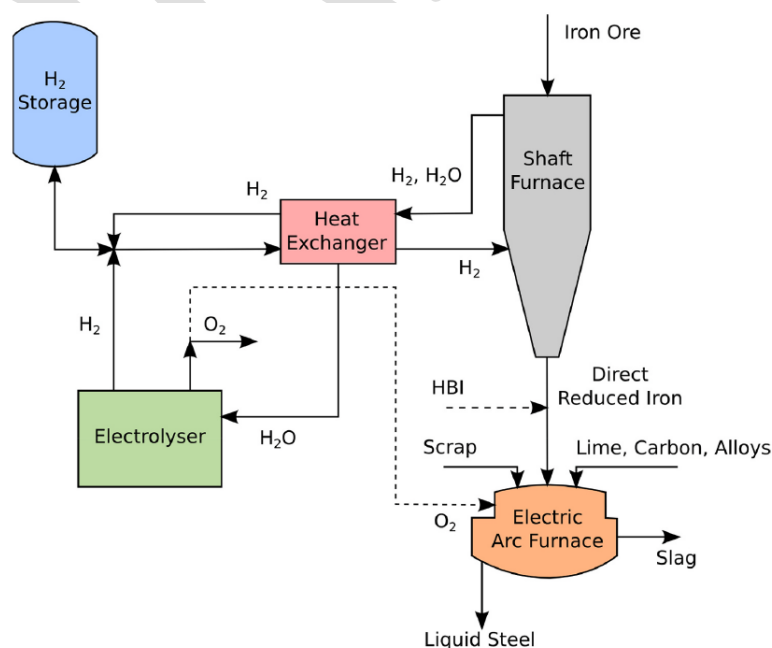
3.1.4 H2-DRI-EAF route

The hydrogen-based direct reduction iron route is similar, process-wise, to the classic EAF route described above. In the H2-DRI-EAF, the inputs are mainly DRI in addition to a small quantity of scrap. However, this time, the DRI has been produced with green hydrogen as the reducing agent, instead of natural gas. Green hydrogen is hydrogen produced through renewable electricity and electrolysis. The reactions to produce H2-DRI is roughly the same as the one for NG-DRI, but instead of emitting CO₂ through the reduction of the iron oxides with carbon, these oxides react with hydrogen producing only H₂O as a by-product, and no greenhouse gases:



In comparison to the traditional BF-BOF route, this alternative route offers a significant reduction in CO₂ emissions, with a potential decrease of 97-99% achievable through the utilisation of 100% hydrogen alongside renewable energy sources. Figure 92 illustrates a simplified flow diagram of this technology route.

Figure 92. Process flow diagram for H2-DRI-EAF



Source: Energy system requirements of fossil-free steelmaking using hydrogen direct reduction (Pimm, Cockerill, and Gale, 2021)

This route has a technology readiness level (TRL) of 8-9 (Agora Industry, Wuppertal Institute, and Lund University, 2024). While no steel site in the EU is currently producing steel at industrial scale from the H2-DRI EAF route, several projects such as H₂ Green Steel and Hybrit are ongoing, hoping to commercialize green steel in the coming years or decade. Many future scenarios for the steel industry in 2030 and 2050 anticipate rapid development of this technology. However, a drawback of this route is the current high prices and limited availability of hydrogen, which may pose challenges for industries. Some are addressing this issue by implementing a transition step, developing NG-DRI plants to gradually replace natural gas with H₂ (Draxler et al., 2021).

3.1.5 Overview of production routes

As outlined above, each steelmaking route possesses unique characteristics and implications for sustainability and efficiency. Table 35 presents a summary of these technology routes, detailing their Technology Readiness Level (TRL), CO₂ emissions reduction potential, energy demand, percentage of scrap input, along with their key advantages and drawbacks.

Table 35. Overview of production routes

Characteristic	BF-BOF	EAF	NG-DRI-EAF	H2-DRI-EAF
TRL level in the EU	9	9	9	6-8
Potential CO₂ reduction compared to BF-BOF	baseline	60%	35-67%	97-99%
Energy demand (Gj/ ton of crude steel)	22 Gj/ton	2-3 Gj/ton	12 Gj/ton	10.8 Gj/ton
% of scrap as input	Max 25%-30%	From 60% up to 100%	25%	25%
Advantages	Production of high-quality steel grades	Limited investment cost Flexibility (batch process)	No need for iron to be melted in order to reduce it	Potential decrease of 97-99% in CO ₂ emissions through the utilisation of hydrogen produced with 100% renewable energy
Drawbacks	Very CO ₂ intensive Large-scale operations only (low flexibility)	Quality of output highly dependent on scrap quality	Requires high grade iron ore as an input	Requires high grade iron ore as an input Current high prices and limited availability of hydrogen

Source: JRC elaboration based on Low-carbon technologies for the global steel transformation (Agora Industry et al., 2024); Technologies to decarbonise the EU steel industry (Somers, 2022); Hydrogen (H₂)-based ironmaking (World Steel Association, 2022); and The use of green steel in the automotive industry (Ricardo, 2024)

3.2 Analysis grid for the selection of intermediate steel representative products

In this section, Base Cases (BCs) for iron and steel products are proposed. The base cases are used as reference for modelling the stock of products together with their environmental and economic impacts and the available improvement design options (Task 4 and 5). The BCs reflect the average EU products, constituting an abstraction of the reality, a virtual product and not a real specific one.

As a first step to build the bases cases in the framework of this preparatory study, the selection of 5 representative products out of the 20 categories of the consolidated list of finished steel products presented in Task 2 have been made.

3.1.1 Goal and scope

For this preparatory study, a “base case” representing the average product of the EU must be defined. This “base case” will serve as the foundation for most of the environmental (Life Cycle Assessment) and economic (Life Cycle Cost) analyses conducted throughout the study (Task 4). Given the broad scope of steel as an

intermediate product in the ESPR context, as presented in Task 1 and Task 2, a targeted approach is necessary. **To streamline the assessment, a selection of five representative products (RPs) will be made,** prioritising the analysis and quantitative assessment on the most relevant products.

The list of finished steel products presented in Table 36 comes from the “Market analysis” (Task 2) with data developed by Fraunhofer based on a harmonisation of intermediate steel products from (EUROFER, 2023) and (Dworak, Rechberger, and Fellner, 2022).

Table 36. List of potential representative products

#	Product
1	Hot rolled plate
2	Electrical sheet
3	Hot rolled coil
4	Hot rolled narrow strip
5	Cold rolled coil
6	Cold rolled coil tinned
7	Hot rolled coil galvanised
8	Cold rolled coil galvanised
9	Cold rolled coil organic coating
10	Railway track material
11	Heavy sections
12	Light sections
13	Hot rolled bars
14	Concrete reinforced bars
15	Wire rod
16	Welded tube
17	Seamless tube
18	Casted steel products
19	Casted iron products
20	Stainless steel

Source: JRC elaboration

These intermediate products are situated at different steps of the steelmaking value chain, and can be classified as shown in Table 37. Notably, **this classification comprises flat and long products, casted as well as stainless steel categories.** No distinction is made between low-alloys and high alloys content.

To prevent double counting, each intermediate product is treated as a standalone entity, unrelated to others. (E.g. cold rolled coil tinned/galvanised/organic coated/strips are not included in the “cold rolled coil” category, the latter representing in fact “other cold rolled coil”). Additionally, stainless steel is covered as flat product, in the form of steel coils. Finally, “electrical sheets” refers more broadly to *electrical steel strips and sheets* with the majority of majority of the volumes being transferred between the manufacturer and the purchaser as coils. However, to keep consistency with EUROFER product categories, we grouped all the kind of electrical steels, including grain-oriented and non-grain oriented under the denomination “*electrical sheets*”.

Table 37. Intermediate steel product classification by steps of the value chain

Casted steel products	
Hot rolled	
Flat products	
Hot rolled plate	
Quarto plate	
Strips/coils	
Hot rolled coil	
Hot rolled coil galvanised	
Hot rolled narrow strip	
Cold rolled	
Cold rolled coil	
Cold rolled coil tinned	
Cold rolled coil galvanised	
Cold rolled coil organic coating	
Cold rolled strips	
Electrical sheet	
Welded tube*	
Long products	
Wire rod	
Railway track material	
Concrete reinforced bars	
Heavy sections	
Light sections	
Hot rolled bars	
Seamless tube*	
Casted iron products	(*tubes may be further hot or cold rolled)

Legend:

List of candidate for
Representative
Products

Source: JRC elaboration

3.1.2 Methodology and criteria for the selection of Representative Products (RPs)

The list of intermediate products includes 20 items that capture the diversity of the steel market. To rank and select the most relevant products, structured analysis framework was developed, featuring a multi-criteria evaluation grid. This grid evaluates each product based on several criteria, providing a current overview of steel products in the EU market.

EU market data from Task 2 served as the initial basis for capturing current EU consumption, production, and market dynamics for the year 2022. To gain a more comprehensive understanding, two additional dimensions were added: the CO₂ footprint of intermediate steel products and their economic importance. These considerations led to the development of five key criteria to evaluate each product:

Criteria ①: EU apparent **consumption**

Criteria ②: EU **production**

Criteria ③: EU supply dependencies (calculated *via* the product's **import reliance**)

Criteria ④: Estimated **CO₂ footprint** per tonne of steel products

Criteria ⑤: **Economic importance** related to steel end-use sectors

A ranking of the RP candidates was developed based on specific calculations and analysis for each criterion, as detailed below. For each of the five criteria, a proportional relevance score (ranging from 0 to 1) was assigned to each potential RP. The highest-ranked candidate received a score of 1, with subsequent candidates receiving proportional scores relative to the maximum value. If a value was negative (as in the case of import reliance), a score of 0 was assigned, ensuring scores ranged from 0 to 1 for each criterion per candidate.

Equal importance was given to all five criteria, with no weighting applied. The final score for each product was determined by summing the scores obtained across all criteria.

The product with the highest total score will be chosen as RP and prioritized for the later stages of this preparatory study.

3.1.2.1 Criteria ① – EU apparent consumption

In the context of the ESPR, a consumption-based approach is adopted, meaning that the potential policy intervention should target each product placed on the EU market. Therefore, the EU apparent consumption appears as the most relevant parameter to consider within the scope of this assessment.

The apparent consumption of a material can be expressed in kilotonnes (kt) per year, and refers to the estimated total amount of that material used within a given country or region, in this case the EU27, over the course of a year. This means taking into account both the material produced locally and obtained through international trade, while excluding the material sent out of the EU. Thus, the apparent consumption is calculated using the following formula:

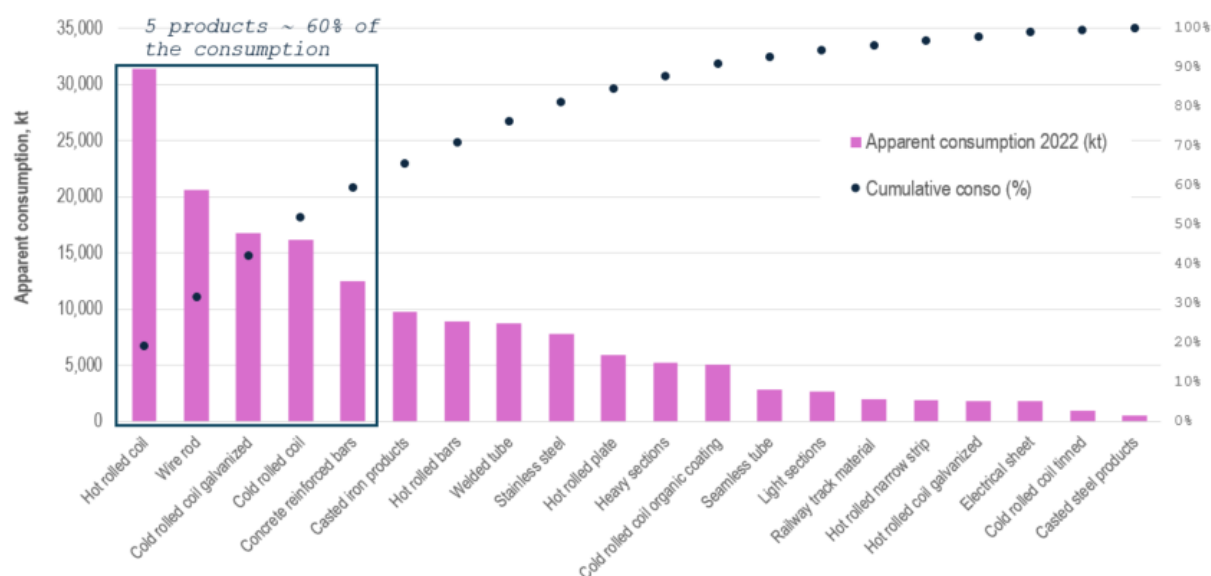
$$\text{Apparent Consumption} = \text{Production} + \text{Imports} - \text{Exports}$$

The data used for this calculation are retrieved from market analysis datasets, and cover the year 2022 as the reference year. The value are expressed in kilo tonnes (kt) per year.

Figure 93 shows the EU apparent consumption values for the 20 steel product categories, along with the corresponding cumulative percentage of consumption. The data indicate that consumption is predominantly concentrated among a limited number of products, with Hot rolled coiled (31,422 kt), Rod and wire rod (20,594 kt), and Galvanised Cold Rolled coil (16,788 kt) collectively accounting for 40% of the EU's steel consumption. Furthermore, the top nine intermediate steel products represent approximately 80% of the total apparent consumption.

Table 38 summarises the scores obtained for each product, as calculated according to the methodology outlined above.

Figure 93. Apparent consumption of candidate for representative products (bars) expressed in kt per year, for an average period 2020-22 and associated cumulative percentage (dots)



Source: JRC elaboration based on data from market analysis

Table 38. Ranking of the 20 intermediate steel products according to criteria 1

	CRIT 1 - EU Consumption score
Hot rolled Coil	1.00
Rod and wire rod	0.66
Galvanised Cold rolled coil	0.53
Cold rolled coil	0.51
Bars for reinforcement of concrete	0.40
Hot rolled plate (e.g. Quarto plate)	0.31
Casted iron products	0.28
Other hot rolled bars (e.g. merchant bars)	0.28
Welded tube	0.25
Heavy sections and mining frames	0.19
Organically coated and other coated products	0.17
Stainless steel products	0.16
Seamless tube	0.09
Light sections	0.09
Railway track material and other long products	0.06
Galvanised Hot rolled coil	0.06
Electrical sheet	0.06
HRC Narrow strip	0.06
Tinplate and tinned products	0.03
Casted steel products	0.02

Source: JRC elaboration

3.1.2.2 Criteria ② – EU production

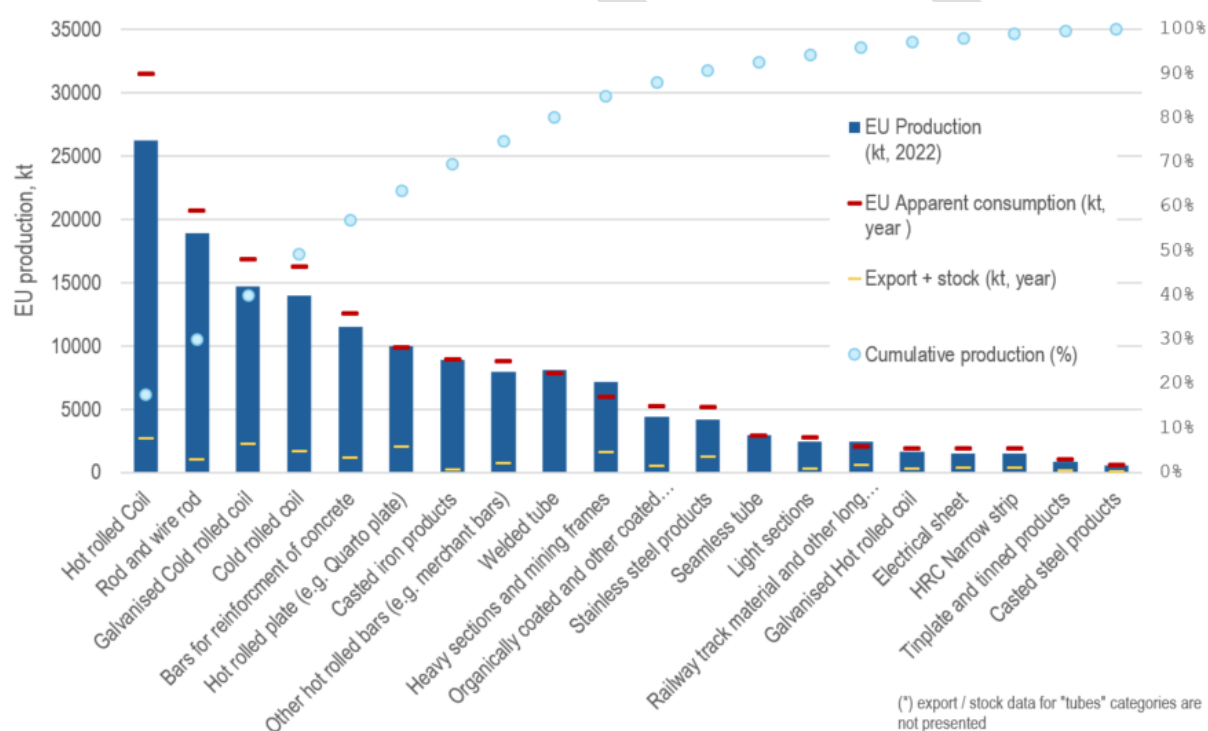
The EU disposes of important production capacities for iron and steel across Member States. In 2023, the production of steel represented 126 Million tonnes or 7% of the global production. Steel production in the EU ranks second worldwide, trailing behind China but surpassing the production levels of North America (109.6 Mt), Russia or the Middle East.

Data are retrieved from the market analysis using figures and shares of steel production, imports and consumption in EU from (Harvey, 2022), (Dworak, Rechberger, and Fellner, 2022), Worldsteel, Worldstainless and World Foundry Association (WFO).

Figure 94 presents the EU production values for the 20 steel product categories, along with the cumulative percentage of production. Similarly to consumption figures, the data reveal a concentrated production profile, with 9 products accounting for 80% of the total production. Notably, hot rolled coil, rod and wire rod, galvanised cold rolled, and Cold rolled coil collectively represent nearly 50% of the EU's steel production. For comparison purposes, apparent consumption figures from the previous analysis are also displayed in the figure, along with the estimated "EU exports and stocks".

Table 39 summarises the scores obtained for each product, as calculated according to the methodology outlined above.

Figure 94. EU production of candidate for representative products (bars) expressed in kilotonnes per year, for an average period 2020-22 and associated cumulative percentage (dots). EU apparent consumption and exports (stacks) are displayed for comparison purpose



Source: JRC elaboration based on data from market analysis

Table 39. Ranking of the 20 intermediate steel products according to criteria 1

	CRIT 1 – EU Consumption Score	CRIT 2 – EU production score
Hot rolled Coil	1.00	1.00
Rod and wire rod	0.66	0.72
Galvanised Cold rolled coil	0.53	0.56
Cold rolled coil	0.51	0.53
Bars for reinforcement of concrete	0.40	0.44
Hot rolled plate (e.g. Quarto plate)	0.31	0.38
Casted iron products	0.28	0.34
Other hot rolled bars (e.g. merchant bars)	0.28	0.30
Welded tube	0.25	0.31
Heavy sections and mining frames	0.19	0.27
Organically coated, other coated products	0.17	0.17
Stainless steel products	0.16	0.16
Seamless tube	0.09	0.11
Light sections	0.09	0.09
Railway track material, other long products	0.06	0.09
Galvanised Hot rolled coil	0.06	0.06
Electrical sheet	0.06	0.06
HRC Narrow strip	0.06	0.06
Tinplate and tinned products	0.03	0.03
Casted steel products	0.02	0.02

Source: JRC elaboration

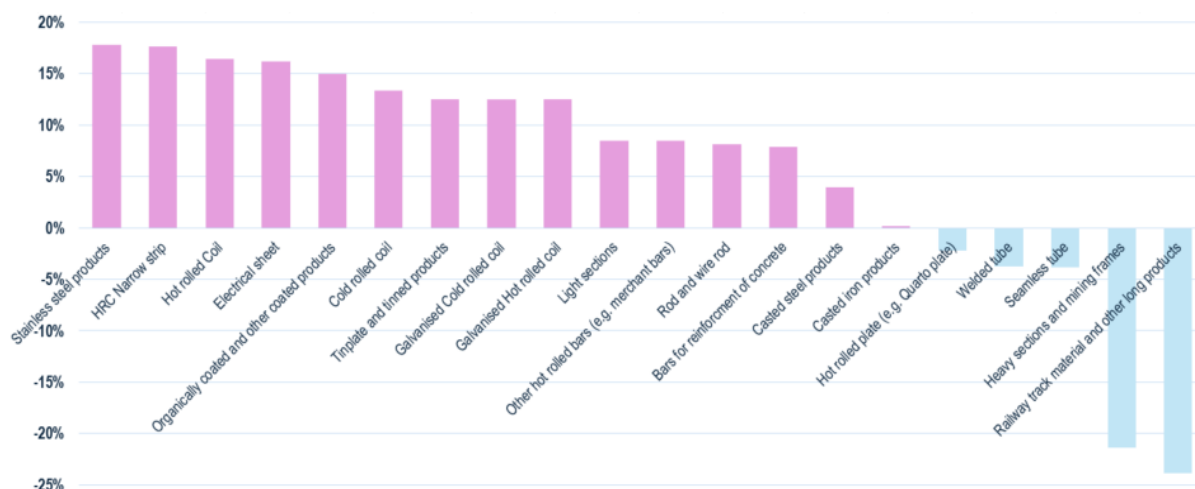
3.1.2.3 Criteria ③ – EU supply dependencies

Criteria 3 relates to EU iron and steel products supply dependency, which is calculated following the import reliance (%) formula:

$$IR = \frac{(\text{Imports} - \text{Exports})}{(\text{Domestic Production} + \text{Imports} - \text{Exports})}$$

The import reliance is one of the main parameters used to compute the supply risk of (critical) raw materials and illustrate the supply dependencies the EU is facing for steel intermediate products. The import reliance is calculated and displayed in Figure 95.

Figure 95. Import reliance for each candidate RP. Negative values means that the EU is net exporter and does not face supply dependencies



Source: JRC elaboration

Notably, the EU has shifted from being a net exporter in previous decades to a net importer for 14 out of 20 products in recent years. However, the import reliance remains generally limited, with most products exhibiting a reliance of less than 12%. The highest values are observed for stainless steel products, and narrow strips each with an import reliance of around 18%, followed by hot rolled coil and electrical sheets with around 16%.

In contrast, hot rolled plates, tubes, heavy sections, and railways track exhibit negative import reliance values, indicating that the EU's exports surpass its imports. Although the overall consumption of these products in kilotonnes (kt) is limited.

Table 40 presents the scores obtained for each product according to the methodology defined above.

Table 40. Ranking of the 20 intermediate steel products according to criteria 3

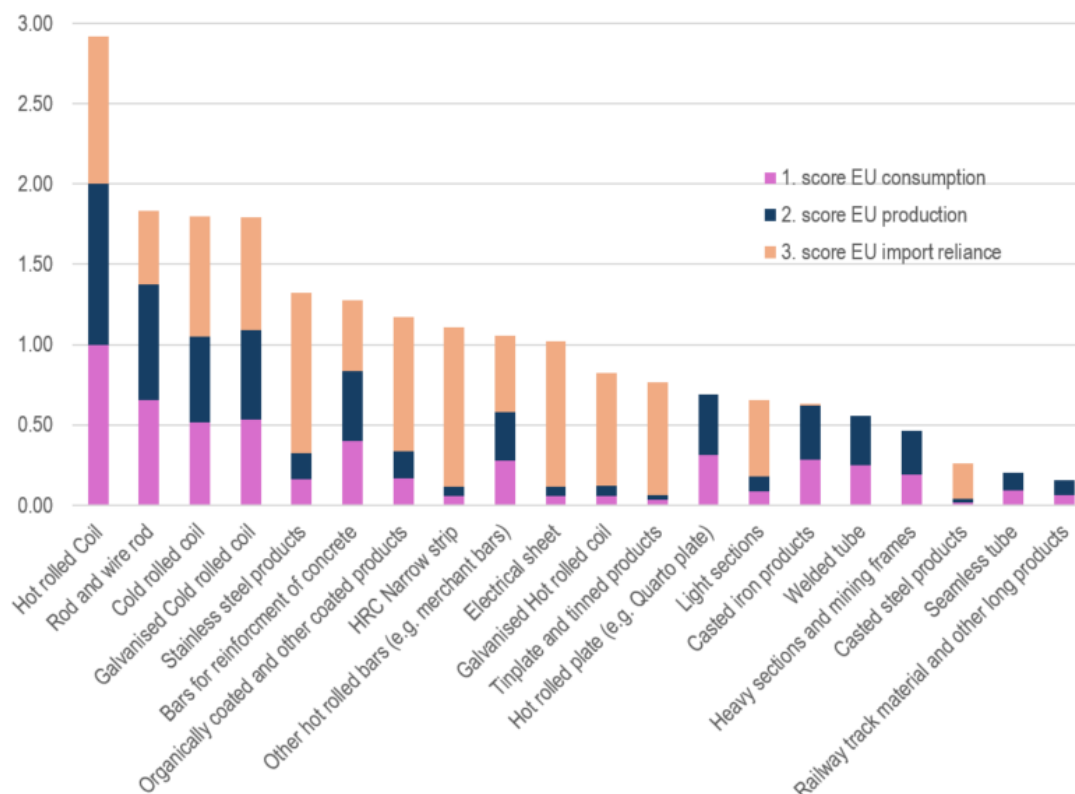
	CRIT 1 – EU Consumption score	CRIT 2 – EU Production score	CRIT 3 – EU Import reliance score
Stainless steel products	0.16	0.16	1.00
HRC Narrow strip	0.06	0.06	0.99
Hot rolled Coil	1.00	1.00	0.92
Electrical sheet	0.06	0.06	0.91
Organically coated, other coated products	0.17	0.17	0.84
Cold rolled coil	0.51	0.53	0.75
Tinplate and tinned products	0.03	0.03	0.70
Galvanised Cold rolled coil	0.53	0.56	0.70
Galvanised Hot rolled coil	0.06	0.06	0.70
Light sections	0.09	0.09	0.48
Other hot rolled bars (e.g. merchant bars)	0.28	0.30	0.48
Rod and wire rod	0.66	0.72	0.46
Bars for reinforcement of concrete	0.40	0.44	0.44
Casted steel products	0.02	0.02	0.22
Casted iron products	0.28	0.34	0.01
Hot rolled plate (e.g. Quarto plate)	0.31	0.38	0.00
Welded tube	0.25	0.31	0.00
Heavy sections and mining frames	0.19	0.27	0.00
Seamless tube	0.09	0.11	0.00
Railway track and other long products	0.06	0.09	0.00

Source: JRC elaboration

The scores obtained for the first 3 criteria are aggregated and displayed in Figure 96. The interim ranking reveals that the top product groups are: *Hot rolled Coil*, *Rod and wire rod*, *Galvanised cold rolled coil*, *Cold rolled coil* and *Stainless steel products*.

Notably, the top four product groups combine high EU consumption and production levels with significant or non-negligible import reliance. In contrast, Stainless steel products rank fifth despite lower volumes of consumption and production, due to their high import reliance.

Figure 96. Interim ranking based on market analysis datasets for criteria 1, 2, and 3.



Source: JRC elaboration

3.1.2.4 Criteria ④ – Estimated CO₂ footprints

Beyond market dynamics, having a first glance at the CO₂ impacts to rank the candidates for RPs appear relevant when considering the necessary path to decarbonised steel production as well as the implementation of CO₂-based trade mechanisms such CBAMs or ETS.

If each product is manufactured to meet specific performance requirements, the fact that some products have a higher CO₂ footprint than others may indicated a higher untapped potential for decarbonisation. Hence, this criteria is important to consider to select the best representative products, which will undergo further analysis through a comprehensive LCA-based environmental footprint.

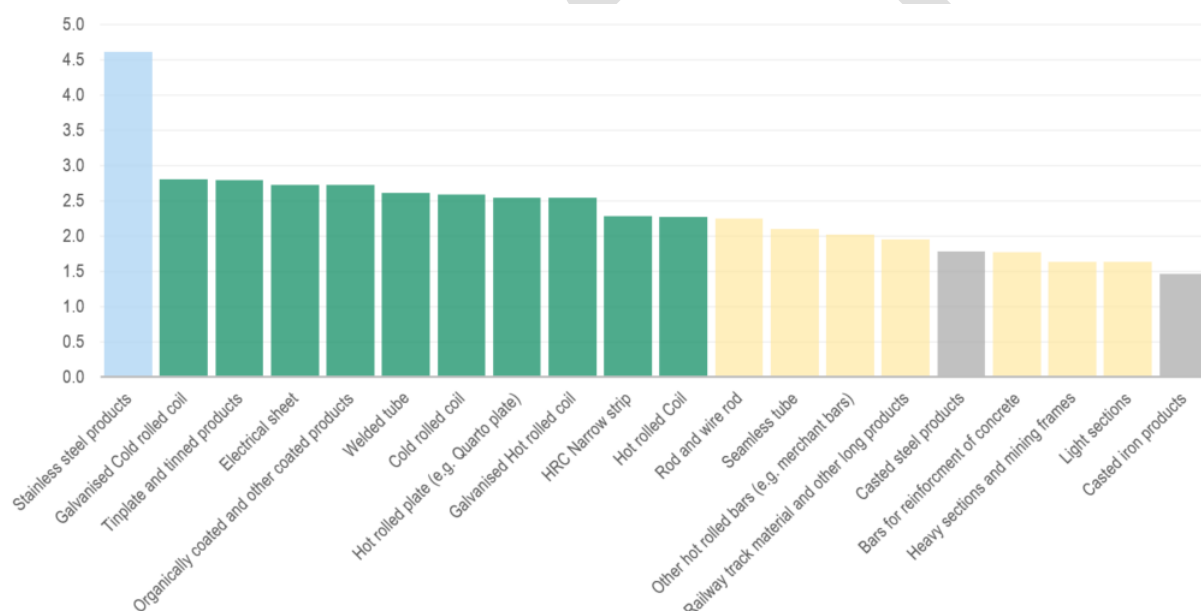
A literature review was conducted to compile CO₂ footprint of the intermediate steel products, under scopes 1 to 3. Several references such as the Environmental Footprint (EF) datasets, Worldsteel LCI datasets, ecoinvent database or the CBAM defaults values have been consulted. Unfortunately, none of the sources cover the 20 product categories in a homogeneous way. The EF datasets recommended by the EC are available for only a limited number of products. The geographical scope is mostly global, depending on the references used and listed in **Error! Reference source not found.** The use of global data is justified since the focus of ESPR is on the EU market (consumption approach), where products from third countries are also imported. It should be noted that these estimates provide an initial order of magnitude for ranking the products, but they are not exhaustive and may involve some approximations. A comprehensive analysis of the environmental profile will be conducted in Task 4 for the selected representative products.

While the scope of CBAMs is broader and covers 70 different trade codes, the use of a “default value” would penalised the EAF production routes by applying only values from primary routes. This approach would be acceptable for flat products, but most long products manufactured via EAF would be unfairly penalised. The most comprehensive datasets come from world steel and provide average profiles covering global production, which is particularly representative of the steel product imported into the EU markets. When dataset were lacking, values from the ecoinvent LCA database or literature were considered.

Figure 97 shows the compiled CO₂ footprint for each product category. Due to the carbon intensity of its alloying elements (nickel and chromium), stainless steel has the highest CO₂ footprint, followed by products typically manufactured via the BF/BOF production routes (flat products). The CO₂ footprint for flat products ranges from 2.3 tonnes of CO₂ per tonne of product (hot rolled coil) to 2.8 tonnes of CO₂ per tonne of product when additional surface treatment is needed (e.g. galvanised cold rolled coil). Long products, often manufactured using the EAF route, present less carbon-intensive profiles, mostly ranging between 1.6 and 2.2 tonnes of CO₂ per tonne of product. These results should be interpreted with caution as they are derived from diverse sources reported in Table 41 and high-level models. A more detailed assessment will be conducted for the most relevant products in Task 4 of this preparatory study.

Finally multiplying the estimated consumption footprint by the apparent consumption, we obtained the estimated EU consumption footprint for each of the products under assessment (see Figure 98).

Figure 97. Carbon footprint of candidate for representative products expressed in tonnes of CO₂ per tonne of product (scope: global production average)



Note. Various data sources were used to cover the entire list of candidates. Colour code: light blue: stainless steel, green: flat products, yellow: long products, grey: cast products

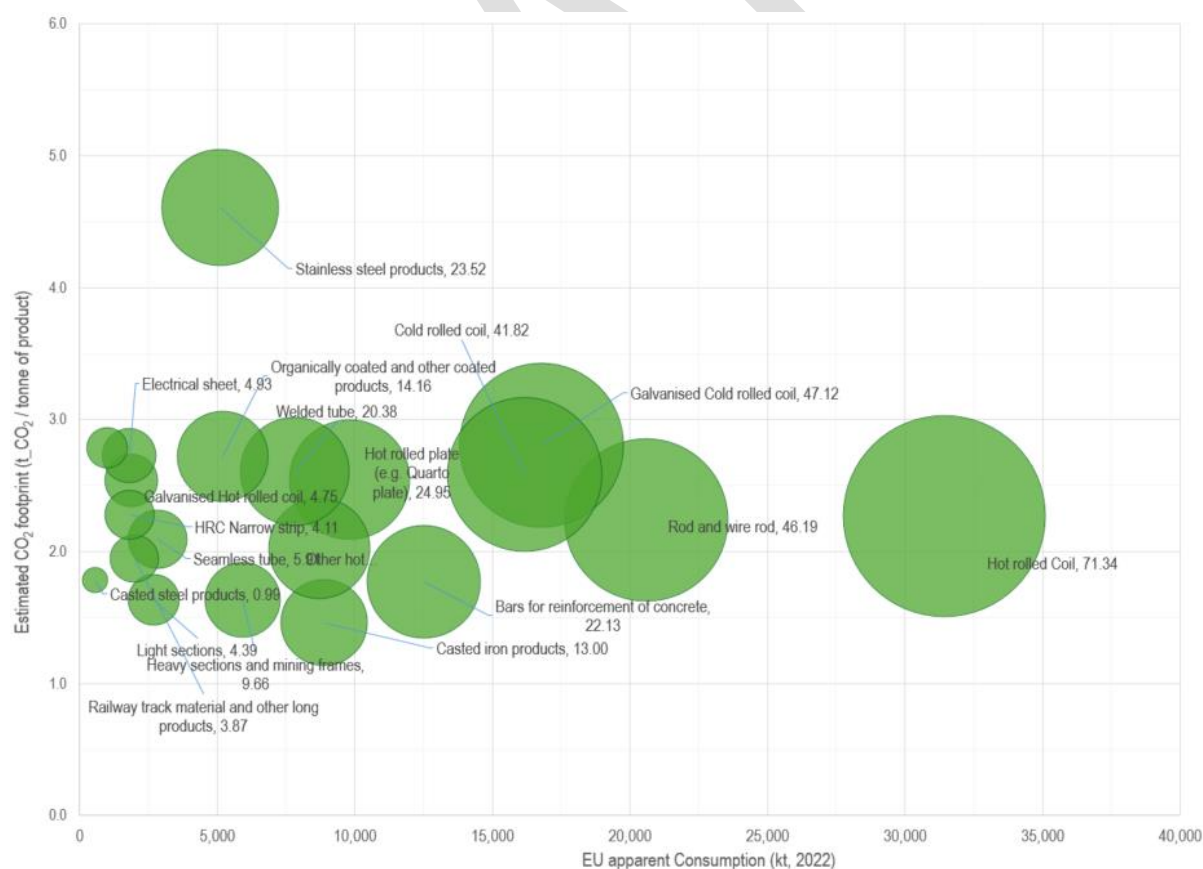
Source: JRC elaboration based on Worldsteel LCI datasets, CBAMs, EF datasets, Nordelof et al. (2017) and ecoinvent database

Table 41. Ranking of the 20 intermediate steel products according to criteria 4

		t.CO2/t _{product}	CRIT 4 - CO ₂ footprint score	
stainless	Stainless steel products	4.61	Ecoinvent	1.00
flat	Galvanised cold rolled coil	2.81	EF dataset	0.61
flat	Tinplate and tinned products	2.79	Worldsteel 2020	0.60
flat	Electrical sheet (non-grain oriented)	2.73	(Nordelof et al.)	0.59
flat	Organically coated and other coated products	2.73	Worldsteel 2020	0.59
long	Welded tube	2.61	Worldsteel 2020	0.57
flat	Cold rolled coil	2.59	Worldsteel 2020	0.56
flat	Hot rolled plate (e.g. quarto plate)	2.55	Worldsteel 2020	0.55
flat	Galvanised hot rolled coil	2.54	Worldsteel 2020	0.55
flat	HRC narrow strip	2.28	CBAM, 2023	0.49
flat	Hot rolled coil	2.27	Worldsteel 2020	0.49
long	Rod and wire rod	2.24	Worldsteel 2020	0.49
long	Seamless tube	2.10	Worldsteel 2020	0.45
long	Other hot rolled bars (e.g. merchant bars)	2.02	Worldsteel 2020	0.44
long	Railway track material and other long products	1.95	CBAM, 2023	0.42
cast	Casted steel products	1.79	EF dataset	0.39
long	Bars for reinforcement of concrete	1.77	Worldsteel 2020	0.38
long	Heavy sections and mining frames	1.64	Worldsteel 2020	0.35
long	Light sections	1.64	Worldsteel 2020	0.35
cast	Casted iron products	1.46	EF dataset	0.32

Source: JRC elaboration

Figure 98. Estimated EU consumption footprint (CO₂) for intermediate steel categories



Note. The Size of the bubble represents the amount of CO₂ eq. emissions for yearly EU consumption of each candidate representative products, and is obtained multiplying apparent consumption by estimated CO₂ footprint per tonne of product consumed.

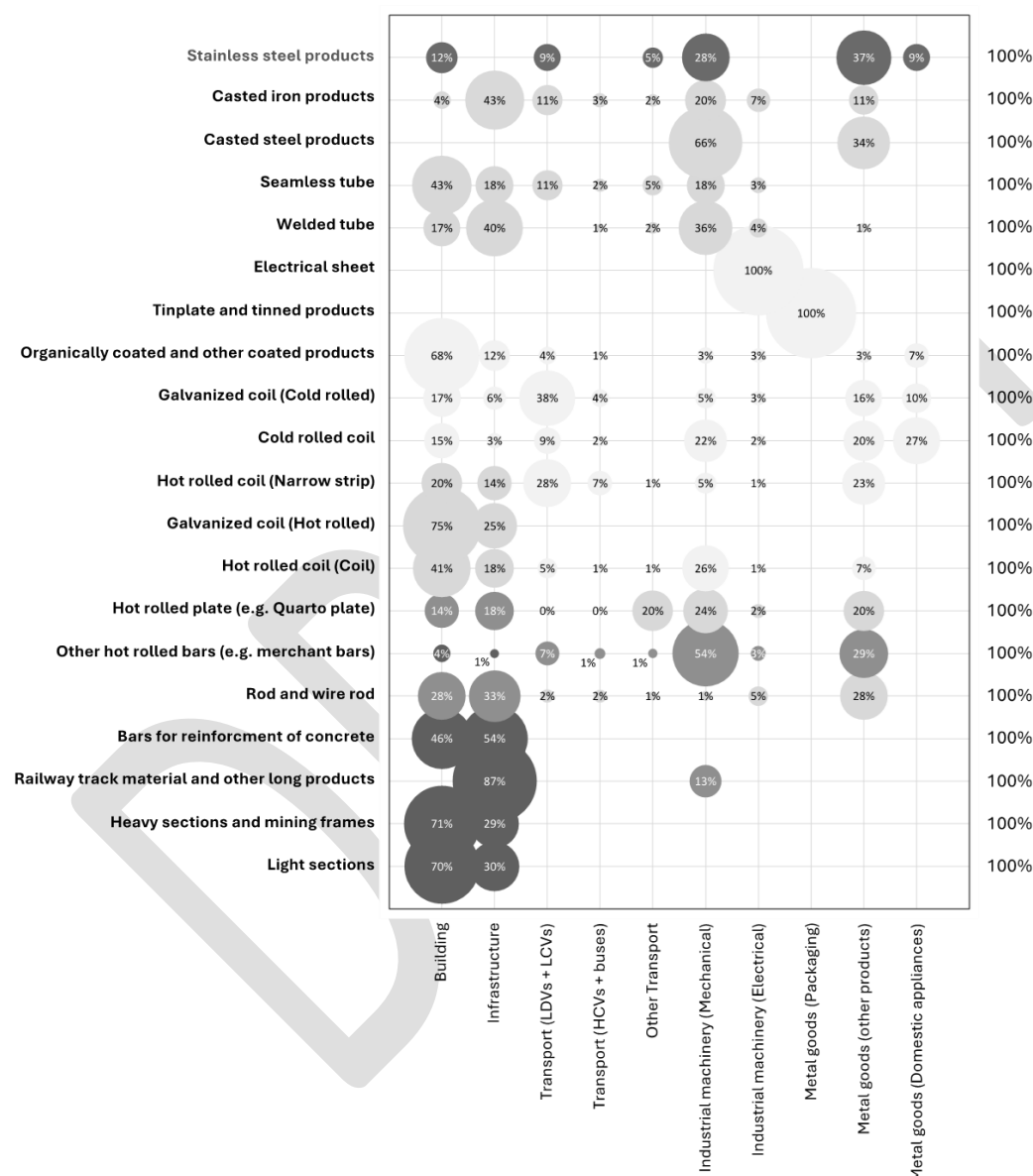
Source: JRC own elaboration

3.1.2.5 Criteria ⑤ – Targeted economic sectors

The last criteria selected for this study focus on the end-use sectors where each product is used. The main objective of this criteria is to capture the strategic nature of the end-use sectors where steel is used by estimating the economic importance (EI) of each sector.

By analogy with the Critical raw material list methodology (Blengini et al. 2017), we propose to link end-use shares of steel products with manufacturing sectors. Figure 99 illustrates the distribution of each RPs across various end-use sectors. For example, 38% of *cold rolled coil galvanised* is used in the sector “Light transport” sector, while 27% of the *cold rolled coiled* is used in “Domestic appliance” sector.

Figure 99. Shares of the end-use sectors in which each intermediate products is used



Source: Ciacchi, 2024 for JRC based on initial model from (Dworak, Rechberger, and Fellner, 2022)

The economic importance of each sector varies significantly. Building upon previous assessment for CRMs list (European Commission, 2023), more particularly for iron ores, the NACE codes are used to estimate the gross Value Added (VA) for each end-use sector (see Table 42).

Table 42. Gross Value Added (VA) value of each sector distributed according to the NACE code

End-use sectors	NACE-2 code	Value Added (M€)	Description
Metal goods (Domestic appliances)	C28	204,200	Manufacture of machinery and equipment n.e.c.
Metal goods (Other)	C24	64,561	Manufacture of basic metals
Metal goods (Packaging)	C24	64,561	Manufacture of basic metals
Industrial machinery (Electrical)	C27	89,422	Manufacture of electrical equipment
Industrial Machinery (Mechanical)	C28	204,200	Manufacture of machinery and equipment n.e.c.
Transport (Other)	C30	55,777	Manufacture of other transport equipment
Transport (HCV + buses)	C29	194,448	Manufacture of motor vehicles, trailers and semi-trailers
Transport (LDV + LCV)	C29	194,448	Manufacture of motor vehicles, trailers and semi-trailers
Infrastructure	C25	163,568	Manufacture of fabricated metal products, except machinery and equipment
Buildings	C25	163,568	Manufacture of fabricated metal products, except machinery and equipment

Source: JRC elaboration based on Study on the critical raw materials for the EU 2023 – Final report (EC, 2023)

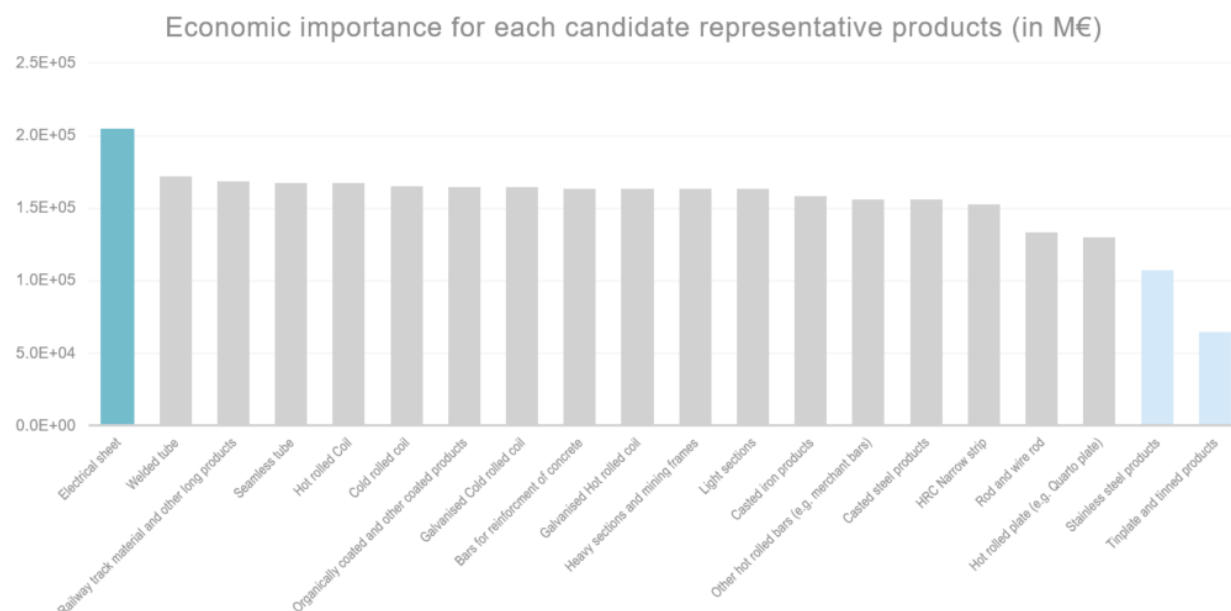
The economic importance (EI) calculation formula is then used as follows (substitution potential of the materials is disregarded):

$$EI_{\text{steel Representative Product}} = \sum (As * Qs)$$

Where:

- EI stands for economic importance for one candidate for representative products (e.g. cold rolled coil);
- As is the share of a certain steel RP (%) used in a NACE Rev. 2 2-digit level sector (taken from Figure 99);
- Qs is the Value Added of the 2-digit-level NACE Rev. 2 sector (taken from Table 42).

Figure 100. Estimated economic importance (EI) expressed in gross value added (M€) for end-sectors where steel candidate for representative products are used



Source: JRC elaboration

The results reveal that the highest values are driven by the VA of industrial machinery (C28), Transport (C29) and to a lesser extent Buildings and Infrastructures (C25). For instance, electrical sheet is used in C28 sector which encompasses wind turbine generators (subcategory C2811). However, the variability of the results are rather limited, the assessment allowing to discriminate only few candidate RPs: the ones going to sectors with a lower value added, mainly packaging (for stainless steel and tinned products), and the one going to high value added like industrial machinery (for electrical sheet).

The ranking according to criteria 5 is presented in Table 43.

Table 43. Ranking of the 20 intermediate steel products according to criteria 5 on Economic Importance (EI) of end-sectors

	Economic Importance (EI) in M€	CRIT 5 – EI score of the end-sectors
Electrical sheet (non-grain oriented)	2.0E+05	1.00
Welded tube	1.7E+05	0.84
Railway track material, other long products	1.7E+05	0.83
Seamless tube	1.7E+05	0.82
Hot rolled Coil	1.7E+05	0.82
Cold rolled coil	1.6E+05	0.81
Organically coated and other coated products	1.6E+05	0.81
Galvanised Cold rolled coil	1.6E+05	0.81
Bars for reinforcement of concrete	1.6E+05	0.80
Galvanised Hot rolled coil	1.6E+05	0.80
Heavy sections and mining frames	1.6E+05	0.80
Light sections	1.6E+05	0.80
Casted iron products	1.6E+05	0.78
Other hot rolled bars (e.g. merchant bars)	1.6E+05	0.77
Casted steel products	1.6E+05	0.77
HRC Narrow strip	1.5E+05	0.75
Rod and wire rod	1.3E+05	0.65
Hot rolled plate (e.g. Quarto plate)	1.3E+05	0.64
Stainless steel products	1.1E+05	0.53
Tinplate and tinned products	6.5E+04	0.32

Source: JRC elaboration

3.1.3 Final ranking of intermediate products

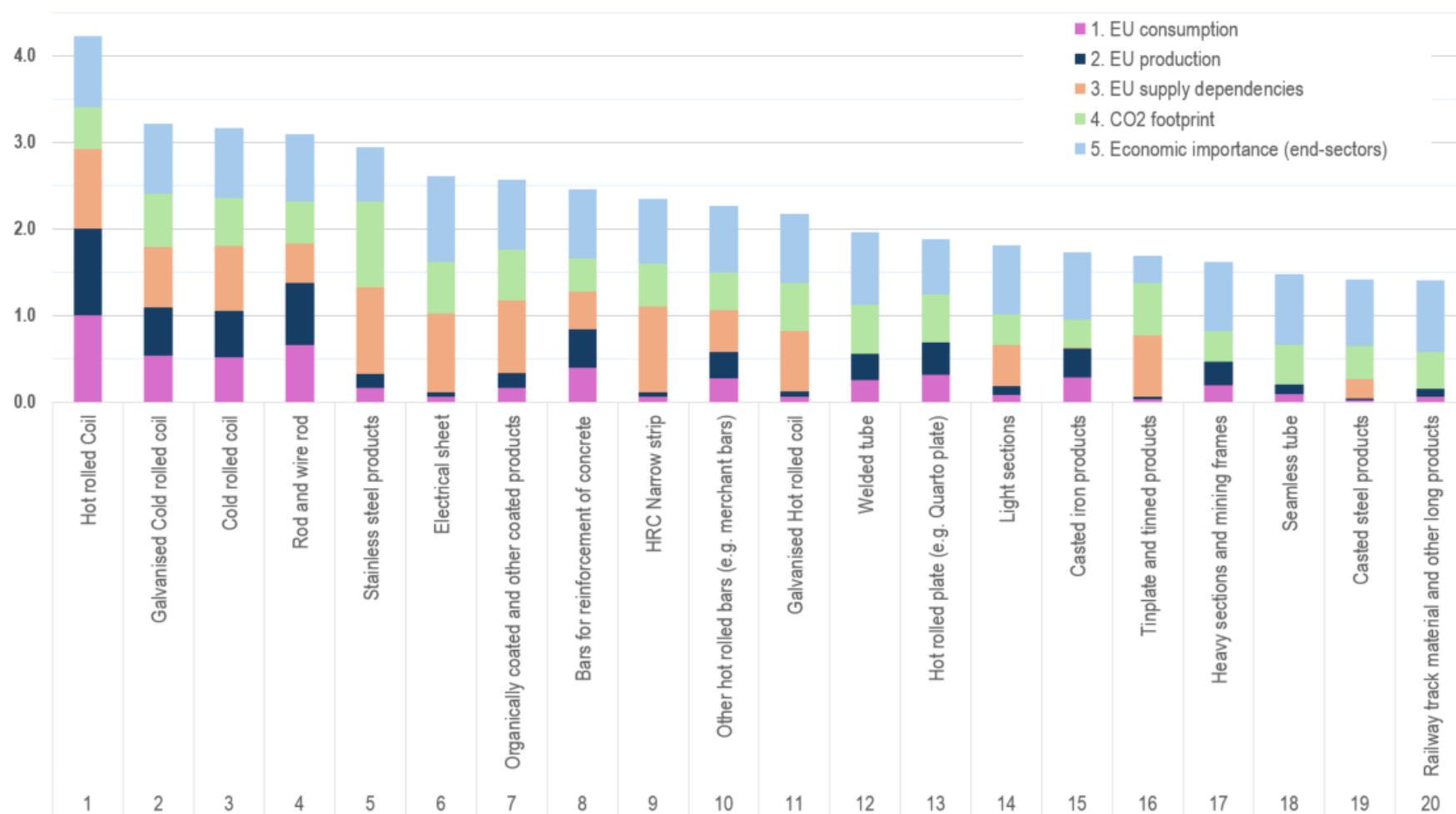
The final ranking of the RP is presented in Table 44 and Figure 101. With the ranking from 0 to 1 of each candidate for RP for each of the 5 criteria, which is determined by aggregating the scores from each of the 5 criteria. The total score is calculated by summing these individual scores.

Table 44. Final ranking of the 20 candidates for RP

rank		EU Consumption score	EU Production score	EU Import reliance score	CO2 footprint score	Economic Importance score	Total
1	Hot rolled Coil	1.00	1.00	0.92	0.49	0.82	4.23
2	Galvanised Cold rolled coil	0.53	0.56	0.70	0.61	0.81	3.21
3	Cold rolled coil	0.51	0.53	0.75	0.56	0.81	3.17
4	Rod and wire rod	0.66	0.72	0.46	0.49	0.77	3.09
5	Stainless steel products	0.16	0.16	1.00	1.00	0.62	2.94
6	Electrical sheet	0.06	0.06	0.91	0.59	1.00	2.61
7	Organically coated, other coated products	0.17	0.17	0.84	0.59	0.81	2.57
8	Bars for reinforcement of concrete	0.40	0.44	0.44	0.38	0.80	2.46
9	HRC Narrow strip	0.06	0.06	0.99	0.49	0.75	2.35
10	Other hot rolled bars (e.g. merchant bars)	0.28	0.30	0.48	0.44	0.77	2.27
11	Galvanised Hot rolled coil	0.06	0.06	0.70	0.55	0.80	2.17
12	Welded tube	0.25	0.31	0.00	0.57	0.84	1.96
13	Hot rolled plate (e.g. Quarto plate)	0.31	0.38	0.00	0.55	0.64	1.89
14	Light sections	0.09	0.09	0.48	0.35	0.80	1.81
15	Casted iron products	0.28	0.34	0.01	0.32	0.78	1.73
16	Tinplate and tinned products	0.03	0.03	0.70	0.60	0.32	1.69
17	Heavy sections and mining frames	0.19	0.27	0.00	0.35	0.80	1.62
18	Seamless tube	0.09	0.11	0.00	0.45	0.82	1.48
19	Casted steel products	0.02	0.02	0.22	0.39	0.77	1.42
20	Railway track material, other long products	0.06	0.09	0.00	0.42	0.83	1.41

Source: JRC elaboration

Figure 101. Final ranking according to the 5 selected criteria



Source: JRC own elaboration

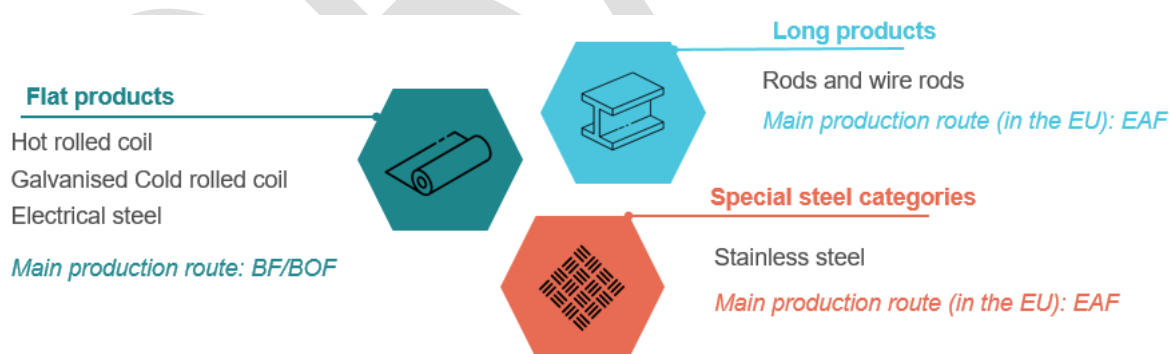
Table 44 shows that the intermediate steel products having the highest scores are: **Hot rolled coil, Galvanised cold rolled coil, Cold rolled coil, Rod and Wire rod and Stainless steel**. Additionally, the choice of the final 5 RP takes more reflexions into account, such as the followings:

- Cold rolled coil (CRC) and galvanised cold rolled coil (gCRC) differ only in the hot-dip galvanisation of the product. The potential intervention on the Cold rolled coil would by definition also apply to the downstream steps of the value chain, including the cold rolled coil galvanised. The choice between the two intermediate steel products has been to keep the cold rolled coil galvanised as a representative product.
- Stainless steel faces a more limited EU production than the products ranking first. This leads to a substantial import reliance (18%). In terms of volume, its associated CO₂ consumption footprint is of the same order of magnitude as the most consumed products (as displayed in Figure 98).
- The development of renewable energy as well as electric cars in the coming decades highlight the growing need for **electrical steel**. Indeed, “Electrical steel is considered a critical material for its role in modernising electrical grid infrastructure and transitioning the EU to clean energy.” (Carrara et al., 2023). The demand for “electrical sheet” is expected to increase significantly in the coming decades in strategic sectors. Hence, due to its strategic nature electrical sheet has been chosen as a RP, despite its ranking driven by a current low production and consumption in the EU.

In conclusion, the five representative products selected through this grid analysis and further studied in the LCA analysis are:

- ❖ **Hot rolled coil**
- ❖ **Cold rolled coil galvanised**
- ❖ **Rod and Wire rod**
- ❖ **Stainless steel**
- ❖ **Electrical sheet**

Figure 102. Final selection of the five representative products for the ESPR preparatory study on iron and steel



Source: JRC elaboration

The list of representative products will support the scoping of the further assessment. The base cases scenarios developed in the next section (see Section 3.3) are based on these five main products.

Additionally, **data collection** will focus on these 5 five representative products to establish “bill of materials” and base cases for the EU industry current situation and forecast for 2035 and associated inventory datasets. This will serve as the foundation for the **environmental footprint assessment** (Task 4), including the identification of associated hotspots and the evaluation of **design options** that could inform potential future policy interventions.

3.3 Definition of the Base Cases

In an EU Ecodesign preparatory study, the base case refers to a reference scenario or baseline that represents the current state of a product's design, manufacturing, and use. It is used as a benchmark to evaluate and compare potential design improvements and leading to potential future policy interventions.

The present study focuses on steel intermediate products placed on the EU market. For each representative product selected in the previous section **Error! Reference source not found.**, a quantitative analysis is performed. These base cases will serve for the modelling of the environmental (Life Cycle Assessment) and economic (Life Cycle Costing) analyses conducted throughout the study (Task 4).

3.3.1 Selected representative products

3.3.1.1 Hot rolled coil

Hot Rolled Coil (HRC) is a type of flat steel product produced from semi-finished slabs. It is manufactured by re-rolling semi-finished slabs at high temperatures (typically above 1000°C) either in plate mills (producing quarto plates) or in hot strip mills (producing strips and coils). The resulting HRC, with a thickness starting from 1 mm and above, can be cut to length or shipped directly to steel users and downstream processors.

HRC offers higher yield strength compared to other steel types, making it suitable for high-stress applications. It also exhibits excellent abrasion resistance, making it ideal for environments prone to wear and tear. Additionally, HRC possesses good welding and shaping properties, facilitating easy fabrication into parts and components. HRC is widely used in the construction and manufacturing industries as a fundamental material for producing steel pipes, wires, sheets, and other products (Eurofer, 2024; Stavian, 2024).

3.3.1.2 Cold rolled coil galvanised

Cold Rolled Coil (CRC) is a type of flat steel produced from semi-finished slabs that have undergone a hot rolling process. CRC is created by cold rolling hot-rolled sheets in a cold rolling mill at room temperature, which compresses and shapes the material into thinner and more precise coils. Before cold rolling, the hot-rolled steel is pickled to remove surface scale and impurities. This process enables tighter tolerances and the production of thinner gauges of steel (Stavian, 2024).

Galvanisation involves applying a protective zinc coating to steel to prevent rusting, with hot-dip galvanising being the most common method. In this process, a cold rolled coil of steel is unrolled at the entry of the galvanising line. The steel threads through a welder, where the tail end of one coil is joined to the head end of the next coil to ensure continuous operation. The strip then passes through an accumulator system before traveling into a molten zinc bath, where it receives a zinc alloy coating. Afterward, the strip proceeds through a cooler to harden the coating (AIST, 2024). Galvanised steel is particularly suited for applications requiring enhanced corrosion resistance (Eurofer, 2024).

Cold rolled, galvanised steel combines the benefits of both cold rolling and galvanising. It is widely used in automotive applications (e.g., cars, buses, trucks, and trains), shipbuilding, white goods (e.g., appliances), consumer durables, and for further processing into coated sheet products. Its enhanced corrosion resistance makes it ideal for environments requiring superior durability (Eurofer, 2024).

3.3.1.3 Wire rod

A rod is a hot rolled long steel product having a nominal size generally of 5mm or above, and wound into irregular coils. The cross section may be round, oval, square, rectangular, hexagonal, octagonal, half round or of any similar shape. Its surface is smooth. Rod is generally intended to undergo further processing, although it may also be used without further processing (EN10079).

A wire is a product of constant full cross section along its length, obtained by cold drawing rod through a reducing die or passing under pressure between rollers and rewinding the drawn product. The cross section is generally round, although it can also be found in other shapes (EN10079).

A wire rod is therefore a type of long steel product, the result of a hot rolled rod mill, generally with a diameter range of 5.5 to 20.6 mm (AIST, 2024). It is produced from semi-finished billets, usually wound up in coils.

The mechanical properties and potential applications of wire rods will depend on the different grades and alloys of wire rods. They are normally used to produce steel wires and, with further processing, bright bars (Eurofer, 2024). The use of wire is extremely wide, ranging from cords for belted tires to cables for suspension bridges.

3.3.1.4 Electrical sheet

Electrical steel is a type of flat steel product, characterised by its magnetic properties. It is often supplied in the form of cold rolled sheet or strip, generally less than 2 mm thick and of width up to and 1500 mm. There are also certain hot rolled flat products in thicknesses of 1.5 mm up to 5 mm with specified mechanical and magnetic properties (EN10079).

Electromagnets for alternating current are always made by laminating many thin sheets, which are insulated in order to minimize the flow of eddy currents and thereby reduce current losses and heat generation. A further improvement is achieved by adding up to 4.5% silicon, which imparts high electrical resistance. For electric transformers, grain-oriented sheets are often used; these contain about 3.5% silicon and are rolled and annealed in such a way that the edges of the unit cubes are oriented parallel to the direction of rolling. This improves the magnetic flux density by about 30% (Britannica, 2024).

Electrical sheet is intended for use in magnetic circuits in electrical machines (EN10079), for the generation and transmission of electrical power.

3.3.1.5 Stainless steel

Stainless steel receives its stainless characteristics from an invisible, self-healing chromium oxide film that forms when chromium is added at concentrations greater than 10.5%. There are three major groups within this type of steel: austenitic, ferritic, and martensitic.

The best corrosion resistance is obtained in austenitic stainless steels. These steels contain 16–26% chromium and up to 35% nickel, which, like manganese, is a strong austeniser. Austenitic steels cannot be hardened by heat treatment; they are also nonmagnetic. The most common type is the 18/8 or 304 grade, which contains 18% chromium and 8% nickel.

Figure 103. Pictures of representative product

Hot rolled coil  Source: EUROMETAL (link)	Cold rolled coil galvanised  Source: EUROMETAL (link)	Wire rod  Source: EUROMETAL (link)
Electrical sheet  Source: ArcelorMittal (link)	Stainless steel  Source: Thyssenkrupp (link)	

3.3.2 Bill of Materials (BOM)

In addition to the definition of the base cases conducted in this part of the study, a Bill of Materials (BOM) for each step of the value chain of the steelmaking processes was developed, in order to capture the process flows for all the material and energy inputs, outputs, direct emissions, wastes, etc. To conduct this exercise, various references were used such as the BREF document for iron and steel production (Joint Research Centre et al., 2013), specific databases such as ecoinvent¹²¹ or CRU platform¹²², as well as scientific literature ((Andrea Arguillarena et al., 2021), (Olmez et al., 2016), (Bhaskar, Assadi, and Somehsaraei, 2020), (Ruth, 2004), (Muscolino et al., 2016), (Gyllenram and Wei, 2022), (Nordelof and Tillman, 2018)). These Bill of Materials (BoM) compiling large datasets are used as a support to develop the Life Cycle Inventory for the LCA in Task 4. A deep comparison exercise has been conducted with industrial data from EUROFER and Worldsteel in order to increase the robustness of the BoM. Table 45 presents the main manufacturing stages considered within scope of the BOM built for the 5 representative products selected in section 3.2.

Table 45. Main manufacturing stages considered in scope of the BOM developed for the steel representative products

Main manufacturing stages							
Pre-processing	Coke oven	Iron making	DRI (from natural gas)	Steel making	Basic Oxygen Furnace	Steel manufacturing	Hot rolling
	Sinter plant		DRI (from hydrogen)				Cold rolling (incl. annealing)
	Pellet plant		Blast Furnace				Galvanising
							Stainless steel (EAF+ AOD)

Source: JRC elaboration

3.3.3 EU iron and steel sector current and 2035 base cases

The development of different policy design options requires the assessment of their potential improvements in comparison with a well-defined baseline called “base cases”. Over the years, the steel industry has developed several technologies and improvements to their production routes. These technologies aim to increase, both their production efficiency and their sustainability. As it will be explained in section 3.4 some of these technologies are not yet fully implemented and/or they are under development. To fully grasp the benefits of different improvements e.g. in terms of their environmental performance it is important to define base case scenarios for comparison.

With this objective in mind, the JRC developed two base case scenarios that intend to depict the steel industry status in 2022 (data refers to 2022) and the potential developments until 2035 (considering that no designed policy is implemented by then). These base cases are translated in Table 46, Table 47 and Table 70, and Figure 104,

Figure 105 and

Figure 106. They present several parameters that translate the situation of the European production, consumption, recycling and trade of steel across its value chain, from extraction to the production of the five representative products (selected in section 3.2). Additionally, Figure 109 to

¹²¹ <https://ecoinvent.org/database/>

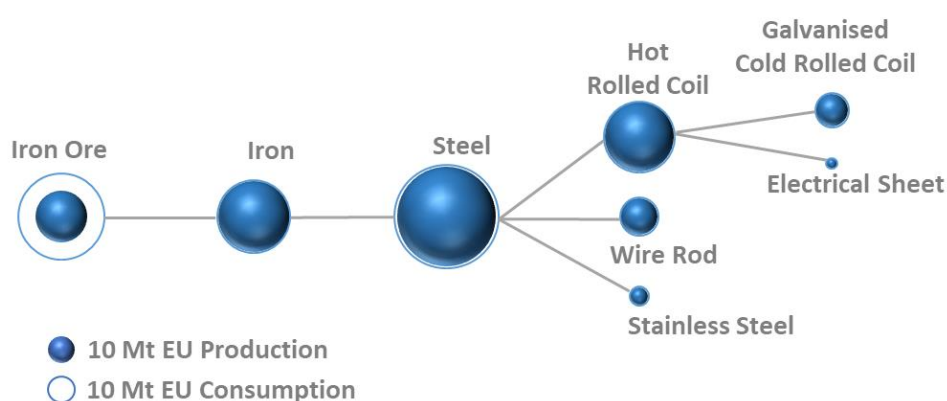
¹²² <https://www.cruigroup.com/en/data/>

Figure 113 illustrate the EU supply mix from EXTRA EU countries, showing imports and exports of the five representative products broke down by production technology (BOF, EAF and DRI-EAF). The methodology followed to develop the base cases is described in Annex 6.

The information presented in the Table 46 was used in the LCA and LCC analyses (Task 4). The data gathered in these base case scenarios will be also crucial for the impact assessment study.

Figure 104 presents a comparison of EU production and consumption of various iron and steel products, as analysed in Task 2. It is evident that the EU is heavily reliant on imports of iron ore and ferro-alloys. However, the EU production is sufficient to meet between 60-87% of the demand for intermediate steel representative products.

Figure 104. Comparison between EU production and EU consumption of iron and steel in 2022



Source: JRC own elaboration

According to EUROFER (EUROFER, 2024), EU crude steel production declined by 20% (30.8 Mt) between 2014 and 2023, with a 12% decrease (7.7 Mt) in the EAF route and 25% decrease (23.3 Mt) in the BF-BOF. This decline led to a 14% reduction (12 Mt) in EU scrap consumption, resulting in a significant portion being directly exported. The total scrap consumption decreased from 87 Mt in 2014 to 75.1 Mt in 2023. Notably, EU net scrap exports doubled from 7 Mt in 2014 to 15 Mt in 2023, while imports decreased from 5.1 Mt to 3.9 Mt.

The EU has a relatively high amount of end-of-life (EoL) scrap available compared to other geographical areas. However, the demand for high-quality steel products in EU, particularly for automotive applications, requires the use of the best quality scrap. Steel contaminants (also known as “tramp elements” such as copper) limit scrap uptake and larger deployment of secondary route in Europe.

The average copper content for scrap in OECD countries is estimated to 0.25%. For products manufactured via the EAF route, the concentration is 0.15% for product manufactured via the EAF (Material Economics, 2020). This lower concentration is achieved through the process of “diluting” the copper content. In the short term, contamination can be partially managed by trading scrap globally for use in tolerant applications or by blending it with primary iron sources. As estimates by [Daehn, Cabrera Serrenho, and Allwood \(2017\)](#) suggest, in the longer term quantities of copper arising from conventional scrap preparation can be managed in the global steel system until 2050 assuming perfectly coordinated trade and extensive dilution.

However, this strategy will become increasingly impractical due to the accumulation of tramp elements in the recovered scrap. Table 46, Table 47 and (*) Copper tolerance limit are displayed in italic with "<", specific grades of electrical steel may have different impurity tolerances; (**) Considering ferro-alloys have 0% recycled content.

Source: JRC elaboration

Table 48 present the average concentration of copper in steel scrap alongside the maximum allowable thresholds for each representative products. Currently, scrap content in the representative products is limited due to the role played by the primary route in Europe, except for long products like wire rod and for the specific case of very high alloys steel as it is the case for stainless steel, which have a higher share of EAF in their production process.

The following sub-sections and associated tables compiled the main elements used for the definition of the base case for each of the 5 representative products selected in section 3.2. Each table depicts the shares of BF-BOF, DRI-EAF and EAF used in the production of intermediate steel products, the associated use of inputs materials, as well as an estimation of scrap content highlighting specifically post-consumer sources:

- **Table 46** compiles the current base case (2022 as a reference year) based on current crude steel production (138 Mt) and consumption (151 Mt) as present in the market analysis (Chapter 2) and the section 3.2.
- **Table 47 and Table 48** compile the estimates of the steel EU industry in 2035, assuming that no policy design options were implemented by that date. These base case scenarios will help to understand the near-future impact of applying any improvements and regulations to the steel sector. This tables were developed using information available in EU and global forecast reports and stakeholders' consultation. The presented estimates are in line also with the literature (Su and Assous, 2022), (EUROFER AISB, 2019), (EuRIC, 2024), (Material Economics Sverige AB, 2018), (EuRIC, 2023), (Dworak, Rechberger, and Fellner, 2022) and forecasts announced by stakeholders. The scenarios definition is available in Annex 6.
 - Table 47 depicts a forecasted base case (2035 reference year) considering an increased production (164 Mt) and consumption (190 Mt).
 - Table 48 depicts a forecasted base case (2035 reference year) considering a stable production (138 Mt) and consumption (151 Mt).
 - Both tables present two different assumptions for the development of steelmaking technologies in the EU (shares of Bf-BOF, DRI-EAF and EAF) (as further explained in Annex 6).

These base case tables present a split between alloyed and low-alloyed steel products within the section on "EU-27 production", to depict the volumes of steel in the EU sector that contain other important metals for the EU industry as e.g. critical raw materials. However, as detailed by stakeholders the current distinction between alloyed and non-alloyed is defined according with EN 10020 (see detailed classification in the Annex 1) which allows for the inclusion in the classification of alloyed steel grades with relatively small alloy content. Therefore, estimating the accurate composition of alloyed elements and the associated consumption of ferro allows materials in the EU steel is particularly challenging. This represents a significant data gap that requires further investigation in future studies. Also, the share of each production routes is compiled. As it can be seen, currently production via DRI-EAF is very low or non-existing. In fact, as far of authors' knowledge, there is only a demonstration facility in Hamburg from ArcelorMittal with a DRI capacity of 600 kt (Global Energy Monitor, s.d.).

Several elements, such as trade, alloying and non-alloying shares and division of electrical sheet between oriented and non-oriented grain were considered unchanged between 2022 and 2035 base case scenarios. The focus of the development of the 2035 forecasts was on the definition of production technology shares and recycle content throughout the steel life cycle. In total 4 scenarios were developed (see scenario 2 and 3 in the Annex 1) considering: (i). an increase in steel production/demand (scenario 1 and 3) to 164 Mt (according with the projections of (Su and Assous, 2022)), (ii). EU production/demand stayed constant at 137 Mt (scenario 2 and 4) and (iii). two different assumptions for the development of steelmaking technologies in the EU (shares of Bf-BOF, DRI-EAF and EAF).

Regarding the steel production volumes allocated to Bf-BOF, DRI-EAF and EAF, the shares reported in Table 47 were obtained by correcting the 2022 values with information on the announced and under construction facilities and forecasted capacity for 2035 (Global Energy Monitor, s.d.). This distribution is quite conservative,

as it assumes no reduction in EU BF-BOF capacity until 2035. Notably, a substantial portion of BF-BOF facilities in the EU were constructed during the 1950s and 1960s and will require replacement or retrofitting within the next decade. According to (Su and Assous, 2022), 74% of BF-BOF furnace capacity in the EU will need retrofits or replacements with high investments associated to ensure their economic viability. This could lead to facility closures, especially in comparison to the already announced EAF and DRI-EAF projects facing lower carbon pricing. These assumptions led to the construction of new 2035 scenarios, in Table 48 and Table 80 of Annex 6. Additionally, according with information on facilities capacities from Global Energy Monitor, the announced DRI and EAF projects will surpass the capacities of the facilities with more than 50 years old, which supports the assumption taken for the constriction of scenarios 3 and 4.

DRAFT

Table 46. 2022 base case representing the state of play of the EU Iron and Steel sector according with production, consumption, recycling and trade from extraction to the production of the five representative products. The value are expressed in Mt for the reference year 2022

	Fe materials			Steel		Selected representative products						References	
	Iron ore	Iron	Ferrous scrap	Ferro-Alloys	Carbon Steel	HOT ROLLED COIL	GALVANISED COLD ROLLED COIL	WIRE ROD	ELECTRICAL SHEET		STAINLESS STEEL		
									Grain orientated	Non-Grain orientated			
EU 27 Aparent consumption (Mt)	107	76.6	75.2	3.37	150.9	31.4	16.17	20.59	0.39	1.42	5.1	[1,3,5,7,8]	
Share of Imports in the EU cons. (%)	77%	7%	5%	74%	21%	26%	27%	13%	35%	32%	40%		
% Import reliance	64%	7%	-20%	65%	9%	16%	13%	8%	-11%	24%	18%		[1,2]
EU 27 Production (Mt)	38	71.4	90.2	1.2	137.7	26.3	14.7	18.9	0.43	1.08	4.190	[1,3,5,7]	
share of alloys steel (%)	-	-	-	-	19%	20%	40%	16%	100%	60%	0%	[1,5,6,7]	
share of non-alloys (%)	-	-	-	-	77%	80%	60%	84%	0%	40%	0%		
share of stainless(%)	-	-	-	-	5%	-	-	-	-	-	100%		
Production routes (EU27)													
DRI-EAF	-	1%	-	-	2%	-	-	4%				[1,2]	
BF-BOF	-	99%	-	-	58%	95%	96%	35%	100%	100%	0%	[1,3,4,9]	
EAF	-	-	-	-	40%	5%	4%	61%	0%	0%	100%		
Scrap use / recycled content in EU production													
Scrap content (BF-BOF contribution)	-	-	17.6	-	18-25%	13%	13%	6%	15-16%	15-16%	n/a		
Scrap content (EAF contribution)	-	-	58.3	-	92-100%	5%	5%	60%	n/a	n/a	80-95%**		
average scrap content in product	-	-	100%	-	55%	18%	18%	66%	15-16%	15-16%	80-95%**		
Post consumer (up to)													
Post-consumers scrap content (BF-BOF contribtion)	-	-	4.4	-	6-7%	6%	6%	3%	6-7%	6-7%	n/a	[8]	
Post-consumers scrap content (EAF contribution)	-	-	46.2	-	75-85%	4%	4%	49%	n/a	n/a	60-80%		
average post-consumer scrap content in product	-	-	100%	-	37%	10%	10%	52%	6-7%	6-7%	60-80%		
Estimated average Cu-concentration*	n/a	0.01%	0.25%	n/a	0.08%	<0.1%	<0.08%	<0.25%	<0.1%	<0.1%	n/a		

[1] Market analyses of chapter 2; [2] RMIS – Raw Materials Information System; [3] CRU platform; [4] EUROFER Bill Of Materials; [5] Prodcum and Comext databases; [6] (Georgitzikis, 2023); [7] Eurofer, European steel in Figures (2024); [8] (Dworak, Rechberger, and Fellner, 2022) and (Daehn, Cabrera Serrenho, and Allwood, 2017); [9] (Systemiq and (commissioned by) Thyssenkrupp Steel Europe, 2023)

(*) Copper tolerance limit are displayed in italic with "<", specific grades of electrical steel may have different impurity tolerances; (**) Considering ferro-alloys have 0% recycled content.

Source: JRC elaboration

Table 47. 2035 scenario 1 Baseline representing estimates of a near future perspective of the EU Iron and Steel sector according to production, consumption, recycling and trade from extraction to the production of the five representative products. Value are expressed in Mt for 2035 reference year

	Fe materials			Steel		Selected representative products						Reference	
	Iron ore	Iron	Ferrous scrap	Ferro-Alloys	Carbon Steel	HOT ROLLED COIL	GALVANISED COLD ROLLED COIL	WIRE ROD	ELECTRICAL SHEET		STAINLESS S STEEL		
	Grain orientated	Non -Grain orientated											
EU 27 Aparent consumption (Mt)	107	76.6	101.7	4.1	189.7	36.8	20.1	24.5	0.5	1.7	6.1	[1,3,5,7]	
Share of imports in the EU cons. (%)	77%	7%	4%	74%	27%	27%	21%	11%	35%	32%	34%		
% Import reliance	64%	7%	-15%	65%	14%	15%	13%	8%	-11%	24%	18%		[1,2]
EU 27 Production (Mt)	38	71.4	120	1.4	164	31.3	17.5	22.5	0.51	1.29	5.0	[1,3,5,7,10,11]	
share of alloys steel (%)	-	-	-	-	19%	20%	40%	16%	100%	60%	0%	[1,5,6,7]	
share of non-alloys (%)	-	-	-	-	77%	80%	60%	84%	0%	40%	0%		
share of stainless(%)	-	-	-	-	5%	-	-	-	-	-	100%		
Production routes (EU27)													
DRI-EAF	-	30%	-	-	16%	34%	20%	8%	-	-	-	[1,3,4,9,12]	
BF-BOF	-	70%	-	-	40%	51%	68%	15%	99%	99%	0%		
EAF	-	-	-	-	44%	15%	12%	77%	1%	1%	100%		
Scrap use / recycled content in EU production													
Scrap content coming from BF-BOF	-	-	18.22	-	25%	13%	17%	4%	24%	24%	-		
Scrap content coming from EAF	-	-	76.17	-	95%	14%	11%	73%	1%	1%	80-95%**		
Scrap content in DRI-EAF	-	-	7.29	-	25%	9%	5%	2%	n/a	-	-		
average scrap content in product	-	-	100%	-	62%	36%	33%	79%	25%	25%	80-95%**		
Post consumer (up to)													
Post-consumers scrap content (BF-BOF)	-	-	10.93	-	15%	8%	10%	2%	14%	14%	-		
Post-consumers scrap content (EAF)	-	-	60.13	-	75%	11%	9%	58%	8%	8%	60-80%		
Post-consumers scrap content (DRI)	-	-	7.29	-	15%	5%	3%	1%	-	-	-		
average post-consumer scrap content in product	-	-	100%	-	48%	24%	22%	61%	22%	22%	60-80%		
Estimated average Cu-concentration*	n/a	0.01%	0.30%	n/a	0.1%	<0.1%	<0.08%	<0.25%	<0.1%	<0.1%	n/a	[8]	

[1] Market analyses of chapter 2; [2] RMIS – Raw Materials Information System; [3] CRU platform; [4] EUROFER Bill Of Materials; [5] Prodcum and Comext databases; [6] (Georgitzikis, 2023); [7] Eurofer, European steel in Figures (2024); [8] (Dworak, Rechberger, and Fellner, 2022) and (Daehn, Cabrera Serrenho, and Allwood, 2017); [9] (Systemiq and (commissioned by) Thyssenkrupp Steel Europe, 2023); [10] (Su and Assous, 2022); [11] (EUROFER AISB, 2019) and stakeholders information; [12] (Global Energy Monitor, s.d.).

(*) Copper tolerance limit are displayed in italic with "<", specific grades of electrical steel may have different impurity tolerances; (**) Considering ferro-alloys have 0% recycled content.

Source: JRC elaboration

Table 48. 2035 scenario 4 Baseline representing estimates of a near future perspective of the EU Iron and Steel sector according with production, consumption, recycling and trade from extraction to the production of the five representative products. Value are expressed in Mt for 2035 reference year

	Fe materials			Steel		Selected representative products						Reference	
	Iron ore	Iron	Ferrous scrap	Ferro-Alloys	Carbon Steel	HOT ROLLED COIL	GALVANISED COLD ROLLED COIL	WIRE ROD	ELECTRICAL SHEET		STAINLES S STEEL		
	Grain orientated	Non -Grain orientated											
EU 27 Aparent consumption (Mt)	107	57.9	91.8	3.4	150.9	31.4	16.2	20.6	0.4	1.4	5.1	[1,3,5,7]	
Share of Imports in the EU cons. (%)	77%	7%	4%	74%	27%	32%	27%	13%	35%	32%	40%		
% Import reliance	64%	-23%	-21%	65%	14%	15%	13%	8%	-11%	24%	18%		[1,2]
EU 27 Production (Mt)	38	71.4	120	1.2	137.70	26.27	14.70	18.90	0.43	1.08	4.19	[1,3,5,7,10,11]	
share of alloys steel (%)	-	-	-	-	19%	20%	40%	16%	100%	60%	0%	[1,5,6,7]	
share of non-alloys (%)	-	-	-	-	77%	80%	60%	84%	0%	40%	0%		
share of stainless(%)	-	-	-	-	5%	-	-	-	-	-	100%		
Production routes (EU27)													
DRI-EAF	-	38%	-	-	18%	50%	27%	9%	-	-		[1,3,4,9,12]	
BF-BOF	-	62%	-	-	32%	22%	57%	9%	99%	99%	0%		
EAF	-	-	-	-	50%	28%	16%	82%	5%	5%	100%		
Scrap use / recycled content in EU production													
Scrap content coming from BF-BOF	-	-	12.24	-	25%	6%	15%	3%	24%	24%	-		
Scrap content coming from EAF	-	-	72.68	-	95%	14%	11%	73%	5%	5%	80-95%**		
Scrap content in DRI-EAF	-	-	6.89	-	25%	15%	7%	3%	n/a	-	-		
average scrap content in product	-	-	100%	-	67%	36%	33%	79%	29%	29%	80-95%**		
Post consumer (up to)													
Post-consumers scrap content (BF-BOF)	-	-	7.34	-	15%	4%	9%	2%	14%	14%	-		
Post-consumers scrap content (EAF)	-	-	57.38	-	75%	11%	9%	58%	8%	8%	60-80%		
Post-consumers scrap content (DRI)	-	-	6.89	-	15%	9%	4%	2%	-	-	-		
average post-consumer scrap content in product	-	-	100%	-	52%	24%	22%	61%	22%	22%	60-80%		
Estimated average Cu-concentration *	n/a	0.01%	0.30%	n/a	0.1%	<0.1%	<0.08%	<0.25%	<0.1%	<0.1%	n/a	[8]	

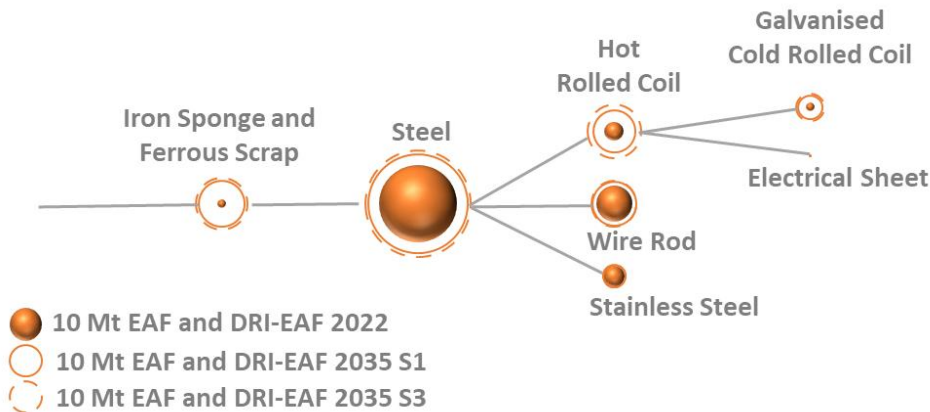
[1] Market analyses of chapter 2; [2] RMIS – Raw Materials Information System; [3] CRU platform; [4] EUROFER Bill Of Materials; [5] Prodcorn and Comext databases; [6] (Georgitzikis, 2023); [7] Eurofer, European steel in Figures (2024); [8] (Dworak, Rechberger, and Fellner, 2022) and (Daehn, Cabrera Serrenho, and Allwood, 2017); [9] (Systemiq and (commissioned by) Thyssenkrupp Steel Europe, 2023); [10] (Su and Assous, 2022); [11] stakeholders' consultation; [12] (Global Energy Monitor, s.d.).

(*) Copper tolerance limit are displayed in italic with "<", specific grades of electrical steel may have different impurity tolerances; (**) Considering ferro-alloys have 0% recycled content.

Source: JRC elaboration

Figure 105 illustrates the expected increase in the use of EAF and DRI-EAF technologies in the EU. Significant variations are anticipated in the production of hot rolled coil and galvanized cold rolled coil. According to the Global Energy Tracker (Global Energy Monitor, s.d.), new DRI-EAF facilities are expected to become operational in the EU by 2035. By that year, DRI-EAF is projected to significantly expand its role in EU steelmaking, increasing its production share from 1% to 16% in Scenario 1 and from 1% to 18% in Scenario 4. Additionally, stand-alone EAF capacity is also predicted to grow substantially, with its production share rising from 44% to 50%.

Figure 105. Comparison of EU iron and steel production using EAF and DRI-EAF technologies in 2022 and 2035.

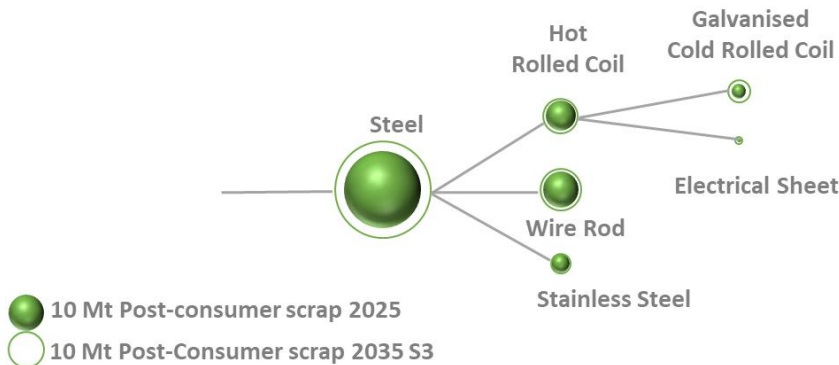


Source: JRC own elaboration

Currently, the the scrap ratio per unit of crude steel produced in the EU is 55% (see Table 46, share of scrap consumed in the overall crude steel production), this is in line with the estimates from Euric (EuRIC, 2024). With the projected increase in the EAF share (as outlined in scenario 4, Table 49), the overall scrap ratio is expected to rise to 61% per unit of crude steel by 2035. This shift would result in substantial scrap content in flat products, such as hot rolled coil (6%), galvanized cold rolled coil (15%), and electrical sheet (24%), as depicted in

Figure 106. In Europe, these flat products are currently produced primarily through routes with low scrap content, except in rare instances.

Figure 106. Comparison of the use of Post-consumer scrap between 2022 and 2035



Source: JRC own elaboration

3.3.4 EU trade dynamics for representative product (export and supply mix)

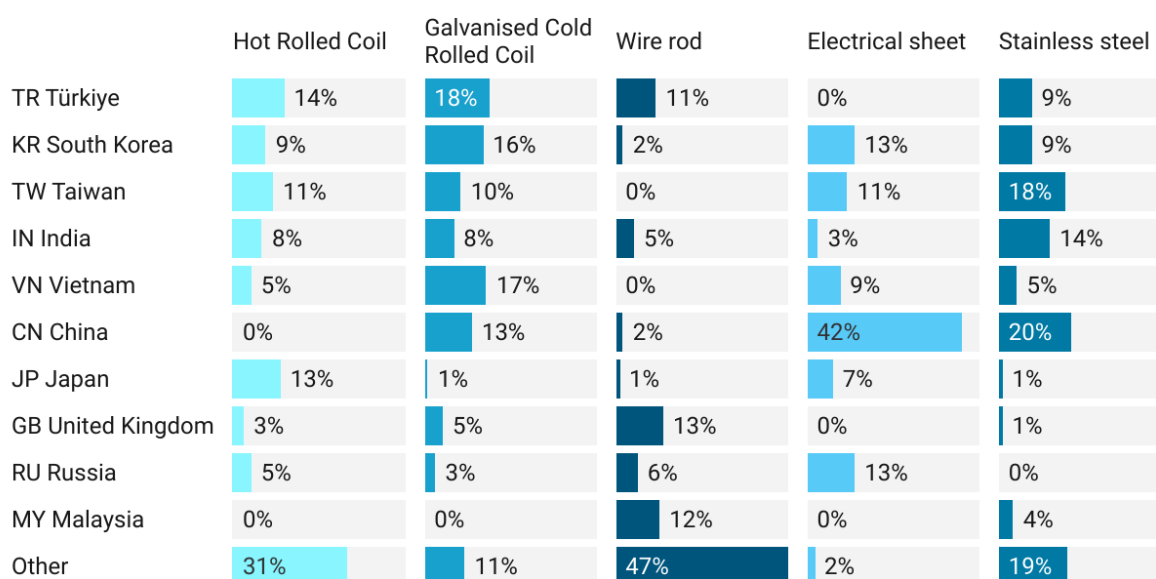
Figure 107 and Figure 108 illustrate the EU supply mix from extra EU countries, based on 2022 data. This section details the EU's imports and exports for each representative product. The values of imported materials are categorised according to their production shares using BF-BOF, EAF, and DRI-EAF for each country of origin.

These estimates provide insights into the production routes of the imported products, offering valuable information for assessing the impact of potential future policy designs. Such analysis is crucial for evaluating the carbon footprint of the current EU supply mix and proposing alternative strategies for improvement.

Figure 107 shows the top three countries supplying all five representative products to the EU, whereas Figure 108 shows the top three countries receiving steel representative products from the EU. Turkey, Japan, and Taiwan are the primary suppliers of hot rolled coil to the EU, while the EU exports this product mainly to Turkey, the United States, and the United Kingdom. Galvanized cold rolled coil is predominantly imported from Türkiye, Vietnam, and South Korea, with the EU exporting it to the United Kingdom, Türkiye, and Mexico. The EU's main sources for wire rod are the United Kingdom, Malaysia, and Türkiye, and its primary export destinations for wire rod are Türkiye, the United Kingdom, and the United States. For stainless steel, China, Taiwan, and India are the leading suppliers to the EU, while key export markets include Türkiye, the United Kingdom, and the United States. Lastly, the EU imports electrical sheet primarily from China, Russia, and South Korea, and exports it to India, Switzerland, and Türkiye.

Figure 107. Country of origin for EU imports of the five representative products

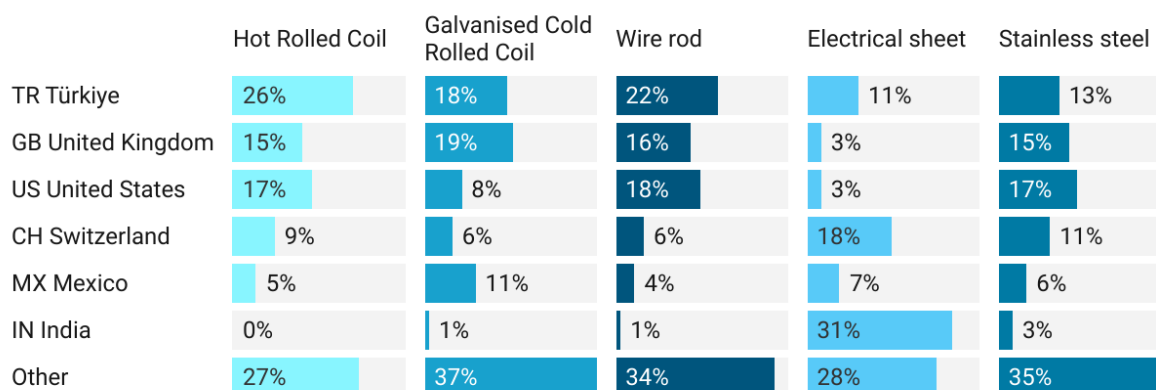
EU Imports



Source: JRC own elaboration

Figure 108. Countries of destination of EU exports of the five representative products.

EU Exports



Source: JRC own elaboration

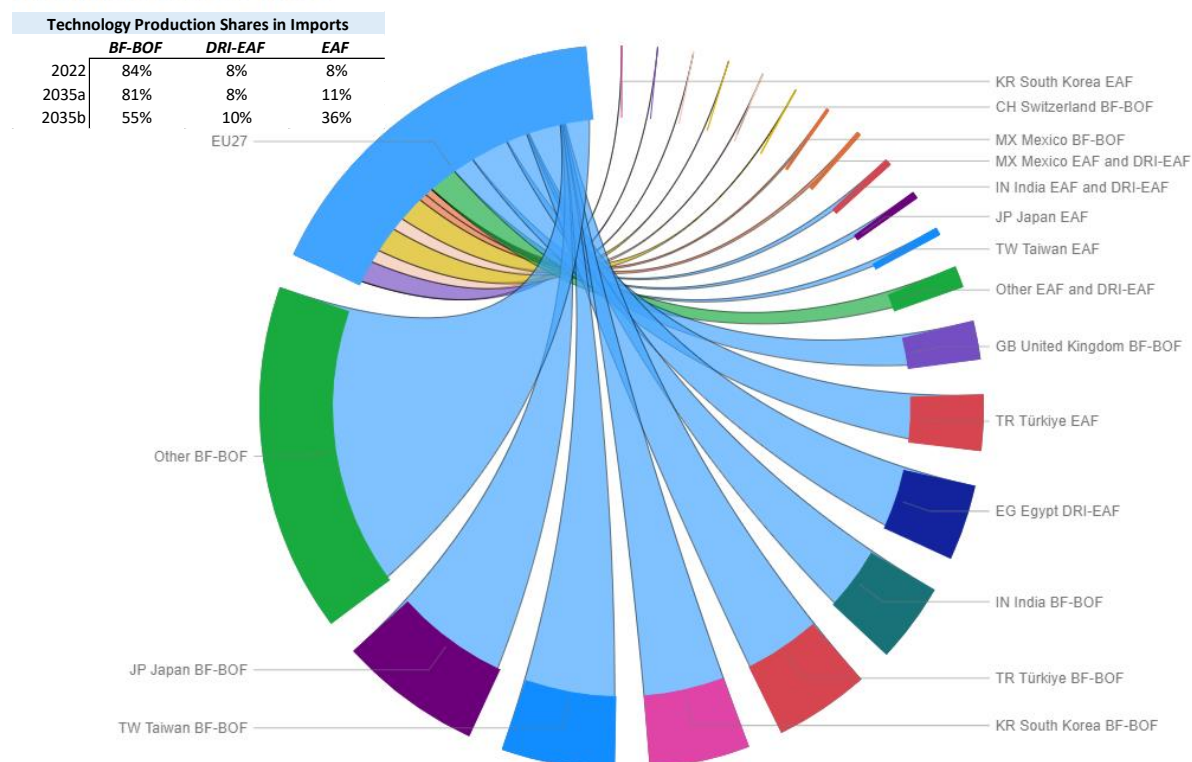
Figure 109 to

Figure 113 show the EU supply mix from extra non-EU countries categorised by the production technologies used to produce the five representative products. The imports of each product into the EU are analysed based on their countries of origin and the production methods used, specifically BF-BOF, DRI-EAF, and EAF. Furthermore, weighted average production technology shares were calculated for each representative product.

In order to project a near-future supply mix from extra non-EU countries, the JRC also estimated the production shares for 203. This estimation was based on information about facilities announced and under construction up to that date across all countries supplying the targeted products to the EU. The complete dataset, detailing imports, exports, and the production technologies utilized by each country for the five representative products, is provided in the Annex 6. Additionally, similar to the EU base cases, production technology shares were estimated under the assumption that facilities operating for more than 50 years would be decommissioned.

Figure 109. Hot rolled coil EU supply mix considering imports and exports from EU trade partners and the shares of production using technologies BF-BOF, DRI-EAF and EAF

Hot Rolled Coil EU Trade



Notes:

- The small table translates the weighted average of the technology production shares in imports of hot rolled coil, in 2022 and estimates for 2035 considering no change in the BF-BOF capacity (a) and conversion or closure of BF-BOF facilities with more than 50 years (b).
- Estimates made using data from CRU platform¹²³ on production volumes and producing facilities for each country supplying to the EU, corrections made with global tracker (Global Energy Monitor, s.d.), data on EU trade was retrieved from COMEXT (the only missing information relates to the exact facilities supplying to the EU). The JRC believes these estimates to be close to the reality.

Source: JRC own elaboration

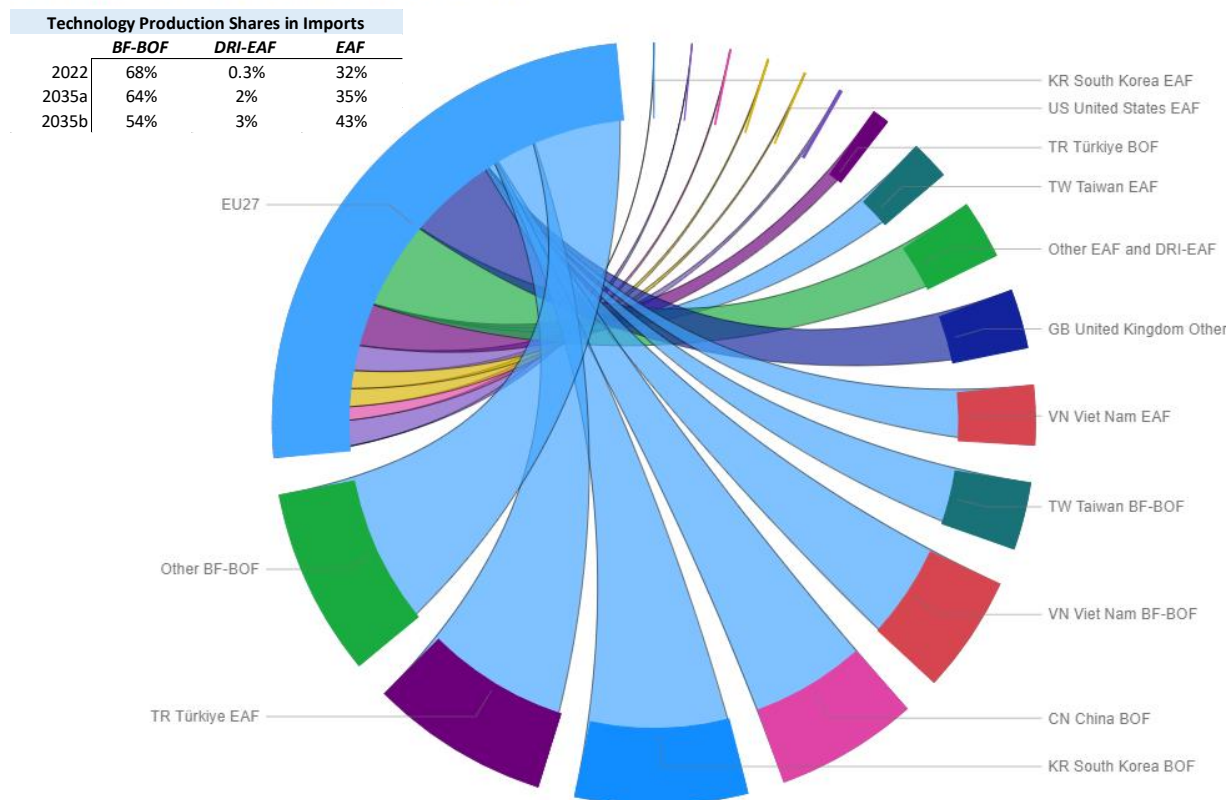
As is the case for EU production of hot rolled coil production, Figure 109 indicates that the supply mix predominantly depends on imports produced via the BF-BOF route. The estimated weighted average production shares for the EU supply mix are 84% BF-BOF, 8% DRI-EAF and 8% EAF. Compared to the EU, the proportion of production from the DRI-EAF route is higher in the non-EU supply mix, largely due to the installed capacity of this technology in countries like India and Egypt.

Projected estimates for the 2035 non-EU supply mix suggest a composition of 81% BF-BOF, 8% DRI-EAF, and 11% EAF, indicating a modest increase in production capacity utilising EAF. Should the current BF-BOF production capacity for hot rolled coil diminish, production via EAF and DRI-EAF is likely to experience significant growth. However, the expansion of the scrap-EAF production route may be constrained by the quality of available scrap in countries supplying the EU. This limitation warrants further investigation to fully understand its impact on scrap usage for flat steel product production. Nevertheless, a shift towards DRI-EAF for primary material processing appears evident, driven by the urgent need to decarbonize steel production and the anticipated rise in carbon pricing in the coming years.

Figure 110. Galvanised cold rolled coil 2022 EU supply mix considering imports and exports from EU trade partners and the shares of production using technologies BF-BOF, DRI-EAF and EAF

¹²³ [CRU Group: Home - CRU Group](#)

Galvanized Cold Rolled Coil EU Trade



Notes:

- The small table translates the weighted average of the technology production shares in imports of galvanized cold rolled coil, in 2022 and estimates for 2035 considering no change in the BF-BOF capacity (a) and conversion or closure of BF-BOF facilities with more than 50 years (b).
- Estimates made using data from CRU platform¹²⁴ on production and producing facilities for each country supplying to the EU. However, CRU platform coverage was limited for this material as data was missing from several countries. Corrections made with global tracker (Global Energy Monitor, s.d.), data on EU trade was retrieved from COMEXT (the missing information relates to the exact facilities supplying to the EU and better coverage of production of countries supplying to the EU). Assumption: To address the missing data, the hypothesis adopted is to adopt the same distribution for the production routes than in the EU.

Source: JRC own elaboration

In Turkey, the primary production route for galvanized cold rolled coil is the EAF process, which contributes to a higher share of EAF production in the EU supply mix. However, specific production volume data for galvanized cold rolled coil from Vietnam and other supplying countries is unavailable. Consequently, the estimates for Vietnam's technology shares are based on its overall steel production profile. The weighted average production shares for the EU supply mix of galvanized cold rolled coil are estimated to be 68% BF-BOF, 0.3% DRI-EAF, and 32% EAF.

Similar to the analysis for hot rolled coil, the JRC estimates that the weighted average production shares for the EU supply mix will be 64% BF-BOF, 2% DRI-EAF, and 35% EAF. These figures indicate an increase in production capacity utilizing EAF and DRI-EAF. As seen with hot rolled coil, a reduction in the current BF-BOF production capacity would likely lead to an increase in production through EAF and DRI-EAF methods.

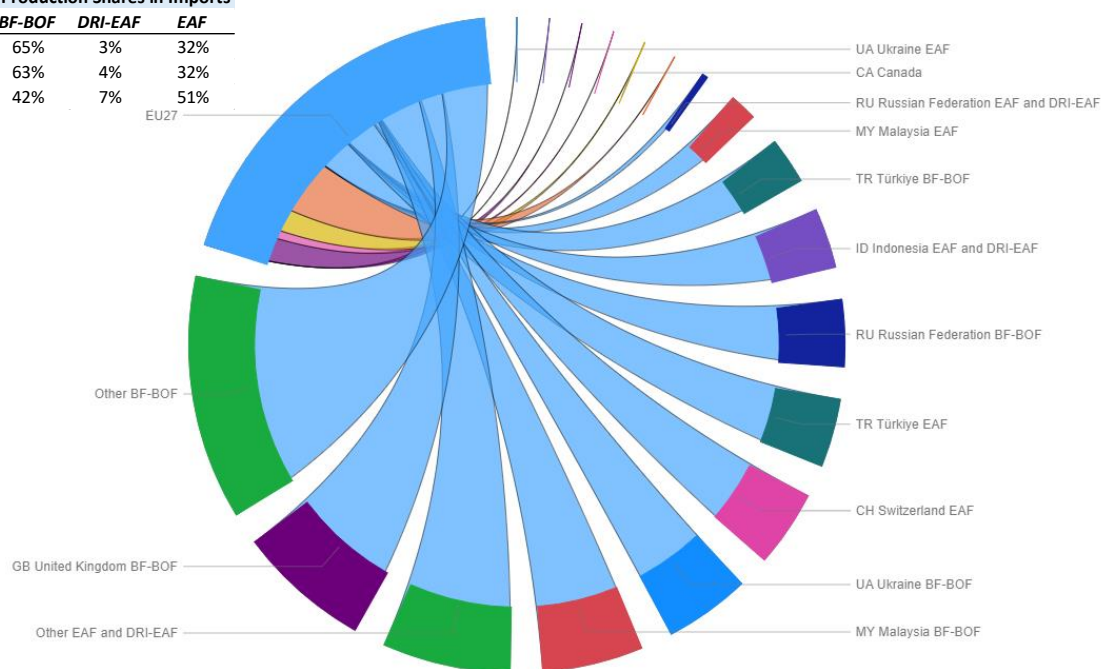
Figure 111. Wire rod 2022 EU supply mix considering imports and exports from EU trade partners and the shares of production using technologies BF-BOF, DRI-EAF and EAF.

¹²⁴ [CRU Group: Home - CRU Group](#)

Wire Rod EU Trade

Technology Production Shares in Imports

	BF-BOF	DRI-EAF	EAF
2022	65%	3%	32%
2035a	63%	4%	32%
2035b	42%	7%	51%



Notes:

- The small table translates the weighted average of the technology production shares in imports of wire rod, in 2022 and estimates for 2035 considering no change in the BF-BOF capacity (a) and conversion or closure of BF-BOF facilities with more than 50 years (b).
- Estimates made using data from CRU platform¹²⁵ on production and producing facilities, corrections made with global taker (Global Energy Monitor, s.d.), data on EU trade was retrieved from COMEXT (the only missing information relates to the exact facilities supplying to the EU). The JRC believes these estimates to be close to the reality.

Source: JRC own elaboration

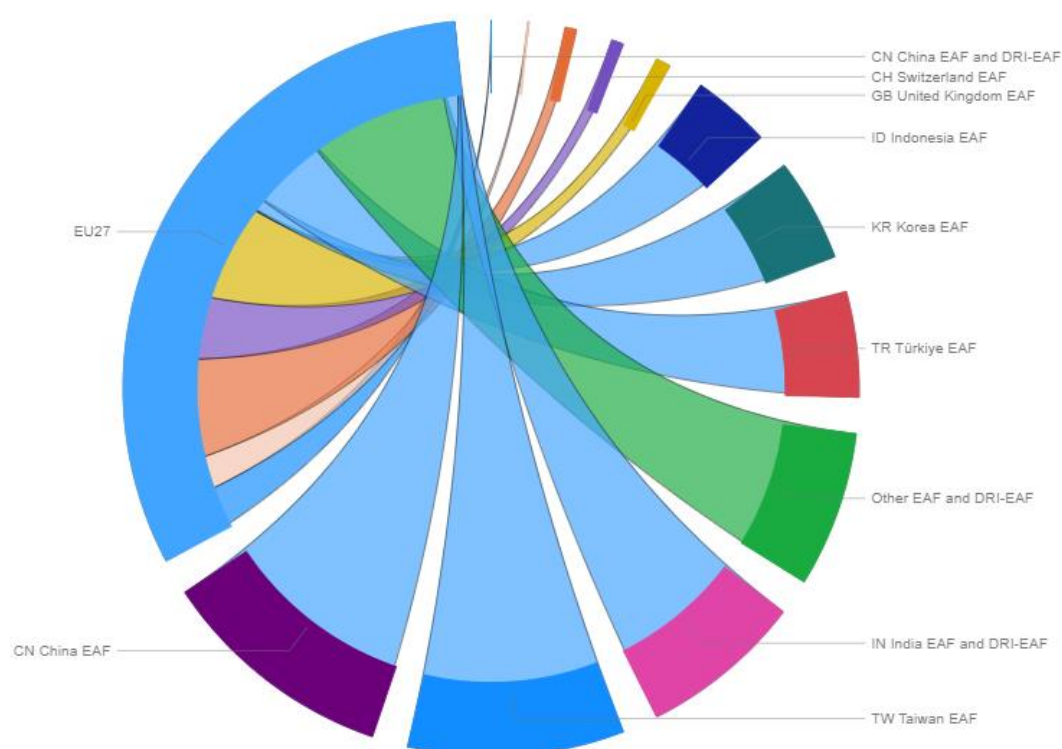
In contrast to the technological approach employed for wire rod production within the EU, Figure 111 reveals that the EU supply mix relies more heavily on BF-BOF facilities than on EAF. The weighted average estimates are essentially reversed compared to those reported for EU production, with the EU supply mix comprising 65% BF-BOF, 3% DRI-EAF, and 32% EAF.

Projections for the EU supply mix in 2035 indicate production shares of 63% BF-BOF, 4% DRI-EAF, and 32% EAF, reflecting an increase in production capacity utilizing DRI-EAF. If BF-BOF capacity decreases in the future (as hypothesized in Scenario 2035b), it can be easily compensated by the expansion of EAF, given that the quality restrictions related to scrap are less stringent for long products. Consequently, Scenario 2035b suggests that the scrap-EAF process will become the predominant production route in the future EU supply mix.

Figure 112. Stainless steel 2025 EU supply mix considering imports and exports from EU trade partners and the shares of production using technologies BF-BOF, DRI-EAF and EAF.

¹²⁵ [CRU Group: Home - CRU Group](#)

Stainless Steel EU Trade



Note: No data was available from CRU platform¹²⁶ on production volumes and producing facilities, as for the EU EAF route was consider the main production route for stainless steel, data on EU trade was retrieved from COMEXT.

Source: JRC own elaboration

No specific information was available regarding facilities producing stainless steel outside the EU. However, it is assumed that all imported stainless steel is produced via the EAF route. The primary distinction between various production methods for stainless steel lies in the alloying material used in the EAF. In Europe, ferronickel, which is high in purity and nickel content, is typically used. In contrast, most production in China and Indonesia relies on Nickel Pig Iron (NPI), which is of lower quality and has a lower nickel content.

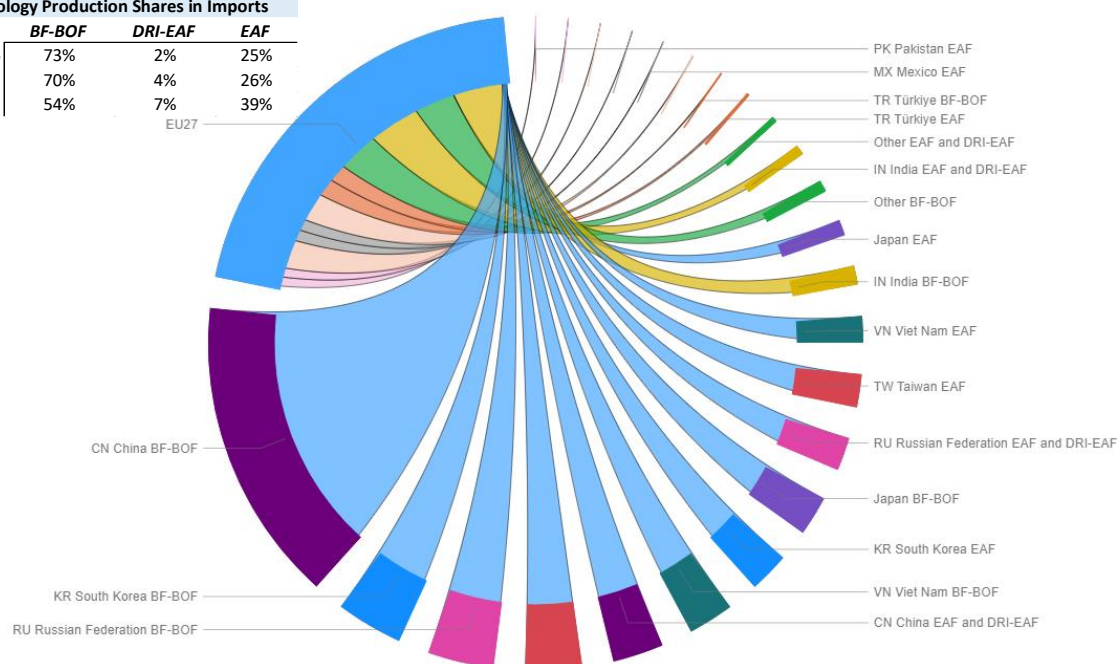
Figure 113. Electrical sheet 2022 EU supply mix considering imports and exports from EU trade partners and the shares of production using technologies BF-BOF, DRI-EAF and EAF

¹²⁶ [CRU Group: Home - CRU Group](https://www.cru.com/)

Electrical Sheet

Technology Production Shares in Imports

	BF-BOF	DRI-EAF	EAF
2025	73%	2%	25%
2035a	70%	4%	26%
2035b	54%	7%	39%



Note:

- No data was available from CRU on production and producing facilities, EAF and BOF shares were made with info from each country global steel production using global taker, data on EU trade was retrieved from COMEXT.
- The small table translates the weighted average of the technology production shares in imports of electrical sheet.

Source: JRC own elaboration

No specific information is available regarding facilities producing electrical sheet outside the EU. The estimates for each country are based on their overall steel production landscape. The estimated weighted average production shares for the EU supply mix are 73% BF-BOF, 2% DRI-EAF, and 25% EAF.

Looking ahead to 2035, the weighted average shares are projected to be 70% BF-BOF, 4% DRI-EAF, and 26% EAF. These projections indicate an increase in the DRI-EAF production share within the EU supply mix, driven by expanding DRI-EAF capacity in Russia and South Korea. The scenario for 2035b is similar to the one for hot rolled coil and galvanized cold rolled coil, where BF-BOF remains the dominant production route. The growth of scrap-EAF production for electrical sheet is likely to be limited by the high-quality requirements of this steel product. Consequently, the actual share of EAF production might be lower than the estimates presented here.

3.4 Technical aspects affecting the environmental performance of iron and steel products

This section evaluates the key technical aspects that influence the environmental performance of iron and steel products from a cradle-to-gate perspective. Based on literature review and stakeholder consultation, the authors have identified the following key technical aspects (as illustrated in Figure 114): use of scrap, preparation of precursors, energy improvements on the preparation of precursors, the consideration of the energy sources used, the processes optimisation, water management, increased utilisation of scrap and carbon capture, storage and utilisation (CCSU)

Figure 114. Technical aspects affecting environmental performance of iron and steel products



Source: JRC elaboration

Subsequent sections will delve into the discussion of **best available technologies (BAT) and not best available technologies (NBAT)** for each of the drivers included in Figure 114. In Task 5, these BAT will be utilised as the foundation for proposing design options with the potential to improve the environmental performance the bases cases defined in section 3.3.

3.4.1 Use of scrap

Producing steel from iron ore consumes almost 3 times more energy and generates 2.5 times more emissions per ton of crude steel than the production from recycled scrap. These emissions could be avoided if virgin iron ore is replaced by scrap (Draxler et al., 2021).

The use of scrap in steel making is an opportunity to decrease EU import dependency of primary iron materials, internalise all scrap supply chain within the EU and also decrease the environmental impacts of iron and steel making. However, complete circularity of steel flows is hindered by several challenges such as: 1) poor collection and sorting, 2) presence of tramp elements that prevent the use of scrap, 3) correct traceability and quality control. Scrap may be generated during manufacturing stage of steel and steel goods (pre-consumer scrap) or it can be generated after the End of Life (EoL) of products containing steel (post-consumer scrap). The recycling of steel requires several stages, such as: collection of steel products, dismantling, depolluting, shredding, sorting and remitting (at steel plant).

This section was developed based on the findings of Ciacci, 2024 report prepared for the joint research centre. With approximately 3500 different grades of steel achieving specific target compositions, using recycled scrap as feedstock becomes increasingly challenging. This discrepancy between the variety of available scrap and the

limited categories it can fit into, results in mismatches, particularly when scrap characteristics fall between established classes, which ultimately result in trade-offs that prioritize economic considerations over resource efficiency. Scrap materials, indeed, are rarely homogeneous and may require significant preparation before remelting to ensure they possess the necessary physical and chemical properties for their intended use. To achieve circularity in the steel EU cycle, better separation of post-consumer scrap appears to be fundamental: however, increasing EoL recycling rates may not be sufficient to tackle the issue of lower quality of steel scrap expected in the coming years. As of today, Cu likely causes the main issue to secondary steelmakers as its content is found to be more than double (approximately, 0.4% on average) of that of the other alloying elements and because Cu is highly detrimental to most of steel properties (although Cu contamination seems not to have similar detrimental effects on cast iron and steel products): for instance, Cu concentrations over 0.1% can cause hot shortness, leading to surface cracking in hot rolling and forming.

The adoption of manufacturing design options that promote resource efficiency (i.e., promoting disassembly and recycling since product planning and development stages) is also of crucial importance to increase the use of scrap in steelmaking. Improved design options will promote the efficiency of scrap sorting and increase the quality and tractability of scrap products.

Table 50. BAT & BNAT for reducing environmental impacts through scrap us

BAT	BNAT
- Hammer shredding and magnetic separation - Eddy current metal separators - Heavy media separators - Origin control and visual scrap sorting and separation - Remelting by EAF - Choice of material, surface coating and options for joining (manufacturing stage) Design for circularity.	- Advanced sorting (X-ray fluorescence, Optical emission spectroscopy, Prompt gamma neutron activation analysis, artificial intelligence) - Chemical treatments during remelting - Solid state recycling

Source: JRC elaboration

3.4.1.1 Best Available Technologies

Hammer shredding and magnetic separation

Current hammer shredding followed by magnetic separation is a consolidated technology at very competitive costs for scrap processing. Magnetic separation is available in different configurations for extracting ferrous materials from waste streams: for instance, there are scrap lift magnet attached to excavator arms, primary drum magnets, magnetic ballistic separators (MBSs), over-belt magnets pulling up residual ferrous from non-ferrous stream conveyor belt or magnetic head pulling residual ferrous down. Secondary drum magnets can be installed to refine the separated ferrous stream from residual non-ferrous materials. MBSs are typically installed at the shredding plant and enable almost complete removal of pieces tangled to Cu. Magnetic separators are commercialised with either electromagnets or permanent magnets. Rare earth magnet separators have much higher magnetic strength and can attract and separate weakly magnetic, even slightly magnetic austenitic stainless steel, or very fine iron contaminants from the main material stream. Among limitations, magnetic separators cannot deal with multi-material assemblies such as EoL vehicles (ELVs) and waste electrical and electronic equipment (WEEE) but require a series combination of different magnetic units to liberate material pieces to be separated.

Although, magnetic separation is only partially selective as part of Cu wiring may get tangled up to steel scrap (i.e., “Cu meatballs”), requiring further refining (Reijnders, 2016). The increase of scrap quality after shredding is also dependent on separation, dismantling and depollution stages. Mandatory removal of certain components prior to shredding to reach high quality and quantity of recovered steel are not existing to date. However, depending on the needs of secondary steelmaking plants, waste input flows can be separated before pre-processing and shredding.

Eddy current metal separators

Eddy current-based metal detectors and separators are used in metal separation since long time. However, they can be further developed to differentiate between stainless steel and ferromagnetic steel. These separators make use of eddy currents principle to separate non-ferrous metals from the waste stream.

Heavy media separators

Heavy media separation employs liquid bath of water with a specific gravity or water containing fine suspension of magnetite or ferrosilicon, the quantity of which can be adjusted to induce separation of materials by density. Although, heavy media separation may work well to separate ferrous from nonferrous pieces, this method seems not be effective with different steel grades. In addition, it implies the issue of adding contamination to the outflows and high cost of maintaining density constant in the slurry.

Origin control and visual scrap sorting and separation

While scrap quality has traditionally focused on chemical composition and physical condition, industry actors also consider factors like shape, size, and homogeneity in their operations and transactions. Within this complex landscape, industry stakeholders implicitly rely on their knowledge of specific steel products embedded in their origin or production methods. Upon arrival at the mill scrapyard, visual inspections and control checks are conducted to verify compliance with quality specifications outlined in supply contracts. The sampling plan, as well as the method and frequency of characterisation, depend on the delivery's size and source. Despite educated estimates, the lack of comprehensive information poses challenges, particularly during the decision-making process regarding which scrap piles to prioritise for transportation and production. The need to quickly move material due to limited yard space, coupled with weather considerations, adds also to the complexities of scrap management. Furthermore, since scrap steel is a traded commodity its processing might occur in locations distant from its origin. Scrap pricing influences mixing practices, with dealers potentially selling scrap to locations with lower quality demands or less stringent regulations if local sales are challenging (Compañero, Feldmann, and Tilliander, 2021).

Remelting by EAF

Steel scrap is remelted in EAF facilities. These facilities generally process mixes of scrap from different scrap batches and need to find a trade-off between material costs and the achievement of standard steel product specifications. Recipes determine the selection and quantity of scrap fed into the furnace for melting, ideally matching the steel being produced. However, limitations in availability as well as scrap piles containing compositionally diverse material often restrict the ideal recipe match. Dilution with DRI and input of ferroalloys are often the only commercially viable solutions to adjust chemical composition of steel products (Compañero, Feldmann, and Tilliander, 2021). In any case, the elemental composition of steel products from EAFs is measured and declared to meet existing national and EU standards for specific products (e.g., rebars). For certain end-use applications such as automotive steel, these compositional standards are based on own requirements of private companies (i.e., car manufacturers), (information from stakeholders consultation).

In the majority of the cases, tramp elements are non-functional to steel products, i.e. when a neutral effect on metal performance occur. Reijnders, 2016 discussed options for preserving functionality of critical and strategical raw materials added to steel alloys, which typically undergo non-functional recycling in steel life cycles. These metals may include Cr, Co, Cu, Mn, Mo, niobium (Nb), Ni, lead (Pb), Sn, V, and tungsten (W): their presence may either related to natural concentrations in ores or presence in steel scrap, but also occur in combination with other elements such as oxygen, Fe and Si (e.g., MoO_3 , FeV, FeNi, SiMn). These metals may be also intentionally used for steel surface treatment aimed at increasing resistance to corrosion and wear, or in soldering and brazing materials. Some alloying elements at certain concentrations exceeding standard requirements can be also detrimental to steel properties, with synergic effects onto dysfunctionality such as in the case of Sn, Sb, and arsenic (As) with surface defects, reduced hot plasticity and ductility induced by Cu. The potential concentration limits for cold-rolled steel have been determined at 0.04% and 0.02% for Cu and Sn. For rebar steel, instead, Cu and Sn tolerances are in the order of 0.4%. Eckelman et al., 2012 estimated about 22% of Ni inflow to use at the global level in 2005 may end up as non-functional element in carbon steel. Because concentrations of non-functional and dysfunctional elements in steel are expected to increase, measures to increase tolerance to elements such as Cu may be needed. While the increase of Ni in carbon steel

may buffer detrimental effects induced by Cu, increasing tolerance of steels, this would determine a loss of functional Cu and Ni from the economy. Some EAFs produce steels with high technical requirements (including flat products), but most have steel intermediates (e.g., rebars) with higher tolerance towards the presence of alloying elements in their product portfolio. This is due to challenges of exploiting the content of alloying elements, with perhaps the exception of austenitic stainless steel which is nonmagnetic and can be easily separated from other steels. A consequence is that EAF output is not suitable for high-demanding applications such as packaging and body parts of vehicles, due to thermodynamics. While some alloying elements such as Mn, Cr, and V tend to oxidize into slag, other elements such as Ni, Mo, Co, W, and Cu tend to remain in the metal phase and to accumulate along with repeated recycles.

Better information about scrap and scrap sorting by alloy type can lead to more efficient resource allocation, enabling EAF steelmakers to expand their production recipes to include a wider variety of scrap types, thereby stimulating demand in the secondary market. Scrap dealers are motivated to make cost-effective deliveries, as higher-quality scrap is easier to sell. For EAF steelmakers, success involves loading the furnace with the correct qualities and quantities of scrap to minimize or eliminate the need for virgin materials and maximize the use of specific elements already present in the scrap, while maintaining product tolerances.

Some steel products produced in European EAFs have already a significant recycled content: for instance, about 97% of rebars in the EU is produced from scrap, and 85% of sections, wired rods, and merchant bars come from EAFs. However, the majority of high-quality steel products is produced from primary steel route, which also counts on scrap inputs but typically limited to 20% on mass.

Choice of material, surface coating and options for joining

Minimizing steel variety in products can reduce cross-contamination and recycling costs (Bartl, 2014), while facilitating source separation of steel varieties at EoL. Preferring widely used alloys to promote closed-loop recycling could promote steel product recovery and re-employment for similar applications as those for which they were initially designed (Gesing and Wolanski, 2001), (Graedel et al., 2011). The choice of material can be also relevant for selecting of alloying elements for specific functions: for instance, Cu can be replaced by Co and Ni for stabilizing austenite steels (Yellishetty et al., 2011).

Steel surface coating may cover a variety of elements including Zn and Ni for corrosion resistance, Co, Cr, Ni, and W for hardness and wear resistance, although also colouring purposes may drive the choice of top coatings. The selection of substitutes for current surface coatings should take into account recycling technologies and their requirements. For instance, metal and mineral coatings could be replaced by non-metallic materials (e.g., Sn coated steel for beverage cans by PET and HDPE, Zn-coated steel with Al and Si alloys). Similarly, polymeric organic coatings or hybrid inorganic-organic coatings, in particular sol-gel derived coatings based on silanes, could in principle avoid metal and mineral based coatings. Although, their removal can be relatively easier than that of metal and mineral based coatings (e.g., through burn off, use of blasting agents, treatment with molten salts and laser ablation), the main issues seem to relate to the overall adhesion and performance of alternative coatings.

About joining, friction stir welding can be applied to join the same steel varieties, while mechanical fasteners or shape memory alloys and polymers activated by external riggers may be employed when the joining materials and the joined materials differ in composition.

3.4.1.2 Best Not Available Technologies

Advanced sorting

Potential improvement in the management of steel scrap may be possible by incorporating material characterization in scrap sorting.

The most promising detecting and sorting technologies applicable to sort Cu-rich fragments as well as to determine, the composition of a scrap batch before remelting include: X-ray fluorescence, laser-induced breakdown spectroscopy, neutron activation analysis, and image processing. Dual energy X-ray transmission (DE-XRT) sensors have been historically employed for airport luggage inspection and medical uses. DE-XRT sensors provide an image of the materials and the colour and intensity give a relationship between the atomic number or density and the inspected material. DE-XRT sensors can be used in metal recycling for sorting light metal scrap (e.g., Al and Mg) from dense nonferrous ones (e.g., Zn, Cu, brass) as well as for guiding separation of non-metallic fractions in car fluff. However, the current DE-XRT sensors on the market seems not to have adequate resolution to distinguish between dense metals and alloys.

X-ray fluorescence (XRF) spectrometry exploits electron transition induced on a material and its relationship with the characteristics of the material itself. The energy difference measured provides information of the identity of the material, while the intensity of fluorescent photons quantifies the concentration of the elements embedded. XRF can provide real-time qualitative elemental analysis, but also quantitative characterization through calibration techniques. XRF can penetrate through materials and XRF-based technologies are usually distinguished between energy dispersive (ED-XRF) and wavelength dispersive (WD-XRF) spectrometers. ED-XRF can be both handheld and in belt configurations and it has been commercialized to find Cu parts in shredded steel scrap outflows. Current limitations include lack of precision, challenges to find hidden materials, relative slow read times and the need of safety precautions to contrast radiation emitted from XRF instruments.

Optical emission spectroscopy (OES) is a plasma technique for elemental characterization. It relies on the detection of photons emitted during the de-excitation of the energetic particles in the plasma. OES is non-intrusive, non-disturbing, and it enables real-time results (in the order of nanoseconds). The plasma can be generated through different configurations, among which the arc spark-OES, which employs an electric arc to melt, vaporize and ionize a small metal sample, is the most used analytical method for metal alloys characterization in industrial plants. An emerging configuration couples OES with a laser source (i.e., laser induced breakdown spectroscopy, LIBS). LIBS is considered a micro-analysis, surface technique because only small distances into the surface of a metal sample can be penetrated by pulse lasers. Although pre-cleaning is generally incorporated into the analyser, this adds supplemental time for the analysis. LIBS results can be affected by surface coatings, which may induce inaccurate bulk measurements, with efficiency that may decrease when Cu is wound to steel or hidden by other metals.

Prompt gamma neutron activation analysis (PGNAA) enables qualitative identification and quantitative determination by exploiting the prompt gamma rays emitted during the decay of radioactive products for elemental analysis. PGNAA techniques can penetrate a large depth of metal scrap and permit real time analysis of scrap layer conveyed on the belt, eliminating the issue of sampling: all materials can be characterized, and no limitations exist for belt speed nor particle size. PGNAA sensors can be combined to separators for diverting contaminated materials from the stream, enabling scrap source-separation by composition, ultimately improving the quality of scrap batches in EAF. Gamma Tech (GammaTech.us) has done significant work on the application of PGNAA cross belt analysers for scrap metal. PGNAA cross belt analysers do not have as much resolution as other optical analysers, but they can provide accurate average elemental composition results of scrap conveyed by belt so that the target cannot be individual particles, but rather the diverting of a defined portion of a stream that contains impurities. Current research is aimed at developing less expensive neutron sources employed in the first generations of PGNAA techniques as they are priced to be economically competitive.

Brooks and Gaustad (Brooks and Gaustad, 2021) demonstrated that surface coatings and interferences with scrap may induce fluctuations in chemical composition, making designation by alloy grade useless. These authors assessed different handled analysers on scrap characterization and highlighted how their response is generally reliable, although exact chemical composition determination is usually out of their reach. However, their use in combination with visual inspection and trained operators may be of great value for relatively good matching. Handled analysers are becoming a relative low risk, low cost and high reward instrument. As of today, technologies for advanced automatic sorting are hence theoretically available but, in several cases, unlikely to be fully scaled-up in the short term because of different speed rate for processing steel scrap and inefficiency due to potential hidden of Cu.

Additionally, artificial intelligence (AI) recognition software could improve the efficiency of inspections, sensors, diversion, separation, and sorting technologies.

Chemical treatments during remelting

Several strategies may also be undertaken at EAF steelmaking for enhancing removal of undesired tramp elements or increasing tolerance to them. For example, vaporization or formation of Cu sulfide or chloride have been proposed as incoming mechanism to remove Cu from the melt. In contrast, shortening oxidation times and surface quenching may increase Cu tolerance in steel intermediates, while addition of Ni or Si may overcome hot shortness (Daehn, Cabrera Serrenho, and Allwood, 2017).

Increasing tolerance to tramp elements could also reduce losses of critical materials in EAF steelmaking. For instance, addition of Al_2O_3 and SiO_2 may reduce losses of Cr in EAF, while Mn losses can be reduced by increased slag basicity in EAF, or losses of Mo can be decreased by substituting MoO_3 by Fe_2MoO_4 . Nb loss in EAF is found

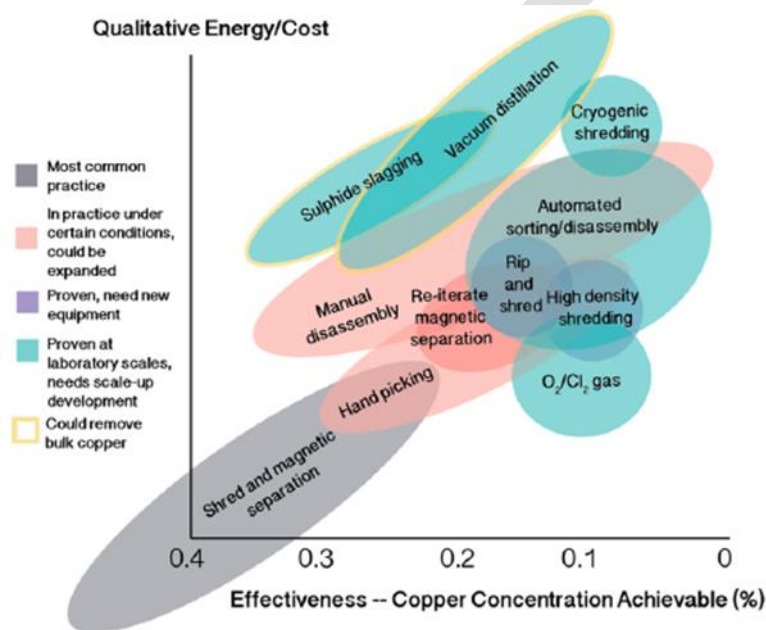
to be dependent on temperature equilibration time and basicity. Further improvement could be also pursued by stochastic optimization routines for scrap mixes recipes.

These technologies may be selective only for specific elements and no industrial, full-scale applications have been reported yet. Furthermore, the environmental implications of these options might be all but not negligible.

Solid state recycling

Solid state recycling (SSR) may constitute an alternative route to remelting of scrap, which is in turn affected by potential functionality loss of tramp elements. SSR relies on plastic deformation processes, to which cleaning, reduction of oxides, hot forming, extrusion, annealing and sintering stages may be preparatory treatments. In the case of old scrap SSR, prior removal of coatings and contaminants is required.

Figure 115. Estimated copper concentration achievable and qualitative energy/cost comparison for selected sorting, separation, and recycling technologies.



Source: Daehn et al., 2017

Figure 115 compares several sorting, separation and recycling technologies in terms of their efficiency in removing tramp elements such as copper and their requirements in terms of energy and costs.

No single technology can prevent steel downcycling by itself. A combination of methods such as a better source separation, better sorting technologies, can improve steel scrap quality (Ciacci, 2024)

3.4.2 Preparation of precursors

In the production of iron and steel, the preparation of raw materials is crucial for optimising process efficiency, minimising resource waste, and reducing environmental impacts. Key areas of focus include managing fines generated during mining and processing, ensuring consistent and effective blast furnace charging, and utilizing raw materials with high iron content.

3.4.2.1 Best Available Technologies

Screening out sinter, pellets, and coke fines

The formation of fines may occur during iron ore mining and grinding. These fines represent fractions with less than 10 mm, which decrease the efficiency of the iron making process, because they hinder heat and gas permeability into the ferrous materials and inhibit the reduction process. To avoid the presence of this fines during steel making process, sinter and palletisation processes are applied to agglomerate the fine iron particles into sizes that can be use in the iron making process.

Vibration screens are usually used to separate the fines from larger particles of ferrous materials, which can go directly to iron making, while the fines will undergo further sintering and/or palletization processes.

Also coal mining and coke production creates fines which clog pipes of the iron making processes, decreasing their overall efficiency. Different measures are used to avoid the negative effects of such fines: 1) use of moist coal in the coke process, 2) separation of the fines through hot air separator, 3) after separation the fines can be granulated to increase their size and they can then be used in the iron making process (M. H. T. Team, 2023).

Correct charging of the blast furnace

The efficiency of a blast furnace is highly interlinked with the reliability of the raw materials charging system. Being a continuous process any disturbance on the raw materials feeding may require a decrease in the blast furnace operation speed or even its complete shutdown.

There are several ways and technologies to charge a blast furnace, however modern and more efficient ones use a conveyor system and chute-type cone-less top chargers. According to M HEAVY TECHNOLOGY when compared with other systems these modern charging technologies may reduce coke consumption to 20 kg/t of hot metal, which significantly decreases CO₂ emissions (M. H. T. Team, 2024).

Using raw materials with high iron content

Iron ore may be classified by its content in iron in high or low grade depending if its iron content is above or below 35%, respectively. While high grade may be used directly in steel production, low grade ores need to pass to several beneficiation and concentration processes to increase the iron percentage in the ore (M. H. T. Team, 2024).

3.4.2.2 Best Not Available Technologies

Iron bath reactor smelting reduction (IBRSR)

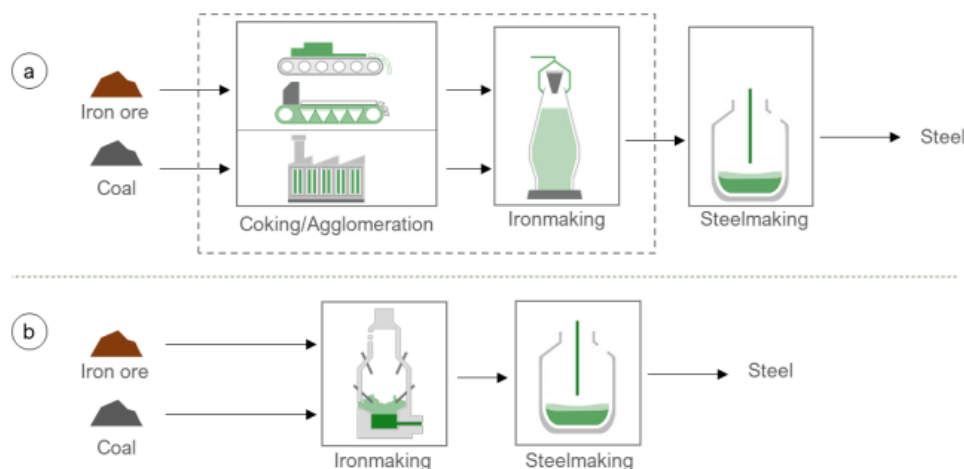
Hlsarna® is an ironmaking process that replaces the BF with a simplified carbon-based smelting reduction process, eliminating the iron ore agglomeration steps (from sinter or pelletising plants) and the use of coke (from the coke oven), see Figure 116. The technology produces hot metal directly from raw materials and requires less stringent conditions on the quality of the raw materials used. The technology's readiness level (TRL) is currently uncertain, and it is unclear whether Tata Steel is actively developing this route and if it will be available in the 2030s (Draxler et al., 2021; Agora Industry, Wuppertal Institute, and Lund University, 2024).

Table 51. Main characteristics of the IBRSR technology

Features	IBRSR
Production route	BOF
TRL level	TRL 5-6
Expected market readiness	2030-2035
Country (Geographical scope) of implementation	The Netherlands and India
Potential CO₂ reduction compared to BF-BOF	38%
Energy Needs	15-17 GJ/ton
Potential of other environmental aspects	- Reduces emissions of nitrogen, sulphur oxides and nitrogen oxides - Avoidance of dumping slags and dusts - Reduction of raw material extraction - Water savings - Energy savings
Levelised cost 2030	584 EUR/ton
Projects	Hisarna (Tata Steel Europe IJmuiden plant) and Hisarna Jamshedpur Tata Steel India)

Source: JRC elaboration based on Climate-neutral steelmaking in Europe (European Commission, European Research Executive Agency, et al., 2022); Low-carbon technologies for the global steel transformation (Agora Industry, Wuppertal Institute, and Lund University, 2024); Hisarna. Building a sustainable steel industry (Tata Steel, 2020); and Net-zero emission opportunities for the Iron and Steel industry at a global scale (Desport, Andrade, and Selosse, 2022).

Figure 116. Comparison of BF-BOF route (a) and the Hisarna® technology (b)



Source: Green Steel for Europe. Technology Assessment and Roadmapping (Draxler et al., 2021)

3.4.3 Energy sources

The iron and steel industry is highly energy-intensive, with coal accounting for approximately 75% of the energy inputs in steelmaking (IEA, 2020). Fuel switching from coal to renewable or less carbon intensive fuels and electrification may be applied to reduce environmental impacts of iron and steelmaking. With several of these practices it is possible to change the energy source and substitute fossil-based reduction agent. The reduction technologies are described in more detail in section 3.3.6.

Alternative fuels like gas or biomass, with lower carbon intensity, could potentially replace or partially replace coal, which generally requires less process chain modifications, although its use is constrained by the availability of gas and biomass resources (IEA, 2020). Electrification, in particular of industrial heat offers the opportunity to increase energy efficiency and to make use of renewable energy, such as solar and wind. Hydrogen will be important for future iron and steel making, while utilizing 100% renewable electricity in the EAF and electrifying ancillary process equipment can be implemented already today. However, the direct electrification of the initial steps in processing steel (like steel rolling) faces severe limitations due to their high energy-intensive processes and needs for major redesigns of rolling lines. Therefore, electrification of industrial e.g. heating is hindered by initial costs for new equipment and infrastructure upgrades, as well as the technological maturity of some electric technologies (in particular in terms of capacity) (Fraunhofer ISI, 2024), (AFID, 2024).

Table 52. BAT & BNAT for reducing environmental impacts through fuel switching and electrification

BAT	BNAT
- Use of 100% renewable electricity in EAF - Substitution of BF-BOF with EAF and DRI-EAF route - Electrification of ancillary process equipment (preheaters and boilers) - Fuel switch from coal to natural gas (DRI route)	- Gas injection into the BF - Change from coal and natural gas to hydrogen (see section 3.4.6) - Change from coal to biomass and biochar (see section 3.4.6) - Use of electrolysis technologies in the reduction stage AEL, MOE and LTE (see section 3.4.6)

Source: JRC elaboration

Reducing agents serve a dual role as both facilitators of chemical reactions and carriers of energy in iron and steel production processes. Their function as energy carriers is crucial because they provide the necessary energy to drive the reduction reactions. Consequently, these technologies will be further explored in Section 3.4.6, which is dedicated to reducing agents, highlighting their significance in energy transfer and overall process efficiency.

3.4.3.1 Best Available Technologies

Use of 100% renewable electricity in the EAF

One way to decrease indirect emissions from steelmaking is by utilising renewable electricity. For instance, in the secondary route using EAF, the use of renewable electricity could remove approximately 0.1-0.2 tCO₂ per tonne of steel (Somers, 2022). The use of 100% renewable energy can also reduce more than 80% CO₂ emissions compared to the integrated BF-BOF route. Several innovative projects developed through Power Purchase Agreements (PPAs) involve retrofitting existing EAF mills to convert them into renewable energy-powered steel plants. A prime example is EVRAZ in the USA, which operates the world's first solar-powered steel mill. Other examples include the Tata Steel and Tata Power project with 41MW solar project in Jharkhand and Odisha, India and Nucor investments on solar and wind energy production projects¹²⁷.

Table 53. Main characteristics of the use of 100% renewable electricity in the EAF

Feature	100% Renewable Energy in EAF
Steelmaking Production route	EAF
TRL level	TRL 9 [1]
Country (Geographical scope) of implementation	USA and India
Potential CO₂ reduction compared to BF-BOF	80% considering comparison with secondary route [2]
Energy Needs (GJ/ ton of crude steel)	2.1 GJ/ton (IEA) , 5.2 GJ/t (Worldsteel)
Potential of other environmental aspects	Energy efficiency
Expected market readiness	Commercially available
Levelised cost 2030	594 - 778 EUR/ton [1]
Projects	EVRAZ (USA), Nucor (USA, Solar and Wind), Tata Steel and Tata Power (IN) [3]

Note. All values are converted to Euro (€) using the average 2023 exchange rate of \$1 = €0.93

Source: JRC elaboration based on Low-carbon technologies for the global steel transformation [1] (Agora Industry, Wuppertal Institute, and Lund University, 2024) ; Technologies to decarbonise the EU steel industry (Somers, 2022) ; [2] (Worldsteel, 2023) and The use of green steel in the automotive industry (Ricardo, 2024), [3] (Worldsteel, 2024)

Substitution of BF-BOF with EAF and DRI-EAF route

Another way to decrease carbon footprint is to increase the use of Electrical Arc Furnaces (EAF) within the steel making industry. Recent data (IEA, 2020) shows that globally, steel production based on the DRI-EAF process, accounts for 10% of primary steel production and EAFs route accounts in total for 28% of the entire steel production, including scrap. EAFs, especially when powered by renewable energy, are more energy-efficient compared to traditional blast furnaces. They also allow for recycling of scrap steel, further reducing energy use and emissions.

Table 54. Main characteristics of the use of increase utilisation of EAF

Feature	Substituting BF-BOF with EAF and DRI-EAF route
Steelmaking Production route	EAF and DRI-EAF
TRL level	TRL 9 [1]
Country (Geographical scope) of implementation	Worldwide
Potential CO₂ reduction compared to BF-BOF	40% (primary route) and 70% (compared with secondary route)* [2]

¹²⁷ Solar energy is fuelling more sustainable steel production <https://worldsteel.org/media/steel-stories/infrastructure/solar-energy-fuels-sustainable-production-of-rails/> & Power plant profile: Western Trail Wind Farm, US ([power-technology.com](https://www.power-technology.com/news/2022/05/10/western-trail-wind-farm-us/))

Energy Needs (GJ/ ton of crude steel)	17.1 GJ/t (IEA), 21.8 GJ/t (Worldsteel) data for NG-DRI with Scrap 2.1 GJ/ton (IEA) , 5.2 GJ/t (Wordsteel) [3]
Potential of other environmental aspects	Energy efficiency
Expected market readiness	Commercially available
Levelised cost 2030	400-600 USD/t [4]
Projects	na

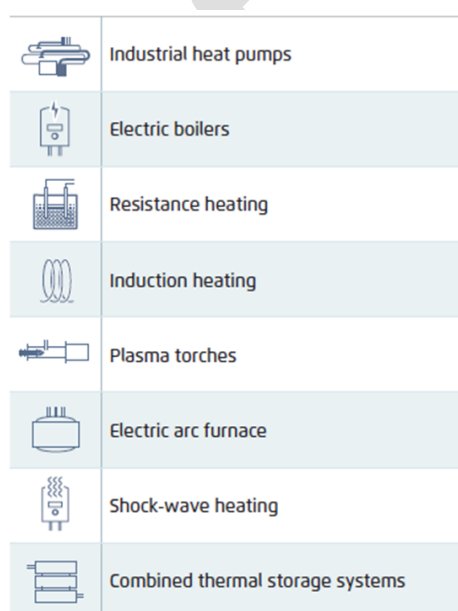
* WorldSteel BF-BOF process, on average, emits 2.33* tons of CO₂ for every ton of steel and DRI-EAF 1.4 and with recycle content 0.7 tons of CO₂ for every ton of steel (Worldsteel, 2023)¹²⁸

Source: JRC elaboration based on Low-carbon technologies for the global steel transformation [1] (Agora Industry, Wuppertal Institute, and Lund University, 2024) ; [2] (Worldsteel, 2023), [3] (IEEFA, 2022), [4] (IEA, 2020).

Electrification of ancillary process equipment

Electrification of industrial processes presents a promising pathway for reducing carbon emissions, enhancing efficiency and lowering operational costs. The Fraunhofer ISI report (2024) provides a comprehensive review of various heat electrification technologies, which are illustrated in Figure 117.

Figure 117. Heat direct electrification technologies



Fraunhofer ISI (2024), Agora Energiewende (2024)

Source: Direct electrification of industrial process heat An assessment of technologies, potentials and future prospects for the EU (Fraunhofer ISI, 2024)

The application of these technologies in the steel industry is highly dependent on their individual heat capacity and operation temperature.

Summary description of the technologies:

Electric Boilers (TRL 9) use electricity to heat water or generate steam, which can then be used for various industrial processes. They are highly efficient, have low maintenance costs, and produce no direct emissions. Heat capacity up to 75 MW and temperature level up to 500 °C. Barriers for their wider use include high electricity price compared to natural gas in most European countries.

Heat Pumps (TRL 9) transfer heat from a source (like air, water, or ground) to a process that requires heating. They can achieve efficiencies greater than 100% by moving heat rather than generating it, making them very

¹²⁸ [Sustainability Indicators 2023 report - worldsteel.org](https://worldsteel.org/reports/sustainability-indicators-2023-report/)

energy-efficient. They are ideal for lower temperature heat requirements, offering high efficiency and reduced energy consumption.

Resistance Heating (TRL 9) involves passing an electric current through a resistive material to generate heat. It is commonly used in furnaces and ovens for processes like melting, forging, and annealing metals. The main barrier for its wider application in steel making industry is the necessity to make profound changes in the existent infrastructure. Heat capacity up to 200 kW/m² (by 2030) and temperature level up to 2000 °C.

Induction Heating (TRL 9) uses electromagnetic fields to heat electrically conductive materials. It provides rapid, localised heating and is highly efficient, making it suitable for processes like metal hardening and welding. Induction furnaces with a capacity of 42 MW are used today in the steel industry.

Plasma torches (TRL 7-9) use electrically charged gases (plasma) to generate high temperatures (up to 5000°C). It is used in precision heat applications such as plasma cutting and arc welding. Future perspectives for this technology include increasing plasma torches lifetime, reducing power input and reducing cooling needs, making them alternatives to fossil fuel burners.

Shock-Wave Heating (TRL 6) uses high pressure waves to rapidly heat a fluid. It is still in the experimental stage but shows promise for high-temperature industrial processes. It is foreseen that in 2030 this technology could potentially achieve temperatures of 1 500 °C (currently achieves 700 to 1000 °C) and power inputs of 50 MW, for now the heat capacity stays near 1 MW. In steel industry it can be applied in surface treatment processes.

Combined Thermal Storage Technologies (TRL 7-9) store thermal energy for later use, allowing for more flexible and efficient energy management. Typically, heat is generated by electrical resistance and then transfer to a storage medium such as molten salt or bricks. They are currently being used in small scale applications, but research indicates that by 2035 they may be able to produce and store heat at temperatures of up to 2 000 °C.

In summary, apart from the EAF the currently available technologies for electrification of heat in steel industry are limited to application in ancillary process like pre-heaters, boilers, and metal working processes.

Table 55. Summary of technologies for electrification of ancillary process equipment

Features	Electric Boilers	Heat Pumps	Resistance Heating	Induction Heating	Plasma torches	Shock-Wave Heating	Combined Thermal Storage
Production route	BF-BOF and EAF	BF-BOF and EAF	BF-BOF and EAF	BF-BOF and EAF	BF-BOF and EAF	BF-BOF and EAF	BF-BOF and EAF
TRL level	TRL 9	TRL 9	TRL 9	TRL 9	TRL 7-9	TRL 6	TRL 7-9
Expected Market Readiness	Currently available	Currently available	Currently available	Currently available	2035	2035	2035
Geographical scope	Worldwide	Worldwide	Worldwide	Worldwide	Worldwide	Worldwide	Worldwide
Potential CO ₂ reduction compared to fossil alternative	up to 100%*	up to 100%*	up to 100%*	up to 100%*	up to 100%*	up to 100%*	up to 100%*
Energy capacity	Up to 75 MW	Current 20MW	80 kW/m ² of material surface	42 MW	8 MW (current)	1 MW (current)	18 MW (current)
Max temperature	500°C	200-300°C	1850°C	3000°C	5000°C	700-1000°C	1000°C
Potential reduction of other environmental aspects	- Energy savings Efficiency 99%	- Energy savings Efficiency 150-600%	- Energy savings Efficiency 99%	- Energy savings Efficiency 90%	- Energy savings Efficiency 50-90%	- Energy savings Efficiency 90%	- Energy savings Efficiency 90%
Levelised cost 2030 (EUR/ton)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Projects	N/A	N/A	N/A	N/A	N/A	N/A	N/A

*depending on electrical mix, N/A not applicable

Fuel switch from coal to natural gas

In the BF-BOF route coke derived from coal is used, while for the DRI-EAF route natural gas is used as energy source and reduction agent. The transition from coal to natural gas is facilitated using natural gas blast furnaces and increase use of DRI furnaces (AFID, 2024). These changes are limited to geographical areas where natural gas is available at high volumes and competitive prices (IEA, 2020). While the use of natural gas is already commercially available (detailed information on this technology is given in section 3.3.6), the use injection of natural gas in blast furnaces (NG-BF) is not yet fully commercially available with a TRL of 5-8 therefore, NG-BF was classified in this report as BNAT (see section 3.4.3.2).

3.4.5.3 Best Not Available Technologies

Fuel switch from coal to biomass or biochar

The substitution of coal by biomass or biochar is almost direct with minimal or no modifications needed in the BF systems. Countries with substantial biomass resources, such as Brazil, are already incorporating biomass in small-scale blast furnaces. However, the widespread use of biomass is hindered by challenges related to its availability and cost.

Fuel switch from coal and natural gas to hydrogen

The use of hydrogen, in particular green hydrogen, can decrease carbon footprint by 20% in a BF operation. This change can be implemented in the existent furnaces by modifying the injection system. The extent of such fuel change adoption depends on the costs of green hydrogen production.

Table 56. Summary of NBAT for substitution of fossil energy carriers with NG in BF and biomass

Features	Gas injection into the BF	Substitution of fossil energy carriers by biomass
Production route	BF-BOF	BF-BOF & EAF
TRL level	TRL 5-8	TRL 2-7
Expected market readiness	2035	2035
Country (Geographical scope) of implementation	Europe	Brazil, Europe
Potential CO₂ reduction compared to BF-BOF	20%	30%
Energy Needs	The same as BF	The same as BF
Potential of other environmental aspects	Prevent resource depletion	Prevent resource depletion
Levelised cost 2030	na	na
Projects	ULCOS (Tata Steel Ijmuiden, Netherlands)	GreenEAF2 (Italy) and RETROFEED (Romania)

Source: JRC elaboration based on AFID, 2024 and Agora Industry, 2024

3.4.4 Process optimisation

Process optimisation, intensification, monitoring and control promotes energy and material efficiency, lowering the costs of steel making. The integrated primary production routes have been optimised over the years and according with the JRC, the implementation of further optimisation best practices is estimated to reduce CO₂ emissions by about 10% (JRC, 2020) on existent infrastructures. This section will summarise the existent best practices on process optimisation in three main categories: energy efficiency, material efficiency and process control measures.

The steel industry is one of the most energy-intensive industries in the world and it has much potential for improving **energy efficiency** of its processes. There are several measures that increase process performance, most of them highly implemented in the steel industry to reduce energy consumption and optimise fuel use, they include: 1) the use of high efficiency burners; 2) waste heat and off-gas recovery; 3) pre-heat input streams (raw materials and combustion air) and ladle; and 4) produce electricity from waste heat recovery. Excess heat

and electricity can be used within the steel production facility and/or exported to nearby district heating or electricity grids. Deployment of waste heat recovery can reduce the average energy intensity of crude steel production via the BF-BOF route by up to 2% (IEA, 2020). Several technologies can be applied to guaranty waste heat recovery and minimise energy losses, some examples include: 1) waste heat boilers, 2) heat exchangers; 3) heat pumps, 4) BF top recovery turbines; 5) coke dry quenching (CDQ) 6) other heat to power technologies.

Table 57. BAT & BNAT for reducing environmental impacts through process optimisation

Source: JRC elaboration

Technologies to promote energy efficiency

Figure 118. Waste heat sources in the steelmaking process

Source: (Fives, 2017)

The largest waste heat sources relate to the ironmaking processes followed by steelmaking, sintering, rolling and coking processes.

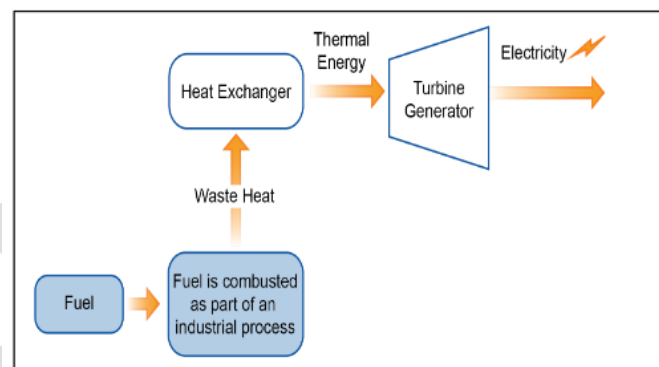
Waste heat boilers can recover waste heat from different processes and use it to generate steam. They can be installed near large furnaces and produce steam for several applications, such as process heating or electricity production.

Heat exchangers are used to transfer heat between two or more fluids at different temperatures. They play a crucial role in steel industry by recovering and utilising waste heat. There are several heat exchanger types and their use depend on the final application and waste heat fluid: scraped surface heat exchangers, shell and tube heat exchanger, plate and frame heat exchanger and heat pipes.

Heat pumps (already disused in section 3.4.3) can extract heat from waste heat, amplify its temperature, and transport it to the final application. They are extremely efficient, and the output energy is typically several times higher than the energy (normally electricity) necessary to power them. They can be used in different processes in the steel industry and to recover heat from cooling of steel products.

Waste heat to power technologies allow to recover the energy existent in the waste heat to produce electrical energy (CHP - EPA, 2022). Figure 119 shows a schematic representation of the electricity production from waste heat. Several technologies can be used for this purpose, and their application depends on the temperature. While there are several options for the recovery of heat from medium to high heat temperatures, the recovery of electricity from low temperature heat is more limited the TRL of available technologies is lower.

Figure 119. Waste Heat to Power Diagram



Source: (CHP - EPA, 2022)

Heat to power technologies that use waste heat streams to generate electricity. Include technologies based on classic thermodynamic cycles, which involves a phase change of a working fluid to produce electricity with a turbine: steam Rankine cycle (SRC); Organic Rankine cycles; Kalina cycle (BNAT) and supercritical CO₂ cycle (BNAT).

The steam Rankine cycle (SRC) (STEAM turbine) is the most used system for power generation from waste heat. The process needs a waste heat boiler that produces steam, which then drives a steam turbine and producing electricity. Steam turbines operate thermodynamically as a Rankine Cycle and are one of the oldest and most versatile technologies. In this cycle water is the fluid used to recover and transport the energy from the waste heat and the turbine.

Organic Rankine Cycle (ORC) similarly to steam turbines running on steam Rankine cycle the organic Rankine cycle technologies can utilize waste heat to generate electricity, but instead of using water as the working fluid, they use organic fluids with lower boiling points. In this way ORC technologies can recover low temperatures heat streams to electricity, increasing the opportunities to recover waste heat (CHP - EPA, 2022).

Top-pressure recovery turbines this system generates electrical power by using the heat and pressure of blast furnace top gas to drive a turbine generator. Blast furnace gas (BFG) has a pressure of 0.2-0.236 MPa (2-2.41 kg/cm²) and temperature of approx. 200 C at the furnace top (UN CTCN, 2017). The electricity generated by this method can yield around 30-40 kWh of electricity for each tonne of pig iron produced when using typical wet de-dusting of top gases (it can be increased to 50- 60 kWh if using dry de-dusting) (IEA, 2020).

Coke dry quenching (CDQ) is a technology that uses the latent heat from the hot coke and uses it to generate electricity. The system uses the hot coke from coke ovens (with a temperature of approximately 1,000°C), that is cooled with an inert gas, the waste heat resultant from this cooling process is recovery in a boiler to produce steam which is then used to generate electricity.

Table 58. Summary of technologies for waste heat recovery

Features	Waste heat boilers	Heat Pumps	Heat exchangers	Steam turbine	Organic Rankine Cycle	Top-pressure recovery turbines	Coke dry quenching (CDQ)
Production route	BF-BOF and EAF	BF-BOF and EAF	BF-BOF and EAF	BF-BOF and EAF	BF-BOF and EAF	BF-BOF and EAF	BF-BOF and EAF
TRL level	TRL 9	TRL 9	TRL 9	TRL 9	TRL 7-9	TRL 9	TRL 9
Expected Market Readiness	Currently available	Currently available	Currently available	Currently available	Currently available	Currently available	Currently available
Geographical scope	Worldwide	Worldwide	Worldwide	Worldwide	Worldwide	Worldwide	Worldwide
Potential CO ₂ reduction compared to fossil alternative	-0.015 to -0.051 tCO ₂ /t	-0.015 to -0.051 tCO ₂ /t	-0.015 to -0.051 tCO ₂ /t	N/A	1-3% direct emissions	-0.014 tCO ₂ /t (1-2%)	-0.085 tCO ₂ /t (2-3%)
Energy Needs*	-0.2 to -1 GJ/t	-0.2 to -1 GJ/t	-0.2 to -1 GJ/t	N/A	N/A	-0.3 GJ/t	-1.6 GJ/t
Potential reduction of other environmental aspects	- Energy savings	- Energy savings	- Energy savings	- Energy savings	- Energy savings	- Energy savings	- Energy savings
Levelised cost 2030 (EUR/ton)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Projects	N/A	N/A	N/A	N/A	N/A	N/A	N/A

*compared with the conventional process, N/A not applicable

Source: JRC elaboration based on (Xu and Cang, 2010), (Pardo et al., 2015), (CHP - EPA, 2022), (IEA, 2020), (AFID, 2024)

Technologies enabling material efficiency

This section focuses on detailing technologies that increase the efficiency of the Iron and steel making processes by preventing material losses through the steel value chain. Therefore, this section also translates solutions for **waste management and waste reduction**.

Steel slag is a by-product of the steelmaking process. It is produced when liquid steel is separated from impurities in BOF and EAF. Slag is composed of impurities such as silicates and metal oxides. Slag represents around 10–15 % of crude steel production. According with the BREF documents the use of this material in other industries and its recovery within the steel making process allows to decrease waste, improve energy efficiency and increase the economic feasibility of the steel industry.

Slag as a co-product: Slag can be processed and sold to external industries, such as the cement, agriculture, construction industries and functional fillers, where it is used as a valuable raw material. Depending on the final uses, slag recovery may decrease CO₂ emissions by roughly 340–826 kg/t (Fan et al., 2024).

Internal slag recycling: Slag can be recycled within the steel plant. For example, it can be used as a fluxing agent in the blast furnace or as a raw material in sinter plants.

Recovery of slag entrained metal: Techniques are employed to recover any metal entrained in the slag. This recovered metal can be reused in the steelmaking process. This recovery can yield reductions of CO₂ in the steel making process up to 15%.

Like explained before heat can be recovered from several streams in the steel making process, including from hot slag coming out of the BOF or the EAF processes. Studies show that waste heat recovery from steel slag can indirectly lower CO₂ emissions by approximately 73.15–111.8 kg/t (Fan et al., 2024).

Additionally, direct carbon sequestration using steel slag can reduce CO₂ emissions by about 77.5–258.5 kg/t (Fan et al., 2024).

Other best available technologies applied to avoid the disposal of wastes include: recovery of metal wastes or scrap (other than slag), recovery of metal oxides and recovery of oily sludge and hydroxide sludge (Aries et al., 2022).

The advantages and challenges of scrap use are detail in section 3.4.1. The use of scrap within iron and steel industry has a high potential to optimise steel production and decrease EU import dependency of primary iron materials.

Reuse of mill scale: after treatment to remove contaminations such as oil, mill scale can be reused internally, within iron and steel production or sold to as by-product to be used in the cement industry (Aries et al., 2022).

Recycled of metal and metal oxides from waste gas: after dry waste gas cleaning the metal particles trapped in e.g. the filters are separated from the rest of the waste by e.g. sieves and/or magnetic techniques and reused within the iron and steel making processes (Aries et al., 2022).

Reuse of drawing lubricants or other sludge oils: after treatment and drying the oils containing low water content may be reused for material or energy recovery. Their treatment may be performed outside the iron and steel production facility (Joint Research Centre et al., 2013), (Aries et al., 2022).

Treatment of hydroxide sludge for material recovery: Hydroxide sludge coming from the recovery of mixed acids may be thermal treated to produce a stream reach in calcium fluoride, which can be applied in argon oxygen decarburization processes (Aries et al., 2022). Lack of space in the steel making facility may block the use of such material recovery technique (Aries et al., 2022).

Process control and monitoring is essential to guaranty energy and material efficiency. Enhanced digitalisation of process controls can use real-time data and dynamic models to adjust the process parameters for optimal performance. Additionally, the integration of artificial intelligence increases the predictive power of process control and planning.

3.4.5.4 Best Not Available Technologies

Technologies to promote energy efficiency

Like for the Rankine cycle Kalina cycle technologies are based on thermodynamic cycle that uses a phase change to recover energy from waste heat and convert that energy in electricity. However, in this case the working fluid is a mixture of water and ammonia. This mixture allows for a more efficient energy conversion than the conventional Rankine cycle. While the Kalina Cycle has been widely adopted in various industries, its application in the steel industry within Europe is still emerging. However, there are ongoing projects and studies exploring its potential. Kashima Steel Works had proved the advantages of the use of such technology (Global Geothermal, 2024).

Supercritical carbon dioxide cycle it is a thermodynamic cycle that uses super-critical carbon dioxide (sCO₂) as the working fluid. It takes advantage of the behaviour of CO₂ at temperatures and pressures above its critical point, where it acts is a gas but it has the density of a liquid. At this critical point the density of CO₂ is highlight sensitive to small changings in temperature and pressure, allowing for an effective transfer of heat when it is used as working fluid to generate heat and/or power. This technology and its application in steel industry is however, at is early stage, being the subject of several R&D-funded programmes in the United States and Europe (AFID, 2024), (US Department of Energy).

3.4.5 Water and wastewater management

Water requirements and use

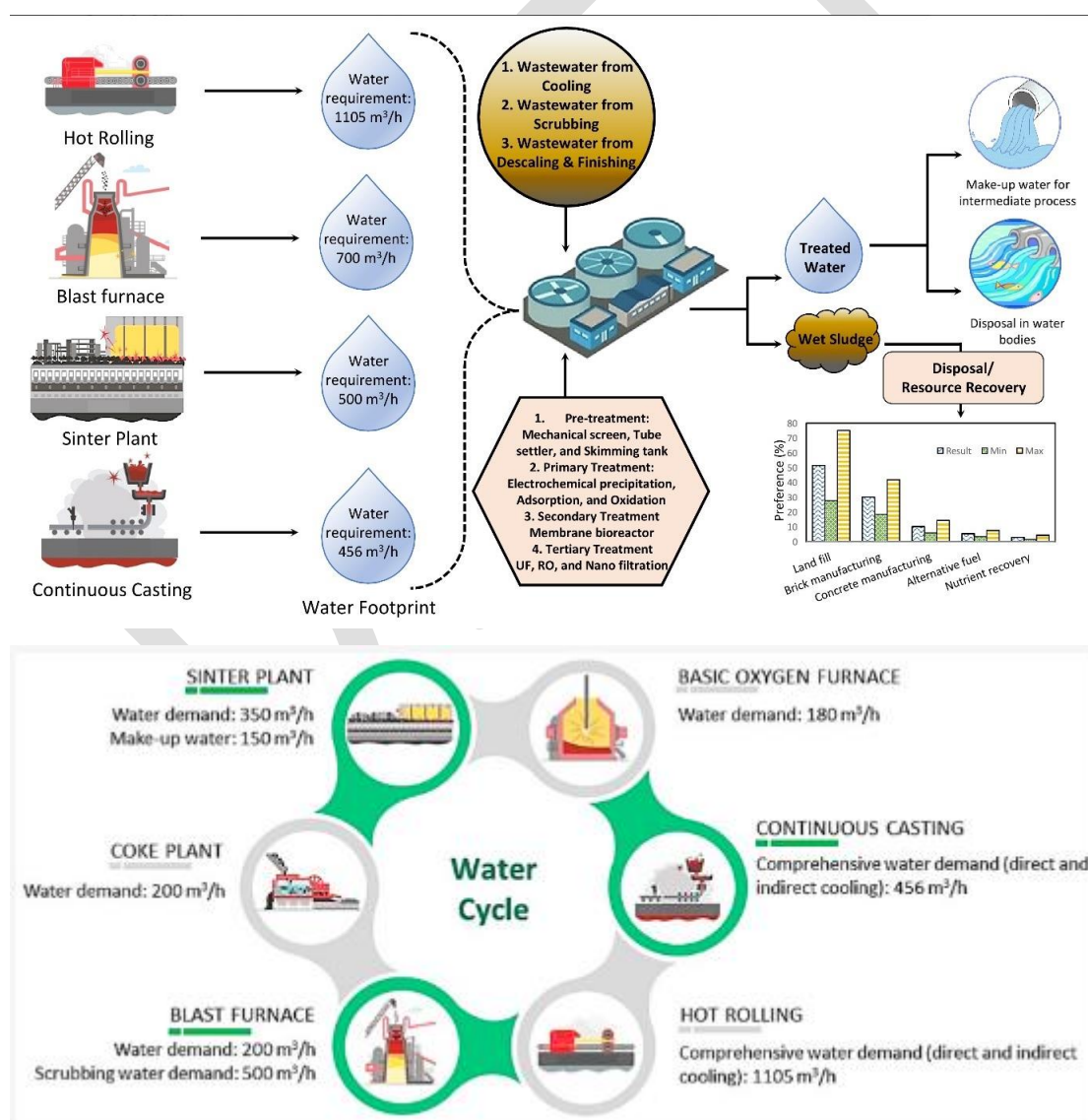
The manufacturing of iron and steel products is a water-intensive process, necessitating substantial amounts of water throughout various stages of the process for diverse purposes. Water plays a vital role in several key activities within the steelmaking process, including **cooling** systems that regulate temperature and prevent overheating in equipment and machinery, such as BF, converters, and continuous casting. Water is also essential for **cleaning** processes, including the removal of pollutants and contaminants from BF, BOF, and coke ovens.

Furthermore, water is used for **descaling purposes** in hot rolling and continuous casting operations, as well as for **chemical and electrochemical treatments**, such as tin-plating and galvanization. Additionally, water is utilised for **scrubbing** processes to suppress particulate matter emissions. The integration of water management strategies is crucial to minimize water consumption, reduce wastewater generation, and mitigate the environmental impacts associated with the steelmaking process (Choudhury et al., 2023).

The iron and steel industry uses diverse sources of water, including **fresh water, sea water and brackish**⁶. Fresh water is predominantly used for applications where it comes into direct contact with materials and equipment, such as process purposes and cooling systems, due to its superior quality and lower risk of corrosion. In contrast, sea water is used for once-through cooling systems, where it does not come into direct contact with materials and equipment, thereby minimising the risk of corrosion and scaling (World Steel Association, 2020). This approach enables the industry to conserve freshwater resources for more critical applications

In regions where freshwater is scarce, steelmakers have implemented innovative strategies to minimize water intake, including the creation of zero-effluent sites that recirculate 100% of freshwater. These sites employ advanced water treatment and recycling technologies to maximize water reuse and minimize wastewater generation (World Steel Association, 2011).

Figure 120. Water requirements for unit operations involved in steelmaking



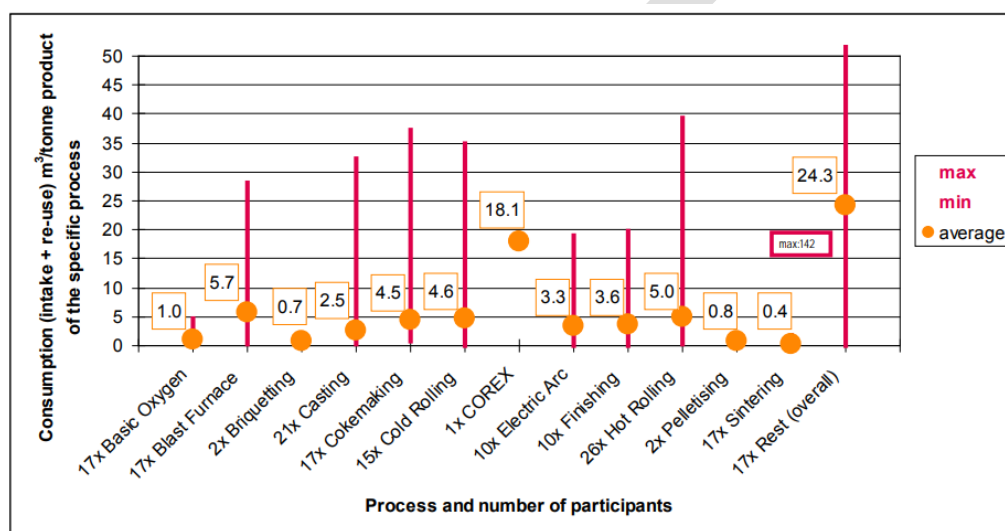
Note. Make-up water to compensate for evaporation and mechanical loss in operations such as quenching, BF shell, continuous casting, and hot rolling

Source: Ascertaining and Optimizing the Water Footprint and Sludge Management Practice in Steel Industries (Choudhury et al., 2023)

Figure 120 provides a comprehensive overview of the water requirements for each unit operation via the BF-BOF route, highlighting the significant water demands of various processes. Notably, the coke oven is the most water-intensive operation unit, requiring approximately 0.4 m³ of water per ton of steel produced (Rawat et al., 2023). However, hot rolling is the most water-intensive process in steelmaking, accounting for a substantial portion of the industry's total water footprint.

Despite the high water requirements in the steel industry, advancements in water treatment technologies have enabled the recovery and reuse of a significant portion of the water used in the steelmaking process. A combination of treatment methods, including electrolytic precipitation, biological digestion, and membrane filtration, allows for the return of approximately **90% of the water used in process** to its source, resulting in a relatively low water consumption of between 3 – 4 m³ (the rest is lost due to evaporation) (Choudhury et al., 2023).

Figure 121. Water use (intake+reuse) by process and number of respondents



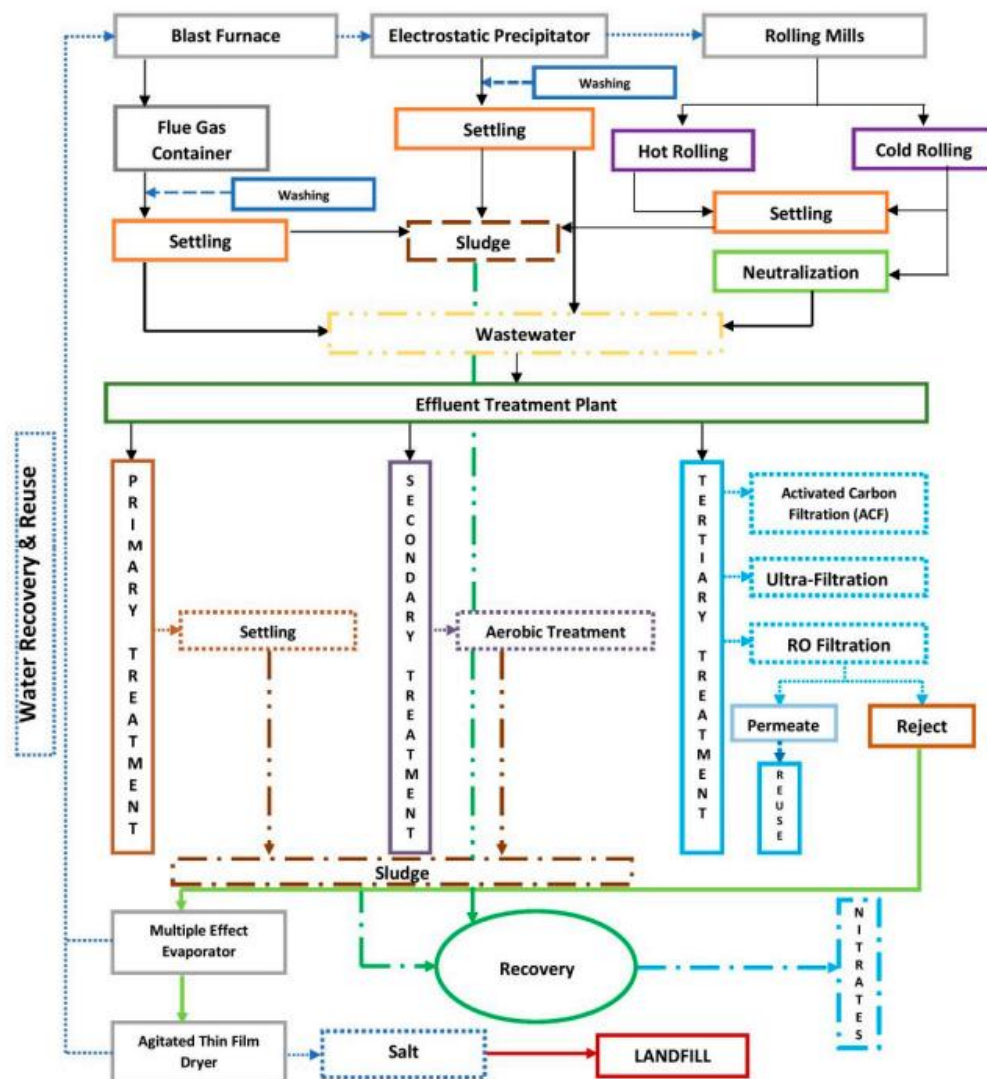
Source: Water Management in steel industry (World Steel Association, 2011)

A comprehensive survey conducted in 2011 by the World Steel Association provided valuable insights into the water intake and discharge patterns of integrated steel plants employing the BF-BOF route and electric arc furnace (EAF) route. The survey revealed that integrated plants (BF-BOF) have an average water intake of 28.6 m³ per ton of steel produced, with an average discharge of 25.3 m³ per ton of steel. In contrast, the EAF route reported an average water intake of 28.1 m³ per ton of steel, with an average discharge of 26.5 m³ per ton of steel (World Steel Association, 2020). However, it is essential to note that these average figures mask significant variations in water consumption patterns among different facilities. A closer examination of the data, as presented in Figure 121 reveals substantial differences in water use (intake + reuse) by process. For instance, the water used in the BF operations ranged from a low 0.2m³/ton of hot metal to a high 28.2m³/ton of hot metal.

Wastewater generation and associated environmental impacts

While significant progress has been made in reducing water consumption in the iron and steel industry, the sector still generates substantial amounts of wastewater that can have detrimental environmental impacts if not properly managed. The major process sources of wastewater in the iron and steel industry are the coke oven, BF, and rolling mills, which account for the largest volumes of wastewater generation (Choudhury et al., 2023; Rawat et al., 2023; Tong et al., 2018). These process wastewaters encompass a range of streams, such as slag quenching wastewater, equipment washing wastewater, air pollution control device rinse water, and contaminated cooling water. Due to their high contaminant levels, these wastewaters necessitate treatment prior to discharge (Rawat et al., 2023). A comprehensive flow diagram illustrating the iron and steelmaking process, along with water streams, wastewater discharge, and water recovery and reuse, is presented in Figure 122.

Figure 122. Process and water flows diagram in steelmaking

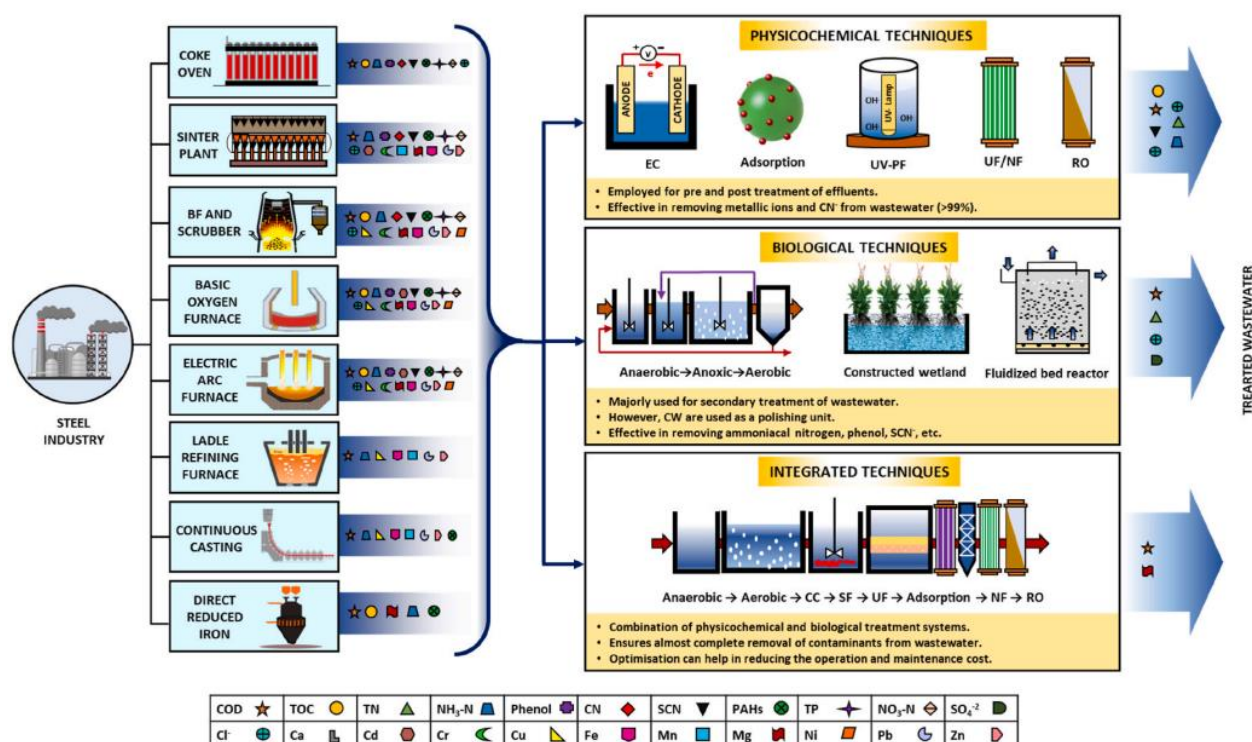


Source: Ascertaining and Optimizing the Water Footprint and Sludge Management Practice in Steel Industries (Choudhury et al., 2023)

The discharge water from various iron and steelmaking processes exhibits varying levels of pollutants. A comprehensive inventory of the primary contaminants associated with each operational unit of iron and steelmaking is presented in Figure 123, providing a detailed understanding of the pollution profiles of different processes. Notably, the coke oven operation stands out as the most water-intensive unit, resulting in the highest wastewater generation and pollution contribution, with coking wastewater being particularly contaminated with hazardous chemicals (Tong et al., 2018). Coking wastewater is particularly contaminated with hazardous chemicals, including ammonia, phenol, thiocyanate (SCN^-), cyanide (CN^-), benzene, and sulfide, at elevated levels.

In addition, the air pollution control systems (APCS) employed in both BF-BOF and EAF routes are significant sources of wastewater, primarily due to the cleaning of scrubbers for gas cleaning and electrostatic precipitators. In the BF-BOF route, the primary source of wastewater is the high-energy scrubbers and gas cooling systems, which generate characteristic pollutants such as ammonia, suspended solids (SS), phenol, and CN^- (Rawat et al., 2023). These pollutants can have detrimental effects on aquatic ecosystems and human health, underscoring the need for effective wastewater treatment and management strategies to mitigate their environmental impacts.

Figure 123. Source of contaminants from iron and steel making and their subsequent treatment



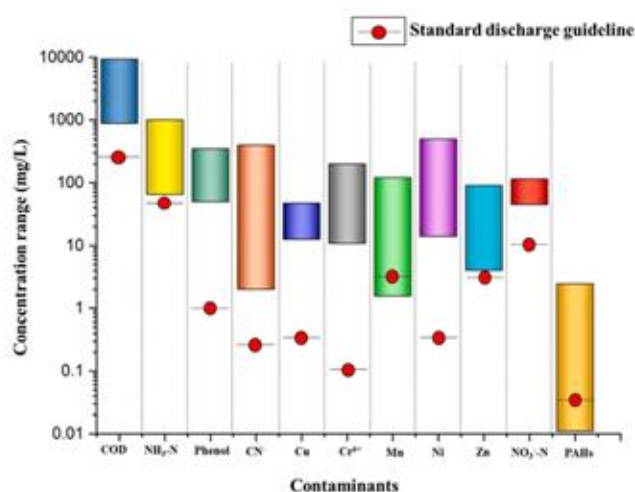
Source: Technological advancements for the treatment of steel industry wastewater: Effluent management and sustainable treatment strategies (Rawat et al., 2023)

The rolling mill process is another significant contributor to industrial wastewater, these unit operations involve processes like pickling and rinsing which utilizes extensive amounts of acid at elevated temperatures, resulting in the generation of corrosive process residues. The effluents produced during rolling, continuous casting, and pickling operations are contaminated with a range of pollutants, including polycyclic aromatic hydrocarbons (PAH), cyanide, ammonia, non-consumed acids, benzene, toluene, xylene, oil, and grease. Furthermore, the combined wastewater from these processes contains high concentrations of metallic oxides, toxic elements, oil, nitrogen, and heavy metals such as zinc, nickel, and chromium. The treatment of this wastewater also generates hazardous sludge, which is typically landfilled (Choudhury et al., 2023).

Figure 124 presents the concentration ranges of typical contaminants released across different unit operations, alongside benchmark values based on general guidelines for discharge of pollutants. As shown, the ranges of these contaminants generally exceed the standard guidelines, indicating that the wastewater requires treatment before it can be reused or returned to the environment.

The environmental and health impacts of the contaminants discussed in this section can be mitigated through the implementation of effective wastewater treatment technologies. The next subsection presents an overview of the best available technologies (BATs) for addressing the contaminants associated with the iron and steel industry. These BATs include advanced treatment processes, such as physical and chemical treatment, biological treatment, and membrane technologies, which can effectively remove pollutants and contaminants from wastewater.

Figure 124. Contaminants released during steelmaking process and general discharge guidelines



Source: Technological advancements for the treatment of steel industry wastewater: Effluent management and sustainable treatment strategies (Rawat et al., 2023)

In conclusion, this section has highlighted the significant water requirements and wastewater generation associated with the iron and steel industry. To provide a comprehensive understanding of the water management practices in the industry, Table 61 present the maximum and minimum intake and water discharge for different operation units of iron and steel based on data from the Best Available Techniques (BAT) Reference Document for Iron and Steel, and Table 60 presents the water consumption across several operation units of ferrous metal processing industries, based on data and the Best Available Techniques (BAT) Reference Document for the Ferrous Metals Processing Industry.

Table 59. Summary table of water intake and discharge by operation unit in iron and steelmaking

Operation Unit	Unit	Intake		Water discharge	
		Maximum	Minimum	Maximum	Minimum
Sinter plant	m ³ /t sinter	0.35	0.01	0.06	0.03
Pelletisation plant	m ³ /t	1.25	0.11	-	-
Coke Oven ¹²⁹	m ³ /t coke	0.24	1.5	0.5	0.1
Blast Furnace (Cooling water ¹³⁰)	m ³ /t hot metal	22.9	0.37	13.736	0.096
Blast Furnace (Process water)	m ³ /t hot metal	13	0.28		
Basic Oxygen Furnace	m ³ /t liquid steel	41.7	0.8	6	0.3
Direct Reduction Iron (DRI)	-	-	-	-	-
Continuous casting	m ³ /t liquid steel	35	5	2	-
EAFF	m ³ /t liquid steel	42.8	1	-	-

Source: JRC elaboration based on Best Available Techniques (BAT) Reference Document for Iron and Steel Production (Joint Research Centre et al., 2013)

Table 60. Summary table of water consumption by operation unit in ferrous metals processing industry

Operation Unit	Unit	Consumption	
		Maximum	Minimum
Hot Rolling Mill	m ³ /t	30	0.01
Cold Rolling Mill	m ³ /t	74	0.0002
Wire Drawing Mill	m ³ /t	6	0.1
Hot Dip Coating Plants	m ³ /t	13.9 ¹³¹	0.08

¹²⁹ Between 0.5 and 1 m³ used for quenching are lost due to evaporation.

¹³⁰ Differences between maximum and minimum intake and water discharge reflect once-through systems and closed systems.

¹³¹ This value is considered to be an outlier. 15 out of 17 plants where data was collected reported consumption below 5 m³/t (Aries et al., 2022)

3.4.6.3 Best Available Technologies

The iron and steel industry is a significant consumer of water, and the management of water resources is crucial to minimize its environmental impact. Various water management approaches can be employed in steelworks, including completely closed, semi-closed, or open circuits. However, the choice of approach is often influenced by local conditions, such as the availability and quality of freshwater resources, as well as regulatory requirements. To reduce water intake and minimize wastewater discharge, several techniques can be employed, as outlined by the Joint Research Centre et al. (2013). These include:

- avoiding the use of potable water for production lines
- increasing the number and/or capacity, and efficiency of water circulating systems when building new plants or modernising/revamping existing plants
- centralising the distribution of incoming fresh water
- semi-closed and closed-loop water circuits
- using the water in cascades until single parameters reach their legal or technical limits
- using the water in other plants if only single parameters of the water are affected and further usage is possible
- keeping treated and untreated waste water separated; by this measure it is possible to dispose of waste water in different ways at a reasonable cost
- using rainwater whenever possible
- automated control systems
- using magnetic drive pumps instead of mechanical pumps to reduce water consumption and prevent leakages of fluids.

The adoption of new technologies and initiatives can significantly improve water efficiency in the iron and steel industry. For example, ArcelorMittal has achieved recirculation rates of up to 98% at several facilities through the use of technologies such as tailings thickening and water recirculation (ArcelorMittal, s.d.). Hot rolling mills, which are the most water-intensive operational unit in steelmaking, offer a high potential for reduction in water consumption and minimization of wastewater generation through the application of closed-loop cooling systems and water systems that facilitate the collection and reuse of water flows.

However, it is essential to consider potential trade-offs when implementing water-saving technologies. These may include increased energy requirements for water cooling and desalination, the generation of by-product salts during desalination that require management, and the possibility of heightened water evaporation due to recirculation.

Table 35. Overview of production routes Table 61 presents a selection of the most effective BATs for reducing water consumption, minimizing wastewater generation, and improving effluent treatment across various operational units in the iron and steel industry. By adopting these technologies, steel producers can significantly decrease their water footprint and reduce their environmental impact

Table 61. Best Available Technologies for water management and wastewater reduction and treatment in the iron and steel industry

Operation Unit	Water Use and wastewater Generation	BAT for reducing water consumption	BAT for wastewater reduction and treatment
Raw Material Handling	- Rainwater runoff from open stocking areas (scrapyard will contain suspended solids. E.g. Runoff water coal storage and blending)		- Appropriate storage techniques and handling systems (impermeable surface and appropriate drainage systems, including an interceptor trap prior to discharge)
Sinter plant	- Rising water		- Rinsing water cleaning techniques - Sedimentation in the recirculation circuit
	- Waste gas treatment (scrubbing)		- Neutralisation, sedimentation and flocculation - Biological Treatment Plant - Waste water sinter treatment plant
	- Cooling of the ignition hoods, fans and sinter machines.	- Cooling towers - Closed-loop cooling systems	- Closed-loop cooling systems
Pelletisation plant	- Off-gas scrubbing	- Low-pressure drop system - Recycling scrubbing water after treatment	- The waste water from scrubbers undergoes neutralisation before it enters a circulation basin. From there it is further processed to settling tanks. The sedimentation properties of the sludge are very often improved by dosage of flocculation agents. - Sand filtration - Heavy metal precipitation - If there is presence of arsenic compounds, the waste water from the water treatment pellet plant is treated also in an arsenic removal plant
Coke oven plants	- Water flows drained from the steam or ammonia liquor used in the goosenecks ¹³² for the suction		<u>Pre-treatment:</u> - Ammonia strippers to reduce the ammonia concentration before discharging the water to a waste water treatment plant or environment - The ammonia-still effluent normally contains no tar but when it does this tar has an adverse effect on the operation of the biological waste water treatment. The tar can be removed by adding chemicals and subsequent separation using techniques as gravitational sedimentation sometimes followed by filtration, centrifuging the coal water, flotation and sand filtration.
	- Water from quenching	- Dry coke quenching	- Dry coke quenching
	- Cooling of charging gases		<u>Treatment:</u> - Aerobic system with activated sludge - Nitrification

¹³² A gooseneck is a curved pipe or duct that connects the top of the coke oven to the gas collection system. Its main functions are to collect and channel the volatile gases released during the coking process, regulate gas pressure, and prevent leaks. The gooseneck directs the gases to the main gas handling system for further processing, ensuring efficient and safe transport.

Operation Unit	Water Use and wastewater Generation	BAT for reducing water consumption	BAT for wastewater reduction and treatment
	- Chemical Water formed during the coking process - Washing water - Water from wet dedusting		- Nitrification-denitrification <u>Other:</u> - Dry coke quenching reduce wastewater generation - Heat recovery coking technology avoids the use of water treatment plant
Blast Furnace	- BF gas scrubbing	- Low-pressure drop system - Recycling scrubbing water after treatment	- Low-pressure drop system for scrubbing (reduce waste water) - Flocculation and sedimentation, which allows recirculating the water treated. - Sludge dewatered with rotary vacuum filters, chamber filter presses or centrifuges
	- Waste water from slag granulation - Blowdown from cooling water circuits	- Frequently controlled pumps - Closed-loop cooling systems	- Closed-loop cooling systems
Basic Oxygen Furnace	- Cooling oxygen lance	- Closed-loop cooling systems - Cooling towers	
	- BOF gas treatment - Scrubbing water from the wet dedusting of desulphurisation	- Recycling scrubbing water after treatment	- Separation of coarse particles, sedimentation and electrical flocculation 2 stages sedimentation - Hydrocyclonage and/or precipitation - Treating the bleed by means of sedimentation and filtration
Direct Reduction Iron Plant	- Cooling purpose	- Closed-loop cooling systems - Cooling towers	- Closed-loop cooling systems
	- Off-gas scrubbing	- Recycling scrubbing water after treatment	- Low-pressure drop system for scrubbing (reduce waste water) - Flocculation and sedimentation, which allows recirculating the water treated. - Sludge dewatered with rotary vacuum filters, chamber filter presses or centrifuges
Electric Arc Furnace	- EAF cooling - Slag cooling with water sprays	- Closed-loop cooling systems - Cooling towers	
	- Rapid quenching of the hot off-gases		- The used water is mostly evaporated
	- Scrubbing water from the wet dedusting	- Recycling scrubbing water after treatment	- Low-pressure drop system for scrubbing (reduce waste water) - Flocculation and sedimentation, which allows recirculating the water treated. - Sludge dewatered with rotary vacuum filters, chamber filter presses or centrifuges
Continuous Casting	- Direct cooling of the slabs, blooms and billets	- Closed-loop cooling systems	- Usually this waste water is treated together with other streams from the rolling mill - Treating the bleed by means of flocculation, sedimentation and filtration - Removal of oil in skimming tanks - The spray water is precipitated by sand filtration

Operation Unit	Water Use and wastewater Generation	BAT for reducing water consumption	BAT for wastewater reduction and treatment
	- Water from vacuum generation	- Recirculation of cooling water and water from vacuum generation as much as possible	- Usually this waste water is treated together with other streams from the rolling mill
Hot Rolling Mill ¹³³	- Cooling purpose	- Cooling towers - Semi-closed and closed-loop cooling systems - Reduction of energy loss through feedstock transportation device (46% reduction in water use) - Optimised water pumps for laminar flows	- Semi-closed cooling systems - Closed-loop cooling systems and cooling towers
	- Scale breaking, flushing scale and scale transport - High-pressure water for descaling	- Water spray descaling triggered by sensors	<u>Pre-treatment:</u> - Skimming devices to eliminate oil pollutants - Scale pits, settling tanks, cyclones, filtration, - Gravel filters <u>Treatment:</u> - Sedimentation
Cold Rolling Mill	- Lubrication - Cooling purpose - Clean surface of rolling stocks - Removal of Fe particles	- Closed-loop water system - Cooling towers - Magnetic pumps (low-alloy and alloy steel)	<u>Reduce wastewater:</u> - Hydrochloric acid regeneration by spray roasting - Hydrochloric acid regeneration by fluidised bed - Effluent-free HCl strip pickling plant (no water discharge) - Sulphuric acid recovery by crystallisation - Cleaning and reuse of the rolling emulsion <u>Treatment:</u> - Neutralisation with agents, such as calcium hydroxide or sodium hydroxide - Sedimentation techniques including clarification or filtration - Flocculation - Chromium (VI) reduction processes using sodium bisulphite or iron (II) compounds
Wire Mill	- Rinsing water - Off-gas scrubbing - Cooling purpose - Quenching	- Closed-loop cooling system - Dry drawing - Reuse quenching water after treatment	<u>Treatment:</u> - Neutralisation with lime milk followed by precipitation of Fe and Pb and other heavy metals, followed by decantation and filter-pressing of the cake

¹³³ In some cases, the hot rolling mill water circuit is coupled with the water feeding systems of other iron and steel production units, like for example continuous casting. The reasons for this connection are the similarity of the waste water contents and the proximity of the installations (Aries et al., 2022)

Operation Unit	Water Use and wastewater Generation	BAT for reducing water consumption	BAT for wastewater reduction and treatment
Hot dip coating plants	- Cooling purpose - Chemical strip treatment sections - Rising and brushing operations - Spray water for the skin-pass mill	- Closed-loop water system - Air extraction as close as possible to the source from chemical baths or tanks in post-treatment (i.e. phosphating and passivation) (reduction of water demand)	<u>Reduce wastewater:</u> - Reverse cascade degreasing - Cleaning and reuse of the degreasing solution - Squeeze rolls - Cleaning and reuse of the phosphating or passivation solution <u>Pre-treatment:</u> - Clarified and filtered <u>Treatment:</u> - Skin-pass mill water treatment plant - Water pants that consist of three circuits: chromic water line, oily water line and the general waste water line. Techniques use are oil separation, flotation, chromium reduction, neutralisation, flocculation, sedimentation, activated carbon filtration and ion exchange
Effluent Treatment Plant	- Water streams coming from different operational units		- Electrocoagulation combined with photo-Fenton Anodic oxidation - Advanced oxidation processes (AOPs) - Electrochemical oxidation - Ozonation - Adsorption - Biological techniques (e.g. aerobic treatment, constructed wetlands and bioremediation) - Membrane technologies, including ultrafiltration (UF), reverse osmosis (RO), electrodialysis (ED), and electrodialysis reversal (EDR) - Coagulation, flocculation and precipitation - Tertiary treatment units, such as pressure sand filter (PSF), activated carbon filter (ACF), micron filter (MF), ultra-filtration (UF), reverse osmosis (RO)

Source: JRC elaboration based on Water Management in steel industry (World Steel Association, 2011), and Ascertaining and Optimizing the Water Footprint and Sludge Management Practice in Steel Industries (Choudhury et al., 2023), Best Available Techniques (BAT) Reference Document for Iron and Steel Production (Joint Research Centre et al., 2013), and Best Available Techniques (BAT) Reference Document for the Ferrous Metals Processing Industry (Aries et al., 2022)

Table 62. BAT-associated environmental performance levels (BAT-AEPLs) for specific water consumption

Operation Unit	Unit	BAT-AEPL
Hot Rolling Mill	m ³ /t	0.5 – 5
Cold Rolling Mill	m ³ /t	0.5 - 10
Wire Drawing Mill	m ³ /t	0.5 - 5
Hot Dip Coating Plants	m ³ /t	0.5 - 5

Source: Best Available Techniques (BAT) Reference Document for the Ferrous Metals Processing Industry (Aries et al., 2022)

3.4.6.4 Best Not Yet Available Technologies

After conducting a thorough mapping of not yet available technologies, it was found that no information on innovative technologies in phase of development (with low TRL level) specifically related to water and wastewater management in the iron and steel industry could be identified.

3.4.6 Reducing agents

Iron ore reduction is a critical process in the iron/steel-making industry, where iron ore is reduced to metallic iron, usually with carbon (from coal and coke). The exploration of alternative reducing agents is a key driver in the decarbonisation of steel industry. Substituting carbon with natural gas, green hydrogen, or electricity can yield significant CO₂ savings (Draxler et al., 2021). This section of the preparatory study introduces BAT and BNAT (see Table 63) that enhance environmental performance of iron & steel products by replacing carbon as reducing agent.

Table 63. BAT and BNAT for substitution of reducing agents

BAT	BNAT
- Natural gas-based direct reduction iron (NG-DRI)	- Hydrogen-based direct reduction iron (H ₂ -DRI) - Hydrogen-based direct reduction iron smelter (H ₂ -DRI-SMELT) - Hydrogen plasma smelting reduction (HPSR) - Alkaline electrolysis or electrowinning (AEL) - Molten oxide electrolysis (MOE)

Source: JRC elaboration

3.4.6.1 Best Available Technologies

Natural Gas-based Direct Reduction Iron (NG-DRI)

This technology has been previously presented in section 3.1.3 of the preparatory study, as it a commercial available technology that accounts for approximately 5% of global steel production (World Steel Association, 2022). The current process, with the potential to reduce emissions by 35-67% compared to the BF-BOF (Ricardo, 2024), uses natural gas for reducing iron pellets in the shaft furnace of the DRI plans. The resulting DRI is processed into crude steel in an EAF, where steel scrap can be used as an additional feed (Agora Industry, Wuppertal Institute, and Lund University, 2024). A summary of the main characteristics of this technology is provided in Table 64.

Table 64. Main characteristics of the NG-DRI technology

Feature	NG-DRI
Steelmaking Production route	EAF
TRL level	TRL 9
Country (Geographical scope) of implementation	USA, Russia, Middle East, parts of Central and Wouth America
Potential CO₂ reduction compared to BF-BOF	35-67% [a]
Energy Needs (GJ/ ton of crude steel)	12 GJ/ton
Potential of other environmental aspects	Energy efficiency, resource efficiency
Expected market readiness	Commercially available
Levelised cost 2030	560 - 658 EUR/ton

Note. All values i are converted to Euro (€) using the average 2023 exchange rate of \$1 = €0.93

Source: JRC elaboration based on Low-carbon technologies for the global steel transformation (Agora Industry, Wuppertal Institute, and Lund University, 2024) ; Technologies to decarbonise the EU steel industry (Somers, 2022) ; and The use of green steel in the automotive industry (Ricardo, 2024)

3.4.6.2 Best Not yet Available Technologies

The NBAT included in the analysis and presented in Table 65 use green hydrogen, electricity or hydrogen plasma as reducing agents. A critical consideration for the adoption of these technologies is their levelised cost, which is related to the costs of the electricity used. While the potential for CO₂ reduction compared to the BF-BOF route is approximately to 100%, this outcome will heavily depend on the emissions intensity of the electricity

mix utilised. Presently, there is substantial uncertainty regarding the timeline and associated costs for the availability of these technologies.

Table 65. Not Best Available Technologies based on reducing agent substitution

Features	H2-DRI	H2-DRI-SMELT	HPSR	AEL	MOE
Production route	EAF	BOF	EAF	EAF	None
Reducing Agent	Green hydrogen	Green hydrogen	Hydrogen Plasma	Electricity	Electricity
TRL level	TRL 6-8	TRL 7-8 [1]	TRL 5-7	TRL 6	TRL 2
Expected Market Readiness	2025-2030	2026	2050	2040-2045	2035-2040
Geographical scope	Europe	Europe (Germany)	Europe (Austria) and USA	Europe (France)	USA
Potential CO ₂ reduction compared to BF-BOF	97-99%	98%	95%	99.6%	95%
Energy needs	10.8 GJ/ton	11 GJ/ton	15.12 GJ/ton	13.7 GJ/ton	12.4-14.8 GJ/ton
Potential reduction of other environmental aspects	- Energy savings	- Energy savings	- Lower slag generation - Lower dust emissions	- Energy savings	- Energy savings
Levelised cost 2030 (EUR/ton)	676 - 797 [b]	674 - 810 [b]	n/a	568 - 795	541 - 712
Projects	Hybrit (SSAB), H2 Green Steel	Thyssenkrupp	H2PasmaRed Horizon Europe, SuSteel (Voestalpine), University of Utha (USA)	SIDERWIN (ArcelorMittal)	Boston Metals

Note. All values are converted to Euro (€) using the average 2023 exchange rate of \$1 = €0.93

Source: JRC elaboration based on Low-carbon technologies for the global steel transformation (Agora Industry, Wuppertal Institute, and Lund University, 2024); Making Net-Zero Steel Possible (Mission Possible Partnership, 2022); Climate-neutral steelmaking in Europe : decarbonisation pathways, investment needs, policy conditions, recommendations (European Commission, European Research Executive Agency, et al., 2022); Iron and Steel Technology Roadmap (IEA, 2020); The use of green steel in the automotive industry (Ricardo, 2024); and Carbon-free steel production. Cost reduction options and usage of existing gas infrastructure (Van Woensel and García Higuera, 2021)

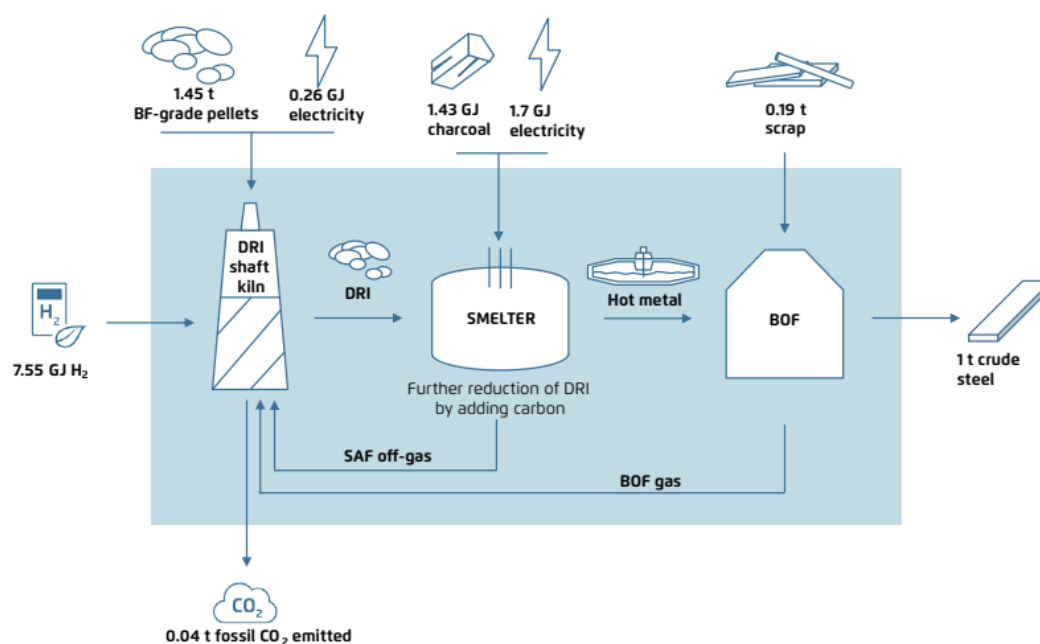
Hydrogen-based direct reduction (H2-DRI)

The H2-DRI-EAF technology route has been already presented in the section 3.1.4, as it constitutes the most scalable low-carbon primary steelmaking technology before 2030, particularly in the EU. (Agora Industry, 2023). In this process, hydrogen is used as the reductant in the DRI plant. When renewable electricity is used for hydrogen production, this technology route has the potential to achieve nearly 100% CO₂ emissions reduction. However, this technology faces some constraints, it requires high-grade DRI for the EAF, and to produce high grade DRI a high quality iron ore is needed, which is in short supply. As a result, this technology alone is unlikely to serve as the sole solution for decarbonising the entire steel industry (CRU Consulting, 2024).

Hydrogen-based direct reduction iron smelter (H2-DRI-SMELT)

The H2-DRI-SMELT route is based on the continued use of the basic oxygen furnace (BOF). This requires an intermediate smelting stage so that the solid DRI leaving the shaft furnace is fed into the BOF in a liquid state. There are two similar smelter technologies on the market: the submerged arc furnace (SAF) and the open slag bath furnace (OSBF). In addition to the potential reduction of approximately 98% of CO₂ emissions, a key benefit of this technology is that lower-grade iron pellets can be used to produce high-quality steel. Given the limitations of the DRI-EAF, which requires pellets made from high-grade iron ores and its insufficient availability could prove a serious bottleneck to the global steel transformation, the H2-DRI-SMELT could be a possible solution (Agora Industry, Wuppertal Institute, and Lund University, 2024).

Figure 125. Schematic and simplified view of H₂-DRI-SMELT-BOF

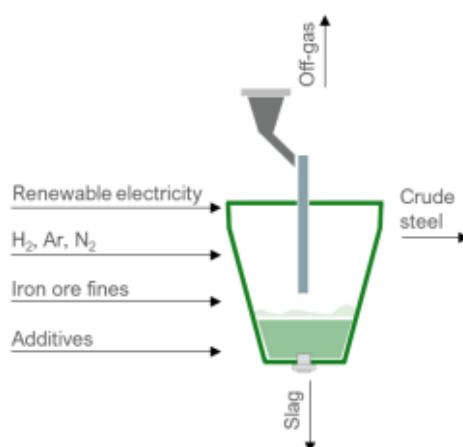


Source: Low-carbon technologies for the global steel transformation (Agora Industry, Wuppertal Institute, and Lund University, 2024)

Hydrogen plasma smelting reduction (HPSR)

Hydrogen Plasma Smelting Reduction technology (HPSR) is a direct process that transforms iron oxides into liquid steel through the use of ionised hydrogen (H⁺, hydrogen plasma). Unlike the conventional BF-BOF route, HPSR eliminates the need for pre-treatment of the ore. The process involves generating plasma by passing an electric current through a gas, which acts as a reducing agent and provides the necessary energy to melt metallic iron. During the procedure, a plasma arc is ignited between a hollow graphite electrode (cathode) and the melted pool (anode), with iron ore fines, hydrogen, and argon introduced via the electrode into the reduction zone. The charged ore is melted in-flight and reduced to metallic iron with excited and ionized hydrogen species producing steam as a by-product. (Draxler et al., 2021; Pauna et al., 2022)

Figure 126. Schematic and simplified view of hydrogen plasma smelting reduction



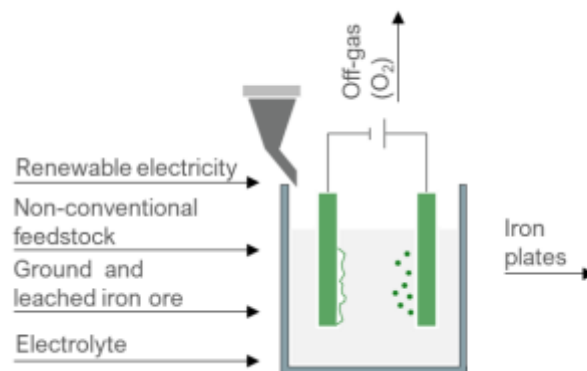
Source: Green Steel for Europe. Technology Assessment and Roadmapping (Draxler et al., 2021)

Alkaline Electrolysis or Electrowinning (AEL)

The low temperature alkaline iron ore electrolysis or electrowinning of iron is based on the decomposition of iron oxides into iron and oxygen by passing an electric current through a highly alkaline aqueous sodium hydroxide solution at about 110 °C. The resulting solid iron (iron plate) is then processed to steel in an EAF. The solution will be available at commercial scale at least until 2030, further research is needed, in particular to

design an electrode that constitutes a good compromise between mechanical strength and open porosity (Draxler et al., 2021; Agora Industry, Wuppertal Institute, and Lund University, 2024).

Figure 127. Schematic and simplified view of alkaline electrolysis

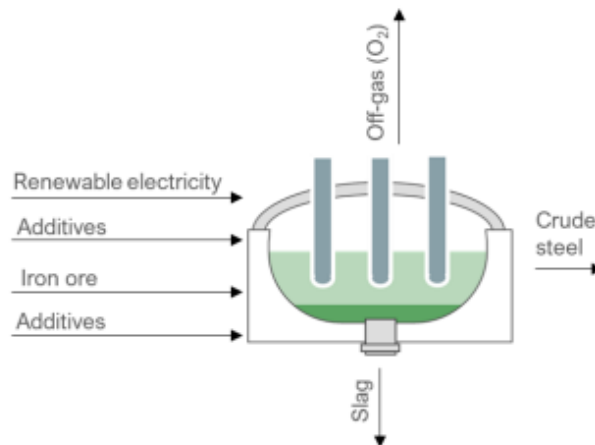


Source: Green Steel for Europe. Technology Assessment and Roadmapping (Draxler et al., 2021)

Molten oxide electrolysis (MOE)

In the molten oxide electrolysis (MOE) iron ore is converted electrochemically to its elementary components oxygen (O_2) and iron (Fe) in an electrolysis cell, using electricity as reducing agent and generating only oxygen as a by-product, which also has a commercial value. The MOE allows producing directly molten metal from iron ore which can be directly fed into the downstream steelmaking process. This technology was developed by MIT Professor Donald Sadoway, who created Boston Metals (MIT News, 2024). However, this technology has not yet been proved at commercial scale and it will not constitute a wide-scale solution for the current industry decarbonisation effort (CRU Consulting, 2024; Agora Industry, Wuppertal Institute, and Lund University, 2024).

Figure 128. Schematic and simplified view of molten oxide electrolysis



Source: Green Steel for Europe. Technology Assessment and Roadmapping (Draxler et al., 2021)

3.4.7 Carbon Capture, Use and Storage (CCUS)

Carbon Capture, Utilization, and Storage (CCUS) encompasses a suite of technologies aimed at capturing carbon dioxide (CO₂) emissions from sources like power generation and industrial processes, preventing CO₂ from entering the atmosphere. According to the IEA¹³⁴, this set of technologies will be essential to decarbonise the industry and meet the climate targets. Similarly, the Intergovernmental Panel on Climate Change (IPCC) highlights CCUS as a key set of technologies for reducing emissions from large-scale fossil-based energy and industrial activities (IPCC, 2024).

The CCUS process involves several steps: capturing CO₂ emissions, separating CO₂ from other gases, compressing it, transporting it to a storage site, and either utilizing it in other industrial applications or storing it underground. The effectiveness and relevance of CCUS vary across different applications and industries.

According to (Sandbag, 2024) most CCS/U projects in the iron and steel sector are still in their early stages, with only a few pilot or demonstration projects in operation. Estimates for future CCS/U applications in iron and steelmaking from the (IEA, 2020) show that these applications are not expected to contribute to any significant emission reductions before at least 2040.

3.4.3.1 Best Available Technologies

Chemical Absorption Applied to Natural Gas-Based Direct Reduced Iron (NG-DRI) with CCUS

The NG-DRI with CCS is already commercially deployed in the Al Reyadah CCS facility is the world's first commercial carbon capture steel project, and the CO₂ is used for enhanced oil recovery. According to (Agora Industry, Wuppertal Institute, and Lund University, 2024), The CCUS facility has been operating since 2016, with an announced capture potential of up to 0.8 MtCO₂ /year, which is used for enhanced oil recovery (EOR) in nearby oil fields.

The GHG intensity of NG-DRI without CCUS stands at approximately 0.415 tCO₂ per tonne of DRI. Incorporating CCUS can reduce this intensity by 62%, bringing it down to 0.159 tCO₂ per tonne of DRI (Sandbag, 2024).

A practical application of this technology involves capturing CO₂ for Enhanced Oil Recovery (EOR), where the captured CO₂ is utilized to increase oil extraction efficiency. While this provides a use for the captured CO₂, the overall climate benefits depend on the lifecycle emissions of the extracted oil.

Transportation of the CO₂

The infrastructure for CO₂ transportation is well-established, employing various modalities such as highways, railroads, pipelines (both onshore and offshore), and ships. The choice of transportation method depends on factors like distance, volume, and geographic considerations. However, in the case of CO₂ storage, the limited availability of suitable sites in Europe (mainly concentrated in the North Sea) also pose challenge about the distance and associated cost that separates potential CO₂ storage sites from industrial facilities from which captured CO₂ emissions will be transported.

3.4.3.2 Best Not yet Available Technologies

Chemical absorption applied to BF-BOF (separation step)

Applying CCUS to the traditional BF-BOF steelmaking route involves retrofitting facilities with chemical absorption systems, typically amine-based, to capture CO₂ emissions from major sources such as on-site power plants, coking plants, and blast furnace hot stoves.

However, this approach faces several challenges according to (Agora Industry, Wuppertal Institute, and Lund University, 2024):

¹³⁴ <https://www.iea.org/energy-system/carbon-capture-utilisation-and-storage>

- **Residual Emissions:** Even with CCUS, the BF-BOF route may still emit significant CO₂, undermining deep decarbonization efforts.
- **Upstream Methane Emissions:** Coal mining associated with BF-BOF operations contributes to methane emissions, further impacting the overall GHG footprint.
- **Commercial Viability:** The anticipated availability of this technology is post-2030, with projected costs comparable to NG-DRI. Given the declining costs of alternative technologies like hydrogen-based direct reduction (H₂-DRI), CCUS in the BF-BOF route may be less competitive

The global steel industry, responsible for 8% of global CO₂ emissions, is exploring low-carbon technologies to reduce its carbon footprint. Key technologies, particularly those based on direct reduction of iron (DRI), are expected to be commercially available this decade.

DRI technology offers flexibility, allowing operation with a gradually increasing share of hydrogen and decoupling iron and steelmaking stages. Additionally, Enhancing scrap quality and utilizing renewable electricity can facilitate high-quality steel production via electric arc furnace (EAF) routes, further contributing to emission reductions. In contrast, retrofitting blast furnace–basic oxygen furnace (BF-BOF) plants with carbon capture and storage (CCS) is considered a less viable strategy (Agora Industry, Wuppertal Institute, and Lund University, 2024)(Amand et al. 2024). Challenges include residual emissions, substantial infrastructure requirements, upstream methane emissions from coal mining, and diminishing commercial attractiveness due to declining hydrogen costs and rising CO₂ prices.

Consequently, while integrating CCS in the BF-BOF route may offer some emission reduction potential, its large scale adoption do not seem promising. Instead, deploying CCS in the iron and steel industry should be confined to scenarios where CO₂ emissions cannot be otherwise abated, such as in DRI plants not operating with green hydrogen.

3.5 Conclusions and perspectives

Task 3 aimed to perform a technical analysis of the current iron and steel products available on the EU market and provide insights for **defining the Base Cases**, which represent a virtual average product sold on the EU market in terms of functional performance and environmental profile.

Given the broad scope of steel as an intermediate product in the ESPR context, as presented in Task 1 and Task 2, a targeted approach is necessary. To streamline the assessment, a five representative products (RPs) were selected based on 5 criteria: 1) EU apparent consumption; 2) EU production; 3) EU supply dependencies (calculated *via* the product's import reliance); 4) Estimated CO₂ footprint per tonne of steel products and 5) Economic importance related to steel end-use sectors.

To streamline the assessment within the broad scope of steel as an intermediate product, five representative products were selected based on criteria such as EU consumption, production, supply dependencies, CO₂ footprint, and economic importance. The selected products include Hot Rolled Coil, Cold Rolled Coil Galvanized, Rod and Wire Rod, Stainless Steel, and Electrical Sheet. Base Cases for these products were developed for 2022 and projected for 2035. These Base Cases serve as benchmarks for evaluating potential design improvements and informing future policy interventions, with a detailed quantitative analysis revealing the following key insights:

- **Import dependencies:** the EU heavily relies on imports for iron ore and ferro-alloys. However, domestic production satisfies 60-87% of steel and intermediate product demand.
- **Scrap availability and associated quality:** the EU has a significant amount of post-consumer scrap available round 90 million tonnes, with 19 million tonnes exported. High-quality steel production, especially for automotive applications, demands the best quality scrap. Contaminants in steel, known as "tramp elements" like copper, limit scrap usage and the expansion of secondary routes in Europe.
- **Forecasted technologies uptake:** by 2035, both in the EU and in third country exporting to the EU market (EU extra supply mix), a significant increase in the adoption of EAF and DRI-EAF technologies is expected. Notable differences are anticipated in the production of hot rolled coil and galvanized cold rolled coil. However, the expansion of the scrap-EAF production route may be constrained by the quality of available scrap. The transition towards DRI-EAF for primary material processing appears to be a clear trend, driven by the urgent need to decarbonize steel production and the anticipated rise in carbon prices in the coming years. According to one of the forecast scenarios designed by the JRC, consumption is

expected to remain stable at 151 million tonnes, with EU production at 138 million tonnes. This production is projected to be distributed as follows: DRI-EAF (18%), BF-BOF (32%), and scrap-EAF (50%).

- **Current and projected scrap ratios:** Presently, the scrap ratio per unit of crude steel produced in the EU is 55%. With an anticipated increase in the share of EAF production, this ratio is expected to rise to 61% by 2035. This shift would result in significant scrap content in the production of hot rolled coil (6%), galvanized cold rolled coil (15%), and electrical sheet (24%). In Europe, these flat products are typically manufactured using primary routes with low scrap content, except in rare instances.
- **DRI-EAF production in the EU:** currently, production via DRI-EAF in the EU is minimal or virtually nonexistent. As far as the authors are aware, ArcelorMittal operates the only demonstration facility in Hamburg, with a DRI capacity of 600 kt. Despite this limited current capacity, a shift towards DRI-EAF for primary material processing appears imminent. This transition is driven by the urgent need to decarbonize steel production and the anticipated rise in carbon prices in the coming years.
- **Extra-EU Supply Mix:** The analysis reveals minor differences between imported steel products and EU domestic production. For hot rolled coil, the import scenario closely mirrors the EU's, with the BF-BOF route constituting over 85% of production. In contrast, wire rod imports are primarily sourced from BF-BOF routes, differing from the EU's production methods. Furthermore, a larger proportion of galvanized cold rolled coil imports are produced via EAF, primarily due to significant imports from Türkiye, which boasts a substantial EAF production share.

Lastly, this chapter examined the technical aspects influencing the environmental performance of iron and steel products, along with the best techniques and technologies associated with them. Best available techniques and technologies are today already mature to ensure substantial decarbonisation of the European steel sectors as well as reducing other environmental impacts of these products. While some of these solutions require substantial upfront investments and involve restructuring the iron and steelmaking technology route used for production (such as substituting reducing agents or implementing CCUS technologies) others focus on process optimization and resource efficiency, which are easier to implement but offer limited environmental benefits compared to comprehensive decarbonisation of the industry.

The base cases developed in this task provides the foundation for Task 4: LCA & LCC. Additionally, the assessment of best available techniques and technologies constitutes the groundwork for Task 5: Design Options. In Task 5, based on the results of the hotspot analysis conducted in Task 4, design options aimed at enhancing the environmental performance of the base cases will be proposed. To ensure the feasibility of these options, current best practices, techniques, and technologies in the industry will be used as benchmarks

Annex 1. Existing definitions and categories for steel

Steel can be subject to classification using numerous diverse systems. This section presents an overview of existing definitions of steel products using the most common categorisation based on physical and chemical properties, products characteristics, and economic statistics:

- Steel grades, designated according to the chemical composition, mechanical and physical properties (Section 1.1.2.1)
- Application and mechanical or physical properties (Section 1.1.2.2)
- Shape and surface conditions (Section 1.1.2.3)
- European Statistics (Section 1.1.2.4)
- Combined Nomenclature (Section 1.1.2.5)
- Other categorisations (Section 1.1.2.6)

1.1. Steel grades

The steel grade is linked to the chemical composition, the structure and the crystallography of the product. According to the standard EN 10020:2000 “Definition and classification of grades of steel” (CEN, 2000), steel can be classified into:

- Non alloy, stainless steel and other alloy steels by chemical composition
- Main quality classes defined by main property or application characteristics for non-alloy, stainless and other alloy steels.

In addition, a classification is also provided by the European Commission JRC study “Best Available Techniques (BAT). Reference Document for the Ferrous Metals Processing Industry” (Aries et al., 2022). Steel grades is defined in four principal categories:

- Carbon steel non alloy, for which the content of alloy elements is lower than 0.4%, except for silicon (< 0.6%) and manganese (<1.65%);
- Carbon steel low alloy, for which the content of each alloy element is lower than 5%;
- Stainless steel, which is also considered as a high alloy steel;
- High alloy steels, other than stainless steel for which the content of some alloy elements is over 5%.

Electrical steels and high strength steel should also be considered as special steel grades.

Finally, the European Emissions Trading System (EU ETS), also provides a definition for EAF carbon steel and EAF high alloy steel (European Commission, 2019).

1.2.1. Classification based on standard EN 10020:2000

Classification by chemical composition

a) Non-alloy steel

Non-alloy steels are steel grades in which the mass fraction of an alloying element remains below a certain limited value. These critical values, which set the boundaries between non-alloy and alloy steels are established by the standard EN 10020:2006. See Figure 129..

Figure 129. Boundary between non-alloy and alloy steels (ladle analysis)

Specified element		Limit value % by mass
Al	Aluminium	0,30
B	Boron	0,0008
Bi	Bismuth	0,10
Co	Cobalt	0,30
Cr	Chromium	0,30
Cu	Copper	0,40
La	Lanthanides (each)	0,10
Mn	Manganese	1,65 ^{a)}
Mo	Molybdenum	0,08
Nb	Niobium	0,06
Ni	Nickel	0,30
Pb	Lead	0,40
Se	Selenium	0,10
Si	Silicon	0,60
Te	Tellurium	0,10
Ti	Titanium	0,05
V	Vanadium	0,10
W	Tungsten	0,30
Zr	Zirconium	0,05
Others (except carbon, phosphorus, sulphur, nitrogen) (each)		0,10

a) Where manganese is specified only as a maximum the limit value is 1,80 % and the 70 % rule (see 3.1.2) does not apply.

Source: EN 10020:2000 – Definition and classification of grades of steel (CEN, 2000)

b) Stainless steel

Stainless steel is a generic term for a group of corrosion-resistant steel with at least 10.5 % of chromium and max.1.2% of carbon.

c) Other alloy steel

Other alloy steels are steel grades not complying with the definition of stainless steels in which at least one of the limit values in Figure 129. is reached by the contents. In the case of the other alloy steels, alloy quality steels comprise steel grades required to meet certain requirements in relation to, e.g. fracture toughness, grain size and/or formability.

Classification of main quality classes

The standard EN 10020:2000 (CEN, 2000, p.100) classifies steel grades into main quality classes defined by main property or application characteristics for non-alloy, stainless and other alloy steels. Classification is as follows:

1. Non-alloy quality steels

Non alloy quality steels are steel grades which, in general, have specified property requirements such as, for example, toughness, grain size control and/or formability.

Non alloy electrical steels are defined as: non alloy quality steels with specified requirements for maximum values of specific total loss or minimum values of magnetic induction, polarisation or permeability.

2. Non-alloy special steels

Non alloy special steels are steel grades, which comply with one or more of the following requirements:

- specified minimum impact strength in the quenched and tempered condition ;
- specified hardness penetration depth or surface hardness in the quenched, quenched and tempered or surface hardened condition ;
- particularly low contents of non-metallic inclusions are specified ;
- specified maximum phosphorus and sulphur content :
 - for ladle analysis $\leq 0,020$ % ;
 - for product analysis $\leq 0,025$ %.

- specified minimum impact strength on Charpy-V-notch test pieces at - 50 °C greater than 27 J for test pieces taken in the longitudinal direction or greater than 16 J for test pieces taken in the transverse direction¹³⁵
- steels for nuclear reactors having the following specified elements simultaneously restricted on product analysis to :
 - copper ≤ 0,10 %, cobalt ≤ 0,05 %, vanadium ≤ 0,05 % ;
- specified electrical conductivity ≤ 9 Sm/mm² ;
- precipitation hardening steels with minimum specified carbon contents of 0,25 % or more in the ladle analysis and a ferritic/pearlitic microstructure, containing one or more micro alloy elements such as niobium or vanadium with contents less than the limit values for alloy steels. Precipitation hardening is generally achieved by controlled cooling from the hot forming temperature ;
- prestressing steels.

3. Stainless steel

Stainless steel is a generic term for a group of corrosion-resistant steel with at least 10.5 % of chromium and max.1.2% of carbon.

They are subdivided further according to the following criteria:

- by nickel content into :
 - nickel less than 2,5 % ;
 - nickel 2,5 % or more.
- by main property into :
 - corrosion resisting ;
 - heat resisting ;
 - creep resisting.

4. Alloy quality steels

Alloy quality steels are steel grades for which requirements exist with regard to, for example, toughness, grain size control and/or formability. Alloy quality steels are not generally intended for quenching and tempering or for surface hardening.

- Weldable fine grained structural steels, including steels for pressure vessels and tubes, which meet all the following conditions :
 - specified minimum yield strength < 380 N/mm² for thickness ≤ 16 mm ;
 - alloy contents are less than the limit values given in Figure 130.;
 - specified minimum impact strength on Charpy-V-notch test pieces at - 50 °C ≤ 27 J for test pieces taken in the longitudinal direction or ≤ 16 J for test pieces taken in the transverse direction¹³⁶

¹³⁵ If no impact value is specified at - 50 °C the value specified between - 50 °C and - 60 °C shall be used.

¹³⁶ If no impact value is specified at - 50°C the value specified between - 50°C and -60°C shall be used.

Figure 130. Weldable fine grained alloy steels. Chemical composition boundary between quality steels and special steels

Specified element		Limit value % by mass
Cr	Chromium	0,50
Cu	Copper	0,50
Mn	Manganese	1,80
Mo	Molybdenum	0,10
Nb	Niobium	0,08
Ni	Nickel	0,50
Ti	Titanium	0,12
V	Vanadium	0,12
Zr	Zirconium	0,12

Source: EN 10020:2000 – Definition and classification of grades of steel (CEN, 2000)

- Alloy steels for rails, sheet piling and mining frames.
- Alloy steels for hot or cold rolled flat products for severe cold forming applications¹³⁷ containing grain refining elements such as boron, niobium, titanium, vanadium and/or zirconium or dual phase steels.¹³⁸
- Alloy steels in which copper is the only specified alloy element.
- Alloy electrical steels are steels mainly containing silicon or silicon and aluminium as alloying elements to meet specified requirement

5. Alloy special steels

This class includes steel grades other than stainless steels for which are characterized by precise control of chemical composition and particular conditions of manufacture and process control to ensure improved properties which are frequently specified in combination and within closely controlled limits.

Alloy special steels include alloy engineering steels and alloy steels for pressure vessels, bearing steels, tool steels, high-speed steels, and steels with special physical properties such as ferritic-nickel steels with controlled expansion coefficient, or electrical resistance steels.

1.2.2. Classification provided in the BAT reference document for the ferrous metals processing industry

Carbon steel non alloy: This first category covers a wide range of utilisation for long and flat products. It includes the ultra-low carbon steel, medium carbon steel (0.3% < carbon content <0.6%) and high carbon steel (carbon content >0.6%) as well as the microalloy such as High Strength Low Alloy (HSLA) steels and advanced high strength steels. In the flat product category, the main cold rolled carbon steels are: steels for packaging (very thin gages) and sheets for construction, general industry, automotive, appliances. In the category long products, the main carbon steel products are rails and improved machinability and free-cutting steels. For this type of product, a lead content (up to 0.35%) is used for easier machining, but sulfur can also be used (up to 0.4%).

Carbon steel low alloy: The main alloy elements (each element <5%) are Cr, Co, Cu, Mn, Mo, Nb, Ni, Ti, W, V, P and Si. These types of steel have improved quenching and improved mechanical properties and includes spring steels, bearings steels and wear resistant steels. Electrical steels, which contain Si from 1% up to 3%, also belong to this category. Some HSLA, High Strength and Advanced High Strength Steels, especially for flat products can also belong to this category.

¹³⁷ Excluding steels for pressure vessels or tubes

¹³⁸ Dual phases steels have a microstructure which is essentially ferritic, with about 10% to 35 % of martensite in small isolated areas uniformly dispersed throughout.

Stainless steel: There are four main classes of stainless steels: ferritic, austenitic, martensitic and duplex (combination of austenitic and ferritic stainless steels). All of them contain chromium (10-23%). Austenitic steel also contains around 8-10% of nickel. In hot rolling, the process of production of stainless is very similar to the production process of carbon steels, even if there are different heating patterns. For flat products, however, the processes are quite different, compared to the carbon steels. In addition, both for hot and cold processes, stainless steels have a high level of yield strength for the deformation and specific annealing cycles.

Other high alloy steels: The main alloy elements (> 5% for some of them) are Cr, Mo, W, V, Mn, Ni. This category contains, for example, the following type of products: high speed, moulds, tools, refractory and maraging (martensitic-aging) steels.

Special steel grade classification

Electrical steels: Electrical steels of flat products are produced following the hot rolling route followed by cold rolling. They are tailored to get specific properties which includes: low electrical power losses and high magnetic permeability. Electrical steels are usually produced with a silicon content within the range 2% to 3.2%. In general, they can be classified as part in the category of carbon steel low alloy. Usually, two main families of electrical steel are distinguished:

- Electrical steels with standard or improved electrical properties, designated as 'non grain oriented' (NGRO), or more usually named 'non oriented' (NONO) steel. The production of this type of electrical steel does not require any specific processing steps in hot rolling, however in cold rolling, it is sometimes necessary to carry out a first annealing (batch annealing) step before pickling and cold rolling. Also, in the case of laminated sheets which may be directly used by the customer, the final process of the cold plant includes a continuous annealing step (with temperatures up to 1050°C) followed by a varnish coating step.
- Electrical steels with optimised electrical properties, named 'grain oriented' (GO), with a silicon content of 3%. These steels are processed in such a way that the optimised properties are developed in the rolling direction. This needs a tight control of the crystal orientation during the production process. In hot rolling, it is necessary to operate at very high reheating temperatures (at around 1400°C), with specific devices in the furnace in order to recover the liquid scale, and very specific rolling patterns with high temperatures, both in the roughing and the finishing mill. In a cold rolling plant, very specific process steps are also required such as: annealing and pickling before cold rolling; cold rolling using a reverse multi-rolls stand (similar to stainless steel); continuous decarburising and annealing line with a final batch annealing step.

High-Strength Steels: These types of steel have improved or high mechanical characteristics related to strength such as yield strength (Re) and tensile strength (Rm). These high strength steels are mainly developed for automotive applications. High strength steels generally exhibit a yield strength value > 420 MPa and includes the following products HSLA (High Strength Low Alloy), FB (ferrite-Bainite), TRIP (TRAnsformation Induced Plasticity), DP (Dual Phase), CP (Complex Phase), MS (Martensite Steel), TWIP (TWIn Induced Plasticity), AHSS (Advanced High Strength Steels), Mn-B+HF (Manganese and Boron + Hot Forming). All these high strength steels are more difficult to be rolled and processed in a cold rolling plant, even if for some of them (TRIP, TWIN, MnB+HF) the final mechanical properties are obtained after transformation by the customer. Compared to the classical carbon mild steel, all these high strength steels requires higher energy consumption for the cold deformation, and very specific annealing thermal cycles.

1.2.3. EU ETS classification

The Regulation (EU) 2019/331 determining rules for harmonised free allocation of emission allowances (European Commission, 2019) defines two categories of steel products, one for **carbon steel** and another for **high alloy steel** (see Table 66). These are also the two categories considered by the German Steel Association in its classification system for green steel (WV Stahl, 2024), with the caveat that terms used are structural and reinforcing steel for carbon steel and quality steel for high-alloy steel.

Table 66. Categories of steel according to EU-ETS classification

EU – ETS Term	WV Stahl Term	Definition
EAF carbon steel	Structural and reinforcing steel	Steel containing less than 8 % metallic alloying elements and tramp elements to such levels limiting the use to those applications where no high surface quality and processability is required and if none of the criteria for the content of the metal alloying elements and the steel quality for high alloy steel are met
EAF high alloy steel	Quality steel (QST)	Steel containing 8 % or more metallic alloying elements or where high surface quality and processability is required. Steel is considered to be QST if at least one of the following criteria is met: 7. A maximum hydrogen content of 0.0003 per cent 8. A maximum sulphur content of 0.003 per cent 9. A maximum phosphorus content of 0.01 per cent 10. Micro-unit: f) K3 (oxide) < 40; K4 < 50 according to DIN 50602 (or any equivalent international standard) g) Sulphide: A/thin 2.0; A/thick 1.5 according to ISO 4967 h) Oxide: B/thin 1.5; B/thick 0.5 according to ISO 4967 i) ASTM E45: Procedure B, C, D maximum 2 j) SEP 1920: ultrasonic testing: core test – KSR maximum 2 mm 11. Macro-unit: blue brittleness break test: maximum 2.5 mm/dm² It can also be assumed that QST has a high surface quality and processing capability if 12. One of the following non-destructive technological tests is required for more than 10 per cent of the outgoing product stream: e) Ultrasonic testing according to ASTM E213 or EN 10246-6, 7.14 f) Magnetic particle testing according to ASTM E709 or EN 10246-12 g) Dye penetrant inspection according to ASTM E165 h) Electromagnetic testing i. Eddy current, ASTM E309: ii. Flux leakage, ASTM E570:

Source: Guidance Document n°9 on the harmonised free allocation methodology for the EU-ETS post 2020 (European Commission, 2019) and Proposal for a rule book relating to the classification system for green steel (WV Stahl, 2024).

In addition, Table 67 shows the correspondence between the categories defined by the EU-ETS and the PRODCOM codes.

Table 67. Correspondence between EU-ETS categories and PRODCOM categories

EU – ETS Term	PRODCOM code	PRODCOM description
EAF carbon steel	24.10.21.10	Flat semi-finished products (of non-alloy steel)
	24.10.21.21	Ingots, other primary forms and long semi-finished products for seamless tubes (of non-alloy steel)
	24.10.21.22	Other ingots, primary forms and long semi-finished products including blanks (of non-alloy steel)
EAF high alloy steel	24.10.23.10	Flat semi-finished products (of alloy steel other than of stainless steel)
	24.10.23.21	Ingots, other primary forms and long semi-finished products for seamless tubes (of alloy steel other than of stainless steel)
	24.10.23.22	Other ingots, primary forms and long semi-finished products (of alloy steel other than of stainless steel)
	24.10.22.10	Flat semi-finished products (slabs) (of stainless steel)
	24.10.22.21	Ingots, other primary forms and long semi-finished products for seamless tubes (of stainless steel)
	24.10.22.22	Other ingots, primary forms and long semi-finished products (of stainless steel)

Source: Guidance Document n°9 on the harmonised free allocation methodology for the EU-ETS post 2020 (European Commission, 2019)

1.2. Classification according to application

1.2.1. EN 10027-1:2016

An additional approach to the classification of steel products involves grouping them based on their application, considering their mechanical and physical properties. The standard **EN 10027-1:2016 – Designation systems for steels – Part 1: Steel names**, outlines specific rules for designating steels by means of symbolic letters and numbers to express application and principal characteristics.

Information required:

- An intended application, coded with a letter (e.g. S for structural steel, see Table 68)
- A relevant mechanical property, reported in MPa (e.g. Minimum Yield Strength, see Table 68)
- Optional mechanical properties or delivery conditions (e.g. thermal treatment or impact resistance)

Table 68 shows the eleven categories according to steel application, the letter used to designate this class of steel and the relevant mechanical property associated to it. There are eleven distinct application classes, with each class governed by specific set of rules to specify the properties of the steel. For instance, the code “S355J0CM” describes a structural steel (S) with Minimum Yield Strength of 355 MPa and an impact resistance of 27 J at 0 °C. Letters C and M express additional characteristics, in this case they refer to special cold forming and thermomechanically rolled, respectively. To illustrate, Figure 131. offers a simplified example of classification for steel used in structural steel. A short definition of all classes of applications is reported below.

Table 68. Classes of application and their relevant mechanical properties. Mandatory requirement for nomenclature according to EN 10027-1:2016

Application	Letter	Relevant mechanical property
Structural steel	S	Minimum Yield Strength
Pressure purposes	P	Minimum Yield Strength
Line pipe	L	Minimum Yield Strength
Engineering Steel	E	Minimum Yield Strength
Steel for reinforcing concrete	B	Characteristic yield strength
Steels for prestressing concrete	Y	Nominal tensile strength
Steel for or in the form of rails	R	Specified minimum Brinell Hardness (HBW)
Flat products for cold forming	D	Symbol for Hot/cold rolling
High strength steel flat products for cold forming	H	
Tin mill products (steel products for packaging)	T	Nominal yield strength
Electrical steel	M	Max specified loss

Source: Advancing on comparability aspects for product and organisation environmental footprint (European Commission, Joint Research Centre, et al., 2022)

Figure 131. Designation system for structural steel according to EN 10027-1:2016

1			2			3			
G S n n n			an			+an +an			
Principal symbols			Additional symbols						
Letter	Mechanical property	For steel						For steel product	
		Group 1 b			Group 2 c d				
G = steel casting (where necessary) S = structural steel	nnn = specified minimum yield strength ^e in MPa ^f for the smallest thickness range	Impact property Energy Joules (J)		Test temperature		C = Special cold forming			Tables 16, 17 and 18
		27J	40J	60J	°C	D = Hot dip coating			
		JR	KR	LR	20	E = Enamelling			
		J0	K0	L0	0	F = Forgings			
		J2	K2	L2	-20	H = Hollow section			
		J3	K3	L3	-30	L = Low temperature			
		J4	K4	L4	-40	M = Thermomechanically rolled			
		J5	K5	L5	-50	N = Normalized or normalized rolled			
		J6	K6	L6	-60	P = Sheet piling			
				A = Precipitation hardening				Q = Quenched and tempered	
		M = Thermomechanically rolled				S = Ship building			
		N = Normalized or normalized rolled				T = Tubes			
		Q = Quenched and tempered				W = Weather resistant			
		G = Other characteristics followed, where necessary by 1 or 2 digits				an = Chemical symbol of specified additional elements, e.g. Cu, together, where necessary, with a single digit representing 10 × the average (rounded to 0.1%) of that specified range of the content of that element			

Source: EN 10027-1:2016 - Designation systems for steels - Part1: Steel Names

Hereafter a concise overview of specific properties, typical sectors of application, and the most prevalent steel grades is provided for each category included in Table 68.

Structural Steel (S):

Structural steel is a type of steel specifically designed for use in construction and engineering applications where strength and load-bearing capacity are crucial. There exist various different shapes and grades, depending on the needs for the specific application. Structural steels are classified by the shape of their cross-sections, such as the most frequently used I, T, C shapes. Additionally, the mechanical properties of the steel are directly influenced by its grade. Therefore, different grades of structural steel must be chosen according to different design requirements (Yena Engineering).

It is commonly used in buildings, bridges, towers, and other infrastructure projects. Structural steel is known for its high strength-to-weight ratio, making it ideal for supporting heavy loads without excessive bulk. Common grades of structural steel include S235, S275, S355, and higher-strength variants like S460 (Gilbert, 2023).

Steel for Pressure Purposes (P):

Pressure purposes refer to applications where materials, including steel, are used to contain or handle fluids or gases under pressure. Examples of pressure purposes include pressure vessels and boilers. For this application, the steel must possess good tensile strength, impact resistance, and resistance to high and low-temperature environments. Common grades of steel for pressure purpose include P235GH, P265GH, P355GH, and P460GH.

Line Pipe (L):

Line pipes are used to transport fluids, such as oil, gas, or water, over long distances. They play a crucial role in the oil and gas industry, connecting drilling sites to refineries and distribution centres. Line pipes are typically made from high-strength carbon steel or low-alloy steel to withstand the high pressures and environmental conditions encountered during transportation (Liu, 2023). Common grades of steel for pipelines include L245, L290, L360, L450, L485 and L555.

Engineering Steel (E):

Engineering steel is a broad category that encompasses various steels used in mechanical and allied engineering applications. This application requires critical and often stringent levels of elasticity, strength, ductility, toughness and fatigue resistance. Additionally, it may also require resistance to high or low temperatures, corrosive and other aggressive environments (World Auto Steel, 2023). The utility of engineering steel spans across numerous industries, including shipbuilding, automotive, architecture, and machinery building. Examples of engineering steels include carbon steels (e.g., E295, E360), alloy steels (e.g., E335, E360), and stainless steels (e.g., E295).

Steel for Reinforcing Concrete (B):

Reinforcing steel, often referred to as rebar, is used to strengthen and reinforce concrete structures, such as buildings, bridges, and highways. It provides additional tensile strength to counteract the weak tensile nature of concrete (The Constructor, 2023). Reinforcing steel bars come in various grades, and the choice of grade depends on the specific application and engineering requirements. Some examples are B500A, B500B and B500C.

Steels for Prestressing Concrete (Y):

Prestressed concrete is a construction technique where steel tendons or cables are pre-tensioned or post-tensioned to induce compressive stresses in the concrete. This process helps counteract the tensile stresses that concrete may experience during its service life. The steel used for prestressing applications is typically high-strength steel, such as high-tensile strands or bars (Portland Cement Association, 2023) (Obinna, 2022). Some grades examples are Y1860, Y1960 and Y1860H.

Steel for or in the Form of Rails (R):

Rail steel is a specific type of steel used in the manufacturing of railway tracks. It must possess excellent wear resistance, high strength, and durability to withstand the heavy loads and repetitive stresses induced by trains passing over them. The properties of rail steel are carefully tailored to ensure a long service life and safety for train transportation (Federal Steel Supply, 2022). Common grades of steel for railway purpose are R220, R260, R350HT, R320Cr, R370HT and R400HT.

Flat Products for Cold Forming (D):

Flat products for cold forming refer to steel products that undergo a cold forming process at or near room temperature to achieve specific mechanical properties suitable for shaping and deformation. The cold forming process involves reducing the thickness of the steel through a series of rolling passes while maintaining a low working temperature. This ensures that the material remains below its recrystallization temperature, where the grain structure is retained, leading to enhanced formability and improved surface finish. The most common grades is DC01. This grade is a low-carbon, cold-rolled steel known for its excellent cold formability, making it ideal for a wide range of applications, including automotive parts, household appliances, and construction components.

Flat Products of High Strength for Cold Forming (H):

These are flat steel products with even higher strength suitable for cold forming applications. They are designed to provide superior strength and formability for applications requiring more substantial mechanical performance and load-bearing capabilities. They withstand substantial deformation and forming stress while providing reduced weight in structural components. They are used in applications that demand high strength and formability, such as in the construction of load-bearing structures, heavy machinery, and components subjected to higher mechanical loads. Examples include high-strength low-alloy (HSLA) steel and dual-phase steel. Common grades are H220PD, H260PD, H300LAD and H340LAD.

Tin Mill Products (Steel Products for Packaging) (T):

Tin mill products, also known as electrolytic tinplate or tin-free steel, are flat-rolled steel products that have been coated with a thin layer of tin or chromium to prevent corrosion. These materials are commonly used in the manufacturing of food and beverage cans, containers, and various other packaging applications (Berlin Metals, 2023). Some common designations are T1, T2, T3, T4, T5, T6 and T7.

Electrical Steel (M):

Electrical steel, also known as silicon steel or lamination steel, is a specialized type of steel with low carbon content and added silicon. It exhibits low core losses and high magnetic permeability, making it ideal for electrical applications. Electrical steel is classified into two types: non-oriented (NO) electrical steel and grain-oriented (GO) electrical steel. NO steel is widely used in rotating electrical machines (e.g., motors and generators) in which the magnetization direction is rotated in the sheet plane. GO steel is mainly used as a core material for static equipment (e.g., transformers), in which the magnetization direction is unidirectional (Hayakawa, 2022).

1.2.1. Sub-committees under CEN/TC 459

Other option to classify steel products based on their application, it is using as categories (application groups) the sub-committees under CEN/TC 459, which are in charge of standardising a wide range of steel products with many different applications. These product categories are as follow:

- Structural steels other than reinforcements
- Concrete reinforcing and prestressing steels
- Steels for heat treatment, alloy steels, free-cutting steels and stainless steels
- Wire rod and wires
- Steels for pressure purposes
- Steel sheet and strip for electrical applications
- Coated and uncoated flat products to be used for cold forming
- Steel tubes, and iron and steel fittings
- Steel castings and forgings

1.3. Steel products grouped by shape and surface conditions

1.3.1. Standard EN 10079:2007

The standard EN 10079:2007 also provides a comprehensive framework for classifying steel products. This classification system takes into account various aspects of steel products, such as their shape, dimensions (e.g. flat products, long products, and heavy sections) appearance, and surface conditions (e.g. coated flat products, bright products). Table 69 presents the different categories included in the standard.

Table 69. Steel products categories according to EN 10079:2007

Term	Definition
1 Liquid Steel	
Liquid steel	Steel in the liquid state ready for pouring and obtained from the melting of raw materials or scrap
2. Ingots and semi-finished products	
2.1. Ingots	Products obtained by pouring liquid steel into moulds of a shape appropriate to the subsequent processing ¹³⁹ into semi-finished products, or flat or long products, generally by hot rolling or forging
2.2. Semi-finished products	They are products obtained by: • continuous casting that may or may not be followed by rolling, forging or cutting; • pressure casting; • rolling, forging or cutting of ingots or large section continuous cast products and generally intended for conversion into flat or long products by hot rolling or forging, or for the manufacture of forgings

¹³⁹ In the case of ingots remelted by the vacuum arc of electroslag process, the products are obtained by melting, in a mould of appropriate shape, steel electrodes that have been previously cast, forged or rolled

2.2.1. Semi-finished products of square cross section	Semi-finished products with sides of 50mm or over, generally described as blooms if the sides are greater than 200mm, or as billets if smaller
2.2.2. Semi-finished products of rectangular cross section	Semi-finished products of cross section area $\geq 500 \text{ mm}^2$ or over of width up to twice the thickness, generally described as blooms if the cross section area is greater than $40\,000 \text{ mm}^2$, or as billets if smaller
2.2.3. Flat semi-finished products	Products of thickness generally 50 mm or over of width twice the thickness or over, generally described as slabs
2.2.4. Round semi-finished products	Continuously cast or forged semi-finished products of circular cross section
2.2.5. Blanks for sections	Semi-finished products intended for the manufacture of sections that have been preformed for that purpose
3. Flat products	
3.1. General flat products	Products having almost rectangular cross sections, the width being much greater than the thickness
3.2. Uncoated flat products	Flat products without any coating or surface treatment
3.2.1. Hot rolled uncoated flat products	Flat products manufactured by hot rolling semi-finished products, more rarely by hot rolling ingots. They include wide flat, plate and sheets, and strip
3.2.2. Cold rolled uncoated flat products	Uncoated flat products that have undergone a reduction in cross-section of 25 % or over by cold rolling. They include plate and sheet, and strip
3.3. Electrical steels	Steels characterised by their magnetic properties, that are intended for use in magnetic circuits in electrical machines
3.3.1. non oriented grain electrical steels	Non alloy steels and steels alloyed with silicon or silicon and aluminium that are essentially isotropic in their magnetic properties; i.e. the magnetic properties are similar both in the direction of rolling and in the transverse direction
3.3.2. grain oriented electrical steel	Steels alloyed with silicon that are anisotropic in that they possess a metallurgical structure that gives preferential magnetic properties in the direction of rolling
3.4. Tin mill and allied products for packaging	
3.4.1. Blackplate	Non alloy, low carbon steel supplied in strip or sheet form that has been single or double cold reduced
3.4.2. Tinplate	Non alloy, low carbon steel supplied in strip or sheet form that has been single or double cold reduced, and coated on both surfaces with tin in a continuous electrolytic process
3.4.3. Tinned sheet and strip	Non alloy, low carbon steel supplied in strip or sheet form of a thickness of 0,50 mm or over and tin coated on both surfaces
3.4.4. Electrolytic chromium/chromium oxide coated steel (ECCS)	Non alloy, low carbon steel supplied in strip or sheet form that may have been single or double cold reduced; coated on both surfaces by a cathodic process with a duplex film of metallic chromium adjacent to the steel substrate with an outer layer of hydrated chromium oxide or hydroxide
3.5. Coated hot or cold rolled flat products	Hot or cold rolled products with a permanent coating
3.5.1. Metal coated sheet and strip	They include hot dipped metal coated sheet and strip and electrolytically metal coated sheet and strip
3.5.2. Sheet and strip with organic coatings	Uncoated or metal coated (e.g. zinc coated) sheet and strip, subsequently coated with an organic material or a mixture of metal powder and organic material
3.5.3. Sheet and strip with miscellaneous inorganic coatings	Sheet and strip coated with an inorganic material, e.g. vitreous enamel
3.6. Profiled sheet	Sheet usually manufactured from coated sheet, but also from uncoated sheet, with a width much greater than the height of the profile
3.7. Composite products	Products comprising: a) plate, sheet and strip clad with steels or alloys to resist for example wear, chemical corrosion or heat distortion. Bonding is usually achieved by rolling, spraying, welding or explosion; b) sandwich steel sheet formed from two sheets bonded together by means of a synthetic sound insulating plastic layer; c) sandwich panels fabricated from two ribbed sheets bonded by an insulating layer
4. Long Products	
4.1. General long products	Those products that do not conform to the definition of flat products

4.2. Rod	Hot rolled long product having a nominal size generally of 5 mm or above and wound into irregular coils
4.3. Wire	Product of constant full cross section along its length, obtained by cold drawing rod through a reducing die or passing under pressure between rollers and rewinding the drawn product
4.4. Hot finished bars	Products supplied in straight lengths but never in coils, thus differentiating them from rod
4.4.1. Hot rolled bars	Hot rolled products in straight lengths of constant transverse section having a solid (convex) cross section. They include rounds, squares, hexagons, and octagons, flats, and bars of special shapes.
4.4.2. Forged bars	Products obtained by forging and that do not undergo subsequent hot conversion
4.4.3. Hollow mining drill bars	Bars with an internal hollow of any cross section shape, suitable for the manufacture of drill bits, with a maximum external cross section dimension over 15 mm up to and including 52 mm that is at least twice the maximum dimension of the cross section of the internal hollow.
4.5. Bright products	
4.5.1. Drawn products	Products of various cross section shapes obtained, after descaling, by drawing of hot rolled bars or rod on a draw bench (cold deformation without removing material)
4.5.2. Turned products	Round bars produced by turning on a lathe where the product can be further processed by straightening and polishing
4.5.3. Ground products	Drawn or turned round bars given an improved surface quality and dimensional accuracy by grinding or grinding and polishing
4.6. Deformed products for reinforcement and prestressing of concrete	Products with a cross section that is round or almost round, with crenelated or ribbed surfaces, for reinforcement and prestressing of concrete and supplied in the forms of rod, wire and bars
4.7. Hot rolled sections	
4.7.1. Railways material	Products used in the construction of railway tracks and other systems of rails
4.7.2. Piling	They include sheet piling, fabricated bearing piling and tubular fabricated bearing piling
4.7.3. Mining frame section	Products with cross sections resembling the letter I or the Greek capital letter Omega (Ω)
4.7.4. Heavy sections	Hot rolled products with cross sections resembling the letters I, H, U
4.7.5. Other sections	They include, small U, I and H sections, angles, T sections with equal flanges, bulb flats and special sections.
4.8. Welded sections	Long products of open cross sections that have cross section shapes similar to the products defined in 4.7.4 And 4.7.5, but instead of being obtained directly by hot rolling, are made up by welding together combinations of hot rolled long products, hot rolled flat products or cold rolled flat products
4.9. Cold formed sections	Cold formed long products having various cross section shapes, either open or with edges abutting, constant along their length
4.10. Tubular products	
4.10.1. Tubes	Hollow long products, open at both ends, of round or polygonal cross-section
4.10.2. Seamless tubes	Tubes made by piercing a solid product to obtain a tube hollow, which is further processed either hot or cold, into its final dimensions
4.10.3. Welded tubes	Tubes made by forming a hollow profile from a flat product and welding adjacent edges together
4.10.4. Hollow sections	Seamless or welded tubes of circular, square, rectangular or elliptical cross-section used in construction, e.g. of structural steelwork, cranes, vehicle chassis, handrails etc.
4.10.5. Hollow bars	Circular seamless tubes, intended for the manufacture of engineering components by machining, e.g. by removing shavings with a tool
5. Other products	
5.1. Open die forgings	Products obtained by forming steel at a suitable temperature by impact or pressure, using an open die to produce approximate shapes that do not require further hot deformation
5.2. Closed die forgings and stampings	Products obtained by forming steels at a specified temperature in a closed die that determines the required shape and volume of the product.
5.3. Castings	Products whose shapes and final dimensions, apart from any dressing or machining, are obtained directly by the solidification of liquid steel cast in sand moulds, fire clay or other refractory materials, or more rarely in metal or graphite permanent moulds
5.4. Powder metallurgy products	

5.4.1. Steel powder	Collection of steel particles of dimensions generally up to 1 mm
5.4.2. Sintered products	Products manufactured from powder by pressing and sintering and sometimes by re-pressing
5.4.3. Full density products	Products manufactured from powder by joint use of temperature and pressure (hot isostatic compression, extrusion, etc.)

Source: EN 10079:2007

In terms of the correspondence between classification systems, definitions outlined in the standard EN 10079:2007 exhibit a significant alignment with those found in the Combined Nomenclature (CN, section 1.1.1.5). However, it is important to acknowledge that certain limitations exist due to the nature of both systems:

- CN encompasses every product in trade and provides headings for processes or further manufactures beyond the scope of this European Standard;
- CN definitions classify products with more emphasis on shape than on manufacturing process or application;
- CN scope of product names and their descriptions may be different.

1.3.1. World Steel Association Statistics

Table 70 presents the categories used by World Steel Association in their statistics. The categorisation responds to a shape criteria but it is also partly based on the intended use (e.g. concrete reinforcing bars).

Table 70. Categories of steel products according to World Steel Association statistics

Product Category
Crude steel
Semi-finished products
Ingots
Continuous-cast steel
Liquid steel for castings
Hot rolled products
Hot rolled long products
Railway track materials
Heavy sections (≥ 80 mm)
Light sections (< 80 mm)
Concrete reinforcing bars
Hot rolled bars (other than concrete reinforcing bars)
Wire rod
Hot rolled flat products
Hot rolled plate (≥ 3 mm)
Hot rolled coil, sheet, and strip (< 3 mm)
Electrical Sheet and strip
Tinmill products
Tubes
Seamless tubes
Welded tubes

Source: Steel Statistical Yearbook 2022 (World Steel Association, 2022)

1.3.1. EUROFER classification

EUROFER gathers statistical data from its members using questionnaires and compiles this information to produce its statistics. EUROFER publishes numbers for European production and European deliveries of finished steel goods. The respectively used product classification schemes are similar but not identical.

The product categories considered are outlined in Table 71. The categories for finished steel products are determined based on shape criteria but for certain categories also take into account the intended use of the steel product and its surface conditions.

Table 71. Overview of the product categories used by EUROFER for finished steel products

Product categories - Production	Product categories - Deliveries
Hot rolled flat products	Total Flat Products
Hot rolled wide strip	Hot Rolled Flat Products
Cold rolled flat products	Cold Rolled Sheets
Hot dipped metal coated sheets	Total Coated Sheets
Organically coated and other coated sheets	<i>Organic Coated Sheets</i>
Other hot dipped metal coated sheets	<i>Hot Dipped Metal Coated Sheets</i>
Other rolled flat products	<i>Tin plate, tinned sheet and ECCS</i>
Other rolled white strip	<i>Electrolytically Metal Coated Sheets</i>
	Electrical sheets
	<i>Grain Oriented</i>
	<i>Grain Non-Oriented</i>
	<i>Not Finally Annealed</i>
	<i>Black Plate, Uncoated</i>
Plate and sheet rolled in other mills (Quarto plate)	Quarto Plate
Other flat products	
Hot rolled long products	Total Long Products
Rod	Wire Rod - Deformed Rod
Bars for reinforcement of concrete	Reinforcing Bars + Deformed Rod
Other bars and flats and sections (merchant bars)	Merchant Bars
Heavy sections and mining frames	
	Beams
Other rolled long products	Other rolled long products

Note: Bold categories indicate that there are available of data for this product group Italic categories are not reported. Categories neither bold nor italic are product groups that can be calculated from the superordinate category by subtracting all other subgroups.

Source: Steel in Figures 2023 (EUROFER, 2023)

1.4. European statistics

The EU's industrial production statistics play a pivotal role in tracking and analysing economic activities within the EU. These statistics are compiled through the PRODCOM¹⁴⁰ (PRODUCTION COMMunautaire) survey and also in the EUROPROD database, which incorporates external trade statistics. The PRODCOM survey captures data on various economic activities, which are then classified according to the Statistical Classification of Economic Activity (NACE).

Within this comprehensive framework, iron and steel products are specifically categorised under subsectors of NACE Sector C24 (Manufacture of basic metals, Table 72):

- C24.1 Manufacture of basic iron and steel and of ferro-alloy
 - 24.10 Manufacture of basic iron and steel and of ferro-alloy
- C24.2 Manufacture of tubes, pipes, hollow profiles and related fittings, of steel
 - 24.20 Manufacture of tubes, pipes, hollow profiles and related fittings, of steel
- C24.3 Manufacture of other products of first processing of steel
 - 24.31 Cold drawing of bars

¹⁴⁰ <https://ec.europa.eu/eurostat/web/metadata/classifications>

- 24.32 Cold rolling of narrow strip
- 24.33 Cold forming or folding
- 24.34 Cold drawing of wire
- C24.5 Casting of metals
 - 24.51 Casting of iron
 - 24.52 Casting of steel

Table 72. PRODCOM classification for steel

Nace code	Category
24.10	Manufacture of basic iron and steel and of ferro-alloys
24.10.11	Pig iron and spiegeleisen in pigs, blocks or other primary forms
24.10.12	Ferro-alloys
24.10.13	Ferrous products obtained by direct reduction of iron ore and other spongy ferrous products, in lumps, pellets or similar forms; iron having a minimum purity by weight of 99,94%, in lumps, pellets or similar forms
24.10.14	Granules and powders, of pig iron and spiegeleisen, or steel
24.10.21	Non alloy steel in ingots or other primary forms and semi-finished products of non alloy steel
24.10.22	Stainless steel in ingots or other primary forms and semi-finished products of stainless steel
24.10.23	Other alloy steel in ingots or other primary forms and semi-finished products of other alloy steel
24.10.31	Flat rolled products of non alloy steel, not further worked than hot rolled, of a width of ≥ 600 mm
24.10.32	Flat rolled products of non alloy steel, not further worked than hot rolled, of a width of < 600 mm
24.10.33	Flat rolled products of stainless steel, not further worked than hot rolled, of a width ≥ 600 mm
24.10.34	Flat rolled products of stainless steel, not further worked than hot rolled, of a width of < 600 mm
24.10.35	Flat rolled products of other alloy steel, not further worked than hot rolled, of a width of ≥ 600 mm
24.10.36	Flat rolled products of other alloy steel, not further worked than hot rolled, of a width of < 600 mm (except products of silicon-electrical steel)
24.10.41	Flat rolled products of non alloy steel, not further worked than cold rolled, of a width of ≥ 600 mm
24.10.42	Flat rolled products of stainless steel, not further worked than cold rolled, of a width of ≥ 600 mm
24.10.43	Flat rolled products of other-alloy steel, not further worked than cold rolled, of a width of ≥ 600 mm
24.10.51	Flat rolled products of non alloy steel, of a width of ≥ 600 mm, clad, plated or coated
24.10.52	Flat rolled products of other alloy steel, of a width of ≥ 600 mm, clad, plated or coated
24.10.53	Flat rolled products of silicon-electrical steel, of a width of ≥ 600 mm
24.10.54	Flat rolled products of silicon-electrical steel, of a width of < 600 mm
24.10.55	Flat rolled products of high speed steel, of a width of < 600 mm
24.10.61	Bars and rods, hot rolled, in irregularly wound coils, of non alloy steel
24.10.62	Other bars and rods of steel, not further worked than forged, hot rolled, hot-drawn or extruded, but including those twisted after rolling
24.10.63	Bars and rods, hot rolled, in irregularly wound coils, of stainless steel
24.10.64	Other bars and rods of stainless steel, not further worked than forged, hot rolled, hot-drawn or extruded, but including those twisted after rolling
24.10.65	Bars and rods, hot rolled, in irregularly wound coils, of other alloy steel
24.10.66	Other bars and rods of other alloy steel, not further worked than forged, hot rolled, hot-drawn or extruded, but including those twisted after rolling
24.10.67	Hollow drill bars and rods
24.10.71	Open sections, not further worked than hot rolled, hot-drawn or extruded, of non alloy steel
24.10.72	Open sections, not further worked than hot rolled, hot-drawn or extruded, of stainless steel
24.10.73	Open sections, not further worked than hot rolled, hot-drawn or extruded, of other alloy steel
24.10.74	Sheet piling, of steel and welded open sections of steel
24.10.75	Railway or tramway track construction material of steel
24.20	Manufacture of tubes, pipes, hollow profiles and related fittings, of steel

24.20.11	Line pipe of a kind used for oil or gas pipelines, seamless, of steel
24.20.12	Casing, tubing and drill pipe, of a kind used in the drilling for oil or gas, seamless, of steel
24.20.13	Other tubes and pipes, of circular cross section, of steel
24.20.14	Tubes and pipes, of non-circular cross-section and hollow profiles, of steel
24.20.21	Line pipe of a kind used for oil or gas pipelines, welded, of an external diameter of > 406,4 mm, of steel
24.20.22	Casing and tubing of a kind used in drilling for oil or gas, welded, of an external diameter of > 406,4 mm, of steel
24.20.23	Other tubes and pipes, of circular cross section, welded, of an external diameter of > 406,4 mm, of steel
24.20.24	Other tubes and pipes, of circular cross section, such as open seam, riveted or similarly closed, of an external diameter of > 406,4 mm, of steel
24.20.31	Line pipe of a kind used for oil or gas pipelines, welded, of an external diameter of ≤ 406,4 mm, of steel
24.20.32	Casing and tubing, of a kind used in drilling for oil or gas, welded, of an external diameter of ≤ 406,4 mm, of steel
24.20.33	Other tubes and pipes, of circular cross section, welded, of an external diameter of ≤ 406,4 mm, of steel
24.20.34	Tubes and pipes, of non-circular cross-section, welded, of an external diameter of ≤ 406,4 mm, of steel
24.20.35	Other tubes and pipes, such as open seam, riveted or similarly closed, of an external diameter of ≤ 406,4 mm, of steel
24.20.40	Tube or pipe fittings of steel, not cast
24.31	Cold drawing of bars
24.31.10	Cold drawn bars and solid profiles of non alloy steel
24.31.20	Cold drawn bars and solid profiles of alloy steel, other than stainless steel
24.31.30	Cold drawn bars and solid profiles of stainless steel
24.32	Cold rolling of narrow strip
24.32.10	Flat cold rolled steel products, uncoated, of a width of < 600 mm
24.32.20	Flat cold rolled steel products, clad, plated or coated, of a width of < 600 mm
24.33	Cold forming or folding
24.33.11	Open sections cold formed or folded of non alloy steel
24.33.12	Open sections cold formed or folded of stainless steel
24.33.20	Ribbed sheets of non alloy steel
24.33.30	Sandwich panels of coated steel sheet
24.34	Cold drawing of wire
24.34.11	Cold drawn wire of non alloy steel
24.34.12	Cold drawn wire of stainless steel
24.34.13	Cold drawn wire of other alloy steel
24.51	Casting of iron
24.51.11	Casting services of malleable cast iron
24.51.12	Casting services of spheroidal cast iron
24.51.13	Casting services of grey cast iron
24.51.20	Tubes, pipes and hollow profiles of cast-iron
24.51.30	Tube or pipe fittings, of cast-iron
24.52	Casting of steel
24.52.10	Casting services of steel
24.52.30	Tube or pipe fittings, of cast-steel

Source: Statistical classification of economic activities (NACE)¹⁴¹

¹⁴¹ <https://ec.europa.eu/eurostat/web/products-manuals-and-guidelines/-/ks-ra-07-015>

1.5. Combined Nomenclature

The **Combined Nomenclature (CN)** is a comprehensive system used for the classification of goods in accordance with both the Common Customs Tariff and the external trade statistics of the European Union. It represents a notable extension of the Harmonised System (HS) nomenclature, tailored to encompass specific EU-related subdivisions, employing eight-digit codes for precise classification.

Within the CN framework, iron and steel products find their classification in two primary chapters: Chapter 72, dedicated to iron and steel, and Chapter 73, focusing on articles made from iron and steel. Table 73 presents HS headings at four-digit level of the main categories of iron and steel products.

Table 73. Harmonised System

CN Code	Description
7201	Pig iron and spiegeleisen, in pigs, blocks or other primary forms
7202	Ferro-alloys
7203	Ferrous products obtained by direct reduction of iron ore and other spongy ferrous products, in lumps, pellets or similar forms; iron having a minimum purity by weight of 99,94%, in lumps, pellets or similar forms
7204	Ferrous waste and scrap; remelting scrap ingots of iron or steel (excl. slag, scale and other waste from the production of iron or steel; radioactive waste and scrap; fragments of pigs, blocks or other primary forms of pig iron or spiegeleisen)
7205	Granules and powders of pig iron, spiegeleisen, iron or steel (excl. granules and powders of ferro-alloys, turnings and filings of iron or steel, radioactive iron powders "isotopes" and certain low-calibre, substandard balls for ballbearings)
7206	Iron and non-alloy steel in ingots or other primary forms (excl. remelting scrap ingots, products obtained by continuous casting and iron of heading 7203)
7207	Semi-finished products of iron or non-alloy steel
7208	Flat-rolled products of iron or non-alloy steel, of a width $\geq$ 600 mm, hot-rolled, not clad, plated or coated
7209	Flat-rolled products of iron or non-alloy steel, of a width of $\geq$ 600 mm, cold-rolled "cold-reduced", not clad, plated or coated
7210	Flat-rolled products of iron or non-alloy steel, of a width $\geq$ 600 mm, hot-rolled or cold-rolled "cold-reduced", clad, plated or coated
7211	Flat-rolled products of iron or non-alloy steel, of a width of $<$ 600 mm, hot-rolled or cold-rolled "cold-reduced", not clad, plated or coated
7212	Flat-rolled products of iron or non-alloy steel, of a width of $<$ 600 mm, hot-rolled or cold-rolled "cold-reduced", clad, plated or coated
7213	Bars and rods of iron or non-alloy steel, hot-rolled, in irregularly wound coils
7214	Bars and rods, of iron or non-alloy steel, not further worked than forged, hot-rolled, hot-drawn or hot-extruded, but incl. those twisted after rolling (excl. in irregularly wound coils)
7215	Bars and rods, of iron or non-alloy steel, cold-formed or cold-finished, whether or not further worked, or hot-formed and further worked, n.e.s.
7216	Angles, shapes and sections of iron or non-alloy steel, n.e.s.
7217	Wire of iron or non-alloy steel, in coils (excl. bars and rods)
7218	Stainless steel in ingots or other primary forms (excl. remelting scrap ingots and products obtained by continuous casting); semi-finished products of stainless steel
7219	Flat-rolled products of stainless steel, of a width of $\geq$ 600 mm, hot-rolled or cold-rolled "cold-reduced"
7220	Flat-rolled products of stainless steel, of a width of $<$ 600 mm, hot-rolled or cold-rolled "cold-reduced"
7221	Bars and rods of stainless steel, hot-rolled, in irregularly wound coils
7222	Other bars and rods of stainless steel; angles, shapes and sections of stainless steel, n.e.s.
7223	Wire of stainless steel, in coils (excl. bars and rods)
7224	Steel, alloy, other than stainless, in ingots or other primary forms, semi-finished products of alloy steel other than stainless (excl. waste and scrap in ingot form, and products obtained by continuous casting)
7225	Flat-rolled products of alloy steel other than stainless, of a width of $\geq$ 600 mm, hot-rolled or cold-rolled "cold-reduced"
7226	Flat-rolled products of alloy steel other than stainless, of a width of $<$ 600 mm, hot-rolled or cold-rolled "cold-reduced"
7227	Bars and rods of alloy steel other than stainless, hot-rolled, in irregularly wound coils
7228	Other bars and rods of alloy steel other than stainless, angles, shapes and sections of alloy steel other than stainless, n.e.s.; hollow drill bars and rods, of alloy or non-alloy steel
7229	Wire of alloy steel other than stainless, in coils (excl. bars and rods)

7301	Sheet piling of iron or steel, whether or not drilled, punched or made from assembled elements; welded angles, shapes and sections, of iron or steel
7302	Railway or tramway track construction material of iron or steel, the following : rails, check-rails and rack rails, switch blades, crossing frogs, point rods and other crossing pieces, sleepers "cross-ties", fish-plates, chairs, chair wedges, sole plates "base plates", rail clips, bedplates, ties and other material specialised for jointing or fixing rails
7303	Tubes, pipes and hollow profiles, of cast iron
7304	Tubes, pipes and hollow profiles, seamless, of iron or steel (excl. products of cast iron)
7305	Tubes and pipes, having circular cross-sections and an external diameter of > 406,4 mm, of flat-rolled products of iron or steel "e.g., welded, riveted or similarly closed"
7306	Tubes, pipes and hollow profiles "e.g., open seam or welded, riveted or similarly closed", of iron or steel (excl. of cast iron, seamless tubes and pipes and tubes having internal and external circular cross-sections and an external diameter of > 406,4 mm)
7307	Tube or pipe fittings "e.g. couplings, elbows, sleeves", of iron or steel
7308	Structures and parts of structures "e.g., bridges and bridge-sections, lock-gates, towers, lattice masts, roofs, roofing frameworks, doors and windows and their frames and thresholds for doors, shutters, balustrades, pillars and columns", of iron or steel; plates, rods, angles, shapes, sections, tubes and the like, prepared for use in structures, of iron or steel (excl. prefabricated buildings of heading 9406)
7309	Reservoirs, tanks, vats and similar containers, of iron or steel, for any material "other than compressed or liquefied gas", of a capacity of > 300 l, not fitted with mechanical or thermal equipment, whether or not lined or heat-insulated (excl. containers specifically constructed or equipped for one or more types of transport)
7310	Tanks, casks, drums, cans, boxes and similar containers, of iron or steel, for any material "other than compressed or liquefied gas", of a capacity of ≤ 300 l, not fitted with mechanical or thermal equipment, whether or not lined or heat-insulated, n.e.s.
7311	Containers of iron or steel, for compressed or liquefied gas (excl. containers specifically constructed or equipped for one or more types of transport)
7312	Stranded wire, ropes, cables, plaited bands, slings and the like, of iron or steel (excl. electrically insulated products and twisted fencing wire and barbed wire)
7313	Barbed wire of iron or steel; twisted hoop or single flat wire, barbed or not, and loosely twisted double wire, of a kind used for fencing, of iron or steel
7314	Cloth, incl. endless bands, grill, netting and fencing, of iron or steel wire, expanded metal of iron or steel (excl. woven products of metal fibres of a kind used for cladding, lining or similar purposes)
7315	Chain and parts thereof, of iron or steel (excl. watch chains, necklace chains and the like, cutting and saw chain, skid chain, scraper chain for conveyors, toothed chain for textile machinery and the like, safety devices with chains for securing doors, measuring chains)
7316	Anchors, grapnels and parts thereof, of iron or steel
7317	Nails, tacks, drawing pins, corrugated nails, staples and similar articles of iron or steel, whether or not with heads of other material (excl. such articles with heads of copper and staples in strips)
7318	Screws, bolts, nuts, coach screws, screw hooks, rivets, cotters, cotter pins, washers, incl. spring washers, and similar articles, of iron or steel (excl. lag screws, stoppers, plugs and the like, threaded)
7319	Sewing needles, knitting needles, bodkins, crochet hoods, embroidery stiletos and similar articles, for use in the hand, of iron or steel; safety pins and other pins of iron or steel, n.e.s.
7320	Springs and leaves for springs, of iron or steel (excl. clock and watch springs, springs for sticks and handles of umbrellas or parasols, shock absorbers and torque rod or torsion bar springs of Section 17)
7321	Stoves, ranges, grates, cookers, incl. those with subsidiary boilers for central heating, barbecues, braziers, gas rings, plate warmers and similar non-electric domestic appliances, and parts thereof of iron or steel (excl. boilers and radiators for central heating, geysers and hot water cylinders)
7322	Radiators for central heating, non-electrically heated, and parts thereof, of iron or steel; air heaters and hot-air distributors, incl. distributors which can also distribute fresh or conditioned air, non-electrically heated, incorporating a motor-driven fan or blower, and parts thereof, of iron or steel
7323	Table, kitchen or other household articles, and parts thereof, of iron or steel; iron or steel wool; pot scourers and scouring or polishing pads, gloves and the like, of iron or steel (excl. cans, boxes and similar containers of heading 7310; waste baskets; shovels, corkscrews and other articles of the nature of a work implement; articles of cutlery, spoons, ladles, forks etc. of heading 8211 to 8215; ornamental articles; sanitary ware)
7324	Sanitary ware, and parts thereof, of iron or steel (excl. cans, boxes and similar containers of heading 7310, small wall cabinets for medical supplies or toiletries and other furniture of chapter 94, and fittings)
7325	Articles of iron or steel, cast, n.e.s.
7326	Articles of iron or steel, n.e.s. (excl. cast articles)

1.6. Other categorisations for steel products

1.6.1. World Steel LCI Study

The World Steel Association also presents an additional classification in its Worldsteel Life cycle inventory (LCI) study (World Steel Association, 2021). The LCA profiles include 17 categories (see Table 74) of finished steel products that cover almost the whole steel market ($\geq 95\%$). Based on this scheme, intermediate steel products belonging to the same category may be considered as a class of comparable products. Since, the classification of these products is partly based on the intended use of the material (e.g. engineering steel) and partly on the shape of the intermediate product (e.g. cold rolled coil) it is possible to create a partial correspondence of these profiles, but still not sufficient (European Commission, Joint Research Centre, et al., 2022).

Table 74. Description of steel products covered by the LCI Study Report

Product Category	Product	Product Description
Long Products	Sections	A steel section rolled on a hot rolling mill. Steel Sections include I-beams, H-beams, wide-flange beams, and sheet piling. This product is used in construction, multi-story buildings, industrial buildings, bridge trusses, vertical highway supports, and riverbank reinforcement.
	Rebar	A steel reinforcing bar is rolled on a hot rolling mill; can be further processed. This product is used to strengthen concrete in highway and building construction also as primary product for the wire rod process.
	Wire rod	Wire rod is a rolled steel product, produced from a semi and having a round, rectangular or other cross-section. Particularly fine cross-sections may be achieved by subsequent cold forming (drawing). Wire rod is wound into coils and transported in this form.
	Engineering steel (Tool steel)	Engineering Steel is rolled on a Hot Rolling mill. It can be found on the market and is further processed into finished products by the manufacturers. This steel is used in the manufacture of tools, dies, components for engines, drives, equipment, transmissions, etc.
Flat Products	Plate	A flat steel sheet rolled on a hot rolling mill; can be further processed. Includes use in the following sectors: structural steels, shipbuilding, pipes, pressure vessels, boilers, heavy metal structures, offshore structures etc. Typical thickness between 2 to 20 mm. The maximum width is 1860 mm.
	Hot rolled coil	Steel coil rolled on a hot-strip mill; can be further processed. Applications in virtually all sectors of industry: transport, construction, shipbuilding, gas containers, pressure vessels, energy pipelines, etc. Typical thickness between 2 - 7 mm. Typical width between 600 - 2100 mm
	Pickled hot rolled coil	Hot rolled steel from which the iron oxides present at the surface have been removed in a pickling process; can be further processed. Applications in virtually all sectors of industry: transport, construction, shipbuilding, gas containers, pressure vessels, energy pipelines, etc. Typical thickness between 2 - 7 mm. Typical width between 600 - 2100 mm
	Cold rolled coil	Obtained by a further thickness reduction of a pickled hot rolled coil. This step is achieved at low temperature in a cold-reduction mill; can be further processed. Used as primary material for finished cold rolled coils and coated coils. Typical thickness between 0.15 - 3 mm. Typical width between 600 - 2100 mm

Finished cold rolled coil	Obtained by heat treatment (annealing) and strain-hardening of cold rolled steel in a way to achieve final mechanical properties making the steel suitable for further uses (forming and bending); can be further processed. Classified into the following: formable steels, high strength formable steels, weathering structural steels, structural steels, hardenable steels. They have excellent forming properties, electromagnetic properties, paintability, weldability, and are suitable for fabrication by forming, pressing and bending. Applications include domestic applications, automotive applications, lighting fixtures, electrical components (stators, rotors) and various kinds of sections roofing applications, profiled sheets, wall elements, etc. Typical thickness between 0.3 - 3 mm. Typical width between 600 - 2100 mm.
Hot-dip galvanized steel	Obtained by passing cold rolled coil through a molten zinc bath, in order to coat the steel with a thin layer of zinc to provide corrosion resistance; can be further processed. They have excellent forming properties, paintability, weldability, and are suitable for fabrication by forming, pressing and bending. Applications include domestic applications, building applications (e.g. wall elements, roofing applications), automotive applications (e.g. body in white for vehicles underbody auto parts), lighting fixtures, drums and various kinds of sections applications, profiled sheets, etc. Typical thickness between 0.3 - 3 mm. Typical width between 600 - 2100 mm.
Electrogalvanized steel	Obtained by electro-plating finished cold rolled steel with a thin layer of zinc or zincnickel to provide corrosion resistance; can be further processed. They have excellent forming properties, paintability, weldability, and are suitable for fabrication by forming, pressing and bending. Applications include domestic applications, building applications (e.g. wall elements, roofing applications), automotive applications (e.g. body in white for vehicles underbody auto parts), lighting fixtures, drums and various kinds of sections applications, profiled sheets, etc. Typical thickness between 0.3 - 3 mm. Typical width between 600 - 2100 mm.
Welded pipe	A flat plate steel coil that is bended and welded into a tube. It can be found on the market for final use. A heavy-wall pipe is technically used to transport fluids (e.g. oil, gases, water, chemicals).
UO pipe	UO pipe is usually large in diameter and produced one piece at a time by forming plates. The plate is first pressed into a U shape by the U-press, and then into an O shape by the O-press. Because relatively thick material is used for making UO pipes, submerged arc welding is used for joining. UO pipe is mainly used as line pipe for transporting petroleum and natural gas in large quantity over long distances.
Seamless pipe	The seamless pipe is manufactured using a process called "extrusion". During this process a solid steel bar is pierced through the centre using a die, turning the solid steel into a tube which is processed into the correct shape and dimensions. They have an advantage in aggressive environments as there is no weld.
Tinplate	Obtained by electro plating a thin finished cold rolled coil with a thin layer of tin. It can be found on the market in coil or in sheets and is further processed into finished products by the manufacturers. Tin plated steel is used primarily in food cans, industrial packaging (e.g. small drums)

		Typical thickness between 0.13 - 0.49 mm. Typical width between 600 - 1100 mm.
	Tin-free (ECCS)	Also known as Electrolytic Chrome Coated Steel (ECCS). Obtained by electro plating a thin finished cold rolled coil with a thin layer of chrome. It can be found on the market in coil or in sheets and is further processed into finished products by the manufacturers. ECCS is used primarily in food cans, industrial packaging (e.g. small drums). Typical thickness between 0.13 - 0.49 mm. Typical width between 600 - 1100 mm.
	Organic coated	Obtained by coating a steel substrate with organic layers such as paint or laminated film. The substrate is mainly hot-dip galvanized coil but may also be electrogalvanized coil, finished cold rolled coil or tin-free steel. It can be found on the market in coil or in sheets and is further processed into finished products by the manufacturers. Used in all activity sectors e.g. construction (roof, wall and ceiling claddings, lighting, radiators etc.), general industry (e.g. office furniture, heating, ventilating, air conditioning), domestic appliances (refrigerators, washing machines, small kitchen appliances, computer casings & DVD casings, etc.) and packaging. Typical thickness between 0.15 - 1.5 mm. Typical width between 600 - 1300 mm.

Note: shadowed products are not included in the 2020 LCI Study (most updated version of the Study

Source: Worldsteel Life cycle inventory (LCI) study (World Steel Association, 2021).

Annex 2. European Steel Standards

There are hundreds of steel standards currently in use. Unfortunately, there is no international standard on steel in the world. Many SDOs have established standards on steel that are not equivalent between them¹⁴². Table 75 compiles the European Standards (including EN and EN ISO) related to steel, blue-shaded cells correspond to standards on testing (more details in Section 1.3.4).

Table 75. EN and EN ISO Steel Standards List

Code	Title
EN 39:2001	Loose steel tubes for tube and coupler scaffolds - Technical delivery conditions
EN 502:2013	Roofing products from metal sheet - Specification for fully supported roofing products of stainless steel sheet
EN 505:2013	Roofing products from metal sheet - Specification for fully supported roofing products of steel sheet
EN 508-1:2014	Roofing and cladding products from metal sheet - Specification for self-supporting of steel, aluminium or stainless steel sheet - Part 1: Steel
EN 508-2:2008	Roofing products from metal sheet - Specification for self-supporting products of steel, aluminium or stainless steel sheet - Part 2: Aluminium
EN 508-3:2008	Roofing products from metal sheet - Specification for self-supporting products of steel, aluminium or stainless steel sheet - Part 3: Stainless steel
EN 523:2003	Steel strip sheaths for prestressing tendons - Terminology, requirements, quality control
EN 524-1:1997	Steel strip sheaths for prestressing tendons - Test methods - Part 1: Determination of shape and dimensions
EN 524-2:1997	Steel strip sheaths for prestressing tendons - Test methods - Part 2: Determination of flexural behaviour
EN 524-3:1997	Steel strip sheaths for prestressing tendons - Methods of test - Part 3: To-and-fro bending test
EN 524-4:1997	Steel strip sheaths for prestressing tendons - Test methods - Part 4: Determination of lateral load resistance
EN 524-5:1997	Steel strip sheaths for prestressing tendons - Test methods - Part 5: Determination of tensile load resistance
EN 524-6:1997	Steel strip sheaths for prestressing tendons - Test methods - Part 6: Determination of leaktightness (determination of water loss)
EN 10138-1	Prestressing Steel. General requirements
EN 10138-2	Prestressing Steel. Wire
EN 10138-3	Prestressing Steel. Strand
EN 10138-4	Prestressing Steel. Bars
EN 544:2011	Bitumen shingles with mineral and/or synthetic reinforcements - Product specification and test methods
EN 1123-1:1999	Pipes and fittings of longitudinally welded hot-dip galvanized steel pipes with spigot and socket for waste water systems - Part 1: Requirements, testing, quality control
EN 1123-2:2006	Pipes and fittings of longitudinally welded hot-dip galvanized steel tube with spigot and socket for waste water systems - Part 2: Dimensions
EN 1123-3:2004	Pipes and fittings of longitudinally welded hot-dip galvanized steel pipes with spigot and socket for waste water systems - Part 3: Dimensions and special requirements for vacuum drainage systems and for drainage systems in ship-building
EN 1124-1:1999	Pipes and fittings of longitudinally welded stainless steel pipes with spigot and socket for waste water systems - Part 1: Requirements, testing, quality control
EN 1124-2:2014	Pipes and fittings of longitudinally welded stainless steel pipes with spigot and socket for waste water systems - Part 2: System S, forms and dimensions
EN 1124-3:2008	Pipes and fittings of longitudinally welded stainless steel pipes with spigot and socket for waste water systems - Part 3: System X - Dimensions
EN 1124-4:2013	Pipes and fittings of longitudinally welded stainless steel pipes with spigot and socket for wastewater systems - Part 4: Components for vacuum drainage systems and for drainage systems on ships
EN 1370:2011	Founding - Examination of surface condition
EN 1503-1:2000	Valves - Materials for bodies, bonnets and covers - Part 1: Steels specified in European Standards
EN 1503-2:2000	Valves - Materials for bodies, bonnets and covers - Part 2: Steels other than those specified in European Standards
EN 1503-3:2000	Valves - Materials for bodies, bonnets and covers - Part 3: Cast irons specified in European Standards
EN 1503-4:2002	Valves - Materials for bodies, bonnets and covers - Part 4: Copper alloys specified in European Standards
EN 1559-1:2011	Founding - Technical conditions of delivery - Part 1: General
EN 1559-2:2014	Founding - Technical conditions of delivery - Part 2: Additional requirements for steel castings
EN 1559-3:2011	Founding - Technical conditions of delivery - Part 3: Additional requirements for iron castings
EN 1559-4:2015	Founding - Technical conditions of delivery - Part 4: Additional requirements for aluminium alloy castings
EN 1559-5:1997	Founding - Technical conditions of delivery - Part 5: Additional requirements for magnesium alloy castings
EN 1559-6:1998	Founding - Technical conditions of delivery - Part 6: Additional requirements for zinc alloy castings

¹⁴² For a comprehensive list of standards on steel from different Standards Development Organizations (SDOs) and their comparison see *Handbook of Comparative World Steel Standards, 5th Edition* (Bringas, 2016)

EN 1561:2011	Founding - Grey cast irons
EN 1562:2012	Founding - Malleable cast irons
EN 1563:2011	Founding - Spheroidal graphite cast irons
EN 1677-1:2000	Components for slings - Safety - Part 1: Forged steel components, Grade 8
EN 1677-2:2000	Components for slings - Safety - Part 2: Forged steel lifting hooks with latch, Grade 8
EN 1677-3:2001	Components for slings - Safety - Part 3: Forged steel self-locking hooks - Grade 8
EN 1677-4:2000	Components for slings - Safety - Part 4: Links, Grade 8
EN 1677-5:2001	Components for slings - Safety - Part 5: Forged steel lifting hooks with latch - Grade 4
EN 1677-6:2001	Components for slings - Safety - Part 6: Links - Grade 4
EN 1759-1:2004	Flanges and their joints - Circular flanges for pipes, valves, fittings and accessories, Class designated - Part 1: Steel flanges, NPS 1/2 to 24
EN 2515:1990	Rod ends, adjustable single fork and threaded shank - Dimensions and loads
EN 3305:1996	Screws, 100° countersunk reduced head, offset cruciform recess, close tolerance normal shank, short thread, in alloy steel, cadmium plated - Classification: 1 100 MPa (at ambient temperature)/235 °C
EN 10024:1995	Hot rolled taper flange I sections - Tolerances on shape and dimensions
EN 10025-1:2004	Hot rolled products of structural steels - Part 1: General technical delivery conditions
EN 10025-2:2004	Hot rolled products of structural steels - Part 2: Technical delivery conditions for non-alloy structural steels
EN 10025-3:2004	Hot rolled products of structural steels - Part 3: Technical delivery conditions for normalized/normalized rolled weldable fine grain structural steels
EN 10025-4:2004	Hot rolled products of structural steels - Part 4: Technical delivery conditions for thermomechanical rolled weldable fine grain structural steels
EN 10025-5:2004	Hot rolled products of structural steels - Part 5: Technical delivery conditions for structural steels with improved atmospheric corrosion resistance
EN 10025-6:2004	Hot rolled products of structural steels - Part 6: Technical delivery conditions for flat products of high yield strength structural steels in the quenched and tempered condition
EN 10027-1:2005	Designation systems for steels - Part 1: Steel names
EN 10027-2:2015	Designation systems for steels - Part 2: Numerical system
EN 10028-1:2007	Flat products made of steels for pressure purposes - Part 1: General requirements
EN 10028-2:2009	Flat products made of steels for pressure purposes - Part 2: Non-alloy and alloy steels with specified elevated temperature properties
EN 10028-3:2009	Flat products made of steels for pressure purposes - Part 3: Weldable fine grain steels, normalized
EN 10028-4:2009	Flat products made of steels for pressure purposes - Part 4: Nickel alloy steels with specified low temperature properties
EN 10028-5:2009	Flat products made of steels for pressure purposes - Part 5: Weldable fine grain steels, thermomechanically rolled
EN 10028-6:2009	Flat products made of steels for pressure purposes - Part 6: Weldable fine grain steels, quenched and tempered
EN 10028-7:2007	Flat products made of steels for pressure purposes - Part 7: Stainless steels
EN 10029:2010	Hot-rolled steel plates 3 mm thick or above - Tolerances on dimensions and shape
EN 10034:1993	Structural steel I and H sections - Tolerances on shape and dimensions
EN 10048:1996	Hot rolled narrow steel strip - Tolerances on dimensions and shape
EN 10051:2010	Continuously hot-rolled strip and plate/sheet cut from wide strip of non-alloy and alloy steels - Tolerances on dimensions and shape
EN 10055:1995	Hot rolled steel equal flange tees with radiused root and toes - Dimensions and tolerances on shape and dimensions
EN 10056-1:1998	Structural steel equal and unequal leg angles - Part 1: Dimensions
EN 10056-2:1993	Structural steel equal and unequal leg angles - Part 2: Tolerances on shape and dimensions
EN 10067:1996	Hot rolled bulb flats - Dimensions and tolerances on shape, dimensions and mass
EN 10079:2007	Definition of steel products
EN 10080:2005	Steel for the reinforcement of concrete - Weldable reinforcing steel - General
EN 10083-1:2006	Steels for quenching and tempering - Part 1: General technical delivery conditions
EN 10083-2:2006	Steels for quenching and tempering - Part 2: Technical delivery conditions for non-alloy steels
EN 10083-3:2006	Steels for quenching and tempering - Part 3: Technical delivery conditions for alloy steels
EN 10084:2008	Case hardening steels - Technical delivery conditions
EN 10085:2001	Nitriding steel - Technical delivery conditions
EN 10087:1998	Free cutting steels - Technical delivery conditions for semi-finished products, hot-rolled bars and rods
EN 10088-1:2014	Stainless steels - Part 1: List of stainless steels
EN 10088-2:2014	Stainless steels - Part 2: Technical delivery conditions for sheet/plate and strip of corrosion resisting steels for general purposes
EN 10088-3:2014	Stainless steels - Part 3: Technical delivery conditions for semi-finished products, bars, rods, wire, sections and bright products of corrosion resisting steels for general purposes
EN 10088-4:2009	Stainless steels - Part 4: Technical delivery conditions for sheet/plate and strip of corrosion resisting steels for construction purposes

EN 10088-5:2009	Stainless steels - Part 5: Technical delivery conditions for bars, rods, wire, sections and bright products of corrosion resisting steels for construction purposes
EN 10089:2002	Hot-rolled steels for quenched and tempered springs - Technical delivery conditions
EN 10090:1998	Valve steels and alloys for internal combustion engines
EN 10095:1999	Heat resisting steel and nickel alloys
EN 10106:2007	Cold rolled non-oriented electrical steel sheet and strip delivered in the fully processed state
EN 10107:2014	Grain-oriented electrical steel strip and sheet delivered in the fully processed state
EN 10111:2008	Continuously hot rolled low carbon steel sheet and strip for cold forming - Technical delivery conditions
EN 10120:1996	Steel sheet and strip for welded gas cylinders
EN 10130:2006	Cold rolled low carbon steel flat products for cold forming - Technical delivery conditions
EN 10131:2006	Cold rolled uncoated and zinc or zinc-nickel electrolytically coated low carbon and high yield strength steel flat products for cold forming - Tolerances on dimensions and shape
EN 10132-1:2000	Cold rolled narrow steel strip for heat treatment - Technical delivery conditions - Part 1: General
EN 10132-2:2000	Cold rolled narrow steel strip for heat treatment - Technical delivery conditions - Part 2: Case hardening steels
EN 10132-3:2000	Cold rolled narrow steel strip for heat treatment - Technical delivery conditions - Part 3: Steels for quenching and tempering
EN 10132-4:2000	Cold rolled narrow steel strip for heat treatment - Technical delivery conditions - Part 4: Spring steels and other applications
EN 10139:1997	Cold rolled uncoated mild steel narrow strip for cold forming - Technical delivery conditions
EN 10140:2006	Cold rolled narrow steel strip - Tolerances on dimensions and shape
EN 10143:2006	Continuously hot-dip metal coated steel sheet and strip - Tolerances on dimensions and shape
EN 10149-1:2013	Hot rolled flat products made of high yield strength steels for cold forming - Part 1: General technical delivery conditions
EN 10149-2:2013	Hot rolled flat products made of high yield strength steels for cold forming - Part 2: Technical delivery conditions for thermomechanically rolled steels
EN 10149-3:2013	Hot rolled flat products made of high yield strength steels for cold forming - Part 3: Technical delivery conditions for normalized or normalized rolled steels
EN 10151:2002	Stainless steel strip for springs - Technical delivery conditions
EN 10152:2009	Electrolytically zinc coated cold rolled steel flat products for cold forming - Technical delivery conditions
EN 10160:1999	Ultrasonic testing of steel flat product of thickness equal or greater than 6 mm (reflection method)
EN 10162:2003	Cold rolled steel sections - Technical delivery conditions - Dimensions and cross-sectional tolerances
EN 10163-1:2004	Delivery requirements for surface condition of hot-rolled steel plates, wide flats and sections - Part 1: General requirements
EN 10163-2:2004	Delivery requirements for surface condition of hot-rolled steel plates, wide flats and sections - Part 2: Plate and wide flats
EN 10163-3:2004	Delivery requirements for surface condition of hot-rolled steel plates, wide flats and sections - Part 3: Sections
EN 10164:2004	Steel products with improved deformation properties perpendicular to the surface of the product - Technical delivery conditions
EN 10169:2010	Continuously organic coated (coil coated) steel flat products - Technical delivery conditions
EN 10202:2001	Cold reduced tinmill products - Electrolytic tinplate and electrolytic chromium/chromium oxide coated steel
EN 10204:2004	Metallic products - Types of inspection documents
EN 10205:1991	Cold reduced blackplate in coil form for the production of tinplate or electrolytic chromium/chromium oxide coated steel
EN 10207:2005	Steels for simple pressure vessels - Technical delivery requirements for plates, strips and bars
EN 10209:2013	Cold rolled low carbon steel flat products for vitreous enamelling - Technical delivery conditions
EN 10210-1:2006	Hot finished structural hollow sections of non-alloy and fine grain steels - Part 1: Technical delivery conditions
EN 10210-2:2006	Hot finished structural hollow sections of non-alloy and fine grain steels - Part 2: Tolerances, dimensions and sectional properties
EN 10213:2007	Steel castings for pressure purposes
EN 10216-1:2013	Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
EN 10216-2:2013	Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 2: Non-alloy and alloy steel tubes with specified elevated temperature properties
EN 10216-3:2013	Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 3: Alloy fine grain steel tubes
EN 10216-4:2013	Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 4: Non-alloy and alloy steel tubes with specified low temperature properties
EN 10216-5:2013	Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 5: Stainless steel tubes
EN 10217-1:2002	Welded steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
EN 10217-2:2002	Welded steel tubes for pressure purposes - Technical delivery conditions - Part 2: Electric welded non-alloy and alloy steel tubes with specified elevated temperature properties

EN 10217-3:2002	Welded steel tubes for pressure purposes - Technical delivery conditions - Part 3: Alloy fine grain steel tubes
EN 10217-4:2002	Welded steel tubes for pressure purposes - Technical delivery conditions - Part 4: Electric welded non-alloy steel tubes with specified low temperature properties
EN 10217-5:2002	Welded steel tubes for pressure purposes - Technical delivery conditions - Part 5: Submerged arc welded non-alloy and alloy steel tubes with specified elevated temperature properties
EN 10217-6:2002	Welded steel tubes for pressure purposes - Technical delivery conditions - Part 6: Submerged arc welded non-alloy steel tubes with specified low temperature properties
EN 10217-7:2014	Welded steel tubes for pressure purposes - Technical delivery conditions - Part 7: Stainless steel tubes
EN 10218-1:2012	Steel wire and wire products - General - Part 1: Test methods
EN 10218-2:2012	Steel wire and wire products - General - Part 2: Wire dimensions and tolerances
EN 10219-1:2006	Cold formed welded structural hollow sections of non-alloy and fine grain steels - Part 1: Technical delivery conditions
EN 10219-2:2006	Cold formed welded structural hollow sections of non-alloy and fine grain steels - Part 2: Tolerances, dimensions and sectional properties
EN 10221:1995	Surface quality classes for hot-rolled bars and rods - Technical delivery conditions
EN 10222-1:1998	Steel forgings for pressure purposes - Part 1: General requirements for open die forgings
EN 10222-2:1999	Steel forgings for pressure purposes - Part 2: Ferritic and martensitic steels with specified elevated temperature properties
EN 10222-3:1998	Steel forgings for pressure purposes - Part 3: Nickel steels with specified low-temperature properties
EN 10222-4:1998	Steel forgings for pressure purposes - Part 4: Weldable fine-grain steels with high proof strength
EN 10222-5:1999	Steel forgings for pressure purposes - Part 5: Martensitic, austenitic and austenitic-ferritic stainless steels
EN 10223-1:2012	Steel wire and wire products for fencing and netting - Part 1: Zinc and zinc-alloy coated steel barbed wire
EN 10223-2:2012	Steel wire and wire products for fencing and netting - Part 2: Hexagonal steel wire netting for agricultural, insulation and fencing purposes
EN 10223-3:2013	Steel wire and wire products for fencing and netting - Part 3: Hexagonal steel wire mesh products for civil engineering purposes
EN 10223-4:2012	Steel wire and wire products for fencing and netting - Part 4: Steel wire welded mesh fencing
EN 10223-5:2012	Steel wire and wire products for fencing and netting - Part 5: Steel wire woven hinged joint and knotted mesh fencing
EN 10223-6:2012	Steel wire and wire products for fencing and netting - Part 6: Steel wire chain link fencing
EN 10223-7:2012	Steel wire and wire products for fencing and netting - Part 7: Steel wire welded panels for fencing
EN 10224:2002	Non-alloy steel tubes and fittings for the conveyance of aqueous liquids including water for human consumption - Technical delivery conditions
EN 10225:2009	Weldable structural steels for fixed offshore structures - Technical delivery conditions
EN 10228-1:1999	Non-destructive testing of steel forgings - Part 1: Magnetic particle inspection
EN 10228-2:1998	Non-destructive testing of steel forgings - Part 2: Penetrant testing
EN 10228-3:1998	Non-destructive testing of steel forgings - Part 3: Ultrasonic testing of ferritic or martensitic steel forgings
EN 10228-4:1999	Non-destructive testing of steel forgings - Part 4: Ultrasonic testing of austenitic and austenitic-ferritic stainless steel forgings
EN 10238:2009	Automatically blast-cleaned and automatically prefabrication primed structural steel products
EN 10240:1997	Internal and/or external protective coatings for steel tubes - Specification for hot dip galvanized coatings applied in automatic plants
EN 10241:2000	Steel threaded pipe fittings
EN 10243-1:1999	Steel die forgings - Tolerances on dimensions - Part 1: Drop and vertical press forgings
EN 10243-2:1999	Steel die forgings - Tolerances on dimensions - Part 2: Upset forgings made on horizontal forging machines
EN 10244-1:2009	Steel wire and wire products - Non-ferrous metallic coatings on steel wire - Part 1: General principles
EN 10244-2:2009	Steel wire and wire products - Non-ferrous metallic coatings on steel wire - Part 2: Zinc or zinc alloy coatings
EN 10244-3:2001	Steel wire and wire products - Non-ferrous metallic coatings on steel wire - Part 3: Aluminium coatings
EN 10244-4:2001	Steel wire and wire products - Non-ferrous metallic coatings on steel wire - Part 4: Tin coatings
EN 10244-5:2001	Steel wire and wire products - Non-ferrous metallic coatings on steel wire - Part 5: Nickel coatings
EN 10244-6:2001	Steel wire and wire products - Non-ferrous metallic coatings on steel wire - Part 6: Copper, bronze or brass coatings
EN 10245-1:2011	Steel wire and wire products - Organic coatings on steel wire - Part 1: General rules
EN 10245-2:2011	Steel wire and wire products - Organic coatings on steel wire - Part 2: PVC finished wire
EN 10245-3:2011	Steel wire and wire products - Organic coatings on steel wire - Part 3: PE coated wire
EN 10245-4:2011	Steel wire and wire products - Organic coatings on steel wire - Part 4: Polyester coated wire
EN 10245-5:2011	Steel wire and wire products - Organic coatings on steel wire - Part 5: Polyamide coated wire
EN 10248-1:1995	Hot rolled sheet piling of non alloy steels - Part 1: Technical delivery conditions
EN 10248-2:1995	Hot rolled sheet piling of non alloy steels - Part 2: Tolerances on shape and dimensions
EN 10249-1:1995	Cold formed sheet piling of non alloy steels - Part 1: Technical delivery conditions
EN 10249-2:1995	Cold formed sheet piling of non alloy steels - Part 2: Tolerances on shape and dimensions
EN 10250-1:1999	Open die steel forgings for general engineering purposes - Part 1: General requirements
EN 10250-2:1999	Open die steel forgings for general engineering purposes - Part 2: Non-alloy quality and special steels
EN 10250-3:1999	Open die steel forgings for general engineering purposes - Part 3: Alloy special steels

EN 10250-4:1999	Open die steel forgings for general engineering purposes - Part 4: Stainless steels
EN 10251:1997	Magnetic materials - Methods of determination of the geometrical characteristics of electrical steel sheet and strip
EN 10252:1997	Magnetic materials - Methods of measurement of magnetic properties of magnetic steel sheet and strip at medium frequencies
EN 10253-1:1999	Butt-welding pipe fittings - Part 1: Wrought carbon steel for general use and without specific inspection requirements
EN 10253-2:2007	Butt-welding pipe fittings - Part 2: Non-alloy and ferritic alloy steels with specific inspection requirements
EN 10253-3:2008	Butt-welding pipe fittings - Part 3: Wrought austenitic and austenitic-ferritic (duplex) stainless steels without specific inspection requirements
EN 10253-4:2008	Butt-welding pipe fittings - Part 4: Wrought austenitic and austenitic-ferritic (duplex) stainless steels with specific inspection requirements
EN 10254:1999	Steel closed die forgings - General technical delivery conditions
EN 10255:2004	Non-alloy steel tubes suitable for welding and threading - Technical delivery conditions
EN 10257-1:2011	Zinc or zinc alloy coated non-alloy steel wire for armouring either power cables or telecommunication cables - Part 1: Land cables
EN 10257-2:2011	Zinc or zinc alloy coated non-alloy steel wire for armouring either power cables or telecommunication cables - Part 2: Submarine cables
EN 10263-1:2001	Steel rod, bars and wire for cold heading and cold extrusion - Part 1: General technical delivery conditions
EN 10263-2:2001	Steel rod, bars and wire for cold heading and cold extrusion - Part 2: Technical delivery conditions for steels not intended for heat treatment after cold working
EN 10263-3:2001	Steel rod, bars and wire for cold heading and cold extrusion - Part 3: Technical delivery conditions for case hardening steels
EN 10263-4:2001	Steel rod, bars and wire for cold heading and cold extrusion - Part 4: Technical delivery conditions for steels for quenching and tempering
EN 10263-5:2001	Steel rod, bars and wire for cold heading and cold extrusion - Part 5: Technical delivery conditions for stainless steels
EN 10264-1:2012	Steel wire and wire products - Steel wire for ropes - Part 1: General requirements
EN 10264-2:2012	Steel wire and wire products - Steel wire for ropes - Part 2: Cold drawn non alloy steel wire for ropes for general applications
EN 10264-3:2012	Steel wire and wire products - Steel wire for ropes - Part 3: Round and shaped non alloyed steel wire for high duty applications
EN 10264-4:2012	Steel wire and wire products - Steel wire for ropes - Part 4: Stainless steel wire
EN 10267:1998	Ferritic-pearlitic steels for precipitation hardening from hot-working temperatures
EN 10268:2006	Cold rolled steel flat products with high yield strength for cold forming - Technical delivery conditions
EN 10269:2013	Steels and nickel alloys for fasteners with specified elevated and/or low temperature properties
EN 10270-1:2011	Steel wire for mechanical springs - Part 1: Patented cold drawn unalloyed spring steel wire
EN 10270-2:2011	Steel wire for mechanical springs - Part 2: Oil hardened and tempered spring steel wire
EN 10270-3:2011	Steel wire for mechanical springs - Part 3: Stainless spring steel wire
EN 10271:1998	Electrolytically zinc-nickel (ZN) coated steel flat products - Technical delivery conditions
EN 10272:2007	Stainless steel bars for pressure purposes
EN 10273:2007	Hot rolled weldable steel bars for pressure purposes with specified elevated temperature properties
EN 10277-1:2008	Bright steel products - Technical delivery conditions - Part 1: General
EN 10277-2:2008	Bright steel products - Technical delivery conditions - Part 2: Steels for general engineering purposes
EN 10277-3:2008	Bright steel products - Technical delivery conditions - Part 3: Free-cutting steels
EN 10277-4:2008	Bright steel products - Technical delivery conditions - Part 4: Case-hardening steels
EN 10277-5:2008	Bright steel products - Technical delivery conditions - Part 5: Steels for quenching and tempering
EN 10278:1999	Dimensions and tolerances of bright steel products
EN 10279:2000	Hot rolled steel channels - Tolerances on shape, dimensions and mass
EN 10283:2010	Corrosion resistant steel castings
EN 10293:2015	Steel castings - Steel castings for general engineering uses
EN 10294-1:2005	Hollow bars for machining - Technical delivery conditions - Part 1: Non alloy and alloy steels
EN 10294-2:2012	Hollow bars for machining - Technical delivery conditions - Part 2: Stainless steels with specified machinability properties
EN 10295:2002	Heat resistant steel castings
EN 10296-1:2003	Welded circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions - Part 1: Non-alloy and alloy steel tubes
EN 10296-2:2005	Welded circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions - Part 2: Stainless steel
EN 10297-1:2003	Seamless circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions - Part 1: Non-alloy and alloy steel tubes
EN 10297-2:2005	Seamless circular steel tubes for mechanical and general engineering purposes - Technical delivery conditions - Part 2: Stainless steel
EN 10302:2008	Creep resisting steels, nickel and cobalt alloys

EN 10303:2001	Thin magnetic steel sheet and strip for use at medium frequencies
EN 10305-1:2010	Steel tubes for precision applications - Technical delivery conditions - Part 1: Seamless cold drawn tubes
EN 10305-2:2010	Steel tubes for precision applications - Technical delivery conditions - Part 2: Welded cold drawn tubes
EN 10305-3:2010	Steel tubes for precision applications - Technical delivery conditions - Part 3: Welded cold sized tubes
EN 10305-4:2011	Steel tubes for precision applications - Technical delivery conditions - Part 4: Seamless cold drawn tubes for hydraulic and pneumatic power systems
EN 10305-5:2010	Steel tubes for precision applications - Technical delivery conditions - Part 5: Welded cold sized square and rectangular tubes
EN 10305-6:2005	Steel tubes for precision applications - Technical delivery conditions - Part 6: Welded cold drawn tubes for hydraulic and pneumatic power systems
EN 10312:2002	Welded stainless steel tubes for the conveyance of aqueous liquids including water for human consumption - Technical delivery conditions
EN 10324:2004	Steel wire and wire products - Hose reinforcement wire
EN 10325:2006	Steel - Determination of yield strength increase by the effect of heat treatment [Bake-Hardening-Index]
EN 10340:2007	Steel castings for structural uses
EN 10341:2006	Cold rolled electrical non-alloy and alloy steel sheet and strip delivered in the semi-processed state
EN 10343:2009	Steels for quenching and tempering for construction purposes - Technical delivery conditions
EN 10346:2015	Continuously hot-dip coated steel flat products for cold forming - Technical delivery conditions
EN 10347:2006	Guidance for forming of structural steels in processing
EN 12007-1:2012	Gas infrastructure - Pipelines for maximum operating pressure up to and including 16 bar - Part 1: General functional requirements
EN 12007-2:2012	Gas infrastructure - Pipelines for maximum operating pressure up to and including 16 bar - Part 2: Specific functional requirements for polyethylene (MOP up to and including 10 bar)
EN 12007-3:2015	Gas infrastructure - Pipelines for maximum operating pressure up to and including 16 bar - Part 3: Specific functional requirements for steel
EN 12007-4:2012	Gas infrastructure - Pipelines for maximum operating pressure up to and including 16 bar - Part 4: Specific functional requirements for renovation
EN 12007-5:2014	Gas infrastructure - Pipelines for maximum operating pressure up to and including 16 bar - Part 5: Service lines - Specific functional requirements
EN 12269-1:2000	Determination of the bond behaviour between reinforcing steel and autoclaved aerated concrete by the "beam test" - Part 1: Short term test
EN 12269-2:2010	Determination of the bond behaviour between reinforcing steel and autoclaved aerated concrete by the beam test - Part 2: Long term test
EN 12385-1:2002	Steel wire ropes - Safety - Part 1: General requirements
EN 12385-2:2002	Steel wire ropes - Safety - Part 2: Definitions, designation and classification
EN 12385-3:2004	Steel wire ropes - Safety - Part 3: Information for use and maintenance
EN 12385-4:2002	Steel wire ropes - Safety - Part 4: Stranded ropes for general lifting applications
EN 12385-5:2002	Steel wire ropes - Safety - Part 5: Stranded ropes for lifts
EN 12385-6:2004	Steel wire ropes - Safety - Part 6: Stranded ropes for mine shafts
EN 12385-7:2002	Steel wire ropes - Safety - Part 7: Locked coil ropes for mine shafts
EN 12385-8:2002	Steel wire ropes - Safety - Part 8: Stranded hauling and carrying-hauling ropes for cableway installations designed to carry persons
EN 12385-9:2002	Steel wire ropes - Safety - Part 9: Locked coil carrying ropes for cableway installations designed to carry persons
EN 12385-10:2003	Steel wire ropes - Safety - Part 10: Spiral ropes for general structural applications
EN 12513:2011	Founding - Abrasion resistant cast irons
EN 12536:2000	Welding consumables - Rods for gas welding of non alloy and creep-resisting steels - Classification
EN 13262:2004	Railway applications - Wheelsets and bogies - Wheels - Product requirement
EN 13411-1:2002	Terminations for steel wire ropes - Safety - Part 1: Thimbles for steel wire rope slings
EN 13411-2:2001	Terminations for steel wire ropes - Safety - Part 2: Splicing of eyes for wire rope slings
EN 13411-3:2004	Terminations for steel wire ropes - Safety - Part 3: Ferrules and ferrule-securing
EN 13411-4:2011	Terminations for steel wire ropes - Safety - Part 4: Metal and resin socketing
EN 13411-5:2003	Terminations for steel wire ropes - Safety - Part 5: U-bolt wire rope grips
EN 13411-6:2004	Terminations for steel wire ropes - Safety - Part 6: Asymmetric wedge socket
EN 13674-1:2011	Railway applications - Track - Rail - Part 1: Vignole railway rails 46 kg/m and above
EN 13674-2:2006	Railway applications - Track - Rail - Part 2: Switch and crossing rails used in conjunction with Vignole railway rails 46 kg/m and above
EN 13674-3:2006	Railway applications - Track - Rail - Part 3: Check rails
EN 13674-4:2006	Railway applications - Track - Rail - Part 4: Vignole railway rails from 27 kg/m to, but excluding 46 kg/m
EN 13835:2012	Founding - Austenitic cast irons
EN 14195:2014	Metal framing components for gypsum board systems - Definitions, requirements and test methods
EN 14783:2013	Fully supported metal sheet and strip for roofing, external cladding and internal lining - Product specification and requirements
EN 10202:20022	Cold-reduced tinplate and electrolytic chromium/chromium oxide coated steel

EN 50446:2006	Straight thermocouple assembly with metal or ceramic protection tube and accessories
EN ISO 636:2008	Welding consumables - Rods, wires and deposits for tungsten inert gas welding of non-alloy and fine-grain steels - Classification
EN ISO 683-17:2014	Heat-treated steels, alloy steels and free-cutting steels - Part 17: Ball and roller bearing steels
EN ISO 1127:1996	Stainless steel tubes - Dimensions, tolerances and conventional masses per unit length
EN ISO 3183:2012	Petroleum and natural gas industries - Steel pipe for pipeline transportation systems
EN ISO 3580:2011	Welding consumables - Covered electrodes for manual metal arc welding of creep-resisting steels - Classification
EN ISO 3581:2012	Welding consumables - Covered electrodes for manual metal arc welding of stainless and heat-resisting steels - Classification
EN ISO 3766:2003	Construction drawings - Simplified representation of concrete reinforcement
EN ISO 4957:1999	Tool steels
EN ISO 7153-1:2000	Surgical instruments - Metallic Materials - Part 1: Stainless steel
EN ISO 9445-1:2010	Continuously cold-rolled stainless steel - Tolerances on dimensions and form - Part 1: Narrow strip and cut lengths
EN ISO 9445-2:2010	Continuously cold-rolled stainless steel - Tolerances on dimensions and form - Part 2: Wide strip and plate/sheet
EN ISO 10893-1:2011	Non-destructive testing of steel tubes - Part 1: Automated electromagnetic testing of seamless and welded (except submerged arc-welded) steel tubes for the verification of hydraulic leaktightness
EN ISO 10893-2:2011	Non-destructive testing of steel tubes - Part 2: Automated eddy current testing of seamless and welded (except submerged arc-welded) steel tubes for the detection of imperfections
EN ISO 10893-3:2011	Non-destructive testing of steel tubes - Part 3: Automated full peripheral flux leakage testing of seamless and welded (except submerged arc-welded) ferromagnetic steel tubes for the detection of longitudinal and/or transverse imperfections
EN ISO 10893-4:2011	Non-destructive testing of steel tubes - Part 4: Liquid penetrant inspection of seamless and welded steel tubes for the detection of surface imperfections
EN ISO 10893-5:2011	Non-destructive testing of steel tubes - Part 5: Magnetic particle inspection of seamless and welded ferromagnetic steel tubes for the detection of surface imperfections
EN ISO 10893-6:2011	Non-destructive testing of steel tubes - Part 6: Radiographic testing of the weld seam of welded steel tubes for the detection of imperfections
EN ISO 10893-7:2011	Non-destructive testing of steel tubes - Part 7: Digital radiographic testing of the weld seam of welded steel tubes for the detection of imperfections
EN ISO 10893-8:2011	Non-destructive testing of steel tubes - Part 8: Automated ultrasonic testing of seamless and welded steel tubes for the detection of laminar imperfections
EN ISO 10893-9:2011	Non-destructive testing of steel tubes - Part 9: Automated ultrasonic testing for the detection of laminar imperfections in strip/plate used for the manufacture of welded steel tubes
EN ISO 10893-10:2011	Non-destructive testing of steel tubes - Part 10: Automated full peripheral ultrasonic testing of seamless and welded (except submerged arc-welded) steel tubes for the detection of longitudinal and/or transverse imperfections
EN ISO 10893-11:2011	Non-destructive testing of steel tubes - Part 11: Automated ultrasonic testing of the weld seam of welded steel tubes for the detection of longitudinal and/or transverse imperfections
EN ISO 10893-12:2011	Non-destructive testing of steel tubes - Part 12: Automated full peripheral ultrasonic thickness testing of seamless and welded (except submerged arc-welded) steel tubes
EN ISO 11960:2014	Petroleum and natural gas industries - Steel pipes for use as casing or tubing for wells
EN ISO 14171:2010	Welding consumables - Solid wire electrodes, tubular cored electrodes and electrode/flux combinations for submerged arc welding of non alloy and fine grain steels - Classification
EN ISO 14341:2011	Welding consumables - Wire electrodes and weld deposits for gas shielded metal arc welding of non alloy and fine grain steels - Classification
EN ISO 14343:2009	Welding consumables - Wire electrodes, strip electrodes, wires and rods for arc welding of stainless and heat resisting steels - Classification
EN ISO 16120-1:2011	Non-alloy steel wire rod for conversion to wire - Part 1: General requirements
EN ISO 16120-2:2011	Non-alloy steel wire rod for conversion to wire - Part 2: Specific requirements for general-purpose wire rod
EN ISO 16120-3:2011	Non-alloy steel wire rod for conversion to wire - Part 3: Specific requirements for rimmed and rimmed substitute, low- carbon steel wire rod
EN ISO 16120-4:2011	Non-alloy steel wire rod for conversion to wire - Part 4: Specific requirements for wire rod for special applications
EN ISO 17632:2008	Welding consumables - Tubular cored electrodes for gas shielded and non-gas shielded metal arc welding of non- alloy and fine grain steels - Classification
EN ISO 21952:2012	Welding consumables - Wire electrodes, wires, rods and deposits for gas shielded arc welding of creep-resisting steels - Classification
BS EN ISO 17660-1:2006	Welding. Welding of reinforcing steel - Load-bearing welded joints
BS EN ISO 17660-2:2006	Welding. Welding of reinforcing steel - Non load-bearing welded joints

Note. Blue-shaded cells correspond to standards on testing
Source: Handbook of Comparative World Steel Standards. 5th Edition (Bringas, 2016)

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Annex 3. Data availability & correspondence matrixes

PART A: Ex-ante data availability results

The ex-ante data availability file consist information on data sources in the following stages of the product life:

- Iron ore
- Iron
- Ferroalloys
- Crude steel
- Finished steel goods
- Scrap and Recycling
- Trade

In additional the review of existing literature is included that targets future scenarios and LCA.



Steel data
availability.xlsx

PART B: Correlation between end-use sector, products' application (EN 10027) and consolidated product classification

This annex examines the correlation between the end-use sector demanding steel intermediate product, classified according to European Steel in Figures (EUROFER, 2023), and the corresponding categories of end-use sectors as defined by Dworak et al. (2022). The analysis also draws on the standard EN 10027 categories for steel applications and the consolidated product classification. While the list of products is not exhaustive, it provides a comprehensive overview of the diverse applications of intermediate steel products within each end-use sector.

Table 76. Correlation between end-use sector, standard EN 10027 and consolidated product classification

EUROFER (end-use sector)	Dworak et al.	EN10027	Consolidated product classification
Construction Steel in Figures 2023: all qualities = 37% of total steel demand [NACE codes Rev.2: 41, 42, 43, 25.1, 25.2]	Buildings	S Structural steel	Hot rolled plate (Quarto plate)
			Other hot rolled bars (merchant bars)
			Rod and wire rod
		L Steels for line pipe	Hot rolled plate (Quarto plate)
			Hot rolled coil
			Welded tubes
			Seamless tube
			Cold rolled coil
			Galvanized coil
			Organically coated and other coated products
		B Steels for reinforcing concrete	Bars for reinforcement of concrete
		Y Steels for prestressing concrete	Rod and wire rod
	Infrastructure	NA NA	Heavy sections and mining frames
			Light sections
			Stainless steel
			Casted iron products
		S Structural steel	Hot rolled plate (Quarto plate)
			Other hot rolled bars (merchant bars)
			Rod and wire rod
		L Steels for line pipe	Hot rolled plate (Quarto plate)
			Hot rolled coil
			Welded tubes
			Seamless tube
			Cold rolled coil
			Galvanized coil
			Organically coated and other coated products
		B Steels for reinforcing concrete	Bars for reinforcement of concrete
		Y Steels for prestressing concrete	Rod and wire rod
		R Steel for or in the form of rails	Railway track material and other long products
Automotive 2,784,000	Cars	NA NA	Light sections
			Heavy sections and mining frames
			Stainless steel
			Casted iron products
		L Steels for line pipe	Hot rolled plate (Quarto plate)

t/month = 16% [NACE codes Rev.2: 29]	Trucks			Hot rolled coil
				Seamless tube
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		E	Engineering steels	Hot rolled plate (Quarto plate)
				Hot rolled coil
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
				Other hot rolled bars (merchant bars)
				Rod and wire rod
		D	Flat products for cold forming	Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		H	Flat products of high strength for cold forming	Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		NA	NA	Stainless steel
				Casted iron products
	Trucks	G	Steel casting	
				Hot rolled plate (Quarto plate)
				Hot rolled coil
				Welded tubes
				Seamless tube
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
				Hot rolled plate (Quarto plate)
				Hot rolled coil
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
				Other hot rolled bars (merchant bars)
				Rod and wire rod
		D	Flat products for cold forming	Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		H	Flat products of high strength for cold forming	Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		NA	NA	Stainless steel
				Casted iron products
Other transport 2% [NACE code Rev.2: 30, 30.1,	Other Transport	G	Steel casting	
		E	Engineering steels	Hot rolled plate (Quarto plate)
				Hot rolled coil

30.2, 30.91]: shipbuilding and other transportation equipment				Other hot rolled bars (merchant bars)
				Rod and wire rod
		L	Steels for line pipe	Hot rolled plate (Quarto plate)
				Hot rolled coil
				Welded tubes
				Seamless tube
		NA	NA	Casted iron products
				Stainless steel
Mechanical Engineering 15% [NACE codes Rev.2: 28, 27.1 25.3, 26]	Mechanical Engineering	G	Steel casting	Casted steel products
				Casted iron products
		P	Steels for pressure purpose	Hot rolled plate (Quarto plate)
				Hot rolled coil
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		L	Steels for line pipe	Hot rolled plate (Quarto plate)
				Hot rolled coil
				Welded tubes
				Seamless tube
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		E	Engineering steels	Hot rolled plate (Quarto plate)
				Hot rolled coil
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
				Other hot rolled bars (merchant bars)
		R	Steel for or in the form of rails	Railway track material and other long products
				Stainless steel
		NA	NA	Stainless steel
	Electrical Engineering	L	Steels for line pipe	Hot rolled plate (Quarto plate)
				Hot rolled coil
				Welded tubes
				Seamless tube
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		E	Engineering steels	Hot rolled plate (Quarto plate)
				Hot rolled coil
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		M	Electrical steel	Other hot rolled bars (merchant bars)
				Rod and wire rod
				Electrical sheet

		NA	NA	Stainless steel
				Casted iron products
Domestic Appliances 2% [NACE code Rev.2: 27.51]	Domestic appliances	M	Electrical steel	Electrical sheet
		E	Engineering steels	Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		NA	NA	Stainless steel
Metalware 14% [NACE code Rev.2: 25 Manufacture of fabricated metal products excluding 25.1-25.2-25.3]; Miscellaneous 2%	Other Metal Goods	G	Steel casting	Casted steel products
		E	Engineering steels	Hot rolled plate (Quarto plate)
				Hot rolled coil
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
				Other hot rolled bars (merchant bars)
				Rod and wire rod
		L	Steels for line pipe	Hot rolled plate (Quarto plate)
				Hot rolled coil
				Welded tubes
				Seamless tube
				Cold rolled coil
				Galvanized coil
				Organically coated and other coated products
		NA	NA	Stainless steel
				Casted iron products
	Packaging	T	Tin mill products (steel products for packaging)	Tinplate and tinned products
		NA	NA	Stainless steel

Source: Fraunhofer elaboration

PART C: Correspondence between finished steel product categories and CN codes

Table 77. Correspondence between product categories and CN codes

Consolidated product categories	CN Code			
Hot rolled flat products				
Hot rolled plate (Quarto plate)	7208	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, hot-rolled, not clad, plated or coated:
	7208	non-alloy	72085120	Other, not in coils, not further worked than hot-rolled: Of a thickness exceeding 10 mm: Of a thickness exceeding 15 mm
	7208	non-alloy	72085191	Other, not in coils, not further worked than hot-rolled: Of a thickness exceeding 10 mm but not exceeding 15 mm, of a width of: 2 050 mm or more
	7208	non-alloy	72085198	Other, not in coils, not further worked than hot-rolled: Of a thickness exceeding 10 mm but not exceeding 15 mm, of a width of: Less than 2 050 mm
	7208	non-alloy	72085220	/*not listed*/
	7208	non-alloy	72085291	Other, not in coils, not further worked than hot-rolled: Of a thickness of 4,75 mm or more but not exceeding 10 mm: Other, of a width of: 2 050 mm or more
	7208	non-alloy	72089000	Other:
	7208	non-alloy	72089020	Other: Perforated
	7208	non-alloy	72089080	Other: Other
	7208	non-alloy	72084000	Not in coils, not further worked than hot-rolled, with patterns in relief
	7208	non-alloy	72085210	Other, not in coils, not further worked than hot-rolled: Of a thickness of 4,75 mm or more but not exceeding 10 mm: Rolled on four faces or in a closed box pass, of a width not exceeding 1 250 mm
	7208	non-alloy	72085280	/*not listed*/
	7208	non-alloy	72085299	Other, not in coils, not further worked than hot-rolled: Of a thickness of 4,75 mm or more but not exceeding 10 mm: Other, of a width of: Less than 2 050 mm
	7208	non-alloy	72085300	Other, not in coils, not further worked than hot-rolled: Of a thickness of 3 mm or more but less than 4,75 mm:
	7208	non-alloy	72085310	Other, not in coils, not further worked than hot-rolled: Of a thickness of 3 mm or more but less than 4,75 mm: Rolled on four faces or in a closed box pass, of a width not exceeding 1 250 mm and of a thickness of 4 mm or more
	7208	non-alloy	72085390	Other, not in coils, not further worked than hot-rolled: Of a thickness of 3 mm or more but less than 4,75 mm: Other
	7208	non-alloy	72085400	Other, not in coils, not further worked than hot-rolled: Of a thickness of less than 3 mm
	7210	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, clad, plated or coated:
	7210	non-alloy	72109030	Other: Clad
	7225	other alloy		Flat-rolled products of other alloy steel, of a width of 600 mm or more:
	7225	other alloy	72254040	Other, not further worked than hot-rolled, not in coils: Other: Of a thickness exceeding 10 mm
	7225	other alloy	72254060	Other, not further worked than hot-rolled, not in coils: Other: Of a thickness of 4,75 mm or more but not exceeding 10 mm
	7225	other alloy	72254012	Other, not further worked than hot-rolled, not in coils: Of tool steel
	7225	other alloy	72254015	Other, not further worked than hot-rolled, not in coils: Of high-speed steel
	7225	other alloy	72254090	Other, not further worked than hot-rolled, not in coils: Other: Of a thickness of less than 4,75 mm
	7211	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of less than 600 mm, not clad, plated or coated
	7211	non-alloy	72111300	Not further worked than hot-rolled: Rolled on four faces or in a closed box pass, of a width exceeding 150 mm and a thickness of not less than 4 mm, not in coils and without patterns in relief
Electrical sheet	7225	other alloy		Flat-rolled products of other alloy steel, of a width of 600 mm or more:
	7225	other alloy	72251100	Of silicon-electrical steel: Grain-oriented

	7226	other alloy	72251990	Of silicon-electrical steel: Other: Cold-rolled
	7226	other alloy		Flat-rolled products of other alloy steel, of a width of less than 600 mm:
	7226	other alloy	72261100	Of silicon-electrical steel: Grain-oriented
	7226	other alloy	72261980	Of silicon-electrical steel: Other: Other
	7209	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, cold-rolled (cold-reduced), not clad, plated or coated:
	7209	non-alloy	72091610	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness exceeding 1 mm but less than 3 mm: 'Electrical'
	7209	non-alloy	72091710	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness of 0,5 mm or more but not exceeding 1 mm: 'Electrical'
	7209	non-alloy	72091810	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness of less than 0,5 mm: 'Electrical'
	7209	non-alloy	72092610	Not in coils, not further worked than cold-rolled (cold-reduced): Of a thickness exceeding 1 mm but less than 3 mm: 'Electrical'
	7209	non-alloy	72092710	Not in coils, not further worked than cold-rolled (cold-reduced): Of a thickness of 0,5 mm or more but not exceeding 1 mm: 'Electrical'
	7209	non-alloy	72092810	Not in coils, not further worked than cold-rolled (cold-reduced): Of a thickness of less than 0,5 mm: 'Electrical'
	7211	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of less than 600 mm, not clad, plated or coated:
	7211	non-alloy	72112320	Not further worked than cold-rolled (cold-reduced): Containing by weight less than 0,25 % of carbon: 'Electrical'
Hot rolled coil	7208	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, hot-rolled, not clad, plated or coated:
	7208	non-alloy	72081000	In coils, not further worked than hot-rolled, with patterns in relief
	7208	non-alloy	72082500	Other, in coils, not further worked than hot-rolled, pickled: Of a thickness of 4,75 mm or more
	7208	non-alloy	72082600	Other, in coils, not further worked than hot-rolled, pickled: Of a thickness of 3 mm or more but less than 4,75 mm
	7208	non-alloy	72082700	Other, in coils, not further worked than hot-rolled, pickled: Of a thickness of less than 3 mm
	7208	non-alloy	72083600	Other, in coils, not further worked than hot-rolled: Of a thickness exceeding 10 mm
	7208	non-alloy	72083700	Other, in coils, not further worked than hot-rolled: Of a thickness of 4,75 mm or more but not exceeding 10 mm
	7208	non-alloy	72083800	Other, in coils, not further worked than hot-rolled: Of a thickness of 3 mm or more but less than 4,75 mm
	7208	non-alloy	72083900	Other, in coils, not further worked than hot-rolled: Of a thickness of less than 3 mm
	7211	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of less than 600 mm, not clad, plated or coated:
	7211	non-alloy	72111400	Not further worked than hot-rolled: Other, of a thickness of 4,75 mm or more
	7211	non-alloy	72111900	Not further worked than hot-rolled: Other
	7212	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of less than 600 mm, clad, plated or coated:
	7212	non-alloy	72126000	Clad
	7225	other alloy		Flat-rolled products of other alloy steel, of a width of 600 mm or more:
	7225	other alloy	72251910	Of silicon-electrical steel: Other: Hot-rolled
	7225	other alloy	72253010	Other, not further worked than hot-rolled, in coils: Of tool steel
	7225	other alloy	72253030	Other, not further worked than hot-rolled, in coils: Of high-speed steel
	7225	other alloy	72253090	Other, not further worked than hot-rolled, in coils: Other
	7226	other alloy		Flat-rolled products of other alloy steel, of a width of less than 600 mm:
	7226	other alloy	72261910	Of silicon-electrical steel: Other: Not further worked than hot-rolled
	7226	other alloy	72269120	Other: Not further worked than hot-rolled: Of tool steel
	7226	other alloy	72269191	Other: Not further worked than hot-rolled: Other: Of a thickness of 4,75 mm or more
	7226	other alloy	72269199	Other: Not further worked than hot-rolled: Other: Of a thickness of less than 4,75 mm
Cold rolled coil	7209	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, cold-rolled (cold-reduced), not clad, plated or coated:

	7209	non-alloy	72091500	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness of 3 mm or more
	7209	non-alloy	72091690	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness exceeding 1 mm but less than 3 mm: Other
	7209	non-alloy	72091790	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness of 0,5 mm or more but not exceeding 1 mm: Other
	7209	non-alloy	72091891	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness of less than 0,5 mm: Other: Of a thickness of 0,35 mm or more but less than 0,5 mm
	7209	non-alloy	72092500	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness of 3 mm or more
	7209	non-alloy	72092690	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness exceeding 1 mm but less than 3 mm: Other
	7209	non-alloy	72092790	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness of 0,5 mm or more but not exceeding 1 mm: Other
	7209	non-alloy	72092890	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness of less than 0,5 mm: Other
	7209	non-alloy	72099000	Other:
	7209	non-alloy	72099020	Other: Perforated
	7209	non-alloy	72099080	Other: Other
	7211	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of less than 600 mm, not clad, plated or coated:
	7211	non-alloy	72112330	Not further worked than cold-rolled (cold-reduced): Containing by weight less than 0,25 % of carbon: Other: Of a thickness of 0,35 mm or more
	7211	non-alloy	72112900	Not further worked than cold-rolled (cold-reduced): Other
	7211	non-alloy	72119000	Other:
	7211	non-alloy	72119020	Other: Perforated
	7211	non-alloy	72119080	Other: Other
	7225	other alloy		Flat-rolled products of other alloy steel, of a width of 600 mm or more:
	7225	other alloy	72252000	/*not listed*/
	7225	other alloy	72255000	Other, not further worked than cold-rolled (cold-reduced):
	7225	other alloy	72255020	Other, not further worked than cold-rolled (cold-reduced): Of high-speed steel
	7225	other alloy	72255080	Other, not further worked than cold-rolled (cold-reduced): Other
	7226	other alloy		Flat-rolled products of other alloy steel, of a width of less than 600 mm:
	7226	other alloy	72262000	Of high-speed steel
	7226	other alloy	72269200	Other: Not further worked than cold-rolled (cold-reduced)
	7226	other alloy	72269910	Flat-rolled products of alloy steel other than stainless, of a width of < 600 mm, hot-rolled or cold-rolled and electrolytically plated or coated with zinc (excluding products of high-speed steel or silicon-electrical steel)
Tinplate and tinned products	7209	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, cold-rolled (cold-reduced), not clad, plated or coated:
	7209	non-alloy	72091899	In coils, not further worked than cold-rolled (cold-reduced): Of a thickness of less than 0,5 mm: Other: Of a thickness of less than 0,35 mm
	7210	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, clad, plated or coated:
	7210	non-alloy	72101100	Plated or coated with tin: Of a thickness of 0,5 mm or more
	7210	non-alloy	72101220	Plated or coated with tin: Of a thickness of less than 0,5 mm: Tinplate
	7210	non-alloy	72101280	Plated or coated with tin: Of a thickness of less than 0,5 mm: Other
	7210	non-alloy	72105000	Plated or coated with chromium oxides or with chromium and chromium oxides
	7210	non-alloy	72107010	Painted, varnished or coated with plastics: Tinplate, varnished; products, plated or coated with chromium oxides or with chromium and chromium oxides, varnished
	7210	non-alloy	72109040	Other: Tinned and printed
	7211	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of less than 600 mm, not clad, plated or coated:
	7211	non-alloy	72112380	Not further worked than cold-rolled (cold-reduced): Containing by weight less than 0,25 % of carbon: Other: Of a thickness of less than 0,35 mm

	7212	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of less than 600 mm, clad, plated or coated:
	7212	non-alloy	72121010	Plated or coated with tin: Tinplate, not further worked than surface-treated
	7212	non-alloy	72121090	Plated or coated with tin: Other
	7212	non-alloy	72124020	Painted, varnished or coated with plastics: Tinplate, not further worked than varnished; products, plated or coated with chromium oxides or with chromium and chromium oxides, varnished
	7212	non-alloy	72125020	Otherwise plated or coated: Plated or coated with chromium oxides or with chromium and chromium oxides
Galvanized coil	7210	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, clad, plated or coated:
	7210	non-alloy	72102000	Plated or coated with lead, including terne-plate
	7210	non-alloy	72104100	Otherwise plated or coated with zinc: Corrugated
	7210	non-alloy	72104900	Otherwise plated or coated with zinc: Other
	7210	non-alloy	72106100	Plated or coated with aluminium: Plated or coated with aluminium-zinc alloys
	7210	non-alloy	72106900	Plated or coated with aluminium: Other
	7210	non-alloy	72109080	Other: Other
	7210	non-alloy	72103000	Electrolytically plated or coated with zinc
	7212	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of less than 600 mm, clad, plated or coated:
	7212	non-alloy	72123000	Otherwise plated or coated with zinc
	7212	non-alloy	72125061	Otherwise plated or coated: Plated or coated with aluminium: Plated or coated with aluminium-zinc alloys
	7212	non-alloy	72125030	Otherwise plated or coated: Plated or coated with chromium or nickel
	7212	non-alloy	72125040	Otherwise plated or coated: Plated or coated with copper
	7212	non-alloy	72125069	Otherwise plated or coated: Plated or coated with aluminium: Other
	7212	non-alloy	72125090	Otherwise plated or coated: Other
	7212	non-alloy	72122000	Electrolytically plated or coated with zinc
	7225	other alloy		Flat-rolled products of other alloy steel, of a width of 600 mm or more:
	7225	other alloy	72259200	Other: Otherwise plated or coated with zinc
	7225	other alloy	72259900	Other: Other
	7225	other alloy	72259100	Other: Electrolytically plated or coated with zinc
	7226	other alloy		Flat-rolled products of other alloy steel, of a width of less than 600 mm:
	7226	other alloy	72269400	/*not listed*/
	7226	other alloy	72269900	Other: Other
	7226	other alloy	72269930	Other: Other: Otherwise plated or coated with zinc
	7226	other alloy	72269970	Other: Other: Other
	7226	other alloy	72269300	/*not listed*/
Organically coated and other coated products	7210	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, clad, plated or coated:
	7210	non-alloy	72107080	Painted, varnished or coated with plastics: Other
	7212	non-alloy		Flat-rolled products of iron or non-alloy steel, of a width of less than 600 mm, clad, plated or coated:
	7212	non-alloy	72124080	Painted, varnished or coated with plastics: Other
Hot rolled long products				
Railway track material and other long products	7302	non-alloy		Railway or tramway track construction material of iron or steel, the following: rails, check-rails and rack rails, switch blades, crossing frogs, point rods and other crossing pieces, sleepers (cross-ties), fish-plates, chairs, chair wedges, sole plates (base plates), rail clips, bedplates, ties and other material specialised for jointing or fixing rails:

	7302	non-alloy	73021010	Current-conducting rails of iron or steel, with parts of non-ferrous metal, for railway or tramway track (excl. check-rails)
	7302	non-alloy	73021021	/*not listed*/
	7302	non-alloy	73021022	Rails: Other: New: Vignole rails: Of a weight per metre of 36 kg or more
	7302	non-alloy	73021023	/*not listed*/
	7302	non-alloy	73021028	Rails: Other: New: Vignole rails: Of a weight per metre of less than 36 kg
	7302	non-alloy	73021029	/*not listed*/
	7302	non-alloy	73021040	Rails: Other: New: Grooved rails
	7302	non-alloy	73021050	Rails: Other: New: Other
	7302	non-alloy	73021090	Rails of iron or steel, for railway or tramway track, used (excl. current-conducting rails with parts of non-ferrous metal)
	7302	non-alloy	73023000	Switch blades, crossing frogs, point rods and other crossing pieces, for railway or tramway track, of iron or steel
	7302	non-alloy	73024000	Fish-plates and sole plates
	7302	non-alloy	73029000	Sleepers "cross-ties", check-rails, rack rails, chairs, chair wedges, rail clips, bedplates and ties and other specialised material for the jointing or fixing of railway or tramway track, of iron or steel (excl. rails, switch blades, crossing frogs, point rods and other crossing pieces, and fish-plates and sole plates)
Heavy sections and mining frames	7216	non-alloy		Angles, shapes and sections of iron or non-alloy steel:
	7216	non-alloy	72163110	U, I or H sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of 80 mm or more: U sections: Of a height exceeding 220 mm
	7216	non-alloy	72163190	U, I or H sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of 80 mm or more: U sections: Of a height of 80 mm or more but not exceeding 220 mm
	7216	non-alloy	72163211	U, I or H sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of 80 mm or more: I sections: Of a height of 80 mm or more but not exceeding 220 mm: With parallel flange faces
	7216	non-alloy	72163219	U, I or H sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of 80 mm or more: I sections: Of a height of 80 mm or more but not exceeding 220 mm: Other
	7216	non-alloy	72163291	U, I or H sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of 80 mm or more: I sections: Of a height exceeding 220 mm: With parallel flange faces
	7216	non-alloy	72163299	U, I or H sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of 80 mm or more: I sections: Of a height exceeding 220 mm: Other
	7216	non-alloy	72163310	U, I or H sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of 80 mm or more: H sections: Of a height of 80 mm or more but not exceeding 180 mm
	7216	non-alloy	72163390	U, I or H sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of 80 mm or more: H sections: Of a height exceeding 180 mm
	7216	non-alloy	72166110	Cold-formed sections, obtained from flat products, of non-alloy steel, not coated
	7301	non-alloy		Sheet piling of iron or steel, whether or not drilled, punched or made from assembled elements; welded angles, shapes and sections, of iron or steel:
	7301	non-alloy	73011000	Sheet piling
	7301	non-alloy	73012000	Angles, shapes and sections
Light sections				
Other hot rolled bars (merchant bars)	7214	non-alloy		Other bars and rods of iron or non-alloy steel, not further worked than forged, hot-rolled, hot-drawn or hot-extruded, but including those twisted after rolling:
	7214	non-alloy	72143000	Other, of free-cutting steel
	7214	non-alloy	72149110	Other: Of rectangular (other than square) cross-section: Containing by weight less than 0,25 % of carbon
	7214	non-alloy	72149190	Other: Of rectangular (other than square) cross-section: Containing by weight 0,25 % or more of carbon
	7214	non-alloy	72149931	Other: Other: Containing by weight less than 0,25 % of carbon: Other, of circular cross-section measuring in diameter: 80 mm or more
	7214	non-alloy	72149939	Other: Other: Containing by weight less than 0,25 % of carbon: Other, of circular cross-section measuring in diameter: Less than 80 mm
	7214	non-alloy	72149950	Other: Other: Containing by weight less than 0,25 % of carbon: Other

	7214	non-alloy	72149971	Other: Other: Containing by weight 0,25 % or more of carbon: Of a circular cross-section measuring in diameter: 80 mm or more
	7214	non-alloy	72149979	Other: Other: Containing by weight 0,25 % or more of carbon: Of a circular cross-section measuring in diameter: Less than 80 mm
	7214	non-alloy	72149995	Other: Other: Containing by weight 0,25 % or more of carbon: Other
	7216	non-alloy		Angles, shapes and sections of iron or non-alloy steel:
	7216	non-alloy	72161000	U, I or H sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of less than 80 mm
	7216	non-alloy	72162100	L or T sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of less than 80 mm: L sections
	7216	non-alloy	72162200	L or T sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of less than 80 mm: T sections
	7216	non-alloy	72164010	L or T sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of 80 mm or more: L sections
	7216	non-alloy	72164090	L or T sections, not further worked than hot-rolled, hot-drawn or extruded, of a height of 80 mm or more: T sections
	7216	non-alloy	72165010	Other angles, shapes and sections, not further worked than hot-rolled, hot-drawn or extruded: With a cross-section which is capable of being enclosed in a square the side of which is 80 mm
	7216	non-alloy	72165091	Other angles, shapes and sections, not further worked than hot-rolled, hot-drawn or extruded: Other: Bulb flats
	7216	non-alloy	72165099	Other angles, shapes and sections, not further worked than hot-rolled, hot-drawn or extruded: Other: Other
	7216	non-alloy	72166900	Angles, shapes and sections, of iron or non-alloy steel, not further worked than cold-formed or cold-finished (e.g. by cold-drawing) (excluding profiled sheet)
	7216	non-alloy	72169110	Cold-profiled (ribbed) sheets, of non-alloy steel
	7216	non-alloy	72169180	Cold-formed sections, obtained from flat products, of non-alloy steel, coated with zinc
	7216	non-alloy	72169900	Other: Other
	7228	other alloy		Other bars and rods of other alloy steel; angles, shapes and sections, of other alloy steel; hollow drill bars and rods, of alloy or non-alloy steel:
	7228	other alloy	72281020	Bars and rods, of high-speed steel: Not further worked than hot-rolled, hot-drawn or extruded; hot-rolled, hot-drawn or extruded, not further worked than clad
	7228	other alloy	72281090	Bars and rods, of high-speed steel, not further worked than cold-formed or cold-finished, even further worked, or hot-formed and further worked (excluding forged, semi-finished or flat-rolled products, hot-rolled bars and rods in irregularly wound coils); bars and rods, of silico-manganese steel, only cold-formed or cold-finished or hot-rolled, including further worked (excluding hot-rolled, hot-drawn or extruded, just clad, semi-finished products, flat-rolled products, hot-rolled bars and rods in irregularly wound coils)
	7228	other alloy	72282010	Bars and rods, of silico-manganese steel: Of rectangular (other than square) cross-section, hot-rolled on four faces
	7228	other alloy	72282091	Bars and rods, of silico-manganese steel: Other: Not further worked than hot-rolled, hot-drawn or extruded; hot-rolled, hot-drawn or extruded, not further worked than clad
	7228	other alloy	72283020	Other bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Of tool steel
	7228	other alloy	72283041	Other bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Containing by weight 0,9 % or more but not more than 1,15 % of carbon, 0,5 % or more but not more than 2 % of chromium and, if present, not more than 0,5 % of molybdenum: Of circular cross-section of a diameter of 80 mm or more
	7228	other alloy	72283049	Other bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Containing by weight 0,9 % or more but not more than 1,15 % of carbon, 0,5 % or more but not more than 2 % of chromium and, if present, not more than 0,5 % of molybdenum: Other
	7228	other alloy	72283061	Other bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Other: Of circular cross-section, of a diameter of: 80 mm or more
	7228	other alloy	72283069	Other bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Other: Of circular cross-section, of a diameter of: Less than 80 mm
	7228	other alloy	72283070	Other bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Other: Of rectangular (other than square) cross-section, hot-rolled on four faces
	7228	other alloy	72283089	Other bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Other: Other
	7228	other alloy	72285020	Bars and rods of tool steel, only cold-formed or cold-finished (e.g. by cold-drawing) (excluding semi-finished products, flat-rolled products and hot-rolled bars and rods in irregularly wound coils)

	7228	other alloy	72285061	Bars and rods, of alloy steel, not further worked than cold-formed or cold-finished (e.g. by cold-drawing) (excluding stainless steel, high-speed steel, silico-manganese steel, alloy bearing steel, tool steel)
	7228	other alloy	72286080	Bars and rods, of alloy steel, cold-formed or cold-finished (e.g. by cold-drawing), painted, coated, clad or further worked (excluding stainless steel)
	7228	other alloy	72287090	Sections, of alloy steel other than stainless, cold-finished or cold-formed (e.g. by cold-drawing)
	7228	other alloy	72287010	Angles, shapes and sections: Not further worked than hot-rolled, hot-drawn or extruded
	7228	non-alloy		Other bars and rods of other alloy steel; angles, shapes and sections, of other alloy steel; hollow drill bars and rods, of alloy or non-alloy steel:
	7228	non-alloy	72288000	Hollow drill bars and rods
Bars for reinforcement of concrete	7214	non-alloy		Other bars and rods of iron or non-alloy steel, not further worked than forged, hot-rolled, hot-drawn or hot-extruded, but including those twisted after rolling:
	7214	non-alloy	72142000	Containing indentations, ribs, grooves or other deformations produced during the rolling process or twisted after rolling
	7214	non-alloy	72149910	Other: Other: Containing by weight less than 0,25 % of carbon: Of a type used for concrete reinforcement
	7213	non-alloy		Bars and rods, hot-rolled, in irregularly wound coils, of iron or non-alloy steel:
	7213	non-alloy	72131000	Containing indentations, ribs, grooves or other deformations produced during the rolling process
Rod and wire rod	7213	non-alloy		Bars and rods, hot-rolled, in irregularly wound coils, of iron or non-alloy steel:
	7213	non-alloy	72131000	Containing indentations, ribs, grooves or other deformations produced during the rolling process
	7213	non-alloy	72132000	Other, of free-cutting steel
	7213	non-alloy	72139110	Other: Of circular cross-section measuring less than 14 mm in diameter: Of a type used for concrete reinforcement
	7213	non-alloy	72139120	Other: Of circular cross-section measuring less than 14 mm in diameter: Of a type used for tyre cord
	7213	non-alloy	72139141	Other: Of circular cross-section measuring less than 14 mm in diameter: Other: Containing by weight 0,06 % or less of carbon
	7213	non-alloy	72139149	Other: Of circular cross-section measuring less than 14 mm in diameter: Other: Containing by weight more than 0,06 % but less than 0,25 % of carbon
	7213	non-alloy	72139170	Other: Of circular cross-section measuring less than 14 mm in diameter: Other: Containing by weight 0,25 % or more but not more than 0,75 % of carbon
	7213	non-alloy	72139190	Other: Of circular cross-section measuring less than 14 mm in diameter: Other: Containing by weight more than 0,75 % of carbon
	7213	non-alloy	72139910	Other: Other: Containing by weight less than 0,25 % of carbon
	7213	non-alloy	72139990	Other: Other: Containing by weight 0,25 % or more of carbon
	7217	non-alloy		Wire of iron or non-alloy steel
	7217	non-alloy	72171050	Iron or non-alloy steel wire containing < 0,25 % of carbon including crimping wire excluding stranded wire, barbed wire used for fencing - duplex wire - saw-tooth wire, insulated electric wire
	7217	non-alloy	72171010	Iron or non-alloy steel wire containing < 0,25 % of carbon including crimping wire excluding stranded wire, barbed wire used for fencing - duplex wire - saw-tooth wire, insulated electric wire
	7227	other alloy		Bars and rods, hot-rolled, in irregularly wound coils, of other alloy steel:
	7227	other alloy	72271000	Of high-speed steel
	7227	other alloy	72272000	Of silico-manganese steel
	7227	other alloy	72279010	Other: Containing by weight 0,0008 % or more of boron with any other element less than the minimum content referred to in note 1(f) to this chapter
	7227	other alloy	72279050	Other: Containing by weight 0,9 % or more but not more than 1,15 % of carbon, 0,5 % or more but not more than 2 % of chromium and, if present, not more than 0,5 % of molybdenum
	7227	other alloy	72279095	Other: Other
	7229	other alloy		Wire of alloy steel other than stainless, in coils (excl. bars and rods)
	7229	other alloy	72292000	Alloy steel wire (excluding stranded wire, barbed wire of a kind used for fencing, duplex wire, saw-tooth wire, insulated electric wire, of stainless steel)

Tubular Products				
Seamless tube	7304			Tubes, pipes and hollow profiles, seamless, of iron (other than cast iron) or steel
	7304		73041910	Line pipe of a kind used for oil or gas pipelines, seamless, of iron or steel, of an external diameter of $\leq 168,3$ mm (excl. products of stainless steel or of cast iron)
	7304		73041930	Line pipe of a kind used for oil or gas pipelines, seamless, of iron or steel, of an external diameter of $> 168,3$ mm but $\leq 406,4$ mm (excl. products of stainless steel or of cast iron)
	7304		73041990	Line pipe of a kind used for oil or gas pipelines, seamless, of iron or steel, of an external diameter of $> 406,4$ mm (excl. products of stainless steel or of cast iron)
	7304		73042300	Casing, tubing and drill pipe, of a kind used in the drilling for oil or gas, seamless, of steel other than stainless steel
	7304		73042400	Casing and tubing, seamless, of a kind used for drilling for oil or gas, of stainless steel
	7304		73042910	Casing and tubing of a kind used for drilling for oil or gas, seamless, of iron or steel, of an external diameter $\leq 168,3$ mm (excl. products of cast iron)
	7304		73042930	Casing and tubing of a kind used for drilling for oil or gas, seamless, of iron or steel, of an external diameter $> 168,3$ mm, but $\leq 406,4$ mm (excl. products of cast iron)
	7304		73042990	Casing and tubing of a kind used for drilling for oil or gas, seamless, of iron or steel, of an external diameter $> 406,4$ mm (excl. products of cast iron)
	7304		73043120	Precision tubes and pipes, of circular cross-section, cold-drawn or cold-rolled, seamless, of steel other than stainless steel
	7304		73043180	Tubes and pipes, of circular cross-section, cold-drawn or cold-rolled, seamless, of steel other than stainless steel (excluding precision tubes and pipes)
	7304		73043910	Tubes and pipes, of circular cross-section, hot-finished, seamless, of steel other than stainless steel (excluding line pipe of a kind used for oil or gas pipelines and casing, tubing and drill-pipe used for oil or gas drilling)
	7304		73043950	Threaded or threadable tubes "gas pipe", seamless, of iron or non-alloy steel (excl. of cast iron)
	7304		73043982	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of iron or non-alloy steel, of an external diameter of $\leq 168,3$ mm (excl. cold-drawn or cold-rolled, of cast iron, line pipe of a kind used for oil or gas pipelines, casing, tubing and drill pipe of a kind used in drilling for oil or gas and tubes, and gas pipes of subheading 73043950)
	7304		73043983	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of iron or non-alloy steel, of an external diameter of $> 168,3$ mm but $\leq 406,4$ mm (excl. cold-drawn or cold-rolled, of cast iron, line pipe of a kind used for oil or gas pipelines, casing, tubing and drill pipe of a kind used in drilling for oil or gas and tubes, and gas pipes of subheading 73043950)
	7304		73043988	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of iron or non-alloy steel, of an external diameter of $> 406,4$ mm (excl. cold-drawn or cold-rolled, of cast iron, line pipe of a kind used for oil or gas pipelines, casing, tubing and drill pipe of a kind used in drilling for oil or gas and tubes, and gas pipes of subheading 73043950)
	7304		73045110	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of alloy steel other than stainless, cold-drawn or cold-rolled "cold-reduced", straight and of uniform wall-thickness, containing by weight $\geq 0,9\%$ but $\leq 1,15\%$ carbon and $\geq 0,5\%$ but $\leq 2\%$ chromium, whether or not containing by weight $\leq 0,5\%$ molybdenum (excl. tubes, pipes and hollow profiles of subheadings 730419 to 730429)
	7304		73045181	Precision tubes, seamless, of circular cross-section, of alloy steel other than stainless, cold-drawn or cold-rolled "cold-reduced" (excl. line pipe of a kind used for oil or gas pipelines, casing and tubing of a kind used for drilling for oil and tubes, and pipes and hollow profiles, straight and of uniform wall-thickness, containing by weight $\geq 0,9\%$ but $\leq 1,15\%$ carbon and $\geq 0,5\%$ but $\leq 2\%$ chrome, whether or not containing by weight $\leq 0,5\%$ molybdenum)
	7304		73045189	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of alloy steel other than stainless, not cold-drawn or cold-rolled "cold-reduced" (excl. line pipe of a kind used for oil or gas pipelines, casing and tubing of a kind used for drilling for oil, precision tubes, and , pipes and hollow profiles, straight and of uniform wall-thickness, containing by weight $\geq 0,9\%$ but $\leq 1,15\%$ carbon and $\geq 0,5\%$ but $\leq 2\%$ chrome, whether or not containing by weight $\leq 0,5\%$ molybdenum)
	7304		73045930	Tubes, pipes and hollow profiles of alloy steel (excl. stainless), seamless, of circular cross-section (not cold-drawn or cold-rolled), straight and of uniform wall-thickness, containing by weight $\geq 0,9\%$ but $\leq 1,15\%$ carbon and $\geq 0,5\%$ but $\leq 2\%$ chromium, whether or not containing by weight $\leq 0,5\%$ molybdenum (excl. tubes, pipes and hollow profiles of subheadings 730419 to 730429)
	7304		73045982	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of alloy steel other than stainless, of an external diameter of $\leq 168,3$ mm (excl. cold-drawn or cold-rolled, line pipe

				of a kind used for oil or gas pipelines, casing and tubing of a kind used for drilling for oil or gas, and products of subheading 73045930)
	7304		73045983	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of alloy steel other than stainless, of an external diameter of > 168,3 mm but ≤ 406,4 mm (excl. cold-drawn or cold-rolled, line pipe of a kind used for oil or gas pipelines, casing and tubing of a kind used for drilling for oil or gas, and products of subheading 73045930)
	7304		73045989	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of alloy steel other than stainless, of an external diameter of > 406,4 mm (excl. cold-drawn or cold-rolled, line pipe of a kind used for oil or gas pipelines, casing and tubing of a kind used for drilling for oil or gas, and products of subheading 73045930)
	7304		73049000	Tubes and pipes, of non-circular cross-section, seamless, and hollow profiles, seamless, of steel
Welded tube	7305			Iron or steel (excluding cast iron); tubes and pipes (e.g. welded, riveted or similarly closed), having circular cross-sections, external diameter of which exceeds 406.4mm, not seamless
	7305		73051100	Line pipe, of a kind used for oil or gas pipelines, longitudinally welded, of an external diameter > 406,4 mm, of steel
	7305		73051200	Line pipe of a kind used for oil or gas pipelines, having circular cross-sections and an external diameter of > 406,4 mm, of iron or steel, longitudinally arc welded (excl. products longitudinally submerged arc welded)
	7305		73051900	Line pipe, of a kind used for oil or gas pipelines, other than longitudinally welded, of an external diameter > 406,4 mm, of steel
	7305		73052000	Casing, of a kind used in drilling for oil or gas, welded, of an external diameter > 406,4 mm, of steel
	7305		73053100	Tubes and pipes, welded, of an external diameter > 406,4 mm, of steel (excluding line pipe of a kind used for oil or gas pipelines and casing used for oil or gas drilling)
	7305		73053900	Tubes and pipes having circular cross-sections and an external diameter of > 406,4 mm, of iron or steel, welded (excl. products longitudinally welded or of a kind used for oil or gas pipelines or of a kind used in drilling for oil or gas)
	7305		73059000	Tubes and pipes, riveted or similarly closed, of an external diameter > 406,4 mm, of steel (excluding line pipe for oil and gas pipelines, casing used for oil or gas drilling and welded tubes and pipes)
	7306			Iron or steel (excluding cast iron); tubes, pipes and hollow profiles (not seamless), n.e.c. in chapter 73
			73061900	Line pipe of a kind used for oil or gas pipelines, welded, of flat-rolled products of iron or steel, of an external diameter of ≤ 406,4 mm (excl. products of stainless steel or of cast iron)
	7306		73061910	Line pipe, of a kind used for oil or gas pipelines, longitudinally or spirally welded, of an external diameter ≤ 406,4 mm, of steel other than stainless steel
	7306		73062900	Casing and tubing, of a kind used in drilling for oil or gas, welded, of an external diameter ≤ 406,4 mm, of steel other than stainless steel
	7306		73063011	Precision tubes and pipes, of circular cross-section, welded, of an external diameter ≤ 406,4 mm, of steel other than stainless steel
	7306		73063012	Precision tubes, welded, of circular cross-section, of iron or non-alloy steel, cold-drawn or cold-rolled "cold-reduced"
	7306		73063018	Precision tubes, welded, of circular cross-section, of iron or non-alloy steel (excl. cold-drawn or cold-rolled)
	7306		73063041	Threaded or threadable tubes "gas pipe", welded, of circular cross-section, of iron or non-alloy steel, plated or coated with zinc
	7306		73063049	Threaded or threadable tubes "gas pipe", welded, of circular cross-section, of iron or non-alloy steel (excl. products plated or coated with zinc)
	7306		73063072	Other tubes, pipes and hollow profiles, welded, of circular cross-section, of iron or non-alloy steel, of an external diameter of ≤ 168,3 mm, plated or coated with zinc (excl. line pipe of a kind used for oil or gas pipelines or casing and tubing of a kind used in drilling for oil or gas)
	7306		73063077	Other tubes, pipes and hollow profiles, welded, of circular cross-section, of iron or non-alloy steel of an external diameter of ≤ 168,3 mm (excl. plated or coated with zinc and line pipe of a kind used for oil or gas pipelines, casing and tubing of a kind used in drilling for oil or gas, precision tubes and threaded or threadable tubes "gas pipe")
	7306		73063080	Tubes, pipes and hollow profiles, welded, having a circular cross-section, of iron or steel, of an external diameter of > 168,3 mm but ≤ 406,4 mm (excl. line pipe of a kind used for oil or gas pipelines or casing and tubing of a kind used in drilling for oil or gas, or precision steel tubes, electrical conduit tubes or threaded or threadable tubes "gas pipe")
	7306		73065021	Precision steel tubes, welded, of circular cross-section, of alloy steel other than stainless, cold-drawn or cold-rolled "cold-reduced"

	7306		73065029	Precision steel tubes, welded, of circular cross-section, of alloy steel other than stainless (excl. cold-drawn or cold-rolled)
	7306		73065080	Tubes, pipes and hollow profiles, welded, of circular cross-section, of alloy steel other than stainless (excl. tubes and pipes having internal and external circular cross-sections and an external diameter of > 406,4 mm, and line pipe of a kind used for oil or gas pipelines or casing and tubing of a kind used in drilling for oil or gas, and precision steel tubes)
	7306		73066192	Tubes and pipes, of square or rectangular cross-section, of a wall thickness ≤ 2 mm, hot- or cold-formed and welded, of steel other than stainless steel
	7306		73066199	Tubes and pipes, of square or rectangular cross-section, of a wall thickness > 2 mm, hot- or cold-formed and welded, of steel other than stainless steel
	7306		73066910	Tubes, pipes and hollow profiles, welded, of non-circular cross-section, of stainless steel (excl. tubes and pipes having internal and external circular cross-sections and an external diameter of > 406,4 mm, line pipe of a kind used for oil or gas pipelines or casing and tubing of a kind used in drilling for oil or gas, and tubes and pipes and hollow profiles of square or rectangular cross-section)
	7306		73066990	Tubes and pipes, of other non-circular cross-section than square or rectangular, hot- or cold-formed and welded, of steel other than stainless steel
	7306		73069000	Tubes and pipes, open seam, riveted or similarly closed, of steel (excluding line pipe for oil and gas pipelines, casing and tubing used for oil or gas drilling and other welded tubes and pipes)
Tube fittings	7307			Tubes or pipe fittings (e.g. couplings, elbows, sleeves), of iron or steel
	7307		73072100	Flanges, of steel (excluding cast fittings)
	7307		73072210	Sleeves, of stainless steel, threaded (excl. cast products)
	7307		73072290	Elbows and bends, of stainless steel, threaded (excl. cast products)
	7307			
	7307		73072310	Butt welding elbows and bends, for tubes or pipes, of steel (excluding cast fittings)
	7307		73072390	Butt welding tube or pipe fittings, other than elbows and bends, of steel (excluding cast fittings)
	7307		73072910	Threaded tube or pipe fittings of stainless steel (excl. cast products, flanges, elbows, bends and sleeves)
	7307		73072980	Tube or pipe fittings of stainless steel (excl. cast, threaded, butt welding fittings and flanges)
	7307		73079100	Flanges of iron or steel (excl. cast or stainless products)
	7307		73079210	Sleeves of iron or steel, threaded (excl. cast or of stainless steel)
Stainless steel				
	7307		73079290	Elbows and bends, of iron or steel, threaded (excl. cast or of stainless steel)
	7307		73079311	Butt welding elbows and bends, of iron or steel, with greatest external diameter ≤ 609,6 mm (excl. cast iron or stainless steel products)
	7307		73079319	Butt welding fittings of iron or steel, with greatest external diameter ≤ 609,6 mm (excl. cast iron or stainless steel products, elbows, bends and flanges)
	7307		73079391	Butt welding elbows and bends, of iron or steel, with greatest external diameter > 609,6 mm (excl. cast iron or stainless steel products)
	7307		73079399	Butt welding fittings of iron or steel, with greatest external diameter > 609,6 mm (excl. cast iron or stainless steel products, elbows, bends and flanges)
	7307		73079910	Threaded tube or pipe fittings, of iron or steel (excl. cast iron or stainless steel products, flanges, elbows, bends and sleeves)
	7307		73079980	Tube or pipe fittings, of iron or steel (excl. of cast iron or stainless steel, threaded, butt welding fittings, and flanges)
Stainless steel products	7219	stainless		Flat-rolled products of stainless steel, of a width of 600 mm or more:
	7219	stainless	72191100	Not further worked than hot-rolled, in coils: Of a thickness exceeding 10 mm
	7219	stainless	72191210	Of a thickness of 4,75 mm or more but not exceeding 10 mm: Containing by weight 2,5 % or more of nickel
	7219	stainless	72191290	Of a thickness of 4,75 mm or more but not exceeding 10 mm: Containing by weight less than 2,5 % of nickel
	7219	stainless	72191310	Of a thickness of 3 mm or more but less than 4,75 mm: Containing by weight 2,5 % or more of nickel
	7219	stainless	72191390	Of a thickness of 3 mm or more but less than 4,75 mm: Containing by weight less than 2,5 % of nickel
	7219	stainless	72191410	Of a thickness of less than 3 mm: Containing by weight 2,5 % or more of nickel

	7219	stainless	72191490	Of a thickness of less than 3 mm: Containing by weight less than 2,5 % of nickel
	7219	stainless	72192110	Plates and sheets produced on a reversing mill (quarto) of a width of 600 mm or more and wide flats, of stainless steel
	7219	stainless	72192210	Not further worked than hot-rolled, not in coils: Of a thickness of 4,75 mm or more but not exceeding 10 mm: Containing by weight 2,5 % or more of nickel
	7219	stainless	72192290	Not further worked than hot-rolled, not in coils: Of a thickness of 4,75 mm or more but not exceeding 10 mm: Containing by weight less than 2,5 % of nickel
	7219	stainless	72192300	Not further worked than hot-rolled, not in coils: Of a thickness of 3 mm or more but less than 4,75 mm
	7219	stainless	72192400	Not further worked than hot-rolled, not in coils: Of a thickness of less than 3 mm
	7219	stainless	72193100	Not further worked than cold-rolled (cold-reduced): Of a thickness of 4,75 mm or more
	7219	stainless	72193210	Not further worked than cold-rolled (cold-reduced): Of a thickness of 3 mm or more but less than 4,75 mm: Containing by weight 2,5 % or more of nickel
	7219	stainless	72193290	Not further worked than cold-rolled (cold-reduced): Of a thickness of 3 mm or more but less than 4,75 mm: Containing by weight less than 2,5 % of nickel
	7219	stainless	72193310	Not further worked than cold-rolled (cold-reduced): Of a thickness exceeding 1 mm but less than 3 mm: Containing by weight 2,5 % or more of nickel
	7219	stainless	72193390	Not further worked than cold-rolled (cold-reduced): Of a thickness exceeding 1 mm but less than 3 mm: Containing by weight less than 2,5 % of nickel
	7219	stainless	72193410	Not further worked than cold-rolled (cold-reduced): Of a thickness of 0,5 mm or more but not exceeding 1 mm: Containing by weight 2,5 % or more of nickel
	7219	stainless	72193490	Not further worked than cold-rolled (cold-reduced): Of a thickness of 0,5 mm or more but not exceeding 1 mm: Containing by weight less than 2,5 % of nickel
	7219	stainless	72193510	Not further worked than cold-rolled (cold-reduced): Of a thickness of less than 0,5 mm: Containing by weight 2,5 % or more of nickel
	7219	stainless	72193590	Not further worked than cold-rolled (cold-reduced): Of a thickness of less than 0,5 mm: Containing by weight less than 2,5 % of nickel
	7219	stainless	72199000	Other:
	7219	stainless	72199020	Other: Perforated
	7219	stainless	72199080	Other: Other
	7220	stainless		Flat-rolled products of stainless steel, of a width of less than 600 mm:
	7220	stainless	72202021	Not further worked than cold-rolled (cold-reduced): Of a thickness of 3 mm or more, containing by weight: 2,5 % or more of nickel
	7220	stainless	72202029	Not further worked than cold-rolled (cold-reduced): Of a thickness of 3 mm or more, containing by weight: Less than 2,5 % of nickel
	7220	stainless	72202041	Not further worked than cold-rolled (cold-reduced): Of a thickness exceeding 0,35 mm but of less than 3 mm, containing by weight: 2,5 % or more of nickel
	7220	stainless	72202049	Not further worked than cold-rolled (cold-reduced): Of a thickness exceeding 0,35 mm but of less than 3 mm, containing by weight: Less than 2,5 % of nickel
	7220	stainless	72202081	Not further worked than cold-rolled (cold-reduced): Of a thickness not exceeding 0,35 mm, containing by weight: 2,5 % or more of nickel
	7220	stainless	72202089	Not further worked than cold-rolled (cold-reduced): Of a thickness not exceeding 0,35 mm, containing by weight: Less than 2,5 % of nickel
	7220	stainless	72209000	Other:
	7220	stainless	72209020	Other: Perforated
	7220	stainless	72209080	Other: Other
	7220	stainless	72201100	Not further worked than hot-rolled: Of a thickness of 4,75 mm or more
	7220	stainless	72201200	Not further worked than hot-rolled: Of a thickness of less than 4,75 mm
	7221	stainless		Bars and rods, hot-rolled, in irregularly wound coils, of stainless steel:
	7221	stainless	72210010	Containing by weight 2,5 % or more of nickel
	7221	stainless	72210090	Containing by weight less than 2,5 % of nickel
	7222	stainless		Other bars and rods of stainless steel; angles, shapes and sections of stainless steel:
	7222	stainless	72221111	Bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Of circular cross-section: Of a diameter of 80 mm or more, containing by weight: 2,5 % or more of nickel
	7222	stainless	72221119	Bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Of circular cross-section: Of a diameter of 80 mm or more, containing by weight: Less than 2,5 % of nickel
	7222	stainless	72221181	Bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Of circular cross-section: Of a diameter of less than 80 mm, containing by weight: 2,5 % or more of nickel

	7222	stainless	72221189	Bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Of circular cross-section: Of a diameter of less than 80 mm, containing by weight: Less than 2,5 % of nickel
	7222	stainless	72221910	Bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Other: Containing by weight 2,5 % or more of nickel
	7222	stainless	72221990	Bars and rods, not further worked than hot-rolled, hot-drawn or extruded: Other: Containing by weight less than 2,5 % of nickel
	7222	stainless	72222011	Cold-drawn bars and solid profiles of stainless steel
	7222	stainless	72223097	Bars and rods of stainless steel, cold-formed or cold-finished and further worked, or hot-formed and further worked, n.e.s. (excluding forged products)
	7222	stainless	72224010	Angles, shapes and sections: Not further worked than hot-rolled, hot-drawn or extruded
	7222	stainless	72224050	Cold-formed sections, obtained from flat products, of stainless steel
	7223	stainless		Wire of stainless steel, in coils (excl. bars and rods)
	7223	stainless	72230011	Stainless steel wire (excluding very fine sterile stainless wire used for surgical sutures)
	7304	stainless		Tubes, pipes and hollow profiles, seamless, of iron (other than cast iron) or steel
	7304	stainless	73044100	Tubes and pipes, of circular cross-section, seamless, of stainless steel (excluding line pipe of a kind used for oil or gas pipelines and casing, tubing and drill pipe used for oil or gas drilling)
	7304	stainless	73041100	Line pipe, of a kind used for oil or gas pipelines, seamless, of stainless steel
	7304	stainless	73042200	Casing, tubing and drill pipe, of a kind used in the drilling for oil or gas, seamless, of stainless steel
	7304	stainless	73044983	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of stainless steel, of an external diameter of $\leq 168,3$ mm (excl. cold-drawn or cold-rolled, line pipe of a kind used for oil or gas pipelines, and casing and tubing of a kind used for drilling for oil or gas and tubes)
	7304	stainless	73044985	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of stainless steel, of an external diameter of $> 168,3$ mm but $\leq 406,4$ mm (excl. cold-drawn or cold-rolled, line pipe of a kind used for oil or gas pipelines, and casing and tubing of a kind used for drilling for oil or gas and tubes)
	7304	stainless	73044989	Tubes, pipes and hollow profiles, seamless, of circular cross-section, of stainless steel, of an external diameter of $> 406,4$ mm (excl. cold-drawn or cold-rolled, line pipe of a kind used for oil or gas pipelines, and casing and tubing of a kind used for drilling for oil or gas and tubes)
	7306	stainless		Iron or steel (excluding cast iron); tubes, pipes and hollow profiles (not seamless), n.e.c. in chapter 73
	7306	stainless	73061110	Line pipe, of a kind used for oil or gas pipelines, longitudinally or spirally welded, of an external diameter $\leq 406,4$ mm, of stainless steel
	7306		73061100	Line pipe of a kind used for oil or gas pipelines, welded, of flat-rolled products of stainless steel, of an external diameter of $\leq 406,4$ mm
	7306	stainless	73062100	Casing and tubing, of a kind used in drilling for oil or gas, welded, of an external diameter $\leq 406,4$ mm, of stainless steel
	7306	stainless	73064020	Tubes, pipes and hollow profiles, welded, of circular cross-section, of stainless steel, cold-drawn or cold-rolled "cold-reduced" (excl. products having internal and external circular cross-sections and an external diameter of $> 406,4$ mm, and line pipe of a kind used for oil or gas pipelines or casing and tubing of a kind used in drilling for oil or gas)
	7306		73064080	Tubes, pipes and hollow profiles, welded, of circular cross-section, of stainless steel (excl. products cold-drawn or cold-rolled "cold-reduced", tubes and pipes having internal and external circular cross-sections and an external diameter of $> 406,4$ mm, and line pipe of a kind used for oil or gas pipelines or casing and tubing of a kind used in drilling for oil or gas)
	7306	stainless	73066110	Tubes and pipes, of non-circular cross-section, hot- or cold-formed and welded, of stainless steel
Foundry products				
Cast steel products	7325	castings		Other cast articles of iron or steel
	7325	castings	73259100	Grinding balls and similar articles for mills
	7325	castings	73259990	Other: other
	7307	castings		Tubes or pipe fittings (e.g. couplings, elbows, sleeves), of iron or steel
	7307	castings	73071990	Cast tube or pipe fittings of steel
Cast iron products	7303	castings		Tubes, pipes and hollow profiles, of cast iron

	7303	castings	73030010	Tubes, pipes and hollow profiles, of cast iron
	7307	castings		Tubes or pipe fittings (e.g. couplings, elbows, sleeves), of iron or steel
	7307	castings	73071110	Tube or pipe fittings of non-malleable cast iron, of a kind used in pressure systems
	7308	castings	73071190	Tube or pipe fittings of non-malleable cast iron (excl. products of a kind used in pressure systems)
	7307	castings	73071910	Tube or pipe fittings of cast iron (excl. of non-malleable)
	7322	castings		Radiators for central heating, not electrically heated and parts thereof, of iron or steel; air heaters, hot air distributors not electrically heated, with motor fan or blower
	7322	castings	73221100	Radiators and parts thereof: of cast iron
	7323	castings		Table, kitchen, other household articles and parts, of iron or steel; iron or steel wool; pot scourers and scouring or polishing pads, gloves and the like, of iron or steel
	7323	castings	73239100	Other: of cast iron, not enamelled
	7323	castings	73239200	Other: of cast iron, enamelled
	7324	castings		Sanitary ware and parts thereof, of iron or steel
	7324	castings	73242100	Baths: of cast iron, whether or not enamelled
	7325	castings		Other cast articles of iron or steel
	7325	castings	73251010	Of non-malleable cast iron: for sewage, water, etc., systems
	7325	castings	73251090	Of non-malleable cast iron: other
	7325	castings	73259100	Grinding balls and similar articles for mills
	7325	castings	73259910	Articles of cast iron, n.e.s. (excl. of non-malleable cast iron, and grinding balls and similar articles for mills)

Note. CN codes highlighted in red indicate duplicates, as they can be classified under multiple categories.

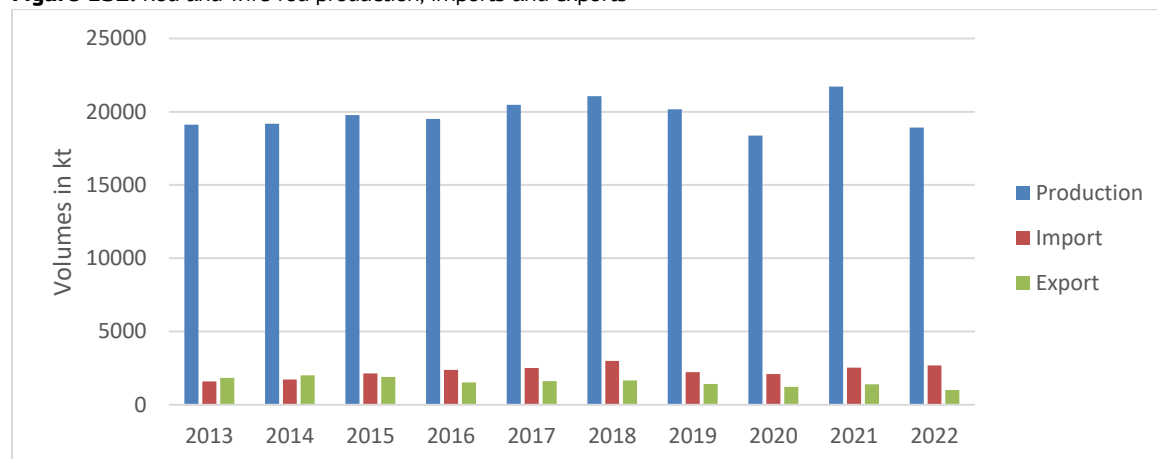
Source: EUROFER and Fraunhofer elaboration

Annex 4. Market analysis

PART A: Finished steel products production, imports and exports over time

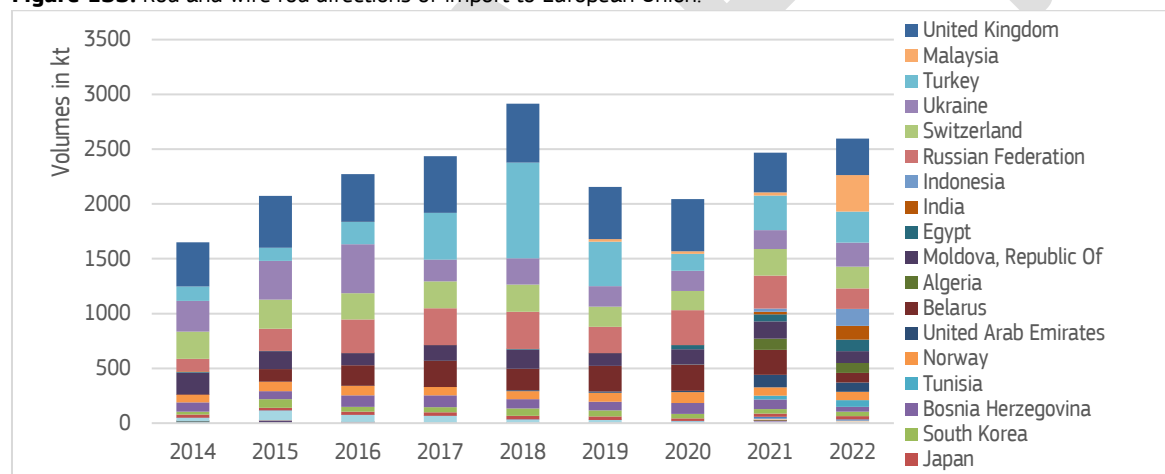
4.1 Rod and wire rod

Figure 132. Rod and wire rod production, imports and exports



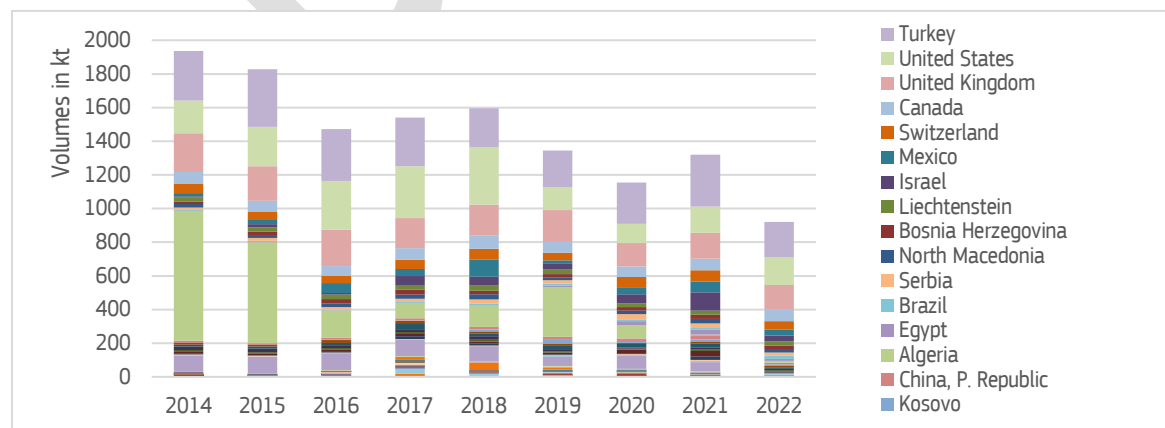
Source: (EUROFER 2023)

Figure 133. Rod and wire rod directions of import to European Union.



Source: Comext (eurostat, 2024)

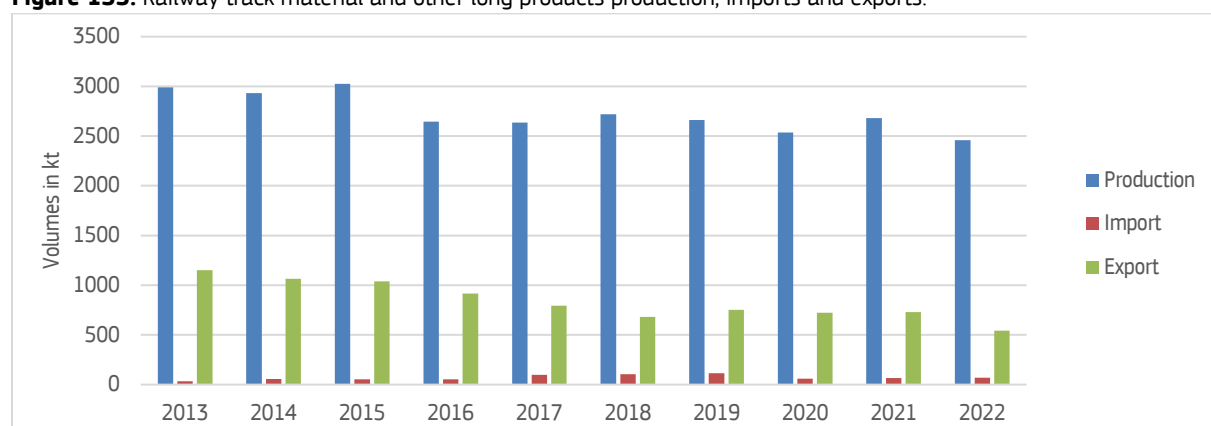
Figure 134. Rod and wire rod directions of export from European Union.



Source: Comext (eurostat, 2024)

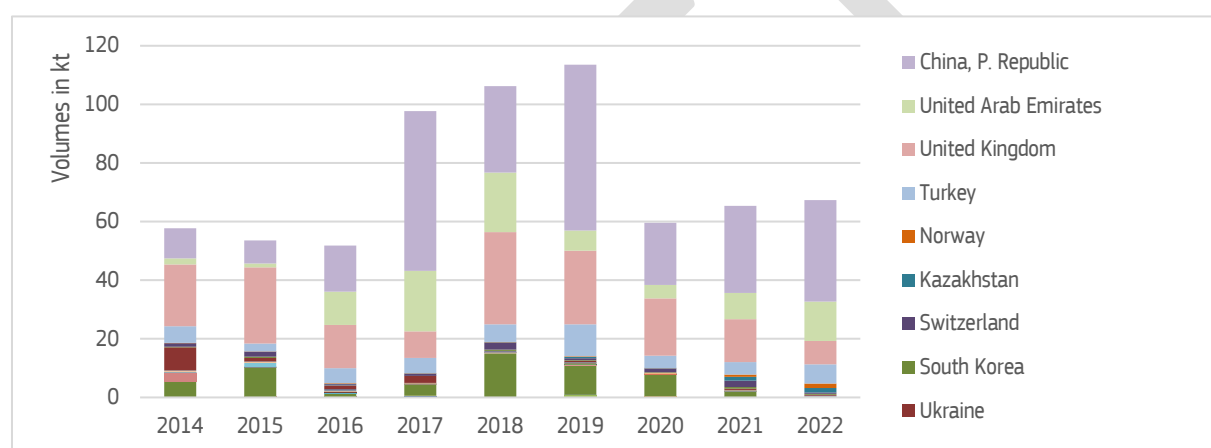
4.2 Railway track material and other long products

Figure 135. Railway track material and other long products production, imports and exports.



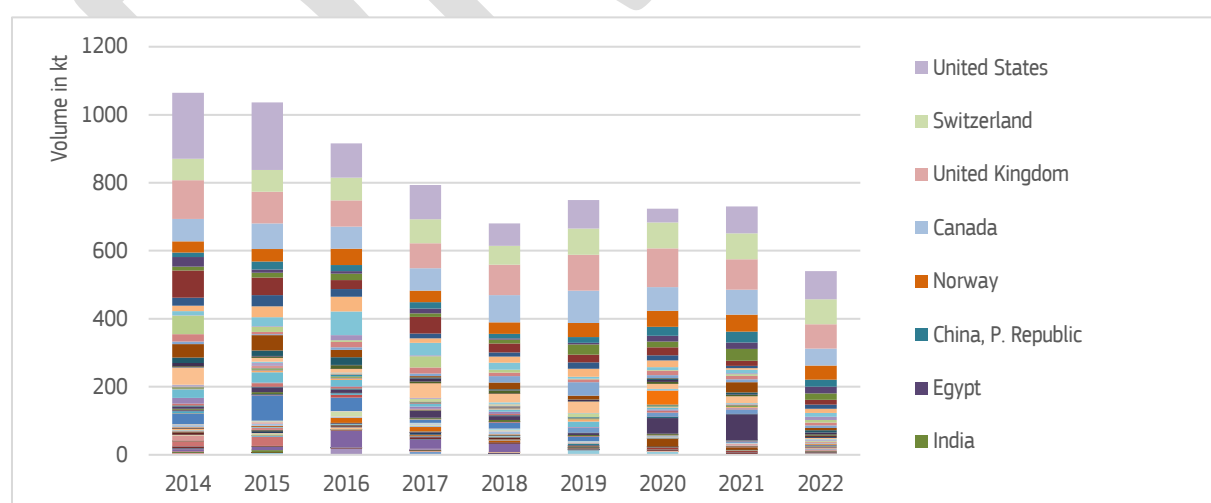
Source: (EUROFER 2023).

Figure 136. Railway track material and other long products directions of import to European Union



Source: Comext (eurostat, 2024)

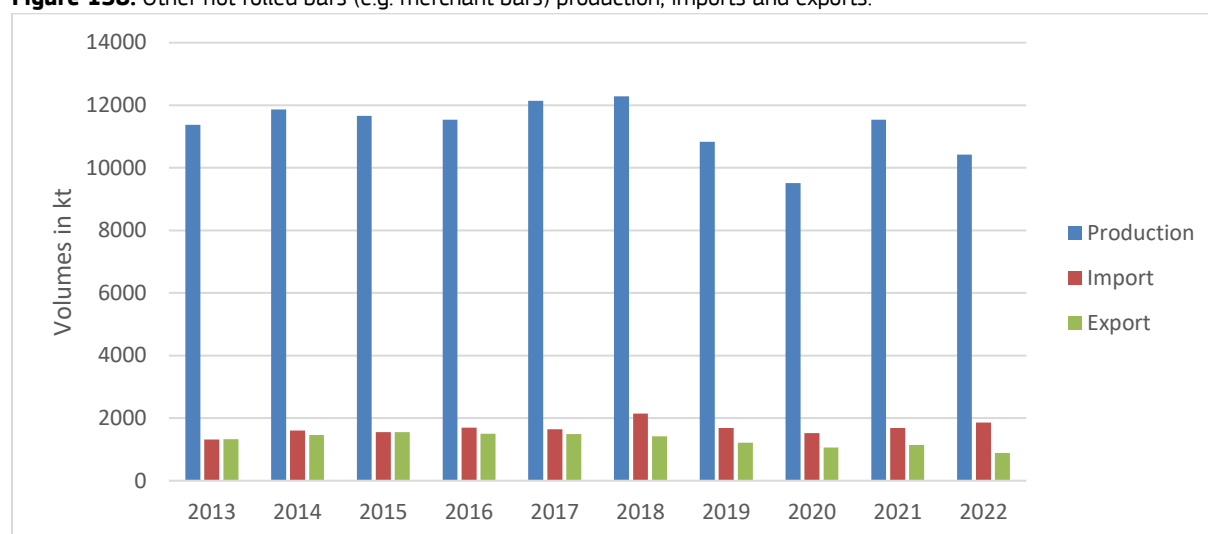
Figure 137. Railway track material and other long products directions of export from European Union.



Source: Comext (eurostat, 2024)

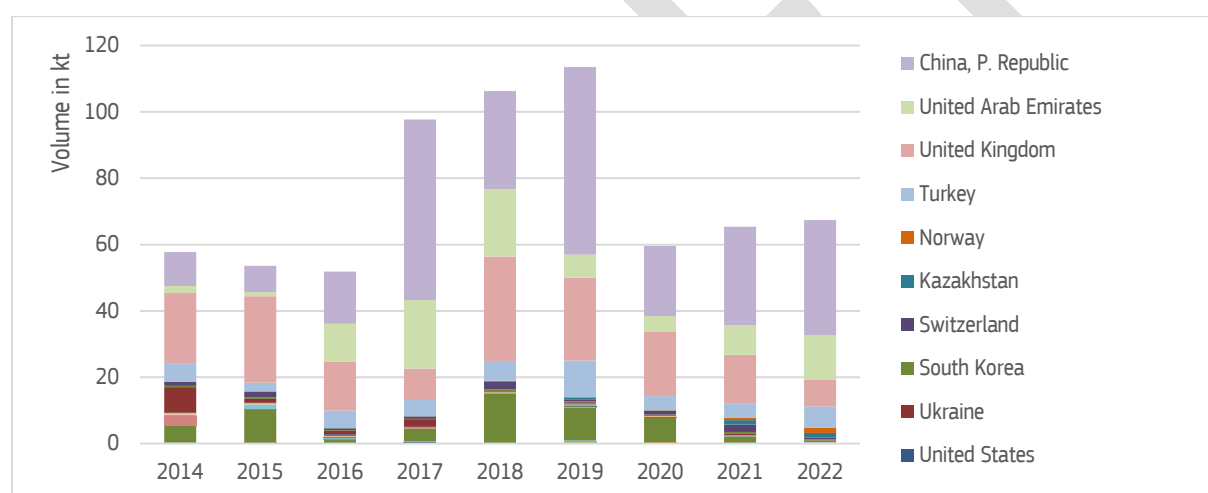
4.3 Other hot rolled bars (e.g. merchant bars)

Figure 138. Other hot rolled bars (e.g. merchant bars) production, imports and exports.



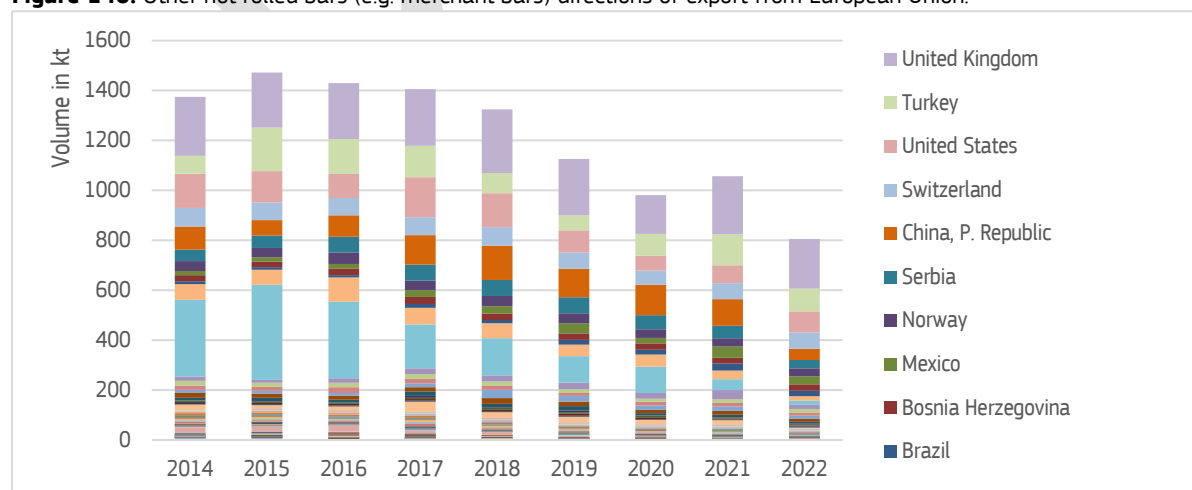
Source: (EUROFER 2023).

Figure 139. Other hot rolled bars (e.g. merchant bars) directions of import to European Union.



Source: Comext (eurostat, 2024)

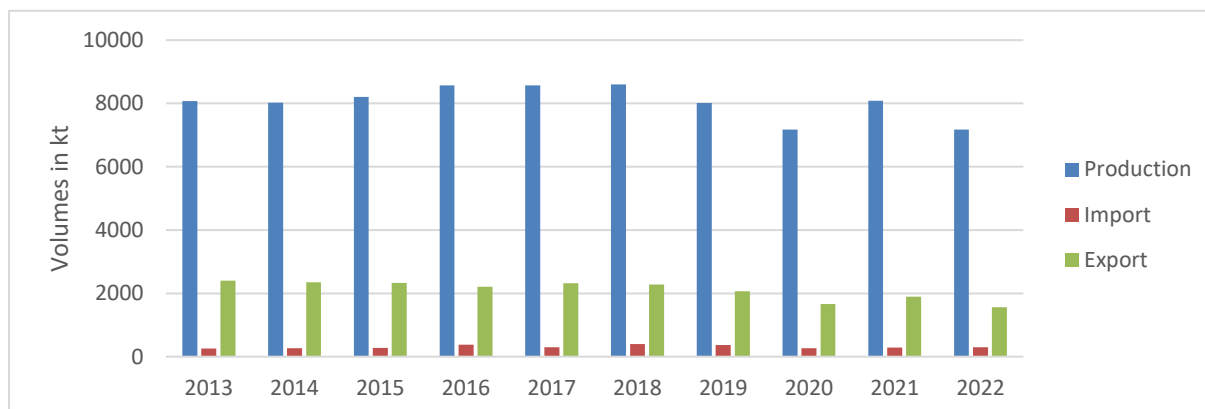
Figure 140. Other hot rolled bars (e.g. merchant bars) directions of export from European Union.



Source: Comext (eurostat, 2024)

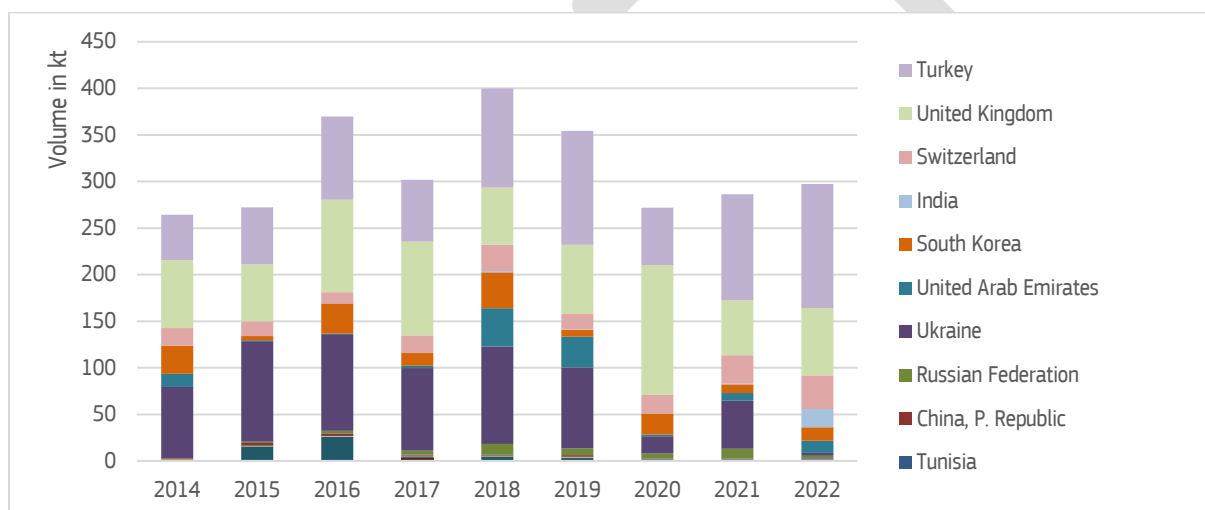
4.4 Heavy sections and mining frames

Figure 141. Heavy sections and mining frames production imports and exports.



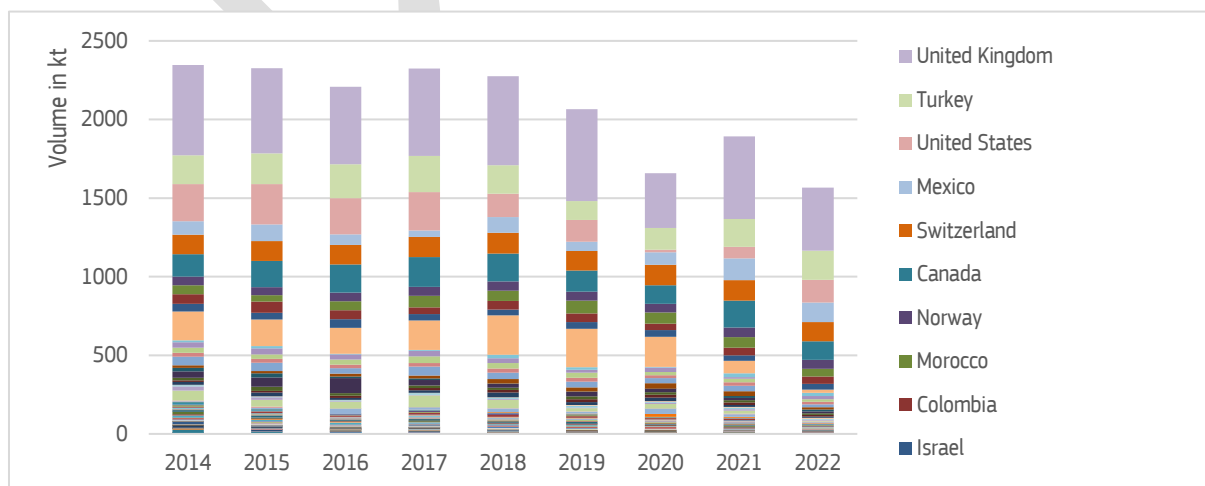
Source: (EUROFER 2023)

Figure 142. Heavy sections and mining frames directions of import to European Union.



Source: Comext (eurostat, 2024)

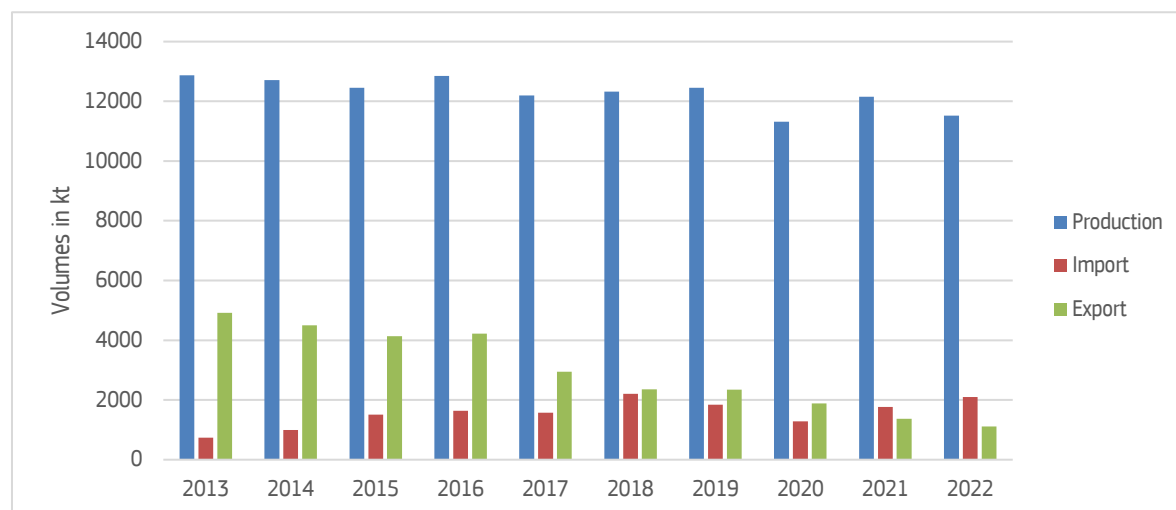
Figure 143. Heavy sections and mining frames directions of export from European Union.



Source: Comext (eurostat, 2024)

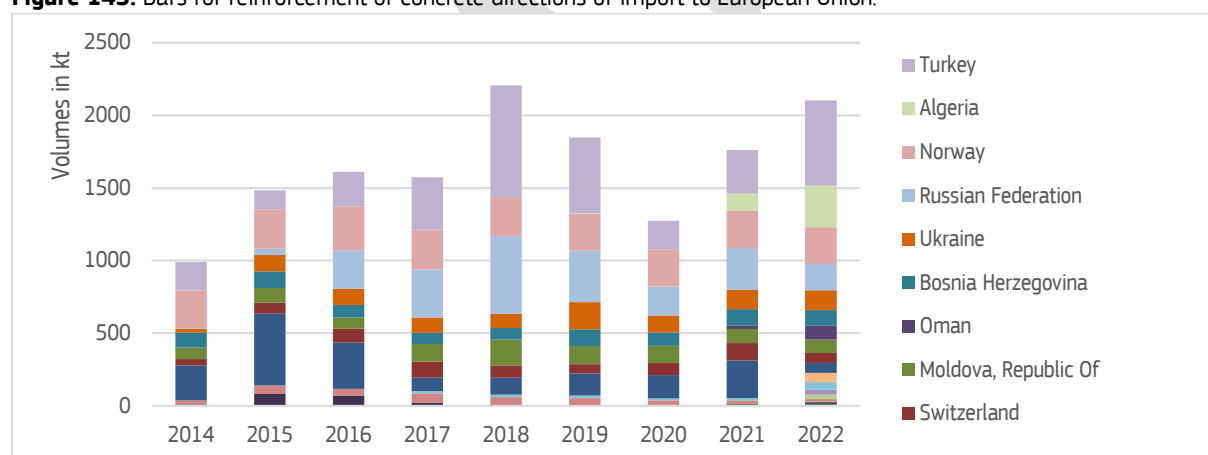
4.5 Bars for reinforcement of concrete

Figure 144. Bars for reinforcement of concrete production, imports and exports.



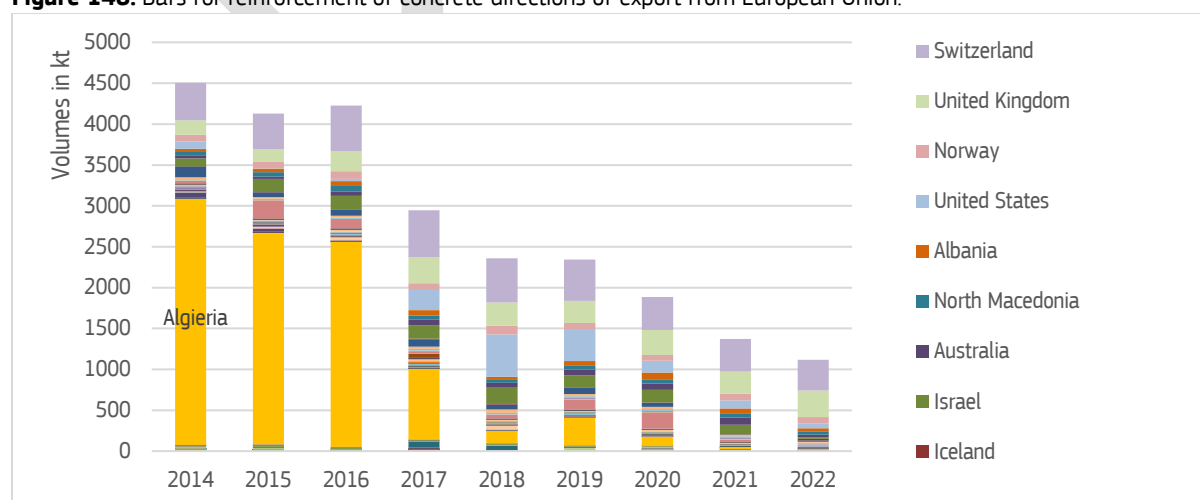
Source: (EUROFER 2023)

Figure 145. Bars for reinforcement of concrete directions of import to European Union.



Source: Comext (eurostat, 2024)

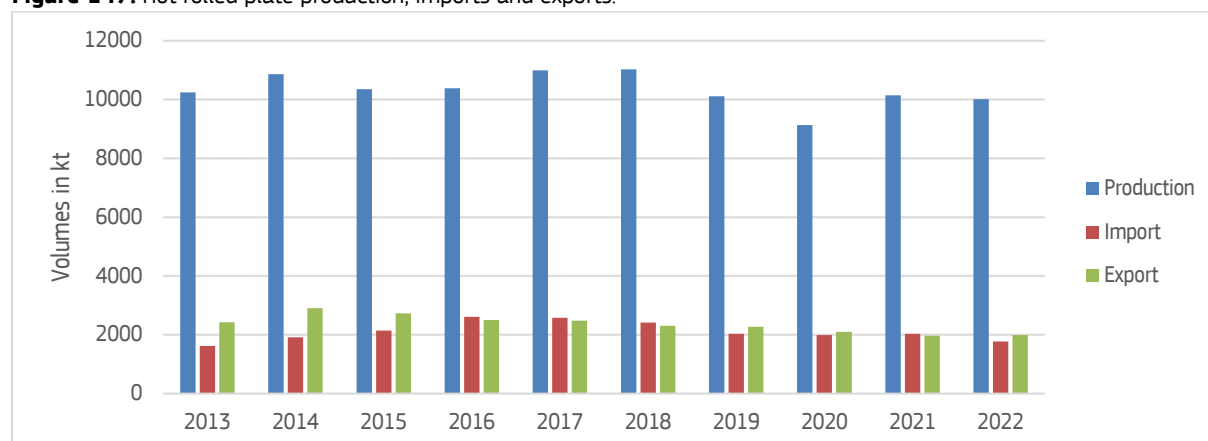
Figure 146. Bars for reinforcement of concrete directions of export from European Union.



Source: Comext (eurostat, 2024)

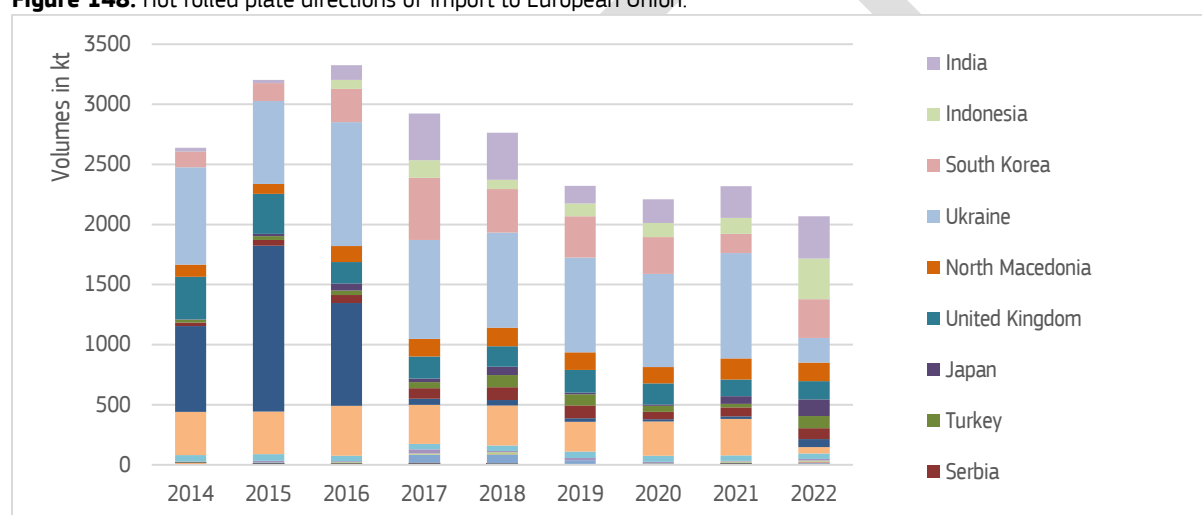
4.6 Hot rolled plate (e.g. Quarto plate)

Figure 147. Hot rolled plate production, imports and exports.



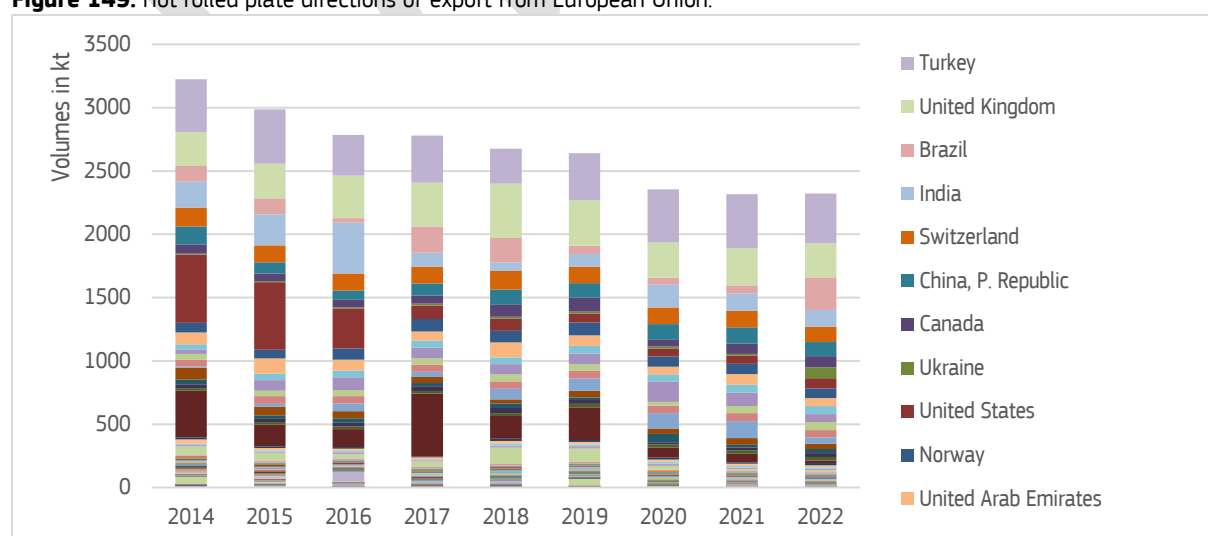
Source: (EUROFER 2023).

Figure 148. Hot rolled plate directions of import to European Union.



Source: Comext (eurostat, 2024)

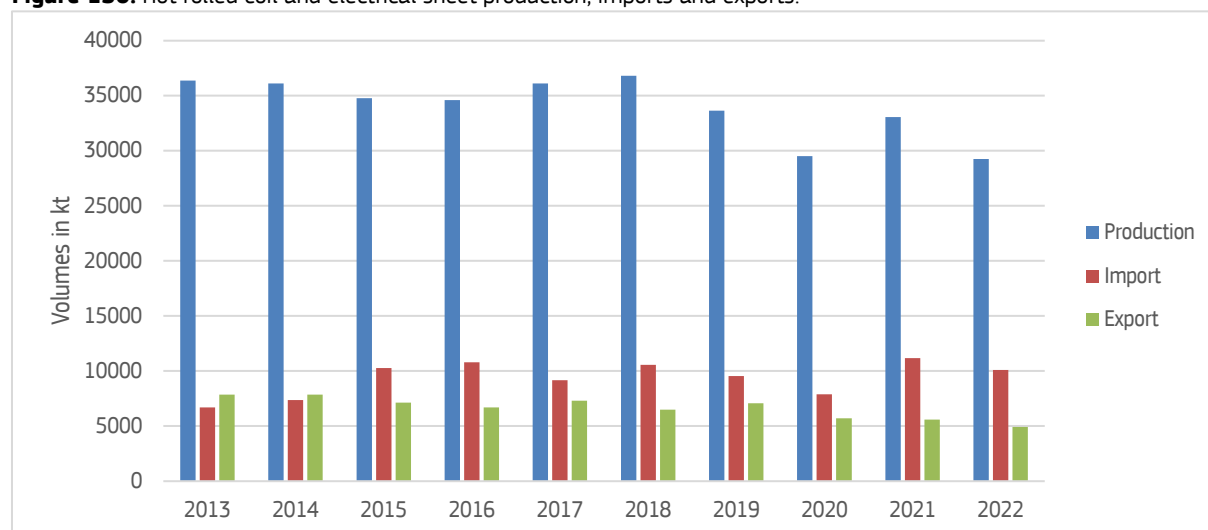
Figure 149. Hot rolled plate directions of export from European Union.



Source: Comext (eurostat, 2024)

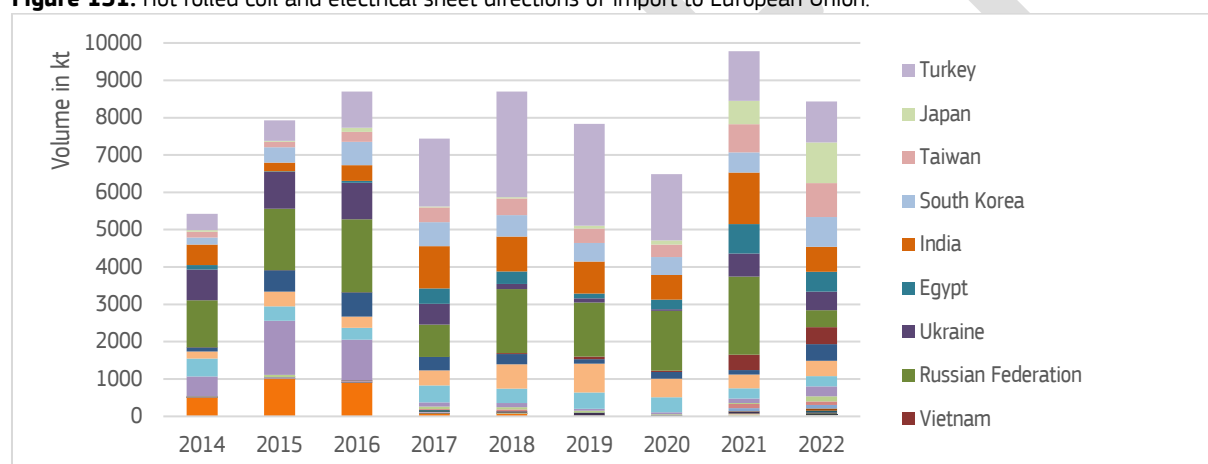
4.7 Hot rolled coil and electrical sheet production

Figure 150. Hot rolled coil and electrical sheet production, imports and exports.



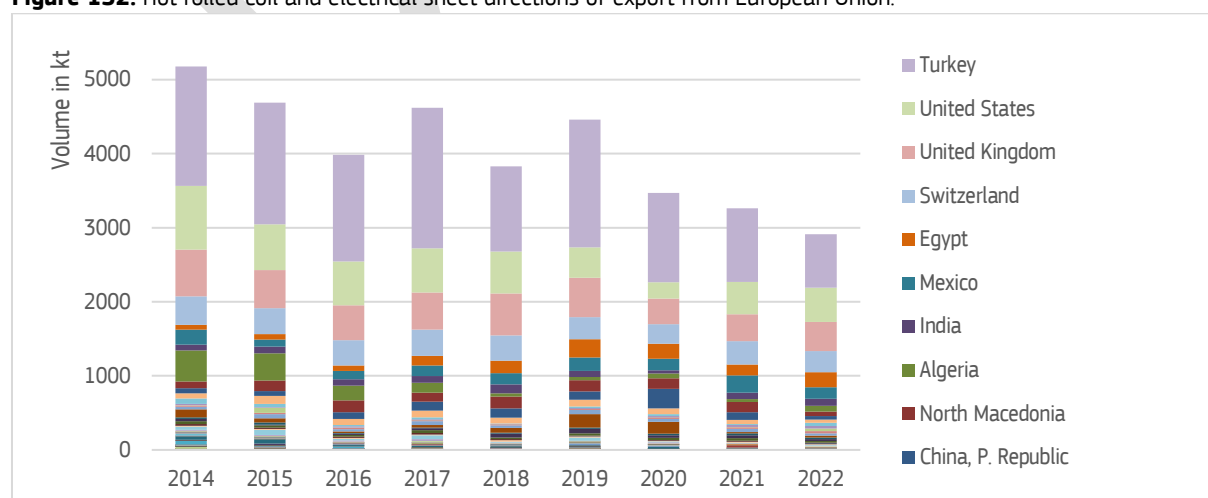
Source: (EUROFER 2023).

Figure 151. Hot rolled coil and electrical sheet directions of import to European Union.



Source: Comext (eurostat, 2024)

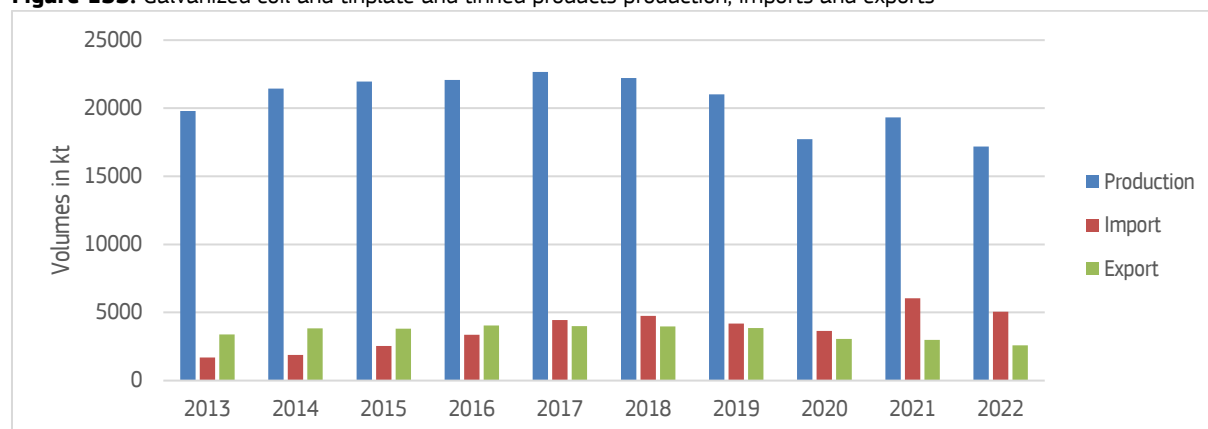
Figure 152. Hot rolled coil and electrical sheet directions of export from European Union.



Source: Comext (eurostat, 2024)

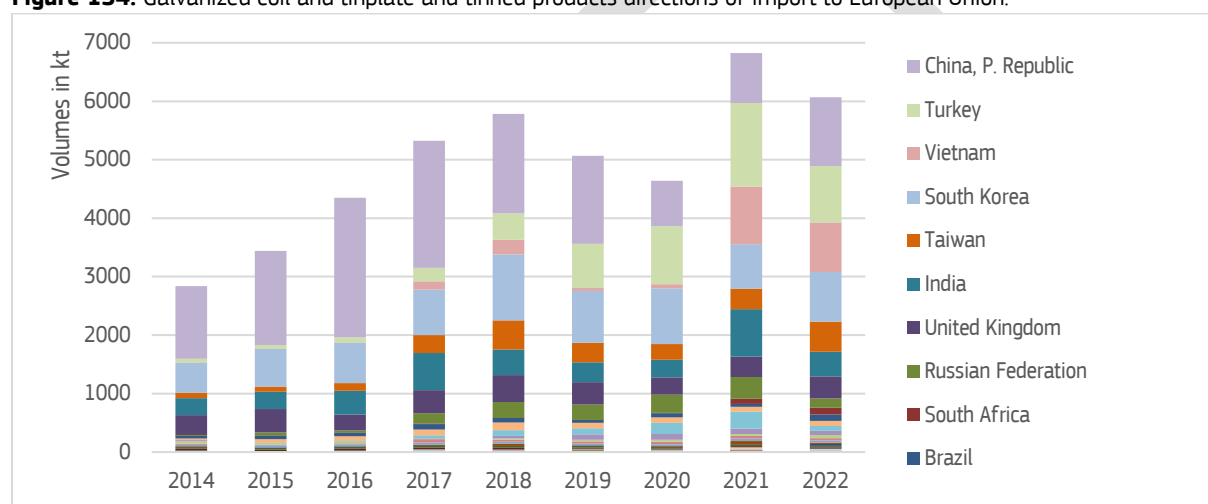
4.8 Galvanized coil, tinplate and tinned products

Figure 153. Galvanized coil and tinplate and tinned products production, imports and exports



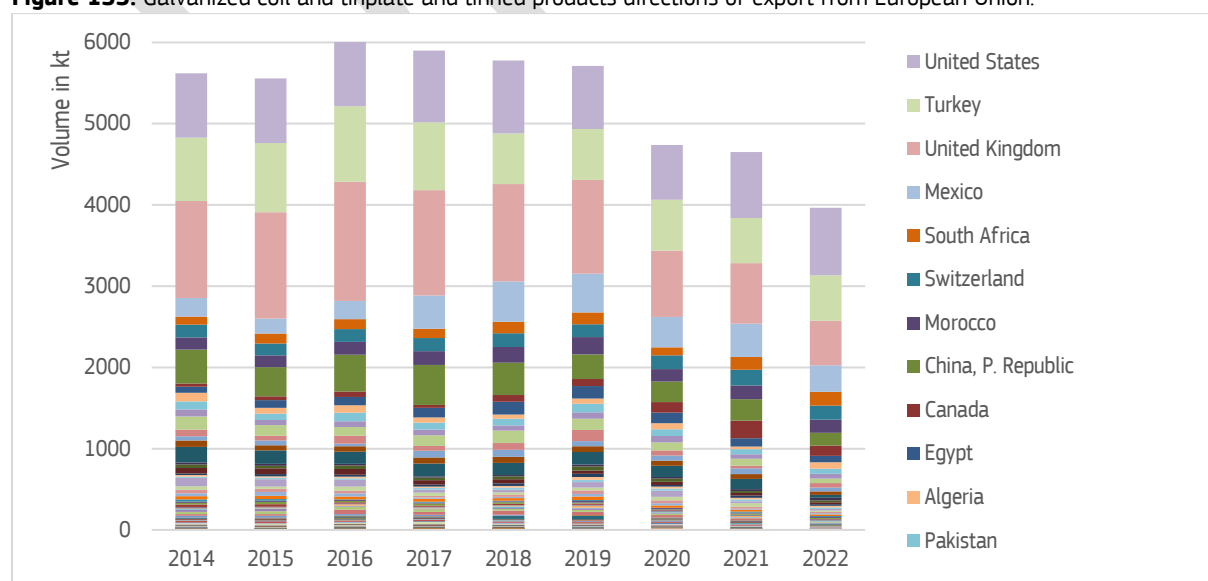
Source: (EUROFER 2023)

Figure 154. Galvanized coil and tinplate and tinned products directions of import to European Union.



Source: Comext (eurostat, 2024)

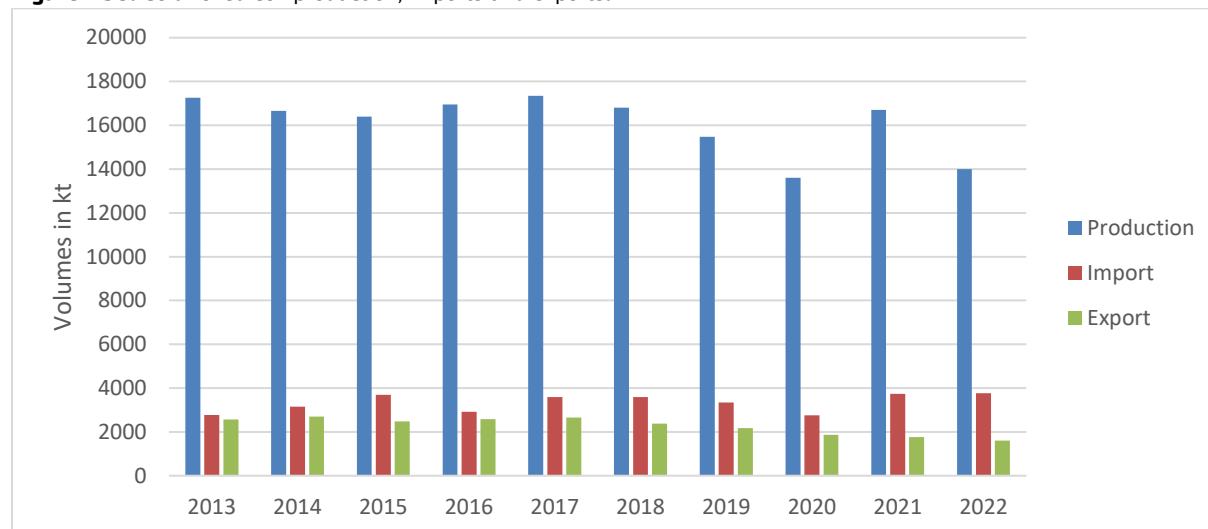
Figure 155. Galvanized coil and tinplate and tinned products directions of export from European Union.



Source: Comext (eurostat, 2024)

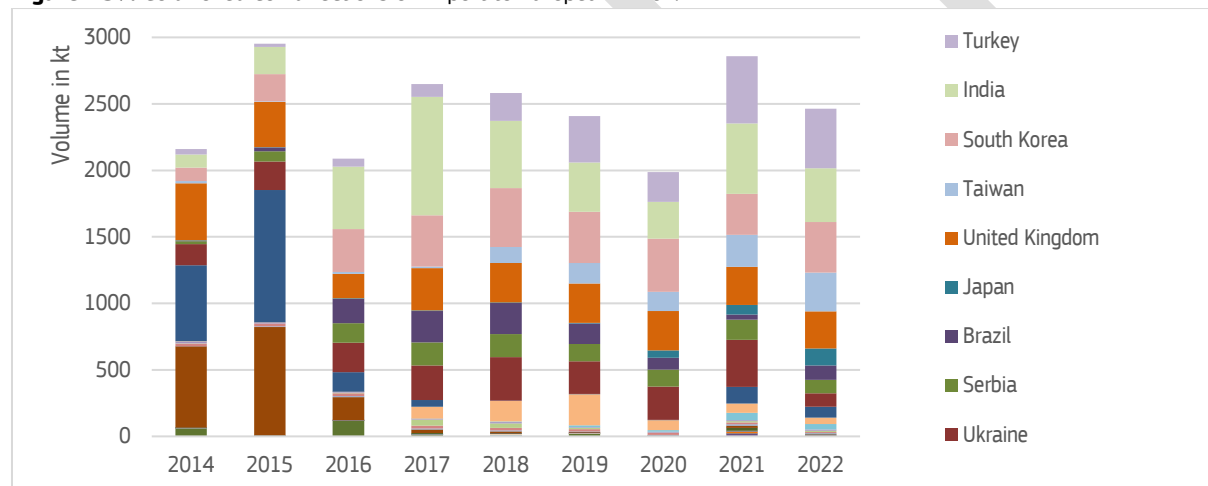
4.9 Cold rolled coil

Figure 156. Cold rolled coil production, imports and exports.



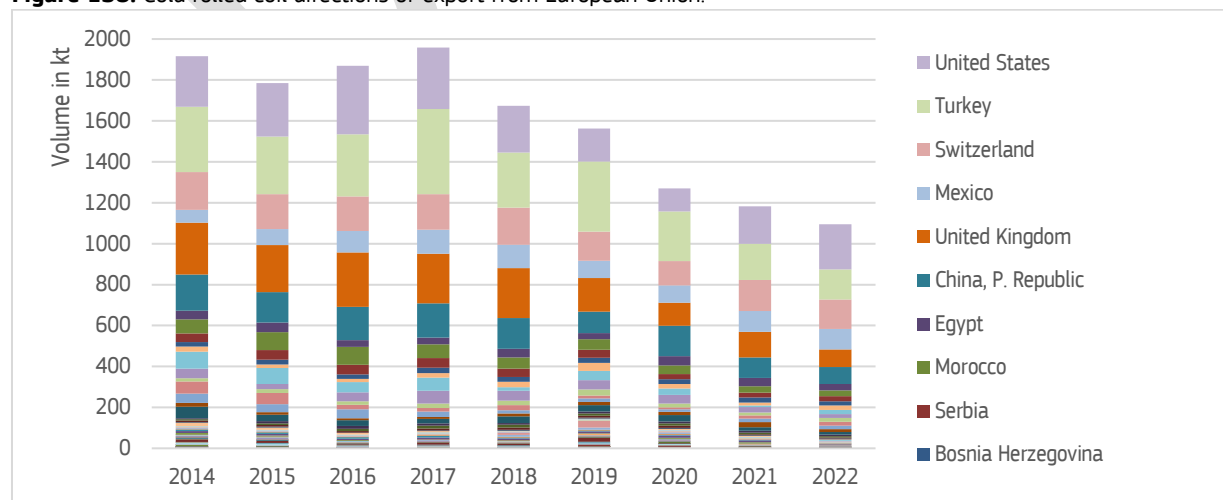
Source: (EUROFER 2023).

Figure 157. Cold rolled coil directions of import to European Union.



Source: Comext (eurostat, 2024)

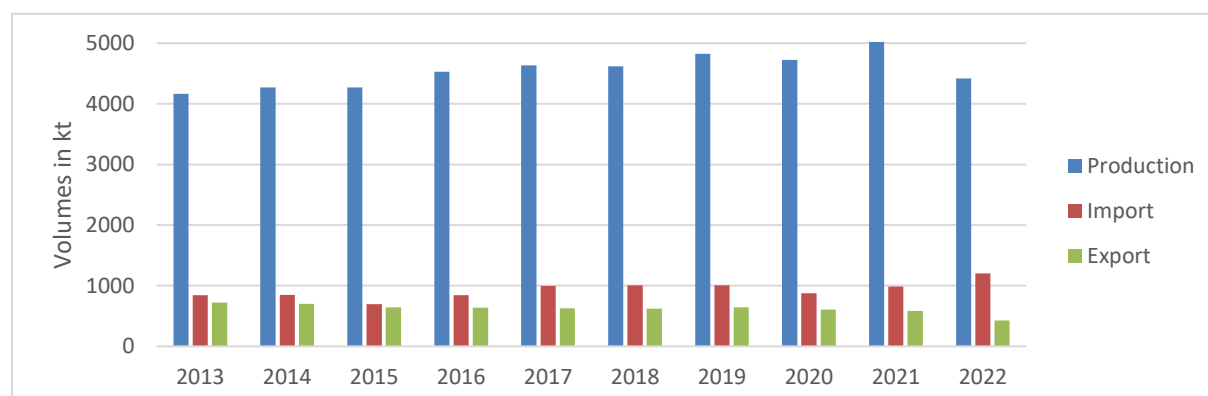
Figure 158. Cold rolled coil directions of export from European Union.



Source: Comext (eurostat, 2024)

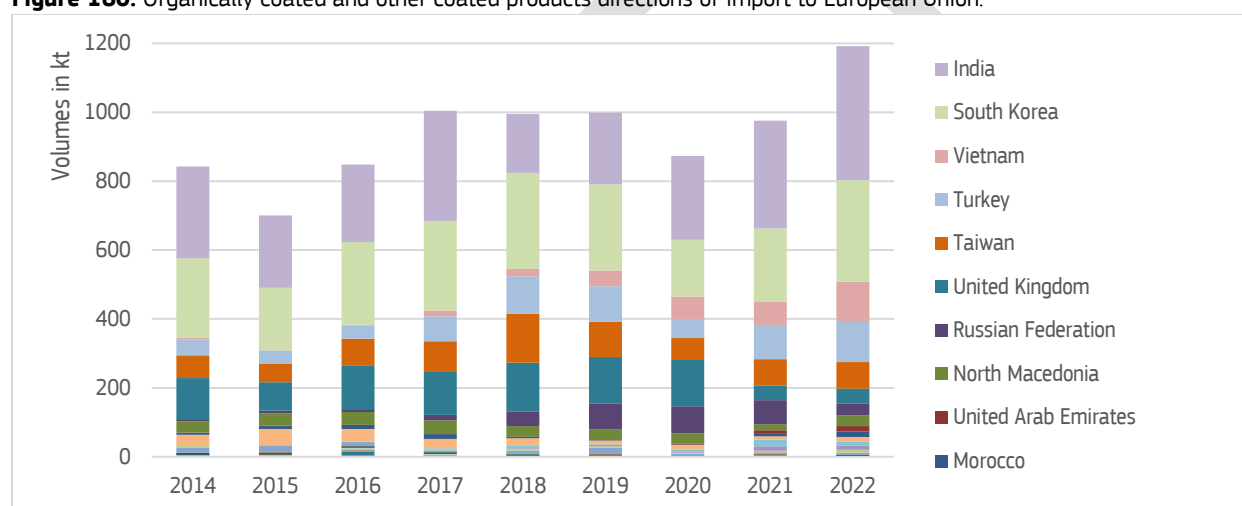
4.10 Organically coated and other coated products

Figure 159. Organically coated and other coated products production, imports and exports.



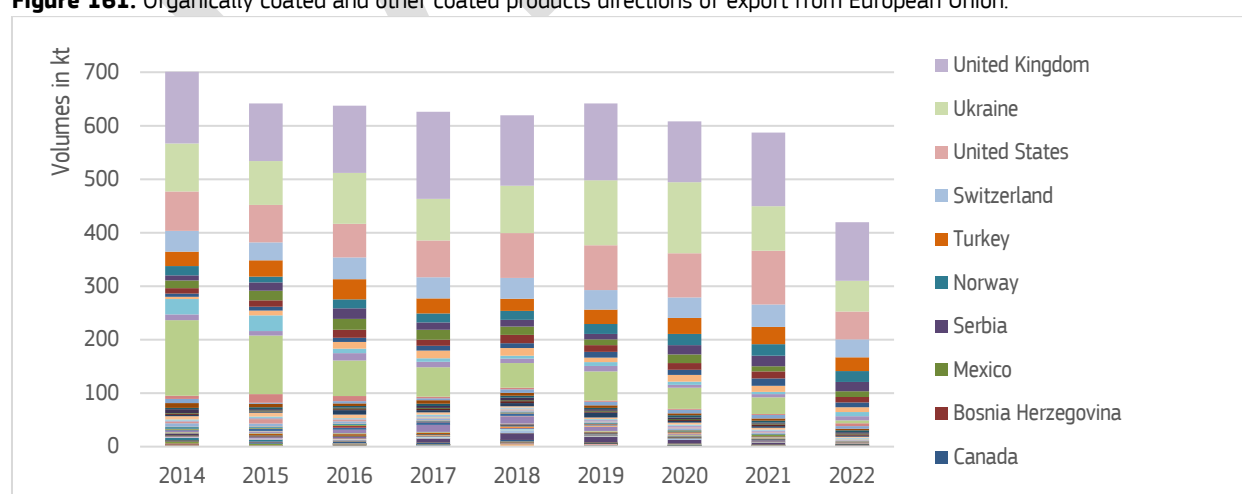
Source: (EUROFER 2023)

Figure 160. Organically coated and other coated products directions of import to European Union.



Source: Comext (eurostat, 2024)

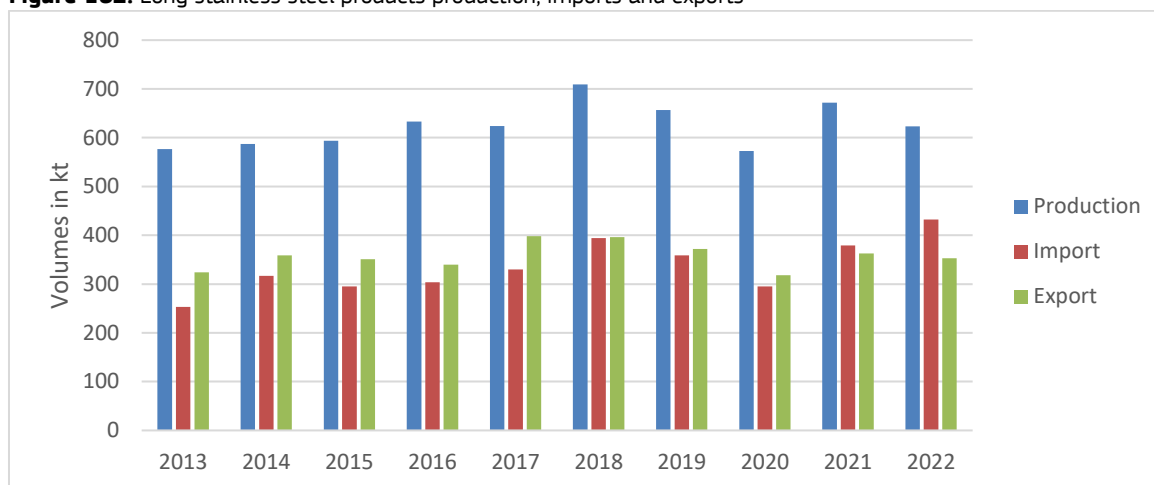
Figure 161. Organically coated and other coated products directions of export from European Union.



Source: Comext (eurostat, 2024)

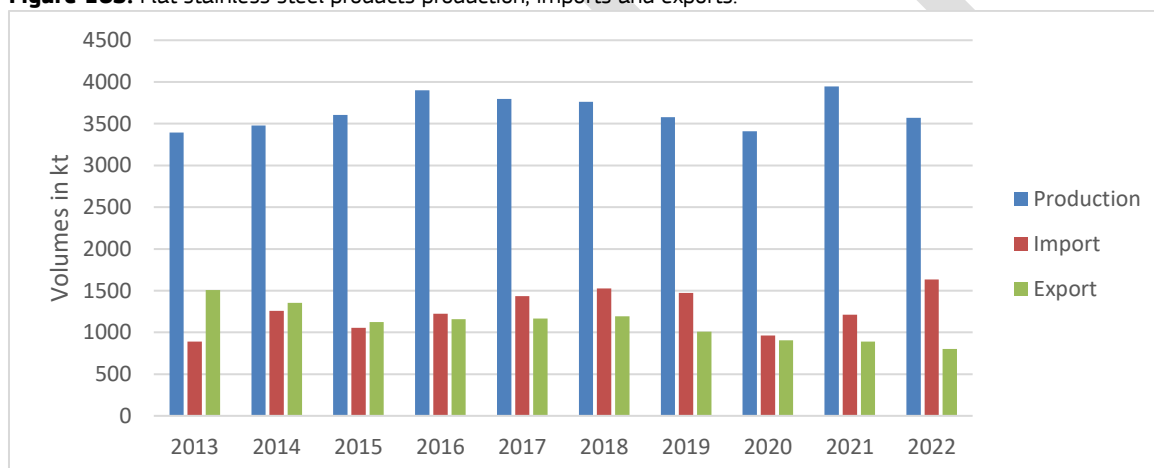
4.11 Stainless steel

Figure 162. Long stainless steel products production, imports and exports



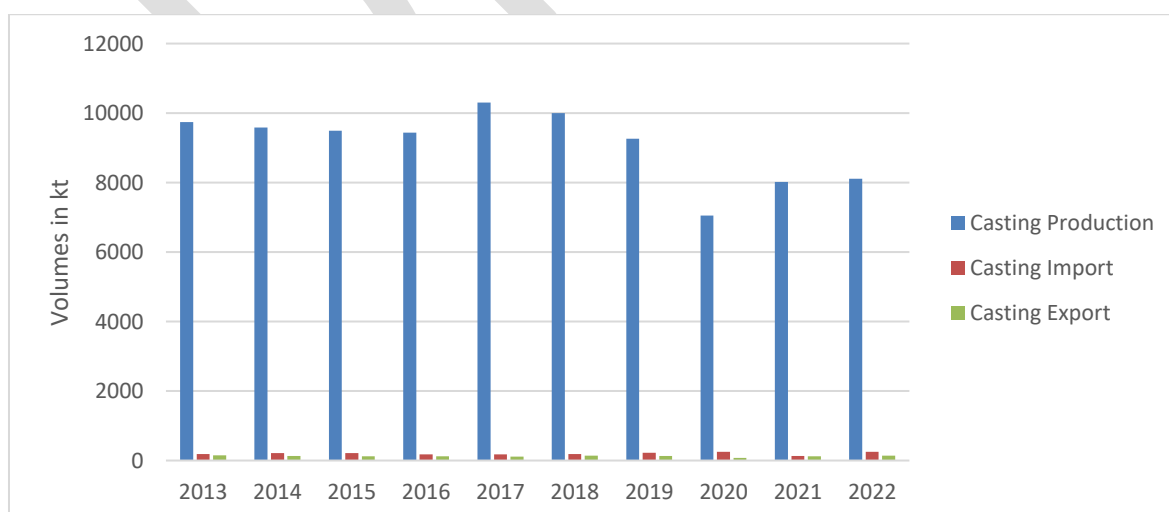
Source: (EUROFER 2023)

Figure 163. Flat stainless steel products production, imports and exports.



Source: (EUROFER 2023)

Figure 164. Castings (both iron and steel castings) production, imports and exports.



Part B: Accounting matrix of finished iron and steel products to end-use sectors

This part presents the outcome of the market data analysis to support the selection of the representative products for the preparatory study with data on the production, trade and consumption of finished steel products and their use in end use sectors.

As a starting point, the work by Dworak et al. (2021) and Harvey (2022) on distribution matrices of finished iron and steel products to end-use sectors was used. The matrix needed to be adjusted to reflect the consolidated list of product categories as well as contain the appropriate and timely data sources.

The work process was as follows:

- The finished products categories of the matrix were matched to the consolidated list of product categories.
- Production data for foundry was added (WFO 2021).
- Production data for tubular products is reported by World Steel Association 2021 and eurostat 2024. Welded tubes are assumed to be made from other finished products. 20% of the amount of welded tubes produced was therefore subtracted from hot rolled plates and 80% were removed from hot rolled coil, sheet, and strip.
- Stainless steel products were not included in the original matrix. To complete the picture, it was added as finished good category. Data for stainless steel was taken from EUROFER 2023. Production data was not available so European deliveries had to be used. The use of stainless steel products in end use sectors was taken from worldstainless 2023.
- The finished steel products categories were then matched to the EUROFER categories again using production data (EUROFER 2023).
- By combining the information of the worldsteel and the EUROFER categories, the EUROFER production data was split into more detailed categories. This did not allow the disaggregation of all product categories from the matrix.
- In a next step trade data was added. Data on imports and exports was taken from EUROFER 2023; World Steel Association 2021 (tubular products). Shown are imports and European consumption (production + imports - exports)¹⁴³. For foundry, PRODCOM was queried for trade data since no data was available from WFO or CAEF (eurostat 2024).
- Harvey 2022 provided shares for steel demand per sector. These are global estimates. However, the original work by Dworak et al. 2022 is also based on global estimates for steel use. With the percentages in the table and the calculated number for overall consumption, tonnages for the use of finished products in the sectors were calculated.
- Since the combination of EUROFER and worldsteel data did not allow to disaggregate the EUROFER/WFO/worldstainless data to the same level as the matrix, the EUROFER data was disaggregated using the tonnages calculated for the matrix from Dworak et al. 2022 and Harvey 2022. This means that not also the matrix part but also the finished steel products statistics at the bottom of the graphs are not purely data-based but involve a combination of data, MFA study results and assumptions.

The resulting tonnages and percentages of this exercise should be used with caution! They are not meant to be used as individual numbers but as rough estimates to draw the overall picture of steel use. The combination of so many data sources with a considerable number of data transformation steps produces considerable uncertainties.

¹⁴³ Consumption numbers contain some amount of double counting. Some finished goods are imported and processed again by finished goods producers. They show up in production and import statistics. EUROFER provides data for European deliveries without those so called receipts. However, those numbers are reported in different product categories. It was decided to keep the error of receipt and avoid the error of another matching with a third product classification system.

However, during the exercise numbers were continuously compared and transformation steps were questioned for their validity. In our opinion, the result is valuable and can be used for the task at hand which is understanding the importance of products and end use sectors in the context of steel use in the EU.

Figure 165. Accounting matrix of finished iron and steel products to end-use sectors.

Metal goods (Domestic appliances)	0%	0%	0%	0%	27%	0%	0%	10%	7%	0%	0%	0%	0%	0%	0%	0%	0%	9%	0%	0%	5,373		
Metal goods (Other)	20%	0%	7%	22%	20%	0%	0%	16%	3%	0%	0%	0%	29%	0%	28%	1%	0%	37%	34%	11%	20,975		
Metal goods (Packaging)	0%	0%	0%	0%	0%	100%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	37%	0%	0%	1,047		
Industrial machinery (Electrical)	2%	100%	1%	1%	2%	0%	0%	3%	3%	0%	0%	0%	3%	0%	5%	4%	3%	28%	0%	7%	6,774		
Industrial Machinery (Mechanical)	24%	0%	26%	5%	22%	0%	0%	5%	3%	13%	0%	0%	54%	0%	1%	36%	18%	66%	20%		24,240		
Transport (Other)	20%	0%	1%	1%	0%	0%	0%	0%	0%	0%	0%	0%	1%	0%	1%	2%	5%		0%	2%	3,401		
Transport (HCV + buses)	0%	0%	1%	7%	2%	0%	0%	4%	1%	0%	0%	0%	1%	0%	2%	1%	2%	5%	0%	3%	2,095		
Transport (LDV + LCV)	0%	0%	5%	28%	9%	0%	0%	38%	4%	0%	0%	0%	7%	0%	2%	0%	11%	9%	0%	11%	12,555		
Infrastructure	18%	0%	18%	14%	3%	0%	25%	6%	12%	87%	29%	30%	1%	54%	33%	40%	18%	12%	0%	43%	37,489		
Buildings	14%	0%	41%	20%	15%	0%	75%	17%	68%	0%	71%	70%	4%	46%	28%	17%	43%	12%	0%	4%	43,841		
	Hot rolled plate (e.g. Quarto plate)	Electrical sheet	Hot rolled coil	Coil	Narrow strip	Cold rolled coil	Tinplate and tinned products	Galvanized coil	Hot rolled	Cold rolled	Organically coated and other coated products	Railway track material and other long products	Heavy sections and mining frames	Light sections	Other hot rolled bars (e.g. merchant bars)	Bars for reinforcement of concrete	Rod and wire rod	Welded tube	Seamless tube	Stainless steel products	Casted steel products	Casted iron products	157,789
			Flat products						Long products					Tubular products					Foundry				
Production	6%	1%	14%	1%	10%	1%	1%	10%	3%	2%	5%	2%	6%	8%	13%	6%	2%	4%	0%	6%		143,791	
Imports	4%	1%	21%	2%	9%	1%	1%	10%	3%	0%	1%	1%	3%	5%	6%	20%	7%	5%	0%	0%		42,698	
Consumption	5%	1%	16%	1%	10%	1%	1%	11%	3%	1%	4%	2%	6%	8%	13%	5%	2%	5%	0%	6%		157,789	

Sources: (Dworak et al. 2021; EUROFER 2023; eurostat 2024b; Harvey 2022; WFO 2021; World Steel Association 2022; worldstainless 2023b).

Annex 5. Initiatives and standards for the definition of “green steel”

First Mover Coalition (FMC) definition of near emission steel

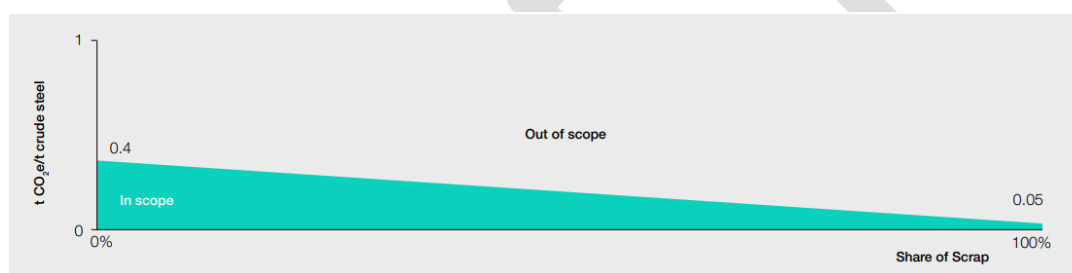
The FMC comprises companies across different sectors including automotive industry which aim to mobilise demand for emerging technologies that have the potential to decarbonise hard-to-abate sectors such as steel (Urban, Nipius, and Egenhofer, 2024).

The FMC is fully aligned with the IEA proposal and has set a product level CO₂ threshold per tonne of crude steel that includes Scope 1 and 2 emissions as defined by the GHG Protocol.

Members commit to procuring 10% of their steel as near-zero-emissions steel according to the following criteria:

- The crude steel from breakthrough technology production facilities. Per IEA guidance, the steel should emit <0.4t (0% scrap inputs) to <0.05t (100% scrap input) of CO₂eq per tonne of crude steel.
- The analytical boundary for emissions is cradle-to-gate, in alignment with IEA guidance in the report Achieving Net Zero Heavy Industry Sectors in G7 Members (IEA, 2022).

Figure 166. FCM definition of near emission steel



Source: First Movers Coalition. Steel commitment (WEF, 2023)

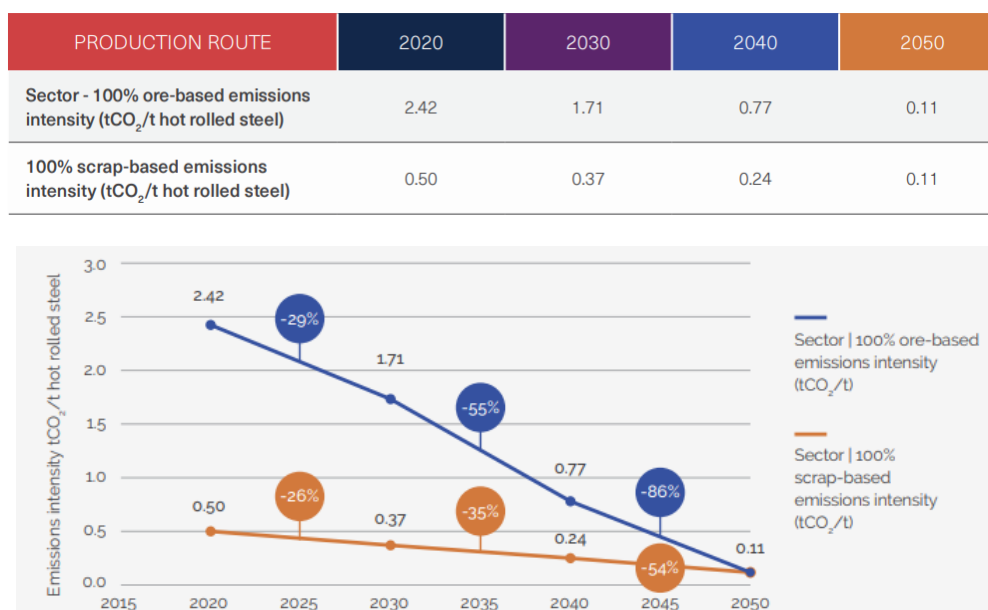
Science Based Target Initiative for Steel

The Science Based Targets initiative (SBTi), has developed a specific **Steel Science-Based Target- Setting Guidance** (SBTi, 2023), which aims to reduce GHG emissions in alignment with the goals of Paris Agreement. This specific guidance is the world's first framework for companies in the steel sector and potential users of steel, such as the automotive industry, to establish **science-based targets at a corporate level**. Each company decide whether to set a near-term target only, or a net-zero target, which includes a near- and a long-term target.

As illustrated in Figure 167., the framework establishes **two decarbonisation pathways** with different targets (thresholds) at the **corporate level**, one for ore-based and another for scrap-based production. Therefore, this approach differs from other initiatives that assess companies' performance today or which evaluates individual plants or products.

According to the SBTi, the primary and secondary steelmaking routes should be considered as separate industries due to their different processes and carbon footprints. The majority of production and emissions come from ore-based steelmaking, so it must decarbonise faster to meet the 1.5°C budget. On the other hand, scrap-based producers have the potential to decarbonise faster by switching to renewable energy sources, as their production is mainly electricity-based (SBTi, 2023).

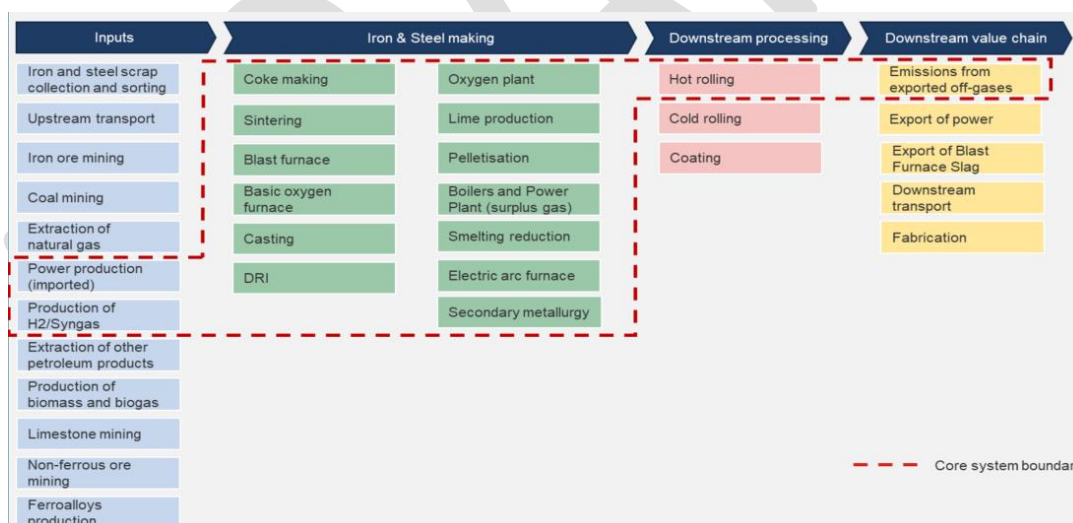
Figure 167. Emission intensity for 1.5°C pathway



Source: Steel Science-Based Target-Setting Guidance (SBTi, 2023)

The core system boundaries for GHG accounting is presented in Figure 168. According to the guidelines, the targets shall cover company-wide scope 1 and scope 2 emissions, as defined by the **GHG Protocol Corporate Standard**. For scope 3 emissions outside the boundary, a company has to include them in their near-term targets if relevant Scope 3 emissions are 40% or more of total Scope 1, 2 and 3 emissions (SBTi, 2024).

Figure 168. Core system boundary of SBTi



Source: SBTi Steel Target Setting Tool (SBTi, n.d.)

This methodology has several limitations regarding the comparability of products' environmental performance. Firstly, decarbonisation targets are set at the corporate level rather than for individual products. Additionally, companies have significant latitude to decide whether to set near-term or net-zero targets, as well as to determine the scopes and methods. As a result, while SBTi's steel sector pathways provide a valuable benchmark for decarbonisation trajectory, it does not offer a comprehensive scheme for comparison of products or companies' emissions intensity. Science-based targets are expressed as a relative reduction in emissions over a timeframe, and are individual to each company based on their starting emissions and activity growth (SBTi, 2023).

Indian Ministry of Steel

The Indian Ministry of Steel has launched in December 2024 the Green Steel Taxonomy¹⁴⁴. This taxonomy provides a framework for defining green steel, which is produced with a CO₂ equivalent emission intensity of less than 2.2 tonnes per tonne of finished steel.

The scheme introduces a star-rating system which is as follows:

- Five-star green-rated steel (emission intensity lower than 1.6 tonnes of CO₂e/t of finished steel.
- Four-star green-rated steel (emission intensity between 1.6 and 2.0 tonnes of CO₂e/t of finished steel.
- Three-star green-rated steel (emission intensity between 2.0 and 2.2 tonnes of CO₂e/t of finished steel.

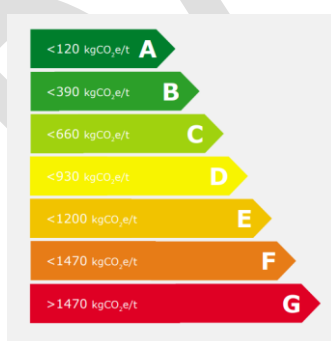
The threshold limit for defining the star rating of Green Steel will be reviewed every three years to ensure it remains aligned with the country's sustainability goals. The taxonomy also outlines the scope of emissions to be considered, including Scope 1, Scope 2, and limited Scope 3 emissions up to finished steel production. The National Institute of Secondary Steel Technology (NISST) will serve as the nodal agency for measurement, reporting, and verification (MRV) and will issue greenness certificates and star ratings on a yearly basis, with the option for more frequent MRV and certification if desired by the steel plants.

Construction Leadership Council of UK

In early 2024, the Construction Leadership Council published the "Five Client Carbon Commitments" to promote the use of green public procurement criteria. This proposal seeks to establish a strong push towards achieving NetZero in the construction sector. Objective 4 specifically targets the gradual reduction and eventual elimination of the most carbon intensive steel products in construction, such as reinforcement and structural steel.

As shown in Figure 169., the proposal includes a performance scale with seven bands. The scope of the carbon calculation includes the quarrying, mining and transport of raw materials, and the manufacture of hot rolled steel. Coatings, fabrication, transport to site and assembly are excluded from the system boundary considered. It is worth noting that offsetting is not permitted.

Figure 169. Performance scale for steel products



Source: Five Client Carbon Commitments Setting a path to Net Zero construction (Construction Leadership Council, 2024)

Other conceptual proposals

This section introduces the conceptual proposals developed by the Italian and Spanish Steel Federation, which share numerous similarities with some of the initiatives and standards presented thus far. However, it is important to note that these proposals do not constitute a science-based methodology.

Federacciai (Italian Steel Federation) Proposal

The Italian Steel Federation has proposed a dual system that combines a **carbon footprint index** and a **decarbonisation effort index** as a “conceptual exercise”.¹⁴⁵ The proposal defines green steel based on the

¹⁴⁴ <https://pib.gov.in/PressReleasePage.aspx?PRID=2083839>

¹⁴⁵ The technical methodology for this proposal has not been fully developed at this time. Therefore, it should be considered as a conceptual exercise rather than a fully detailed plan.

carbon footprint index (kgCO₂/ton steel) and introduces the decarbonisation effort index to track progresses towards green steel (percentage of achievement of decarbonisation objectives).

The **carbon footprint index** (Figure 170.) refers to the specific emissions of the steel product without considering production routes or technologies. The Italian Steel Federation suggests calculating it according to international standard methodologies, with specific detailed rules. It also proposes the possibility of defining carbon footprint indexes for different steel products, although it does not specify how these products should be categorized, such as based on grades of steel, application, functionality, etc.

Figure 170. Carbon footprint index proposed by Italian Steel Federation

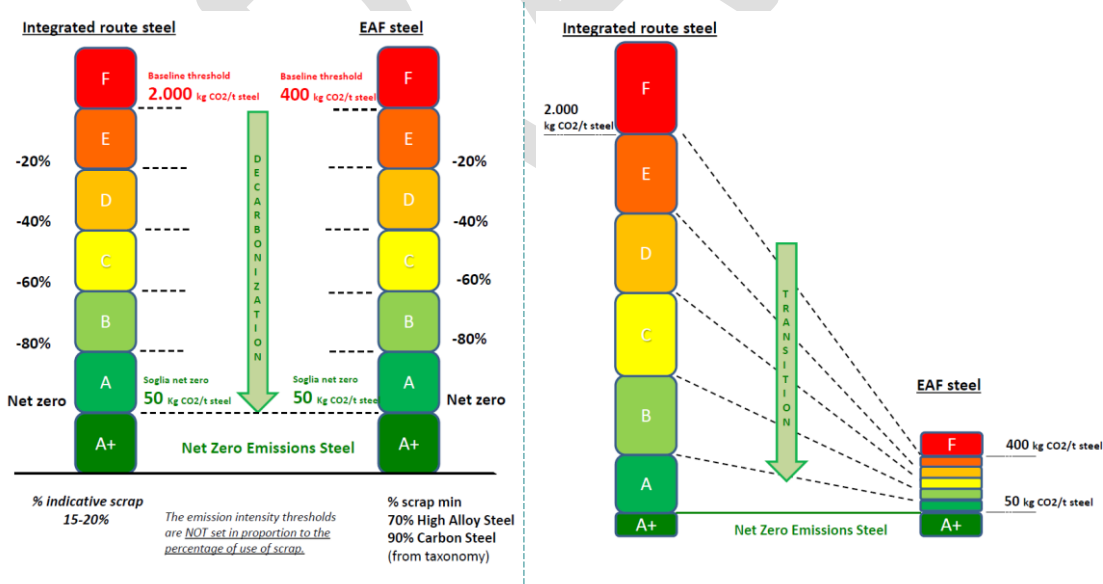


Note: purely indicative values for illustrative purpose
Source: Italian Steel Federation

This index is intended for **green procurement** purpose and to provide product information on environmental footprint to downstream user sectors. In contrast, the **decarbonisation effort index** is meant to be used to get incentives and access to sustainable funds, as well as to track progress towards climate neutrality and make it visible to the market.

As shown in Figure 171., the decarbonisation effort index presents different classes, which are established on the basis of the percentage of achievement of the climate neutrality objective (net zero threshold) and can be calibrated differently for integrated route (which includes also DRI or DRI/EAF technologies) and EAF according to the low carbon technologies applicable to each route. Emission intensity thresholds that define the classes are set at the EU sector level, with a **baseline threshold** representing the emission intensity of basic technology with "zero interventions" of decarbonisation. The **absolute threshold** of emissions intensity to achieve class A+ (net zero) is equivalent for both technology routes

Figure 171. Decarbonisation effort index by Italian Steel Federation



Note: purely indicative values for illustrative purpose
Source: Italian Steel Federation

Finally, an example of the proposed level is presented in Figure 172., which includes both index, the carbon footprint and the recycled content.

Figure 172. Example of label including index of the decarbonisation effort and carbon footprint index



Note: purely indicative values for illustrative purpose
Source: Italian Steel Federation

UNESID (Spanish Steel Federation) Proposal

The Spanish Steel Federation has recognised the imperative to transition towards a climate-neutral steel industry and has formulated a proposal for classifying steel products according to their environmental performance. The definition of classes of environmental performance refers exclusively to the CO₂e emissions, thus overlooking other environmental impacts and sustainable aspects such as social and governance factors.

UNESID's proposal entails a **dual approach**, distinguishing between different steel intermediate products. It establish two product families based on the presence of absence of competition between the BF-BOF and EAF routes for producing specific steel products. Table 78 presents the two family types.

Table 78. Families of products and methodological approach

Family Type	Definition	Products*	Methodological approach
Type 1	- Reinforcing and structural steel (long products) - EAF HA – fine “C” steel where a competence between the BF-BOF and EAF route exist.	Carbon hot rolled coil Seamless tubes Drawing Steel Railway material	Sliding scale (carbon footprint & recycling content)
Type 2	EAF HA – fine “C” steel that are produced through EAF route in the EU and do not compete with the integrated route.	Stainless steel High-alloy steel Specialty steels	Column scale (carbon footprint)

* Not exhaustive list, they are examples of products within these two families

Note. EAF HA – fine “C” steel definition is equivalent to the definition of EAF high alloy steel in Guidance Document n°9 on the harmonised free allocation methodology for the EU-ETS post 2020 (European Commission, 2019)

Source: UNESID

For products falling under family type 1, UNESID proposes a “sliding scale” methodology that classifies GHG emissions from the production phase as a function of the **metal recycled content**. This approach differs from methodologies developed by other organisations such as the IEA, FMC, Responsible Steel and WV Stahl, which consider the **share of scrap** in their calculation. The distinction between metal recycled and scrap is made because the production of one tonne of steel needs more than one tonne of scrap input. Conversely, for family type 2 of products, UNESID advocates for establishing thresholds based on a “column scale” that considers the product carbon footprint without factoring in the recycle content.

UNESID asserts that this differentiation is essential because products belonging to family type 2 are predominantly manufactured via the EAF route. Therefore, applying a sliding scale for these type of products

could result in steel items produced through the integrated route being classified in the same performance category as products with significantly lower carbon footprints. In this case there is no competence between the route and the available technology allows the production via EAF route.

Similar to VWStahl's proposal, this method follows a bottom-up approach based on virtual references plants. However, UNESID suggests modelling the reference plants as the average of the EU steel mills and consider the EU average electricity mix when calculating Scope 2 emissions (average European emission used to define the sectoral references within the EU ETS). These adjustments will result in different threshold values, which have not been yet calculated.

Regarding emissions under scope and system boundaries, the classification system adopts a cradle-to-gate approach, excluding downstream emissions and considering hot rolled steel level of processing or other hot thermomechanical transformation.

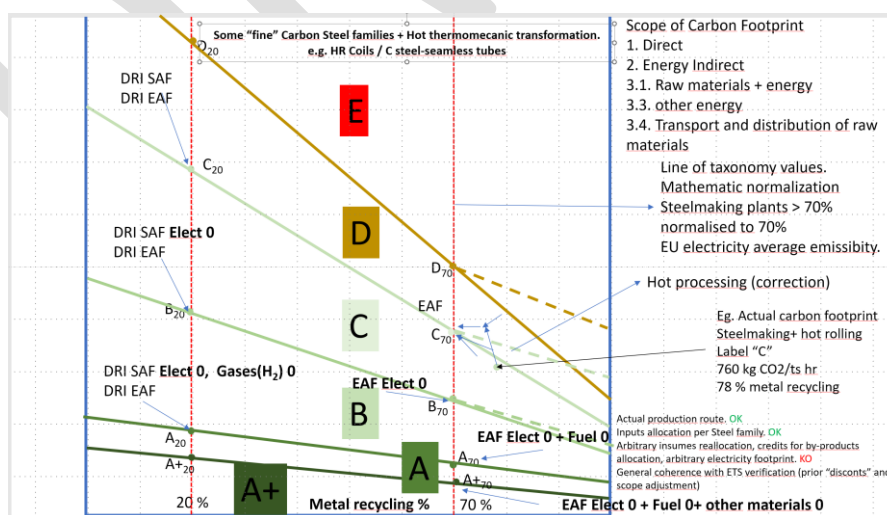
While emissions under scope are the same as in VWStahl's proposal, UNESID permits carbon offsetting by purchasing emissions rights but does not allow the application of credits for substances and energies that are dispensed beyond the accounting limits (slag, use of domed gases for generation of electricity or heat and energy supplied to third parties). This is to prevent double counting of emissions and claimed reductions across multiple sectors. For instance, the cement industry could also claim the credit for the use of slag.

UNESID argues that this could discourage the cement industry from purchasing slag, as the manufacturer would not be able to claim the credit, thereby undermining the pursuit of environmental objectives. Therefore, unless accounting methodologies are harmonized across sectors, they proposed not to apply credits.

As shown in Figure 173., the sliding scale methodology proposed by UNESID includes specific adjustments to WStahl's proposal, notably:

- Correction of the sliding with regard to GHG emissions from hot rolling. In the integrated route, co-product gases from the coke oven, blast furnace (BF) and basic oxygen furnace (BOF) are used, saving on additional fossil fuel and energy resources for the hot rolling phase. This does not occur when producing steel via the EAF necessitating extra energy sources for the hot rolling process and resulting in higher CO₂ emissions. Consequently, the regression is modified to consider it.
- Inclusion of a higher performance class "A+". There is biogenic CO₂ inherent in the inputs essential for the steelmaking process, such as electrodes and foamers. Currently, there is no available technology to eliminate CO₂ emissions associated with process inputs, thus the carbon footprint cannot be reduced to zero. Theoretically, this could be resolved through carbon offsetting, thereby achieving an A+ environmental performance.

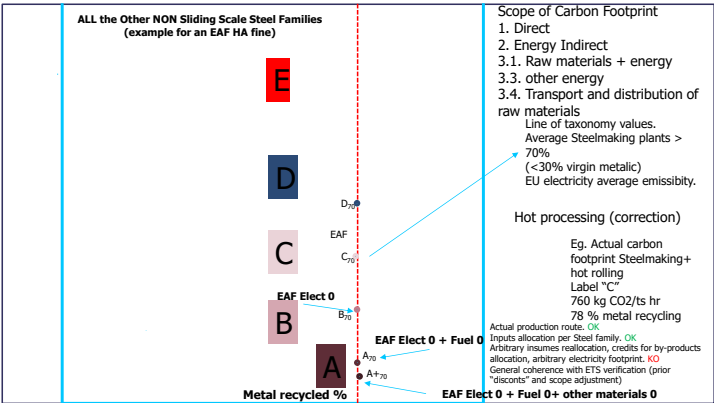
Figure 173. Conceptual categorisation proposed by UNESID for family of products type 1



Source: UNESID definition of "green steel"

The categorisation for family of products type 2 follows the same methodological approach, however only considers the carbon footprint of steel products when setting the thresholds. As shown in Figure 174. the thresholds are determined based on the emissions calculated for reference plants, with 70% of metal recycling content and 30% of virgin material.

Figure 174. Conceptual categorisation proposed by UNESID for family of products type 2



Source: UNESID definition of "green steel"

In a subsequent phase, UNESID proposes to expand the method to cover downstream processing, thereby broadening the system boundaries to include steel finished products. To achieve this, new families of steel intermediate products with reasonably similar emissions profiles, meaning that they undergo similar thermomechanical treatments, need to be established.

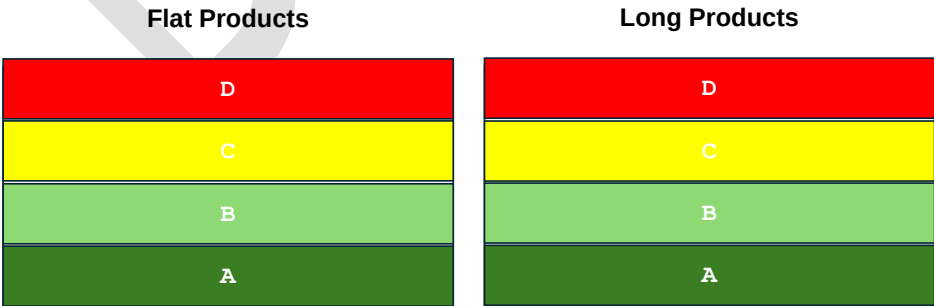
Sandbag Proposal

Sandbag is a non-profit climate change think tan that has recently published a report called "From niche to mainstream: Shaping demand for green steel" (Zaccaro, 2024), which address the issue of "green steel definition" and raise concerns on the limitations of the sliding scale approach. As part of this report, Zaccaro (2024) presents an alternative conceptual proposal to the sliding scale, with two distinct tiered sets of green steel classification for flat and long products, focusing solely on embedded emissions. The author suggest that this approach would allow steel production routes to openly compete for lead markets, allowing the most cost-competitive to emerge as the most viable for sectoral decarbonisation.

The proposal argues that the sliding scale discriminate against scrap, which should be seen as a complementary tool for decarbonisation. It also recognises that flat products are predominantly produced via BF-BOF, resulting in higher carbon intensity compared to scrap-based EAF long products. Therefore, implementing a distinct scale for flat and long products could incentivize the decarbonisation of both routes and allow for fair product comparability in the EU market.

Therefore, the approach also introduce pressure under EAF long products producers. This encouragement becomes crucial as, in the EU, long steel producers already employ 100% scrap and electricity, leaving limited avenues for further emission reduction. However, there are opportunities for additional abatement within the EAF route, depending on the electricity procurement strategy of individual steel plants.

Figure 175. Sandbag Proposal



Source: JRC elaboration based on Sandbag

Company Initiatives

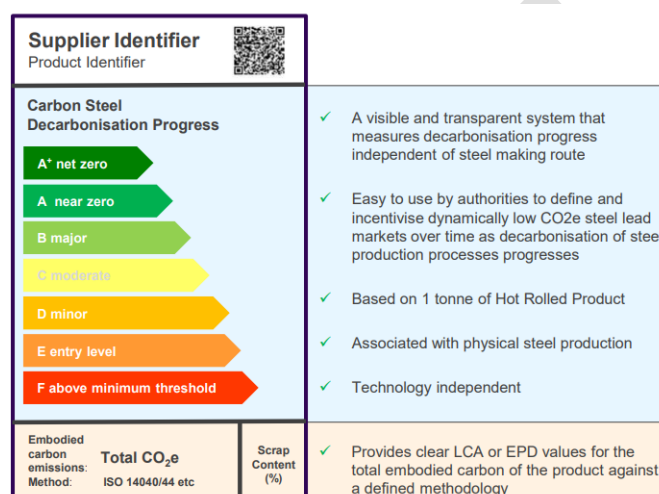
The following section offers a **non-exhaustive** compilation of initiatives undertaken by private companies, who are actively developing their own green steel brands to provide clients with options that meet their growing environmental demands.

Arcelor Mittal Low-Carbon Emissions Steel Proposal

The **ArcelorMittal Low Carbon Emissions Steel Standard** is a conceptual proposal that adopts a product-level approach, measuring the carbon footprint of hot rolled steel produced. This proposal features a dual score system comprising a LCA value for finished products (EPD for construction products) (see Figure 176.) as well as a decarbonisation rating system (see Figure 177.) that categorises low and near-zero carbon emissions per tonne of hot rolled steel, offering incentives to producers as they progress in their decarbonisation efforts.

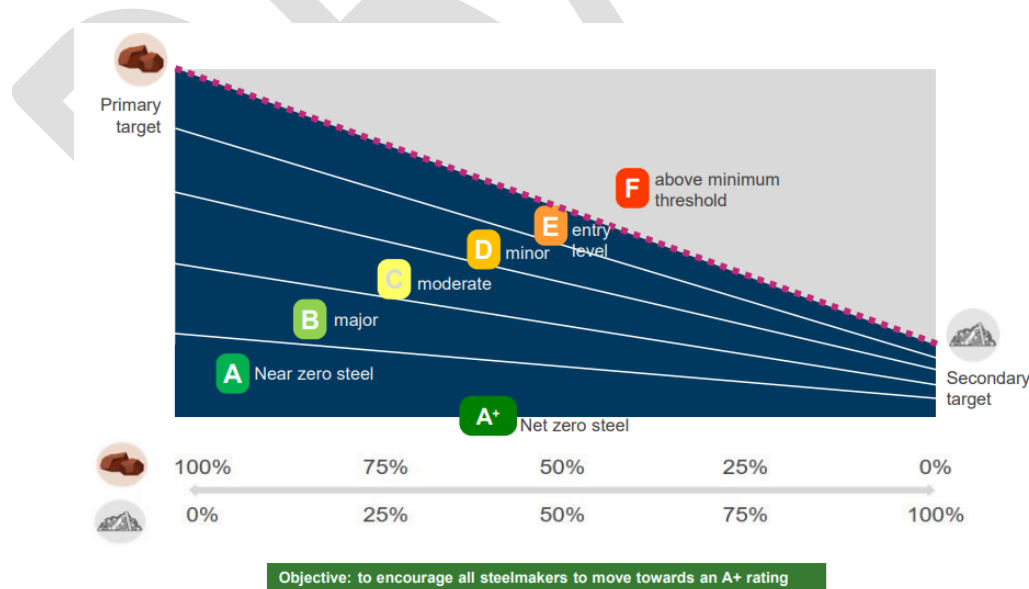
Of paramount importance, the proposal is designed to incentivize the decarbonisation of all methods of steel production through technological shifts, rather than solely relying on increasing scrap rates using existing technology. This is achieved through the implementation of a sliding scale based on the percentage of scrap used in production, a system that aligns with the principles of the ResponsibleSteel™ and IEA proposals.

Figure 176. Label proposed by ArcelorMittal



Source: Establishing a global standard for low-carbon emissions steelmaking (ArcelorMittal, 2022)

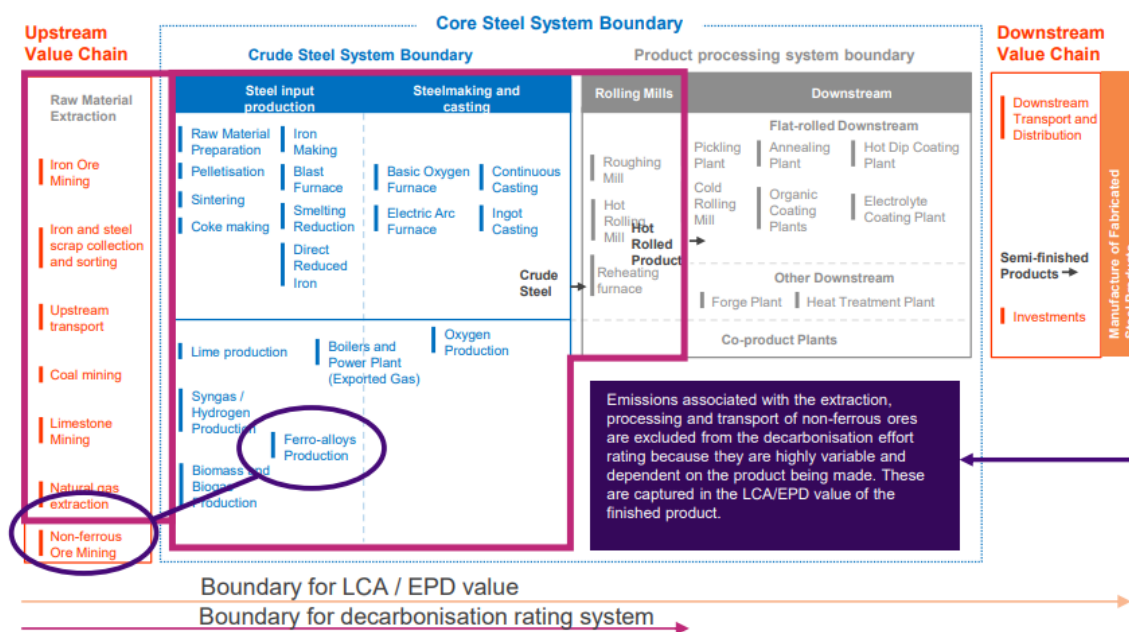
Figure 177. Scoring system proposed by ArcelorMittal



Source: Establishing a global standard for low-carbon emissions steelmaking (ArcelorMittal, 2022).

As shown in Figure 178., the conceptual proposal includes a defined system boundary for accounting CO₂e emissions. In order to simplify the analysis, the proposal excludes emissions associated with the extraction, processing and transport of ferroalloys, making therefore all steel products directly comparable up to this boundaries.

Figure 178. System boundary for hot rolled product



Source: Establishing a global standard for low-carbon emissions steelmaking (ArcelorMittal, 2022)

In addition, ArcelorMittal introduced the **Xcarb™** in 2020, which is the umbrella brand that brings together all of ArcelorMittal's reduced, low and zero-carbon products and steelmaking activities, as well as wider initiatives such as the **Xcarb™ Innovation Fund Accelerator Programme**.

As part of this portfolio, the company offers **Xcarb™ green steel certificates**, which enables customers to report an equivalent reduction in their Scope 3 emissions of their flat products, in accordance with the GHG Protocol. These certificates are based on CO₂ savings initiatives that have been independently verified by DNV, and the savings are calculated according to the GHG Protocol for Project Accounting. Customers who purchase Xcarb™ green steel certificates receive the corresponding CO₂ savings.

Furthermore, ArcelorMittal's **XCarb™ recycled and renewably** produced label is designated for steel produced in an EAF using high levels of scrap and 100% renewable electricity. The electricity used is independently verified and has a 'Guarantee of Origin' ensured by the purchase of 'Renewable Energy Certificates'. The company's long products business was the first to offer low-carbon emissions steel in 2021, starting with sections and merchant bars, followed by sheet piles and rebar. As of March 2022, ArcelorMittal Europe – Flat Products has also commenced producing XCarb® recycled and renewably produced steel using an EAF and renewable electricity. This label has been applied to specific hot rolled coils, organic coated steels, Magnelis®, and press hardenable steel (PHS) grades.

Klöckner & co

The German company, a leading player in the global steel industry, introduced its CO₂-reduced steel brand **Nexigen®** in 2022. Under the Nexigen® brand, the company offers carbon-reduced solutions in three categories of materials, steel, stainless steel and aluminium.

To facilitate the comparison of their products and provide a definition of green steel, the company has developed **Nexigen® Categorization**. This system classifies CO₂-reduced steel products into five categories based on the product's actual carbon footprint from resource extraction and production to the point where they leave the warehouse.

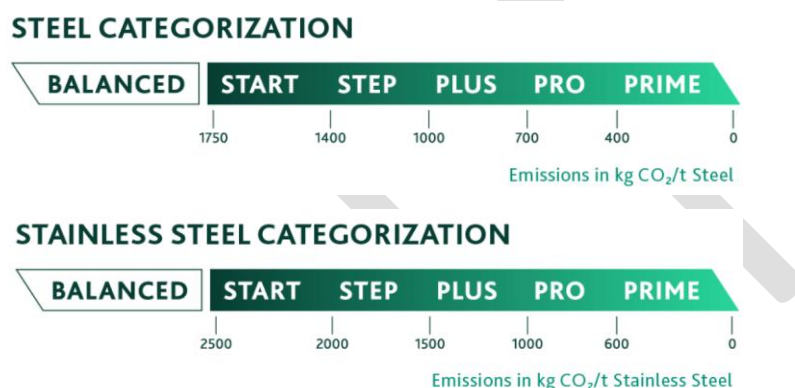
The system boundaries for CO₂ accounting considers all the value chain steps, from extracting raw materials to the steelmaking process itself (Scope 1, Scope 2 and Scope 3 upstream). This cradle-to-gate assessment enables comparisons among intermediate products at the hot rolled steel level of processing. The methodology for setting the threshold values in the categorisation system are determined based on studies conducted by the Boston Consulting Group, which have modelled the emission intensity of different production technologies for steelmaking decarbonisation (BCG, 2023), similar to the virtual reference plants modelled by the German Steel

Association. However, the company does not disclose detailed information on how threshold values are calculated, failing to provide a transparent methodology despite it is validated by TÜV SÜD.

As shown in Figure 179., the categorisation considers two categories of steel, labelled “steel” and “stainless steel”. The introduction of a different categorisation scale for stainless steel is due to the differing emission intensity of this range of steel grades compared to non-alloy steel products.

For the “steel” category, classes of performance range from START (with a footprint from 1,750 to 1,400 kg CO₂ per metric ton) to PRIME (400 to 0 kg CO₂ per metric ton). Similarly, for the category of stainless steel, classes of performance range from START (with a footprint from 2,500 to 2,000 kg CO₂ per metric ton) to PRIME (600 to 0 kg CO₂ per metric ton) (see Figure 179.). The classification also introduces the category “balanced”. Balancing involves first taking the facility’s total CO₂ emissions into account and then calculating the average carbon emissions per metric ton of the manufactured product. Reduced emissions are thus allocated to limited volumes, which reduces the carbon footprint balance.

Figure 179. Nexigen® Categorisations for steel and stainless steel by Klöckner & co



Source: Product Categorizations (Klöckner & co, n.d.)

The company has also developed its own methodology for calculating product carbon footprints (PCFs), known as the **Nexigen® PCF Algorithm**. This calculation method has been validated by TÜV SÜD and is in conformity with the applicable requirements of ISO 14067, ref. 14040 and 14044 standards, as well as the GHG Emissions Protocol. However, a certified PCF is not issued for each steel intermediate product, unless it is requested by the client.

SSAB Zero™ and SSAB Fossil-free™

In Sweden, SSAB has taken the lead as the first steel producer to introduce a scrap-based steel product that could meet various near-zero emission steel production thresholds and has the aim to deliver fossil-free steel to the market in 2026.

The company is actively advancing the HYBRIT project together with partners LKAB and Vattenfall. HYBRIT replaces the traditional coking coal used in iron ore-based steelmaking with fossil-free electricity and hydrogen, resulting in steel production with virtually no carbon footprint.

SSAB launched **SSAB Zero™** at a commercial scale in 2023 and plans to make **SSAB Fossil-free™** steel available on the market in 2026. Both SSAB Zero™ and SSAB Fossil-free™ steel are produced with very low carbon emissions, utilising fossil-free electricity and biogas. **SSAB Zero** is based on scrap, while **SSAB Fossil-free** steel is based on iron-ore produced without fossil fuels, using HYBRIT® technology, which uses hydrogen instead of coal in the iron ore reduction process, emitting water instead of CO₂.

Both SSAB Zero™ and SSAB Fossil-free™ steel are considered emission-free steels in terms of own operations (Scope 1 and 2). When considering end-of-life, the carbon footprint of SSAB Fossil-free steel is significantly lower than that of SSAB Zero. The carbon emissions of both steels are less than 0.05 kg CO₂e per kg/steel (Scope 1 and 2). When considering also Scope 3 upstream (including raw materials) the carbon footprint target value is 0.3 kg of CO₂e per kg/steel.

Stahlo Classification Level

Stahlo, one of the largest plant-independent steel service centres in Germany, has developed a **classification label** to inform their consumers about the environmental performance of its product portfolio. This label is designed to facilitate product comparability and aid clients in decision-making processes, rather than being a scientific-based methodology for establishing environmental performance classes.

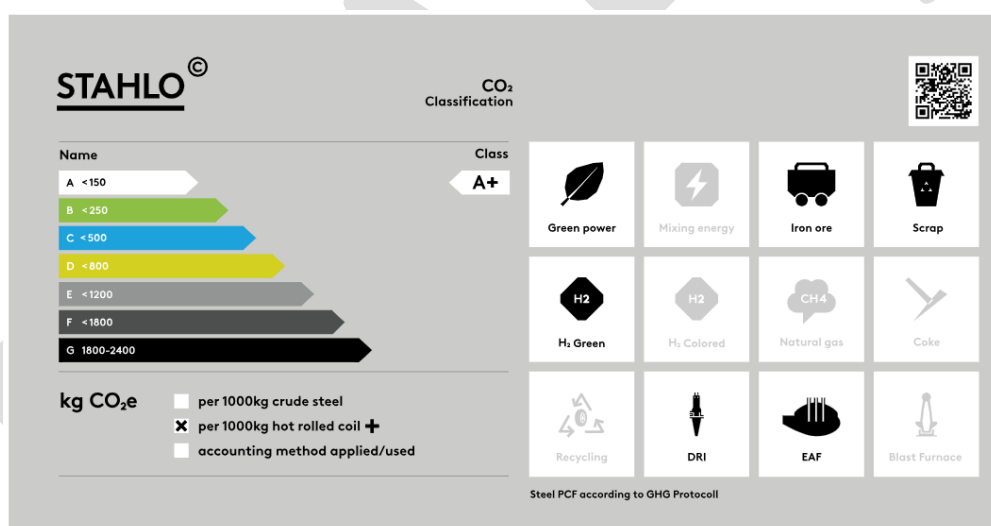
The label includes a CO₂ classification and indicates the inputs used for steelmaking, as shown in the right side of Figure 180.. The classification system consists of seven classes, ranging from G to A, that categorise products based on their carbon footprint. Threshold values for each performance class have been established using survey data provided by the steelmakers supplying the company, serving as a virtual reference based on their product portfolio.

The classification system allows comparison at the **crude steel** processing level and considers Scope 1 and Scope 2 GHG emissions. In addition, the method assigns a "+" (plus) to the class for products that undergo additional processing, such as hot rolled coil.

It is essential to emphasise that the system is specifically designed for comparing carbon steel non-alloy products, which are the exclusive grades distributed by Stahlo. These products include hot rolled coils, cold rolled coils, hot-dip galvanized coils, and electrolytically galvanized coils.

In addition to the classification system, Stahlo also provides clients with **information on the carbon footprint** of their products, calculated according to international standards such as GHG Protocol and EN ISO 14067:2018, accounting for Scope 1, Scope 2, and partially Scope 3 (upstream) emissions from a cradle-to-customer gate approach.

Figure 180. Stahlo's Classification Label



Source: Classification Label (Stahlo, n.d.)

1.4.6.1 Swiss Steel Group – Green Steel Portfolio

The Swiss Steel Group has established a comprehensive portfolio of green steel products, catering to the growing demand for sustainable steel solutions. This portfolio comprises three distinct categories: **Green Steel**, **Green Steel Climate+** and **Green Steel Stainless+**. As depicted in Figure 181., these products showcase varying levels of ambition in reducing greenhouse gas emissions, with corresponding carbon footprint reductions. For instance, the carbon footprint of a bearing steel and a rolled bar can be significantly reduced when using 100% renewable electricity, with decreases of 33% and 31%, respectively.

Figure 181. Swiss Steel Group's green steel product portfolio

Product-categories	Current offer			Offer under development	
	 Green Steel	 Green Steel Climate+	 Green Steel Stainless+	 Green Steel CLIMATE+ H2	 Green Steel NET-ZERO
Description	Swiss Steel Group produces steel exclusively with EAF technology and promotes recycling and the circular economy. The result is a total product offering with significantly lower carbon emissions compared to the industry average.	Green Steel Climate+ comprises products with a further reduced carbon footprint. These products are produced exclusively with electricity from renewable sources, such as wind, water or solar.	Green Steel Stainless+ is aligned with our stainless steel portfolio. It offers our customers a stainless steel product produced with a scrap content of at least 95%. ¹	Green Steel Climate + H2 includes not only products produced exclusively with energy from renewable sources and green hydrogen. ²	The Green Steel Net Zero product offers customers steel with fully compensated carbon emissions.

Scope 1+2+3 emissions of exemplary grades kg CO2 per ton finished product			
INDICATIVE	 Green Steel	 Green Steel Climate+	 Green Steel Stainless+
Bearing steel 100Cr6	~1050	~700	-
Rolled bar 1.7003	~320	~220	-
Stainless 4441	~8700	~8500	~1000

Source: Green Steel World – Decarbonisation in a European context (Swiss Steel Group, 2023)

thyssenkrupp – bluemintTM Steel

In 2021, thyssenkrupp has introduced its **bluemint[®] Steel brand**, offering high-quality flat steel products with a reduced environmental impact. The basis for today's bluemint[®] Steel products are CO₂ reductions in the existing production environment. The production of **bluemint[®] recycled** involves the use of 100% processed high-quality steel scrap, replacing a portion of iron ore, coke and coal in the blast furnace. bluemint[®] recycled can save up to 64% of specific CO₂ emissions compared to conventionally produces steel products.

Additionally, the company has launched a further product under this brand, the **bluemint[®] pure**. By either using sponge iron (HBI) in the blast furnace process or replacing a part of natural gas with biomethane in the hot rolling process, bluemint[®] pure can save 1.5 tonnes of CO₂ compared to conventionally produced steel. Both products take a balance sheet approach, which means that CO₂ savings in the production processes are allocated to a portion of steel production. For both products, the emissions reductions are externally verified and certified by TÜV SÜD and DNV, respectively.

Annex 6. Current and 2035 base cases definition methodology

2022 Base cases:

The definition of the 2022 base case required several steps of data gathering and assumptions. The next paragraphs summarise the procedure followed to estimate each parameter presented in the table:

- 1. Production:** EU production figures were derived from the market analysis (Task 2). Several databases were used to complement the production values: CRU platform data, Eurofer statistics, World Steel statistics and Prodcum data. All the values refer to the year 2022 and it was assumed that this year would be representative of the current base case. At the time of the analyses performed in Task 2 this was the year with the most complete dataset.
- 2. Shares of imports and apparent consumption:** Share of imports and apparent consumption were calculated based on Task 2 corrected with few updates from Prodcum and Comext data bases;
- 3. The shares of alloy and non-alloy steel** were calculated taking into account prodcom data available for each of the intermediate products. However, as highlighted by stakeholders the current distinction between alloyed and non-alloyed is defined according with EN 10020 which allows for the inclusion in the classification of alloyed steel grades with relatively small alloy content. Non-alloy are steel grades in which none of the following limit values is reached by any specified element as defined in according to EN 10020:

1.65% manganese	0.08% molybdenum
0.60% silicon	0.06% niobium
0.40% copper or lead	0.0008% boron
0.30% chromium, nickel, aluminium, cobalt or tungsten	0.10% of any other alloy except carbon, nitrogen, phosphorus or sulphur
0.10% bismuth, selenium, tellurium, vanadium	

Therefore, estimating the composition of alloyed elements in the steel EU consumption is very challenging. This represents an important data gap that needs dedicated work in future studies.

- 4. Shares of EU production, divided between DRI, BF-BOF and EAF** were estimated based on several sources: 1) EUROFER data supplied to the JRC; 2) Information gathered from CRU platform on RP production volumes, which includes information on the facilities producing RP; 3) Confirmation and update of the estimates based on the global energy tracker, which allowed to improve the accuracy of the information collected from sources 1 and 2.
- 5. Ferrous scrap:** information was collected from 2024 EUROFER statistics.
- 6. The recycled content** was estimated based on the estimated shares of EAF/BOF and the data provided by EUROFER.
- 7. Post-consumer recycled content** was estimated based on the shares of EAF/BOF and the data supplied by EUROFER. These values overestimated the post-consumer sources as they compile material flows that include both post-consumer scrap and pre-consumer scrap. This flows are used as input to the production from an external source (not home-scrap). According to several stakeholders consulted, the split between pre and post-consumer may be captured by the following order of magnitude: 1) EAF scrap content: 1/3 pre and 2/3 post and 2) BF-BOF scrap content: 3/4 pre and 1/4 post. These figures aims to provide first estimates despite potential uncertainties linked to this.
- 8. Shares of grain and non-grain oriented electrical sheet** were calculated based on Prodcum and Comext data sources. Using the CN codes 72251100 and 72261100 for grain oriented and the codes 72091610, 72091710, 72091810, 72092610, 72092710, 72092810, 72112320, 72251990, 72261980 for non-grain oriented. According with stakeholders the increase relevance of electrical sheet products created the necessity of developing new classification and new CN codes will be implemented in the future.
- 9. EU supply mix,** was estimated considering both imports and exports of the EU27 from and to external partners. Data was collected from Comext database for the selection of representative products CN codes (selection done for the market analysis presented in section 2, Task 2).

10. **BOF/EAF shares in the imports** of the representative products, were calculated based the weighted average of the amount of imports (data from COMEXT using the 8 digit CN codes) and the production of each RP in the EU trade partners (information on production facilities came from CRU platform and, as before, it was updated for each production facility using global energy tracker).

2035 Base cases:

The next paragraphs summarise the procedure followed for the development of the 2035 base cases:

1. **Production and apparent consumption:** Two scenarios of steel production figures were considered in the development of the 2035 base case: 1) increase of steel production to 164 Mt following the forecast for 2030 presented in the Sandbag forecast scenario for 2030 (Su and Assous, 2022) (Scenario 1 Table 47 and Scenario 3 Table 80); 2) constant production equal to the 2022 (Scenario 2 Table 79 and Scenario 4 (*)) Copper tolerance limit are displayed in italic with "<", specific grades of electrical steel may have different impurity tolerances; (**) Considering ferro-alloys have 0% recycled content.

Source: JRC elaboration

2. Table 48). The representative products production values, presented in the tables, were calculated assuming a production growth promotional to steelmaking. In the case of Iron ore and Iron making it was assumed that there was no change between the 2022 value and the 2035 value, because there was no change on EU extraction since several years. For **Ferrous scrap** the value was obtain from estimates presented by Eurofer (EUROFER AISB, 2019). Scrap value doesn't represent production but scrap availability, consumption of scrap was calculated based on the recycled content shares of BOF, EAF and DRI-EAF production assuming a conversion yield of 90%. Iron consumption was calculated based on the total steel production and scrap use, assuming also a yield of 90%.
3. **Shares of imports, alloy and non-alloy steel and grain and non-grain oriented electrical sheet** were assumed to stay at the same values of 2022 for the majority of the steel materials except for Iron and ferrous scrap. In the case of these products the shares of imports were calculated based on production and consumption values.
4. Shares of EU production, divided between **DRI, BF-BOF and EAF** were estimated based on EU values of 2022 and corrected with information of announced and under construction facilities, which will be ready by 2035. Two scenarios were considered for the determination of these production shares: 1) considering that all existing facilities would continue to operate without changes in the capacity of BF-BOF (Scenario 1 Table 47 and Scenario 2 Table 79); 2) considering that EU BF-BOF facilities with more than 50 years would be closed before 2035 (Scenario 3 Table 80 and Scenario 4 Table 48).
5. **Recycled content:** The starting point of the estimates was the assumption of a maximum of 25% of recycled content in BOF and DRI-EAF production. The estimates of recycled content for each representative products were calculated based on the shares of DRI, BF-BOF and EAF used in their production (described in point 3) and the recycled content in the slabs. The average scrap ratio per unit of steel produced is the total scrap consumed divided by the total crude steel production per year. This ratio overlooked potential losses due to process efficiencies which are consider equal for scrap inputs and other input flows, estimated over 10% for both EAF and BF-BOF.
6. **Post-consumer recycled content.** The estimates of post-consumer recycled content for each representative product were calculated based on the shares of DRI, BF-BOF and EAF (described in point 3) used in their production and the post-consumer recycled content in the slabs.
7. **EU supply Extra EU** Estimates of production shares between DRI, BF-BOF and EAF were corrected for the 2035 base case scenario using information from announced and under construction facilities (Global Energy Monitor, s.d.). This estimates assumed that all existing facilities will still be in operation in 2035 (scenario 2035a in the supply mix section 3.3.1.1). As for the EU production base case a scenario was constructed considering the closure of all BF-BOF facilities with more than 50 years (scenario 2035a in the supply mix section 3.3.1.1)

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Table 79. 2035 scenario 2 Baseline representing estimates of a near future perspective of the EU Iron and Steel sector according with production, consumption, recycling and trade from extraction to the production of the five representative products.

	Fe materials			Steel		Selected representative products						Reference	
	Iron ore	Iron	Ferrous scrap	Ferro-Alloys	Carbon Steel	HOT ROLLED COIL	GALVANISED COLD ROLLED COIL	WIRE ROD	ELECTRICAL SHEET		STAINLESS S STEEL		
	Grain orientated	Non -Grain orientated											
EU 27 Aparent consumption (Mt)	107	64.3	88.7	3.4	150.9	31.4	16.2	20.6	0.4	1.4	5.1	[1,3,5,7]	
Share of Imports in the EU cons. (%)	77%	7%	4%	74%	27%	32%	27%	13%	35%	32%	40%		
% Import reliance	64%	-11%	-21%	65%	14%	15%	13%	8%	-11%	24%	18%		[1,2]
EU 27 Production (Mt)	38	71.4	120	1.2	137.7	26.3	14.7	18.9	0.43	1.08	4.2	1,3,5,7,10,11	
share of alloys steel (%)	-	-	-	-	19%	20%	40%	16%	100%	60%	0%	[1,5,6,7]	
share of non-alloys (%)	-	-	-	-	77%	80%	60%	84%	0%	40%	0%		
share of stainless(%)	-	-	-	-	5%	-	-	-	-	-	100%		
Production routes (EU27)													
DRI-EAF	-	30%	-	-	16%	34%	20%	8%	-	-	-	[1,3,4,9,12]	
BF-BOF	-	70%	-	-	40%	51%	68%	15%	99%	99%	0%		
EAF	-	-	-	-	44%	15%	12%	77%	1%	1%	100%		
Scrap use / recycled content in EU production													
Scrap content coming from BF-BOF	-	-	15.3	-	25%	13%	17%	4%	24%	24%	-		
Scrap content coming from EAF	-	-	67.3	-	100%	14%	11%	73%	1%	1%	80-95%**		
Scrap content in DRI-EAF	-	-	6.1	-	25%	9%	5%	2%	-	-	-		
average scrap content in product	-	-	100%	-	64%	36%	33%	79%	25%	25%	80-95%**		
Post consumer (up to)													
Scrap content coming from BF-BOF (Post consumer)	-	-	9.2	-	15%	8%	10%	2%	14%	14%	-		
Scrap content coming from EAF (Post consumer)	-	-	50.5	-	75%	11%	9%	58%	8%	8%	60-80%		
Scrap content in DRI-EAF Post consumer	-	-	6.1	-	15%	5%	3%	1%	-	-	-		
average post-consumer scrap content in product	-	-	100%	-	48%	24%	22%	61%	22%	22%	60-80%		
Estimated average Cu-concentration*	n/a	0.01%	0.30%	<0.1%	<0.08%	<0.1%	<0.08%	<0.25%	<0.1%	<0.1%	n/a	[8]	

[1] Market analyses of chapter 2; [2] RMIS – Raw Materials Information System; [3] CRU platform; [4] EUROFER Bill Of Materials; [5] Prodcum and Comext databases; [6] (Georgitzikis, 2023); [7] Eurofer, European steel in Figures (2024); [8] (Dworak, Rechberger, and Fellner, 2022) and (Daehn, Cabrera Serrenho, and Allwood, 2017); [9] (Systemiq and (commissioned by) Thyssenkrupp Steel Europe, 2023); [10] (Su and Assous, 2022); [11] targeted stakeholders consultation; [12] (Global Energy Monitor, s.d.).

* Cu tolerance limit are displayed in italic with "<"; ** Considering alloys have 0% recycled content.

Table 80. 2035 scenario 3 Baseline representing estimates of a near future perspective of the EU Iron and Steel sector according with production, consumption, recycling and trade from extraction to the production of the five representative products.

	Fe materials			Steel		Selected representative products						Reference	
	Iron ore	Iron	Ferrous scrap	Ferro-Alloys	Carbon Steel	HOT ROLLED COIL	GALVANISED COLD ROLLED COIL	WIRE ROD	ELECTRICAL SHEET		STAINLES S STEEL		
	Grain orientated	Non -Grain orientated											
EU 27 Aparent consumption (Mt)	107	68.9	109.3	4.1	189.7	36.8	20.1	24.5	0.5	1.7	6.1	[1,3,5,7]	
Share of Imports in the EU cons. (%)	77%	7%	4%	74%	27%	27%	21%	11%	35%	32%	34%		
% import reliance	64%	-4%	-17%	65%	14%	15%	13%	8%	-11%	24%	18%		[1,2]
EU 27 Production (Mt)	38	71.4	120	1.4	164	31.3	17.5	22.5	0.51	1.29	5.0	[1,3,5,7,10,11]	
share of alloys steel (%)	-	-	-	-	19%	20%	40%	16%	100%	60%	0%	[1,5,6,7]	
share of non-alloys (%)	-	-	-	-	77%	80%	60%	84%	0%	40%	0%		
share of stainless(%)	-	-	-	-	5%	-	-	-	-	-	100%		
Production routes (EU27)													
DRI-EAF	-	38%	-	-	18%	50%	27%	9%	-	-	-	[1,3,4,9,12]	
BF-BOF	-	62%	-	-	32%	22%	57%	9%	99%	99%	0%		
EAF	-	-	-	-	50%	28%	16%	82%	5%	5%	100%		
Scrap use / recycled content in EU production													
Scrap content coming from BF-BOF	-	-	14.6	-	25%	6%	15%	3%	24%	24%	-	[1,3,4,9,12]	
Scrap content coming from EAF	-	-	86.6	-	95%	14%	11%	73%	5%	5%	80-95%**		
Scrap content in DRI-EAF	-	-	8.2	-	25%	15%	7%	3%	n/a	n/a	-		
average scrap content in product	-	-	100%	-	67%	36%	33%	79%	29%	29%	80-95%**		
Post consumer (up to)													
Scrap content coming from BF-BOF (Post consumer)	-	-	8.7	-	15%	4%	9%	2%	14%	14%	-	[8]	
Scrap content coming from EAF (Post consumer)	-	-	68.3	-	75%	11%	9%	58%	8%	8%	60-80%		
Scrap content in DRI-EAF Post consumer	-	-	8.2	-	15%	9%	4%	2%	-	-	-		
average post-consumer scrap content in product	-	-	100%	-	52%	24%	22%	61%	22%	22%	60-80%		
Estimated average Cu-concentration*	n/a	0.01%	0.30%	n/a	0.1%	<0.1%	<0.08%	<0.25%	<0.1%	<0.1%	n/a		

[1] Market analyses of chapter 2; [2] RMIS – Raw Materials Information System; [3] CRU platform; [4] EUROFER Bill Of Materials; [5] Prodcum and Comext databases; [6] (Georgitzikis, 2023); [7] Eurofer, European steel in Figures (2024); [8] (Dworak, Rechberger, and Fellner, 2022) and (Daehn, Cabrera Serrenho, and Allwood, 2017); [9] (Systemiq and (commissioned by) Thyssenkrupp Steel Europe, 2023); [10] (Su and Assous, 2022); [11] targeted stakeholders consultation; [12] (Global Energy Monitor, s.d.).

* Cu tolerance limit are displayed in italic with "<"; ** Considering alloys have 0% recycled content.

Table 81. Hot rolled coil 2022 EU supply mix and shares of production using technologies BF-BOF, DRI-EAF and EAF in each of the EU trade partners. These shares were estimated based on actual hot rolled coil production in each of the countries.

Country	Supply mix	Share BF-BOF	Share DRI-EAF	Share EAF
Imports				
JP Japan	1049	96%	0%	4%
TR Türkiye	1056	63%	0%	37%
TW Taiwan	845	94%	0%	6%
KR Korea	705	100%	0%	0%
IN India	652	85%	6%	0%
EG Egypt	529	0%	100%	0%
UA Ukraine	494	100%	0%	0%
Other	2449			
Total	7779			
Exports				
TR Türkiye	682			
US United States	456			
GB United Kingdom	386			
CH Switzerland	232			
EG Egypt	203			
MX Mexico	136			
IN India	6			
Other	515			
Total	2616			

Table 82. Hot rolled coil 2035 EU supply mix considering no changes from 2022 imports and exports. Shares of production using technologies BF-BOF, DRI-EAF and EAF were updated considering announced and under construction facilities in each EU trade partner and assuming that there are no facilities closing until 2035.

Country	Share BF-BOF	Share DRI-EAF	Share EAF
Imports			
JP Japan	94%	0%	6%
TR Türkiye	59%	0%	41%
TW Taiwan	90%	0%	10%
KR Korea	95%	5%	0%
IN India	100%	0%	0%
EG Egypt	0%	94%	6%
UA Ukraine	100%	0%	0%
Other			

Table 83. Galvanised cold rolled coil 2022 EU supply mix and shares of production using technologies BF-BOF, DRI-EAF and EAF in each of the EU trade partners. These shares were estimated based on available data on galvanised cold rolled coil production in each of the countries.

Country	Supply mix (kt)	Share BF-BOF	Share DRI-EAF	Share EAF
Imports				
TR Türkiye	901	8%	0%	92%
VN Viet Nam	846	64%	0%	32%
KR Korea	795	100%	0%	0%
CN China	659	99%	0%	1%
TW Taiwan	487	64%	0%	36%
GB United Kingdom	271	0%	0%	0%
RU Russia	162	100%	0%	0%
Others	987			
Total	5108			
Exports				

GB United Kingdom	523	
TR Türkiye	499	
MX Mexico	294	
US United States	209	
CN China	155	
VN Viet Nam	5	
KR South Korea	6	
Other	1020	
Total	2713	

* estimated based on the country global steel production

Table 84. Galvanised cold rolled coil 2035 EU supply mix considering, assuming no changes from 2022 imports and exports. Shares of production using technologies BF-BOF, DRI-EAF and EAF were updated considering announced and under construction facilities in each EU trade partner and assuming that there are no facilities closing until 2035.

Country	Share BF-BOF	Share DRI-EAF	Share EAF
Imports			
TR Turkey	4%	0%	96%
VN Viet Nam	64%	0%	36%
KR Korea	95%	5%	0%
CN China	97%	0%	3%
TW Taiwan	60%	0%	40%
GB United Kingdom	75%	0%	25%
RU Russian Federation	91%	9%	0%
Others			

Table 85. Wire rod 2022 Eu supply mix and shares of production using technologies BF-BOF, DRI-EAF and EAF in each of the EU trade partners. These shares were estimated based on actual wire rod production in each of the countries.

Country	Supply mix	Share BF-BOF	Share DRI-EAF	Share EAF
Imports				
TR Türkiye	297	40%	0%	60%
GB United Kingdom	343	100%	0%	0%
MY Malaysia	331	79%	0%	21%
UA Ukraine	222	100%	0%	0%
CH Switzerland	198	0%	0%	100%
RU Russia	173	100%	8%	0%
ID Indonesia	155	0%	17%	83%
Other	965			
Total	2685			
Exports				
CH Switzerland	54			
GB United Kingdom	154			
TR Türkiye	212			
US United States	178			
CA Canada	73			
RU Russia	3			
MY Malaysia	2			
Other	297			
Total	972			

Table 86. Wire rod 2035 EU supply mix considering, assuming no changes from 2022 imports and exports. Shares of production using technologies BF-BOF, DRI-EAF and EAF were updated considering announced and under construction facilities in each EU trade partner and assuming that there are no facilities closing until 2035.

Country	Share BF-BOF	Share DRI-EAF	Share EAF
Imports			
TR Turkey	36%	0%	64%
GB United Kingdom	75%	0%	25%
MY Malaysia	93%	7%	0%
UA Ukraine	100%	0%	0%
CH Switzerland	0%	0%	100%
RU Russian Federation	83%	17%	0%
ID Indonesia	19%	17%	64%
Other			

Table 87. Stainless steel 2022 EU supply mix and shares of production using technologies BF-BOF, DRI-EAF and EAF in each of the EU trade partners. These shares were estimated based on each country global steel production situation, no data was available no stainless steel production outside the EU.

Country	Supply mix	Share BF-BOF	Share DRI-EAF	Share EAF
Imports				
CN China	389	86%	0%	14%
TW Taiwan	355	64%	0%	36%
IN India	268	66%	31%	3%
TR Türkiye	173	23%	0%	77%
KR South Korea	169	63%	0%	37%
ID Indonesia	128	62%	0%	38%
ZA South Africa	123	82%	8%	11%
Other	319			
Total	1925			
Exports				
US United States	198			
GB United Kingdom	167			
TR Türkiye	151			
CH Switzerland	127			
CN China	78			
MX Mexico	66			
IN India	34			
Other	316			
Total	1137			

Table 88 Stainless steel 2035 EU supply mix considering, assuming no changes from 2022 imports and exports. Shares of production using technologies BF-BOF, DRI-EAF and EAF were updated considering announced and under construction facilities in each EU trade partner and assuming that there are no facilities closing until 2035.

Country	Share BF-BOF	Share DRI-EAF	Share EAF
Imports			
CN China	83%	0%	17%
TW Taiwan	60%	0%	40%
IN India	78%	21%	1%
TR Turkey	19%	0%	81%
KR South Korea	58%	5%	37%
ID Indonesia	78%	0%	22%
ZA South Africa	56%	12%	32%

Other			
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Table 89. Electrical sheet 2022 EU supply mix and shares of production using technologies BF-BOF, DRI-EAF and EAF in each of the EU trade partners. These shares were estimated based on each country global steel production situation, no data was available no stainless steel production outside the EU.

Country	Supply mix	Share BF-BOF	Share DRI-EAF	Share EAF
Imports				
CN China	250	86%	0%	14%
RU Russia	77	61%	9%	30%
KR South Korea	76	63%	0%	37%
TW Taiwan	64	64%	0%	36%
VN Viet Nam	51	64%	0%	36%
JP Japan	39	72%	0%	28%
IN India	20	66%	31%	3%
Other	14			
Total	590			
Exports				
IN India	92			
CH Switzerland	52			
TR Türkiye	33			
MX Mexico	21			
PK Pakistan	17			
BR Brazil	11			
US United States	8			
Other	65			
Total	298			

Table 90. Electrical sheet 2035 EU supply mix considering, assuming no changes from 2022 imports and exports. Shares of production using technologies BF-BOF, DRI-EAF and EAF were updated considering announced and under construction facilities in each EU trade partner and assuming that there are no facilities closing until 2035.

Country	Share BF-BOF	Share DRI-EAF	Share EAF
Imports			
CN China	83%	0%	17%
RU Russian Federation	57%	18%	25%
KR South Korea	58%	5%	37%
TW Taiwan	60%	0%	40%
VN Viet Nam	60%	0%	40%
JP Japan	70%	0%	30%
IN India	78%	21%	1%
Other			

Annex 7. Definitions of product aspects in ESPR

In this section, the authors have proposed definitions for the 16 product aspects of Article 5 of ESPR (Table 91). The hierarchy applied for selecting the most appropriate source for each definition was the following: the current legislation, standards and scientific literature. In some cases, the authors have adapted the definition from those sources. These definitions are fully in line with the ESPR Methodology under development (Rodríguez-Manotas et al., 2025-in preparation).

Table 91. Definitions of product aspects

Product aspect	Definition	Source
Durability	The ability of a product to maintain over time its function and performance under specified conditions of use, maintenance and repair;	ESPR
Reliability	The probability that a product functions as required under given conditions for a given duration without an occurrence which results in a primary or secondary function of the product no longer being performed	ESPR
Reusability	Ability of a product or component that is not waste to be used again for the same purpose for which it was conceived.	JRC (adapted based on Directive 2008/98/EC definition for 'Re-use')
Upgradability	Ability of a product to be accessible for implementing actions to enhance its functionality, performance, capacity, safety or aesthetics of a product.	JRC (adapted based on ESPR definition for 'Upgrading')
Reparability	Ability of a defective product or waste to return to a condition where it fulfils its intended use.	JRC (adapted based on ESPR definition for 'Repair')
Possibility of maintenance	Possibility of maintenance: ability of a product to be kept in a condition where it is able to fulfil its intended purpose through one or more actions.	JRC (adapted based on ESPR definition for 'maintenance')
Possibility of refurbishment	Possibility of refurbishment: ability of a product or a discarded product to be prepared, cleaned, tested, serviced and, where necessary repaired to restore its performance or functionality within the intended use and range of performance originally conceived at the design stage at the time of the placing of the product on the market.	JRC (adapted based on ESPR definition for 'refurbishment')
Presence of substances of concern	The presence of substances that: - meets the criteria laid down in Article 57 of Regulation (EC) No 1907/2006 and is identified in accordance with Article 59(1) of that Regulation - is classified in Part 3 of Annex VI to Regulation (EC) No 1272/2008 in one of the following hazard classes or hazard categories - is regulated under Regulation (EU) 2019/1021; or - negatively affects the reuse and recycling of materials in the product in which it is present	ESPR
Energy use and energy efficiency	<u>Energy use</u> : Use of energy in all lifecycle stages of a product <u>Energy efficiency</u> : the ratio of output of performance, service, goods or energy to input of energy.	JRC JRC
Water use and water efficiency	<u>Water use</u> : Use of water in all lifecycle stages. <u>Water efficiency</u> : The ratio of output of performance, service, goods to input of water.	JRC JRC
Resource use and resource efficiency	<u>Resource use</u> : Use of raw materials, mainly abiotic (minerals, metals, fossils), in all lifecycle stages. It can also include other biotic resources such as land, air, ecosystems. The use of natural resources can be accounted for as the volumes of resources consumed (materials) or used (land, air, ecosystems), or the impacts derived from the use of resources. Water and energy are not considered within resources under the scope of ESPR. <u>Resource efficiency</u> : The ratio of output of performance, service, goods to input of resources, raw materials, air, land, soil and ecosystem services.	JRC and (BIO Intelligence Service, DG for Enterprise and Industry (EC), 2013)
Recycled content	Proportion, by mass, of recycled material, from pre- and post-consumer waste, in a product or packaging.	ISO 14021

Possibility of remanufacturing	Possibility of producing through actions a new product from objects that are waste, products or components and through which at least one change is made that may affects the performance, purpose or type of the product.	JRC (adapted based on ESPR definition for ' <i>Remanufacturing</i> ')
Recyclability	Ability of product's waste materials to be reprocessed into products, materials or substances whether for the original or other purposes. It includes the reprocessing of organic material but does not include energy recovery and the reprocessing into materials that are to be used as fuels or for backfilling operations.	JRC (adapted based on Directive 2008/98/EC definition for ' <i>recycling</i> ')
Possibility of recovery of materials	Ability of the product waste generated to be recovered through any recovery operation, other than energy recovery and the reprocessing into materials that are to be used as fuels or other means to generate energy. It includes, inter alia, preparing for re-use, recycling and backfilling.	JRC (adapted based on Directive 2008/98/EC definition for ' <i>material recovery</i> ')
Environmental impacts	Any change to the environment, whether adverse or beneficial, wholly or partially resulting from a product during its life cycle	ESPR
Expected generation of waste	Generation of any substance, or object that the holder discards or intends or is required to discard.	JRC (adapted based on Directive 2008/98/EC definition for ' <i>waste</i> ' and ' <i>hazardous waste</i> ')

Annex 8. Baseline scenario

Demand projections

DRAFT

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